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Lucian Bădescu · Ettore Carletti

Lectures on Geometry

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Lectures on Geometry

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To my family
Miruna and Ciprian
Matei and Eliza
Cosmin and Alexandra
Cornel and Iuliana
Mihaela

Lucian Badescu

To my nephews
Francesco and Anna
Beatrice and Chiara

Ettore Carletti

Preface

Μηδείς ἀγεωμέτρητος εἰσίτω

The aim of this book is to offer some fundamentals of the affine, euclidean and projective geometries which are indispensable for any course in mathematics and, in general, useful for the students of scientific faculties. We are also strongly convinced that a comparative study of these geometries is essential for a true understanding of their concepts and methods. Nevertheless, one can use the first five chapters as a course in affine and euclidean geometry independently from the remaining eight chapters. As a companion to this book, we recommend [12], which is a very good collection of exercises in projective geometry.

The first ten chapters give an outline of the affine, euclidean and projective geometries a little wider than usually requested by the academic programs. Base fields have been chosen arbitrarily (not only $\mathbb{R}$ and $\mathbb{C}$) in order to have a more complete point of view, unless otherwise required. We have followed an axiomatic-synthetic approach as well as an analytic one which permit a deeper comprehension and more essential and elegant exposition of various subjects.

The first chapter is devoted to explaining those concepts and results from linear algebra absolutely necessary for the following chapters.

In Chap. 2, we study symmetric bilinear forms defined on vector spaces over arbitrary fields of characteristic $\neq 2$. Sylvester's theorem on real and complex symmetric bilinear forms is the core of this chapter. This important result is essential for classifying real and complex hyperquadrics. We have concluded this chapter by introducing Gram determinants and cross-products which are indispensable tools in euclidean geometry of dimension ≥ 3.

Chapter 3 is devoted to affine spaces of finite dimension over an arbitrary field, presented from a synthetic as well as an analytic point of view. Barycentric coordinates play a central role in this exposition; classical theorems by Menelaus, Ceva and Thales are proved by means of them. A careful treatment of affine morphisms

ends with the structure theorem of the group of automorphisms $\mathrm{Aut}_K(\mathscr{A})$ of an affine space $\mathscr{A}$ (Proposition 3.57).

In Chap. 4, some euclidean geometry in dimension $n \geq 2$ is presented (also using the cross-product defined in Chap. 2). The core of this chapter is the structure theorem of the group of isometries $\mathrm{O}(\mathscr{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. of a euclidean space $(\mathscr{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ (Theorem 4.35).

After a brief introduction to the affine hypersurfaces of $\mathbb{A}^n(K)$, Chap. 5 is devoted to the study of hyperquadrics of $\mathbb{A}^n(K)$, where K is an arbitrary field of characteristic $\neq 2$. A reduced form for the hyperquadrics is given by Theorem 5.37, followed by the euclidean classification of the hyperquadric of $\mathbb{E}^n(\mathbb{R})$. At the end of this chapter, an affine version of Bézout's theorem is proved and used to demonstrate the affine Pascal's theorem.

Chapters 6, 7 and 8 are mainly devoted to the study of projective spaces associated with a vector space over an arbitrary field. Some synthetic projective geometry is developed in order to introduce the crucial Desargues' axiom (the entire Chap. 7) and Pappus' theorem (Chap. 8). The group $\mathrm{Aut}(\mathbb{P}^n(K))$ of projective automorphisms of $\mathbb{P}^n(K)$, and in particular, the structure of the general linear group $\mathrm{PGL}_n(K)$ is treated in Chap. 8. At the end of Chap. 8, we give a geometric characterization of the base field K by means of two classes of general linear automorphisms: the projective translations and the projective homotheties. They are used in Chap. 9 for a synthetic presentation of the cross-ratio.

In the first part of Chap. 9, the basic links between affine geometry and projective geometry are outlined. Then the fundamental concept of cross-ratio is introduced from a synthetic and analytic point of view. At the end of this chapter is proved a classical theorem by von Staudt that puts the importance of harmonic cross-ratio in evidence.

Projective hypersurfaces of $\mathbb{P}^n(K)$ are briefly introduced in Chap. 10, which is mainly concerned with the projective hyperquadrics of $\mathbb{P}^n(K)$, where $\mathrm{char}(K) \neq 2$. The projective classification of hyperquadrics is given by Theorems 10.53, 10.55 and 10.63 for an arbitrary field K ($\mathrm{char}(K) \neq 2$), for the complex field $\mathbb{C}$ and for the real field $\mathbb{R}$ respectively. Polarity with respect to a hyperquadric of $\mathbb{P}^n(K)$ is a main theme of this chapter. The projective point of view of various concepts of the affine and euclidean theory of hyperquadrics (in particular conics and quadrics) is shown to be very fruitful by various examples also in the exercises section.

Bézout's theorem for projective curves of $\mathbb{P}^2(K)$, where K is an arbitrary algebraically closed field, is proved in a particular case and in its general form in Chap. 11. We have adopted the axiomatic definition of intersection multiplicity of two curves as given in [19] in order to keep the necessary commutative algebra to a minimum. Several important geometric applications such as Pascal's and Brianchon's theorems, the inflection points of a curve of degree ≥ 3, Legendre form of cubic curves, and linear systems of conics passing through a finite number of points are exposed in a detailed form.

Absolute planes are axiomatically introduced and studied in Chap. 12. An important example is the Poincaré hyperbolic plane which is built at the end of this chapter. The "projective nature" of this geometry is proved by the relation (12.13)

which expresses the hyperbolic distance by means of a cross-ratio. This link between projective geometry and non-euclidean geometries is the central theme of Chap. 13 which is concerned with so-called Cayley–Klein geometries. Euclidean, hyperbolic and elliptic geometries are Cayley–Klein geometries, but other geometries like Minkowskian geometry can be "projectively" generated. We give some details on all projective geometries of $\mathbb{P}^1(\mathbb{R})$ and of $\mathbb{P}^2(\mathbb{R})$ and also a brief account on some geometries of $\mathbb{P}^3(\mathbb{R})$. A generalization of these geometries to $\mathbb{P}^n(\mathbb{R})$ according to [37] is mentioned at the end this chapter. This exposition is obviously by far incomplete, but its scope is to stir the reader's interest in this topic. Classical references on Cayley–Klein geometries are [10], [20], [33], but we would also like to mention [9], [27], [36], [35] and [37].

We wish to thank our friend Giacomo Monti Bragadin for his contributions to various chapters of this book.

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The second author wishes to pay a heartfelt tribute to the memory of Lucian Badescu, dearest friend and master.

Genoa, Italy L. Bădescu
April 2023 E. Carletti

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Chapter 1
Linear Algebra



1.1 Introductory Notions

Notation i.e. (from latin *id est*) means "namely"; e.g. (from latin *exempli gratia*) means "for instance".

We will denote the sets of the natural numbers $\{1, 2, \dots, \}$, the integers, the rationals, the reals and the complex numbers by $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ respectively.

Let X be a non-empty set. The cartesian product $\underbrace{X \times \cdots \times X}_{n \text{ times}}$ will be denoted by X^n. We shall denote the set of all parts of a set X by $\mathcal{P}(X)$ and the cardinal number of X (i.e. "the number of elements" of X) by $\#(X)$. We shall not be concerned with cardinal arithmetic but we have only to recall some basic definitions. A set X is *finite* and $\#(X) = n$ if there is a bijective map $X \to \{1, \dots, n\}$ for some natural n. The empty set $\varnothing$ is also considered finite with $\#(\varnothing) = 0$. Set that is not finite will be called *infinite*. We simply write $\#(X) = \infty$ to say that X is infinite and $\#(X) < \infty$ when X is finite. A set X is *countable* and $\#(X) = \aleph_0$ if there is a bijective map $X \to \mathbb{Z}$. An infinite set is *uncountable* if it is not countable, e.g. $\mathbb{R}$ and $\mathbb{C}$.

Definition 1.1 (Equivalence Relations) Let X be a non-empty-set. A non-empty subset $R \subset X \times X$ is called a *binary relation* in X. If $x, y \in X$ and $(x, y) \in R$ it is customary to write xRy and say that x is *in the relation R to y*. For instance $R := \{(x, x) | x \in X\}$ (the diagonal of $X \times X$) is the relation of equality in X.

A binary relation R in a set X is an *equivalence relation* if the following conditions are satisfied:

(i) xRx for all $x \in X$ (Reflexivity).
(ii) xRy implies yRx for all $x, y \in X$ (Symmetry).
(iii) xRy and yRz imply xRz for all $x, y, z \in X$ (Transitivity).

If R is an equivalence relation and xRy we will say that x and y are *equivalent*. The *equivalence class* or *coset* of an element $x \in X$ is the subset $\widehat{x} := \{y \in X \mid xRy\}$.

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Equivalence classes determine the equivalence relation uniquely and they satisfy the following properties:

(i) $x \in \widehat{x}$ for every $x \in X$.
(ii) $X = \bigcup_{x \in X} \widehat{x}$.
(iii) For every $x, y \in X$, $\widehat{x} \cap \widehat{y} \neq \varnothing$ if and only if $\widehat{x} = \widehat{y}$, i.e. if and only if xRy.

Thus the collection of distinct equivalence classes is a partition of X. Conversely, to any partition $\{U_i : i \in I\}$ of a set X we can associate the equivalence relation in X

$$xRy \iff x, y \in U_i \quad \text{for some index } i \in I.$$

The set of equivalence classes $X/R := \{\widehat{x} \,|\, x \in X\}$ is called the *quotient set* of X under the relation R (or of X mod R) . The well-defined map $\pi : X \to X/R$, $\pi(x) = \widehat{x}$ is called a *quotient map*. A *complete system of representatives of* X mod R is a family $\{x_i\}_{i \in I}$ of elements of X such that:

(i) For each $(i, j) \in I \times I$, $x_i R x_j$ if and only if $i = j$.
(ii) $X = \bigcup_{i \in I} \widehat{x_i}$.

An equivalence relation R is most frequently denoted by $\sim$ so we will write $x \sim y$ instead of xRy and $X/\sim$ instead of X/R.

Examples 1.2

(1) Given a natural number $n \geq 2$ and two integers a and b, we say that a is *congruent b modulo n*, and we write

$$a \equiv b \pmod{n} \tag{1.1}$$

is $a - b = kn$ for some $k \in \mathbb{Z}$. The congruenc//e (1.1) is the equivalent relation R_n in $\mathbb{Z}$ defined by

$$R_n = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \,|\, x - y \in n\mathbb{Z}\}.$$

The set $\{0, 1, \ldots, n-1\}$ is a complete system of representatives of $\mathbb{Z}$ mod R_n. We will denote the quotient set $\mathbb{Z}/R_n$ by $\mathbb{Z}_n$.

(2) It easy to see that $R = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \,|\, x - y \in \mathbb{Z}\}$ represents an equivalent relation in $\mathbb{Q}$. The set $\{x \in \mathbb{Q} \,|\, 0 \leq x < 1\}$ is a complete system of representatives of $\mathbb{Q}$ mod R.

The next two subsections are devoted to recalling some basic definitions and results concerning groups, rings and fields. Very good references are [6], [7], [22] and [39].

Groups

A *binary operation* in a non-empty set G is a map $G \times G \to G$. It is customary to denote the image of the pair $(g, h) \in G \times G$ by $g \cdot h$ (or simply gh), "the

multiplicative notation", or $g + h$, "the additive notation". We will write $(G, \cdot)$, $(G, +)$ to show our choice.

Definition 1.3 (Groups) A non-empty set G is called a *group* if there is binary operation $\cdot$ in G such that the following conditions are satisfied:

1. If $g, h, k \in G$, then $(gh)k = g(hk)$ (associative law).
2. There is $u \in G$ such that $ug = gu = g$ for every $g \in G$. The element u is unique and it is called the *identity* of G (denoted by 1 (or 1_G). Other names are *neutral element*, *unity* and *zero* (denoted by 0 (or 0_G), when the operation is denoted additively).
3. For every $g \in G$ there is $x_g \in G$ such that $gx_g = x_g g = 1$. The element x_g, which is unique, is called the *inverse of* g and denoted by g^{-1} (in additive notation it is the *opposite* of g, denoted by $-g$). We define the nth-power of $g \in G$ for $n \in \mathbb{Z}$ as follows:

$$g^n = \underbrace{g \cdots g}_{n \text{ times}}, \quad \text{if } n > 0, \qquad g^n = \underbrace{g^{-1} \cdots g^{-1}}_{-n \text{ times}} \quad \text{if } n < 0, \qquad g^0 = 1_G.$$

Therefore one has $g^{n+m} = g^n g^m$ and $g^{nm} = (g^n)^m$ for $n, m \in \mathbb{Z}$. The Reader can formulate an analogue definition for additive groups. We define the *order* or the *period* of an element $g \in G$ as the smallest positive integer n such that $g^n = 1_G$ if this integer n exists, otherwise we say that g has *infinite order*.

The cardinal number $\#(G)$ of G will be called the *order* of G and denoted also by $|G|$. The *centre* of a group G is the subset $\mathcal{C}(G)$ consisting of all elements $g \in G$ such that $gh = hg$ for all $h \in G$. It is obvious that $\mathcal{C}(G)$ is a subgroup of G. A group G is *commutative* or *abelian* if $\mathcal{C}(G) = G$ i.e. $gh = hg$ for every $g, h \in G$. The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are abelian groups with respect to the usual sum while $\mathbb{Q} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$ are abelian groups with respect to the usual product. If X has more than 2 elements the set of bijective maps $X \to X$ is a non-commutative group with respect to the composition of applications.

Let G be a group and S a subset of G. We shall say that S *generates* G (and write $G = \langle S \rangle$) if for every $g \in G$, $g = s_1 \cdots s_r$, where $s_i \in S$ or $s_i^{-1} \in S$ for $i = 1, \ldots, r$. We shall call *cyclic* a group G if there exists an element $a \in G$ such that every element $g \in G$ can be written in the form a^n for some $n \in \mathbb{Z}$ i.e $G = \langle \{a\} \rangle$. A cyclic group is obviously abelian.

A non-empty set $H \subset G$ is a *subgroup* of G if $gh^{-1} \in H$ whenever $g, h \in H$. We can associate to every subgroup H two equivalence relations defined by

$$g \sim_{\mathrm{r}} g' \iff g(g')^{-1} \in H, \qquad g \sim_{\mathrm{l}} g' \iff g^{-1}g' \in H.$$

The equivalence classes of $\sim_{\mathrm{r}}$ are the sets $Hg := \{hg : h \in H\}$ and they are called *right cosets*, while $gH := \{gh : h \in H\}$ are the equivalence classes of $\sim_{\mathrm{l}}$ and they are called *left cosets*. The quotient spaces will be denoted by $H\backslash G$ and G/H respectively. The natural map $H\backslash G \to G/H$, $Hg \to gH$, is bijective so that

$\#(H\backslash G) = \#(G/H)$. Thus we call $\#(G/H)$ the *index* of H in G and denote it by $[G : H]$. It can be proved that

$$|G| = [G : H]|H|.$$

This is immediate if G is a finite group i.e. $|G| < \infty$.

Definition 1.4 A subgroup H is *normal* if $gH = Hg$ for every $g \in G$. Therefore $H\backslash G = G/H$ and there is a well-defined binary operation in G/H

$$gH \cdot g'H = gg'H, \qquad \forall\, g, g' \in G,$$

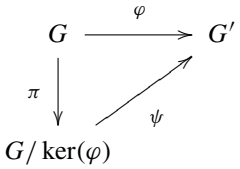
so that G/H is a group, called *the quotient group of G with respect to H* (or also *the factor group of G by H*). If $G = \mathbb{Z}$ and $H = n\mathbb{Z} := \{nm \mid m \in \mathbb{Z}\}$, the additive group G/H will be denoted by $\mathbb{Z}_n$. The centre $\mathcal{C}(G)$ is a normal subgroup of G.

Let G and G' be two groups. A map $\varphi\colon G \to G'$ is a *homomorphism of groups* (or a *group-homomorphism*) if $\varphi(gh) = \varphi(g)\varphi(h)$ for all $g, h \in G$.

The following statements are very easy to prove.

(a) $\varphi(1_G) = 1_{G'}$ and $\varphi(g^{-1}) = (\varphi(g))^{-1}$ for all $g \in G$.
(b) If H is a subgroup of G then $\varphi(H)$ is a subgroup of G'.
(c) If L is a subgroup of G' then $\varphi^{-1}(L)$ is a subgroup of G; in particular $\ker(\varphi) := \varphi^{-1}(\{1_{G'}\})$ is a normal subgroup of G called the *kernel* of φ. We can deduce at once that φ is injective if and only if $\ker(\varphi) = \{1_G\}$.
(d) If H is a normal subgroup of G, the quotient map $\pi\colon G \to G/H$ is a homomorphism of groups.
(e) If $\varphi\colon G \to G'$ is bijective homomorphism of groups, its inverse map $\varphi^{-1}\colon G' \to G$ is a homomorphism of groups. A bijective homomorphism of groups is called an *isomorphism of groups*.

Theorem 1.5 (First Isomorphism Theorem) *Let $\varphi\colon G \to G'$ be a homomorphism of groups, then there is a unique homomorphism of groups $\psi\colon G/\ker(\varphi) \to G'$ such that the diagram*



is commutative and $\psi\colon G/\ker(\varphi) \to \varphi(G)$ is an isomorphism.

Definition 1.6 (Transformation Groups) Let G be a group and X a non-empty set. We say that G *acts on the left* on X if there is a map $G \times X \to X$ which, denoting by $g \odot x$ the image of (g, x), satisfies the following conditions:

(i) For every $g, g' \in G$ and $x \in X$, $g'g \odot x = g' \odot (g \odot x)$.
(ii) For every $x \in X$, $1_G \odot x = x$.

For every $g \in G$ the map $X \to X$, defined by $x \to gx$, is bijective, thus we call the triple $(G, X, \odot)$ *transformation groups*. In a very similar manner an *action on the right* of a group on a set is defined. There is a bijective correspondence between the left actions and the right actions of a group G on a set X: $g \odot x \longleftrightarrow x \odot g^{-1}$.

The set $\mathrm{Orb}(x) := \{g \odot x \mid g \in G\}$ is called the *orbit* of x. For every $x \in X$ the subset $O(x) := \{g \in G \mid ; g \odot x = x\}$ is a subgroup of G which is called an *isotropy subgroup* (or *stabilizer*) of x. For any fixed $x \in X$ the map $f_x : G \to \mathrm{Orb}(x)$, defined by $f_x(g) = g \odot x$, is surjective and $f_x(g) = f_x(h)$ if and only if $(h^{-1}g) \odot x = x$, i.e. $h^{-1}g \in O(x)$. Then f_x induces the natural bijective map $F_x : G/O(x) \to \mathrm{Orb}(x)$, $F_x(\widehat{g}) := g \odot x$ with $\widehat{g} := gO(x)$. Every action of G on X induces an equivalence relation in X:

$$x \sim y \quad \Longleftrightarrow \quad y \in \mathrm{Orb}(x), \qquad \forall\, x, y \in X.$$

The equivalence class of x is nothing else than $\mathrm{Orb}(x)$ and the quotient space $X/\sim$ is the set of all orbits.

It is worthwhile to observe that an action of a group G on a set X induces an action of any subgroup H of G on X.

Rings and Fields

Definition 1.7 (Rings) Let A be a non-empty set in which two binary operations, $+$ (addition) and $\cdot$ (multiplication), are given is called a *ring* if the following conditions are satisfied:

1. A is an abelian group with respect to addition.
2. If $a, b, c \in A$, then $a(bc) = (ab)c$ (associative law).
3. If $a, b, c \in A$, then $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ (distributive laws).

A ring A is *unitary* if there is an element $\neq 0_A$, denoted by 1 or 1_A, such that $a \cdot 1 = 1 \cdot a = a$ for every $a \in A$; such an element is unique and is called the *identity* of A. The set of integers $\mathbb{Z}$ is an abelian unitary ring, as well as the "ring of residue classes $\pmod n$" $\mathbb{Z}_n$. If A is a commutative unitary ring, the set of polynomials in $n \geq 1$ indeterminates $A[X_1, \ldots, X_n]$ is a commutative unitary ring with respect the usual operations on polynomials. We shall see that the set $\mathcal{M}_n(R)$ of square matrices of order $n \geq 2$ (Definition 1.14) with entries in a commutative unitary ring R is a non-abelian unitary ring.

From Now on, Ring Will be Synonymous with Unitary Ring

The *centre* of a ring A is the subset $\mathcal{C}(A)$ consisting of all elements of $a \in A$ such that $ax = xa$ for all $x \in A$. It is obvious that $\mathcal{C}(A)$ is a subring of A. A ring A is *commutative* if $\mathcal{C}(A) = A$ i.e. $ab = ba$ for every $a, b \in A$.

A subgroup B of a ring A containing 1_A is called a *subring* of A if $xy \in B$ whenever $x, y \in B$. An *left-ideal* $\mathfrak{A}$ of A is a subgroup of A such that $A\mathfrak{A} := \{xy :$

$x \in A,\ y \in \mathfrak{A}\} \subset \mathfrak{A}$. To define a *right-ideal* we require $\mathfrak{A}A \subset \mathfrak{A}$. A *two-sided ideal* is a subset which is both a left and a right ideal. If A is commutative every ideal is two-sided, thus we call it simply ideal.

An element $a \in A$ is a *unit* (or *invertible*) if there exists $b \in A$ such that $ab = ba = 1$; this element b is unique and it is denoted by a^{-1}, the *inverse* of a. The set $A^* := \{a \in A\ :\ a \text{ is invertible}\}$ is a group with respect to product. If $A^* = A \setminus \{0\}$, A is called a *division ring*. A commutative division ring is called a *field* while a non-commutative division ring is called a *skew-field*. Familiar examples of fields are $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{Z}_p$ if p is a prime number.

Let A be a ring and $\mathfrak{A}$ a two-sided ideal of A. Since $\mathfrak{A}$ is a normal subgroup of A, $A/\mathfrak{A}$ is an abelian group. Furthermore there is a well-defined product in $A/\mathfrak{A}$:

$$\widehat{x} \cdot \widehat{y} := \widehat{xy}, \qquad x, y \in A.$$

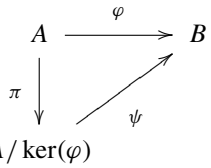
Thus $A/\mathfrak{A}$ is a unitary ring called *the factor ring of A by $\mathfrak{A}$*. If A is commutative $A/\mathfrak{A}$ is commutative.

Let A and B be two rings and $\psi\colon A \to B$ a map. We say that ψ is a *homomorphism of rings* if it is a homomorphism of additive groups, $\psi(aa') = \psi(a)\psi(a')$, for every $a, a' \in A$ and $\psi(1_A) = 1_B$.

One can easily prove the following statements.

(a) If C is a subring of A then $\psi(C)$ is a subring of B.
(b) If $\mathfrak{B}$ is a left-ideal (right-ideal, subring) of B then $\psi^{-1}(\mathfrak{B})$ is an left-ideal (right-ideal, subring) of A.
(c) We call the set $\ker(\psi) := \psi^{-1}(0_B)$ the *kernel* of ψ . The kernel of a homomorphism of rings is a two-sided ideal and $\ker(\psi) = \{0_A\}$ if and only if ψ is injective.
(d) If $\mathfrak{A}$ is a two-sided ideal of A, the quotient map $\pi\colon A \to A/\mathfrak{A}$ is a homomorphism of rings.
(e) If $\varphi\colon A \to B$ is a bijective homomorphism of rings, its inverse map $\varphi^{-1}\colon B \to A$ is a homomorphism of rings. A bijective homomorphim of rings is called an *isomorphism of rings*.

Theorem 1.8 (First Isomorphism Theorem) *Let $\varphi\colon A \to B$ be a homomorphism of rings, then there is a unique homomorphism of rings $\psi\colon A/\ker(\varphi) \to B$ such that the diagram*



is commutative and $\psi\colon A/\ker(\varphi) \to \varphi(A)$ is an isomorphism.

Let $\mathfrak{A}$ be a left-ideal of a ring A. A set $S \subset \mathfrak{A}$ *generates* $\mathfrak{A}$ if all elements of $\mathfrak{A}$ can be written in the form

$$\sum_{i=1}^{r} a_i s_i, \qquad a_i \in A,\ s_i \in S, \quad i = 1, \ldots, r. \tag{1.2}$$

A left-ideal $\mathfrak{A}$ is *principal* if it is generated by an element of A, i.e there exists $s \in A$ such that $\mathfrak{A} = As := \{as : a \in A\}$. For right-ideals we have to put the elements a_i of A in (1.2) on the right. From elementary mathematics we see that the only ideals of $\mathbb{Z}$ are principal of the form $n\mathbb{Z}$ with $n \geq 0$.

Up to the End of this Subsection We Suppose that A Is a Commutative Ring
If $a \neq 0$ and $b \neq 0$ are elements of A, we say that b *divides* a if there exists $c \in A$ such that $a = bc$; we write $a \mid b$. We say also that b is a *divisor* or a *factor* of a.

A *zero-divisor* (or 0-*divisor*) of A is an element $a \in A$ such that there is $b \in A$, $b \neq 0$, with $ab = 0$; a 0-divisor $a \neq 0$ is called a *proper* 0-divisor. If A does not contain proper zero-divisors it is called an *integral domain* or simply a *domain*. An non-unit element $a \in A$, $a \neq 0$,

is *irreducible* if $a = bc$ implies b or c is a unit;
is *prime* if from $a \mid bc$ it follows that $a \mid b$ or $a \mid c$.

Every prime element is irreducible but the converse is not always true. Two elements a and b are *associates* if $a = ub$, where u is a unit.

An integral domain A is a *unique factorization domain* (*UFD*) or a *factorial domain* if for every non-unit $a \in A$ there exist uniquely determined irreducible elements $p_1, \ldots, p_r$, such that

$$a = p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}, \qquad n_j \in \mathbb{N},\ = 1, \ldots, r.,$$

where p_i and p_j are not associates if $i \neq j$. The above factorization of a is unique up to order and unit factors. The only divisors of a are of the form

$$p_1^{m_1} p_2^{n_2} \cdots p_r^{m_r}, \qquad 0 \leq m_j \leq n_j,\ j = 1, \ldots, r.,$$

The ring of rational integers $\mathbb{Z}$ is a unique factorization domain by the fundamental theorem of arithmetics and every field is trivially a unique factorization domain. We observe that in a factorial domain every irreducible element is prime. Two elements a and b of a factorial domain A,

$$a = p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}, \qquad b = q_1^{m_1} q_2^{m_2} \cdots q_s^{m_s},$$

are *coprime* if they do not have common irreducible divisors, i.e. if each irreducible factor p_i of a is not associated to any irreducible factor q_j of b. In a factorial domain every irreducible element is prime.

The following two theorems are crucial for this chapter and for Chap. 10. Their proofs can be found in [39].

Theorem 1.9 (Gauss) *If A is a factorial domain also $A[X_1, \ldots, X_n]$ is a factorial domain for every $n \geq 1$. In other words, every non-constant polynomial F is factorized in a unique way up to the order and unit factors in the form*

$$F = F_1^{m_1} \cdots F_r^{m_r} \tag{1.3}$$

where $F_1, \ldots, F_r$ are pairwise coprime irreducible polynomials and $m_i \in \mathbb{N}$, $i = 1, \ldots, r$. Furthermore, F is homogeneous if and only if $F_1, \ldots, F_r$ are homogeneous polynomials.

If in (1.3) we have $m_1 = m_2 = \cdots = m_r = 1$, F is called *reduced*.

Theorem 1.10 *Let A be an integral domain. Let $f(X)$, $g(X) \in A[X]$ be two polynomials of respective degrees m and n. If a is the leading coefficient of $g(X)$ there exist polynomials $q(X)$, $r(X) \in A[X]$ such that*

$$a^k f(X) = q(X)g(X) + r(X),$$

where $k = \max(m - n + 1, 0)$ and $r(X)$ is either of degree less than n or is the zero polynomial. Furthermore, $g(X)$ and $r(X)$ are uniquely determined. In particular $b \in A$ is a root of $f(X)$ if and only if $X - b$ divides $f(X)$. Hence a non-zero polynomial $f(X)$ of degree m can have at most m distinct roots.

Definition 1.11 (Characteristic of a Field) Let K be a field. The map $\varphi\colon \mathbb{Z} \to K$ defined by $\varphi(n) = n \cdot 1_K$, where

$$n \cdot 1_K := \begin{cases} \underbrace{1_K + \cdots + 1_K}_{n \text{ times}} & \text{if } n > 0 \\ 0_K & \text{if } n = 0 \\ \underbrace{(-1_K) + \cdots + (-1_K)}_{-n \text{ times}} & \text{if } n < 0 \end{cases}$$

is a homomorphism of rings. Then $\mathrm{Ker}(\varphi)$ is an ideal of $\mathbb{Z}$, say $\mathrm{Ker}(\varphi) = p\mathbb{Z}$ with $p \geq 0$.

If $p = 0$, i.e. φ is injective, we say that K has *characteristic* 0. The homomorphism $\varphi\colon \mathbb{Z} \to K$ can be uniquely continued to a homomorphism $\widetilde{\varphi}\colon \mathbb{Q} \to K$ defined by

$$\widetilde{\varphi}\left(\frac{m}{n}\right) = \frac{\varphi(m)}{\varphi(n)},$$

which is injective. Thus every field K of characteristic 0 contains a subfield H isomorphic to the field of rational numbers $\mathbb{Q}$ and H is the intersection of all subfields of K.

If $p > 0$, then p is the smallest positive integer such that $p \cdot 1_K = 0_K$ and we say that K has *characteristic* p. Moreover p is a prime number, otherwise if $p = mn$, with $1 < n < p, 1 < m < p$, then $(m \cdot 1_K)(n \cdot 1_K) = p \cdot 1_K = 0_K$ that would imply $m \cdot 1_K = 0_K$ or $n \cdot 1_K = 0_K$ contradicting the assumption that p is the smallest positive integer such that $p \cdot 1_K = 0_K$. By Theorem 1.8 K contains a subfield H isomorphic to the field $\mathbb{Z}_p$. This field H is the intersection of all subfields of K.

In both cases, we denote the characteristic of K by $\text{char}(K)$ and we call the smallest field contained in K the *prime field* in K.

Definition 1.12 A field K is *algebraically closed* if every polynomial $P(X) \in K[X]$ of degree ≥ 1 has at least a root in K. The fundamental theorem of algebra asserts that the field $\mathbb{C}$ of complex numbers is algebraically closed whereas the field $\mathbb{R}$ of real numbers is not algebraically closed because the polynomial $X^2 + 1$ has no real roots. Moreover, for every field K there is an algebraically closed field $\overline{K}$, which is an algebraic extension of K, i.e. every element of $\overline{K}$ is a root of a polynomial $f(X) \in K[X]$. This field $\overline{K}$ is unique up to K-isomorphism and it is called the *algebraic closure* of K (see Chapters V and VI of [22]); for instance the field $\mathbb{C}$ is the algebraic closure of $\mathbb{R}$.

Lemma 1.13 *Every algebraically closed field is infinite.*

Proof Assume that $K = \{a_1 = 0, a_2 = 1, a_3, \ldots, a_m\}$. Then the polynomial

$$P(X) = (X - a_1)(X - a_2) \cdots (X - a_m) + 1$$

has no roots in K since $P(a_i) = 1 \neq 0$ for $i = 1, 2, \ldots, m$, contradicting our hypothesis. □

1.1.1 Matrices and Linear Systems

In this subsection we recall some basic notions concerning matrices, determinants and linear systems that will be frequently used. We have thought it convenient to consider matrices with entries in a commutative ring (and not only in a field) in view of concepts like the characteristic polynomial of a matrix (Definition 1.116) and the resultant of two polynomials (Chap. 11). As a matter of fact all proofs of the here presented statements do not change if the entries of a matrix belong to a field or to a commutative ring. The reader can find all proofs (for fields) in [14] and [15]. We suggest [6–8] and [24] for advanced treatments of linear algebra over a commutative ring.

Definition 1.14 Let R be a commutative ring with identity 1. A rectangular array

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

of mn scalars (i.e. elements of R) is called a *matrix of m rows and n columns* or, in brief, an $m \times n$*-matrix*. The scalars a_{ij} are called the *entries*. In the following we shall also use the abbreviated form $A = (a_{ij})_{\substack{i=1,\ldots,m \\ j=1,\ldots,n}}$ or simply (a_{ij}). The set of all $m \times n$-matrices with entries in R will be denoted by $\mathcal{M}_{m,n}(R)$.

An $n \times n$-matrix is called a *square matrix*. The set of $n \times n$-matrices will be denoted by $\mathcal{M}_n(R)$. A square matrix $A = (a_{ij})_{i,j} \in \mathcal{M}_n(R)$ is

- *diagonal* if $a_{ij} = 0, \forall\, i \neq j, i, j = 1, \ldots, n$;
- *upper triangular* if $a_{ij} = 0, \forall\, i > j, i, j = 1, \ldots, n$;
- *lower triangular* if $a_{ij} = 0, \forall\, i < j, i, j = 1, \ldots, n$.

We shall denote a diagonal matrix $A = (a_{ij})_{i,j} \in \mathcal{M}_n(R)$ by $\operatorname{diag}(d_1, \ldots, d_n)$ where $d_i = a_{ii}$, $i = 1, \ldots, n$. The *main diagonal* of a square matrix $A = (a_{ij})_{i,j} \in \mathcal{M}_n(R)$ is the list of entries $a_{11}, a_{22}, \ldots, a_{nn}$.

We define the *sum* of two matrices $A = (a_{ij})_{i,j}$ and $B = (b_{ij})_{i,j} \in \mathcal{M}_{m,n}(R)$ as the matrix

$$A + B = (a_{ij})_{i,j} + (b_{ij})_{i,j} := (a_{ij} + b_{ij})_{i,j} \in \mathcal{M}_{m,n}(R)$$

and the *product of a scalar $r \in R$ and a matrix $A = (a_{ij})_{i,j} \in \mathcal{M}_{m,n}(R)$* as the matrix

$$rA := (ra_{ij})_{i,j} \in \mathcal{M}_{m,n}(R).$$

The reader can easily prove the following properties which hold for every A, B, $C \in \mathcal{M}_{m,n}(K)$, $h, k \in R$:

(1)	$A + B = B + A$	commutativity,
(2)	$A + (B + C) = (A + B) + C$	associativity,
(3)	$k(hA) = (kh)A$	compatibility,
(4)	$k(A + B) = kA + kB$	distributivity,
(5)	$(k + h)A = kA + hA$	distributivity,
(6)	$\mathbf{0}_{m,n} + A = A$	neutral element,
(7)	$1A = A$	unitarity,
(8)	$A + (-A) = \mathbf{0}_{m,n}$,	opposite element,
(9)	$0A = \mathbf{0}_{m,n}, \quad (0 \in R)$,	
(10)	$k\mathbf{0}_{m,n} = \mathbf{0}_{m,n}$,	

where $\mathbf{0}_{m,n} = (a_{ij})_{i,j}$ with $a_{ij} = 0$, $1 \le i \le m$, $1 \le j \le n$, and $-A = (-a_{ij})_{i,j}$ if $A = (a_{ij})_{i,j}$. Hence $\mathcal{M}_{m,n}(R)$ is an abelian group with respect to sum (as a matter of fact $\mathcal{M}_{m,n}(R)$ is an R-module (see Remark 1.32)).

Interchanging rows and columns of a matrix $A = (a_{ij})_{i,j} \in \mathcal{M}_{m,n}(R)$ we obtain from A the *transposed matrix*

$$A^t := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix} \in \mathcal{M}_{n,m}(R).$$

For all $A, B \in \mathcal{M}_{m,n}(R)$ and $r \in R$, we have immediately

$$(A + B)^t = A^t + B^t, \qquad (rA)^t = rA^t.$$

Definition 1.15 The *product* of the $m \times n$-matrix $A = (a_{ij})_{i,j}$ and the $n \times p$-matrix $B = (b_{jk})_{i,j}$ is the $m \times p$-matrix $(c_{ik})_{i,k}$, denoted by AB, where

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk} \quad i = 1, \ldots, m,\ k = 1, \ldots, p.$$

The matrix-multiplication has the following properties which hold for every $A, B \in \mathcal{M}_{m,n}(R)$ and $C, D \in \mathcal{M}_{n,p}(R)$, $k \in R$:

(1) $(A + B)C = AC + BC$ distributivity,
(2) $A(C + D) = AC + AD$ distributivity,
(3) $A(kC) = k(AC) = (kA)C$ homogeneity,
(4) $AI_n = A$, $I_n C = C$ unitarity,

where $I_n = (\delta_{ij})_{i,j} \in \mathcal{M}_n(R)$ is the *unit matrix*. Furthermore, if $E \in \mathrm{M}_{m,n}(R)$, $F \in \mathrm{M}_{n,p}(R)$ and $G \in \mathrm{M}_{p,s}(R)$ we have

(5) $(EF)G = E(FG)$ associativity,
(6) $(EF)^t = F^t E^t$.

Hence $\mathcal{M}_n(R)$ is a unitary ring (not commutative if $n \ge 2$) with respect to the sum and to the product introduced above.

We define the *trace* of a matrix $A \in \mathcal{M}_n(R)$ as the scalar $\mathrm{tr}(A) := \sum_{i=1}^{n} a_{ii}$. The following properties are immediate:

$$\mathrm{tr}(kA) = k\,\mathrm{tr}(A), \quad \mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B), \quad \forall\, k \in R,\ \forall\, A, B \in \mathcal{M}_n(R). \tag{1.4}$$

Thanks to (1.4) we can say that $\mathrm{tr}\colon \mathcal{M}_n(R) \to R$ is a homomorphism of R-modules (see Remark 1.67). Moreover we have

$$\mathrm{tr}(A^t) = \mathrm{tr}(A), \qquad \mathrm{tr}(AB) = \mathrm{tr}(BA), \qquad \forall\, A, B \in \mathcal{M}_n(R). \tag{1.5}$$

Definition 1.16 Let $X = \{1, 2, \cdots, n\}$. A bijective map $p\colon X \to X$ is called a *permutation* of X. It is customary to represent a permutation p by a matrix like this:

$$\begin{pmatrix} 1 & 2 & \dots & n \\ p(1) & p(2) & \dots & p(n) \end{pmatrix}.$$

The set $\mathcal{S}_n$ of all permutations of X is a group (with respect to composition of permutations) of order $n!$. There is a canonical action of $\mathbb{Z}$ on X (see Definition 1.6) associated to $p \in \mathcal{S}_n$,

$$\mathbb{Z} \times X \to X, \qquad (n, x) \to p^n(x), \tag{1.6}$$

where

$$p^n = \begin{cases} \underbrace{p \circ \cdots \circ p}_{n \text{ times}}, & \text{if } n > 0, \\ \underbrace{p^{-1} \circ \cdots \circ p^{-1}}_{-n \text{ times}} & \text{if } n < 0, \\ \mathrm{id}_X, & \text{if } n = 0. \end{cases}$$

Let m_p be the number of orbits of the action (1.6). We call the number $\varepsilon(p) = (-1)^{n-m_p}$ the *signature* of the permutation p. For instance if p is the permutation

$$\begin{pmatrix} 1\ 2\ 3\ 4 \\ 1\ 3\ 2\ 4 \end{pmatrix}$$

then $\varepsilon(p) = -1$ since p has 3 orbits: $\{1\},\ \{2, 3\},\ \{4\}$.

Definition 1.17 Let $A = (a_{ij})_{i,j} \in \mathcal{M}_n(R)$; the *determinant* of A is the scalar defined by

$$\det(A) := \sum_{p \in \mathcal{S}_n} \varepsilon(p) a_{1p(1)} a_{2p(2)} \dots a_{np(n)}.$$

The determinant of $A = (a_{ij})_{i,j}$ is also written in the form

$$\begin{vmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}.$$

For instance

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}, \quad \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

Definition 1.18 Let $A = (a_{ij})_{i,j} \in \mathcal{M}_{m,n}(R)$. For every set of indices

$$1 \le i_1 < \cdots < i_r \le m, \qquad 1 \le j_1 < \cdots < j_r \le n \tag{1.7}$$

denote by $A^{j_1,\dots,j_r}_{i_1,\dots,i_r}$ the $r \times r$-matrix consisting of the rows $i_1, \dots, i_r$ and the columns $j_1, \dots, j_r$ of A. We call $A^{j_1,\dots,j_r}_{i_1,\dots,i_r}$ a *submatrix* of A. The number of submatrices of A is clearly $\binom{m}{p}\binom{n}{p}$.
The determinant $\left|A^{j_1,\dots,j_r}_{i_1,\dots,i_r}\right| := \det\left(A^{j_1,\dots,j_r}_{i_1,\dots,i_r}\right)$ is called a *minor of order* r of the matrix A.

Let $A = (a_{ij})_{i,j} \in \mathcal{M}_n(R)$. To every set of indices

$$1 \le i_1 < \cdots < i_r \le n, \qquad 1 \le j_1 < \cdots < j_r \le n \tag{1.8}$$

we associate the $(n-m) \times (n-m)$-matrix obtained from A deleting the rows $i_1, \dots, i_m$ and the columns $j_1, \dots, j_m$ and denote it by $A^{\widehat{j_1,\dots,j_m}}_{\widehat{i_1,\dots,i_m}}$. We define the *cofactor of size* $n-m$ as the scalar

$$\left|A^{\widehat{j_1,\dots,j_m}}_{\widehat{i_1,\dots,i_m}}\right| := (-1)^{i_1+\cdots+i_m+j_1+\cdots+j_m} \det\left(A^{\widehat{j_1,\dots,j_m}}_{\widehat{i_1,\dots,i_m}}\right). \tag{1.9}$$

If $m = 1$ (and $i_1 = i$, $j_1 = j$) instead of $\left|A^{\widehat{j}}_{\widehat{i}}\right|$ we adopt the simpler notation A_{ij} and we call A_{ij} the *cofactor of the element* a_{ij}. The $n \times n$-matrix of cofactors $(A_{ij})_{i,j}$ is called the *adjoint matrix* of A and denoted by $\mathrm{adj}(A)$.

The following important result holds.

Theorem 1.19 (Laplace) *Let $A \in \mathcal{M}_n(R)$. We have the Laplace expansion by cofactors along row i*

$$\sum_{k=1}^{n} a_{ik}A_{jk} = \det(A)\delta_{ij}, \tag{1.10}$$

and the Laplace expansion by cofactors along column j

$$\sum_{k=1}^{n} a_{ki} A_{kj} = \det(A)\delta_{ij}. \tag{1.11}$$

Hence we obtain

$$\operatorname{adj}(A)\, A = A \operatorname{adj}(A) = \det(A) I_n. \tag{1.12}$$

More generally, for every integer $1 \le m \le n$, fixing the rows $i_1, \ldots, i_m$ we have

$$\det(A) = \sum_{1 \le j_1 < \cdots < j_m \le n} \left| A^{j_1,\ldots,j_r}_{i_1,\ldots,i_r} \right| \left| A^{\widehat{j_1,\ldots,j_m}}_{\widehat{i_1,\ldots,i_m}} \right|,$$

and if we fix the columns $j_1, \ldots, j_m$ then

$$\det(A) = \sum_{1 \le i_1 < \cdots < i_m \le n} \left| A^{j_1,\ldots,j_r}_{i_1,\ldots,i_r} \right| \left| A^{\widehat{j_1,\ldots,j_m}}_{\widehat{i_1,\ldots,i_m}} \right|.$$

From Theorem 1.19 we can deduce the following proposition.

Proposition 1.20 *Let A and $B \in \mathcal{M}_n(R)$.*

1. *If $A = (a_{ij})_{i,j}$ is an upper or lower triangular matrix then $\det(A) = a_{11}a_{22}\cdots a_{nn}$; in particular $\det(I_n) = 1$.*
2. *Interchanging two rows or two columns of A we obtain a matrix C such that $\det(C) = -\det(A)$.*
3. *For every $k \in R$ we have $\det(kA) = k^n \det(A)$.*
4. *$\det(A^t) = \det(A)$.*
5. *$\det(AB) = \det(A)\det(B)$. Binet's theorem.*

Definition 1.21 A matrix $A \in \mathcal{M}_n(R)$ is called *nonsingular* or *invertible* if there is a matrix $B \in \mathcal{M}_n(R)$ such that $AB = BA = I_n$. The matrix B is unique and is called the *inverse* of A, and the notation A^{-1} is used to designate B. Here we list some immediate facts about nonsingular matrices.

1. $(A^{-1})^{-1} = A$.
2. $(AB)^{-1} = B^{-1}A^{-1}$.
3. If A is nonsingular then A^t is nonsingular and $(A^{-1})^t = (A^t)^{-1}$.
4. If A is nonsingular and $P \in \mathcal{M}_n(R)$ then $\operatorname{tr}(APA^{-1}) = \operatorname{tr}(P)$.

The multiplicative group $\mathcal{M}_n(R)^*$ of all invertible matrices of $\mathcal{M}_n(R)$ will be denoted by $\mathrm{GL}_n(R)$ and called the *general linear group of order n on R*.

From Proposition 1.20 (Binet's theorem) and from Theorem 1.19 we deduce immediately the following characterization of nonsingular matrices.

Proposition 1.22 *Let $A \in \mathcal{M}_n(R)$. If A is nonsingular then* $\det(A)$ *is a unit of R and*

$$\det(A^{-1}) = (\det(A))^{-1}.$$

Conversely, if $\det(A)$ *is a unit of R then A is nonsingular and*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)^t.$$

In particular, if R is a field A is nonsingular if and only if $\det(A) \neq 0$.

By Propositions 1.20 and 1.22 the set $\mathrm{GL}_n^+(\mathbb{R}) := \{A \in \mathrm{GL}_n(\mathbb{R}) : \det(A) > 0\}$ is a subgroup of $\mathrm{GL}_n(\mathbb{R})$.

Definition 1.23 Two matrices A and $B \in \mathcal{M}_n(R)$ are called

Similar if there is a nonsingular matrix $P \in \mathcal{M}_n(R)$ such that $A = P^{-1}BP$. If A and B are similar then $\det(A) = \det(B)$ and $\operatorname{tr}(A) = \operatorname{tr}(B)$.

Congruent if there is a nonsingular matrix $P \in \mathcal{M}_n(R)$ such that $A = P^tBP$. If A and B are congruent then $\det(A) = \det(P)^2 \det(B)$.

Similarity and congruence of matrices are equivalence relations on $\mathcal{M}_n(R)$.

Definition 1.24 For our purposes it is enough to introduce the notion of rank of a matrix with entries in a field. The reader can find the general treatment for matrices with entries in a commutative ring in [8] or in [24].

If $A \in \mathcal{M}_{m,n}(K)$, then the *rank* of A is the size of the largest nonvanishing minor of A. The rank of A will be denoted by $\operatorname{rank}(A)$. The following properties hold.

1. The rank of a matrix is invariant with respect to interchanging rows or columns.
2. $\operatorname{rank}(A) = \operatorname{rank}(A^t)$.
3. If $B \in \mathcal{M}_{n,p}(K)$, then $\operatorname{rank}(AB) \leq \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$.
4. If $B \in \mathcal{M}_{m,n}(K)$ then $\operatorname{rank}(A + B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$.
5. If $A = (a_{ij})_{i,j} \in \mathcal{M}_{m,n}(\mathbb{C})$ and $\overline{A} = (\overline{a_{ij}})_{i,j}$ ($\overline{a_{ij}}$ is the complex conjugate of a_{ij}), then $\operatorname{rank}(A) = \operatorname{rank}(\overline{A})$.

Let P, Q be two nonsingular matrices of $\mathcal{M}_n(K)$ and $A \in \mathcal{M}_n(K)$, then we have

$$\operatorname{rank}(PAQ) = \operatorname{rank}(A). \tag{1.13}$$

In particular, if $A \in \mathcal{M}_n(K)$ is similar or congruent to $B \in \mathcal{M}_n(K)$ then $\operatorname{rank}(A) = \operatorname{rank}(B)$.

The following result is useful to determine the rank of a matrix.

Proposition 1.25 (Kronecker's Rule) *Let $A \in \mathcal{M}_{m,n}(K)$. Assume that there is a submatrix $B \in \mathcal{M}_p(K)$ of A such that* $\det(B) \neq 0$. *If every submatrix $C \in \mathcal{M}_{p+1}(K)$ of A containing B as a submatrix has a vanishing determinant then* $\operatorname{rank}(A) = p$.

Linear Systems Let us consider the system of m linear equations in n unknowns $x_1, x_2, \ldots, x_n$ with coefficients a_{ij} belonging to a field K

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = b_1 \\ \qquad\qquad \vdots & \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = b_m. \end{cases} \tag{1.14}$$

If $b_1 = \cdots = b_m = 0$ the system (1.14) is called *homogeneous* and every its solution $\neq (0, \ldots, 0)$ is called *non-trivial*.

In the case $m = n$, we say that the system (1.14) is *square*. If there is at least one solution, the linear system is *consistent*; if there is no solution, the linear system is *inconsistent*.

We associate to the system (1.14) the *coefficient matrix* A and the *augmented matrix* (A, b)

$$A := \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \qquad (A, b) := \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{pmatrix}.$$

For square linear systems with nonsingular coefficient matrix we have:

Proposition 1.26 (Cramer's Rule) *If the system* (1.14) *is square and if* $\det(A) \neq 0$ *then there is one and only one solution given by the ordered* n*-tuple* $(x_1, x_2, \ldots, x_n)$ *with*

$$x_j = \frac{\begin{vmatrix} a_{11} & \ldots & b_1 & \ldots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \ldots & b_n & \ldots & a_{nn} \end{vmatrix}}{\det(A)}, \qquad 1 \leq j \leq n,$$

where the column $\begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ *takes the place of the column* $\begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix}$.

The following theorem contains the desired criterion for an arbitrary linear system.

Theorem 1.27 (Rouchè–Capelli) *The linear system* (1.14) *is consistent if and only if*

$$\operatorname{rank}(A) = \operatorname{rank}((A, b)).$$

Remark 1.28 Assume that $\operatorname{rank}(A) = \operatorname{rank}((A, b)) = p$ and that $\left|A_{1,\dots,p}^{1,\dots,p}\right| \neq 0$ (this can always be achieved by a suitable renumbering of the n unknowns and a rearrangement of the equations). All the solutions of the system (1.14) are of the form $(y_1, \dots, y_p, z_{p+1}, \dots, z_n)$ where $z_{p+1}, \dots, z_n$ are arbitrarily chosen and $(y_1, \cdots, y_p)$ is the (unique) solution of Cramer's system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1p}x_p & = a_{1p+1}z_{p+1} + \cdots + a_{1n}z_n + b_1 \\ \quad\vdots & \qquad\qquad\vdots \\ a_{p1}x_1 + \cdots + a_{pp}x_p & = a_{pp+1}z_{p+1} + \cdots + a_{pn}z_n + b_p. \end{cases}$$

We say that the system (1.14) has ∞^{n-p} solutions since they depend on the choice of $n - p$ parameters.
If the system is also homogeneous there is a non-trivial solution if and only if $\det(A) = 0$.

Remark 1.29 Cramer's rule has the following generalization (see [6], p. 535).

Proposition 1.30 *Assume that the system* (1.14) *is square with coefficients in a commutative unitary ring R. Then it admits at least a solution if and only if* $\det(A)$ *is a unit of R; moreover the solution is unique.*
If the system is also homogeneous there is a non-trivial solution if and only if $\det(A)$ *is a* 0*-divisor.*

1.2 Vector Spaces

Definition 1.31 Let K be a field and $(V, +)$ an abelian group. Let V be equipped with a *scalar multiplication*

$$K \times V \to V, \qquad (\lambda, v) \to \lambda v$$

satisfying the following axioms for every $v, w \in V$ and $\lambda, \mu \in K$:

$$\lambda(v + w) = \lambda v + \lambda w, \quad \text{distributivity,} \tag{1.15}$$

$$(\lambda + \mu)v = \lambda v + \mu v, \quad \text{distributivity,} \tag{1.16}$$

$$\lambda(\mu v) = (\lambda\mu)v, \quad \text{compatibility,} \tag{1.17}$$

$$1_K\, v = v, \quad \text{unitarity.} \tag{1.18}$$

We shall call V a *vector space on K* (or *K-vector space*), the elements of V *vectors* and the elements of K *scalars*. Because there is no chance of misunderstanding, we

can use the same symbol $+$ to denote the sum in K and V as well as for the product in K and the scalar multiplication.

Here are some very simple consequences of Definition 1.31:

$$0_K v = 0_V, \qquad \lambda 0_V = 0_V, \qquad (-\lambda)v = \lambda(-v) = (-\lambda)v, \qquad \forall\, v \in V,\ \forall\, \lambda \in K.$$

Moreover,

$$\lambda\, v = 0_V \quad \Longrightarrow \quad \lambda = 0_K \quad \text{or} \quad v = 0_V. \tag{1.19}$$

Indeed, if $\lambda \neq 0_K$, we have $0_V = \lambda^{-1}0_V = \lambda^{-1}(\lambda v) = (\lambda^{-1}\lambda)v = 1_K v = v$.

Remark 1.32 If in Definition 1.31 K is only a unitary commutative ring, we call V a *K-module*. In particular a unitary commutative ring R is a module on itself with the multiplication of the ring. We observe that (1.19) is not true in general. For instance $2\,\widehat{3} = 0$ in the $\mathbb{Z}$-module $\mathbb{Z}_6$ (where the scalar multiplication is $r\,\widehat{m} = \widehat{rm}$, $\forall\, r \in \mathbb{Z}$, $\forall\, \widehat{m} \in \mathbb{Z}_6$).

Examples 1.33

1. For every field K and for every integer $n \geq 1$, we put on $V := K^n$ the following operations:

$$\begin{aligned}(a_1, \ldots, a_n) + (b_1, \ldots, b_n) &:= (a_1 + b_1, \ldots, a_n + b_n),\\ &\quad \forall\, (a_1, \ldots, a_n), (b_1, \ldots, b_n) \in V,\\ \lambda(a_1, \ldots, a_n) &:= (\lambda a_1, \ldots, \lambda a_n), \quad \forall\, \lambda \in K,\\ &\quad \forall\, (a_1, \ldots, a_n) \in V.\end{aligned}$$

   With these operations V becames a K-vector space called the *standard vector space on K of dimension n*.
2. The ring of polynomials $V := K[X_1, \ldots, X_n]$ in $n \geq 1$ indeterminates $X_1, \ldots, X_n$ is a K-vector space with respect to the usual addition of polynomials and multiplication of a polynomial by a scalar. Let $m \geq 1$ be an integer.

   (a) $K[X_1, \ldots, X_n]_{\leq m} = \{f \in V : \deg(f) \leq m\}$;
   (b) $K[X_1, \ldots, X_n]_m = \{f \in V : \deg(f) = m\}$;
   (c) ${}_{\mathrm{h}}K[X_1, \ldots, X_n]_m = \{f \in K[X_1, \ldots, X_n]_m : f \text{ is homogeneous}\} \cup \{0\}$;
   (d) ${}_{\mathrm{h}}K[X_1, \ldots, X_n]^*_m = \{f \in K[X_1, \ldots, X_n]_m : f \text{ is homogeneous}\}$.

   With the same operations the subsets (a) and (c) of V are K-vector spaces:
3. Let A be a ring and K a field, $K \subset \mathcal{C}(A)$, the centre of A. Then the addition in A and the multiplication in A restricted to $K \times A$ make A a vector space on K. In particular, $\mathbb{R}$ is a $\mathbb{Q}$-vector space while $\mathbb{C}$ is a vector space on $\mathbb{Q}$ as well as on $\mathbb{R}$.
4. Let $k \in \{0, 1, \ldots, \infty\}$. The set $C^k([a, b]; \mathbb{R})$ of all k-times continuously differentiable functions $f : [a, b] \to \mathbb{R}$ on the real interval $[a, b]$ $(a < b)$,

equipped with the operations

$$(f+g)(x) := f(x)+g(x), \qquad \forall\, f,g \in C([a,b];\mathbb{R}),\ \forall\, x \in [a,b],$$
$$(\lambda f)(x) := \lambda f(x), \qquad \forall\, f \in C([a,b];\mathbb{R}),\ \forall\, \lambda \in \mathbb{R},\ \forall\, x \in [a,b],$$

is a vector space on $\mathbb{R}$.

5. The set $\mathcal{M}_{m,n}(K)$ is a K-vector space with respect to the sum and the product by a scalar of Definition 1.14.

Definition 1.34 Let $V_1, \ldots, V_r$ be $r \geq 2$ vector spaces on K. The set $V := V_1 \times V_2 \cdots \times V_r$ equipped with the following operations

$$(v_1, v_2, \ldots, v_r) + (w_1, w_2, \ldots, w_r) := (v_1 + w_1, v_2 + w_2, \ldots, v_r + w_r),$$
$$\forall v_i, w_i \in V_i,\ i = 1, \ldots, r,$$
$$\lambda(v_1, v_2, \ldots, v_r) := (\lambda v_1, \lambda v_2, \ldots, \lambda v_r),$$
$$\forall \lambda \in K,\ \forall v_i \in V_i,\ i = 1, \ldots, r,$$

is a K-vector space called the *direct product* of $V_1, \ldots, V_r$. It follows immediately that $0_V = (0_{V_1}, 0_{V_2}, \ldots, 0_{V_r})$ and the opposite $-(v_1, v_2, \ldots, v_r)$ of $(v_1, v_2, \ldots, v_r)$ is $(-v_1, -v_2, \ldots, -v_r)$.

1.2.1 Vector Subspaces, Generators, Linearly Independent Sets, Bases and Dimension of a Vector Space

Definition 1.35 Let V be a vector space on K. A non-empty subset $W \subset V$ is called a *vector subspace* if the following conditions are satisfied:

(i) If $v_1, v_2 \in W$ then $v_1 + v_2 \in W$.
(ii) If $v \in W$ and $\lambda \in K$ then $\lambda v \in W$.

From the above definition it follows that the operations on V restricted to W make W a K-vector space. Trivial examples of subspaces are $W = V$ and $W = \{0_V\}$. The vector spaces of Examples 1.33, 2 are vector subspaces of $K[X_1, \ldots, X_n]$.

Proposition 1.36 *Let V be a K-vector space and $\{W_i\}_{i\in I}$ a non-empty family of vector subspaces of V. Then the intersection $\bigcap_{i\in I} W_i$ is a vector subspace of V.*

Proof The proof easily follows from Definition 1.35. □

Definition 1.37 Let S be a non-empty subset of a K-vector space V. Let

$$\langle S \rangle := \bigcap_{W \supset S} W$$

where W runs through all vector subspaces of V. If $S = \{v_1, v_2, \ldots, v_n\} \subseteq V$ is a finite set of vectors of V we shall denote $\langle S\rangle$ also by $\langle v_1, \ldots, v_n\rangle$. In order to avoid a misunderstanding we also use the notation $\mathrm{J}(v_1, v_2)$ instead of $\langle v_1, v_2\rangle$ which denotes the inner product of v_1 and v_2 (see Definition 1.138).

From Proposition 1.36 we can deduce the following properties:

(i) $\langle S\rangle$ is a vector subspace of V containing S.
(ii) If W is a vector subspace of V containing S then $\langle S\rangle \subseteq W$.

In other words $\langle S\rangle$ is the smallest vector subspace of V containing S; $\langle S\rangle$ is called *the vector subspace of V generated by S* or *the linear span of S*. We also say that S *generates* the K-vector space V (or that S is a *system of generators* of V) if $\langle S\rangle = V$. If S is a system of generators of V and if $S \subset S' \subset V$, then S' also is a system of generators of V.

Definition 1.38 Let $W_1, W_2, \ldots W_r$ be $r \geq 2$ vector subspaces of a vector space V on K. The subspace

$$W_1 + W_2 + \cdots + W_r := \langle W_1 \cup W_2 \cup \cdots \cup W_r\rangle.$$

is called the *sum* of subspaces $W_1, W_2, \ldots W_r$. Then $W_1 + W_2 + \cdots + W_r$ is the smallest vector subspace of V containing $W_1, W_2, \ldots W_r$.

Proposition 1.39 *Under the assumptions of Definition 1.38 we have*

$$W_1 + W_2 + \cdots + W_r = \{v_1 + v_2 + \cdots v_r \mid v_i \in W_i,\ i = 1, \ldots, r\}. \tag{1.20}$$

Proof The right–hand side W' of (1.20) is clearly a vector subspace of V. Since $W_i \subset W'$ for $i = 1, \ldots r$, then we get $W_1 + W_2 + \cdots + W_r \subset W'$. Conversely, if $w \in W'$ then $w = v_1 + v_2 + \cdots v_r$ for some $v_i \in W_i$, $i = 1, \ldots, r$. Thus

$$v_i \in W_1 \cup W_2 \cup \cdots \cup W_r \subset W_1 + W_2 + \cdots + W_r, \qquad i = 1, \ldots, r,$$

and $w = v_1 + v_2 + \cdots v_r \in W_1 + W_2 + \cdots + W_r$. □

Definition 1.40 Let $v_1, v_2, \ldots, v_n$ be $n \geq 1$ vectors of a K-vector space V. A vector v of the form

$$v = a_1 v_1 + \cdots + a_n v_n, \qquad (a_1, \ldots, a_n) \in K^n \tag{1.21}$$

is called a *linear combination with coefficient in K of the vectors $v_1, \ldots, v_n$*. We call a linear combination whose coefficients are all zero *trivial* and otherwise *non-trivial*.

Proposition 1.41 *Let $S = \{v_1, v_2, \ldots, v_n\}$ be a finite subset of a K-vector space V. Then*

$$\langle v_1, \ldots, v_n\rangle = \{a_1 v_1 + a_2 v_2 + \cdots + a_n v_n \mid (a_1, a_2, \ldots, a_n) \in K^n\}. \tag{1.22}$$

If S is an infinite set then

$$\langle S\rangle = \{a_1v_1 + a_2v_2 + \cdots + a_nv_r \,|\, v_i \in S,\ a_i \in K,\ 1 \le i \le r,\ r \in \mathbb{N}\}. \tag{1.23}$$

In other words, the linear span $\langle S\rangle$ is the set of all linear combinations with coefficient in K of elements of S.

Proof Denoting the right–hand side of (1.22) by W we have immediately the inclusion $W \subset \langle S\rangle$. On the other hand $\langle S\rangle \subset W$ since W is a subspace containing S. By a similar argument (1.23) easily follows. □

Corollary 1.42 *One subset S of V is a system of generators of V if and only if for every vector $v \in V$ there are $a_1, a_2, \ldots, a_n \in K$ and $v_1, v_2, \ldots, v_n \in S$ for some $n \ge 1$ such that $v = a_1v_1 + a_2v_2 + \cdots + a_nv_n$.*

Definition 1.43 A K-vector space V which admits a finite system of generators is called a *finitely generated vector space on K*. We shall be mostly concerned with finitely generated vector spaces. A vector space V which is not finitely generated is said to be *infinitely generated*.

Definition 1.44 Let V be a K-vector space and S a non-empty subset of V. We shall say that S is a *linearly independent subset* (or *free*) *on K* if, for every $v_i \in S$, $a_i \in K$, $1 \le i \le r$, $r \in \mathbb{N}$,

$$\sum_{i=1}^{r} a_iv_i = 0_V \qquad \Longrightarrow \qquad a_1 = a_2 = \ldots = a_r = 0,$$

i.e every linear combination of element of S whose value is 0_V is trivial. Every vector space $V \neq \{0_V\}$ has free subsets, for instance if $v \in V$, $v \neq 0_V$ then $\{v\}$ is clearly free (by (1.19)).

Remark 1.45 Let S be a linearly independent subset of a vector space V.

(i) $0_V \notin S$, otherwise if $v_1 = 0_V$ we should have a non-trivial linear combination

$$1_K \cdot v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_n = 0_V.$$

(ii) If $v, w \in S$ then $v \neq w$, otherwise if $v = w$ we should have $1_K \cdot v + (-1) \cdot w + 0 \cdot v_3 + \cdots + 0 \cdot v_n = 0_V$.

(iii) Every subset of S is obviously free.

Proposition 1.46 *Let S be a non-empty subset of a vector space V. Then S is free if and only if*

$$v \notin \langle S \setminus \{v\}\rangle, \qquad \forall\, v \in S.$$

Proof Suppose that S is free and that there is a vector $v_1 \in \langle S \setminus \{v_1\}\rangle$. By Proposition 1.41 for some $n \ge 2$ there would be $a_2, \ldots, a_n \in K$ such that

$v_1 = a_2v_2 + \cdots + a_nv_n$ with $v_i \in S \setminus \{v_1\}$, $i = 2, \ldots, n$. Thus we should have the non-trivial linear combination $(-1_K)v_1 + a_2v_2 + \cdots + a_nv_n = 0_V$, contradicting our assumption. □

Conversely, suppose that $v \notin \langle S \setminus \{v\}\rangle$, $\forall v \in S$ and that S is not free. Hence for some $n \geq 2$ there would be $(a_1, \ldots, a_n) \in K^n \setminus \{(0, \ldots, 0)\}$ and mutually distinct $v_1, \ldots, v_n \in S$ such that $a_1v_1 + \cdots + a_nv_n = 0_V$. We can suppose that $a_1 \neq 0$ so that we have

$$a_1^{-1}(a_1v_1) = a_1^{-1}((-a_2)v_2 + \cdots + (-a_n)v_n), \quad \text{namely}$$
$$v_1 = (-a_1^{-1}a_2)v_2 + \cdots + (-a_1^{-1}a_n)v_n.$$

By Proposition 1.41, we have $v_1 \in \langle S \setminus \{v_1\}\rangle$, and this contradicts our assumption. □

Corollary 1.47 *Let V be a K-vector space, S a non-empty free subset of V and $v \in V$. Then $S \cup \{v\}$ is free if and only if $v \notin \langle S\rangle$.*

Proof If $S \cup \{v\}$ is free then by Proposition 1.46 $v \notin \langle S\rangle$. Conversely, if $v \notin \langle S\rangle$ then $v \notin \langle (S \cup \{v\}) \setminus \{v\}\rangle$. Furthermore, if $w \in S$ and $w \in \langle (S \cup \{v\}) \setminus \{w\}\rangle$ we should have for some integer $r \geq 1$

$$w = \lambda_1w_1 + \cdots \lambda_rw_r + \mu v, \qquad w_i \in S,\ w_i \neq w,\ \lambda_i, \mu \in K,\ i = 1, \ldots, r.$$

Then $\mu \neq 0$, otherwise $w \in \langle S \setminus \{w\}\rangle$ and S would not be free. Hence $v = \mu^{-1}(w - \lambda_1w_1 - \cdots - \lambda_rw_r) \in \langle S\rangle$ contradicting our assumptions. Hence $w \notin \langle (S \cup \{v\}) \setminus \{w\}\rangle$ and by Proposition 1.46 $S \cup \{v\}$ is free. □

Definition 1.48 Let V be a K-vector space and S a non-empty subset of V. We say that S is a *basis of V on K* if S is a system of generators of V which is free. It is convenient to regard the basis of $V = \{0_V\}$ as the empty set.

Theorem 1.49 *Every finite system of generators of a K-vector space contains a basis of V on K. In particular all finitely generated K-vector spaces have at least one basis on K.*

Proof By Definition 1.48 we can suppose that $V \neq \{0_V\}$. Let S be a finite system of generators of V. If there is $v \in S$ such that $v \in \langle S \setminus \{v\}\rangle$ then the set $S \setminus \{v\}$ is still a finite system of generators of V. Indeed, if $S = \{v_1 = v, v_2, \ldots, v_n\}$, there are $a_2, \ldots, a_n \in K$ such that $v_1 = a_2v_2 + \cdots + a_nv_n$ (Proposition 1.41). If $w \in V$ is an arbitrary vector there are $b_1, \ldots, b_n \in K$ such that $w = b_1v_1 + \cdots + b_nv_n$. Hence

$$w = (b_2 + b_1a_2)v_2 + \cdots + (b_n + b_1a_n)v_n$$

so that, by Proposition 1.41, $S \setminus \{v\}$ is a system of generators of V. In this way, after removing all $v \in S$ such that $v \in \langle S \setminus \{v\} \rangle$ we get a non-empty finite set $S' \subset S$ which is free (by Proposition 1.46) and generates V, i.e. S' is a basis of V. □

Remark 1.50 The statement of Theorem 1.49 is also true for infinitely generated vector spaces but its proof requires Zorn's lemma (see [15] pp. 12–13).

Corollary 1.51 *Let V be a K-vector space, S a free subset of V and $S' = \{w_1, \ldots, w_m\}$, $m \geq 1$, a system of generators of V on K. Then S is finite and*

$$\#(S) \leq m.$$

Proof Suppose that $S = \{v_1, \ldots, v_n\}$, $n \geq 1$. By Theorem 1.49 S' contains a basis S'' which is supposed to be $\{w_1, \ldots, w_p\}$ with $p \leq m$ by renumbering $\{w_1, \ldots, w_m\}$. If

$$v_j = \sum_{i=1}^{p} \mu_{ij} w_i, \qquad j = 1, \ldots, n, \qquad \text{and} \qquad \sum_{j=1}^{n} \lambda_j v_j = 0_V,$$

then we have

$$\sum_{i=1}^{p} \left(\sum_{j=1}^{n} \mu_{ij} \lambda_j \right) w_i = 0_V.$$

Since S'' is free, we have the homogeneous linear system in the unknowns $\lambda_1, \ldots, \lambda_n$

$$\begin{cases} \sum_{j=1}^{n} \mu_{1j} \lambda_j & = 0 \\ \cdots\cdots & \cdots \\ \sum_{j=1}^{n} \mu_{pj} \lambda_j & = 0 \end{cases}$$

that admits solutions $(\lambda_1, \ldots, \lambda_n) \neq (0, \ldots, 0)$ if $n > p$ by Theorem 1.27. Since S is free we must have $n \leq p \leq m$. If S is infinite there are free subsets of an arbitrarily large number of elements (e.g. all finite subsets of S) contradicting what we proved above. □

Corollary 1.52 *Let V be a finitely generated K-vector space, and S a finite subset of V. The following assertions are equivalent:*

(i)) *S is a basis of V.*
(ii)) *S is a minimal system of generators of V.*
(iii) *S is a maximal free subset of V.*

Proof (i) $\Rightarrow$ (ii). Let $v \in S$. If $S \setminus \{v\}$ span V then $v \in \langle S \setminus \{v\}\rangle$ so that S is not free by Proposition 1.46.
(ii) $\Rightarrow$ (i). The set S contains a basis S' by Theorem 1.49 so that $S = S'$ by its minimality.
(i) $\Longrightarrow$ (iii). Let $v \notin S$. Since $v \in \langle S\rangle$, by Corollary 1.47 $S \cup \{v\}$ is not free.
(iii) $\Longrightarrow$ (i). If $V \neq \langle S\rangle$ there is $v \in V$ such that $v \notin \langle S\rangle$. By Corollary 1.47 $S \cup \{v\}$ is free, contradicting the maximality of S. □

Corollary 1.53 *Every free subset of a finitely generated K-vector space V is contained in a basis of V.*

Proof Choose a base S' with m elements; by Corollary 1.51 we have $\#(S) \leq m$. If S is not maximal, by adding a finite number ($\leq m - \#(S)$) of vectors to S we obtain a maximal free subset of V, i.e. a basis by Corollary 1.52. □

Remark 1.54 The statements of Corollaries 1.52 and 1.53 are also true for infinitely generated vector spaces (see [15]).

Corollary 1.55 *All bases of a finitely generated K-vector space have the same number of elements.*

Proof Let S and S' be two bases of V. By Corollary 1.51 we have $\#(S) \leq \#(S')$ and $\#(S') \leq \#(S)$. □

Remark 1.56 If V is infinitely generated, all bases of V have the same cardinality (see [15]).

Definition 1.57 Let V be a K-vector space of dimension n. An *ordered basis* of V is an ordered n-tuple $(v_1, \ldots, v_n)$ of vectors for which the set $\{v_1, \ldots, v_n\}$ is a basis of V.

Corollary 1.58 *Let $S = (v_1, \ldots, v_n)$ be an ordered basis of a K-vector space V. For every vector $v \in V$ there is one and only one n-tuple $(a_1, \ldots, a_n) \in K^n$ such that $v = a_1 v_1 + \cdots + a_n v_n$.*

Proof We have only to prove the uniqueness. Since

$$v = a_1 v_1 + \cdots + a_n v_n = b_1 v_1 + \cdots + b_n v_n \quad \Longrightarrow \quad (a_1 - b_1)v_1 + \cdots + (a_n - b_n)v_n = 0,$$

we have $a_i - b_i = 0$, $i = 1, \ldots, n$ because S is free. □

Definition 1.59 Under the assumptions of Corollary 1.58, the scalars $a_1, \ldots, a_n$ are called *coordinates* or *components* of v with respect to the ordered base S. The notation $v_S = (a_1, \ldots, a_n)$ is sometimes used instead of equality $v = a_1 v_1 + \cdots + a_n v_n$. Thus the coordinates of a vector are always associated with an ordered basis.

Definition 1.60 Let V be a finitely generated K-vector space. If $V \neq 0_V$, the number of elements of a basis depends only on V and not on the basis chosen by Corollary 1.55. We call it the *dimension of V on K* and denote it by $\dim_K V$. If

$V = 0_V$ we write $\dim_K V = 0$. If V is infinitely generated we simply say that the dimension of V is *infinite* and write $\dim_K V = \infty$ (see Remark 1.56).

Corollary 1.61 *Let V be a K-vector space of finite dimension and W a subspace of V. Then W is a K-vector space of finite dimension and* $\dim_K W \leq \dim_K V$. *We have* $\dim_K W = \dim_K V$ *if and only if* $W = V$.

Proof A basis S of W is a free subset of V so that $\dim_K W \leq \dim_K V$ by Corollary 1.51. Suppose that $\dim_K W = \dim_K V$ and $W = \langle S \rangle \neq V$, hence there is a vector $v \in V \setminus W$ with $v \notin \langle S \rangle$. By Corollary 1.47 $S \cup \{v\}$ is free and $\#(S \cup \{v\}) > \dim_K V$ and this leads to a contradiction. □

Examples 1.62

(1) Let $V = K^n$ be the standard vector space of Example 1.33, 1. Let

$$e_i = \underbrace{(0, \ldots, 0, 1, 0, \ldots, 0)}_{i-\text{th place}}, \qquad i = 1, \ldots, n.$$

The set $\{e_1, \ldots, e_n\}$ is a basis of K^n so that $\dim_K K^n = n$. The ordered basis $\mathcal{B} := (e_1, \ldots, e_n)$ is called the *canonical basis of* K^n.

(2) The set of matrices

$$E_{hk} = (a_{ij})_{i,j},$$

$$\text{where} \quad \begin{cases} a_{ij} = 1 & \text{if } i = h \text{ and } j = k \\ 0 & \text{otherwise,} \end{cases} \qquad h = 1, \ldots, m, \ k = 1, \ldots, n$$

is a basis of $\mathcal{M}_{m,n}(K)$ so that $\dim \mathcal{M}_{m,n}(K) = mn$.

(3) The field of complex numbers $\mathbb{C}$ is a vector space on $\mathbb{R}$ (Example 1.33, 4)) of dimension 2, $S = \{1, i\}$ being a basis of $\mathbb{C}$ on $\mathbb{R}$. The field of real numbers $\mathbb{R}$ is a $\mathbb{Q}$-vector space of infinite dimension.

(4) Let $V = \mathbb{R}[X]_{\leq n}$ the $\mathbb{R}$-vector space of all polynomials of degree $\leq n$ (with a fixed $n \geq 2$) (see Example 1.33, 2). Then

$$\{1, X, X^2, \ldots, X^n\} \quad \text{and} \quad \left\{1, X - a, \frac{(X-a)^2}{2}, \ldots, \frac{(X-a)^n}{n!}\right\}$$

(with a fixed real number a) are two basis of V.

(5) Let L and K be two fields. If $K \subset L$ then L is a K-vector space (see Example 1.33-3). An *algebraic number field* L is a field $\mathbb{C} \supset L \supset \mathbb{Q}$ such that $\dim_{\mathbb{Q}} L < \infty$. If $\dim_{\mathbb{Q}} L = n$ one can prove (see [28], Ch. 2) that there are an irreducible polynomial $f(X) \in \mathbb{Q}[X]$ of degree n and a complex number $\alpha \in L$ such that $f(\alpha) = 0$ and

$$L = \mathbb{Q}(\alpha) = \{a_0 + a_1\alpha + \cdots a_{n-1}\alpha^{n-1} \ : \ a_i \in \mathbb{Q}, \ i = 0, \ldots, n-1\},$$

i.e. $\{1, \alpha, \alpha^2, \ldots, \alpha^{n-1}\}$ is a basis of L on $\mathbb{Q}$. For instance if $\dim_{\mathbb{Q}} L = 2$ then $L = \mathbb{Q}(\sqrt{m})$ where m is an integer $\neq 0, 1$, which has no square factors.

Proposition 1.63 *Let V and W be two K-vector spaces of finite dimensions. If $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ are bases of V and W respectively, then*

$$\{(v_1, 0_W), \ldots, (v_n, 0_W), (0_V, w_1), \ldots, (0_V, w_m)\}$$

is a basis of the direct product $V \times W$. Hence

$$\dim_K(V \times W) = \dim_K V + \dim_K W.$$

Proof Let $a_1, \ldots, a_n, b_1, \ldots, b_m \in K$ be such that

$$a_1(v_1, 0_W) + \cdots + a_n(v_n, 0_W) + b_1(0_V, w_1) + \cdots + b_m(0_V, w_m) = (0_V, 0_W). \tag{1.24}$$

The equality (1.24) is equivalent to $a_1 v_1 + \cdots + a_n v_n = 0_V$ and $b_1 w_1 + \cdots + b_m w_m = 0_W$, so that, by our assumptions, we have $a_1 = \cdots = a_n = b_1 = \cdots = b_m = 0$.

Let $(v, w) \in V \times W$. By our hypothesis there are $a_1, \ldots, a_n \in K$ and $b_1, \ldots, b_m \in K$ such that $v = a_1 v_1 + \cdots + a_n v_n$ and $w = b_1 w_1 + \cdots + b_m w_m$. Hence $(v, w) = a_1(v_1, 0_W) + \cdots + a_n(v_n, 0_W) + b_1(0_V, w_1) + \cdots + b_m(0_W, w_m)$. This proves our assertion. □

Remark 1.64 Let K be a field. Let $A = (a_{ij})_{i,j} \in \mathcal{M}_{m,n}(K)$.

The i-th *row vector* of A is $v_i := (a_{i1}, \ldots, a_{in}) \in K^n$, $i = 1, \ldots, m$.
The j-th *column vector* of A is $w_j := (a_{1j}, \ldots, a_{mj}) \in K^m$, $j = 1, \ldots, n$.

We can now obtain the rank of A by means of row vectors

$$\operatorname{rank}(A) = \dim_K \langle v_1, \ldots, v_m \rangle$$

or column vectors (since $\operatorname{rank}(A) = \operatorname{rank}(A^t)$)

$$\operatorname{rank}(A) = \dim_K \langle w_1, \ldots, w_n \rangle.$$

Thus $\operatorname{rank}(A) = m$ if and only if $S = \{v_1, \ldots, v_m\}$ is free on K; $\operatorname{rank}(A) = n$ if and only if S is a system of generators of K^n.

Corollary 1.65 *The set of solutions of the homogeneous linear system*

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = 0 \\ \qquad\qquad \vdots & \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = 0 \end{cases} \tag{1.25}$$

is a vector subspace of K^n whose dimension is $n - p$ where $p = \operatorname{rank}(A)$ ($A = (a_{ij})_{i,j}$ is the coefficient matrix). Conversely, every vector subspace of K^n of dimension k is the set of solutions of a homogeneous linear system in n unknowns and $n - k$ equations.

Proof We left the proof of the first statement to the reader as an exercise. Let $\{v_1, \ldots, v_k\}$ be a basis of W, $v_i = (v_{i1}, \ldots, v_{in})$, $i = 1, \ldots, k$. We can suppose that the minor

$$\begin{vmatrix} v_{11} & v_{12} & \ldots & v_{1k} \\ v_{21} & v_{22} & \ldots & v_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ v_{k1} & v_{k2} & \ldots & v_{kk} \end{vmatrix} \neq 0,$$

by modifying the basis of W if necessary. The row vector $(x_1, \ldots, x_n)$ belongs to W if and only if the matrix

$$B = (b_{ij})_{\substack{i=0,\ldots,k \\ j=1,\ldots,n}}, \qquad b_{0j} = x_j, \quad j = 1, \ldots, n,$$
$$b_{ij} = v_{ij}, \quad i = 1, \ldots, k, \ j = 1, \ldots, n,$$

has rank k. Thus By Kronecker's rule we must have

$$\left| B^{1,\ldots,k,j}_{0,\ldots,k} \right| = 0, \qquad j = k+1, \ldots, n$$

which is the desired linear system of $n - k$ equations. □

1.3 Linear Maps

Definition 1.66 Let V and W two K-vector spaces. A map $f : V \to W$ is called a *linear map* or *homomorphism of K-vector spaces* if it preserves the addition and the scalar multiplication:

(i) For every $v_1, v_2 \in V$, $f(v_1 + v_2) = f(v_1) + f(v_2)$.
(ii) For every $v \in V$ and for every $\lambda \in K$, $f(\lambda v) = \lambda f(v)$.

If $W = V$ we call a homomorphism $f : V \to W$ an *endomorphism of V* or *linear operator on V*. A homomorphism of K-vector spaces $f : V \to W$ su K is called an *isomorphism* if there is a homomorphism of K-vector spaces $g : W \to V$ such that $g \circ f = \mathrm{id}_V$ and $f \circ g = \mathrm{id}_W$. We say that two K-vector spaces V and W are *isomorphic* if there is an isomorphism $f : V \to W$. An *automorphism* of a vector space V is an endomorphism which is isomorphism.

Remark 1.67 If K is an unitary commutative ring, V and W two K-modules (see Remark 1.32), a map $f: V \to W$ satisfying (i) and (ii) of Definition 1.66 is called a *homomorphism of K-modules*.

Proposition 1.68 *Let V, W and U be K-vector spaces.*

(i) *For every K-vector space V the identity map* $\mathrm{id}_V : V \to V$ *is an isomorphism.*
(ii) *The composition $g \circ f : V \to U$ of two homomorphisms of K-vector spaces $f : V \to W$ and $g : W \to U$ is a homomorphism of K-vector spaces.*
(iii) *Let $f : V \to W$ be a homomorphism of K-vector spaces. Then f is an isomorphism if and only if f is bijective.*

Proof Exercise for the reader. □

Examples 1.69

1. If V is a K-vector space and W is a vector subspace of V, the inclusion map $i: W \to V$, $i(w) = w$, $\forall\, w \in W$, (sometimes denoted by $W \hookrightarrow V$) is a linear map.
2. Let $V = V_1 \times V_2$ be the direct product of two K-vector spaces V_1 and V_2, then the maps

$$\begin{aligned} &i_1: V_1 \to V,\ i_1(v_1) = (v_1, 0_{V_2}),\ \forall\, v_1 \in V_1;\\ &i_2: V_2 \to V,\ i_2(v_2) = (0_{V_1}, v_2),\ \forall\, v_2 \in V_2;\\ &p_1: V \to V_1,\ p_1(v_1, v_2) = v_1,\ \forall\, v_1 \in V_1,\ v_2 \in V_2;\\ &p_2: V \to V_2,\ p_2(v_1, v_2) = v_2,\ \forall\, v_1 \in V_1,\ v_2 \in V_2, \end{aligned}$$

   are homomorphisms of K-vector spaces. The linear maps i_1 and i_2, which are injective, are called the *canonical inclusion maps* of the direct product $V_1 \times V_2$, whereas p_1 and p_2 are surjective and they are called the *canonical projections* of the direct product $V_1 \times V_2$. Furthermore we have the identities

$$\begin{aligned} p_1 \circ i_1 = \mathrm{id}_{V_1}, \quad p_2 \circ i_2 = \mathrm{id}_{V_2}, \quad p_2 \circ i_1 = 0_{V_2},\\ p_1 \circ i_2 = 0_{V_1}, \quad i_1 \circ p_1 + i_2 \circ p_2 = \mathrm{id}_V\,. \end{aligned}$$

3. We can associate to a matrix $A = (a_{ij})_{i,j} \in \mathcal{M}_{m,n}(K)$ a linear map

$$f_A: K^n \to K^m, \quad (f_A(\mathbf{x}))^t = A\,\mathbf{x}^t, \quad \forall\, \mathbf{x} = (x_1, \ldots, x_n) \in K^n.$$

   If $\mathbf{y} = (y_1, \ldots, y_m) = f_A(\mathbf{x})$ we have

$$y_i = \sum_{j=1}^{n} a_{ij} x_j, \quad i = 1, \ldots, m.$$

4. The set $C^\infty(\mathbb{R},\mathbb{R})$ of infinitely often differentiable functions $f\colon \mathbb{R}\to\mathbb{R}$ is clearly an $\mathbb{R}$-vector space of infinite dimension. The derivative

$$D\colon C^\infty(\mathbb{R},\mathbb{R})\to C^\infty(\mathbb{R},\mathbb{R}),\qquad D(f)(x)=f'(x),\qquad \forall\, x\in\mathbb{R},$$

is a linear map.
5. The *formal derivative* $D\colon \mathbb{R}[X]\to\mathbb{R}[X]$ defined by

$$D(a_0+a_1X+\cdots+a_mX^m)=a_1+2a_2X+\cdots+ma_mX^{m-1}$$

is a linear map.

Definition 1.70 Let V, W be K-vector spaces. The set

$$\mathrm{Hom}_K(V,W):=\{f\colon V\to W\mid f \text{ is homomorphism of } K\text{-vector spaces}\}$$

equipped with the sum: for every $f,g\in\mathrm{Hom}_K(V,W)$,

$$(f+g)(v):=f(v)+g(v),\qquad \forall\, v\in V,$$

and the scalar multiplication: for every $f\in\mathrm{Hom}_K(V,W)$ and for every $\lambda\in K$,

$$(\lambda f)(v):=\lambda f(v),\qquad \forall\, v\in V,$$

is a K-vector space. The neutral element of $\mathrm{Hom}_K(V,W)$ is the null homomorphism $0\colon V\to W$ defined by $0(v):=0_W\ \forall\, v\in V$ and if $f\in\mathrm{Hom}_K(V,W)$ the opposite of f, denoted by $-f$, is the homomorphism defined by $(-f)(v)=-f(v),\ \forall\, v\in V$.

Let V, W, U be three K-vector spaces. By ii) of Proposition 1.68 we have the *natural composition map*

$$\mathrm{Hom}_K(V,W)\times\mathrm{Hom}_K(W,U)\to\mathrm{Hom}_K(V,U),$$
$$(f,g)\to g\circ f,\quad \forall\, f\in\mathrm{Hom}_K(V,W),\ \forall\, g\in\mathrm{Hom}_K(W,U),$$

which satisfies the relations $\forall f,f_1,f_2\in\mathrm{Hom}_K(V,W)$ and $\forall g,F_1,F_2\in\mathrm{Hom}_K(W,U)$,

$$g\circ(f_1+f_2)=g\circ f_1+g\circ f_2,\quad (F_1+F_2)\circ f=F_1\circ f+F_2\circ f.$$

If $W=V$ we shall denote $\mathrm{Hom}_K(V,W)$ by $\mathrm{End}_K(V)$, the set of all endomorphisms of V. The set $\mathrm{End}_K(V)$ equipped with the above–defined operations $+$ and $\circ$ is a unitary ring with $1_{\mathrm{End}_K(V)}=\mathrm{id}_V$ which is non-commutative if $\dim_K V\geq 2$. The multiplicative group of units of $\mathrm{End}_K(V)$ is denoted by $\mathrm{GL}_K(V)$ (instead of $\mathrm{End}_K(V)^*$); $\mathrm{GL}_K(V)$ is the set of all automorphisms of V and it is called the *general linear group of* V.

Definition 1.71 Let $f: V \to W$ be a linear map of K-vector spaces. The *kernel of* f is the subset of V defined by

$$\mathrm{Ker}(f) = \{v \in V \mid f(v) = 0_W\}.$$

The *image of* f is the subset of W

$$\mathrm{Im}(f) := \{w \in W \mid \exists\, v \in V \text{ such that } f(v) = w\}.$$

We leave the easy proofs that $\mathrm{Ker}(f)$ and $\mathrm{Im}(f)$ are subspaces of V and W respectively to the reader.

Proposition 1.72 *A linear map of K-vector spaces $f: V \to W$ is injective if and only if* $\mathrm{Ker}(f) = \{0_V\}$.

Proof We have to observe only that $f(v_1) = f(v_2)$ if and only if $v_1 - v_2 \in \mathrm{Ker}(f)$. □

Theorem 1.73 *Let $f: V \to W$ be a linear map of K-vector spaces. If* $\dim_K V < \infty$, *then* $\dim_K \mathrm{Im}(f) < +\infty$ *and*

$$\dim_K V = \dim_K \mathrm{Ker}(f) + \dim_K \mathrm{Im}(f).$$

Proof If $\mathrm{Ker}(f) = \{0_V\}$ or if $\mathrm{Im}(f) = \{0_W\}$, the proof is trivial, hence assume that $\mathrm{Ker}(f) \neq \{0_V\}$ and $\mathrm{Im}(f) \neq \{0_W\}$.

By Corollary 1.53 a base $\{v_1, \ldots, v_k\}$ of $\mathrm{Ker}(f)$ (because it is a free subset of V) is contained in a basis $\{v_1, \ldots, v_k, v_{k+1}, \ldots, v_n\}$ of V. Thus it is enough to prove that $S := \{w_{k+1}, \ldots, w_n\}$ is a basis of $\mathrm{Im}(f)$, where $w_i := f(v_i)$, $i = k+1, \ldots, n$.

First we prove that S is free. Let $(a_{k+1}, \ldots, a_n) \in K^{n-k}$ such that $a_{k+1}w_{k+1} + \cdots + a_n w_n = 0_W$, i.e

$$a_{k+1}f(v_{k+1}) + \cdots + a_n f(v_n) = f(a_{k+1}v_{k+1} + \cdots + a_n v_n) = 0_W.$$

Hence $a_{k+1}v_{k+1} + \cdots + a_n v_n \in \mathrm{Ker}(f)$. Since $\{v_1, \ldots, v_k\}$ is a basis of $\mathrm{Ker}(f)$, there is $(a_1, \ldots, a_k) \in K^k$ such that $a_{k+1}v_{k+1} + \cdots + a_n v_n = a_1 v_1 + \cdots + a_k v_k$, i.e.

$$(-a_1)v_1 + \cdots + (-a_k)v_k + a_{k+1}v_{k+1} + \cdots + a_n v_n = 0_V.$$

Since $\{v_1, \ldots, v_n\}$ is a basis of V we must have $a_1 = \cdots = a_k = a_{k+1} = \cdots = a_n = 0$, which proves our assertion.

Now we prove that S is a system of generators of $\mathrm{Im}(f)$. Let $w \in \mathrm{Im}(f)$. Then there is $v \in V$ such that $w = f(v)$. Since $\{v_1, \ldots, v_n\}$ is a basis of V there is

$(a_1, \ldots, a_n) \in K^n$ such that $v = a_1 v_1 + \ldots + a_n v_n$. Hence

$$\begin{aligned} w = f(v) = f(a_1 v_1 + \ldots + a_n v_n) &= a_1 f(v_1) + \cdots + a_k f(v_k) \\ &+ a_{k+1} f(v_{k+1}) + \cdots + a_n f(v_n) \\ &= a_{k+1} f(v_{k+1}) + \cdots + a_n f(v_n) \end{aligned}$$

(since $v_1, \ldots, v_k \in \mathrm{Ker}(f)$), thereby proving our assertion. □

Corollary 1.74 *Let $f: V \to W$ be a linear map of K-vector spaces of finite dimensions. If $\dim_K V = \dim_K W$ the following assertions are equivalent:*

(i) *f is bijective.*
(ii) *f is injective.*
(iii) *f is surjective.*

Proof It is an easy consequence of Theorem 1.73 and Corollary 1.61. □

Corollary 1.75 *If V and W are isomorphic K-vector spaces of finite dimensions, then $\dim_K V = \dim_K W$.*

Proof Let $f: V \to W$ be an isomorphism. If $\{v_1, \ldots, v_n\}$ is a basis of V then $\{f(v_1), \ldots, f(v_n)\}$ is a basis of W. □

Corollary 1.76 (Grassmann's Formula) *Let V_1 and V_2 be two vector subspaces of a K-vector space V of finite dimension. Then we have*

$$\dim_K(V_1 + V_2) = \dim_K V_1 + \dim_K V_2 - \dim_K(V_1 \cap V_2). \tag{1.26}$$

Proof Let $V' := V_1 \times V_2$. By Proposition 1.63 we have that $\dim_K V' = \dim_K V_1 + \dim_K V_2$. It easy to see that $f: V' \to V$, defined by $f(v_1, v_2) = v_1 - v_2, \forall\, v_1 \in V_1$ and $\forall\, v_2 \in V_2$, is a linear map. By Proposition 1.39 we have $\mathrm{Im}(f) = V_1 + V_2$; furthermore we get $\mathrm{Ker}(f) = \{(v, v) \mid v \in V_1 \cap V_2\}$. Since $g: V_1 \cap V_2 \to \mathrm{Ker}(f)$, defined by $g(v) := f(v, v)$, is a bijective linear map, it follows from Corollary 1.75 that $\dim_K \mathrm{Ker}(f) = \dim_K(V_1 \cap V_2)$. Our thesis follows by applying Theorem 1.73 to f. □

Definition 1.77 Let $f: V \to W$ be a linear map of K-vector spaces of finite dimensions. The *rank of* f is the non-negative integer

$$\mathrm{rank}(f) := \dim_K \mathrm{Im}(f).$$

By Theorem 1.73 we have the inequality $\mathrm{rank}(f) \leq \min\{\dim_K V, \dim_K W\}$. Referring to Example 1.69,2, we remark that $\mathrm{rank}(f)$ and $\mathrm{rank}(g)$ are the same as the rank of matrix $A = (a_{ij})_{i=1,\ldots,m,\, j=1,\ldots,n}$ given in Definition 1.24.

Definition 1.78 Let V be a K-vector space and W a subspace of V. Since W is a normal subgroup of the additive group V we have a canonical structure of group on the quotient set V/W (see Definition 1.4). We recall that two vectors $u, v \in V$ are

equivalent (mod W) (or *congruent* (mod W)) if $u - v \in W$. For every $v \in V$ the coset $\widehat{v} = v + W := \{v + w \mid w \in W\}$ is called the *equivalence class* or *congruence class* of v (mod W).

The quotient set V/W is equipped with the addition $\widehat{u} + \widehat{v} := \widehat{u + v}$, $\forall\, u, v \in V$ and with the scalar multiplication $a\widehat{v} = \widehat{av}$, $\forall\, a \in K$, $\forall\, v \in V$. Both the operations are well-defined and make V/W a K-vector space as the reader can easily prove; we call V/W the *factor vector space of V by W*. It is easily seen that the quotient map $\pi : V \to V/W$ is a surjective linear map.

Proposition 1.79 *Let V be a K-vector space and W a subspace of V. For every linear map of K-vector spaces $g : V \to V'$ with $W \subset \ker(g)$, there is a unique linear map $\widetilde{g} : V/W \to V'$ such that $\widetilde{g} \circ \pi = g$. Moreover if $W = \ker(g)$, then $\widetilde{g}$ is injective.*

Proof We have to take $\widetilde{g}(\widehat{v}) = g(v)$ which is well-posed and linear. If $\widehat{u} = \widehat{v}$ then $v = u + w$ with $w \in W$ and $g(v) = g(u + w) = g(u) + g(w) = g(u)$ since $W \subset \mathrm{Ker}(g)$ so that $\widetilde{g}(\widehat{v}) = \widetilde{g}(\widehat{u})$. Furthermore, for every $h, k \in K$ and $\widehat{u}$, $\widehat{v} \in V/W$ we have

$$\widetilde{g}(h\widehat{v} + k\widehat{u}) = \widetilde{g}(\widehat{hv + ku}) = g(hv + ku) = hg(v) = kg(u) = h\widetilde{g}(\widehat{v}) + k\widetilde{g}(\widehat{u}).$$

Uniqueness of $\widetilde{g}$ follows from surjectivity of π. Finally, if $\widetilde{g}(\widehat{v}) = \widetilde{g}(\widehat{u})$ then $g(v) = g(u)$ i.e. $u - v \in W$ and $\widehat{u} = \widehat{v}$. □

Corollary 1.80 *Under the assumptions and notation of Proposition 1.79, if $\dim_K V < \infty$, then $\dim_K V/W < \infty$ and the relation $\dim_K V/W = \dim_K V - \dim_K W$ yields.*

Proof Apply Theorem 1.73 to canonical projection $\pi : V \to V/W$. □

Proposition 1.81 *Let $f : V \to W$ be a linear map of K-vector spaces. Suppose that $\dim_K V < \infty$. For every basis S of V let $\mathcal{F}(S, W)$ be the set of all maps $g : S \to W$. Then the application*

$$\varphi : \mathrm{Hom}_K(V, W) \to \mathcal{F}(S, W), \qquad \varphi(f) = f|_S,$$

where $f|_S$ is the restriction of f to S, is bijective. Thus in order to define a linear map $f : V \to W$ it suffices to give the values of f on the elements of a basis of V.

Proof Let $S = \{v_1, \ldots, v_n\}$ be a basis of V. By Corollary 1.58, every vector $v \in V$ can be represented as a unique linear combination $v = a_1 v_1 + \cdots + a_n v_n$ with $(a_1, \ldots, a_n) \in K^n$. By linearity of f we have $f(v) = a_1 f(v_1) + \cdots + a_n f(v_n)$, thus f is uniquely determined by $f|_S$, i.e. φ is injective.

For all $u \in \mathcal{F}(S, W)$ take $f_u(v) := a_1 u(v_1) + \cdots + a_n u(v_n)$ for all $v \in V$. Then $f_u : V \to W$ is a linear map and $\varphi(f_u) = u$, i.e. φ is surjective. □

Remark 1.82 Proposition 1.81 is true also if $\dim_K V = \infty$.

Definition 1.83 Let V and W be two K-vector spaces with $\dim_K V = n$ and $\dim_K W = m$ and $f : V \to W$ be a linear map. In order to represent f in an explicit manner we have to fix an ordered basis $S = (v_1, \ldots, v_n)$ of V and an ordered basis $T = (w_1, \ldots, w_m)$ of W. To show that choice we write $f : (V, S) \to (W, T)$. By Corollary 1.58, for every $j \in \{1, \ldots, n\}$ there is a unique m-tuple $(a_{1j}, \ldots, a_{mj}) \in K^m$ such that

$$f(v_j) = \sum_{i=1}^{m} a_{ij} w_i. \tag{1.27}$$

According to Proposition 1.81, to give f is equivalent to giving the matrix

$$M_f^{S,T} := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathcal{M}_{m,n}(K),$$

whose columns are the components of $f(v_1), \ldots, f(v_n)$ with respect to basis T. Hence the relations (1.27) can be written in the form

$$(w_1, \ldots, w_m)\, M_f^{S,T} = (f(v_1), \ldots, f(v_n)). \tag{1.28}$$

We say that $M_f^{S,T}$ is the *matrix associated with the linear map* $f \in \mathrm{Hom}_K(V, W)$ *relative to the bases* S *and* T. If $(x_1, \ldots, x_n)$ are the coordinates of a vector $v \in V$ with respect to S and $(y_1, \ldots, y_m)$ are the coordinates of $f(v) \in W$ with respect to T, then

$$\begin{aligned} f(v) = f\left(\sum_{j=1}^{n} x_j v_j\right) &= \sum_{j=1}^{n} x_j f(v_j) = \sum_{j=1}^{n} x_j \sum_{i=1}^{m} a_{ij} w_i \\ &= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j\right) w_i = \sum_{i=1}^{m} y_i w_i, \end{aligned}$$

i.e.

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{n2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}. \tag{1.29}$$

Remark 1.84 If $A \in \mathcal{M}_{m,n}(K)$ and $f_A \colon K^n \to K^m$ is the linear map associated with A (see Example 1.69-3) then

$$A = M_{f_A}^{S,T}$$

if S and T are the canonical bases of K^n and K^m respectively.

Theorem 1.85 *Under the above assumptions and notation the correspondence*

$$\mathrm{Hom}_K(V, W) \to \mathcal{M}_{m,n}(K), \qquad f \to M_f^{S,T}, \tag{1.30}$$

is an isomorphism of K-vector spaces.

Proof Our statement follows from Proposition 1.81 by canonically identifying $\mathcal{F}(S, W)$ with $\mathcal{M}_{m,n}(K)$. □

Lemma 1.86 *Let V, W, and U be three K-vector spaces of dimensions n, m and p respectively. Let $f \colon V \to W$ and $g \colon W \to U$ be two linear maps of K-vector spaces. Fix three bases $\mathcal{B}_V = (v_1, \ldots, v_n)$, $\mathcal{B}_W = (w_1, \ldots .w_m)$ and $\mathcal{B}_U = (u_1, \ldots, u_p)$ of V, W, and U respectively. Then we have*

$$M_{g\circ f}^{\mathcal{B}_V,\mathcal{B}_U} = M_g^{\mathcal{B}_W,\mathcal{B}_U} M_f^{\mathcal{B}_V,\mathcal{B}_W}. \tag{1.31}$$

Proof Let $M_f^{\mathcal{B}_V,\mathcal{B}_W} = \left(a_{ij}\right)_{\substack{i=1,\ldots,m\\ j=1,\ldots,n}}$ and $M_g^{\mathcal{B}_W,\mathcal{B}_U} = (b_{hi})_{\substack{h=1,\ldots,p\\ i=1,\ldots,m}}$. Then we have

$$g(w_i) = \sum_{h=1}^{p} b_{hi} u_h, \qquad f(v_j) = \sum_{i=1}^{m} a_{ij} w_i,$$

hence

$$\begin{aligned} g(f(v_j)) = g\left(\sum_{i=1}^{m} a_{ij} w_i\right) &= \sum_{i=1}^{m} a_{ij} g(w_i) = \sum_{i=1}^{m} a_{ij} \sum_{h=1}^{p} b_{hi} u_h \\ &= \sum_{h=1}^{p} \left(\sum_{i=1}^{m} b_{hi} a_{ij}\right) u_h \end{aligned}$$

so that

$$M_{g\circ f}^{\mathcal{B}_V,\mathcal{B}_U} = (c_{hj})_{\substack{h=1,\ldots,p\\ j=1,\ldots,n}} = \left(\sum_{i=1}^{m} b_{hi} a_{ij}\right)_{\substack{h=1,\ldots,p\\ j=1,\ldots,n}}$$

which is (1.31). □

Definition 1.87 If $S = (v_1, \dots, v_n)$ and $T = (w_1, \dots, w_n)$ are two bases of V and $\mathrm{id}_V : (V, T) \to (V, S)$ we have

$$w_j = \sum_{i=1}^{n} a_{ij} v_i, \qquad j = 1, \dots, n. \tag{1.32}$$

The associated matrix

$$\Lambda^{S,T} := M_{\mathrm{id}_V}^{T,S} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \in \mathcal{M}_n(K)$$

is called the *matrix of transition from the basis S to the basis T*. The columns of $\Lambda^{S,T}$ are the components of the vectors $w_1, \dots, w_n$ of T with respect to the basis S.

If $(x_1, \dots, x_n)$ are the coordinates of a vector v with respect to S and $(y_1, \dots, y_n)$ those with respect to T, we get as in (1.29)

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}. \tag{1.33}$$

Let $\Lambda^{T,S} = \left(b_{ij}\right)_{i,j=1,\dots,n} \in \mathcal{M}_n(K)$, so that

$$v_j = \sum_{i=1}^{n} b_{ij} w_i, \qquad j = 1, \dots, n. \tag{1.34}$$

From (1.32) and (1.34) we get

$$v_j = \sum_{i=1}^{n} b_{ij} w_i = \sum_{i=1}^{n} b_{ij} \sum_{k=1}^{n} a_{ki} v_k = \sum_{k=1}^{n} \left(\sum_{i=1}^{n} a_{ki} b_{ij} \right) v_k, \qquad i = j, \dots, n,$$

hence (since S is a basis)

$$\sum_{i=1}^{n} a_{ki} b_{ij} = \delta_{kj}, \qquad j, k = 1, \dots, n. \tag{1.35}$$

The relations (1.35) can be expressed in term of matrices of transition as $\Lambda^{S,T}\Lambda^{T,S} = I_n$. In the same way we get $\Lambda^{T,S}\Lambda^{S,T} = I_n$, thus we have proved the following proposition.

Proposition 1.88 *Let S and T two bases of a K-vector space V of dimension n. Then $\Lambda^{S,T}$ is an invertible matrix of the ring $\mathcal{M}_n(K)$ and $(\Lambda^{S,T})^{-1} = \Lambda^{T,S}$.*

If $V = W$ the isomorphism of K-vector spaces (1.30) is also an isomorphism of K-algebras according to the following definitions.

Definition 1.89 Let K be a field and A a unitary ring (not necessarily commutative). We define *K-algebra* (or an *algebra over K*) as a triple (A, K, u) where $u\colon K \to A$ is a homomorphism of unitary rings such that $u(K)$ is contained in the centre $C(A)$ of A (see Definition 1.7). The reader can prove that u is necessarily injective. Moreover A is a K-vector space: the scalar multiplication is given by $ka := u(k)a, \forall\, k \in K, \forall\, a \in A$.

Let (A, K, u) and (B, K, v) two K-algebras. A homomorphism $\varphi\colon A \to B$ of unitary rings is called a *homomorphism of K-algebras* if $\varphi \circ u = v$; if φ is an isomorphism of unitary rings we call it an *isomorphism of K-algebras*. We give three important examples:

(a) $(\mathrm{End}_K(V), K, u)$ (see Definition 1.70) where $u\colon K \to \mathrm{End}_K(V)$, $u(k) = k\,\mathrm{id}_V, \forall\, k \in K$.
(b) $(\mathcal{M}_n(K), K, u)$ (see Definition 1.15) where $u\colon K \to \mathcal{M}_n(K)$, $u(k) = k\,I_n$, $\forall\, k \in K$.
(c) $(K[T], K, u)$ where $u\colon K \to K[T]$, $u(k) = k, \forall\, k \in K$.

Notation If $f \in \mathrm{End}_K(V)$ and we take $S = T = (v_1, \ldots, v_n)$, the matrix $M_f^{S,T}$ of Definition 1.83 will be denoted by M_f^S.

Corollary 1.90 *Let $S = (v_1, \ldots, v_n)$ be an ordered basis of a K-vector space V. Then the isomorphism of K-vector spaces*

$$\mathrm{End}_K(V) \longrightarrow \mathcal{M}_n(K), \qquad f \longrightarrow M_f^S,$$

is an isomorphism of K-algebras i.e. $\mathrm{id}_V \to I_n$ *and*

$$M_{g\circ f}^S = M_g^S\, M_f^S, \qquad \forall\, f, g \in \mathrm{End}_K(V) \tag{1.36}$$

Furthermore $f \in \mathrm{GL}(V)$ if and only if $M_f^S \in \mathrm{GL}_n(K)$.

Proof It follows immediately from Lemma 1.86. □

Proposition 1.91 *Under the notation of Definition 1.87 let $f \in \mathrm{End}_K(V)$ be an endomorphism of a K-vector space V and $S = (v_1, \ldots, v_n)$, $T = (w_1, \ldots, w_n)$ two ordered bases of V. Then we have*

$$M_f^T = \Lambda^{T,S}\, M_f^S\, \Lambda^{S,T}. \tag{1.37}$$

Proof Let

$$M_f^S = (a_{ki})_{k,i=1,\dots,n}\,, \qquad M_f^T = \big(b_{ij}\big)_{i,j=1,\dots,n}\,, \qquad \Lambda^{S,T} = \big(\gamma_{ij}\big)_{i,j=1,\dots,n}\,,$$

i.e.

$$f(w_j) = \sum_{i=1}^{n} b_{ij} w_i\,, \quad w_j = \sum_{i=1}^{n} \gamma_{ij} v_i\,, \quad f(v_i) = \sum_{k=1}^{n} a_{ki} v_k\,, \quad j = 1, \dots, n.$$

We have for $j = 1, \dots, n$

$$f(w_j) = f\left(\sum_{i=1}^{n} \gamma_{ij} v_i\right) = \sum_{i=1}^{n} \gamma_{ij} f(v_i) = \sum_{i=1}^{n} \gamma_{ij} \sum_{k=1}^{n} a_{ki} v_k = \sum_{k=1}^{n} \left(\sum_{i=1}^{n} a_{ki}\gamma_{ij}\right) v_k,$$

as well as

$$f(w_j) = \sum_{i=1}^{n} b_{ij} w_i = \sum_{i=1}^{n} b_{ij} \sum_{k=1}^{n} \gamma_{ki} v_k = \sum_{k=1}^{n} \left(\sum_{i=1}^{n} \gamma_{ki} b_{ij}\right) v_k.$$

Comparing these identities by Corollary 1.58 we have

$$\sum_{i=1}^{n} a_{ki}\gamma_{ij} = \sum_{i=1}^{n} \gamma_{ki} b_{ij}, \qquad j, k = 1, \dots, n,$$

that is, $M_f^S \Lambda^{S,T} = \Lambda^{S,T} M_f^T$. By Proposition 1.88, $\Lambda^{S,T}$ is an invertible matrix of $\mathcal{M}_n(K)$, so that $M_f^T = (\Lambda^{S,T})^{-1} M_f^S \Lambda^{S,T} = \Lambda^{T,S} M_f^S \Lambda^{S,T}$. □

Definition 1.92 The *derminant* and the *trace* of an endomorphism $f \in \mathrm{End}_K(V)$ can be defined by

$$\det(f) := \det(M_f^S), \qquad \mathrm{tr}(f) := \mathrm{tr}(M_f^S),$$

where S is a basis of V. Thanks to Proposition 1.91 the above definitions are independent of the basis chosen.

1.3.1 The Dual Vector Space

Definition 1.93 Let V be a K-vector space. The vector space $V^* := \mathrm{Hom}_K(V, K)$ is called a *dual vector space of* V and its elements are called *linear functionals on* V.

From real analysis we have the following examples:

Example 1.94

(1) Let $V = C([a, b]; \mathbb{R})$ (Example 1.33, 4), then

$$\varphi : V \to \mathbb{R}, \qquad \varphi(f) = \int_a^b f(t)dt, \quad \forall f \in V,$$

is a linear functional on V, i.e. $\varphi \in V^*$.

(2) Let $V = C^\infty(\mathbb{R}; \mathbb{R})$ (Example 1.69, 4), then, for any fixed point $x_0 \in \mathbb{R}$,

$$\psi_{x_0} : V \to \mathbb{R}, \qquad \psi_{x_0}(f) = f'(x_0), \qquad \forall f \in V,$$

is a linear functional on V.

Definition 1.95 Let V be a K-vector space of finite dimension and $\mathcal{B} := (v_1, \ldots, v_n)$ a basis of V. By Proposition 1.81) for every $i = 1, \ldots, n$ there is a unique linear functional

$$v_i^*(v_j) = \delta_{ij}, \qquad j = 1, \ldots, n. \tag{1.38}$$

Theorem 1.96 *According to above definition and notation the set* $\mathcal{B}^* := (v_1^*, \ldots, v_n^*)$ *is a basis of* V^* *so that* V^* *is a* K*-vector space of finite dimension and* $\dim_K V^* = \dim_K V$.

Proof Let $a_1, \ldots, a_n \in K$ such that $v_1^* a_1 + \cdots + v_n^* a_n = 0_{V^*}$. For any $j \in \{1, \ldots, n\}$ we have

$$0 = (v_1^* a_1 + \cdots + v_n^* a_n)(v_j) = v_j^*(v_j) a_j = a_j,$$

which proves that $\mathcal{B}^*$ is free.

Let $f \in V^*$ be a linear functional on V and $a_i := f(v_i)$, $i = 1, \ldots, n$. We want to prove that $f = v_1^* a_1 + \cdots + v_n^* a_n$. By Proposition 1.81, we have to show only that f and $v_1^* a_1 + \cdots + v_n^* a_n$ coincide on the elements of base $\mathcal{B}$. Indeed

$$(v_1^* a_1 + \cdots + v_n^* a_n)(v_i) = a_i = f(v_i), \qquad i = 1, \ldots, n.$$

This concludes our proof. □

The basis $\mathcal{B}^*$ defined above is called *the dual basis of* $\mathcal{B}$.

Example 1.97 Let $V = K^n$. The functional e_i^*, $i = 1, \ldots, n$, defined by

$$e_i^*(x_1, \ldots, x_n) = x_i, \qquad i = 1, \ldots, n,$$

forms a basis of V^* which is the dual basis of the canonical basis $\{e_1, \ldots, e_n\}$ of K^n (see Example 1.62-1)).

Definition 1.98 Let V be a K-vector space. The *bidual of* V is defined by

$$V^{**} := (V^*)^* = \mathrm{Hom}_K(\mathrm{Hom}_K(V, K), K).$$

There is a *canonical homomorphism*

$$\alpha \colon V \to V^{**}, \qquad \alpha(v)(f) = f(v), \ \forall\, v \in V, \ \forall\, f \in V^*. \tag{1.39}$$

Indeed, $\forall\, v \in V$, $\alpha(v)$ is a linear functional on V^*, i.e.

$$\alpha(v)(f+g) = (f+g)(v) = f(v) + g(v) = \alpha(v)(f) + \alpha(v)(g), \ \forall\, f, \ g \in V^*,$$

$$\alpha(v)(\lambda f) = (\lambda f)(v) = \lambda f(v) = \lambda \alpha(v)(f), \quad \forall\, f \in V^*, \ \forall\, \lambda \in K.$$

Furthermore***, α is a linear map, i.e. $\forall\, f \in V^*$,

$$\begin{aligned} \alpha(v_1 + v_2)(f) &= f(v_1 + v_2) = f(v_1) + f(v_2) \\ &= \alpha(v_1)(f) + \alpha(v_2)(f), \ \forall\, v_1, \ v_2 \in V, \\ &\quad \alpha(v_1)(f) + \alpha(v_2)(f), \quad \forall\, v \in V, \ \forall\, \lambda \in K. \end{aligned}$$

As a consequence of Theorem 1.96 we get the following important result:

Theorem 1.99 *Let V be a K-vector space of finite dimension. Then the canonical homomorphism $\alpha \colon V \to V^{**}$ is an isomorphism.*

Proof Since $\dim_K V = \dim_K V^* = \dim_K V^{**}$ by Proposition 1.74 we have to prove only that α is injective. If $v \in \mathrm{Ker}(\alpha)$ then $f(v) = 0\ \forall\, f \in V^*$. Suppose that $v \neq 0_V$. Since $\{v\}$ is free there is a basis $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ of V which contains v by Corollary 1.53. Suppose $v = v_1$, hence if v_1^* is the linear functional defined in (1.38) we have $v_1^*(v) = v_1^*(v_1) = 1 \neq 0$ and $v \notin \mathrm{Ker}(\alpha)$, so that $\mathrm{Ker}(\alpha) = \{0_V\}$ and α is injective by Proposition 1.72. □

Remark 1.100 Let $V = \mathbb{R}[X]$ be the vector space of polynomials with real coefficients. Since $\{1, X, X^2, \ldots\}$ is a basis of V, $\dim_{\mathbb{R}} V = \aleph_0$, i.e. V has countable dimension. We shall show that V^* and V^{**} contain an uncountable free subset so that they cannot be isomorphic to V. We claim that

$$\Phi = \{\phi_t : t \in \mathbb{R}\}, \qquad \phi_t(f(X)) = f(t) \in \mathbb{R},$$

is a free subset of V^* which is obviously uncountable. Given the linear combination

$$a_1 \phi_{t_1} + \cdots + a_n \phi_{t_n} = 0, \qquad a_1, \ldots, a_n \in \mathbb{R},$$

where $t_1, \dots, t_n$ are distinct real numbers, let us consider the polynomials

$$f_i(X) = \prod_{j \neq i} (X - t_j), \qquad i = 1, \dots, n,$$

which vanish in t_j with $j \neq i$ but $f_I(t_i) \neq 0$. Then for $i = 1, \dots, n$ we have

$$0 = (a_1 \phi_{t_1} + \cdots + a_n \phi_{t_n})(f_i(X)) = a_1 f_i(t_1) + \cdots + a_n f_i(t_n) = a_i f_i(t_i),$$

so that $a_i = 0$ for $i = 1, \dots, n$.

Let B be a basis of V^* which contains Φ. For every $t \in \mathbb{R}$ we put

$$g_t(b) := \begin{cases} 1 & \text{if } b = \phi_t, \\ 0 & \text{if } b \neq \phi_t, \end{cases} \qquad \forall\, b \in B.$$

We can extend g_t to a linear map $V^* \to \mathbb{R}$ still denoted by g_t (see Remark 1.82). Then $\{g_t \in V^{**} : t \in \mathbb{R}\}$ is a uncountable free subset of V^{**}. Reasoning as above, from the equality

$$a_1 g_{t_1} + \cdots + a_m g_{t_m} = 0, \qquad a_1, \dots, a_m \in \mathbb{R},$$

where $t_1, \dots, t_m$ are distinct real numbers, we have

$$0 = (a_1 g_{t_1} + \cdots + a_n g_{t_n})(\varphi_{t_i}) = a_i, \qquad i = 1, \dots, m.$$

Definition 1.101 A linear map $f : V \to W$ of K-vector spaces induces a map (see Proposition 1.68)

$$f^* : W^* \to V^*, \qquad f^*(\alpha) := \alpha \circ f, \qquad \forall\, \alpha \in W^*.$$

Since

$$f^*(\alpha + \beta) = (\alpha + \beta) \circ f = \alpha \circ f + \beta \circ f = f^*(\alpha) + f^*(\beta), \qquad \forall\, \alpha, \beta \in W^*,$$

and

$$f^*(k\alpha) = (k\alpha) \circ f = k(\alpha \circ f) = k\, f^*(\alpha), \qquad \forall \alpha \in W^*,\ k \in K,$$

f^* is a linear map called *the dual* of $f : V \to W$.

Analogously, $f^* : W^* \to V^*$ induces a linear map $f^{**} : V^{**} \to W^{**}$ defined by

$$f^{**}(v^{**}) = v^{**} \circ f^*, \qquad \forall\, v^{**} \in V^{**}.$$

If $\alpha_V : V \to V^{**}$ and $\alpha_W : W \to W^{**}$ are the canonical isomorphisms (1.39), then we get the commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ {\scriptstyle \alpha_V}\downarrow & & \downarrow{\scriptstyle \alpha_W} \\ V^{**} & \xrightarrow[f^{**}]{} & W^{**} \end{array}$$

Lemma 1.102 *Let $f : V \to W$, $g : V \to W$ and $h : W \to U$ be linear maps of K-vector spaces. Then*

(a) $(f+g)^* = f^* + g^*$,
(b) $(kf)^* = k\, f^*$, $\forall k \in K$,
(c) $(h \circ f)^* = f^* \circ h^*$,
(d) $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$; *if* 0 *is the null homomorphism* $V \to W$ *then* 0^* *is the null homomorphism* $W^* \to V^*$.

Proof We leave it to the reader as an easy exercise. □

Proposition 1.103 *Let $f : V \to W$ be a linear map of K-vector spaces of finite dimensions. If $\mathcal{B}$ and $\mathcal{C}$ are bases of V and W respectively, then*

$$M_{f^*}^{\mathcal{C}^*,\mathcal{B}^*} = (M_f^{\mathcal{B},\mathcal{C}})^t,$$

i.e. the matrix associated with f^* relative to dual bases $\mathcal{C}^*$ and $\mathcal{B}^*$ is the transpose of the matrix associated with f relative to bases $\mathcal{B}$ and $\mathcal{C}$.

Proof Put $\mathcal{B} = (v_1, \ldots, v_n)$, $\mathcal{C} = (w_1, \ldots, w_m)$ and $M_f^{\mathcal{B},\mathcal{C}} = \big(a_{ij}\big)_{\substack{i=1,\ldots,m \\ j=1,\ldots,n}} \in \mathcal{M}_{m,n}(K)$. Hence we have the equalities (see (1.27))

$$f(v_j) = \sum_{i=1}^{m} a_{ij} w_i, \qquad j = 1, \ldots, n. \tag{1.40}$$

In the same way, if $\mathcal{C}^* = (w_1^*, \ldots, w_m^*)$, $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ and $M_{f^*}^{\mathcal{C}^*,\mathcal{B}^*} = (c_{ji})_{\substack{j=1,\ldots,n \\ i=1,\ldots,m}} \in \mathcal{M}_{n,m}(K)$ we get

$$f^*(w_i^*) = \sum_{j=1}^{n} c_{ji} v_j^*, \qquad i = 1, \ldots, m, \tag{1.41}$$

and hence

$$f^*(w_i^*)(v_j) = (w_i^* \circ f)(v_j)$$
$$= \sum_{k=1}^{n} c_{ki} v_k^*(v_j) = c_{ji}, \quad i = 1, \ldots, m, \ j = 1, \ldots, n.$$

On the other hand, by (1.40) we have

$$(w_i^* \circ f)(v_j) = w_i^*(f(v_j)) = w_i^* \left(\sum_{k=1}^{m} a_{kj} w_k \right)$$
$$= \sum_{k=1}^{m} a_{kj} w_i^*(w_k) = a_{ij}, \qquad i = 1, \ldots, m, \ j = 1, \ldots, n,$$

so that

$$c_{ji} = a_{ij}, \qquad i = 1, \ldots, m, \ j = 1, \ldots, n,$$

i.e. $M_{f^*}^{\mathcal{C}^*, \mathcal{B}^*} = (M_f^{\mathcal{B}, \mathcal{C}})^t$. □

Lemma 1.104 *Let $f: V \to W$ be a linear map of K-vector spaces of finite dimensions. and $f^*: W^* \to V^*$ the dual homomorphism of f.*

(i) *If f is surjective, then f^* is injective.*
(ii) *If f is injective then f^* is surjective.*
(iii) *If f is an isomorphism then f^* is an isomorphism.*

Proof

(i) If $f^*(\alpha) = f^*(\beta)$ for some $\alpha, \beta \in W^*$ then $\alpha \circ f = \beta \circ f$ and $\alpha = \beta$ since f is surjective.
(ii) Because f is an injective linear map we can suppose that V is a subspace of W and f is the inclusion $V \hookrightarrow W$. Let $\alpha \in V^*$ and suppose $\dim_K V = n$, $\dim_K W = m$. A basis $\{e_1, \ldots, e_n\}$ of V is contained in a basis $\{e_1, \ldots, e_n, e_{n+1}, \ldots, e_m\}$ of W by Corollary 1.53. Then

$$\beta: W \to K, \qquad \beta(e_i) = \begin{cases} \alpha(e_i) & i = 1, \ldots, n, \\ 0 & i = n+1, \ldots, m, \end{cases}$$

is a linear functional on W such that $f^*(\beta) = \alpha$.
(iii) Immediate from (i) and (ii).

□

Corollary 1.105 *Let $f: V \to W$ be a linear map of K-vector spaces of finite dimensions. Then*

$$\operatorname{rank}(f) = \operatorname{rank}(f^*).$$

Proof We can write $f = i \circ g$, where $g: V \to \operatorname{Im}(f)$, $g(v) = f(v)$, $\forall\, v \in V$, and $i: \operatorname{Im}(f) \hookrightarrow W$ the inclusion map. By Lemma 1.102 we have $f^* = g^* \circ i^*$, where $g^*: \operatorname{Im}(f)^* \to V^*$ is injective (g is surjective) and $i^*: W^* \to \operatorname{Im}(f)^*$ is surjective (i is injective). Since $\operatorname{Im}(g^*) = \operatorname{Im}(f^*)$, $g^*: \operatorname{Im}(f)^* \to \operatorname{Im}(f^*)$ is bijective so that $\operatorname{Im}(f)^*$ is isomorphic to $\operatorname{Im}(f^*)$. Hence $\operatorname{rank}(f) = \dim_K \operatorname{Im}(f) = \dim_K \operatorname{Im}(f)^* = \dim_K \operatorname{Im}(f^*) = \operatorname{rank}(f^*)$. □

Definition 1.106 Let W be a subspace of a vector space V. The set $W^0 := \{f \in V^* \mid f(w) = 0,\ \forall\, w \in W\}$ is called the *annihilator* of W. The annihilator W^0 is clearly a vector subspace of V^*.

Corollary 1.107 *Let V be a K-vector space of dimension n. Denote the sets of vector subspaces of V and V^* by $\mathcal{L}(V)$ and $\mathcal{L}(V^*)$ respectively. The correspondence*

$$\mathcal{L}(V) \to \mathcal{L}(V^*), \qquad W \to W^0, \qquad \forall\, W \in \mathcal{L}(V),$$

is bijective and reverses the inclusions, i.e. if $U \subset W$ then $W^0 \subset U^0$, $\forall\, U, W \in \mathcal{L}(V)$. Furthermore

$$\dim_K W + \dim_K W^0 = n.$$

Proof First observe that $W^0 = \operatorname{Ker}(i^*)$, where $i: W \hookrightarrow V$ is the inclusion map. By Lemma 1.104, i^* is surjective so that $\dim_K \operatorname{Ker}(i^*) + \dim_K \operatorname{Im}(i^*) = \dim_K V^*$ by Theorem 1.73. Hence $\dim_K W + \dim_K W^0 = n$ by Corollary 1.105 and Theorem 1.96.

Consider now the correspondence

$$\mathcal{L}(V^*) \to \mathcal{L}(V^{**}), \qquad F \to F^0, \qquad \forall\, F \in \mathcal{L}(V^*).$$

By identifying V^{**} with V by means of the canonical isomorphism $\alpha: V \to V^{**}$ (see Theorem 1.99), we observe that $W \subseteq (W^0)^0$ and, since $\dim_K((W^0)^0) = n - \dim_K W^0 = \dim_K W$, we have $W = (W^0)^0$, for every vector subspace W of V. In this way we have proved that the correspondence $W \to W^0 \to (W^0)^0$ is $\mathrm{id}_{\mathcal{L}(V)}$. In a similar way it can be shown that $F \to F^0 \to (F^0)^0$ is $\mathrm{id}_{\mathcal{L}(V^*)}$. □

1.3.2 Direct Sum of Vector Subspaces

Proposition 1.108 *Let V be a K-vector space and $V_1, \ldots, V_r$ subspaces of V. The following assertions are equivalent:*

(i) $V_1 + \cdots + V_r = V$ *and* $V_i \cap (V_1 + \cdots + V_{i-1} + V_{i+1} + \cdots + V_r) = 0_V$, $i = 1, \ldots, r$.

(ii) *Every vector $v \in V$ can be uniquely represented in the form $v_1 + \cdots + v_r$, with $v_i \in V_i$, $i = 1, \ldots, r$, i.e. if $v = v_1 + \cdots + v_r = v_1' + \cdots + v_r'$, with $v_i, v_i' \in V_i$, $i = 1, \ldots, r$, then $v_i = v_i'$ $i = 1, \ldots, r$.*

(iii) *Every vector $v \in V$ can be represented in the form $v = v_1 + \cdots + v_r$, with $v_i \in V_i$, $i = 1, \ldots, r$; if $0_V = w_1 + \cdots + w_r$, with $w_i \in V_i$ $i = 1, \ldots, r$ then $w_i = 0_V$, $i = 1, \ldots, r$.*

(iv) *The map $f : V_1 \times V_2 \times \cdots \times V_r \to V$, defined by $f(v_1, v_2, \ldots, v_r) := v_1 + v_2 + \cdots + v_r$, is an isomorphism, where $V_1 \times V_2 \times \cdots \times V_r$ is the direct product of $V_1, \ldots, V_r$ according to* Definition 1.34.

Proof (i) $\Longrightarrow$ (ii). By Proposition 1.39 for every $v \in V$ there are $v_i \in V_i$, $i = 1, \ldots, r$, such that $v = v_1 + \cdots + v_r$. If $v = v_1 + \cdots + v_r = v_1' + \cdots + v_r'$, with $v_i, v_i' \in V_i$, $i = 1, \ldots, r$, then $(v_1 - v_1') + \cdots + (v_r - v_r') = 0_V$. Hence we have

$$v_i' - v_i = \sum_{j \neq i} (v_j - v_j'), \qquad i = 1, \ldots, r. \tag{1.42}$$

The left–hand side of (1.42) belongs to V_i, while the right–hand side belongs to $\sum_{j \neq i} V_j$. By our assumption we have $v_i' - v_i = 0_V$, i.e. $v_i = v_i'$, $i = 1, \ldots, r$.

(ii) $\Longrightarrow$ (iii) is obvious.

(iii) $\Longrightarrow$(i). The first statement of (iii) is equivalent to $V = V_1 + \cdots + V_r$ by Proposition 1.39. We have only to prove that $V_i \cap (\sum_{j \neq i} V_j) = 0_V$. We can suppose that $i = 1$ and $v \in V_1 \cap (V_2 + \cdots + V_r)$. There are $w_j \in V_j$, $j = 2, \ldots, r$ such that $v = w_2 + \cdots + w_r$. Let $w_1 := -v$. Thus we obtain $w_1 + w_2 + \cdots + w_r = 0_V$, with $w_i \in V_i$, $i = 1, 2, \ldots, r$, hence $w_i = 0_V$, $i = 1, \ldots, r$ and in particular $v = -w_1 = 0_V$.

(iii) $\Longleftrightarrow$ (iv). We have to remark only that surjectivity of f is equivalent to the first thesis of (iii) and injectivity to the second. Furthermore, f is clearly linear. □

Remark 1.109 Note that the condition (i) of the Proposition 1.108 is stronger than saying simply

$$V_i \cap V_j = 0_V, \qquad \text{for } i \neq j.$$

Definition 1.110 Under the above assumptions and notation, we call V the *direct sum* of subspaces $V_1, \ldots, V_r$ if any of (i), (ii), (iii), (iv) of Proposition 1.108 is satisfied. We write $V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$.

If W is a vector subspace of V, a subspace Z of V such that $V = W \oplus Z$ is called a *complement* of W; W and Z are said to be *complementary*. It is clear that every subspace has a complement.

Corollary 1.111 *Let V be a K-vector space of finite dimension and $V_1, \ldots, V_r$ subspaces of V such that $V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$. Let $\mathcal{B}_i$ a basis of V_i, $i = 1, \ldots, r$. Then $\mathcal{B} := \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$ is a basis of V. In particular we get* $\dim_K V = \dim_K V_1 + \dim_K V_2 + \cdots + \dim_K V_r$.

Proof Let $d_i = \dim V_i$, $i = 1, \ldots, r$. By Proposition 1.108 every $v \in V$ can be represented in the form $v = v_1 + \cdots + v_r$, with $v_i \in V_i$, $i = 1, \ldots, r$. Since $\mathcal{B}_i = \{e_{i1}, \ldots, e_{id_i}\}$ is a basis of V_i

$$v_i = \sum_{j=1}^{d_i} a_{ij} e_{ij}, \qquad i = 1, \ldots, r, \ a_{ij} \in K,$$

so that $\mathcal{B}$ is a system of generators of V. Suppose now

$$\sum_{i=1}^{r} \sum_{j}^{d_i} \lambda_{ij} e_{ij} = 0, \qquad \lambda_{ij} \in K,$$

then from (iii) of Proposition 1.108 it follows that

$$\sum_{j=1}^{d_i} \lambda_{ij} e_{ij} = 0, \qquad i = 1, \ldots, r,$$

which implies $\lambda_{ij} = 0$, $j = 1, \ldots, d_i$, since $\mathcal{B}_i$ is a basis V_i, $i = 1, \ldots, r$. Hence $\mathcal{B}$ is a basis of V. □

Definition 1.112 Let $\sigma \in \mathrm{End}_K(V)$ be an endomorphism of a K-vector space V. A subspace $W \subseteq V$ is an *invariant subspace relative to* σ if $\sigma(W) \subseteq W$.

Proposition 1.113 *Let V be a K-vector space of finite dimension and $\sigma \in \mathrm{End}_K(V)$. Suppose that $V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$ where all W_i are invariant subspaces relative to σ. Let $\mathcal{B}_i$ be a basis of W_i, $i = 1, \ldots, r$, and $\mathcal{B} := \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$. Then the matrix $M_\sigma^{\mathcal{B}}$ is given by*

$$M_\sigma^{\mathcal{B}} = \begin{pmatrix} M_{\sigma_1}^{\mathcal{B}_1} & 0 & \ldots & 0 \\ 0 & M_{\sigma_2}^{\mathcal{B}_2} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & M_{\sigma_m}^{\mathcal{B}_r} \end{pmatrix},$$

where $M_{\sigma_i}^{\mathcal{B}_i}$ is the matrix associated with $\sigma_i := \sigma_{|W_i} \in \mathrm{End}_K(W_i)$.

Proof We have seen in Corollary 1.111 that $\mathcal{B}$ is a basis of V. We consider the case $r = 2$, the general case being equivalent. Let $\mathcal{B}_1 = \{e_1, \ldots, e_p\}$ and $\mathcal{B}_2 = \{e_{p+1}, \ldots, e_n\}$. Since W_1 and W_2 are invariant subspaces of σ we have $\sigma(e_i) = \sum_{k=1}^{p} a_{ki} e_k$, $i = 1, \ldots, p$ and $\sigma(e_j) = \sum_{l=p+1}^{n} a_{lj} e_l$, $j = p+1, \ldots, n$. Hence

$$\begin{aligned}
\sigma(e_1) &= a_{11}e_1 + \cdots + a_{p1}e_p + 0 \cdot e_{p+1} + \cdots + 0 \cdot e_n \\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
\sigma(e_p) &= a_{1p}e_1 + \cdots + a_{pp}e_p + 0 \cdot e_{p+1} + \cdots + 0 \cdot e_n \\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
\sigma(e_{p+1}) &= 0 \cdot e_1 + \cdots + 0 \cdot e_p + a_{p+1,p+1}e_{p+1} + \cdots + a_{n,p+1}e_n \\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
\sigma(e_n) &= 0 \cdot e_1 + \cdots + 0 \cdot e_p + a_{p+1,n}e_{p+1} + \cdots + a_{nn}e_n.
\end{aligned}$$

Our proposition follows from Definition 1.83. □

1.3.3 Eigenvalues and Eigenvectors

Definition 1.114 Let V be a K-vector space and $f \in \mathrm{End}_K(V)$ an endomorphism of V. A vector $v \in V \setminus \{0_V\}$ satisfying the relation $f(v) = \lambda v$ for some $\lambda \in K$ is called an *eigenvector* of f and the scalar λ is called an *eigenvalue* of f. We observe that if $\lambda v = \mu v$ then $\lambda = \mu$ (since $v \neq 0_V$), then the scalar λ is uniquely determined with respect to v. We can associate with an eigenvalue λ the set $V_\lambda := \{v \in V : f(v) = \lambda v\}$ which is a vector subspace of V, called the *eigenspace of the eigenvalue* λ of f. The eigenspace V_λ is clearly an invariant subspace of f.

The definitions given above can be referred to a square matrix $A \in \mathcal{M}_n(K)$. A vector $\mathbf{x} = (x_1, \ldots, x_n) \in K^n \setminus 0_{K^n}$ is called an *eigenvector of* A if $\mathbf{x}$ is an eigenvector of the linear map $f_A \colon K^n \to K^n$ (see Example 1.69-3), i.e if $A\mathbf{x}^t = \lambda \mathbf{x}^t$. The scalar λ is called an *eigenvalue* of A.

The set of all eigenvalues of an endomorphism f or a matrix A is called the *spectrum* of the endomorphism or matrix. We denote the spectrum by Spec (f) or Spec (A).

Lemma 1.115 *Let V be a K-vector space and $f \in \mathrm{End}_K(V)$ an endomorphism of V.*

(i) *Let $v_1, \ldots, v_r \in V$ be r distinct eigenvectors of f corresponding to r distinct eigenvalues $\lambda_1, \ldots, \lambda_r$, then the set $\{v_1, \ldots, v_r\}$ is free.*

(ii) *If $\lambda_1, \ldots, \lambda_r$ are r distinct eigenvalues of f and if B_i is a basis of the eigenspace V_{λ_i}, $i = 1, \ldots, r$, then the set $B_1 \cup \cdots \cup B_r$ is free.*

(iii) $\sum_{i=1}^{r} V_{\lambda_i} = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_r}$.

Proof

(i) We proceed by induction on r. The case $r = 1$ is obvious since $v_1 \neq 0$. Suppose $r \geq 2$ and

$$a_1 v_1 + \cdots + a_r v_r = 0_V, \qquad a_1, \ldots, a_r \in K, \tag{1.43}$$

then $f(a_1 v_1 + \cdots + a_r v_r) = 0_V$, i.e.

$$a_1 \lambda_1 v_1 + a_2 \lambda_2 v_2 + \cdots + a_r \lambda_r v_r = 0_V. \tag{1.44}$$

because $f(v_i) = \lambda_i v_i$, $i = 1, \ldots, r$. Multiplying (1.43) by λ_1 we obtain

$$\lambda_1 a_1 v_1 + \lambda_1 a_2 v_2 + \cdots + \lambda_1 a_r v_r = 0_V. \tag{1.45}$$

Subtracting (1.45) from (1.44) we have

$$a_2(\lambda_2 - \lambda_1)v_2 + \cdots + a_r(\lambda_r - \lambda_1)v_r = 0_V,$$

which implies $a_i(\lambda_i - \lambda_1) = 0 \ \forall i \geq 2$, by inductive hypothesis. Then we get $a_i = 0$ since $\lambda_i - \lambda_1 \neq 0$, $\forall i \geq 2$. Thus the relation (1.44) takes the form $a_1 v_1 = 0_V$, which implies $a_1 = 0$ since $v_1 \neq 0_V$.

(ii) Without loss of generality we can assume $r = 2$. Let $B_1 = \{v_1, \ldots, v_m\}$, $B_2 = \{w_1, \ldots, w_p\}$. Suppose that

$$a_1 v_1 + \cdots + a_m v_m + b_1 w_1 + \cdots + b_p w_p = 0_V, \qquad a_i,\ b_j \in K.$$

The vector $v := a_1 v_1 + \cdots + a_m v_m$ is an eigenvector relative to the eigenvalue λ_1 if $v \neq 0_V$. Similarly, $w := b_1 w_1 + \cdots + b_p w_p$ is an eigenvector relative to the eigenvalue λ_2 if $w \neq 0_V$. Since $v + w = 0_V$, $f(v + w) = 0_V$ so that

$$\lambda_1 v + \lambda_2 w = 0_V. \tag{1.46}$$

Because $\lambda_1 \neq \lambda_2$, the relation (1.46) implies $v = w = 0_V$ by i). Hence $a_1 = \cdots = a_m = b_1 = \cdots = b_p = 0$, since B_i is a basis of V_{λ_i}, $i = 1, 2$.

(iii) Use Proposition 1.108 and (i).

□

Definition 1.116 Let $A = \left(a_{ij}\right)_{i,j=1,\ldots,n} \in \mathcal{M}_n(K)$. We define the *characteristic matrix of* A to be the matrix

$$A(T) = A - TI_n = \left(a_{ij} - T\delta_{ij}\right)_{i,j=1,\ldots,n} = \begin{pmatrix} a_{11} - T & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - T & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - T \end{pmatrix}$$

with entries in the polynomial ring $K[T]$. The determinant

$$P_A(T) := \det(A - TI_n)$$

is a polynomial of $K[T]$ called the *characteristic polynomial* of A.

Lemma 1.117 *If two matrices $A, B \in \mathcal{M}_n(K)$ are similar then*

$$P_A(T) = P_B(T).$$

Proof Let $P \in \mathrm{GL}_n(K)$ such that $B = P^{-1}AP$. Then

$$B - TI_n = P^{-1}AP - TI_n = P^{-1}(A - TI_n)P.$$

Taking the determinants we get (by Proposition 1.20)

$$\det(B - TI_n) = \det\left(P^{-1}(A - TI_n)P\right) = \det(A - TI_n).$$

□

Definition 1.118 Let $f \in \mathrm{End}_K(V)$ be an endomorphism of a K-vector space of dimension n. Fix a basis $\mathcal{B} = \{e_1, \ldots, e_n\}$ of V. We define the *characteristic polynomial of* f to be the polynomial

$$P_f(T) := P_{M_f^{\mathcal{B}}}(T) = \det(M_f^{\mathcal{B}} - TI_n) \in K[T], \tag{1.47}$$

where $M_f^{\mathcal{B}} = \left(a_{ij}\right)_{i,j=1,\ldots,n}$ is the matrix associated with f relative to the basis $\mathcal{B}$.

As a matter of fact the characteristic polynomial $P_f(T)$ is independent of the basis chosen.

If $\mathcal{D}$ is another basis of V, by Proposition 1.91 one has

$$M_f^{\mathcal{D}} = \Lambda^{\mathcal{D},\mathcal{B}} M_f^{\mathcal{B}} \Lambda^{\mathcal{B},\mathcal{D}},$$

i.e. $M_f^{\mathcal{D}}$ is similar to $M_f^{\mathcal{B}}$ so that our thesis follows by Lemma 1.117.

As a consequence, the coefficients of $P_f(T)$ are independent of $\mathcal{B}$ and represent important invariants associated with f. We can write $P_f(T) = T^n + a_1(f)T^{n-1} + \cdots + a_n(f)$ with $n = \dim_K V$. In particular we have the *determinant* of f

$$\det(f) := \det(M_f^{\mathcal{B}}) = (-1)^n a_n(f), \tag{1.48}$$

and the *trace* of f (see Definition 1.15 and Definition 1.21.4)

$$\mathrm{tr}(f) := \mathrm{tr}(M_f^{\mathcal{B}}) = (-1)^{n+1} a_1(f). \tag{1.49}$$

Proposition 1.119 *Let $f \in \mathrm{End}_K(V)$ be an endomorphism of a K-vector space of finite dimension. A scalar $\lambda \in K$ is an eigenvalue of f if and only if λ is a root of the characteristic polynomial $P_f(T)$.*

Proof First we fix a basis $\mathcal{B} = \{e_1, \ldots, e_n\}$ of V. Let $M_f^{\mathcal{B}} = \left(a_{ij}\right)_{i,j=1,\ldots,n}$ and $\mathbf{x} = (x_1, \ldots, x_n)$ be the coordinates of $v \in V$ with respect to $\mathcal{B}$. Then we have the equivalence

$$f(v) = \lambda v \quad \Longleftrightarrow \quad \left(M_f^{\mathcal{B}} - \lambda I_n\right)\mathbf{x}^t = 0_V.$$

We get a solution $v \neq 0_V$ if and only if the homogeneous system of n equations in n unknowns with coefficients in K

$$\begin{cases} (a_{11} - \lambda)\, x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \cdots + a_{2n}x_n & = 0 \\ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + (a_{nn} - \lambda)x_n & = 0 \end{cases}$$

has a non–trivial solution. This condition is equivalent to $\det(M_f^{\mathcal{B}} - \lambda I_n) = 0$ i.e. $P_f(\lambda) = 0$, by Remark 1.28. □

Corollary 1.120 *Let $A \in \mathcal{M}_n(K)$ be a square matrix. A scalar $\lambda \in K$ is an eigenvalue of A if and only if λ is a root of the characteristic polynomial $P_A(T)$.*

Definition 1.121 Let R be a ring. A matrix $A = \left(a_{ij}\right)_{i,j=i,\ldots,n} \in \mathcal{M}_n(R)$ is called *symmetric* if $a_{ij} = a_{ji}$, $i, j = 1, \ldots, n$, i.e. if $A = A^t$. If $A = -A^t$ we call A *skew-symmetric*.

The following result is crucial for the classification of hyperquadrics.

Proposition 1.122 *Let $C \in \mathcal{M}_n(\mathbb{R})$ be a non-zero symmetric matrix. Then all roots of the characteristic polynomial of C are real.*

Proof Since $\mathbb{C}$ is algebraically closed the polynomial $P_C(T) = \det(C - TI_n) \in \mathbb{R}[T]$ has all his roots in $\mathbb{C}$. Let $\lambda \in \mathbb{C}$ be any root of $P_C(T)$. By Proposition 1.119, there is a vector $a = (a_1, \ldots, a_n) \neq 0$ of $\mathbb{C}^n$ such that $Ca^t = \lambda a^t$.

If $b = (b_1, \ldots, b_n) \in \mathbb{C}^n$ and $B = \left(b_{ij}\right) \in \mathcal{M}_n(\mathbb{C})$, denote $(\overline{b_1}, \ldots, \overline{b_n})$ by $\overline{b}$ and $(\overline{b_{ij}})$ by $\overline{B}$, where $\overline{z}$ is the complex conjugate of $z \in \mathbb{C}$. From $Ca^t = \lambda a^t$ we obtain

$$\overline{Ca^t} = \overline{\lambda a^t} \quad \Longleftrightarrow \quad \overline{C}\,\overline{a}^t = \overline{\lambda}\,\overline{a}^t \quad \Longleftrightarrow \quad C\,\overline{a}^t = \overline{\lambda}\,\overline{a}^t \quad \Longleftrightarrow \quad \overline{a}C = \overline{\lambda}\,\overline{a},$$

since C is symmetric with real entries. Hence we get

$$\overline{a}Ca^t = \overline{a}(Ca^t) = \overline{a}(\lambda a^t) = \lambda\,\overline{a}a^t, \qquad \overline{a}Ca^t = (\overline{a}C)a^t = \overline{\lambda}\,\overline{a}a^t,$$

which leads to $\lambda\,\overline{a}a^t = \overline{\lambda}\,\overline{a}a^t$. Since $\overline{a}a^t = |a_1|^2 + \cdots + |a_n|^2 \neq 0$, we have $\lambda = \overline{\lambda}$, i.e. $\lambda \in \mathbb{R}$. □

Definition 1.123 Let $f \in \mathrm{End}_K(V)$ be an endomorphism of a K-vector space V of dimension n and $K[T]$ the K-algebra of polynomials in the indeterminate T with coefficients in K. The map

$$\begin{aligned}\varphi_f \colon K[T] &\to \mathrm{End}_K(V), \quad \varphi_f(a_0 + a_1 T + \cdots + a_m T^m)\\ &:= a_0\,\mathrm{id}_V + a_1 f + \cdots + a_m f^m,\end{aligned}$$

(we recall that $f^m = \underbrace{f \circ \cdots \circ f}_{m \text{ times}}$), i.e. $\varphi_f(F(T)) = F(f)$ for every polynomial $F(T) \in K[T]$, is a homomorphism of K-algebras (Definition 1.89). Thanks to the homomorphism φ_f we can define a scalar multiplication on V:

$$K[T] \times V \to V, \qquad F(T)\,v := F(f)(v), \qquad \forall\, F(T) \in K[T],\ \forall\, v \in V,$$

i.e.

$$(a_0 + a_1 T + \cdots + a_m T^m)\,v := a_0 v + a_1 f(v) + \cdots + a_m f^m(v). \tag{1.50}$$

In this way V is a $K[T]$-module.

Theorem 1.124 (Cayley–Hamilton) *If $P_f(T)$ is the characteristic polynomial of the endomorphism $f \in \mathrm{End}_K(V)$ then one has*

$$P_f(f) = 0.$$

Proof Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of V. Then we have

$$f(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \qquad j = 1, \ldots, n, \qquad M_f^{\mathcal{B}} = \left(a_{ij}\right)_{i,j}.$$

Hence by (1.50) we can write

$$T \cdot v_j = \sum_{i=1}^{n} a_{ij} v_i, \ \ j = 1, \ldots, n. \tag{1.51}$$

If we denote the matrix $M_f^{\mathcal{B}} - T\,I_n \in \mathcal{M}_n(K[T])$ by $B(T)$, the equalities (1.51) can be written in the form

$$(v_1, \ldots, v_n)\,B(T) = (0_V, \ldots, 0_V).$$

Let $\mathrm{adj}(B(T))$ be the adjoint matrix of $B(T)$ (see Definition 1.18), then we get from (1.12)

$$(0_V, \dots, 0_V) = (v_1, \dots, v_n)\, B(T)\, \mathrm{adj}(B(T))$$
$$= (v_1, \dots, v_n)\, \det(B(T))\, I_n = \left(P_f(T)\, v_1, \dots, P_f(T)\, v_n\right),$$

since $\det(B(T)) = P_f(T)$. Hence $P_f(T)\, v_i = P_f(f)(v_i) = 0_V$, $i = 1, \dots, n$, i.e. $P_f(f)$ is the null endomorphism, being null on a basis. □

Corollary 1.125 *Let $A \in \mathcal{M}_n(K)$ be a square matrix with entries in K, then $P_A(A) = 0$.*

Proof It follows at once from Theorem 1.124 when $V = K^n$ and $f \in \mathrm{End}_K(V)$, $f(x_1, \dots, x_n) = A\,(x_1, \dots, x_n)^t$, $\forall\, (x_1, \dots, x_n) \in K^n$. □

Proposition 1.126 *Let V be an $\mathbb{R}$-vector space of dimension $n \geq 1$ and $\sigma \in \mathrm{End}_{\mathbb{R}}(V)$ an endomorphism of V. Then σ has an invariant subspace of dimension 1 or 2.*

Proof The characteristic polynomial $P_\sigma(T)$ has a root $\lambda \in \mathbb{C}$ by the fundamental theorem of algebra (see [22], pp. 272–273).
If $\lambda \in \mathbb{R}$ then λ is an eigenvalues of σ (by Proposition 1.119) and there is an eigenvector $v \neq 0_V$ so that $\sigma(v) = \lambda \cdot v$. Hence $W = \langle v \rangle$ is an invariant subspace of dimension 1.
Suppose now that $\lambda = \alpha + i\beta \in \mathbb{C} \setminus \mathbb{R}$. Let $\mathcal{B} = \{e_1, \dots, e_n\}$ be a basis of V and $(a_{hk})_{h,k=1,\dots,n}$ the matrix associated with σ relative to $\mathcal{B}$. The homogeneous linear system (with complex coefficients)

$$\begin{cases} (a_{11} - \lambda)\, x_1 + a_{12} x_2 + \cdots + a_{1n} x_n & = 0 \\ a_{21} x_1 + (a_{22} - \lambda) x_2 + \cdots + a_{2n} x_n & = 0 \\ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \cdots \\ a_{n1} x_1 + a_{n2} x_2 + \cdots + (a_{nn} - \lambda) x_n & = 0 \end{cases} \tag{1.52}$$

has a complex solution $(\xi_1 + i\eta_1, \dots, \xi_n + i\eta_n) \neq (0, \dots, 0)$, with $\xi_k, \eta_k \in \mathbb{R}$, $k = 1, \dots, n$. If we substitute $(x_1, \dots, x_n)$ with $(\xi_1 + i\eta_1, \dots, \xi_n + i\eta_n)$ in the system (1.52) and we separate real parts and imaginary parts we get

$$\sum_{k=1}^{n} a_{hk}\xi_k = \alpha\xi_h - \beta\eta_h, \qquad \sum_{k=1}^{n} a_{hk}\eta_k = \alpha\eta_h + \beta\xi_h, \qquad h = 1, \dots, n, \tag{1.53}$$

i.e.

$$\sigma(v) = \alpha v - \beta w, \qquad \sigma(w) = \alpha w + \beta v,$$

where $v := \sum_{k=1}^{n} \xi_k e_k$ and $w := \sum_{k=1}^{n} \eta_k e_k$. Hence $W = \langle v, w \rangle$ is an invariant subspace of dimension 2. □

Diagonalizable Endomorphisms and Matrices

Definition 1.127 Let V be a K-vector space of dimension $n \geq 1$, $f \in \mathrm{End}_K(V)$ an endomorphism of V and λ an eigenvalue of f.

(a) The *algebraic multiplicity* of λ is the multiplicity of λ as a root of the characteristic polynomial $P_f(T)$.
(b) The *geometric multiplicity* of λ is the dimension of the eigenspace V_λ.

Proposition 1.128 *Let V be a K-vector space V of dimension $n \geq 1$. The geometric multiplicity of an eigenvalue λ of $f \in \mathrm{End}_K(V)$ is less than or equal to its algebraic multiplicity.*

Proof Let $\mathcal{B} = \{v_1, \ldots, v_r\}$ be a basis of V_λ. By Corollary 1.53 there are $v_{r+1}, \ldots, v_n \in V$ such that $\{v_1, \ldots, v_n\}$ is a basis of V. Then the matrix $M_f^{\mathcal{B}}$ is of the form

$$\begin{pmatrix} \lambda & 0 & \cdots & 0 & a_{1\,r+1} & \cdots & a_{1\,n} \\ 0 & \lambda & \cdots & 0 & a_{2\,r+1} & \cdots & a_{2\,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & a_{r\,r+1} & \cdots & a_{r\,n} \\ 0 & 0 & \cdots & 0 & a_{r+1\,r+1} & \cdots & a_{r+1\,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & a_{n\,r+1} & \cdots & a_{n\,n} \end{pmatrix},$$

so that the characteristic matrix $M_f^{\mathcal{B}} - T I_n$ is

$$\begin{pmatrix} \lambda - T & 0 & \cdots & 0 & a_{1\,r+1} & \cdots & a_{1\,n} \\ 0 & \lambda - T & \cdots & 0 & a_{2\,r+1} & \cdots & a_{2\,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda - T & a_{r\,r+1} & \cdots & a_{r\,n} \\ 0 & 0 & \cdots & 0 & a_{r+1\,r+1} - T & \cdots & a_{r+1\,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & a_{n\,r+1} & \cdots & a_{n\,n} - T \end{pmatrix}.$$

Hence $(T - \lambda)^r$ divides $\det(M_f^{\mathcal{B}} - T I_n) = P_f(T)$. □

Definition 1.129 Let V be a K-vector space V of dimension $n \geq 1$ and $f \in \mathrm{End}_K(V)$ an endomorphism of V. We say that f is *diagonalizable* if there is a basis of V consisting entirely of eigenvectors of f.

Corollary 1.130 *Let V be a K-vector space V of dimension $n \geq 1$ and $f \in \mathrm{End}_K(V)$ an endomorphism of V. Let $\mathrm{Spec}(f) = \{\lambda_1, \ldots, \lambda_r\}$. Then f is diagonalizable if and only if*

$$V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_r}.$$

Proof Easy consequence of Lemma 1.115. □

A square matrix $A \in \mathcal{M}_n(K)$ is said to be *diagonalizable* if it is similar to a diagonal matrix, i.e. there is a nonsingular matrix $P \in \mathrm{GL}_n(K)$ such that

$$P^{-1}AP = D, \tag{1.54}$$

where D is a diagonal matrix. We say that P and D diagonalize A and we call D a *diagonal form* of A. If $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ then $f_D(e_i) = \lambda_i e_i$, where $e_1, \ldots, e_n$ is the canonical basis of K^n. Hence $e_1, \ldots, e_n$ are the eigenvectors of D and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of D. By Lemma 1.117 and Corollary 1.120, $\lambda_1, \ldots, \lambda_n$ are the eigenvalues also of A. Hence if A is diagonalizable its diagonal form D is unique up to the order of eigenvalues of A on the principal diagonal.

Example 1.131 Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the endomorphism defined by $f(x_1, x_2) = (x_1 + x_2, x_2)$. Then 1 is the only eigenvalue of f and it has algebraic multiplicity 2, while the eigenspace $V_1 = \{(a, 0) \mid a \in \mathbb{R}\}$ has dimension 1. Hence f is not diagonalizable.

Example 1.132 Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathcal{M}_2(K)$, where K is any field. Suppose that there is a matrix $P = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det(P) = ad - bc \neq 0$ and a matrix $D = \begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$ such that $PD = AP$, i.e.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} \iff \begin{cases} ae = c \\ bf = d \\ ce = 0 \\ df = 0 \end{cases}$$

which leads to $ad - bc = 0$, contradicting our assumption on P. Then A is not diagonalizable.

Remark 1.133 We denote by $[v_1, \ldots, v_n]$ the matrix whose columns are the vectors $v_1, \ldots, v_n \in K^n$. If $P = [v_1, \ldots, v_n] \in \mathcal{M}_n(K)$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then

$$AP = A[v_1, \ldots, v_n] = [Av_1, \ldots, Av_n],$$

$$[\lambda_1 v_1, \ldots, \lambda_n v_n] = [v_1, \ldots, v_n] \cdot \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} = PD. \tag{1.55}$$

We are now able to show the equivalence between the diagonalizabilities of an endomorphism and of a matrix.

Theorem 1.134 *A matrix $A \in \mathcal{M}_n(K)$ is diagonalizable if and only if the endomorphism*

$$f_A \colon K^n \to K^n, \qquad f_A(\boldsymbol{x}) = A\boldsymbol{x}^t, \qquad \forall \boldsymbol{x} = (x_1, \ldots, x_n) \in K^n,$$

is diagonalizable.

Let V be a K-vector space V of dimension n and $f \in \operatorname{End}_K(V)$ an endomorphism of V. Then f is diagonalizable if and only if $M_f^{\mathcal{B}}$ is a diagonalizable matrix for one basis $\mathcal{B}$ of V.

Proof If A is diagonalizable, i.e. $P^{-1}AP = D$, then we have $AP = PD$. If $P = [v_1, \ldots, v_n]$ we get $Av_i = \lambda_i v_i$, $i = 1, \ldots, n$ by (1.55). Hence v_i is the eigenvector of f_A corresponding to eigenvalue λ_i, $i = 1, \ldots, n$. Furthermore $v_1, \ldots, v_n$ are linearly independent because $\det(P) \neq 0$ so that $\{v_1, \ldots, v_n\}$ is a basis of eigenvectors of f_A and f_A is diagonalizable.

Conversely, if f_A is diagonalizable, i.e. f_A has n linearly independent eigenvectors $v_1, \ldots, v_n \in K^n$ whose corresponding eigenvalues are $\lambda_1, \ldots, \lambda_n$. If we put $P = [v_1, \ldots, v_n]$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ then (1.55) implies $AP = PD$. Since $v_1, \ldots, v_n$ are linearly independent, the matrix P is nonsingular, then $P^{-1}AP = D$ i.e. A is diagonalizable.

If f is diagonalizable and $\mathcal{B}$ is a basis of V consisting entirely of eigenvectors of f, then $M_f^{\mathcal{B}}$ is a diagonal matrix. Conversely, if $M_f^{\mathcal{B}}$ is diagonalizable, we have seen above that there are n linearly independent eigenvectors

$$\mathbf{x}_i = \begin{pmatrix} x_{1i} \\ \vdots \\ x_{ni} \end{pmatrix}, \qquad i = 1, \ldots, n, \qquad M_f^{\mathcal{B}} \mathbf{x}_i = \lambda_i \mathbf{x}_i, \qquad i = 1, \ldots, n.$$

Hence the column vectors

$$(v_i)_{\mathcal{B}} = \begin{pmatrix} x_{1i} \\ \vdots \\ x_{ni} \end{pmatrix}, \qquad i = 1, \dots, n,$$

are linearly independent eigenvectors of f. □

Corollary 1.135 *Let V be a K-vector space V of dimension n. Every endomorphism $f \in \mathrm{End}_K(V)$ (or every matrix $A \in \mathcal{M}_n(K)$) which has n distinct eigenvalues is diagonalizable.*

Proof Let $\lambda_1, \dots, \lambda_n \in K$ be the distinct eigenvalues of f and $v_1, \dots, v_n$ corresponding eigenvectors. By Lemma 1.115, $v_1, \dots, v_n$ are n linearly independent vectors of V so that our thesis follows from Definition 1.129. □

Corollary 1.136 *Let V be a K-vector space V of dimension n and $f \in \mathrm{End}_K(V)$ an endomorphism of V. Then f is if and only if all roots of its characteristic polynomial $P_f(T)$ belong to K and the geometric an algebraic multiplicities of every eigenvalue λ of f coincide.*

A matrix $A \in \mathcal{M}_n(K)$ is diagonalizable if and only if all roots of its characteristic polynomial $P_A(T)$ belong to K and the geometric and algebraic multiplicities of every eigenvalue λ of A coincide.

Proof If all roots of the characteristic polynomial $P_f(T)$ belong to K and the geometric an algebraic multiplicities of every eigenvalue λ of f coincide, by Lemma 1.115, (ii) V has a basis of eigenvectors and hence f is diagonalizable.

Conversely, let f be diagonalizable. If $\lambda_1, \dots, \lambda_r$ are the distinct eigenvalues of f, by Corollary 1.130, we have

$$V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_r}.$$

Denote by m_{λ_i} and ϱ_{λ_i} the geometric and the algebraic multiplicities of λ_i, $i = 1, \dots, r$, respectively,. By Proposition 1.128, we have the implication

$$n = \sum_{i=1}^{r} \varrho_{\lambda_i}, \quad \varrho_{\lambda_i} \le m_{\lambda_i}, \ i = 1, \dots, r,$$

$$\Longrightarrow \qquad n = \sum_{i=1}^{r} m_{\lambda_i}, \quad \varrho_{\lambda_i} = m_{\lambda_i}, \ i = 1, \dots, r,$$

i.e. our thesis. □

Diagonalization Algorithm
INPUT: $A \in \mathcal{M}_n(K)$.

1. Compute the eigenvalues of A: $\lambda_1, \ldots, \lambda_r$ of algebraic multiplicities $\mu_1, \ldots, \mu_r$ respectively.
2. Find a basis $\mathcal{B}_i$ of V_{λ_i}, $i = 1, \ldots, r$, and put $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$
3. If $\#(\mathcal{B}) < n$, then A is not diagonalizable.
4. If $\#(\mathcal{B}) = n$, then A is diagonalizable. Then put $\mathcal{B}_1 = (v_1, \ldots, v_{\mu_1})$, $\mathcal{B}_i = (v_{\mu_1+\cdots+\mu_{i-1}+1}, \ldots, v_{\mu_1+\cdots+\mu_{i-1}+\mu_i})$ for $i = 2, \ldots, r$, and

$$\mathcal{B} = (v_1, \ldots, v_{\mu_1}, v_{\mu_1+1}, \ldots, v_{\mu_1+\mu_2}, \ldots, v_{\mu_1+\cdots+\mu_{r-1}+1} \ldots, v_{\mu_1+\cdots+\mu_r})$$

5. Form the matrices $P = [v_1, \ldots, v_{\mu_1}, v_{\mu_1+1}, \ldots, v_{\mu_1+\mu_2}, \ldots, v_{\mu_1+\cdots+\mu_{r-1}+1} \ldots, v_{\mu_1+\cdots+\mu_r}]$ (see Remark 1.133) and $D = \operatorname{diag}(\underbrace{\lambda_1, \ldots, \lambda_1}_{\mu_1}, \ldots, \underbrace{\lambda_r, \ldots, \lambda_r}_{\mu_r})$.

OUTPUT: P and D diagonalize A.

Two Applications of Diagonalization
Generally speaking, it is very complicated to compute the powers A^m of a matrix $A \in \mathcal{M}_n(K)$ with m a large integer. Suppose that A is diagonalizable, i.e. $A = PDP^{-1}$, with $P \in \mathrm{GL}_n(K)$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. By induction on m we have immediately

$$A^m = PD^mP^{-1}.$$

Since $D^m = \operatorname{diag}(\lambda_1^m, \ldots, \lambda_n^m)$, the computation of A^m is now much simpler.

The following proposition shows how diagonalization is useful in the theory of systems of linear differential equations. The reader can find a very good exposition of the role of linear algebra in the theory of differential equations and dynamical systems in [18].

Proposition 1.137 *Let $A \in \mathcal{M}_n(\mathbb{R})$ be a matrix having n distinct real eigenvalues $\lambda_1, \ldots, \lambda_n$, i.e. $A = PDP^{-1}$, with $P \in \mathrm{GL}_n(\mathbb{R})$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then for all $\boldsymbol{a} = (a_1, \ldots, a_n) \in \mathbb{R}^n$, the linear differential equations system*

$$\boldsymbol{x}' = A\boldsymbol{x}; \quad \boldsymbol{x}(0) = \boldsymbol{a}, \qquad \boldsymbol{x} = (x_1(t), \ldots, x_n(t)), \quad t \in \mathbb{R},$$

has a unique solution.

Proof Introducing the new coordinates $\mathbf{y} = P\mathbf{x}$ in $\mathbb{R}^n$, with $\mathbf{x} = P^{-1}\mathbf{y}$ we find

$$\mathbf{y}' = P\mathbf{x}' = PA\mathbf{x} = PA(P^{-1}\mathbf{y}) = D\mathbf{y}.$$

Hence, if $\mathbf{y} = (y_1(t), \ldots, y_n(t))$, this means

$$y_i'(t) = \lambda_i y_i(t), \qquad i = 1, \ldots, n,$$

which has a unique solution for every initial condition $y_i(0)$

$$y_i(t) = y_i(0)e^{\lambda_i t}, \qquad i = 1, \ldots, n.$$

If we put $y_i(0) = a_i$, $i = 1, \ldots, n$, we have the unique solution

$$\mathbf{x}(t) = P^{-1} \begin{pmatrix} y_1(0)e^{\lambda_1 t} \\ \vdots \\ y_n(0)e^{\lambda_n t} \end{pmatrix}$$

with $\mathbf{x}(0) = \mathbf{a}$. □

1.4 Euclidean Vector Spaces

In this section we are concerned only with $\mathbb{R}$-vector spaces.

Definition 1.138 Let V be a $\mathbb{R}$-vector space. An *inner product* on V is a map

$$V \times V \to \mathbb{R}, \qquad (v, w) \to \langle v, w \rangle, \qquad \forall\, v, w \in V,$$

with the following properties:

(i) For every $v_1, v_2, w \in V$ and for every $a_1, a_2 \in \mathbb{R}$, $\langle a_1 v_1 + a_2 v_2, w \rangle = a_1 \langle v_1, w \rangle + a_2 \langle v_2, w \rangle$.
(ii) For every $v, w \in V$, $\langle v, w \rangle = \langle w, v \rangle$.
(iii) For every $v \in V$, $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ if and only if $v = 0_V$.

The first two axioms imply that $(v, w) \to \langle v, w \rangle$ is a symmetric bilinear map (see Definition 2.1). In particular, the map $(v, w) \to \langle v, w \rangle$ is linear in both variables. A real vector space V, together with an inner product $\langle \cdot, \cdot \rangle$, is called a *euclidean vector space* and denoted by $(V, \langle \cdot, \cdot \rangle)$.

Proposition 1.139

(i) Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space and W a subspace of V. Then the map

$$W \times W \to \mathbb{R}, \qquad (u, w) \to \langle u, w \rangle, \qquad \forall\, u, w \in W,$$

is an inner product on W.

(ii) Let $(V_1, \langle \cdot, \cdot \rangle_1)$ and $(V_2, \langle \cdot, \cdot \rangle_2)$ be two euclidean vector spaces, then an inner product on $V_1 \times V_2$ is defined by

$$\langle (v_1, v_2), (w_1, w_2) \rangle = \langle v_1, w_1 \rangle_1 + \langle v_2, w_2 \rangle_2, \qquad \forall\, (v_1, v_2), w_1, w_2) \in V_1 \times V_2.$$

Proof We leave the proof to the reader as an exercise. □

Examples 1.140

1. The map

$$\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}, \quad \langle v, w\rangle_s := a_1 b_1 + \cdots + a_n b_n,$$
$$\forall\, v = (a_1, \ldots, a_n),\ w = (b_1, \ldots, b_n) \in \mathbb{R}^n,$$

   is an inner product on $\mathbb{R}^n$ called the *standard inner product* on $\mathbb{R}^n$. We shall call $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_s)$ the *standard euclidean vector space* of dimension n.
2. Let $H(t) \in C([a,b];\mathbb{R})$, $H(t) > 0$ for every $t \in [a,b]$. An inner product on $C([a,b];\mathbb{R})$ (Example 1.33, 4) is defined by

$$C([a,b];\mathbb{R}) \times C([a,b];\mathbb{R}) \to \mathbb{R},$$
$$\langle f, g\rangle := \int_a^b f(t)g(t)H(t)dt, \qquad \forall\, f, g \in V. \tag{1.56}$$

3. Since $\mathbb{R}[X]$ and $\mathbb{R}[X]_{\le m}$ (Example 1.33, 2) are subspaces of $C([a,b];\mathbb{R})$, by Proposition 1.139, i), (1.56) induces an inner product on these spaces.
4. On $\mathbb{R}[X]$ (and on $\mathbb{R}[X]_{\le m}$) we can give the inner product defined by

$$\mathbb{R}[X]\times\mathbb{R}[X] \to \mathbb{R}, \qquad \langle f, g\rangle = \int_{-\infty}^{+\infty} f(x)g(x)e^{-x^2}\,dx, \qquad \forall\, f, g \in \mathbb{R}[X].$$

Definition 1.141 If $(V, \langle\cdot,\cdot\rangle)$ is a euclidean vector space, the *norm* or the *length* of a vector v is defined by

$$\|v\| := \sqrt{\langle v, v\rangle}.$$

A vector v is a *unit vector* if $\|v\| = 1$.

Here are the basic properties of the norm.

(i) For every $v \in V$, $\|v\| \ge 0$ and $\|v\| = 0$ if and only if $v = 0_V$.
(ii) For every $a \in \mathbb{R}$ and for every $v \in V$, $\|av\| = |a|\,\|v\|$. (Indeed, $\|av\|^2 = \langle av, av\rangle = a^2\langle v, v\rangle = a^2\|v\|^2$.)
(iii) For every $v, w \in V$,

$$\pm\,\langle v, w\rangle = \frac{1}{2}\left(\|v \pm w\|^2 - \|v\|^2 - \|w\|^2\right). \tag{1.57}$$

(iv) The *polarization identity*: for every $v, w \in V$,

$$\langle v, w\rangle = \frac{1}{4}\left(\|v + w\|^2 - \|v - w\|^2\right). \tag{1.58}$$

Definition 1.142 Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space.

1. Two vectors $v, w \in V$ are *orthogonal*, written $v \perp w$, if $\langle v, w\rangle = 0$.
2. Two subsets W_1 and W_2 of V are *orthogonal*, written $W_1 \perp W_2$ is $\langle v, w\rangle = 0$ for all $v \in W_1$ and $w \in W_2$.
3. The *orthogonal* of a subset $A \subset V$ is the set $A^\perp := \{v \in V : \langle v, w\rangle = 0,\ \forall\, w \in A\}$. It is immediately seen that $A^\perp$ is a vector subspace. Furthermore, the subspace generated by A is contained in $A^{\perp\perp} := (A^\perp)^\perp$.
4. A set A is *orthogonal* if $\langle v, w\rangle = 0$ for every $v, w \in A$, $v \neq w$.
5. An orthogonal set A is *orthonormal* if $\|v\| = 1$ for all $v \in A$. Therefore if $A = \{v_i : i \in I\}$, then $\langle v_i, v_j\rangle = \delta_{ij}, \forall\, i, j \in I$.
6. An orthonormal set A is *complete* if it is not strictly contained in a larger orthonormal set.
7. An *orthogonal basis* (*orthonormal basis*) of V is a basis of V which an orthogonal (orthonormal) set. We shall abbreviate orthonormal basis to on-basis.

   The canonical basis $\{e_1, \ldots, e_n\}$ of $(\mathbb{R}^n, \langle\cdot,\cdot\rangle_s)$ (see Example 1.140-1) is an on-basis.

Lemma 1.143 *An orthonormal subset A of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$ is free. If V has finite dimension, $\#(A) \leq \dim_{\mathbb{R}}(V)$.*

Proof Let $a_1, \ldots, a_n \in \mathbb{R}$ such that $\sum_{i=1}^{n} a_i v_i = 0_V$ with $v_1, \ldots, v_n \in A$. Then

$$0 = \langle \sum_{i=1}^{n} a_i v_i, v_j\rangle = \sum_{i=1}^{n} a_i \langle v_i, v_j\rangle = a_j \|v_j\|^2 = a_j,$$

for $j = 1, \ldots, n$. □

Definition 1.144 Let $A = \{v_1, \ldots, v_n\}$ be an orthonormal subset of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$ and $v \in V$. The scalars

$$a_i := \langle v, v_i\rangle, \qquad i = 1, \ldots, n,$$

are called the *Fourier coefficients* of the vector v with respect to the orthonormal subset A. We also have

$$v - \sum_{i=1}^{n} a_i v_i \in A^\perp. \tag{1.59}$$

Indeed, for $j = 1, \ldots, n$ we have

$$\langle v, v_j\rangle - \sum_{i=1}^{n} a_i \langle v_i, v_j\rangle = a_j - \sum_{i=1}^{n} a_i \delta_{ij} = 0.$$

Theorem 1.145 *Under the assumptions and notation of Definition 1.144 we have Bessel's inequality*

$$\sum_{i=1}^{n} a_i^2 \leq \|v\|^2.$$

Proof These obvious relations

$$\begin{aligned} 0 &\leq \langle v - \sum_{i=1}^{n} a_i v_i, v - \sum_{j=1}^{n} a_j v_j \rangle \\ &= \langle v, v \rangle - \sum_{i=1}^{n} a_i \langle v_i, v \rangle - \sum_{j=1}^{n} a_i \langle v_j, v \rangle + \sum_{i,j=1}^{n} a_i a_j \langle v_i, v_j \rangle \\ &= \|v\|^2 - 2\sum_{i=1}^{n} a_i^2 + \sum_{i,j=1}^{n} a_i a_j \delta_{ij} = \|v\|^2 - \sum_{i=1}^{n} a_i^2. \end{aligned}$$

lead us to our thesis. □

Theorem 1.146 *Let $A = \{v_1, \ldots, v_n\}$ be an orthonormal subset of a euclidean vector space $(V, \langle \cdot, \cdot \rangle)$ and $v \in V$. The following assertions are equivalent:*

(i) *A is complete.*
(ii) *If $v \in V$ satisfies the identities $\langle v, v_i \rangle = 0$, $i = 1, \ldots, n$, then $v = 0_V$.*
(iii) *A is a basis of V.*
(iv) *For every $v \in V$ one has $v = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$.*
(v) *For every pair of vectors $v, w \in V$ Parseval's identity holds:*

$$\langle v, w \rangle = \sum_{i=1}^{n} \langle v, v_i \rangle \langle v_i, w \rangle.$$

(vi) *For every $v \in V$ we have $\|v\|^2 = \sum_{i=1}^{n} \langle v, v_i \rangle^2$.*

Proof (i) $\Longrightarrow$ (ii): Suppose that there is a vector $0_V \neq v \in V$ such that $\langle v, v_i \rangle = 0$, $i = 1, \ldots, n$. Then $v \notin A$ and the set $A \cup \left\{ \dfrac{v}{\|v\|} \right\}$ is an orthonormal subset which strictly contains A, contradicting our assumption.

(ii) $\Longrightarrow$ (iii): Suppose that there is $v \in V \setminus \langle A\rangle$ (we recall that $\langle A\rangle$ is the subspace generated by A). Then the vector $w := v - \sum_{i=1}^{n} \langle v, v_i\rangle v_i$ is $\neq 0_V$ and, by (1.59), $w \in A^{\perp}$, which contradicts ii). By Lemma 1.143 A is free.

(iii) $\Longrightarrow$ (iv): For every $v \in V$ there are $a_1, \ldots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i v_i$. Hence we have

$$\langle v, v_j\rangle = \sum_{i=1}^{n} a_i \langle v_i, v_j\rangle = \sum_{i=1}^{n} a_i \delta_{ij} = a_j, \qquad j = 1, \ldots, n.$$

(iv) $\Longrightarrow$ (v): We can write $v = \sum_{i=1}^{n} \langle v, v_i\rangle v_i$ and $w = \sum_{j=1}^{n} \langle w, v_j\rangle v_j$ so that

$$\langle v, w\rangle = \langle \sum_{i=1}^{n} \langle v, v_i\rangle v_i, \sum_{j=1}^{n} \langle w, v_j\rangle v_j\rangle = \sum_{i,j=1}^{n} \langle v, v_i\rangle \langle w, v_j\rangle \langle v_i, v_j\rangle$$

$$= \sum_{i,j=1}^{n} \langle v, v_i\rangle \langle w, v_j\rangle \delta_{ij} = \sum_{i=1}^{n} \langle v, v_i\rangle \langle w, v_i\rangle.$$

(v) $\Longrightarrow$ (vi): Take $v = w$.

(vi) $\Longrightarrow$ (i): Let $v \in A^{\perp}$. We have $\|v\|^2 = \sum_{i=1}^{n} \langle v, v_i\rangle^2 = 0$, hence $v = 0_V$ which proves (i). □

Theorem 1.147 *Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space. For every $v, w \in V$ we have the Cauchy–Schwarz inequality:*

$$|\langle v, w\rangle| \leq \|v\| \, \|w\|. \tag{1.60}$$

Furthermore, $|\langle v, w\rangle| = \|v\| \, \|w\|$ if and only if v and w are linearly dependent.

Proof If $w = 0_V$ we have $=$ instead of $\leq$. Suppose $w \neq 0$. For every $t \in \mathbb{R}$ we have

$$0 \leq \langle v + tw, v + tw\rangle = t^2 \langle w, w\rangle + 2t \langle v, w\rangle + \langle v, v\rangle.$$

Thus the polynomial of degree 2 is non-negative for all $t \in \mathbb{R}$, hence its discriminant is ≤ 0, i.e.

$$\langle v, w\rangle^2 \leq \langle v, v\rangle \langle w, w\rangle,$$

which is (1.60). If $w = \lambda v$, then $|\langle v, w\rangle| = |\lambda|\,|\langle v, v\rangle| = |\lambda|\,\|v\|\,\|v\| = \|w\|\,\|v\|$. Conversely, if $\lambda v + \mu w = 0_V$, then

$$\begin{aligned} 0 &= \langle \lambda v + \mu, \lambda v + \mu\rangle = \lambda^2\|v\|^2 + \mu^2\|w\|^2 + 2\lambda\mu|\langle v, w\rangle| \\ &= \lambda^2\|v\|^2 + \mu^2\|w\|^2 + 2\lambda\mu\|v\|\,\|w\| = (\lambda\|v\| + \mu\|w\|)^2, \end{aligned}$$

which implies $\lambda = \mu = 0$. □

Remark 1.148 We can deduce the Cauchy–Schwarz inequality from Bessel's inequality. If $w = 0_V$, both the members of (1.60) vanish. If $w \neq 0_V$, $\left\{\dfrac{w}{\|w\|}\right\}$ is an orthonormal set of V, hence by Bessel's inequality we obtain

$$\langle v, \frac{w}{\|w\|}\rangle^2 \leq \|v\|^2, \qquad \forall\, v \in V.$$

Definition 1.149 By the Cauchy–Schwarz inequality we have

$$-1 \leq \frac{\langle v, w\rangle}{\|v\|\,\|w\|} \leq 1.$$

Therefore there exists a unique angle θ, $0 \leq \theta \leq \pi$ for which

$$\cos\theta = \frac{\langle v, w\rangle}{\|v\|\,\|w\|}. \tag{1.61}$$

This angle

$$\sphericalangle\,(v, w) := \theta = \arccos\left(\frac{\langle v, w\rangle}{\|v\|\,\|w\|}\right)$$

is called the *angle* between two vectors $v \neq 0_V$ and $w \neq 0_V$ of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$.

Proposition 1.150 (Pythagoras' Theorem) *Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space. Two vectors v and $w \in V$ are orthogonal if and only if*

$$\|w - v\|^2 = \|v\|^2 + \|w\|^2.$$

Proof It follows immediately from the equalities

$$\|w - v\|^2 = \langle w - v, w - v\rangle = \langle w, w\rangle + \langle v, v\rangle - 2\langle v, w\rangle = \|w\|^2 + \|v\|^2 - 2\langle v, w\rangle.$$

□

Definition 1.151 The Cauchy–Schwarz inequality permits us to introduce a metric on a euclidean vector space. Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space. The *distance*

between two vectors v and $w \in V$ is defined by

$$d(v, w) := \|v - w\| = \sqrt{\langle v - w, v - w \rangle}. \tag{1.62}$$

Proposition 1.152 *The map $(v, w) \to d(v, w)$ satisfies the following properties:*

(i) $d(v, w) \geq 0$, $\forall\, v, w \in V$ *and* $d(v, w) = 0$ *if and only if* $v = w$.
(ii) $d(v, w) = d(w, v)$, $\forall\, v, w \in V$.
(iii) *The triangle inequality holds:*

$$d(u, w) \leq d(u, v) + d(v, w), \qquad \forall\, v, w, u \in V.$$

(iv) $d(v, w) = d(v + u, w + u)$, *for every* $u, v, w \in V$, *i.e. d is invariant under translations.*

Proof Assertions (i), (ii) and (iv) are immediate. We prove (iii). Let $x, y \in V$. From the Cauchy–Schwarz inequality we get

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \leq$$

$$\leq \|x\|^2 + 2\|x\| \cdot \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,$$

hence we have

$$\|x + y\| \leq \|x\| + \|y\|.$$

If we put $x = u - v$ and $y = v - w$, we obtain (iii). □

Definition 1.153 A non-empty set V, equipped with a map $d\colon V \times V \to \mathbb{R}$ satisfying the properties (i), (ii) and (iii) of Proposition 1.152, is called a *metric space* and is denoted by (V, d).
Let (V_1, d_1) and (V_2, d_2) be two metric spaces. A surjective map $f\colon (V_1, d_1) \to (V_2, d_2)$ is called an *isometry* if

$$d_2(f(u), f(v)) = d_1(u, v), \qquad \forall\, u, v \in V.$$

An isometry is necessarily injective so that it is bijective. Two metric spaces (V_1, d_1) and (V_2, d_2) are *isometric* if there exists an isometry $f\colon (V_1, d_1) \to (V_2, d_2)$.

If $f\colon (V_1, d_1) \to (V_2, d_2)$ is an isometry, then $f^{-1}\colon (V_2, d_2) \to (V_1, d_1)$ is also an isometry. Hence the set of all isometries $f\colon (V, d) \to (V, d)$ is a group.

Examples 1.154 The Cauchy–Schwarz inequality relative to $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_s)$ takes the form

$$(a_1 b_1 + \cdots + a_n b_n)^2 \leq (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2),$$

for every $(a_1, \dots, a_n), (b_1, \dots, b_n) \in \mathbb{R}^n$. The associated metric is the *euclidean metric*

$$d(A, B) = \sqrt{\sum_{i=1}^{n} (a_i - b_i)^2},$$

for every $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n) \in \mathbb{R}^n$.
The Cauchy–Schwarz inequality relative to $C([a, b]; \mathbb{R})$ (see Example 1.140, 2) takes the form

$$\left(\int_a^b f(t) g(t) H(t)\, dt \right)^2 \leq \int_a^b f(t)^2 H(t)\, dt \int_a^b g(t)^2 H(t)\, dt,$$

for every $f, g \in C([a, b]; \mathbb{R})$. The associated metric is

$$d(f, g) = \sqrt{\int_a^b (f(t) - g(t))^2 H(t)\, dt},$$

for every $f, g \in C([a, b]; \mathbb{R})$.

Theorem 1.155 *Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space of dimension n. Then there exists an on-basis $\{v_1, \dots, v_n\}$ of V.*

Proof We are able to construct an on-basis starting from any vectorial basis of V. This method is called the *Gram–Schmidt orthogonalization process*.

Let $\{x_1, \dots, x_n\}$ be a basis of V. We proceed by induction on $i \leq n$ by producing an orthonormal set $\{v_1, \dots, v_i\}$ such that v_i is a linear combination of $x_1, \dots, x_i$. If $i = 1$, we put $v_1 = \dfrac{x_1}{\|x_1\|}$ and our orthonormal set will be $\{v_1\}$.

Suppose that $\{v_1, \dots, v_i\}$ is an orthonormal set with $i \leq n - 1$, such that, for every $1 \leq j \leq i$, v_j is a linear combination of $x_1, \dots, x_j$. We define

$$y_{i+1} := x_{i+1} - (a_1 v_1 + \cdots + a_i v_i),$$

where the scalars $a_1, \dots, a_i$ are to be determined. We observe that, for any i-tuple $(a_1, \dots, a_i)$, we have $y_{i+1} \neq 0_V$, otherwise $x_{i+1} = a_1 v_1 + \cdots + a_i v_i$ so that, by our inductive hypothesis, x_{i+1} would be a linear combination of $x_1, \dots, x_i$, contradicting the linear independence of $\{x_1, \dots, x_n\}$. We have

$$\begin{aligned} \langle y_{i+1}, v_j \rangle &= \langle x_{i+1}, v_j \rangle - \sum_{l=1}^{i} a_l \langle v_l, v_j \rangle \\ &= \langle x_{i+1}, v_j \rangle - \sum_{l=1}^{i} a_l \delta_{lj} = \langle x_{i+1}, v_j \rangle - a_j, \quad j = 1, \dots, i. \end{aligned}$$

If we put $a_j = \langle x_{i+1}, v_j \rangle$, $j = 1, \ldots, i$, the non-zero vector y_{i+1} is orthogonal to the set $\{v_1, \ldots, v_i\}$; furthermore y_{i+1} is also a linear combination of $x_1, \ldots, x_{i+1}$.

If we put $v_{i+1} = \dfrac{y_{i+1}}{\|y_{i+1}\|}$, we get the orthonormal set $\{v_1, \ldots, v_{i+1}\}$, where v_j is a linear combination of $x_1, \ldots, x_j$, $j = 1, \ldots, i+1$.

In this way we have constructed an orthonormal set $\{v_1, \ldots, v_n\}$. By Lemma 1.143 this set is free. Since $\dim_K V = n$, $\{v_1, \ldots, v_n\}$ is an on-basis of V. □

Proposition 1.156 *Let W be a subspace of a euclidean vector space $(V, \langle \cdot, \cdot \rangle)$ of finite dimension. Then*

$$V = W \oplus W^{\perp}, \qquad W^{\perp\perp} = W.$$

Proof Let $\dim_K W = m$. By Theorem 1.155 there exists an orthonormal basis $\{v_1, \ldots, v_m\}$ of W. Let $v \in V$. By (1.59) $w := v - \sum_{i=1}^{m} \langle v, v_i \rangle v_i \in W^{\perp}$, hence $v \in W + W^{\perp}$, i.e. $W + W^{\perp} = V$. Furthermore $W \cap W^{\perp} = \{0_V\}$ (if $x \in W \cap W^{\perp}$ then $\langle x, x \rangle = 0$, i.e. $x = 0_V$). Therefore $V = W \oplus W^{\perp}$.

We have the obvious inclusion $W \subset W^{\perp\perp}$. Conversely, let $v \in W^{\perp\perp}$. We have just seen that v can be written in the unique form $v = u + w$, with $u \in W$ and $w \in W^{\perp}$. Hence we have

$$0 = \langle v, w \rangle = \langle u + w, w \rangle = \langle u, w \rangle + \langle w, w \rangle = \langle w, w \rangle = \|w\|^2.$$

Therefore $w = 0_V$, i.e. $v = u \in W$. □

Corollary 1.157 *Under the assumptions of Proposition 1.156, if W' is a subspace of V such that $W \perp W'$ and $V = W + W'$, then $W' = W^{\perp}$.*

Proof The inclusion $W' \subset W^{\perp}$ is obvious. Let $u \in W^{\perp}$, then $u = v + w$ with $v \in W$ and $w \in W'$, because $V = W + W'$. For every $z \in W$ we have $\langle u, z \rangle = 0$ i.e. $0 = \langle v + w, z \rangle = \langle v, z \rangle + \langle w, z \rangle = \langle v, z \rangle$ which implies $v = 0_V$ and $u = w \in W'$. □

Definition 1.158 A euclidean vector space of finite dimension $(V, \langle \cdot, \cdot \rangle)$ is the *orthogonal direct sum* of subspaces W and W' if $V = W \oplus W'$ and $W \perp W'$. We know that $W' = W^{\perp}$ by Corollary 1.157. In this case we write $V = W \perp W'$ and we say that W' is *the orthogonal complement* of W (see Definition 1.142-3).

If $V = W \perp W'$, every $v \in V$ can be written in a unique way as $v = w + w'$, with $w \in W$ and $w' \in W'$. We call w (resp. w') the *orthogonal projection* of v on W (resp. W'). Hence $w' = p_{W,W^{\perp}}(v)$ (see Exercise 1.14) and $p_{W,W^{\perp}}$ is called the *orthogonal projection* on W and denoted simply by p_W.

More generally, V is the *orthogonal direct sum* of the subspaces $W_1, \dots, W_r$ if

$$V = W_1 \oplus \cdots \oplus W_r, \quad W_i \perp W_j \quad \text{for } i \neq j.$$

We shall write $V = W_1 \perp \cdots \perp W_r$.

Proposition 1.159 *Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space of finite dimension. Let $W \subset U$ be two subspaces of V. Then*

$$W^\perp = W_U^\perp \perp U^\perp,$$

where $W_U^\perp$ is the orthogonal complement of W relative to U.

Proof By Proposition 1.156 we have $W \perp W^\perp = V = U \perp U^\perp = W \perp W_U^\perp \perp U^\perp$ so that $W^\perp = W_U^\perp \perp U^\perp$ by Corollary 1.157 (since $U^\perp \perp W_U^\perp$ is orthogonal to W). □

1.4.1 Orthogonal Automorphisms

Definition 1.160 Let $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ be two euclidean vector spaces of finite dimension. A linear map $\varphi \colon (V, \langle \cdot, \cdot \rangle_V) \to W, \langle \cdot, \cdot \rangle_W)$ is called an *orthogonal linear morphism* (or simply *orthogonal morphism*) if

$$\langle \varphi(u), \varphi(v) \rangle_W = \langle u, v \rangle_V, \qquad \forall u, v \in V. \tag{1.63}$$

The identity (1.63) is equivalent to

$$\|\varphi(u)\|_W^2 = \|u\|_V^2 \qquad \forall u \in V. \tag{1.64}$$

If (1.64) holds, starting from the equalities

$$\begin{aligned} &\|\varphi(u)\|_W^2 + \|\varphi(v)\|_W^2 + 2\langle \varphi(u), \varphi(v) \rangle_W \\ &\quad = \langle \varphi(u) + \varphi(v), \varphi(u) + \varphi(v) \rangle_W \\ &\quad = \langle \varphi(u + v), \varphi(u + v) \rangle_W \\ &\quad = \|\varphi(u + v)\|_W^2 = \|u + v\|_V^2 = \langle u + v, u + v \rangle_V = \|u\|_V^2 + \|v\|_V^2 + 2\langle u, v \rangle_V \end{aligned}$$

we deduce (1.63) immediately. In particular, if d_V and d_W are the metrics associated with the inner products $\langle \cdot, \cdot \rangle_V$ and $\langle \cdot, \cdot \rangle_W$ respectively, according to Definition 1.151, an orthogonal linear morphism $\varphi \colon (V, \langle \cdot, \cdot \rangle_V) \to W, \langle \cdot, \cdot \rangle_W)$ preserves the distance, i.e.

$$d_W(\varphi(u), \varphi(v)) = d_V(u, v), \qquad \forall u, v \in V.$$

Thus an orthogonal linear morphism is necessarily injective and $\dim V \leq \dim W$; if it is also surjective then it is an isometry.

A bjiective orthogonal morphism $\varphi\colon (V, \langle\cdot,\cdot\rangle_V) \to W, \langle\cdot,\cdot\rangle_W)$ will be called an *orthogonal linear isomorphism* and if $(V, \langle\cdot,\cdot\rangle_V) = (W, \langle\cdot,\cdot\rangle_W)$, an *orthogonal linear automorphism*. The set $\mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$ of all orthogonal linear automorphism is a subgroup of $\mathrm{GL}_{\mathbb{R}}(V)$ called the *orthogonal linear group* of V. Indeed, if $\varphi \in \mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$, then

$$\langle\varphi^{-1}(v'), \varphi^{-1}(w')\rangle = \langle v, w\rangle = \langle\varphi(v), \varphi(w)\rangle = \langle v', w'\rangle, \qquad \forall\, v', w' \in V.$$

The existence of an orthogonal linear isomorphism $\varphi\colon (V, \langle\cdot,\cdot\rangle_V) \to W, \langle\cdot,\cdot\rangle_W)$ implies that $(V, \langle\cdot,\cdot\rangle_V)$ and $(W, \langle\cdot,\cdot\rangle_W)$ are isometric.

Examples 1.161

(1) Let $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, with $n \leq m$. The map

$$\varphi\colon \mathbb{R}^n \to \mathbb{R}^m, \qquad \varphi(x_1, \ldots, x_n) = (x_1, \ldots, x_n, 0, \ldots, 0)$$

is a non-surjective orthogonal linear morphism.

(2) Let $(V, \langle\cdot,\cdot\rangle_V)$ be a euclidean vector space and W be a subspace of V. The orthogonal projection p_W is not an orthogonal morphism (unless $p_W = \mathrm{id}_V$, i.e. $W = V$). The associated symmetry $s_W = 2p_W - \mathrm{id}_V$ (see Exercise 1.14) is an orthogonal isomorphism called the *orthogonal symmetry* through W. Indeed, if $u = z + z'$ and $v = w + w'$ with $z, w \in W$ and $z', w' \in W^{\perp}$, then we have

$$\begin{aligned}\langle s_W(u), s_W(v)\rangle &= \langle 2z - u, 2w - v\rangle = \langle 2z, 2w\rangle - \langle 2z, v\rangle - \langle u, 2w\rangle + \langle u, v\rangle\\ &= 4\langle z, w\rangle - \langle 2z, w + w'\rangle - \langle z + z', 2w\rangle + \langle u, v\rangle\\ &= 4\langle z, w\rangle - 2\langle z, w\rangle - \langle 2z, w'\rangle - \langle z, 2w\rangle - \langle z', 2w\rangle + \langle u, v\rangle\\ &= \langle u, v\rangle.\end{aligned}$$

Proposition 1.162 *Let $\sigma\colon (V, \langle\cdot,\cdot\rangle_V) \to (W, \langle\cdot,\cdot\rangle_W)$ be an isomorphism of vector spaces of finite dimensions. Let $\{e_1, \ldots, e_n\}$ be an on-basis of V. Then σ is an orthogonal linear isomorphism if and only if $\{\sigma(e_1), \ldots, \sigma(e_n)\}$ is an on-basis of W.*

Proof If σ is an orthogonal linear isomorphism then $\langle\sigma(e_i), \sigma(e_j)\rangle_W = \langle e_i, e_j\rangle_V = \delta_{ij}$, for $i, j = 1, \ldots, n$, so that $\{\sigma(e_1), \ldots, \sigma(e_n)\}$ is an orthonormal

basis of V. Conversely, suppose that $\{\sigma(e_1), \dots, \sigma(e_n)\}$ is an orthonormal basis of V. If $v = \sum_{i=1}^{n} a_i e_i$ and $w = \sum_{j=1}^{n} b_j e_j$ are vectors of V we have

$$\begin{aligned}\langle \sigma(v), \sigma(w)\rangle_W &= \langle \sum_{i=1}^{n} a_i \sigma(e_i), \sum_{j=1}^{n} \sigma(b_j) e_j \rangle \\ &= \sum_{i,j=1}^{n} a_i b_j \langle \sigma(e_i), \sigma(e_j)\rangle \\ &= \sum_{i,j=1}^{n} a_i b_j \delta_{ij} = \sum_{i=1}^{n} a_i b_i .\end{aligned}$$

In a similar way we see that $\langle v, w\rangle_V = \sum_{i=1}^{n} a_i b_i$. □

Proposition 1.163 *Let $(V, \langle\cdot,\cdot\rangle_V)$ be a euclidean vector spaces of finite dimension n. For every on-basis $\{v_1, \dots, v_n\}$ of V there exists one and only one orthogonal linear isomorphism $\varphi\colon (V, \langle\cdot,\cdot\rangle_V) \to (\mathbb{R}^n, \langle\cdot,\cdot\rangle_s)$ such that $\varphi(v_i) = e_i$, $i = 1, \dots, n$, where $\{e_1, \dots, e_n\}$ is the canonical basis of $\mathbb{R}^n$.*

Proof It is easily verified that (by Theorem 1.146)

$$\varphi\colon (V, \langle\cdot,\cdot\rangle_V) \to (\mathbb{R}^n, \langle\cdot,\cdot\rangle_s), \qquad \varphi(v) = (\langle v, v_1\rangle, \dots, \langle v, v_n\rangle), \quad \forall\, v \in V,$$

is an orthogonal linear isomorphism. □

Proposition 1.164 *Let $f\colon (V, \langle\cdot,\cdot\rangle_V) \to (W, \langle\cdot,\cdot\rangle_W)$ be an isometry. Then for every $u_0 \in V$ the map*

$$g_{u_0}\colon (V, \langle\cdot,\cdot\rangle_V) \to (W, \langle\cdot,\cdot\rangle_W), \qquad g_{u_0}(v) = f(v + u_0) - f(u_0), \quad \forall\, v \in V,$$

is an orthogonal linear isomorphism. Conversely, if for some $u_0 \in V$, g_{u_0} is an orthogonal linear isomorphism, then f is an isometry.

Proof First we prove that g_{u_0} preserves the inner product, indeed,

$$\begin{aligned}\langle g_{u_0}(u), g_{u_0}(v)\rangle_W &= \langle f(u + u_0) - f(u_0), f(v + u_0) - f(u_0)\rangle_W \\ &= -\frac{1}{2}\Big(\|f(u + u_0) - f(v + u_0)\|_W^2 - \|f(u + u_0) - f(u_0)\|_W^2 \\ &\quad -\|f(v + u_0) - f(u_0)\|_W^2\Big) \\ &= -\frac{1}{2}\Big(\|u - v)\|_V^2 - \|u\|_V^2 - \|v\|_V^2\Big) = \langle u, v\rangle_V .\end{aligned}$$

We have now to prove that g_{u_0} is linear map. Let $u, v \in V$, $\lambda, \mu \in \mathbb{R}$. For any $w \in V$ we have the relations

$$\begin{aligned}\langle g_{u_0}(\lambda u + \mu v), g_{u_0}(w)\rangle_W &= \langle \lambda u + \mu v, w\rangle_V = \lambda\langle u, w\rangle_V + \mu\langle v, w\rangle_V,\\ \langle \lambda g_{u_0}(u) + \mu g_{u_0}(v), g_{u_0}(w)\rangle_W &= \lambda\langle g_{u_0}(u), g_{u_0}(w)\rangle_W + \mu\langle g_{u_0}(v), g_{u_0}(w)\rangle_W\\ &= \lambda\langle u, w\rangle_V + \mu\langle v, w\rangle_V,\end{aligned}$$

which imply $g_{u_0}(\lambda u + \mu v) = \lambda g_{u_0}(u) + \mu g_{u_0}(v)$ since g_{u_0} is bijective. Conversely, we have

$$\|f(u) - f(v)\|_W^2 = \|g_{u_0}(u - u_0) - g_{u_0}(v - u_0)\|_W^2 = \|u - v\|_V^2,$$

since g_{u_0} is an isometry. □

Definition 1.165 A matrix $A \in \mathcal{M}_n(\mathbb{R})$ is said to be *orthogonal* if

$$A^t A = I_n, \qquad \text{i.e.} \qquad A^t = A^{-1}.$$

A matrix is orthogonal if and only if its column vectors (or its row vectors) constitute an orthonormal basis of $\mathbb{E}^n(\mathbb{R})$. If A is orthogonal, then $\det(A) = \pm 1$.

Corollary 1.166 *Let $\mathcal{B} = \{e_1, \ldots, e_n\}$ be an on-basis of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$. Let $\sigma \in \mathrm{Aut}_{\mathbb{R}}(V)$ be an automorphism of V and $M_\sigma^{\mathcal{B}} = \big(a_{ij}\big)_{i,j=1,\ldots,n}$ the matrix associated with σ with respect to $\mathcal{B}$. The σ is an orthogonal automorphism if and only if*

$$\sum_{k=1}^{n} a_{ki}a_{kj} = \delta_{ij}, \qquad i, j = 1, \ldots, n,$$

i.e. $M_\sigma^{\mathcal{B}}$ is an orthogonal matrix.

Proof We have

$$\langle \sigma(e_i), \sigma(e_j)\rangle = \langle \sum_{k=1}^{n} a_{ki}e_k, \sum_{h=1}^{n} a_{hj}e_h\rangle = \sum_{k=1}^{n}\sum_{h=1}^{n} a_{ki}a_{hj}\delta_{hk} = \sum_{k=1}^{n} a_{ki}a_{kj},$$

so that our thesis follows from Proposition 1.162. □

Corollary 1.167 *If σ is an orthogonal automorphism of $(V, \langle\cdot,\cdot\rangle)$, then* $\det(\sigma) = \pm 1$.

Definition 1.168 Let $\mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$ be the orthogonal linear group of $(V, \langle\cdot,\cdot\rangle_V)$ (Definition 1.160). By Corollary 1.167 we have the map

$$\det\colon \mathrm{O}(V, \langle\cdot,\cdot\rangle_V) \to \{-1, +1\}, \qquad \sigma \to \det(\sigma), \qquad \forall\, \sigma \in \mathrm{O}(V, \langle\cdot,\cdot\rangle_V),$$

which is a surjective homomorphism of groups. The subset

$$\mathrm{SO}(V, \langle\cdot,\cdot\rangle_V) := \mathrm{Ker}(\det) = \{\sigma \in \mathrm{O}(V, \langle\cdot,\cdot\rangle_V) \mid \det(\sigma) = 1\}$$

is a normal subgroup of index 2 in $\mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$, called the *special orthogonal linear group*. The elements of $\mathrm{SO}(V, \langle\cdot,\cdot\rangle_V)$ are called *special orthogonal linear automorphisms* or also *rotations*.

If $(V, \langle\cdot,\cdot\rangle_V) = (\mathbb{R}^n, \langle\cdot,\cdot\rangle_s)$ we shall denote $\mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$ by $\mathrm{O}(n)$ and $\mathrm{SO}(V, \langle\cdot,\cdot\rangle_V)$ by $\mathrm{SO}(n)$.

Proposition 1.169 *Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space, σ an orthogonal linear automorphism of V and W an invariant subspace relative to σ. Then the orthogonal complement $W^\perp$ is still an invariant subspace relative to σ.*

Proof Let $w \in W^\perp$. Since W is invariant and σ is bijective, we have $\sigma(W) = W$. Then for every $v \in W$ there exists $v_1 \in W$ such that $\sigma(v_1) = v$ so that

$$\langle\sigma(w), v\rangle = \langle\sigma(w), \sigma(v_1)\rangle = \langle w, v_1\rangle = 0,$$

hence $\sigma(w) \in W^\perp$. □

Proposition 1.170 *Let σ be an orthogonal automorphism of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$ of dimension $n \leq 2$.*

(i) *If $n = 1$, then $\sigma = \pm\mathrm{id}_V$.*
(ii) *If $n = 2$ and $\det(\sigma) = 1$ there is $\theta \in [0, 2\pi)$ such that*

$$M_\sigma^{\mathcal{B}} = R_\theta := \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \tag{1.65}$$

where $\mathcal{B} = \{e_1, e_2\}$ is any orthonormal basis of V. We call σ a (anti-clockwise) rotation by an angle θ.
(iii) *If $n = 2$ and $\det(\sigma) = -1$ there exists an orthonormal basis $\mathcal{B}' = \{v_1, v_2\}$ of V such that*

$$M_\sigma^{\mathcal{B}'} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.66}$$

Hence σ is an orthogonal symmetry.
(iv) *Every rotation is composition of two orthogonal symmetries.*

Proof

(i) If $\dim_{\mathbb{R}}(V) = 1$, then $\sigma(e) = \lambda e$, with e a unit vector ($\{e\}$ is an orthonormal basis of V) and $\lambda \in \mathbb{R}$. Since $\det(\sigma) = \pm 1$ we have $\lambda = \pm 1$, hence $\sigma = \pm\mathrm{id}_V$.

Suppose $\dim_{\mathbb{R}} V = 2$. Let $\mathcal{B} = \{e_1, e_2\}$ be an orthonormal basis of V. By Corollary 1.166 the matrix $M_\sigma^{\mathcal{B}} = \left(a_{ij}\right)_{i,j=1,2}$ is orthogonal so that we have

$$\begin{cases} a_{11}^2 + a_{21}^2 = 1, \\ a_{11}a_{12} + a_{21}a_{22} = 0, \\ a_{12}^2 + a_{22}^2 = 1. \end{cases} \tag{1.67}$$

For the first and the third equality of (1.67) we can write

$$a_{11} = \cos\theta, \quad a_{21} = \sin\theta, \quad a_{12} = \sin\theta', \quad a_{22} = \cos\theta', \tag{1.68}$$

with $\theta, \theta' \in [0, 2\pi)$.

(ii) Assume that $\det(\sigma) = 1$, i.e. $a_{11}a_{22} - a_{12}a_{21} = 1$. From the second equality of (1.67) and the relation $a_{11}a_{22} - a_{12}a_{21} = 1$ we get

$$\sin(\theta + \theta') = 0, \qquad \cos(\theta + \theta') = 1. \tag{1.69}$$

From (1.69) we obtain $\theta + \theta' = 0$ or $\theta + \theta' = 2\pi$. Both cases give $\sin\theta' = -\sin\theta$ and $\cos\theta' = \cos\theta$, hence we have (1.65).

(iii) Suppose now $\det(\sigma) = -1$. As above we get

$$\sin(\theta + \theta') = 0, \quad \cos(\theta + \theta') = -1, \tag{1.70}$$

i.e. $\theta + \theta' = \pi$ or $\theta + \theta' = 3\pi$, therefore

$$M_\sigma^{\mathcal{B}} = A_\theta := \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}. \tag{1.71}$$

Since $(M_\sigma^{\mathcal{B}})^2 = I_2$ we have $\sigma^2 = \mathrm{id}_V$ by Corollary 1.90. The vectors

$$v_1 := \cos\frac{\theta}{2} \cdot e_1 + \sin\left(\frac{\theta}{2}\right) \cdot e_2, \qquad v_2 := -\sin\frac{\theta}{2} \cdot e_1 + \cos\frac{\theta}{2} \cdot e_2$$

satisfy the relations $\sigma(v_1) = v_1$ and $\sigma(v_2) = -v_2$. Furthermore $\mathcal{B}' = \{v_1, v_2\}$ is an orthonormal basis of V and

$$M_\sigma^{\mathcal{B}'} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.72}$$

(iv) Let σ be a rotation and τ a symmetry. Then $\det(\sigma \circ \tau) = \det(\sigma)\det(\tau) = -1$ so that $\tau' := \sigma \circ \tau$ is a symmetry. Hence $\tau' \circ \tau = \sigma \circ \tau^2 = \sigma$.

□

Remark 1.171 Addition formulas for trigonometric functions prove the following identities:

$$R_\theta R_\varphi = R_{\theta+\varphi} = R_\varphi R_\theta, \quad A_\theta = R_\theta A_0 = A_0 R_{-\theta}, \quad A_\varphi A_\theta = R_{\varphi-\theta}, \quad \forall\, \varphi, \theta \in \mathbb{R}.$$

We now find the simplest form of the matrix of an orthonormal automorphism defined on an euclidean vector space of arbitrary dimension.

Theorem 1.172 *Let σ be an orthogonal automorphism of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$ of dimension n. There exists an on-basis $\mathcal{B} = \{e_1, e_2, \ldots, e_n\}$ such that*

$$M_\sigma^{\mathcal{B}} = \begin{pmatrix} I_p & 0 & 0 & \ldots & 0 \\ 0 & -I_q & 0 & \ldots & 0 \\ 0 & 0 & R_1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & R_k \end{pmatrix}, \tag{1.73}$$

with $p + q + 2k = n$, $p, q, k \geq 0$ (if $p = 0$ or $q = 0$, the corresponding matrix does not occur and if $k = 0$, $R_1, \ldots, R_k$ do not take place) where

$$R_i = R_{\theta_i} = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix}, \tag{1.74}$$

with $\theta_i \in (0, 2\pi)$, $\theta_i \neq \pi$, $i = 1, \ldots, k$. The ± 1 on the principal diagonal correspond to one-dimensional invariant subspaces and the 2×2-matrices R_I correspond to two-dimensional invariant subspaces.

Proof According to Proposition 1.126 there exists an invariant subspace W of V of dimension 1 or 2. By Proposition 1.169 also $W^\perp$ is invariant under σ and $V = W \perp W^\perp$. The matrix of $\sigma_{|W}$ (relative to a suitable orthonormal basis of W) is described by Proposition 1.170. Repeat the same argument for $\sigma_{|W^\perp}$: by Proposition 1.126 there exists a subspace W' of $W^\perp$ invariant under $\sigma_{|W^\perp}$ (and hence under σ) of dimension 1 o 2. If W'' is the orthogonal of W' in $W^\perp$, then W'' is invariant under σ (Proposition 1.169) and from $V = W \perp W^\perp$ and $W^\perp = W' \perp W''$ it follows that $V = W \perp W' \perp W''$. We repeat the same argument and use Proposition 1.170; the matrix associated with $\sigma_{|W'}$ (relative to a suitable orthonormal basis of W') is known. By proceeding in this way we find $s = p + q + k$ subspaces $W_1, \ldots, W_s$ of V, invariant under σ, $p + q$ of dimension 1 and k of dimension 2, such that $V = W_1 \perp W_2 \perp \cdots \perp W_s$. In every W_i there exists an orthonormal basis $\mathcal{B}_i$ whose associated matrix $\sigma_{|W_i}$ is of the form (1.65) o (1.66). By Proposition 1.113 it follows that $\mathcal{B} := \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_s$ is the basis required (after a suitable permutation). □

Definition 1.173 Let σ be an orthogonal automorphism of a euclidean vector space $(V, \langle\cdot,\cdot\rangle)$ of dimension n. Suppose that there exists an orthonormal basis

$\mathcal{B} = \{e_1, \ldots, e_n\}$ such that the associated matrix $M_\sigma^{\mathcal{B}}$ is of the form

$$\begin{pmatrix} 1 & \ldots & 0 & 0 & 0 & \ldots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \ldots & 1 & 0 & 0 & \ldots & 0 \\ 0 & \ldots & 0 & R & 0 & \ldots & 0 \\ 0 & \ldots & 0 & 0 & 1 & \ldots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \ldots & 0 & 0 & 0 & \ldots & 1 \end{pmatrix}. \tag{1.75}$$

If $R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $(\theta \in (0, 2\pi), \theta \neq \pi)$, σ is called a *rotation*. It represents a rotation of a 2-dimensional subspace W (Proposition 1.170) and leaves every vector of the $(n-2)$-dimensional subspace $W^\perp$ fixed.

If $R = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, σ is an orthogonal reflection. It reverses all vectors of 1-dimensional subspace W and leaves every vector of the $(n-1)$-dimensional subspace $W^\perp$ fixed (if -1 is in position ii, with $i \geq 2$, then $W^\perp$ has the equation $x_i = 0$ in the coordinates $(x_1, \ldots, x_n)$ with respect to the on-basis $\mathcal{B}$).

Theorem 1.172 and Proposition 1.170 imply:

Corollary 1.174 *Every orthogonal automorphism of an euclidean vector space $(V, \langle\cdot,\cdot\rangle)$ of dimension $n \geq 2$ can be written as the composition of n orthogonal symmetries at most.*

Proof We have only to remark that the matrix (1.73) (after renumbering the coordinates) is a product of matrices of the form (1.75). □

An immediate consequence of Theorem 1.172 is the following (see also Exercise 1.14):

Corollary 1.175 *Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension 3. If $\sigma \in \mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$, $\sigma \neq \mathrm{id}_V$, then there exists an on-basis $\mathcal{B}$ such that σ takes one of the following forms with respect to $\mathcal{B}$:*

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$ *orthogonal reflection through the plane $x_3 = 0$;*

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$ *orthogonal reflection through the line $x_2 = x_3 = 0$;*

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$ *orthogonal reflection through the origin $(0, 0, 0)$;*

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \textit{ rotation around the line } x_1 = x_2 = 0;$$

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix} \textit{ rotation around around the line } r : x_1 = x_2 = 0 \textit{ followed by}$$

an orthogonal reflection through the plane

$x_3 = 0$ *(which is perpendicular to r).*

1.4.2 Self-Adjoint Operators

Definition 1.176 Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension n and $f \in \mathrm{End}_K(V)$ a linear operator on V. The *adjoint operator* f^* of f is defined by

$$\langle f(v), w\rangle = \langle v, f^*(w)\rangle, \qquad \forall\, v, w \in V. \tag{1.76}$$

Fix an orthonormal basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ and consider the matrices $M_f^{\mathcal{B}} = (a_{ij})_{i,j=1,\ldots,n}$ $M_{f^*}^{\mathcal{B}} = (b_{ij})_{i,j=1,\ldots,n}$ associated with f and f^* relative to $\mathcal{B}$. Then

$$\langle f(v_j), v_i\rangle = \langle \sum_{k=1}^{n} a_{kj} v_k, v_i\rangle = \sum_{k=1}^{n} a_{kj}\langle v_k, v_i\rangle = \sum_{k=1}^{n} a_{kj}\delta_{ki} = a_{ij}.$$

Similarly, we have

$$\langle v_j, f^*(v_i)\rangle = \langle v_j, \sum_{k=1}^{n} b_{ki} v_k\rangle = \sum_{k=1}^{n} b_{ki}\delta_{jk} = b_{ji},$$

Since $\langle f(v_j), v_i\rangle = \langle v_j, f^*(v_i)\rangle$ for $i, j = 1, \ldots, n$ we deduce $a_{ij} = b_{ji}$ so that

$$M_{f^*}^{\mathcal{B}} = (M_f^{\mathcal{B}})^t. \tag{1.77}$$

Conversely, if we define

$$f^*\colon V \to V, \qquad f^*(v_i) = \sum_{k=1}^{n} a_{ik} v_k, \qquad i = 1, \ldots, n,$$

then f^* satisfies (1.76).

We call f *self-adjoint* if $f = f^*$, i.e.

$$\langle f(v), w\rangle = \langle v, f(w)\rangle, \qquad \forall\, v, w \in V. \tag{1.78}$$

Therefore f is self-adjoint if and only if the matrix $M_f^{\mathcal{B}}$ associated to f with respect to an orthonormal basis $\mathcal{B}$ is symmetric.

Lemma 1.177 *Let* $C = \left(c_{ij}\right)_{i,j=1,\dots,n} \in \mathcal{M}_n(\mathbb{R})$. *Then*

$$\langle Cx, y\rangle = \langle x, C^t y\rangle, \qquad \forall\, x = (x_1, \dots, x_n)^t,\ y = (y_1, \dots, y_n)^t \in \mathbb{R}^n.$$

In particular, C is symmetric if and only if $\langle Cx, y\rangle = \langle x, Cy\rangle$, $\forall\, x, y \in \mathbb{R}^n$.

Proof Obvious from the above discussion. □

Theorem 1.178 *Let $C \in \mathcal{M}_n(\mathbb{R})$ a non-zero symmetric matrix. Then there exists an orthogonal matrix $A \in \mathcal{M}_n(\mathbb{R})$ such that $A^t C A$ is diagonal.*

Conversely, if a matrix $C \in \mathcal{M}_n(\mathbb{R})$ is diagonalized by an orthogonal matrix A, i.e. $A^t C A$ is diagonal, then C is symmetric.

Proof We proceed by induction on n. The case $n = 1$ is trivial. Suppose that $n \geq 2$ and our thesis is true for every $m \leq n - 1$. By Proposition 1.122 there exists an eigenvalue d_1 of C and a non-zero column vector $A_1 = (a_1, \dots, a_n)^t$ with real entries such that $CA_1 = d_1 A_1$. Furthermore, multiplying A_1 by a suitable real number we can suppose that $\|A_1^t\| = 1$.

Claim There exists an orthogonal matrix $B \in \mathcal{M}_n(\mathbb{R})$ whose first column is A_1.

Indeed, since the vector $A_1^t \in \mathbb{R}^n$ has norm 1, there exists an orthonormal basis (containing A_1^t) $A_1^t, \dots, A_n^t$ of $\mathbb{E}^n(\mathbb{R})$, by Theorem 1.155, where A_i is a column vector. Then the matrix $B = (A_1, A_2, \dots, A_n)$ (having $A_1, A_2, \dots, A_n$ as columns) is an orthogonal matrix so that the matrix searched is B and our claim is proved.

The first column of CB is the vector $d_1 A_1$ so that the fist column of $B^t C B$ is

$$(A_1^t d_1 A_1, A_2^t d_1 A_1, \dots, A_n^t d_1 A_1)^t = (d_1, 0, \dots, 0)^t$$

(by our hypothesis $A_i^t A_j = \langle A_i^t, A_j^t\rangle = \delta_{ij}$, $i, j = 1, \dots, n$). Since $B^t C B$ is symmetric ($(B^t C B)^t = B^t C^t B = B^t C B$), we have

$$B^t C B = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_{22} & \dots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & d_{n2} & \dots & d_{nn} \end{pmatrix},$$

where the matrix $D := \left(d_{ij}\right)_{i,j=2,\dots,n} \in \mathcal{M}_{n-1}(\mathbb{R})$ is symmetric. By our inductive hypothesis there is an orthogonal matrix $E \in \mathcal{M}_{n-1}(\mathbb{R})$ such that

$$E^t D E = \begin{pmatrix} d_2 & 0 & \dots & 0 \\ 0 & d_3 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{pmatrix}.$$

Hence the matrix

$$B_1 := \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}$$

is orthogonal so that $A := BB_1$ is orthogonal and

$$A^t C A = B_1^t B^t C B B_1 = \begin{pmatrix} 1 & 0 \\ 0 & E^t \end{pmatrix} \begin{pmatrix} d_1 & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}$$

$$= \begin{pmatrix} d_1 & 0 \\ 0 & E^t D E \end{pmatrix} = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{pmatrix}. \tag{1.79}$$

From (1.79) we deduce that the columns of A are eigenvectors of C (see the proof of Theorem 1.134).

Suppose that $A^t C A = \Delta$ with Δ diagonal matrix, then $C = (A^t)^{-1} \Delta A^{-1} = (A^{-1})^t \Delta A^{-1}$ so that $C^t = (A^{-1})^t \Delta A^{-1} = C$. □

Proposition 1.179 *Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space of dimension n and $f \in \mathrm{End}_K(V)$ a self-adjoint operator on V.*

(i) *The eigenvalues of f are all real.*
(ii) *Two eigenspaces V_λ and $V_{\lambda'}$ corresponding to different eigenvalues are orthogonal.*
(iii) *V has a orthonormal basis of eigenvectors of f.*

Proof

(i) The eigenvalues of f are those of $M_f^{\mathcal{B}}$, where $\mathcal{B}$ is an orthonormal basis of V. Since $M_f^{\mathcal{B}}$ is symmetric our thesis follows from Proposition 1.122. We denote the r distinct eigenvalues of f by $\lambda_1, \dots, \lambda_r$.

(ii) Suppose that $f(v) = \lambda v$ and $f(w) = \lambda' w$ with $\lambda \neq \lambda'$. Then

$$\lambda \langle v, w \rangle = \langle \lambda v, w \rangle = \langle f(v), w \rangle = \langle v, f(w) \rangle = \langle v, \lambda' w \rangle = \lambda' \langle v, w \rangle,$$

hence $(\lambda - \lambda')\langle v, w \rangle = 0$, which implies $\langle v, w \rangle = 0$, because $\lambda - \lambda' \neq 0$.

(iii) Since $M_f^{\mathcal{B}}$ is diagonalizable, then f is diagonalizable (Theorem 1.134) and

$$V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_r},$$

by Corollary 1.130. By ii) we have

$$V = V_{\lambda_1} \perp V_{\lambda_2} \perp \cdots \perp V_{\lambda_r}.$$

By Theorem 1.155, V_{λ_i} has an orthonormal basis B_{λ_i}, $i = 1, \ldots, r$. and $B_{\lambda_1} \cup \ldots \cup B_{\lambda_r}$ is an orthonormal basis of V.

□

Orthogonal Diagonalization Algorithm
INPUT: Let $C \in \mathcal{M}_n(\mathbb{R})$ be a symmetric matrix.

1. Compute the eigenvalues of C.
2. If $\lambda_1, \ldots, \lambda_r$ are the distinct eigenvalues of C and $\mu_1, \ldots, \mu_r$ their algebraic multiplicities respectively, find a basis $\mathcal{B}'_{\lambda_i}$ of the eigenspace V_{λ_i}, $i = 1, \ldots, r$.
3. By the Gram–Schmidt orthogonalization process transform $\mathcal{B}'_{\lambda_i}$ into an orthonormal basis $\mathcal{B}_{\lambda_i}$ of V_{λ_i}, $i = 1, \ldots, r$, and put $\mathcal{B} = \mathcal{B}_{\lambda_1} \cup \cdots \cup \mathcal{B}_{\lambda_r}$.
4. Let

$$\mathcal{B} = (v_1, \ldots, v_{\mu_1}, v_{\mu_1+1}, \ldots, v_{\mu_1+\mu_2}, \ldots, v_{\mu_1+\cdots+\mu_{r-1}+1} \ldots, v_{\mu_1+\cdots+\mu_r}).$$

5. Form the matrices $A = [v_1, \ldots, v_{\mu_1}, v_{\mu_1+1}, \ldots, v_{\mu_1+\mu_2}, \ldots, v_{\mu_1+\cdots+\mu_{r-1}+1} \ldots, v_{\mu_1+\cdots+\mu_r}]$ (see Remark 1.133) and $D = \operatorname{diag}(\underbrace{\lambda_1, \ldots, \lambda_1}_{\mu_1}, \ldots, \underbrace{\lambda_r, \ldots, \lambda_r}_{\mu_r})$.

Then A is an orthogonal matrix and

$$A^t C A = \operatorname{diag}(\underbrace{\lambda_1, \ldots, \lambda_1}_{\mu_1}, \ldots, \underbrace{\lambda_r, \ldots, \lambda_r}_{\mu_r}).$$

OUTPUT: $A \in \mathrm{O}(n)$ and $A^t C A$ is diagonal.

Example 1.180 We want to diagonalize the symmetric matrix $C = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$. The characteristic polynomial of C is $-X^3 + 3X + 2 = -(X+1)^2(X-2)$. Hence its roots are $\lambda_1 = -1$ of multiplicity 2 and $\lambda_2 = 2$. The associated eigenspaces are

$$V_{-1} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\} = \mathbb{R}(1, -1, 0) + \mathbb{R}(1, 0, -1).$$

$$V_2 = \{(t, t, t) \mid t \in \mathbb{R}\} = \mathbb{R}(1, 1, 1).$$

The basis $\mathcal{B}'_{-1} = \{(1,-1,0),(1,0,-1)\}$ is not orthogonal, so that we have to apply the Gram–Schmidt orthogonalization process. Put $v_1 := (1,-1,0)$, $v_2 := (1,0,-1)$ and

$$w_1 := \frac{1}{\|v_1\|}v_1 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right).$$

We have to determine $w'_2 = v_2 - aw_1$ such that $\langle w_1, w'_2\rangle = \langle v_2, w_1\rangle - a = 0$. Hence we get $a = \langle v_2, w_1\rangle = \dfrac{1}{\sqrt{2}}$ and $w'_2 = (1,0,-1) - \dfrac{1}{\sqrt{2}}\left(\dfrac{1}{\sqrt{2}}, -\dfrac{1}{\sqrt{2}}, 0\right) = \left(\dfrac{1}{2}, \dfrac{1}{2}, -1\right)$. Consequently the vectors w_1 and $w_2 = \dfrac{1}{\|w_2\|}w'_2 = \left(\dfrac{1}{\sqrt{6}}, \dfrac{1}{\sqrt{6}}, -\dfrac{2}{\sqrt{6}}\right)$ form an orthonormal basis $\mathcal{B}_{-1}$ of V_{-1}. If we put $v_3 = (1,1,1)$ and $w_3 = \left(\dfrac{1}{\sqrt{3}}, \dfrac{1}{\sqrt{3}}, \dfrac{1}{\sqrt{3}}\right)$, then $\{w_3\}$ is an orthonormal basis $\mathcal{B}_2$ of V_2. Therefore

$$\mathcal{B} = \{w_1, w_2, w_3\} = \left\{\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)\right\}$$

is an orthonormal basis of $\mathbb{E}^3(\mathbb{R})$ consisting of eigenvectors of C. Hence

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \qquad \text{and} \qquad A^t C A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

1.5 Exercises

Exercise 1.1 Let $A \in \mathcal{M}_n(K)$ (K is a field) be a triangular matrix commuting with its transpose A^t. Prove that A is diagonal.

Exercise 1.2 Let:
$E_{ij}(\alpha) \in \mathcal{M}_n(K)$ be the matrix obtained from I_n by changing the entry in the position ij ($i \neq j$) from 0 to $\alpha \in K$;
$E_{ii}(\alpha) \in \mathcal{M}_n(K)$ be the matrix obtained from I_n by changing the entry in the position ii from 1 to $\alpha \in K^*$;
$E_{ij} \in \mathcal{M}_n(K)$ be the matrix obtained from I_n by interchanging the i-th row and j-th row.

We refer to $E_{ij}(\alpha)$, $E_{ii}(\alpha)$ and E_{ij} as *elementary matrices*. Note the following (obvious) relations:

$$E_{ij} = E_{ji}, \qquad E_{ij}\,E_{ik}\,E_{ij} = E_{jk},$$
$$E_{ij}\,E_{ii}(\alpha)\,E_{ij} = E_{jj}(\alpha), \qquad E_{ij}\,E_{ik}(\alpha)\,E_{ij} = E_{jk}(\alpha).$$

Prove that elementary matrices are invertible and

$$E_{ij}(\alpha)^{-1} = E_{ij}(-\alpha),\ i \neq j; \qquad E_{ii}(\alpha)^{-1} = E_{ii}(\frac{1}{\alpha}); \qquad E_{ij}^{-1} = E_{ij}.$$

Note that $\{E_{ij}(0), 1 \leq i \leq n, 1 \leq j \leq n, i \neq j; E_{ii}(1), 1 \leq i \leq n\}$ is the canonical basis of $\mathcal{M}_n(K)$ (see Example 1.62-2).

Let $A \in \mathcal{M}_{n,p}(K)$. Prove that a matrix $B \in \mathcal{M}_{n,p}(K)$ is obtained from A by:

changing the i-th row r_i into $r_i + \alpha r_j$ with $j \neq i$ if and only if $B = E_{ij}(\alpha)A$;
changing the i-th row r_i into αr_i with $\alpha \in K^*$ if and only if $B = E_{ii}(\alpha)A$;
interchanging the i-th row and j-th row if and only if $B = E_{ij}A$.

Let $A \in \mathcal{M}_{m,n}(K)$. Prove that a matrix $B \in \mathcal{M}_{m,n}(K)$ is obtained from A by:

changing the i-th column c_i into $c_i + \alpha c_j$ with $j \neq i$ if and only if $B = AE_{ij}(\alpha)$;
changing the i-th column c_i into αc_i with $\alpha \in K^*$ if and only if $B = AE_{ii}(\alpha)$;
interchanging the i-th column and j-th column if and only if $B = AE_{ij}$.

Let $A \in \mathrm{GL}_n(K)$. Prove that there exist k elementary matrices $E_1, \ldots, E_k \in \mathcal{M}_n(K)$ such that

$$E_k E_{k-1} \cdots E_1 A = I_n.$$

Exercise 1.3 Let $A \in \mathrm{GL}_n(K)$ (K is a field). What will happen to the matrix A^{-1} if, in A we:

(a) interchange the i-th row and the j-th rows;
(b) add the j-th row multiplied by $c \in K$ to the i-th one;
(c) multiply the i-row by $c \in K^*$;
(d) apply the operations (a)-(c) to columns?

Exercise 1.4 Prove that the $n \times n$-matrices of rank 2 generate $\mathcal{M}_n(K)$. (Hint: use the elementary matrices.)

Exercise 1.5 Let V be a K-vector space and v_1, v_2, v_3 three linearly independent vectors of V. Is it true that $v_1 + v_2$, $v_1 + v_3$ and $v_2 + v_3$ are linearly independent?

Exercise 1.6 Prove that the set $\mathbb{R}^+ := \{x \in \mathbb{R} \,|\, x > 0\}$ equipped with the operations

$$\mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}^+, \quad (x, y) \to x \boxplus y := xy,$$
$$\mathbb{R} \times \mathbb{R}^+ \to \mathbb{R}^+, \quad (\alpha, x) \to \alpha \boxdot x := x^\alpha,$$

is a $\mathbb{R}$-vector space.

Exercise 1.7 Find an infinite (countable and uncountable) free subset of the $\mathbb{Q}$-vector spaces $\mathbb{R}$ and $\mathbb{C}$.

Exercise 1.8 Let K be a finite field of characteristic p. Show that $\#(K) = p^n$, with $n \geq 1$. (Hint: use Theorem 1.49.)

Exercise 1.9 Let $u : K \to A$ be a homomorphism of unitary rings, where K is a field and A is an unitary ring (not necessarily commutative). Show that u is injective.

Exercise 1.10 If v_1, v_2, v_3 are three linearly dependent vectors of a K-vector space V, is it true that $\langle v_1, v_2\rangle = \langle v_2, v_3\rangle$?

Exercise 1.11 Let V_1, V_2 and V_3 be three vector subspaces of a K-vector space V such that $V_1 \subset V_3$. Prove the equality $(V_1 + V_2) \cap V_3 = V_1 + (V_2 \cap V_3)$.

Exercise 1.12 Find an infinite (countable and uncountable) free subset of the $\mathbb{R}$-vector space $C([a, b]; \mathbb{R})$ (see Example 1.33, 4).

Exercise 1.13 Let W_1, W_2 and W_3 be three subspaces of a K-vector space V. Is it true that $W_2 = W_3$ if $W_1 \oplus W_2 = W_1 \oplus W_3$?

Exercise 1.14 Let V be a K-vector space with $\mathrm{char}(K) \neq 2$. An endomorphism $f : V \to V$ is called:

idempotent if $f^2 := f \circ f = f$,
involutive if $f^2 = \mathrm{id}_V$.

1. Prove that f is idempotent if and only if

$$V = \ker(f - \mathrm{id}_V) \oplus \ker(f).$$

   Note that $\ker(f - \mathrm{id}_V) = \mathrm{Im}(f)$ and $f_{|\mathrm{Im}(f)} = \mathrm{id}_{\mathrm{Im}(f)}$. If $\ker(f) = 0_V$ then $f = \mathrm{id}_V$; if $\ker(f - \mathrm{id}_V) = 0_V$ then $f(v) = 0_V$ for all $v \in V$.
2. Let $V = A \oplus B$; the endomorphism $p_{A,B} : V \to V$, $p_{A,B}(v) = a$ if $v = a + b$, is called the *projection onto A along B*. It is immediate that $p_{A,B}$ is an idempotent endomorphism. if $f : V \to V$ is an idempotent linear endomorphism, then $f = p_{\mathrm{Im}(f),\mathrm{Ker}(f)}$.
3. Prove that f is involutive if and only if

$$V = \ker(f - \mathrm{id}_V) \oplus \ker(f + \mathrm{id}_V).$$

The subspace $\ker(f - \mathrm{id}_V)$ is called the *axis* of f. It may of course happen that one of subspaces $\ker(f - \mathrm{id}_V)$, $\ker(f + \mathrm{id}_V)$ is trivial (i.e. 0_V). If $\ker(f + \mathrm{id}_V) = 0_V$ then $f = \mathrm{id}_V$; if $\ker(f - \mathrm{id}_V) = 0_V$ then $f = -\mathrm{id}_V$ and f is called the *symmetry about the origin*.

4. Let $V = A \oplus B$. The endomorphism $s_{A,B} : V \to V$, $s_{A,B}(v) = a - b$ if $v = a + b$, is called *the reflection* (or *symmetry*) *through* A *along* B. It is easily seen that $s_{A,B}$ is involutive so that $s_{A,B}$ is an isomorphism. if s is involutive, then $s = s_{A,B}$ where $A = \ker(s - \mathrm{id}_V)$ and $B = \ker(s + \mathrm{id}_V)$.
5. Let $s \neq \mathrm{id}_V$ be involutive. Show that $p := \frac{1}{2}s + \frac{1}{2}\mathrm{id}_V$ is idempotent and $p \neq \mathrm{id}_V$. Conversely, if $p \neq \mathrm{id}_V$ is idempotent, then $s := 2p - \mathrm{id}_V$ is involutive and $s \neq \mathrm{id}_V$.
6. Show that a reflection $s_{A,B}$ is an orthogonal linear isomorphism if and only if $B = A^{\perp}$.

Exercise 1.15 Let V_1 be the subset of symmetric matrices and V_2 the subset of skew-symmetric matrices of $V = \mathcal{M}_n(K)$.

(i) Show that V_1 and V_2 are vector subspaces of V and $\dim_K(V_1) = \dfrac{n(n+1)}{2}$.

(ii) If $\mathrm{char}(K) \neq 2$, show that $\dim_K(V_2) = \dfrac{n(n-1)}{2}$.

(iii) If $\mathrm{char}(K) \neq 2$, show that $V = V_1 \oplus V_2$.

Exercise 1.16 Let K be a field with $\mathrm{char}(K) = p > 0$ and $n \geq 1$ a fixed integer. Assume that $p \nmid n$. Prove that $V_1 := \{aI_n \mid a \in K\}$ and $V_2 := \{A \in V \mid \mathrm{tr}(A) = 0\}$ are subspaces of $V = \mathcal{M}_n(K)$ and $V = V_1 \oplus V_2$.

Exercise 1.17 Let $V = \mathbb{R}[X]$. A polynomial $P = P(X) \in \mathbb{R}[X]$ is called *even* if $P(-X) = P(X)$, *odd* if $P(-X) = -P(X)$. Let W be the subset of all even polynomials and W' the subset of all odd polynomials V. Show that W and W' are subspaces of V and $V = W \oplus W'$.

Exercise 1.18 Is the vector space of Exercise 1.6 isomorphic to a finite–dimensional $\mathbb{R}$-vector space?

Exercise 1.19 Let V, W, Z be three K-vector spaces and $f : V \to W$, $g : V \to Z$ two linear maps. Prove that

$$\ker(f) \subset \ker(g) \iff \exists\, h : W \to Z \text{ linear such that } g = h \circ f.$$

If $k : W \to Z$ is a linear maps, prove that

$$\mathrm{Im}(g) \subset \mathrm{Im}(k) \iff \exists\, l : W \to Z \text{ linear such that } g = k \circ l.$$

Exercise 1.20 Let V be a K-vector space and V^* its dual space. Prove the equalities

$$(W_1 + W_2)^0 = W_1^0 \cap W_2^0, \qquad (W_1 \cap W_2)^0 = W_1^0 + W_2^0,$$

for every pair of subspaces W_1 and W_2 of V.

Exercise 1.21 Let $f \in \mathrm{End}_K V$ be an endomorphism of a K-vector space V of dimension $n \geq 2$ such that $f^n = 0$ and $f^{n-1} \neq 0$. Prove that there exists a vector $v \in V$ such that $\mathcal{B} = \{v, f(v), \ldots, f^{n-1}(v)\}$ is a basis V. Find $M_f^{\mathcal{B}}$.

Exercise 1.22 Let $A = \left(a_{ij}\right)_{i,j=1,\ldots,n} \in \mathcal{M}_n(K)$. Show that if $a_{ij} = 0 \; \forall \, i \geq j$ then $A^n = 0$.

Exercise 1.23 Let $A, B, C, D, A', B', C', D' \in \mathcal{M}_n(R)$ (where R is a commutative unitary ring). Show the equality

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \cdot \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix} = \begin{pmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{pmatrix}.$$

Exercise 1.24 Let $A \in \mathcal{M}_n(K)$, $B \in \mathcal{M}_m(K)$, $C \in \mathcal{M}_{n,m}(K)$ and

$$M = \left(\begin{array}{c|c} A & C \\ \hline 0 & B \end{array}\right).$$

Show that the characteristic polynomials $P_A(T)$, $P_B(T)$ and $P_M(T)$ satisfy the following relation:

$$P_M(T) = P_A(T)\, P_B(T).$$

Hint: note that if $A \in \mathcal{M}_n(R)$, $B \in \mathcal{M}_m(R)$, $C \in \mathcal{M}_{n,m}(R)$ (where R is a commutative unitary ring) then we have

$$\left(\begin{array}{c|c} A & C \\ \hline 0_{m,n} & B \end{array}\right) = \left(\begin{array}{c|c} I_n & 0_{n,m} \\ \hline 0_{m,n} & B \end{array}\right) \left(\begin{array}{c|c} A & C \\ \hline 0_{m,n} & I_m \end{array}\right).$$

Exercise 1.25 Let $f, g \in \mathrm{End}_K(V)$ be two endomorphisms of a K-vector space V of finite dimension. Show that the characteristic polynomials of $f \circ g$ and $g \circ f$ coincide. (Hints: From Exercise 1.24 we have the equality

$$\begin{pmatrix} AB - TI_n & -A \\ 0 & -TI_n \end{pmatrix} \cdot \begin{pmatrix} I_n & 0 \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_n & 0 \\ B & I_n \end{pmatrix} \cdot \begin{pmatrix} -TI_n & -A \\ 0 & BA - TI_n \end{pmatrix}, \quad A, B \in \mathcal{M}_n(K).$$

Exercise 1.26 Let $\mathbb{R}[X]_n$ be the $\mathbb{R}$-vector space of all polynomials of degree $\leq n$ (with a fixed $n \geq 2$) (see Example 1.33, 2) and D the formal derivative of Example 1.69-5). Prove that $D_{|\mathbb{R}[X]_n}$ is an endomorphism of $\mathbb{R}[X]_n$ such that $D^{n+1} = 0$ and find its eigenvalues.

Exercise 1.27 Prove that the set of diagonalizable matrices of $\mathcal{M}_m(\mathbb{R})$ generates $\mathcal{M}_m(\mathbb{R})$. (Hint: use the elementary matrices $E_{ii}(1)$ and $\mathrm{diag}(1, 2, \ldots, n)$).

Exercise 1.28 Prove that for all $a \in \mathbb{R}$ the matrix $\begin{pmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & a \end{pmatrix} \in \mathcal{M}_3(\mathbb{R})$ is not diagonalizable.

Exercise 1.29 Applying the diagonalization algorithm in Subsection 1.3.3, diagonalize

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & -2 & 5 \\ 0 & 5 & -2 \end{pmatrix} \in \mathcal{M}_3(\mathbb{R})$$

Exercise 1.30 Show that the characteristic polynomial of a skew-symmetric matrix $A \in \mathcal{M}_n(\mathbb{R})$ has all its roots purely imaginary. (Hint: adapt the proof of Proposition 1.122.)

Exercise 1.31 Let $V = V_1 \oplus V_2$ be a K-vector space direct sum of its subspaces V_1 and V_2. Let $f : V_1 \to V_1$ and $g : V_2 \to V_2$ be two maps. If $h : V \to V$ is defined by $h(v) = f(a) + g(b)$ where $v = a + b$, $a \in V_1$ and $b \in V_2$, prove that:

(i) h is linear if and only if f and g are linear.
(ii) h is diagonalizable if and only if f and g are diagonalizable.

Exercise 1.32 Show that a pair of matrices $A, B \in \mathcal{M}_n(K)$ are *simultaneously diagonalizable* (i.e. there exists a matrix $M \in \mathrm{GL}_n(K)$ such that $M^{-1}AM$ and $M^{-1}BM$ are diagonal) if and only if $AB = BA$.

Exercise 1.33 Let $d_1, \ldots, d_n$ be $n \geq 2$ positive real numbers. Show that

$$\langle (x_1, \ldots, x_n), (y_1, \ldots, y_n) \rangle := d_1 x_1 y_1 + \cdots + d_n x_n y_n$$

is an inner product of $\mathbb{R}^n$.

Exercise 1.34 Consider the euclidean vector space $\mathbb{R}[X]_{\leq 5}$ with inner product defined by

$$\langle f(X), g(X) \rangle = \int_{-1}^{1} f(x) g(x)\, dx.$$

Applying the Gram–Schmidt process orthonormalize the basis $\{1, X, X^2, X^3, X^4, X^5\}$.

Exercise 1.35 Let v and w be two vectors of a euclidean vector space V.

1. If $\|v\| = \|w\|$ show that $v - w$ and $v + w$ are orthogonal and explain the geometric meaning of it.
2. Prove the equality $\|v + w\|^2 + \|v - w\|^2 = 2\|v\|^2 + 2\|w\|^2$ and explain its geometric meaning.

Exercise 1.36 Let $\{v_1, \ldots, v_n\}$ be an on-basis of a euclidean vector space V. For every $i = 1, \ldots, n$ put $x_i := v_1 + \cdots + v_i$. Show that $\{x_1, \ldots, x_n\}$ is a basis of V and orthonormalize it by the Gram–Schmidt process.

Exercise 1.37 Let W and U be two vector subspace of an euclidean vector space V of finite dimension. Show the equalities $(W + U)^\perp = W^\perp \cap U^\perp$ and $(W \cap U)^\perp = W^\perp + U^\perp$.

Exercise 1.38 Let $A \in \mathcal{M}_n(\mathbb{R})$ with $\operatorname{rank}(A) = k$.

(a) Prove that there exist $P \in \mathrm{GL}_n(\mathbb{R})$ and $Q \in \mathrm{O}(n)$ such that $P\,A\,Q = \left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array}\right)$.

(b) Using (a) prove that $\operatorname{rank}(A^t\,A) = \operatorname{rank}(A)$.

Exercise 1.39 Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension n and $f \in \mathrm{End}_K(V)$ a self-adjoint linear operator on V. Prove that f can only have $+1$ and -1 as eigenvalues. If n is odd and $\det(f) = 1$ then f has necessarily $+1$ as eigenvalue.

Exercise 1.40 Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension n and $f \in \mathrm{End}_K(V)$ a linear operator on V. Suppose that

$$\langle v, w\rangle = 0 \quad \Longrightarrow \quad \langle f(v), f(w)\rangle = 0, \qquad \forall\, v, w \in V.$$

Prove that there exist an orthogonal linear automorphism $g\colon (V, \langle\cdot,\cdot\rangle) \to (V, \langle\cdot,\cdot\rangle)$ and a scalar $\lambda \in \mathbb{R}$ such that $f = \lambda g$.

Exercise 1.41 Let $(V, \langle\cdot,\cdot\rangle_V)$ and $(W, \langle\cdot,\cdot\rangle_W)$ be two euclidean spaces of the same dimension. Suppose that a map $f\colon V \to W$ preserves the inner product, i.e. $\langle f(u), f(v)\rangle_W = \langle u, v\rangle_V$ for every $u, v \in V$. Show that f is a linear isomorphism.

Exercise 1.42 Show that every finite subgroup of $\mathrm{SO}(2)$ is cyclic.

Exercise 1.43 Show that for any natural number m one gets $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^m = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix}$.

Exercise 1.44 Let σ and ρ be the orthogonal automorphisms of $\mathbb{E}^2(\mathbb{R})$ defined by the matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \cos\dfrac{2\pi}{n} & -\sin\dfrac{2\pi}{n} \\ \sin\dfrac{2\pi}{n} & \cos\dfrac{2\pi}{n} \end{pmatrix}, \qquad (n \geq 2),$$

respectively. Show that $\sigma^2 = \rho^n = \mathrm{id}$ and $\rho\circ\sigma = \sigma\circ\rho^{n-1}$. Furthermore, prove that the subgroup of $\mathrm{O}(2)$ generated by σ and ρ has $2n$ elements. This group is called the n-th *dihedral subgroup* of $\mathrm{O}(2)$.

Exercise 1.45 Adapting the proof of Theorem 1.178 prove Schur's theorem given below:

Let $C \in \mathcal{M}_n(\mathbb{R})$ be a matrix whose characteristic polynomial has all real roots, then there exists an orthogonal matrix $A \in \mathcal{M}_n(\mathbb{R})$ such that ACA^t is upper triangular.

Which entries does ACA^t have on its main diagonal?

Chapter 2
Bilinear and Quadratic Forms



2.1 Bilinear Maps

In this chapter K denotes a field of characteristic $\neq 2$.

Definition 2.1 Let $V_1, \ldots, V_n, W$ be K-vector spaces. A map

$$f : V_1 \times \cdots \times V_n \to W \tag{2.1}$$

is called *multilinear* if it is linear in each variable separately, i.e. if

$$\begin{aligned} &f(v_1, \ldots, v_{i-1}, \lambda v + \mu v', v_{i+1}, \ldots, v_n) \\ &\quad = \lambda f(v_1, \ldots, v_{i-1}, v, v_{i+1}, \ldots, v_n) + \mu f(v_1, \ldots, v_{i-1}, v', v_{i+1}, \ldots, v_n), \\ &\qquad\qquad \forall\, v, v' \in V_i,\ \forall\, \lambda, \mu \in K, \end{aligned}$$

for all $i = 1, \ldots, n$ and for any fixed $v_j \in V_j$, $j = 1, \ldots, n$, $j \neq i$. If $n = 2$ we use the term *bilinear* instead of multilinear. If $V_1 = \cdots = V_n = V$ we call f an *n-linear map*. If in addition $W = K$, then it is said to be an *n-linear form* on V; if $n = 2$, *bilinear form*.

The set of all bilinear forms on V, denoted by $\mathrm{Bil}(V)$, equipped with the operations

$$\begin{aligned} (f + g)(v, w) &:= f(v, w) + g(v, w), \quad \forall\, v, w \in V, \quad \forall f, g \in \mathrm{Bil}(V), \\ (\lambda\, f)(v, w) &:= \lambda f(v, w), \qquad\qquad \forall\, v, w \in V \quad \forall \lambda \in K, \quad \forall f \in \mathrm{Bil}(V), \end{aligned}$$

is a K-vector space. The neutral element is the *null form* $0 \in \mathrm{Bil}(V)$, $0(v, w) := 0$, $\forall\, v, w \in V$; the opposite of $f \in \mathrm{Bil}(V)$ is the form $-f$, $(-f)(v, w) := -f(v, w)$, $\forall\, v, w \in V$.

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A bilinear form $f \in \mathrm{Bil}(V)$ is said to be *symmetric* if $f(v, w) = f(w, v)$, for all $v, w \in V$; *skew-symmetric* if $f(v, w) = -f(w, v)$, for all $v, w \in V$.

Examples 2.2

1. The product

$$(A, B) \in \mathcal{M}(m, n, K) \times \mathcal{M}(n, p, K) \to AB \in \mathcal{M}(m, p, K),$$

   is a bilinear map.
2. Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space; then $f\colon V \times V \to \mathbb{R}$, $f(v, w) := \langle v, w \rangle$, is a symmetric bilinear form. In particular the standard inner product on $\mathbb{R}^n$ (see Example 1.140, 1) is called the *standard bilinear form* on $\mathbb{R}^n$.
3. The *standard bilinear form* on K^n is given by $f(\mathbf{x}, \mathbf{y}) = x_1 y_1 + \cdots + x_n y_n$ for all $\mathbf{x} = (x_1, \ldots, x_n)$, $\mathbf{y} = (y_1, \ldots, y_n) \in K^n$.
4. If V is a K-vector space of finite dimension, the *natural pairing* of V and V^*

$$\langle \cdot, \cdot \rangle\colon V \times V^* \to K, \qquad \langle v, w^* \rangle = w^*(v),$$

   is a bilinear form.

Definition 2.3 Let $f \in \mathrm{Bil}(V)$ be a bilinear form on a K-vector space V. For every $v \in V$, the following linear functionals on V are defined:

$$f_v\colon V \to K \quad f_v(w) := f(v, w), \quad \forall\, w \in V,$$
$$f'_v\colon V \to K \quad f'_v(w) := f(w, v), \quad \forall\, w \in V.$$

Thus we have the linear maps

$$\delta_f\colon V \to V^* \quad \delta_f(v) := f_v, \quad \forall\, v \in V, \tag{2.2}$$
$$\delta'_f\colon V \to V^* \quad \delta'_f(v) := f'_v, \quad \forall\, v \in V. \tag{2.3}$$

Definition 2.4 Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of a K-vector space V and $f \in \mathrm{Bil}(V)$ a bilinear form on V. We call

$$M_f^{\mathcal{B}} := \left(a_{ij}\right)_{i,j=1,\ldots,n} \in \mathcal{M}_n(K), \qquad a_{ij} := f(v_i, v_j),\ i, j = 1, \ldots, n,$$

the *matrix associated to the bilinear form f with respect to a basis $\mathcal{B}$*. If f is symmetric then $M_f^{\mathcal{B}}$ is obviously symmetric.

Conversely, if with $A = \left(a_{ij}\right)_{i,j=1,\ldots,n} \in \mathcal{M}_n(K)$ we associate the map

$$f_A\colon V \times V \to K, \qquad f_A(v, w) = \sum_{i,j=1}^{n} a_{ij} x_i y_j, \quad v = \sum_{i=1}^{n} x_i v_i,\ w = \sum_{i=1}^{n} y_i v_i, \tag{2.4}$$

then f_A is a bilinear form on V such that $M_{f_A}^{\mathcal{B}} = A$. If A is the null matrix, then f_A is the null form. If A is symmetric, then f_A is symmetric.

Proposition 2.5 *Under the above hypothesis and notation the map*

$$\mathrm{Bil}(V) \to \mathcal{M}_n(K), \qquad f \to M_f^{\mathcal{B}}, \qquad \forall\, f \in \mathrm{Bil}(V),$$

is an isomorphism of K-vector spaces, such that f is symmetric if and only if $M_f^{\mathcal{B}}$ is symmetric.

If $\mathcal{B}'$ is another basis of V and $\Lambda^{\mathcal{B},\mathcal{B}'} \in \mathrm{GL}_n(K)$ is matrix of transition from the basis $\mathcal{B}$ to the basis $\mathcal{B}'$, then we have

$$M_f^{\mathcal{B}'} = (\Lambda^{\mathcal{B},\mathcal{B}'})^t\, M_f^{\mathcal{B}}\, \Lambda^{\mathcal{B},\mathcal{B}'} \tag{2.5}$$

so that $M_f^{\mathcal{B}'}$ and $M_f^{\mathcal{B}}$ are congruent. In particular, $\mathrm{rank}(M_f^{\mathcal{B}}) = \mathrm{rank}(M_f^{\mathcal{B}'})$.

Proof Let $\mathcal{B}' = \{w_1, \ldots, w_n\}$ be another basis of V. If $\Lambda^{\mathcal{B},\mathcal{B}'} = \left(\alpha_{ij}\right)_{i,j=1,\ldots,n}$, then $w_i = \sum_{j=1}^{n} \alpha_{ji} v_j$, $i = 1, \ldots, n$), so that

$$f(w_i, w_j) = f\left(\sum_{k=1}^{n} \alpha_{ki} v_k, \sum_{l=1}^{n} \alpha_{lj} v_l\right) = \sum_{k,l=1}^{n} \alpha_{ki} f(v_k, v_l) \alpha_{lj}.$$

Hence we have $M_f^{\mathcal{B}'} = (\Lambda^{\mathcal{B},\mathcal{B}'})^t\, M_f^{\mathcal{B}}\, \Lambda^{\mathcal{B},\mathcal{B}'}$. □

For instance, the matrix associated with the standard bilinear form on K^n (see Example 2.2, 3) with respect to the canonical basis $e_1 = (1, 0, \ldots, 0), \ldots, e_n = (0, \ldots, 0, 1)$ of K^n is the identity matrix $I_n \in \mathcal{M}_n(K)$.

Definition 2.6 We define the *rank* $\mathrm{rank}(f)$ of a bilinear form $f \in \mathrm{Bil}(V)$ by the relation

$$\mathrm{rank}(f) := \mathrm{rank}(M_f^{\mathcal{B}}),$$

where $\mathcal{B}$ is any basis of V. By Proposition 2.5 $\mathrm{rank}(f)$ is independent of the basis $\mathcal{B}$ chosen. We have the obvious inequality $\mathrm{rank}(f) \leq \dim_K V$. If $\mathrm{rank}(f) < \dim_K V$ we call f a *degenerate* bilinear form, while f is called *non-degenerate* if

$$\mathrm{rank}(f) = \dim_K V.$$

Proposition 2.7 *Let $f \in \mathrm{Bil}(V)$ be a bilinear form on a K-vector space V of finite dimension. The following conditions are equivalent:*

(i) *f is non-degenerate.*
(ii) *For every $v \neq 0_V$ in V there exists $w \in V$ such that $f(v, w) \neq 0$.*
(iii) *For every $w \neq 0_V$ in V there exists $v \in V$ such that $f(v, w) \neq 0$.*
(iv) *The map $\delta_f : V \to V^*$ of (2.2) is an isomorphism of K-vector spaces.*
(v) *The application $\delta'_f : V \to V^*$ of (2.3) is an isomorphism of K-vector spaces.*

Proof (ii) $\iff$ (iv). A vector $v \in V$ is such that $f(v, w) = 0 \ \forall w \in V$ if and only if $v \in \mathrm{Ker}(\delta_f)$, while $\mathrm{Ker}(\delta_f) = 0$ if and only if δ_f is an isomorphism (by Proposition 1.74 and since $\dim_K V = \dim_K V^*$). The equivalence iii) $\iff$ v) is proved in the same way.

Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of V; if $M_f^{\mathcal{B}} = \left(a_{ij}\right)_{i,j=1,\ldots,n}$ and $\mathcal{B}^* = \{e_1^*, \ldots, e_n^*\}$ is the dual basis of V^*, for every $v \in V$ we have

$$\delta_f(v) = \sum_{j=1}^{n} \delta_f(v)(e_j) e_j^* = \sum_{j=1}^{n} f(v, e_j) e_j^*$$

(see the proof of Theorem 1.96). In particular,

$$\delta_f(e_i) = \sum_{j=1}^{n} a_{ij} e_j^*, \qquad i = 1, \ldots, n.$$

Thus

$$M_{\delta_f}^{\mathcal{B}, \mathcal{B}^*} = (M_f^{\mathcal{B}})^t$$

and the equivalence (i) $\iff$ (iv) follows. The equivalence (i) $\iff$ (v) is completely analogous. □

Definition 2.8 Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space V. Two vectors $v, w \in V$ are *f-orthogonal* with respect to f or *orthogonal with respect to* f if $f(v, w) = 0$, written $v \perp_f w$. If $S \subset V$ is a subset of V we define the *f-orthogonal complement* of S with respect to f:

$$S^{\perp_f} := \{w \in V \mid f(v, w) = 0 \quad \forall v \in S\}.$$

The subset $S^{\perp_f}$ is clearly a vector subspace of V. Two vector subspaces V_1 and V_2 are called *f-orthogonal* (or *orthogonal with respect to* f) if $f(v, w) = 0$ for every $v \in V_1$ and $w \in V_2$, written $V_1 \perp V_2$.

The subspace $V^{\perp_f} = \{v \in V \mid f(v, w) = 0 \quad \forall\, w \in V\}$ is called the *radical* of f. By Proposition 2.7 f is non-degenerate if and only if $V^{\perp_f} = 0_V$. It easy to see that

$$\dim_K(V^{\perp_f}) = n - \operatorname{rank}(f).$$

A vector $v \in V$ is called *isotropic with respect to* f if $f(v, v) = 0$. If v is isotropic, then λv is isotropic $\forall\, \lambda \in K$. If v is not isotropic, v is called *non-isotropic*. The only isotropic vector with respect to the bilinear symmetric form $f(v, w) = \langle v, w\rangle$ of the Example 2.2-2 is 0_V.

Let v be a non-isotropic vector of V. The *Fourier coefficient* of $w \in V$ with respect to v relative to a symmetric bilinear form f is the scalar defined by

$$a_v(w) := \frac{f(v, w)}{f(v, v)}. \tag{2.6}$$

Proposition 2.9 *Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space V. If v is a non-isotropic vector then*

$$V = \langle v\rangle \oplus v^{\perp_f},$$

where $\langle v\rangle$ is the subspace generated by v.

Proof From (2.6) we obtain $f(v, w - a_v(w)v) = 0$, i.e. $w - a_v(w)v \in v^{\perp_f}$. Since $w = a_v(w)v + (w - a_v(w)v)$, it follows that $V = \langle v\rangle + v^{\perp_f}$. Furthermore, since v is non-isotropic, $\langle v\rangle \cap v^{\perp_f} = 0_V$ so that our thesis follows. □

Remark 2.10 Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space V. Let W be a subspace of V. In general we have $V \neq W \oplus W^{\perp_f}$. For instance, take $f \in \mathrm{Bil}(\mathbb{R}^2)$, $f((x_1, x_2), (y_1, y_2)) = x_1y_1 - x_2y_2$, and $W = \langle(1, 1)\rangle$, then we have $W = W^{\perp_f}$. Furthermore $W \subsetneq W^{\perp_f\perp_f}$. For instance, take $f \in \mathrm{Bil}(\mathbb{R}^3)$, $f((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1y_1 - x_2y_2$, and $W = \langle(1, 1, 1)\rangle$, then $W^{\perp_f\perp_f} = \{(a, a, b) \mid a, b \in \mathbb{R}\} \supsetneq W$.

Definition 2.11 Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space V. The map

$$q\colon V \to K, \qquad q(v) = f(v, v), \qquad \forall\, v \in V,$$

is called the *quadratic form associated with* f and f is referred to as the bilinear form *polar* to the quadratic form q. The *rank* of q is defined by $\operatorname{rank} q := \operatorname{rank} f$. If A is a symmetric matrix, q_A will denote the quadratic form associated with f_A.

The quadratic form associated with the standard bilinear form on K^n is $q(v) = x_1^2 + \cdots + x_n^2$, $v = (x_1, \ldots, x_n) \in K^n$. This form is called the *standard quadratic form* on K^n.

Proposition 2.12 *Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space V and $q : V \to K$ the quadratic form associated with f. Then*

$$q(\lambda v) = \lambda^2 q(v), \quad 2f(v, w) = q(v + w) - q(v) - q(w), \quad \forall\, \lambda \in K,\ \forall\, v, w \in V.$$

Proof From Definition 2.11 we have

$$\begin{aligned} q(v + w) - q(v) - q(w) &= f(v + w, v + w) - f(v, v) - f(w, w) \\ &= f(v, w) + f(w, v) = 2f(v, w) \end{aligned}$$

and $q(\lambda v) = f(\lambda v, \lambda v) = \lambda^2 f(v, v) = \lambda^2 q(v)$. □

Since $\mathrm{char}(K) \neq 2$, by Proposition 2.12 we can assert that the polar form f is uniquely determined by its quadratic form q.

Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of V and $M_f^{\mathcal{B}} = \left(a_{ij}\right)_{i,j=1,\ldots,n}$ the matrix associated with a symmetric bilinea form $f \in \mathrm{Bil}(V)$ relative to $\mathcal{B}$. If $v = x_1 v_1 + \cdots + x_n v_n$ ($x := (x_1, \ldots, x_n) \in K^n$) then

$$q(v) = f(v, v) = x\, M_f^{\mathcal{B}}\, x^t = \sum_{i,j=1}^{n} a_{ij} x_i x_j.$$

The polynomial

$$Q(x) = x\, M_f^{\mathcal{B}}\, x^t = \sum_{i,j=1}^{n} a_{ij} x_i x_j \in K[x_1, \ldots, x_n] \tag{2.7}$$

is homogeneous of degree 2 and it is said to represent the quadratic form q with respect to the basis $\mathcal{B}$. We observe that any quadratic homogeneous polynomial of $K[x_1, \ldots, x_n]$ can be written in the form (2.7) since $\mathrm{char}(K) \neq 2$.

2.1.1 Diagonalization of Quadratic Forms

Definition 2.13 Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space V and $\mathcal{B} = \{v_1, \ldots, v_n\}$ a basis of V. We call $\mathcal{B}$ *orthogonal* with respect to f or *f-orthogonal* if $f(v_i, v_j) = 0$ for all $i \neq j$. We say that f is *diagonalizable* if there exists an orthogonal basis with respect to f.

If $r = \operatorname{rank}(f)$, eventually renumbering the vectors of $\mathcal{B}$, for every pair of vectors $v = x_1v_1 + \cdots + x_nv_n$ and $w = y_1v_1 + \cdots + y_nv_n$, with $x_i, y_i \in K$, $i = 1, \ldots, n$, we have

$$f(v, w) = a_{11}x_1y_1 + \cdots + a_{rr}x_ry_r, \quad \text{with} \quad a_{ii} = f(v_i, v_i) \neq 0, \quad i = 1, \ldots, r, \tag{2.8}$$

i.e.

$$M_f^{\mathcal{B}} = \begin{pmatrix} a_{11} & 0 & \ldots & 0 & 0 & \ldots & 0 \\ 0 & a_{22} & \ldots & 0 & \vdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & a_{rr} & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 & 0 & \ldots & 0 \end{pmatrix}. \tag{2.9}$$

We say that the quadratic form q associated to the symmetric bilinear form f, with respect to an orthogonal basis, is represented as a *sum of squares*, i.e.

$$q(v) = a_{11}x_1^2 + \cdots + a_{rr}x_r^2 \text{ with } a_{ii} = f(v_i, v_i) \neq 0, \qquad i = 1, \ldots, r. \tag{2.10}$$

Theorem 2.14 (Lagrange) *Any symmetric bilinear form $f \in \operatorname{Bil}(V)$ on a K-vector space V of dimension n is diagonalizable.*

Proof There are several proofs of this fundamental result. We choose a proof based on the concept of a non-isotropic vector. We proceed by induction on n. If $n = 1$ we need not prove anything. Let $n \geq 2$. We can suppose $f \neq 0$. Let $v, w \in V$ be two vectors such that $f(v, w) \neq 0$. Hence at least one among the vectors v, w, $v + w$ is non-isotropic. Indeed, if v and w are both isotropic, then

$$f(v + w, v + w) = f(v, v) + f(w, w) + 2f(v, w) = 2f(v, w) \neq 0,$$

since $\operatorname{char}(K) \neq 2$.

Therefore we can suppose the existence of a vector $e_1 \neq 0$ such that $f(e_1, e_1) \neq 0$. By Proposition 2.9 we have $V = \langle e_1 \rangle \oplus V'$, where $V' := e_1^{\perp_f}$ and $\dim_K V' = n - 1$. By our inductive hypothesis, V' has an orthogonal basis $e_2, \ldots, e_n$ with respect to the symmetric bilinear form

$$f' \colon V' \times V' \to K, \qquad f' := f_{|V' \times V'}.$$

Now we prove that $e_1, e_2, \ldots, e_n$ is an orthogonal basis of V. Let $\lambda_1 e_1 + \lambda_2 e_2 + \cdots + \lambda_n e_n = 0_V$ with $\lambda_i \in K, i = 1, \ldots, n$. If we put $v := \lambda_2 e_2 + \cdots + \lambda_n e_n \in V'$, we get $\lambda_1 e_1 + v = 0_V$, hence

$$0_V = f(\lambda_1 e_1 + v, e_1) = \lambda_1 f(e_1, e_1) + f(v, e_1) = \lambda_1 f(e_1, e_1)$$

(because one has $f(v, e_1) = f(e_1, v) = 0$), which implies $\lambda_1 = 0$ since $f(e_1, e_1) \neq 0$. Therefore $0_V = v = \lambda_2 e_2 + \cdots + \lambda_n e_n$ and $\lambda_2 = \cdots = \lambda_n = 0$ because $e_2, \ldots, e_n$ is a basis of V' su K. Thus our basis $e_1, \ldots, e_n$ is orthogonal with respect to f. □

By Theorem 2.14 and Proposition 2.5 we get:

Corollary 2.15 *Any symmetric matrix of $\mathcal{M}_n(K)$ is congruent to a diagonal matrix.*

Proof Let $A = (a_{ij})_{i.j=1,\ldots n} \in \mathcal{M}_n(K)$ be a symmetric matrix. Then

$$f_A \colon K^n \times K^n \to K, \quad f_A(v, w) = \sum_{i,j=1}^{n} a_{ij} x_i y_j, \quad \forall\, v = (x_1, \ldots, x_n),$$
$$w = (y_1, \ldots, y_n) \in K^n$$

is a symmetric bilinear form on K^n. Then $A = M_{f_A}^{\mathcal{E}}$, where $\mathcal{E}$ is the canonical basis of K^n. By Theorem 2.14 there exists an orthogonal basis $\mathcal{B} = (u_1, \ldots, u_n)$ with $u_i = (z_{i1}, \ldots, z_{in})$, $i = 1, \ldots, n$, such that $M_{f_A}^{\mathcal{B}} = \operatorname{diag}(b_{11}, \ldots, b_{nn})$. By Proposition 2.5 we have

$$M_{f_A}^{\mathcal{B}} = (\Lambda^{\mathcal{E},\mathcal{B}})^t \, A \, \Lambda^{\mathcal{E},\mathcal{B}},$$

where

$$\Lambda^{\mathcal{E},\mathcal{B}} = \begin{pmatrix} z_{11} & z_{21} & \cdots & z_{n1} \\ z_{12} & z_{22} & \cdots & z_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ z_{1n} & z_{2n} & \cdots & z_{nn} \end{pmatrix}$$

is the matrix of transition from the basis $\mathcal{E}$ to the basis $\mathcal{B}$. □

Remark 2.16

(1) We can associate to a symmetric bilinear form f several diagonal representations which depend on the orthogonal bases chosen so that in general we don't have a *canonical diagonal representation*. Let $\operatorname{rank}(f) = r$ and $\mathcal{B} = \{v_1, \ldots, v_r, v_{r+1}, \ldots, v_n\}$ be an orthogonal basis with respect to f

such that $M_f^{\mathcal{B}}$ is defined by the identity 2.9. If $\lambda_1, \ldots, \lambda_r \in K^*$ and $\mathcal{B}' = \{\lambda_1 v_1, \ldots, \lambda_r v_r, v_{r+1}, \ldots, v_n\}$, then $\mathcal{B}'$ is an orthogonal basis with respect to f such that

$$M_f^{\mathcal{B}'} = \begin{pmatrix} \lambda_1^2 a_{11} & 0 & \ldots & 0 & 0 \ldots 0 \\ 0 & \lambda_2^2 a_{22} & \ldots & 0 & \vdots \; \vdots \; \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \; \vdots \; \vdots \\ 0 & 0 & \ldots & \lambda_r^2 a_{rr} & 0 \ldots 0 \\ 0 & 0 & \ldots & 0 & 0 \ldots 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \; \ddots \; \vdots \\ 0 & 0 & \ldots & 0 & 0 \ldots 0 \end{pmatrix}. \tag{2.11}$$

We obviously have many other ways to obtain different diagonal representations: for instance, if $K = \mathbb{Q}$, the quadratic form $2x^2 + 3y^2$ has the diagonal form $5X^2 + 30Y^2$ with respect to the orthogonal basis $\{v_1 = (1, 1), v_2 = (-3, 2)\}$, indeed

$$\begin{pmatrix} 5 & 0 \\ 0 & 30 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}.$$

(2) The matrix $A = \begin{pmatrix} 2 & 3/2 & 2 \\ 3/2 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \in \mathcal{M}_n(\mathbb{Q})$ is congruent to the matrix $\begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & 1/17 \end{pmatrix}$ but A is not similar to any diagonal matrix with entries in $\mathbb{Q}$ since its eigenvalues are $1, \dfrac{3 \pm \sqrt{26}}{2}$.

If K is an algebraically closed field or $K = \mathbb{R}$ we can obtain a *canonical diagonal representation* for a symmetric bilinear form.

Corollary 2.17 *Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space of dimension n, where K is an algebraically closed field. Then there exists a basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V such that*

$$M_f^{\mathcal{B}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

where $r := \mathrm{rank}(f)$ and I_r is the identity matrix of the ring $\mathcal{M}_r(K)$.

Proof By Theorem 2.14 there is an orthogonal basis $w_1, \ldots, w_n$ of V such that (2.8) holds. Since K is algebraically closed, there exist $b_i \in K$ such that $b_i^2 = a_{ii}$,

$i = 1, \ldots, r$. We obtain the required basis if we define $v_i = b_i^{-1} w_i$, $i = 1, \ldots, r$ and $v_j = w_j$, $j = r+1, \ldots, n$. □

Corollary 2.18 (Sylvester's Law of Inertia) *Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a $\mathbb{R}$-vector space V of dimension n. Then there exist a non-negative integer p, which depends only on form f with $0 \leq p \leq r := \mathrm{rank}(f)$ and a basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V such that*

$$M_f^{\mathcal{B}} = \begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_{r-p} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Proof The existence of an orthogonal basis follows from Theorem 2.14. Let $w_1, \ldots, w_n$ be such a basis and $a_{ii} := f(w_i, w_i) \neq 0$ if and only if $i \leq r := \mathrm{rank}(f)$.

Let p be the number of indices i such that $a_{ii} > 0$. Let the vectors w_i be arranged so that

$$a_{ii} > 0 \iff 1 \leq i \leq p, \qquad \text{and} \qquad a_{jj} < 0 \iff p+1 \leq j \leq r.$$

Then there exist $b_1, \ldots, b_r \in \mathbb{R} \setminus \{0\}$ such that $a_{ii} = b_i^2$, $i = 1, \ldots, p$ and $a_{jj} = -b_j^2$, $j = p+1, \ldots, r$. Therefore the required basis consists of the vectors

$$v_i := b_i^{-1} w_i, \quad i = 1, \ldots, r, \qquad \text{and} \qquad v_k := w_k, \quad k = r+1, \ldots, n.$$

It remains to prove that p is an invariant of f. The quadratic form q associated with f with respect to the above basis $(v_1, \ldots, v_n)$ is

$$q(v) = x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2, \quad v = x_1 v_1 + \cdots + x_n v_n. \tag{2.12}$$

Suppose that $(u_1, \ldots, u_n)$ is another basis of V such that

$$q(v) = y_1^2 + \cdots + y_t^2 - y_{t+1}^2 - \cdots - y_r^2, \quad v = y_1 u_1 + \cdots + y_n u_n, \tag{2.13}$$

with $0 \leq t \leq r$. We have to prove the equality $p = t$. Suppose that $p \neq t$, e.g. $t < p$. Let $V_1 := \langle v_1, \ldots, v_p \rangle$ and $V_2 := \langle u_{t+1}, \ldots, u_n \rangle$. Then

$$\dim_{\mathbb{R}}(V_1) + \dim_{\mathbb{R}}(V_2) = p + n - t > n,$$

so that by Grassmann's formula (Corollary 1.76) we have $V_1 \cap V_2 \neq \{0_V\}$. Hence there exists $0_V \neq v \in V_1 \cap V_2$. Let $v = x_1 v_1 + \cdots + x_p v_p = y_{t+1} u_{t+1} + \cdots + y_n u_n$, with $x_i, y_j \in \mathbb{R}$. By (2.12) $q(v) = x_1^2 + \cdots + x_p^2 > 0$ and by (2.13) $q(v) = -y_{t+1}^2 - \cdots - y_r^2 < 0$ but these two inequalities are incompatible, thus $p = t$. □

Definition 2.19 Let q be a quadratic form on an n-dimensional $\mathbb{R}$-vector space V. The equality (2.12) (with $0 \leq p \leq r = \text{rank}(q)$) is called the *canonical representation* of q. It is easy to see that:

- The integer p (called the *positivity index* and denoted by $i_+(q)$) is the maximum dimension of a subspace W such that $q(w) > 0$ for all $w \in W$, $w \neq 0$.
- The integer $r - p$ (called the *negativity index* and denoted by $i_-(q)$) is the maximum dimension of a subspace W such that $q(w) < 0$ for all $w \in W$, $w \neq 0$.
- The integer $n - r$ (called the *nullity index* and denoted by $i_0(q)$) is the maximum dimension of a subspace W such that $q(w) = 0$ for all $w \in W$.

We can obviously extend the above definitions to real symmetric matrices. If A is a real symmetric matrix,

$$i_+(A) := i_+(q_A), \qquad i_-(A) := i_-(q_A), \qquad i_0(A) := i_0(q_A).$$

The pair of integers $\text{Sign}(q) := (i_+(q), i_-(q))$ is called *the signature* of the quadratic form q and $\text{Sign}(A) := (i_+(A), i_-(A))$ is called *the signature* of the real symmetric matrix A. Some authors define the signature of a quadratic form q as the difference $i_+(q) - i_-(q)$.

Corollary 2.20 *Let $A = (a_{ij})_{i.j=1,\ldots n} \in \mathcal{M}_n(\mathbb{R})$ be a symmetric matrix. Then A is congruent to the matrix*

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_{r-p} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{2.14}$$

where $p = i_+(A)$, $r - p = i_-(A)$ and $r = \text{rank}(q_A) = \text{rank}(A)$.

Proof By Theorem 1.178 there exists an orthogonal matrix P such that $P^{-1}AP = \text{diag}(\lambda_1, \ldots, \lambda_r, 0, \ldots, 0)$ where $\lambda_1, \ldots, \lambda_r$ are the non-zero eigenvalues of A (counted with their multiplicities) and $\lambda_1, \ldots, \lambda_p > 0$, while $\lambda_{p+1}, \ldots, \lambda_r < 0$. Let

$$B = \text{diag}(|\lambda_1|^{-1/2}, \ldots, |\lambda_r|^{-1/2}, 1 \ldots, 1).$$

Then $(PB)^t A(PB)$ is the diagonal matrix (2.14). □

Remark 2.21 From Corollary 2.20 it follows that $i_+(A)$ is the number of positive eigenvalues (counted with their multiplicities), $i_-(A)$ of negative eigenvalues of A (counted with their multiplicities) and $i_0(A)$ the multiplicity of the eigenvalue 0.

A quadratic form q on an n-dimensional $\mathbb{R}$-vector space is called:

1. *positive definite* if $\text{Sign}(q) = (n, 0)$, i.e. $q(v) > 0, \forall\, v \in V \setminus \{0_V\}$;
2. *negative definite* if $\text{Sign}(q) = (0, n)$, i.e. $q(v) < 0, \forall\, v \in V \setminus \{0_V\}$;
3. *positive semi-definite* if $\text{Sign}(q) = (r, 0)$, with $0 \leq r \leq n$, i.e. $q(v) \geq 0, \forall\, v \in V$;

4. *negative semi-definite* if $\mathrm{Sign}(q) = (0, r)$, with $0 \leq r \leq n$, i.e. $q(v) \leq 0$, $\forall\, v \in V$;
5. *indefinite* if $\mathrm{Sign}(q) = (p, r - p)$, with $0 < p < r \leq n$).

Examples 2.22

(1) An inner product on a real vector space is a positive definite quadratic form.
(2) An example of non-degenerate indefinite quadratic form (of signature (3, 1)) is

$$q(x_1, x_2, x_3, x_4) = x_1^2 + x_2^2 + x_3^2 - x_4^2.$$

The pair $(\mathbb{R}^4, q)$ is called the *Minkowski space*, which plays an important role in the theory of relativity (see Chapter 13 on Cayley– Klein geometries).

Remark 2.23 To determine the signature of a quadratic form on $\mathbb{R}$ one does not need to find the eigenvalues of the associated symmetric matrix (a very difficult task if the order of the matrix is ≥ 5). The following criterion is very useful (see [29]):

Proposition 2.24 (Descartes' Rule) *Let $f(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots a_1 X + a_0 \in \mathbb{R}[X]$ be a polynomial whose all roots are real and $\neq 0$. The number of the roots > 0 counted with their multiplicities is equal to the number of sign changes in the sequence $a_n, \ldots, a_0$ of non-zero coefficients of f.*

Lemma 2.25 *Let V be a real vector space of dimension n and $b\colon V \times V \to \mathbb{R}$ an indefinite bilinear symmetric form of signature $(p, n - p)$ with $p < n$. Let W be a subspace of V of dimension p such that $b(v, v) > 0$ for all $v \in W$. Then, for every $w \in W^{\perp_b}$, $w \neq 0$, we have*

$$b(w, w) < 0.$$

Similarly, if W is a subspace of V of dimension $n - p$ such that $b(v, v) < 0$ for all $v \in W$, then, for every $w \in W^{\perp_b}$, $w \neq 0$, we have

$$b(w, w) > 0.$$

Proof Suppose that there exists $z \in W^{\perp_b}$ such that $b(z, z) > 0$. Let $N = W \oplus \langle z \rangle$. Every element of N is of the form $w + \lambda z$ with $w \in W$, then, for every $\lambda \in \mathbb{R}$

$$b(w + \lambda z, w + \lambda z) = b(w, w) + \lambda^2 b(z, z) + 2\lambda b(w, z) = b(w, w) + \lambda^2 b(z, z) > 0.$$

Hence $\dim N > \dim W$, contradicting our assumption on W. □

Lemma 2.26 (Cauchy–Schwarz Inequalities) *Let V be a real vector space of dimension n and $b\colon V \times V \to \mathbb{R}$ a bilinear symmetric form.*

(a) *If b positive definite, then for all x and $y \in V$ we have*

$$b(x, y)^2 \leq b(x, x)b(y, y), \tag{2.15}$$

and $b(x, y)^2 = b(x, x)b(y, y)$ if and only if $x = 0_V$ or $y = 0_V$.

(b) *If b is non-degenerate of signature $(n-1, 1)$, $b(x, x) < 0$ and $b(y, y) < 0$ then*

$$b(x, y)^2 > b(x, x)b(y, y). \tag{2.16}$$

Proof

(a) A positive definite bilinear symmetric form is an inner product so that (a) is Theorem 1.147.

(b) The subspace $\langle x \rangle$ generated by x has dimension 1 which is maximum. Take the orthogonal complement $\langle x \rangle^{\perp_b}$, then

$$y = \lambda x + z, \qquad z \in \langle x \rangle^{\perp_b}.$$

By Lemma 2.25 we have $b(z, z) > 0$, hence

$$0 > b(y, y) = \lambda^2 b(x, x) + b(z, z).$$

Thus

$$\begin{aligned} b(x, y)^2 &= b(x, \lambda x + z)^2 = \lambda^2 b(x, x)^2 \\ &= \lambda^2 b(x, x)b(x, x) = [b(y, y) - b(z, z)]b(x, x) > b(y, y)b(x, x), \end{aligned}$$

which is our thesis.

□

Also for finite fields we have a canonical form.

Corollary 2.27 *Let $f \in \mathrm{Bil}(V)$ be a bilinear symmetric form on a K-vector space of dimension n where K is finite. Then there exists a basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V such that $M_f^{\mathcal{B}}$ is the diagonal matrix*

$$\begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & d & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix},$$

where $r := \mathrm{rank}(f)$ and d is unique up to multiplication by a square.

Proof See [32], pp. 290– 291. □

Lagrange's Method
We now show how to select a coordinate system $(\xi_1, \ldots, \xi_n)$ in which the quadratic form on the standard K-vector space K^n

$$q(x_1, \ldots, x_n) = \sum_{i,j=1}^{n} a_{ij} x_i x_j, \qquad a_{ij} = a_{ji}$$

is represented as a sum of squares

$$\lambda_1 \xi_1^2 + \lambda_2 \xi_2^2 + \cdots + \lambda_n \xi_n^2. \tag{2.17}$$

Case 1 There exists a diagonal entry $a_{ii} \neq 0$. By renumbering the variables we can assume that $a_{11} \neq 0$. Then we have

$$\begin{aligned} q(x_1, \ldots, x_n) &= a_{11} x_1^2 + x_1(2a_{12}x_2 + \cdots + 2a_{1n}x_n) + q_1(x_2, \ldots, x_n) \\ &= a_{11}\left(x_1 + \frac{a_{12}}{a_{11}}x_2 + \cdots + \frac{a_{1n}}{a_{11}}x_n\right)^2 + q_2(x_2, \ldots, x_n), \end{aligned} \tag{2.18}$$

where q_1 and q_2 are quadratic forms in $\leq n-1$ variables. By change of variables

$$y_1 = x_1 - \frac{1}{a_{11}}(a_{12}x_2 + \cdots + a_{1n}x_n), \quad y_i = x_i, \quad i = 2, \ldots, n,$$

we get

$$a_{11} y_1^2 + q_2(y_2, \ldots, y_n).$$

The next step is to apply this method to q_2.

Case 2 All diagonal entries a_{ii} are null. By a suitable change of the numbering of the variables we can assume that $a_{12} \neq 0$. Then

$$\begin{aligned} q(x_1, \ldots, x_n) = 2a_{12}x_1x_2 + x_1\ell_1(x_3, \ldots, x_n) + x_1\ell_2(x_3, \ldots, x_n) \\ + q_1(x_3, \ldots, x_n), \end{aligned} \tag{2.19}$$

where ℓ_1 and ℓ_2 are linear forms and q_1 is a quadratic form. Consider the coordinate transformation defined by

$$x_1 = y_1 + y_2, \quad x_2 = y_1 - y_2, , \quad x_i = y_i, \quad i = 3, \ldots, n.$$

Under this transformation we have

$$2a_{12}(y_1^2 - y_2^2) + q_2(y_1, \ldots, y_n),$$

where q_2 is a quadratic form not containing y_1^2 and y_2^2. Thus we are under the assumption of Case 1.

After a finite number of steps of the type just described our quadratic form will finally take the form (2.17).

Example 2.28 We shall now give two examples illustrating Lagrange's method.

(1) Consider the quadratic form $2x_1^2 + 3x_1x_2 + 4x_1x_3 + x_2^2 + x_3^2$. Then

$$\begin{aligned} 2x_1^2 + 3x_1x_2 + 4x_1x_3 + x_2^2 + x_3^2 &= 2x_1^2 + x_1(3x_2 + 4x_3) + x_2^2 + x_3^2 \\ &= 2\left(x_1 + \frac{3}{4}x_2 + x_3\right)^2 - \left(\frac{9}{8}x_2^2 + 3x_2x_3 + 2x_3^2\right) + x_2^2 + x_3^2 \\ &= 2\left(x_1 + \frac{3}{4}x_2 + x_3\right)^2 - \frac{1}{8}x_2^2 - 3x_2x_3 - x_3^2. \end{aligned}$$

Under the change of variables

$$y_1 = x_1 + \frac{3}{4}x_2 + x_3, \quad y_2 = x_2, \quad y_3 = x_3,$$

we obtain

$$2y_1^2 - \frac{1}{8}y_2^2 - 3y_2y_3 - y_3^2.$$

Iterating the above procedure we have

$$\frac{1}{8}y_2^2 + 3y_2y_3 + y_3^2 = \frac{1}{8}(y_2 + 12y_3)^2 - 17y_3^2.$$

If we put

$$z_1 = y_1, \quad z_2 = y_2 + 12y_3, \quad z_3 = y_3$$

then we get

$$2z_1^2 - \frac{1}{8}z_2^2 + 17z_3^2.$$

Thus the change of variables to diagonalize our quadratic form is

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{3}{4} & 8 \\ 0 & 1 & -12 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}.$$

(2) Consider the quadratic form $3x_1x_2+4x_1x_3-x_2x_3$. Under the change of variables

$$x_1 = y_1 + y_2, \quad x_2 = y_1 - y_2, \quad x_3 = y_3,$$

we obtain $3(y_1^2 - y_2^2) + 3y_1y_3 + 5y_2y_3$. As in Case 1 we have

$$3y_1^2 - 3y_2^2 + 3y_1y_3 + 5y_2y_3 = 3\left(y_1 + \frac{1}{2}y_3\right)^2 - 3\left(y_2 - \frac{5}{6}y_3\right)^2 + \frac{4}{3}y_3^2.$$

By the following change of variables

$$z_1 = y_1 + \frac{1}{2}y_3, \quad z_2 = y_2 - \frac{5}{6}y_3, \quad z_3 = y_3$$

we get

$$3z_1^2 - 3z_2^2 + \frac{4}{3}z_3^2.$$

Thus the change of variables to diagonalize our quadratic form is

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & \frac{1}{3} \\ 1 & -1 & -\frac{4}{3} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}.$$

2.2 Cross-Product

Definition 2.29 Let V be a $\mathbb{R}$-vector space of dimension $n \geq 1$. We shall say that two ordered bases S and T are *identically oriented* if $\det(\Lambda^{S,T}) > 0$. It is clear that the set of all oriented bases of V is divided precisely into two classes, consisting of identically oriented bases, while the bases of different classes are oriented differently. The choice of one of these classes is called *the orientation* of the space V.

If $V = \mathbb{R}^n$ is the standard vector space on $\mathbb{R}$ of dimension n we call a basis S *right-handed* or *positive* if identically oriented with the canonical basis; *left-handed* or *negative* in the opposite case.

Definition 2.30 Let V and W be two real vector spaces and $f\colon V^n \to W$ a n-linear map. The group $\mathfrak{S}_n$ of permutation of $\{1, \dots, n\}$ acts on the set $\mathrm{ML}(V^n, W)$ of n-linear maps in such a way that

$$\begin{aligned}\mathfrak{S}_n \times \mathrm{ML}(V^n, W) \to \mathrm{ML}(V^n, W), \quad (\sigma, f) = \sigma \cdot f(v_1, \dots, v_n)\\ := f(v_{\sigma(1)}, \dots, v_{\sigma(n)}).\end{aligned}$$

We shall say that f is *alternating* if for every n-tuple $(v_1, \dots, v_n) \in V^n$ and for every transposition $\tau \in \mathfrak{S}_n$ we have $\tau \cdot f = -f$. We recall that a transposition is a permutation that interchanges two elements and leaves all the others fixed. Since every permutation is a product of transpositions,

$$f(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = \mathrm{sgn}(\sigma) f(v_1, \dots, v_n), \qquad \forall\, \sigma \in \mathfrak{S}_n,$$

where

$$\mathrm{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is product of an even number of transposition} \\ -1 & \text{if } \sigma \text{ is product of an odd number of transposition.} \end{cases}$$

If $W = \mathbb{R}$ we call an alternating n-linear map an *alternating n-linear form*.

Proposition 2.31 *An n-linear map $f\colon V^n \to W$ is alternating if and only if*

$$v_i = v_j \quad \textit{for some} i \neq j \quad \Longrightarrow \quad f(v_1, \dots, v_n) = 0_W.$$

Proof If f is alternating then

$$\begin{aligned} f(v_1, \dots, v_i, \dots, v_j, \dots, v_n) &= -f(v_1, \dots, v_j, \dots, v_i, \dots, v_n) \\ &= -f(v_1, \dots, v_i, \dots, v_j, \dots, v_n), \end{aligned}$$

then $f(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = 0_W$. Conversely, assume the implication: if $v_i = v_j$ for some $i \neq j$ then $f(v_1, \dots, v_n) = 0_W$. Then if τ is any trasposition, $\tau(i) = j$ and $\tau(j) = i$, it follows that

$$\begin{aligned} 0_W &= f(v_1, \dots, v_i + v_j, \dots, v_i + v_j, \dots, v_n) \\ &= f(v_1, \dots, v_i, \dots, v_i + v_j, \dots, v_n) + f(v_1, \dots, v_j, \dots, v_i + v_j, \dots, v_n) \\ &\quad = f(v_1, \dots, v_i \dots, v_i, \dots, v_n) + f(v_1, \dots, v_i, \dots, v_j, \dots, v_n) \\ &\quad + f(v_1, \dots, v_j, \dots, v_i, \dots, v_n) + f(v_1, \dots, v_j, \dots, v_j, \dots, v_n) \end{aligned}$$

$$= f(v_1, \dots, v_i, \dots, v_j, \dots, v_n) + f(v_1, \dots, v_j, \dots, v_i, \dots, v_n),$$

i.e. $\tau \cdot f + f = 0_W$. □

Proposition 2.32 *Let $f\colon V^n \to W$ be an alternating n-linear map. If $v_1, \dots, v_n$ are linearly dependent then $f(v_1, \dots, v_n) = 0_W$.*

Proof Left to the reader as an exercise. □

Examples 2.33

(1) A fundamental example of an alternating n-linear map is the determinant (see [22], Cap. XIII, § 4). Let $\mathbb{R}^n$ be the space of column vectors. Define

$$f\colon \underbrace{\mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{n \text{ times}} \to \mathbb{R}, \quad f(v_1, \dots, v_n) := \det(A),$$

where A is the $n \times n$-matrix whose i-th column is v_i, $i = 1, \dots, n$.

(2) The map $f : \mathcal{M}_n(\mathbb{R}) \times \mathcal{M}_n(\mathbb{R}) \to \mathcal{M}_n(\mathbb{R})$ defined by

$$f(A, B) = AB - BA, \qquad \forall\, A, B \in \mathcal{M}_n(\mathbb{R}),$$

is an alternating bilinear map.

Definition 2.34 Let $(V, \langle \cdot, \cdot \rangle)$ be a euclidean vector space of dimension n and $\mathcal{B} = \{e_1, \dots, e_n\}$ an orthonormal basis of V. Let $\mathbf{v} = (v_1, \dots, v_n) \in V^n$. By Theorem 1.146 we have

$$v_j = \sum_{i=1}^{n} \langle e_i, v_j \rangle e_i, \qquad j = 1, \dots, n.$$

Let $\mathbf{v}\colon V \to V$ be the linear operator such that $\mathbf{v}(e_j) = v_j$, $j = 1, \dots, n$. Then

$$M_{\mathbf{v}}^{\mathcal{B}} = \big(\langle e_i, v_j \rangle\big)_{i,j=1,\dots,n}$$

The map

$$\lambda_{\mathcal{B}}\colon V^n \to \mathbb{R}, \qquad \lambda_{\mathcal{B}}(v_1, \dots, v_n) := \det(M_{\mathbf{v}}^{\mathcal{B}}) = \begin{vmatrix} \langle e_1, v_1 \rangle & \langle e_1, v_2 \rangle & \dots & \langle e_1, v_n \rangle \\ \langle e_2, v_1 \rangle & \langle e_2, v_2 \rangle & \dots & \langle e_2, v_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle e_n, v_1 \rangle & \langle e_n, v_2 \rangle & \dots & \langle e_n, v_n \rangle \end{vmatrix}$$

is called a *volume form associated with the on-basis* $\mathcal{B}$. In particular, if $\mathcal{B}' = (v_1, \dots, v_n)$ is an ordered basis of V, then $\lambda_{\mathcal{B}}(v_1, \dots, v_n) = \det\left(\Lambda^{\mathcal{B},\mathcal{B}'}\right)$.

Proposition 2.35 *Assume the notation of* Definition 2.34. *The volume form* $\lambda_{\mathcal{B}}$ *satisfies the following properties:*

(a) *If* $\mathcal{B}' = \{f_1, \ldots, f_n\}$ *is another orthonormal basis of* V *then*

$$\lambda_{\mathcal{B}}(f_1, \ldots, f_n) = \begin{cases} 1 & \text{if } \mathcal{B}' \text{ and } \mathcal{B} \text{ are identically oriented} \\ -1 & \text{otherwise.} \end{cases}$$

(b) $\lambda_{\mathcal{B}}$ *does not depend on* $\mathcal{B}$ *but only on the orientation class of* $\mathcal{B}$.
(c) $\lambda_{\mathcal{B}}$ *is an alternating* n*-linear form.*
(d) $\lambda_{\mathcal{B}}(v_1, \ldots, v_n) = 0$ *if and only if* $v_1, \ldots, v_n$ *are linearly dependent.*

Proof

(a) The matrix $\Lambda^{\mathcal{B},\mathcal{B}'} = M_{\mathrm{id}_V}^{\mathcal{B}',\mathcal{B}}$ is orthogonal so that $\lambda_{\mathcal{B}}(f_1, \ldots, f_n) = \det\left(\Lambda^{\mathcal{B},\mathcal{B}'}\right) = \pm 1$.

(b) Let $\mathcal{B}' = \{f_1, \ldots, f_n\}$ be another on-basis of V and $\mathbf{v}' \colon V \to V$ be the linear operator such that $\mathbf{v}'(f_j) = v_j$, $j = 1, \ldots, n$. As above,

$$v_j = \sum_{i=1}^{n} \langle f_i, v_j \rangle f_i, \qquad j = 1, \ldots, n,$$

and

$$M_{\mathbf{v}}^{\mathcal{B}} = \left(\langle f_i, v_j \rangle\right)_{i,j=1,\ldots,n}.$$

From (1.28) we have

$$M_{\mathbf{v}}^{\mathcal{B}} = \Lambda^{\mathcal{B},\mathcal{B}'} M_{\mathbf{v}'}^{\mathcal{B}'},$$

so that

$$\lambda_{\mathcal{B}}(v_1, \ldots, v_n) = \det(\Lambda^{\mathcal{B},\mathcal{B}'}) \lambda_{\mathcal{B}'}(v_1, \ldots, v_n),$$

which implies our thesis by (a).

(c) Indeed, $\lambda_{\mathcal{B}}$ is a determinant.

(d) It follows from the equivalences that

$$\sum_{j=1}^{n} \alpha_j v_j = 0_V \iff \langle e_i, \sum_{j=1}^{n} \alpha_j v_j \rangle = 0,\ i = 1, \ldots, n,$$

$$\iff \sum_{j=1}^{n} \alpha_j \langle e_i, v_j \rangle = 0,\ i = 1, \ldots, n$$

$$\iff \sum_{j=1}^{n} \alpha_j \left(\langle e_1, v_j \rangle, \langle e_2, v_j \rangle, \ldots, \langle e_n, v_j \rangle \right) = 0_V,$$

by a basic property of the determinant.

□

Definition 2.36 Let $v_1, \ldots, v_p$ be vectors of a euclidean vector space $(V, \langle \cdot, \cdot \rangle)$. The *Gram determinant* of $v_1, \ldots, v_p$ is defined by

$$\mathrm{Gram}(v_1, \ldots, v_p) := \det((\langle v_i, v_j \rangle)_{i,j}) = \begin{vmatrix} \|v_1\|^2 & \langle v_2, v_1 \rangle & \ldots & \langle v_p, v_1 \rangle \\ \langle v_1, v_2 \rangle & \|v_2\|^2 & \ldots & \langle v_p, v_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle v_1, v_p \rangle & \langle v_2, v_p \rangle & \ldots & \|v_p\|^2 \end{vmatrix}. \tag{2.20}$$

From the basic properties of the inner product and of the determinant we deduce the identity

$$\mathrm{Gram}(v_1, \ldots, v_p) = \mathrm{Gram}\left(v_1, \ldots, v_i + \sum_{j \neq i} \alpha_j v_j, \ldots, v_p \right), \quad \alpha_j \in \mathbb{R}, \tag{2.21}$$

so that if $v_1, \ldots, v_p$ are linearly dependent, $\mathrm{Gram}(v_1, \ldots, v_p) = 0$.

If $v_1, \ldots, v_n$ is an orthonormal basis of V, then

$$\mathrm{Gram}(v_1, \ldots, v_n) = 1.$$

Let $\mathcal{B} = \{e_1, \ldots, e_n\}$ be an orthonormal basis of V. Let $v_1, \ldots, v_n \in V$. By Parseval's identity (Theorem 1.146)

$$\langle v_i, v_k \rangle = \sum_{j=1}^{n} \langle e_j, v_i \rangle \langle e_j, v_k \rangle, \qquad i, k = 1, \ldots, n,$$

we have

$$\begin{pmatrix} \|v_1\|^2 & \langle v_2, v_1\rangle & \dots & \langle v_n, v_1\rangle \\ \langle v_1, v_2\rangle & \|v_2\|^2 & \dots & \langle v_n, v_2\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle v_1, v_n\rangle & \langle v_2, v_n\rangle & \dots & \|v_n\|^2 \end{pmatrix}$$

$$= \begin{pmatrix} \langle e_1, v_1\rangle & \langle e_1, v_2\rangle & \dots & \langle e_1, v_n\rangle \\ \langle e_2, v_1\rangle & \langle e_2, v_2\rangle & \dots & \langle e_2, v_n\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle e_n, v_1\rangle & \langle e_n, v_2\rangle & \dots & \langle e_n, v_n\rangle \end{pmatrix}^t \begin{pmatrix} \langle e_1, v_1\rangle & \langle e_1, v_2\rangle & \dots & \langle e_1, v_n\rangle \\ \langle e_2, v_1\rangle & \langle e_2, v_2\rangle & \dots & \langle e_2, v_n\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle e_n, v_1\rangle & \langle e_n, v_2\rangle & \dots & \langle e_n, v_n\rangle \end{pmatrix}.$$

Hence

$$\mathrm{Gram}(v_1, \dots, v_n) = (\lambda_B(v_1, \dots, v_n))^2. \tag{2.22}$$

Definition 2.37 Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension $n \geq 3$ and $\mathcal{B}$ an on-basis of V. Fixing $n-1$ vectors $v_1, \dots, v_{n-1} \in V$ there exists one and only one $w \in V$ such that

$$\langle w, u\rangle = \lambda_{\mathcal{B}}(v_1, \dots, v_{n-1}, u), \qquad \forall\, u \in V. \tag{2.23}$$

Indeed, the map $V \to V^*$, $w \to w^*$, where $w^*(u) = \langle w, u\rangle$ for every $u \in V$, is an isomorphism (see Proposition 2.7). We call the vector w the *cross-product* of $v_1, \dots, v_{n-1}$ and we denote it by

$$w := v_1 \times v_2 \times \cdots \times v_{n-1}.$$

The cross-product depends only on the orientation class of $\mathcal{B}$.

The following proposition collects the basic properties of the cross-product.

Proposition 2.38 *Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension $n \geq 3$ and $\mathcal{B}$ an on-basis of V.*

(i) *The map*

$$V^{n-1} \to V, \qquad (v_1, \dots, v_{n-1}) \to v_1 \times v_2 \times \cdots \times v_{n-1}$$

is alternating multilinear.

(ii) $v_1 \times v_2 \times \cdots \times v_{n-1} = 0$ *if and only if* $v_1, \dots, v_{n-1}$ *are linearly dependent.*

(iii) $v_1 \times v_2 \times \cdots \times v_{n-1} \in W^{\perp}$, *where* W *is the subspace generated by* $v_1, \dots, v_{n-1}$.

(iv) $\|v_1 \times v_2 \times \cdots \times v_{n-1}\|^2 = \mathrm{Gram}(v_1, \dots, v_{n-1})$.

(v) *If* $v_1, \dots, v_{n-1}$ *are linearly independent, then* $\mathcal{B}' = (v_1, \dots, v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1})$ *is a basis of* V *identically oriented with* $\mathcal{B}$.

Proof

(i) It comes from the fact that $\lambda_{\mathcal{B}}$ is alternating multilinear.

(ii) The implication $\Rightarrow$ is immediate by point (d) of Proposition 2.35. Suppose now that $v_1, \ldots, v_{n-1}$ are linearly independent. Add a vector v_n so that $\{v_1, \ldots, v_{n-1}, v_n\}$ is a basis of V. Then

$$\langle v_1 \times v_2 \times \cdots \times v_{n-1}, v_n \rangle = \lambda_B(v_1, \ldots, v_{n-1}, v_n) \neq 0,$$

from point (d) of Proposition 2.35, hence $v_1 \times v_2 \times \cdots \times v_{n-1} \neq 0$. Again from point (d) of Proposition 2.35 we have

$$\langle v_1 \times v_2 \times \cdots \times v_{n-1}, v_i \rangle = \lambda_B(v_1, \ldots, v_{n-1}, v_i) = 0, \quad i = 1, \ldots, n-1, \tag{2.24}$$

which implies point (iii). By (iii) and Definition 2.36 we have

$$\begin{aligned}
&\mathrm{Gram}(v_1, \ldots, v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1}) \\
&= \begin{vmatrix}
\|v_1\|^2 & \langle v_2, v_1 \rangle & \ldots & \langle v_1 \times v_2 \times \cdots \times v_{n-1}, v_1 \rangle \\
\langle v_1, v_2 \rangle & \|v_2\|^2 & \ldots & \langle v_1 \times v_2 \times \cdots \times v_{n-1}, v_2 \rangle \\
\vdots & \vdots & \ddots & \vdots \\
\langle v_1, v_1 \times v_2 \times \cdots \times v_{n-1} \rangle & \langle v_2, v_1 \times v_2 \times \cdots \times v_{n-1} \rangle & \ldots & \|v_1 \times v_2 \times \cdots \times v_{n-1}\|^2
\end{vmatrix} \\
&= \begin{vmatrix}
\|v_1\|^2 & \langle v_2, v_1 \rangle & \ldots & 0 \\
\langle v_1, v_2 \rangle & \|v_2\|^2 & \ldots & 0 \\
\vdots & \vdots & \ddots & \vdots \\
0 & 0 & \ldots & \|v_1 \times v_2 \times \cdots \times v_{n-1}\|^2
\end{vmatrix} \\
&= \mathrm{Gram}(v_1, \ldots, v_{n-1}) \, \|v_1 \times v_2 \times \cdots \times v_{n-1}\|^2.
\end{aligned} \tag{2.25}$$

From (2.22) and (2.25) it follows that

$$\begin{aligned}
\|v_1 \times v_2 \times \cdots \times v_{n-1}\|^4 &= \langle v_1 \times v_2 \times \cdots \times v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1} \rangle^2 \\
&= (\lambda_B(v_1, \ldots, v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1}))^2 \\
&= \mathrm{Gram}(v_1, \ldots, v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1}) \\
&= \mathrm{Gram}(v_1, \ldots, v_{n-1}) \, \|v_1 \times v_2 \times \cdots \times v_{n-1}\|^2,
\end{aligned}$$

which implies our thesis if $\|v_1 \times v_2 \times \cdots \times v_{n-1}\| \neq 0$. If $\|v_1 \times v_2 \times \cdots \times v_{n-1}\| = 0$, $v_1, \ldots, v_{n-1}$ are linearly dependent by ii) so that $\mathrm{Gram}(v_1, \ldots, v_{n-1}) = 0$ and iv) holds.

(v) By (iii) $\mathcal{B}'$ is a basis. Furthermore

$$\begin{aligned}\lambda_{\mathcal{B}}((v_1, \ldots, v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1}) \\ = \langle v_1 \times v_2 \times \cdots \times v_{n-1}, v_1 \times v_2 \times \cdots \times v_{n-1}\rangle > 0\end{aligned}$$

so that $\mathcal{B}$ and $\mathcal{B}'$ are identically oriented.

□

Now we find the components of $v_1 \times v_2 \times \cdots \times v_{n-1}$ with respect to a frame of V.

Proposition 2.39 *Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension n and $\mathcal{B} = \{e_1, \ldots, e_n\}$ an on-basis of V. Let $v_1, \ldots, v_{n-1} \in V$:*

$$v_j = \sum_{i=1}^{n} \langle e_i, v_j\rangle e_i, \qquad j = 1, \ldots, n-1.$$

Then

$$v_1 \times v_2 \times \cdots \times v_{n-1} = \sum_{k=1}^{n} (-1)^{n+k} \det(A_k)\, e_k,$$

where $A = (\langle e_i, v_j\rangle)_{\substack{i=1,\ldots,n \\ j=1,\ldots,n-1}}$ and A_k is the $(n-1)\times(n-1)$-matrix obtained from A by deleting the k-th row of A.

Proof Let

$$v_1 \times v_2 \times \cdots \times v_{n-1} = \sum_{k=1}^{n} \langle e_k, v_1 \times v_2 \times \cdots \times v_{n-1}\rangle e_k.$$

Then we have

$$\langle e_k, v_1 \times v_2 \times \cdots \times v_{n-1}\rangle = \lambda_B(e_k, v_1, \ldots, v_{n-1}) = \begin{vmatrix} 0 & \langle e_1, v_1\rangle & \ldots & \langle e_1, v_{n-1}\rangle \\ 0 & \langle e_2, v_1\rangle & \ldots & \langle e_2, v_{n-1}\rangle \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \langle e_k, v_1\rangle & \ldots & \langle e_k, v_{n-1}\rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \langle e_n, v_1\rangle & \ldots & \langle e_n, v_{n-1}\rangle \end{vmatrix}$$

Our thesis follows by Laplace expansion of the determinant along column 1. □

The cross-product is not generally invariant by orthogonal automorphisms, indeed we have the following result.

Proposition 2.40 *Let $(V, \langle\cdot,\cdot\rangle)$ be a euclidean vector space of dimension n and $\varphi\colon (V, \langle\cdot,\cdot\rangle) \to (V, \langle\cdot,\cdot\rangle)$ a linear automorphism. Then, for every $(n-1)$-tuple $(v_1, \ldots, v_{n-1}) \in V^{n-1}$ we have*

$$\varphi(v_1) \times \cdots \times \varphi(v_{n-1}) = \det(\varphi)\,(\varphi^*)^{-1}(v_1 \times v_2 \times \cdots \times v_{n-1}). \tag{2.26}$$

where φ^* is the adjoint of φ.

Proof We fix an on-basis $\mathcal{B} = (e_1, \ldots, e_n)$ of V. We recall that $\det(\varphi) = \det(M_\varphi^{\mathcal{B}})$ and $M_{\varphi^*}^{\mathcal{B}}) = (M_\varphi^{\mathcal{B}})^t$. Let u be any vector of V. From Definition 1.176 we have

$$\langle \varphi(v_1) \times \cdots \times \varphi(v_{n-1}), \varphi(u)\rangle = \langle \varphi^*(\varphi(v_1) \times \cdots \times \varphi(v_{n-1})), u\rangle. \tag{2.27}$$

On the other hand, we have

$$\langle \varphi(v_1) \times \cdots \times \varphi(v_{n-1}), \varphi(u)\rangle = \lambda_{\mathcal{B}}(\varphi(v_1), \ldots, \varphi(v_{n-1}, \varphi(u)).$$

Now the columns of the matrix

$$M = \begin{pmatrix} \langle e_1, \varphi(v_1)\rangle & \langle e_1, \varphi(v_2)\rangle & \ldots & \langle e_1, \varphi(v_{n-1})\rangle & \langle e_1, \varphi(u)\rangle \\ \langle e_2, \varphi(v_1)\rangle & \langle e_2, \varphi(v_2)\rangle & \ldots & \langle e_2, \varphi(v_{n-1})\rangle & \langle e_2, \varphi(u)\rangle \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \langle e_n, \varphi(v_1)\rangle & \langle e_n, \varphi(v_2)\rangle & \ldots & \langle e_n, \varphi(v_{n-1})\rangle & \langle e_n, \varphi(u)\rangle \end{pmatrix}$$

are the coordinates of $(\varphi(v_1), \ldots, \varphi(v_{n-1}, \varphi(u))$ with respect to $\mathcal{B}$ so that

$$M = M_\varphi^{\mathcal{B}} N,$$

where N is the matrix whose columns are the coordinates of $(v_1, \ldots, v_{n-1}, u)$ with respect to $\mathcal{B}$. Hence

$$\begin{aligned} \lambda_{\mathcal{B}}(\varphi(v_1), \ldots, \varphi(v_{n-1}, \varphi(u)) &= \det(M) = \det(M_\varphi^{\mathcal{B}})\det(N) \\ &= \det(M_\varphi^{\mathcal{B}})\lambda_B(v_1, \ldots, v_{n-1}, u), \end{aligned}$$

i.e.

$$\langle \varphi(v_1) \times \cdots \times \varphi(v_{n-1}), \varphi(u)\rangle = \det(M_\varphi^{\mathcal{B}})\langle v_1 \times v_2 \times \cdots \times v_{n-1}, u\rangle.$$

From (2.27) we get for all $u \in V$

$$\langle \varphi^*(\varphi(v_1) \times \cdots \times \varphi(v_{n-1})), u\rangle = \det(M_\varphi^{\mathcal{B}})\langle v_1 \times v_2 \times \cdots \times v_{n-1}, u\rangle,$$

which implies

$$\varphi^*(\varphi(v_1) \times \cdots \times \varphi(v_{n-1})) = \det(M_\varphi^{\mathcal{B}})\, v_1 \times v_2 \times \cdots \times v_{n-1},$$

i.e. (2.26) since φ^* is an automorphism. □

2.3 Exercises

Exercise 2.1 Prove that the map

$$f : \mathcal{M}_{m,n}(\mathbb{R}) \times \mathcal{M}_{m,n}(\mathbb{R}) \to \mathbb{R}, \qquad f(A, B) = \operatorname{tr}(A^t B), \quad \forall\, A, B \in \mathcal{M}_{m,n}(\mathbb{R}),$$

is symmetric bilinear.

Exercise 2.2 Prove that the maps

$$f : \mathcal{M}_n(\mathbb{R}) \times \mathcal{M}_n(\mathbb{R}) \to \mathbb{R}, \quad f(A, B) = \operatorname{tr}(AB), \quad \forall\, A, B \in \mathcal{M}_n(\mathbb{R}),$$
$$g : \mathcal{M}_n(\mathbb{R}) \times \mathcal{M}_n(\mathbb{R}) \to \mathbb{R}, \quad g(A, B) = \det(AB), \quad \forall\, A, B \in \mathcal{M}_n(\mathbb{R}),$$

are symmetric bilinear.

(a) Determine $\mathcal{S}_n(\mathbb{R})^\perp$, where $\mathcal{S}_n(\mathbb{R})$ is the subspace of symmetric matrices.
(b) Find the signatures of f and g.

Exercise 2.3 Let $V = \{f \in C^1([a, b]; \mathbb{R}) : f(a) = f(b) = 0\}$. Show that

$$b(f, g) = \int_a^b f g'\, dt, \qquad \forall\, f, g \in V,$$

is a skew-symmetric bilinear form on V.

Exercise 2.4 Let $f \in \mathrm{Bil}(V)$ be a symmetric bilinear form on a n-dimensional $\mathbb{R}$-vector space V. If $b(v, v) \neq 0$ for every $v \in V \setminus \{0_V\}$, show that b is positive definite or negative definite.

Exercise 2.5 Let $f \in \mathrm{Bil}(V)$ be a bilinear form on a n-dimensional $\mathbb{R}$-vector space V. Prove or disprove the following statement:

There exist two linear functionals $f, g \in V^*$ such that $b(v, w) = f(v)g(w), \forall\, v, w \in V$.

Exercise 2.6 Let V be a n-dimensional $\mathbb{R}$-vector space and $v_1, \ldots, v_k$ vectors of V. Show that

$$b(f, g) = \sum_{i=1}^{k} f(v_i) g(v_i), \qquad \forall\, f, g \in V^*,$$

is a symmetric bilinear form on the dual space V^* and determine positivity, negativity and nullity indexes of b.

Exercise 2.7 Determine the signature of the quadratic form on $\mathbb{R}^4$

$$(2+c)x^2 + xy + xz + xt + (2+c)y^2 + yz + yt + (2+c)z^2 + zt + (2+c)t^2$$

when c varies in $\mathbb{R}$. Find the endomorphism which gives the canonical form for $c = 0$.

Exercise 2.8 Diagonalize the following quadratic forms on $\mathbb{Q}$:

$$x^2 + 4xy + 4y^2 + 2xz + z^2 + 2yz, \qquad x^2 + 2xy + 4xz + 3y^2 + yz + 7z^2,$$
$$2x^2 + 3xy + 6y^2, \qquad 8xy + 4y^2.$$

Exercise 2.9 Diagonalize the quadratic form $2x^2 + xy + 3z^2$ on $\mathbb{Z}_5$.

Exercise 2.10 Determine the signature of the quadratic form on $\mathbb{R}$

$$q(x_1, x_2, x_3) = x_2^2 - 3x_3^2 + 2x_1x_2 + 6x_1x_3 + 2x_2x_3.$$

Diagonalize $q(x_1, x_2, x_3)$ on $\mathbb{R}$ and on $\mathbb{Q}$.

Exercise 2.11 (Cross-Product in $(\mathbb{R}^3, \langle\cdot,\cdot\rangle_s)$) Let $v = (v_1, v_2, v_3)$, $w = (w_1, w_2, w_3) \in \mathbb{R}^3$. The cross-product $v \times w \in \mathbb{R}^3$ is defined by the formal determinant (according to Proposition 2.39)

$$\begin{vmatrix} e_1 & e_2 & e_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} e_1 - \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} e_2 + \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} e_3 = \tag{2.28}$$

$$= (v_2w_3 - v_3w_2, v_3w_1 - v_1w_3, v_1w_2 - v_2w_1),$$

with (e_1, e_2, e_3) the canonical basis (or any on-basis) of $\mathbb{R}^3$.

One can verify the following properties from (2.28):

(i) $e_1 \times e_2 = e_3$, $e_2 \times e_3 = e_1$, $e_3 \times e_1 = e_2$,
(ii) $v \times w = -w \times v$, $v \times (w_1 + w_2) = v \times w_1 + v \times w_2$, $\forall\, v, w, w_1, w_2 \in \mathbb{R}^3$,
(iii) $(av) \times w = v \times (aw) = a(v \times w)$, $\forall\, v, w \in \mathbb{R}^3$, $\forall a \in \mathbb{R}$,
(iv) $u \times (v \times w) = \langle u, w\rangle v - \langle u, v\rangle w$, $\forall\, u, v, w \in \mathbb{R}^3$,
(v)

$$\langle u, v \times w\rangle = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}, \qquad \forall\, u = (u_1, u_2, u_3) \in \mathbb{R}^3.$$

In particular, $\langle v, v \times w\rangle = \langle w, v \times w\rangle = 0$, i.e. $v \times w$ is orthogonal to both v and w.

Exercise 2.12 Show that

$$v \times w = (xy' - yx')\, u_1 \times u_2 - (xz' - zx')\, u_1 \times u_3 + (yz' - zy')\, u_2 \times u_3,$$

if $v, w \in \mathbb{R}^3$ have coordinates (x, y, z) and (x', y', z') respectively, with respect to an arbitrary basis.

Exercise 2.13 Verify *Lagrange's identity*

$$\|v \times w\|^2 = \|v\|^2 \cdot \|w\|^2 - \langle v, w\rangle^2, \qquad \forall\, v, w \in \mathbb{R}^3.$$

Exercise 2.14 Let $\theta = \sphericalangle(v, w)$, where v and w are two non-zero vectors of $\mathbb{R}^3$ (see Definition 1.149). Prove the identity

$$\|v \times w\| = \|v\| \cdot \|w\| \sin\theta.$$

In particular the area A of the parallelogram with u, v and w as adjacent sides is given by formula $A = \|v \times w\|$.

Exercise 2.15 Let the vectors u, v and w be the edges from the vertex O of a parallelepiped $\mathcal{P}$ and of a tetrahedron $\mathcal{T}$. Prove that the volumes $V_{\mathcal{P}}$ of $\mathcal{P}$ and $V_{\mathcal{T}}$ of $\mathcal{T}$ are given by formulas $V_{\mathcal{P}} = |\langle u, v \times w\rangle|$ and $V_{\mathcal{T}} = \dfrac{1}{6}|\langle u, v \times w\rangle|$.

Exercise 2.16 Prove *Jacobi's identity*

$$(u \times v) \times w + (v \times w) \times u + (w \times u) \times v = 0_{\mathbb{R}^3}, \qquad \forall\, u, v, w \in \mathbb{R}^3.$$

Chapter 3
Affine Spaces



3.1 Affine Spaces

Definition 3.1 An *affine space* over a field K is a triple $(\mathcal{A}, V, \varphi)$, consisting of a non-empty set $\mathcal{A}$ whose elements are called *points*, a K-vector space V of finite dimension called *director space* or *orienting space*, and a map

$$\varphi : \mathcal{A} \times \mathcal{A} \to V, \qquad (A, B) \to \overrightarrow{AB} := \varphi(A, B), \qquad \forall\, A, B \in \mathcal{A},$$

satisfying the following axioms:

(AFF1) For any three points $A, B, C \in \mathcal{A}$, $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$.
(AFF2) There exists $O \in \mathcal{A}$ such that the map

$$\varphi_O : \mathcal{A} \to V, \qquad \varphi_O(A) := \overrightarrow{OA}, \qquad \forall\, A \in \mathcal{A}, \tag{3.1}$$

is bijective.

The integer $\dim_K \mathcal{A} := \dim_K(V)$ is the *dimension* of $(\mathcal{A}, V, \varphi)$.

Proposition 3.2 *Let $(\mathcal{A}, V, \varphi)$ be an affine space.*

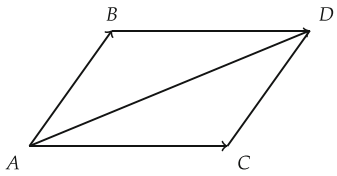
(a) *For every $A, B \in \mathcal{A}$, $\overrightarrow{AA} = 0_V$ and $\overrightarrow{AB} = -\overrightarrow{BA}$.*
(b) *For every point $O' \in \mathcal{A}$, the map $\varphi_{O'} : \mathcal{A} \to V$, defined by $\varphi_{O'}(A) = \overrightarrow{O'A}$, $\forall\, A \in \mathcal{A}$, is bijective.*

Proof (a) follows immediately from axiom (AFF1). From axiom (AFF1) we have $\overrightarrow{O'A} = \overrightarrow{O'O} + \overrightarrow{OA}$, so that $\varphi_{O'} = \psi \circ \varphi_O$ is bijective by axiom AFF2), since $\psi : V \to V$, defined by $\psi(v) = v + \overrightarrow{O'O}$, is bijective. □

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Fig. 3.1 Parallelogram rule



By axiom (AFF2) $\mathcal{A}$ equipped with the operations

$$A + B = C, \qquad \text{where } C \text{ is defined by } \quad \overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}, \quad \forall\, A, B \in \mathcal{A},$$

$$\lambda\, A = B, \qquad \text{where } B \text{ is defined by } \quad \lambda \overrightarrow{OA} = \overrightarrow{OB}, \qquad \forall\, \lambda \in K,\ \forall\, A \in \mathcal{A}.$$

is a K-vector space (whose null vector is the point O) such that the map $\varphi_O : \mathcal{A} \to V$ is an isomorphism of K-vector spaces. This vector space will be denoted by $\overrightarrow{O\mathcal{A}}$. Thanks to (b) of Proposition 3.2 the point O can be arbitrarily chosen.

Lemma 3.3 (Parallelogram Rule) *Let $(\mathcal{A}, V, \varphi)$ be an affine space. Let A, B, C, $D \in \mathcal{A}$ be four points such that $\overrightarrow{AB} = \overrightarrow{CD}$, then $\overrightarrow{AC} = \overrightarrow{BD}$.*

Proof By axiom AFF1) we have $\overrightarrow{AC} + \overrightarrow{CD} = \overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BD}$, so that our assumption $\overrightarrow{AB} = \overrightarrow{CD}$ implies $\overrightarrow{AC} = \overrightarrow{BD}$ (see Fig. 3.1). □

Example 3.4 A basic example of affine space is the *affine space associated with a vector space*. Let V be a K-vector space of finite dimension. Let $\mathcal{A}$ be the set V and $\varphi\colon \mathcal{A} \times \mathcal{A} \to V$ be defined by $\varphi(u, v) = v - u$, where u and v on the left– hand side are two points of set V, while $v - u$ on the right– hand side is the difference between the vectors v and u. It easy to see that $\mathbb{A}(V) = (\mathcal{A} = V, V, \varphi)$ is an affine space which is called the *affine space associated with the vector space V* and that $\dim_K \mathbb{A}(V) = \dim_K V$. If $V = K^n$, with $n \geq 1$, then the affine space $\mathbb{A}(V)$ will be denoted by $\mathbb{A}^n(K)$ and called the *standard affine space of dimension n on K*.

Definition 3.5 Let $A_1, \ldots, A_n$ be any points in an affine space $(\mathcal{A}, V, \varphi)$. For any $a_1, \ldots, a_n \in K$ with the condition $a_1 + \cdots + a_n = 1$ we define the *weighted barycentre* of n-tuple $(A_1, \ldots, A_n)$ with *weights* (or *masses*) $(a_1, \ldots, a_n)$ as the unique point of $\mathcal{A}$

$$A := a_1 A_1 + \cdots + a_n A_n$$

such that

$$\overrightarrow{OA} = a_1 \overrightarrow{OA_1} + \cdots + a_n \overrightarrow{OA_n}$$

for some point O of $\mathcal{A}$. The existence of O and the uniqueness of A follows from axiom AFF2).

As a matter of fact, the weighted barycentre $A \in \mathcal{A}$ does not depend on the choice of point $O \in \mathcal{A}$ but only on the points $A_1, \ldots, A_n$ and the scalars $a_1, \ldots, a_n$ (satisfying $a_1 + \cdots + a_n = 1$). Indeed, let O' any other point of $\mathcal{A}$, then we have to prove that $\overrightarrow{O'A} = a_1\overrightarrow{O'A_1} + \cdots + a_n\overrightarrow{O'A_n}$. By axiom AFF1) of Definition 3.1 we have $\overrightarrow{O'B} = \overrightarrow{O'O} + \overrightarrow{OB}, \forall\, B \in \mathcal{A}$. Then

$$\begin{aligned}\overrightarrow{O'A} = \overrightarrow{O'O} + \overrightarrow{OA} &= \overrightarrow{O'O} + a_1\overrightarrow{OA_1} + \cdots + a_n\overrightarrow{OA_n} \\ &= (a_1 + \cdots + a_n)\overrightarrow{O'O} + a_1\overrightarrow{OA_1} + \cdots + a_n\overrightarrow{OA_n} \\ &= a_1(\overrightarrow{O'O} + \overrightarrow{OA_1}) + a_2(\overrightarrow{O'O} + \overrightarrow{OA_2}) + \cdots + a_n(\overrightarrow{O'O} + \overrightarrow{OA_n}) \\ &= a_1\overrightarrow{O'A_1} + \cdots + a_n\overrightarrow{O'A_n}.\end{aligned}$$

In the affine space $\mathbb{A}^n(\mathbb{R})$ if we assign a mass a_i to A_i, $i = 1, \ldots, n$, then A represents the *physical barycentre* of the ordered system $\{A_1, \ldots, A_n, a_1, \ldots, a_n\}$.

If $\mathrm{char}(K) \nmid n$ and $a_1 = \cdots = a_n = \dfrac{1}{n}$, then the point $A = \dfrac{1}{n}A_1 + \cdots + \dfrac{1}{n}A_n$ is simply called the *barycentre* of the system of points $\{A_1, \ldots, A_n\}$; if $n = 2$, $A = \dfrac{1}{2}A_1 + \dfrac{1}{2}A_2$ is called the *midpoint* of the pair $\{A_1, A_2\}$, and $B := (-1)A_1 + 2A_2$ is the *symmetric* of A_1 with respect to A_2.

Definition 3.6 Let $a_1, \ldots, a_n \in K$ be n scalars such that $a_1 + \cdots + a_n = 0$. Then the vector $a_1\overrightarrow{OA_1} + \cdots + a_n\overrightarrow{OA_n}$ does not depend on the point O for any points $A_1, \ldots, A_n \in \mathcal{A}$. Indeed, we have

$$\begin{aligned}a_1\overrightarrow{O'A_1} + \cdots + a_n\overrightarrow{O'A_n} &= (a_1 + \cdots + a_n)\overrightarrow{O'O} + a_1\overrightarrow{OA_1} + \cdots + a_n\overrightarrow{OA_n} \\ &= a_1\overrightarrow{OA_n} + \cdots + a_n\overrightarrow{OA_n}.\end{aligned}$$

We write $a_1A_1 + \cdots + a_nA_n = \bigcirc$ if there exists a point $O \in \mathcal{A}$ such that $a_1\overrightarrow{OA_1} + \cdots + a_n\overrightarrow{OA_n} = 0_V$.

Remark 3.7 Let $(\mathcal{A}, V, \varphi)$ be an affine space over K. Let $O \in \mathcal{A}$ and $\varphi_O : \mathcal{A} \to V$ as in (3.1). From Definition 3.5 we have

$$\varphi_O(a_1A_1 + \cdots + a_nA_n) = a_1\varphi_O(A_1) + \cdots + a_n\varphi_O(A_n),$$

$\forall\, O, A_1, \ldots, A_n \in \mathcal{A}$ and $\forall\, a_1, \ldots, a_n \in K$ such that $a_1 + \cdots + a_n = 1$.

Definition 3.8 Let $(\mathcal{A}, V, \varphi)$ be an affine space over K and $\mathcal{A}'$ a subset of $\mathcal{A}$. We say that $\mathcal{A}'$ is an *affine subspace* of $(\mathcal{A}, V, \varphi)$ if $\mathcal{A}' = \varnothing$ or if there exists a point $O \in \mathcal{A}'$ such that $\varphi_O(\mathcal{A}')$ $(= \{\overrightarrow{OA} \mid A \in \mathcal{A}'\})$ is a vector subspace of V.

Let $\mathcal{A}'$ be a non-empty affine subspace of $(\mathcal{A}, V, \varphi)$. If $O' \in \mathcal{A}'$ is any other point of $\mathcal{A}'$ then $\varphi_{O'}(\mathcal{A}') = v + \varphi_O(\mathcal{A}')$, with $v = \overrightarrow{O'O} \in \varphi_O(\mathcal{A}')$, so that $\varphi_{O'}(\mathcal{A}') = \varphi_O(\mathcal{A}')$. Hence the vector subspace $\varphi_O(\mathcal{A}')$ of V does not depend on the choice of $O \in \mathcal{A}'$ and it is called the *director subspace* (or *orienting subspace*) of $\mathcal{A}'$ We shall denote $\varphi_O(\mathcal{A}')$ by $D(\mathcal{A}')$. Since the map φ_O is bijective, then we have $\mathcal{A}' = \varphi_O^{-1}(D(\mathcal{A}'))$ for all $O \in \mathcal{A}'$ which will be also denoted by

$$\mathcal{A}' = O + D(\mathcal{A}').$$

Therefore a non-empty subset $\mathcal{A}'$ is an affine subspace of $(\mathcal{A}, V, \varphi)$ if and only if $(\mathcal{A}', D(\mathcal{A}'), \varphi_{|\mathcal{A}' \times \mathcal{A}'})$ is an affine space over K.

Proposition 3.9 *Let $(\mathcal{A}, V, \varphi)$ be an affine space over K and $\mathcal{A}'$ a non-empty subset of $\mathcal{A}$. The following conditions are equivalent:*

(i) *$\mathcal{A}'$ is an affine subspace of $(\mathcal{A}, V, \varphi)$.*
(ii) *For every finite number of points $A_1, \ldots, A_n \in \mathcal{A}'$ and for every $a_1, \ldots, a_n \in K$ such that $a_1 + \cdots + a_n = 1$, we have $a_1 A_1 + \cdots + a_n A_n \in \mathcal{A}'$.*

Proof (i) $\Longrightarrow$(ii): Let $A := \sum_{i=1}^{n} a_i A_i$, with $A_i \in \mathcal{A}'$ and $\sum_{i=1}^{n} a_i = 1$. Then there exists $O \in \mathcal{A}'$ such that $\overrightarrow{OA} = \sum_{i=1}^{n} a_i \overrightarrow{OA_i}$. Since $\varphi_O(\mathcal{A}')$ is a vector subspace of V, $A_i \in \mathcal{A}'$ (i.e. $\overrightarrow{OA_i} \in \varphi_O(\mathcal{A}')$) and $a_i \in K$, $i = 1, \ldots, n$, then we get $\overrightarrow{OA} \in \varphi_O(\mathcal{A}')$, i.e. $A \in \mathcal{A}'$.

(ii) $\Longrightarrow$ (i): We have to show that $\varphi_O(\mathcal{A}')$ is a vector subspace for $O \in \mathcal{A}'$. Let $v_1, v_2 \in \varphi_O(\mathcal{A}')$, $v_i = \overrightarrow{OA_i}$, with $A_i \in \mathcal{A}'$, $i = 1, 2$. If we put $A_3 := O$, $a_1 = a_2 = 1$ and $a_3 = -1$, so that $a_1 + a_2 + a_3 = 1$, by our assumption we obtain that $A := a_1 A_1 + a_2 A_2 + a_3 A_3 = A_1 + A_2 - O \in \mathcal{A}'$, which is equivalent to $v_1 + v_2 = \overrightarrow{OA_1} + \overrightarrow{OA_2} = \overrightarrow{OA_1} + \overrightarrow{OA_2} - \overrightarrow{OO} = \overrightarrow{OA} \in \varphi_O(\mathcal{A}')$. Finally, if $\lambda \in K$ then $B := (1 - \lambda)O + \lambda A_1 \in \mathcal{A}'$, so that $\lambda v_1 = (1 - \lambda)\overrightarrow{OO} + \lambda \overrightarrow{OA_1} = \overrightarrow{OB} \in \varphi_O(\mathcal{A}')$. □

Corollary 3.10 *Let $\{\mathcal{A}_i\}_{i \in I}$ be a non-empty family of affine subspaces of an affine space $(\mathcal{A}, V, \varphi)$. Then the intersection $\bigcap_{i \in I} \mathcal{A}_i$ is an affine subspace (possibly empty) of $(\mathcal{A}, V, \varphi)$.*

Proof It follows Proposition 3.9. □

Definition 3.11 Let $\mathcal{A}'$ be an affine subspace of an affine space $(\mathcal{A}, V, \varphi)$. We define the *dimension* of $\mathcal{A}'$ as the integer

$$\dim_K \mathcal{A}' := \begin{cases} \dim_K (D(\mathcal{A}')) & \text{if } \mathcal{A}' \neq \varnothing, \\ -1 & \text{if } \mathcal{A}' = \varnothing. \end{cases}$$

The *codimension* of $\mathcal{A}'$ in $(\mathcal{A}, V, \varphi)$ is defined by $\operatorname{codim}(\mathcal{A}') := \dim_K \mathcal{A} - \dim_K \mathcal{A}'$.

The points of $(\mathcal{A}, V, \varphi)$ are the affine subspaces of dimension 0. The affine subspaces of dimension 1 are called *affine lines* (or simply *lines*) of $(\mathcal{A}, V, \varphi)$, while those of dimension 2 are called *affine planes* (or simply *planes*). The affine subspaces of codimension 1 are called *affine hyperplanes* (or simply *hyperplanes*) of $(\mathcal{A}, V, \varphi)$.

Proposition 3.12 *Let $\mathcal{A}'$ be an affine subspace of an affine space $(\mathcal{A}, V, \varphi)$ of dimension n. Then $\dim_K \mathcal{A}' \leq n$ and $\dim_K \mathcal{A}' = n$ if and only if $\mathcal{A}' = \mathcal{A}$.*

Proof It follows from Definition 3.11 and from Corollary 1.61. □

Example 3.13 Let $\mathcal{A}'$ be a non-empty subset of the standard affine space $\mathbb{A}^n(K)$. Then $\mathcal{A}'$ is an affine subspace $\mathbb{A}^n(K)$ if and only if there exist a vector subspace $W \subset K^n$ and a vector $v \in K^n$ such that $\mathcal{A}' = v + W := \{v + w \mid w \in W\}$. (If $\mathcal{A}'$is an affine subspace then $W = D(\mathcal{A}')$.)

Definition 3.14 Let $M \subset \mathcal{A}$ be a subset of the affine space $(\mathcal{A}, V, \varphi)$. The *affine span of M* $\langle M \rangle$ is the intersection of all affine subspaces of $(\mathcal{A}, V, \varphi)$ which contain M. From Corollary 3.10 we obtain the following properties:

(i) $\langle M \rangle$ is an affine subspace of $(\mathcal{A}, V, \varphi)$ such that $M \subseteq \langle M \rangle$.
(ii) For every affine subspace $\mathcal{B}$ of $(\mathcal{A}, V, \varphi)$ containing M then $\langle M \rangle \subseteq \mathcal{B}$.
(iii) If $M = \varnothing$ then $\langle M \rangle = \varnothing$.

Proposition 3.15 *Let $\{A_1, \ldots, A_m\} \subseteq \mathcal{A}$ be a finite subset of an affine space $(\mathcal{A}, V, \varphi)$. Then*

$$\langle A_1, \ldots, A_m \rangle = \left\{ \sum_{i=1}^{m} a_i A_i \;\middle|\; \sum_{i=1}^{m} a_i = 1 \right\},$$

i.e. the affine span $\langle A_1, \ldots, A_m \rangle$ is the set of all weighted barycentres and

$$D(\langle A_1, \ldots, A_m \rangle) = \langle \overrightarrow{A_1 A_2}, \ldots, \overrightarrow{A_1 A_m} \rangle.$$

In particular $\dim_K \langle A_1, \ldots, A_m \rangle \leq m - 1$.

Proof Let $\mathcal{A}' := \{\sum_{i=1}^{m} a_i A_i \mid a_i \in K, \sum_{i=1}^{m} a_i = 1\}$ and $O := A_1$. By Proposition 3.9, (ii). we have $\mathcal{A}' \subset \langle A_1, \ldots, A_m \rangle$. Conversely, it will suffice to prove that $\mathcal{A}'$ is an affine subspace of $(\mathcal{A}, V, \varphi)$, since $A_i \in \mathcal{A}'$. So we have to prove that $\varphi_{A_1}(\mathcal{A}')$ is a

vector subspace of V. Then let $v = \overrightarrow{A_1A}$ and $w = \overrightarrow{A_1B}$, where $A = \sum_{i=1}^{m} a_i A_i$ and $B = \sum_{i=1}^{m} b_i A_i$, with $\sum_{i=1}^{m} a_i = \sum_{i=1}^{m} b_i = 1$. Therefore

$$v = \overrightarrow{A_1A} = \sum_{i=2}^{m} a_i \overrightarrow{A_1A_i} \quad \text{and} \quad w = \sum_{i=2}^{n} b_i \overrightarrow{A_1A_i}, \quad \text{which imply}$$

$$v + w = \sum_{i=2}^{m} (a_i + b_i) \overrightarrow{A_1A_i}.$$

If we put $c_i = a_i + b_i$, $i = 2, \ldots, m$ and $c_1 := 1 - \sum_{i=2}^{m} c_i$, we get the point $C := \sum_{i=1}^{n} c_i A_i \in \mathcal{A}'$ such that $v + w = \overrightarrow{A_1C}$, thus $v + w \in \varphi_{A_1}(\mathcal{A}')$. Now we have to prove that $\lambda v \in \varphi_{A_1}(\mathcal{A}')$ for every $\lambda \in K$ and $v \in \varphi_{A_1}(\mathcal{A}')$ as above. The point $D := (1 - \lambda)A_1 + \lambda A = (1 - \lambda)A_1 + \lambda a_1 A_1 + \cdots + \lambda a_m A_m$ belongs to $\mathcal{A}'$ by definition of $\mathcal{A}'$. Hence

$$\lambda v = \lambda \overrightarrow{A_1A} = (1 - \lambda)\overrightarrow{A_1A_1} + \lambda \overrightarrow{A_1A} = \overrightarrow{A_1D} \in \varphi_{A_1}(\mathcal{A}').$$

Therefore $\varphi_{A_1}(\mathcal{A}')$ is a vector subspace of V and $\mathcal{A}'$ is an affine subspace of $(\mathcal{A}, V, \varphi)$ so that $\langle A_1, \ldots, A_m \rangle \subset \mathcal{A}'$ since $\{A_1, \ldots, A_m\} \subset \mathcal{A}'$. The last statement is now obvious. □

Definition 3.16 Let $(\mathcal{A}, V, \varphi)$ be an affine space of dimension n. We call m points $A_1, \ldots, A_m \in \mathcal{A}$ (or the set $\{A_1, \ldots, A_m\}$) *affinely independent* if $\overrightarrow{A_1A_2}, \ldots, \overrightarrow{A_1A_m}$ are linearly independent vectors of V. If $A_1, \ldots, A_m$ are not affine independent we call them *affinely dependent*. Moreover, we say that $A_1, \ldots, A_m$ are *in general position* if $m \leq n$ and they are affinely independent or, if $m > n$, every subset of $\{A_1, \ldots, A_m\}$ consisting of n points is affinely independent. For instance, n distinct points $P_1, \ldots, P_n \in \mathbb{A}^2(K)$, with $n > 2$, are in general position if any subset of three distinct points is not contained in a line.

Corollary 3.17 *Under the assumptions of* Proposition 3.15*, the following conditions are equivalent:*

(i) $\{A_1, \ldots, A_m\}$ *is affinely independent.*
(ii) $\dim_K \langle A_1, \ldots, A_m \rangle = m - 1$.
(iii) *If* $(a_1, \ldots, a_m) \in K^m$ *is such that* $a_1 + \cdots + a_m = 0$ *and* $a_1A_1 + \cdots + a_mA_m = \bigcirc$ (*see* Definition 3.6)*, then we have* $a_1 = \cdots = a_m = 0$.
(iv) *For every* $A \in \langle A_1, \ldots, A_m \rangle$ *there exists a unique* $(a_1, \ldots, a_m) \in K^m$ *such that* $a_1 + \cdots + a_m = 1$ *and* $A = a_1A_1 + \cdots + a_mA_m$.

Proof The equivalence (i) $\Longleftrightarrow$ (ii) follows from Definition 3.16 and from Proposition 3.15.
(i) $\Longrightarrow$ (iii): Let $a_1, \ldots, a_m \in K$ such that $a_1 + \cdots + a_m = 0$ and $a_1 A_1 + \cdots + a_m A_m = \bigcirc$. If we take $O = A_1$, then we have $a_2 \overrightarrow{A_1 A_2} + \cdots + a_m \overrightarrow{A_1 A_m} = 0_V$, so that by i), $a_2 = \cdots = a_m = 0$. Furthermore, we have $a_1 = -(a_2 + \cdots + a_m) = 0$.
(iii) $\Longrightarrow$ (i): We have to show that $\overrightarrow{A_1 A_2}, \ldots, \overrightarrow{A_1 A_m}$ are linearly independent. Let $a_2 \overrightarrow{A_1 A_2} + \cdots + a_m \overrightarrow{A_1 A_m} = 0_V$, with $a_2, \ldots, a_n \in K$. We have just seen that this equality is equivalent to $a_1 A_1 + \cdots + a_m A_m = \bigcirc$, where $a_1 = -(a_2 + \cdots + a_m)$. From (iii) we conclude that $a_1 = a_2 = \cdots = a_m = 0$.
(iii) $\Longrightarrow$ (iv): Let $A \in \langle A_1, \ldots, A_m \rangle$ such that $A = a_1 A_1 + \cdots + a_m A_m = b_1 A_1 + \cdots + b_m A_m$ with $a_1 + \cdots + a_m = b_1 + \cdots + b_m = 1$. Then we have $\sum_{i=2}^{m} a_i \overrightarrow{A_1 A_i} = \sum_{i=2}^{m} b_i \overrightarrow{A_1 A_i}$. Since iii) $\Longleftrightarrow$ i), it follows that $a_i = b_i$, $i = 2, \ldots, m$ from which we obtain $a_1 = 1 - (a_2 + \cdots + a_m) = 1 - (b_2 + \cdots + b_m) = b_1$.
(iv) $\Longrightarrow$ (i): Let $a_2 \overrightarrow{A_1 A_2} + \cdots + a_m \overrightarrow{A_1 A_m} = 0_V$ with $a_2, \ldots, a_n \in K$. Then we have $A_1 = a_1 A_1 + \cdots + a_m A_m$ with $a_1 = 1 - (a_2 + \cdots + a_m)$. Since $A_1 = 1 \cdot A_1 + 0 \cdot A_2 + \cdots + 0 \cdot A_m$, iv) implies $a_i = 0$, $i = 2, \ldots, m$. □

Remark 3.18 As an immediate consequence of Corollary 3.17 we assert that the affine independence of a set $\{A_1, \ldots, A_m\}$ does not depends on the order of $A_1, \ldots, A_m$.

Definition 3.19 We call m points $A_1, \ldots, A_m \in (\mathcal{A}, V, \varphi)$ *collinear* if $\langle A_1, \ldots, A_m \rangle$ is a line, and *coplanar* if $\langle A_1, \ldots, A_m \rangle$ is a plane.

Definition 3.20 Let $(\mathcal{A}, V, \varphi)$ be an affine space of dimension n. If a subset $\{A_1, \ldots, A_m\}$ of V is affine independent, every point $A \in \langle A_1, \ldots, A_m \rangle$ can be written in the unique form $A = a_1 A_1 + \cdots + a_m A_m$ (with $a_1 + \cdots + a_m = 1$) by Corollary 3.17. Thus $(a_1, \ldots, a_m) \in K^m$ can be called the *barycentric coordinates* of A with respect to the ordered affine independent set $\{A_1, \ldots, A_m\}$. Every permutation of the set $\{A_1, \ldots, A_m\}$ determines the same permutation of the coordinates $(a_1, \ldots, a_m)$.

Proposition 3.21 *Let $\{A_1, \ldots, A_m\}$ be a non-empty set of an affine space $(\mathcal{A}, V, \varphi)$. Then $\{A_1, \ldots, A_m\}$ is affinely independent if and only if*

$$A_i \notin \langle A_1, \ldots, A_{i-1}, A_{i+1}, \ldots, A_m \rangle \tag{3.2}$$

for every $i \in \{1, \ldots, m\}$.

Proof If $m = 1$ there is nothing to prove. Let $m \geq 2$. Suppose that $\{A_1, \ldots, A_m\}$ is affinely independent and there is an index i such that $A_i \in \langle A_1, \ldots, A_{i-1}, A_{i+1}, \ldots, A_m\rangle$ After renumbering our points we can suppose $i \neq 1$ so that

$$A_i = \sum_{j\neq i, j=1}^{m} a_j A_j, \quad \text{with} \quad \sum_{j\neq i, j=1}^{m} a_j = 1, \qquad \text{i.e.} \quad \overrightarrow{A_1A_i} = \sum_{j\neq i, j=2}^{m} a_j \overrightarrow{A_1A_j},$$

which contradicts the affine independence of $\{A_1, \ldots, A_m\}$.

Conversely, if $\{A_1, \ldots, A_m\}$ is not affinely independent we can suppose that there is an index $i_0 \neq 1$ such that $\overrightarrow{A_1A_{i_0}} = \sum_{j=2, j\neq i_0}^{m} a_j \overrightarrow{A_1A_j}$ so that

$$A_{i_0} = a_1 A_1 + \sum_{j=2, j\neq i_0}^{m} a_j A_j, \qquad \text{with } a_1 = 1 - \sum_{j=2, j\neq i_0}^{m} a_j,$$

which contradicts (3.2). □

Proposition 3.22 *Let $(\mathcal{A}, V, \varphi)$ be an affine space of dimension n and $\mathcal{A}'$ a non-empty affine subspace of dimension m $(m \leq n)$. There exists an affinely independent subset $\{A_0, A_1, \ldots, A_m\} \subset \mathcal{A}'$ such that $\mathcal{A}' = \langle A_0, A_1, \ldots, A_m\rangle$.*

Proof Let $\{v_1, \ldots, v_m\}$ be a basis of $D(\mathcal{A}')$. Take a point O of $\mathcal{A}'$ and define $A_i := \varphi_O^{-1}(v_i)$ (i.e. $v_i = \overrightarrow{OA_i}$), $i = 1, \ldots, m$. If you put $A_0 := O$, the set $\{A_0, A_1, \ldots, A_m\}$ is affinely independent such that $\mathcal{A}' = \langle A_0, A_1, \ldots, A_m\rangle$. □

Definition 3.23 Let $(\mathcal{A}, V, \varphi)$ be an affine space of dimension $n \geq 1$. We define the *affine frame* of $(\mathcal{A}, V, \varphi)$ as every ordered set of $n + 1$ points $A_0, A_1, \ldots, A_n \in \mathcal{A}$ which are affinely independent. The point A_0 is called the *origin* of the affine frame $A_0, A_1, \ldots, A_n$.

By Proposition 3.22 every affine space $(\mathcal{A}, V, \varphi)$ of dimension $n \geq 1$ has an affine frame at least. For instance $E_0 = (0, 0, \ldots, 0)$, $E_1 = (1, 0, \ldots, 0)$,$\ldots$,$E_n = (0, 0, \ldots, 0, 1)$ is an affine frame of $\mathbb{A}^n(K)$ called the *standard affine frame* of $\mathbb{A}^n(K)$.

Proposition 3.24 *Let $(\mathcal{A}, V, \varphi)$ be an affine space of dimension $n \geq 1$. Giving an affine frame of $(\mathcal{A}, V, \varphi)$ is equivalent to taking a point $O \in \mathcal{A}$ (the origin) and a vector basis of V.*

Proof Let $A_0, A_1, \ldots, A_n$ be an affine frame of $(\mathcal{A}, V, \varphi)$. Take $O := A_0$ and $v_i := \overrightarrow{A_0A_i}$, $i = 1, \ldots, n$. Then the vectors $v_1, \ldots, v_n$ are linearly independent and $\{v_1, \ldots, v_n\}$ is a basis of V because $\dim_K V = n$.

Conversely, given any point $O \in \mathcal{A}$ and an ordered basis $v_1, \ldots, v_n$ of V, by axiom AFF2) and Proposition 3.2, there exists a uniquely determined point $A_i \in \mathcal{A}$

such that $v_i = \overrightarrow{OA_i}$, $i = 1, \ldots, n$. Putting $A_0 := O$, the set $A_0, A_1, \ldots, A_n$ is an affine frame of $(\mathcal{A}, V, \varphi)$. □

An immediate consequence of Corollary 3.17, iv) is the following:

Corollary 3.25 *Let $A_0, A_1, \ldots, A_n$ be an affine frame of $(\mathcal{A}, V, \varphi)$ of dimension n. Then for every point $A \in \mathcal{A}$ there exists a unique $(n+1)$-tuple $(a_0, a_1, \ldots, a_n) \in K^{n+1}$ such that $a_0 + a_1 + \cdots + a_n = 1$ and $A = a_0A_0 + a_1A_1 + \cdots + a_nA_n$.*

Definition 3.26 Under the assumptions and notation of Corollary 3.25, the scalars $a_0, a_1, \ldots, a_n \in K$, with $a_0 + a_1 + \cdots + a_n = 1$, such that $A = a_0A_0 + a_1A_1 + \cdots + a_nA_n$ are called the *barycentric coordinates of A with respect to the affine frame $A_0, A_1, \ldots, A_n$*.

Since $\{\overrightarrow{A_0A_1}, \ldots, \overrightarrow{A_0A_n}\}$ is a basis of V there is a unique n-tuple $(x_1(A), \ldots, x_n(A)) \in K^n$ such that $\overrightarrow{A_0A} = x_1(A)\overrightarrow{A_0A_1} + \cdots + x_n(A)\overrightarrow{A_0A_n}$. We call $x_1(A), \ldots, x_n(A) \in K$ *affine coordinates with respect to the affine frame $A_0, A_1, \ldots, A_n$* or also a *system of affine coordinates*.

If $(a_0, a_1, \ldots, a_n)$ are the barycentric coordinates of A with respect to the affine frame $A_0, A_1, \ldots, A_n$, then the affine coordinates of (with respect to the same frame) are given by $x_i(A) = a_i$, $i = 1, \ldots, n$. Indeed, the equality $A = a_0A_0 + a_1A_1 + \cdots + a_nA_n$ implies the relations

$$\overrightarrow{A_0A} = a_0\overrightarrow{A_0A_0} + a_1\overrightarrow{A_0A_1} + \cdots + a_n\overrightarrow{A_0A_n} = a_1\overrightarrow{A_0A_1} + \cdots + a_n\overrightarrow{A_0A_n}.$$

Conversely, if $(x_1(A), \ldots, x_n(A))$ are the affine coordinates of A, then

$$(a_0, a_1, \ldots, a_n) = (1 - (x_1(A) + \cdots + x_n(A)), x_1(A), \ldots, x_n(A))$$

are the barycentric coordinates of A.

Proposition 3.27 *Let $A_0, A_1, \ldots, A_n$ and $B_0, B_1, \ldots, B_n$ be two affine frames of an affine space $(\mathcal{A}, V, \varphi)$ of dimension $n \geq 1$. Denote the associated vector bases by $S = \{\overrightarrow{A_0A_i}\}_{i=1,\ldots,n}$ and $T = \{\overrightarrow{B_0B_i}\}_{i=1,\ldots,n}$, and the matrix of transition from S to T by $\Lambda^{S,T} = (a_{ij})_{i,j=1,\ldots,n}$ so that*

$$\overrightarrow{B_0B_k} = \sum_{i=1}^{n} a_{ik}\overrightarrow{A_0A_i}, \qquad k = 1, \ldots, n. \tag{3.3}$$

If $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$ are the affine coordinates of any point $A \in \mathcal{A}$ with respect to $A_0, A_1, \ldots, A_n$ and $B_0, B_1, \ldots, B_n$) respectively, then we have

$$x_i = \sum_{k=1}^{n} a_{ik}y_k + b_i, \qquad i = 1, \ldots, n, \tag{3.4}$$

where

$$\overrightarrow{A_0B_0} = \sum_{i=1}^{n} b_i \overrightarrow{A_0A_i}. \tag{3.5}$$

Proof We have from (3.5)

$$\overrightarrow{B_0A} = \overrightarrow{A_0A} - \overrightarrow{A_0B_0} = \sum_{i=1}^{n} x_i \overrightarrow{A_0A_i} - \sum_{i=1}^{n} b_i \overrightarrow{A_0A_i} = \sum_{i=1}^{n} (x_i - b_i) \overrightarrow{A_0A_i}.$$

On the other hand, we have

$$\overrightarrow{B_0A} = \sum_{k=1}^{n} y_k \overrightarrow{B_0B_k} = \sum_{i,k=1}^{n} a_{ik} y_k \overrightarrow{A_0A_i} = \sum_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik} y_k \right) \overrightarrow{A_0A_i}.$$

These two equalities imply (3.4). □

Lines in the Affine Spaces

Let A and B be two distinct points of an affine space $(\mathcal{A}, V, \varphi)$. Then the set $\{A, B\}$ is obviously affinely independent. Denote by $AB := \langle A, B \rangle$ the affine subspace of $\mathcal{A}$ generated by $\{A, B\}$. By Proposition 3.15 we have:

$$AB = \{(1 - t)A + tB \mid t \in K\}.$$

Moreover, by Corollary 3.17, $\dim_K AB) = 1$, then AB is a line according to Definition 3.11. We shall say that AB *joins* A and B. Conversely, every line $\mathcal{B}$ of $(\mathcal{A}, V, \varphi)$ is of the form AB, with $A, B \in \mathcal{B}$, $A \neq B$. Indeed, if we take any two points $A, B \in \mathcal{B}$, $A \neq B$, then $AB \subset \mathcal{B}$ and, by Proposition 3.12, $AB = \mathcal{B}$ (since $\dim_K AB = \dim_K \mathcal{B} = 1$). Thus we have:

Corollary 3.28 *Through two distinct points A and B of an affine space $(\mathcal{A}, V, \varphi)$ there is exacly one line.*

Proposition 3.29 *Let $(\mathcal{A}, V, \varphi)$ be an affine space and $\mathcal{A}'$ a non-empty subset of $\mathcal{A}$. If $\mathcal{A}'$ is an affine subspace of $(\mathcal{A}, V, \varphi)$, then for every pair of distinct points A, $B \in \mathcal{A}'$ the line AB is contained in $\mathcal{A}'$.*

Conversely, if $\mathrm{char}(K) \neq 2$ *and $AB \subseteq \mathcal{A}'$ for every pair of distinct points A, $B \in \mathcal{A}'$, then $\mathcal{A}'$ is an affine subspace of $(\mathcal{A}, V, \varphi)$.*

Proof The first assertion follows from Proposition 3.9, since the line AB is the set of all weighted barycentres of A and B, by definition.

Conversely, assume that $\mathrm{char}(K) \neq 2$ and that $AB \subset \mathcal{A}'$ for every pair of distinct points $A, B \in \mathcal{A}'$, Let O be any point of $\mathcal{A}'$. We have to show that $W := \varphi_O(\mathcal{A}')$ is a vector subspace of V.

Let $v = \overrightarrow{OM} \in W$, with $M \in \mathcal{A}'$ and $\lambda \in K$. Since $OM \subset \mathcal{A}'$, then $N := (1-\lambda)O + \lambda M \in \mathcal{A}'$, i.e. $\lambda v = \lambda\overrightarrow{OM} = (1-\lambda)\overrightarrow{OO} + \lambda\overrightarrow{OM} = \overrightarrow{ON} \in W$.

We have now to prove that $v + v' \in W$ if $v, v' \in W$. Let $M, M' \in \mathcal{A}'$ such that $v = \overrightarrow{OM}$ and $v' = \overrightarrow{OM'}$. Take $N = \frac{1}{2}M + \frac{1}{2}M'$ (we are assuming that $\mathrm{char}(K) \neq 2$!); since $MM' \subset \mathcal{A}'$, then $N \in \mathcal{A}'$ and $\overrightarrow{ON} \in W$. By definition we have $\overrightarrow{ON} = \frac{1}{2}\overrightarrow{OM} + \frac{1}{2}\overrightarrow{OM'} = \frac{1}{2}(v + v')$, hence $\frac{1}{2}(v + v') \in W$ and $v + v' \in W$. □

Remark 3.30 By Corollary 3.17 three points $A, B, C \in \mathcal{A}$ are affinely independent if and only if A, B, C are not collinear. By the same corollary A, B, C are collinear if and only if there exists $(a, b, c) \in K^3 \setminus (0, 0, 0)$ such that $a + b + c = 0$ and $aA + bB + cC = \bigcirc$.

3.1.1 The Affine Ratio and Menelaus' Theorem

Definition 3.31 Let A and B be two distinct points of an affine space $(\mathcal{A}, V, \varphi)$ of dimension ≥ 1 over a field K. The ordered pair (A, B) is an affine frame of the line AB (as affine space of dimension 1). From Corollary 3.25 it follows that, for every $C \in AB$, there is a unique $t \in K$ such that

$$C = (1-t)A + tB. \tag{3.6}$$

Hence the barycentric coordinates of C with respect to the affine frame (A, B) are $(1-t, t)$ and the affine coordinate of C with respect to the same affine frame is t. From (3.6) we get

$$\overrightarrow{AC} = (1-t)\overrightarrow{AA} + t\overrightarrow{AB} = t\overrightarrow{AB}, \tag{3.7}$$

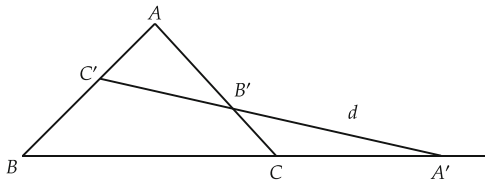
so that we can define

$$\frac{\overrightarrow{AC}}{\overrightarrow{AB}} := t \in K. \tag{3.8}$$

If $C \in AB$ and $C \neq A, B$, the relation (3.6) also implies $A = \frac{t}{t-1}B + \frac{-1}{t-1}C$ from which we get

$$\overrightarrow{CA} = \frac{t}{t-1}\overrightarrow{CB} + \frac{-1}{t-1}\overrightarrow{CC} = \frac{t}{t-1}\overrightarrow{CB}.$$

Fig. 3.2 Menelaus' theorem



These last relations justify the definition of *affine ratio* $\mathrm{ar}(A, B, C)$ of ordered triple (A, B, C)

$$\mathrm{ar}(A, B, C) = \frac{\overrightarrow{CA}}{\overrightarrow{CB}} := \frac{t}{t-1} \in K. \tag{3.9}$$

Definition 3.32 Let A, B and C be three affine independent points of an affine space $(\mathcal{A}, V, \varphi)$ of dimension ≥ 2. We call the triple $\{A, B, C\}$ *triangle* (non-degenerate) and we denote it by $\triangle ABC$. The points A, B and C, which are not collinear, are called the *vertices* of $\triangle ABC$, while the lines AB, AC and BC are the *sides* of $\triangle ABC$.

The affine ratio is a fruitful tool in affine geometry and, to show it, we prove the well-known Menelaus' theorem.

Proposition 3.33 (Menelaus' Theorem) *Let $\triangle ABC$ be a triangle of an affine space $(\mathcal{A}, V, \varphi)$ of dimension 2. Let d be a line such that (see Fig. 3.2)*

$$d \cap BC = A' \neq B, C; \qquad d \cap CA = B' \neq C, A; \qquad d \cap AB = C' \neq A, B.$$

Then

$$\frac{\overrightarrow{A'B}}{\overrightarrow{A'C}} \cdot \frac{\overrightarrow{B'C}}{\overrightarrow{B'A}} \cdot \frac{\overrightarrow{C'A}}{\overrightarrow{C'B}} = 1. \tag{3.10}$$

Conversely, if $A' \in BC \setminus \{B, C\}$, $B' \in CA \setminus \{C, A\}$ and $C' \in AB \setminus \{A, B\}$ are three points such that (3.10) holds, then A', B', C' are collinear.

In other words, if A', B', C' are three points, on the sides of $\triangle ABC$ opposite A, B, C respectively, all different from the vertices, then A', B', C' are collinear if and only if (3.10) holds.

Proof If we put $A' = (1 - t_{A'})B + t_{A'}C$, $B' = (1 - t_{B'})C + t_{B'}A$ and $C' = (1-t_{C'})A+t_{C'}B$, with $t_{A'}, t_{B'}, t_{C'} \in K \setminus \{0, 1\}$, then the equality (3.10) is equivalent to

$$\frac{t_{A'}t_{B'}t_{C'}}{(t_{A'}-1)(t_{B'}-1)(t_{C'}-1)} = 1. \tag{3.11}$$

On the other hand, the equalities $B' = (1-t_{B'})C + t_{B'}A$ and $C' = (1-t_{C'})A + t_{C'}B$ imply

$$\overrightarrow{A'B'} = (1-t_{B'})\overrightarrow{A'C} + t_{B'}\overrightarrow{A'A} \qquad \text{and} \qquad \overrightarrow{A'C'} = (1-t_{C'})\overrightarrow{A'A} + t_{C'}\overrightarrow{A'B}. \tag{3.12}$$

From $A' = (1-t_{A'})B + t_{A'}C$ it follows that $0_V = (1-t_{A'})\overrightarrow{A'B} + t_{A'}\overrightarrow{A'C}$ and

$$\overrightarrow{A'B} = \frac{t_{A'}}{t_{A'}-1}\overrightarrow{A'C}. \tag{3.13}$$

Inserting (3.13) in (3.12) we get

$$\overrightarrow{A'C'} = \frac{t_{C'}t_{A'}}{t_{A'}-1}\overrightarrow{A'C} + (1-t_{C'})\overrightarrow{A'A}. \tag{3.14}$$

Since $\overrightarrow{A'A}$ and $\overrightarrow{A'C}$ are linearly independent vectors the equalities (3.12) and (3.14) imply the equivalences

$$\begin{aligned} A', B', C' \text{ are collinear} &\iff \overrightarrow{A'B'} \text{ and } \overrightarrow{A'C'} \text{ are linearly dependent} \\ &\iff \frac{1-t_{B'}}{\frac{t_{C'}t_{A'}}{t_{A'}-1}} = \frac{t_{B'}}{1-t_{C'}}. \end{aligned}$$

The last equality is equivalent to (3.11), so that the thesis is proved. □

Remark 3.34 Menelaus' theorem obviously holds in any affine space $(\mathcal{A}, V, \varphi)$ of dimension ≥ 2 because its statement concerns only the plane $\langle A, B, C\rangle$ generated by A, B, C.

3.1.2 The Affine Subspaces of $\mathbb{A}^n(K)$

We can now give two analytical descriptions of subspaces of the standard affine space $\mathbb{A}^n(K)$.

Proposition 3.35 *Let $\mathcal{A}$ be any non-empty affine subspace of $\mathbb{A}^n(K)$ of dimension k. Fix a point $O = (a_1, \ldots, a_n) \in \mathcal{A}$ and a basis $v_1, \ldots, v_k$ of $D(\mathcal{A})$ with $v_j = (a_{1j}, \ldots, a_{nj})$, $j = 1, \ldots, k$. Then all points $(x_1, \ldots, x_n) \in \mathcal{A}$ are given by* parametric equations *of the form*

$$x_i = a_i + \sum_{j=1}^{k} a_{ij}t_j, \qquad i = 1, \ldots, n, \tag{3.15}$$

where $(t_1, \ldots, t_k)$ runs in K^k and $\left(a_{ij}\right)_{\substack{i=1,\ldots,n \\ j=1,\ldots,k}}$ is a matrix of (maximum) rank k.

Conversely, every subset of $\mathbb{A}^n(K)$ *described by the Eq.* (3.15) *with* $\left(a_{ij}\right)_{i=1,\dots,n,\, j=1,\dots,k}$ *matrix of rank* k *and* $O := (a_1, \dots, a_n) \in \mathbb{A}^n(K)$ *is an affine subspace* $\mathcal{A}$ *of* $\mathbb{A}^n(K)$ *of dimension* k *such that* $O \in \mathcal{A}$.

Proof Since

$$D(\mathcal{A}) = \left\{ \sum_{j=1}^{k} t_j v_j \;\middle|\; (t_1, \dots, t_k) \in K^k \right\}$$

and a point $X := (x_1, \dots, x_n) \in \mathcal{A}$ if and only if $\overrightarrow{OX} \in D(\mathcal{A})$, we have

$$x_i - a_i = \sum_{j=1}^{k} a_{ij} t_j, \qquad i = 1, \dots, n.$$

Conversely, since the matrix $\left(a_{ij}\right)_{\substack{i=1,\dots,n,\\ j=1,\dots,k}}$ has rank k, the linearly independent column vectors $v_1, \dots, v_k$, where $v_i := (a_{1i}, \dots, a_{ni})$, $i = 1, \dots, k$, generate a vector subspace W of dimension k. Then the Eq. (3.15) define the affine subspace $\mathcal{A} = O + W$ of dimension k. □

Proposition 3.36 *Let* $\mathcal{A}$ *be any non-empty affine subspace of* $\mathbb{A}^n(K)$ *of dimension* k. *Then all points* $X := (x_1, \dots, x_n) \in \mathcal{A}$ *are solutions of a system of linear equations, called* implicit equations *of* $\mathcal{A}$, *of the form*

$$\sum_{j=1}^{n} b_{ij} x_j = c_i, \qquad i = k+1, \dots, n, \tag{3.16}$$

where the matrix $\left(b_{ij}\right)_{\substack{i=k+1,\dots,n\\ j=1,\dots,n}}$ *has rank* $n-k$ *and* $c_{k+1}, \dots, c_n \in K$.

Conversely, every linear system (3.16), *where* $\left(b_{ij}\right)_{\substack{i=k+1,\dots,n\\ j=1,\dots,n}}$ *is a matrix of rank* $n-k$, *defines an affine subspace* $\mathcal{A}$ *of* $\mathbb{A}^n(K)$ *of dimension* k.

Proof Fix a point $O = (a_1, \dots, a_n) \in \mathcal{A}$ and a basis $\mathcal{B} = \{v_1, \dots, v_k\}$ of $D(\mathcal{A})$. By Corollary 1.53, there exists a basis $\mathcal{B}' = \{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ of K^n which contains $\mathcal{B}$. Let $v_i = (a_{1i}, \dots, a_{ni})$, $i = 1, \dots, n$. The non-singular matrix $\left(a_{ij}\right)_{i,j=1,\dots,n} \in \mathcal{M}_n(K)$ coincides with $\Lambda^{\mathcal{E},\mathcal{B}} = \left(a_{ij}\right)_{i,j=1,\dots,n} \in \mathcal{M}_n(K)$, the transition matrix from the canonical basis $\mathcal{E} = \{e_1, \dots, e_n\}$ of K^n (see Example 1.62-1)) to $\mathcal{B}$. Let $B = \left(\Lambda^{\mathcal{E},\mathcal{B}}\right)^{-1} = \left(b_{ij}\right)_{i,j=1,\dots,n}$. If $(y_1, \dots, y_n)$ are the coordinates of $\overrightarrow{OX} = (x_1 - a_1, \dots, x_n - a_n) \in K^n$ with respect to $\mathcal{B}$, we have $(x_1 - a_1, \dots, x_n - a_n) = \sum_{i=1}^{n} y_i v_i$ so that $y_i = \sum_{j=1}^{n} b_{ij}(x_j - a_j)$, $i = 1, \dots, n$.

Furthermore, $(x_1 - a_1, \ldots, x_n - a_n) \in D(\mathcal{A})$ if and only if $y_{k+1} = \ldots = y_n = 0$, hence

$$\mathcal{A} = O + D(\mathcal{A}) = \{(a_1, \ldots, a_n) + (x_1, \ldots, x_n) \in \mathbb{A}^n(K) \mid \sum_{j=1}^{n} b_{ij}(x_j - a_j) = 0,\ i = k+1, \ldots, n\}.$$

Since $\det\left(\left(b_{ij}\right)_{i,j=1,\ldots,n}\right) \neq 0$, then $\left(b_{ij}\right)_{\substack{i=k+1,\ldots,n \\ j=1,\ldots,n}}$ has maximum rank $n-k$. In this way we have obtained a general form of implicit equations that define the subspace $\mathcal{A}$ of $\mathbb{A}^n(K)$ of dimension k:

$$\sum_{j=1}^{n} b_{ij}(x_j - a_j) = 0, \qquad i = k+1, \ldots, n, \tag{3.17}$$

where $O = (a_1, \ldots, a_n)$ and $\operatorname{rank}\left(\left(b_{ij}\right)_{\substack{i=k+1,\ldots,n \\ j=1,\ldots,n}}\right) = n - k$ or

$$\sum_{j=1}^{n} b_{ij}x_j = c_i, \qquad i = k+1, \ldots, n, \tag{3.18}$$

with $c_{k+1}, \ldots, c_n \in K$ if we do not specify the point O.

Conversely, the set of the points $(x_1, \ldots, x_n)$ of $\mathbb{A}^n(K)$ which satisfy (3.16) (where $\operatorname{rank}\left(\left(b_{ij}\right)_{\substack{i=k+1,\ldots \\ n,\, j=1,\ldots,n}}\right) = n-k$) is an affine subspace of $\mathbb{A}^n(K)$ of dimension k whose director space is given by equations

$$\sum_{j=1}^{n} b_{ij}x_j = 0, \qquad i = k+1, \ldots, n, \tag{3.19}$$

where, with abuse of notation, we have denoted the vectors of K^n by $(x_1, \ldots, x_n)$ as the points of $\mathbb{A}^n(K)$. □

Remark 3.37 We want to note that both parametric equations and implicit equations of the affine subspace $\mathcal{A}$ given in Propositions 3.35 and 3.36 depend on the choice of a basis for its director subspace $D(\mathcal{A})$.

Examples 3.38

1. Every hyperplane of $\mathbb{A}^n(K)$ is given by an implicit equation of the form

$$b_1x_1 + \cdots + b_nx_n = c, \quad \text{where} \quad (b_1, \ldots, b_n) \in K^n \setminus \{(0, \ldots, 0)\} \quad \text{and} \quad c \in K. \tag{3.20}$$

2. Parametric equations of a line r containing a point $(a_1, \ldots, a_n) \in \mathbb{A}^n(K)$ have the general form

$$x_j = a_j + l_j t, \qquad j = 1, \ldots, n, \tag{3.21}$$

where the vector $(l_1, \ldots, l_n) \neq (0, \ldots, 0)$ generates the director space of r and the scalars $l_1, \ldots, l_n$ are called *director parameters* of r (which are unique up to a non-zero multiplicative factor). If $A = (a_1, \ldots, a_n)$ and $B = (b_1, \ldots, b_n)$ are two distinct points of r we can take

$$l_i = b_i - a_i, \qquad i = 1, \ldots, n.$$

3. Implicit equations of a line of $\mathbb{A}^n(K)$ are of the form

$$\sum_{j=1}^{n} b_{ij} x_j = c_i, \qquad i = 2, \ldots, n, \tag{3.22}$$

where $\operatorname{rank}\left(\left(b_{ij}\right)_{\substack{i=2,\ldots,n \\ j=1,\ldots,n}}\right) = n - 1$.

In particular if $n = 3$, parametric and implicit equations of a line r containing a point (a_1, a_2, a_3) and having director parameters $(l_1, l_2, l_3) \in K^3 \setminus (0, 0, 0)$ have the form, respectively,

$$\begin{cases} x_1 = a_1 + l_1 t \\ x_2 = a_2 + l_2 t \\ x_3 = a_3 + l_3 t \end{cases} \qquad \forall\, t \in K,$$

$$\begin{cases} a_1 x_1 + a_2 x_2 + a_3 x_3 &= c \\ b_1 x_1 + b_2 x_2 + b_3 x_3 &= d, \end{cases} \qquad a_i, b_i, c, d \in K,\ i = 1, 2, 3$$

such that

$$\operatorname{rank} \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix} = 2.$$

Now we give another way to determine implicit equations of an affine subspace $\mathcal{A}$ of $\mathbb{A}^n(K)$. Fix a point $O = (a_1, \ldots, a_n) \in \mathcal{A}$ and a basis $\{v_1, \ldots, v_k\}$ of $D(\mathcal{A})$ with $v_i = (a_{i1}, \ldots, a_{in})$, $i = 1, \ldots, k$. A point $X := (x_1, \ldots, x_n) \in \mathcal{A}$ if and only if $\overrightarrow{OX} \in D(\mathcal{A})$, i.e.

$$\Omega = \begin{pmatrix} x_1 - a_1 & x_2 - a_2 & \cdots & x_n - a_n \\ a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} \end{pmatrix}$$

has rank k. Since $v_1, \ldots, v_k$ are linearly independent we can find a non-singular $k \times k$-submatrix M of

$$\Omega' = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} \end{pmatrix}.$$

Then we write down $n-k$ implicit equations of $\mathcal{A}$ by vanishing the determinants of the $n-k$ submatrices $N \in \mathcal{M}_{k+1}(K)$ containing M. Indeed, by Kronecker's rule, $\operatorname{rank}(\Omega) = k$ if and only if the $n-k$ submatrices $N \in \mathcal{M}_{k+1}(K)$ containing M are singular.

Lemma 3.39 *Let $A_1, \ldots, A_n$ be n arbitrary points of $\mathbb{A}^n(K)$, with $A_i = (a_{i1}, \ldots, a_{in})$, $i = 1, \ldots, n$. Then $A_1, \ldots, A_n$ are affinely independent if and only if the matrix*

$$\Omega(A_1, \ldots, A_n) := \begin{pmatrix} 1 & a_{11} & \cdots & a_{1n} \\ 1 & a_{21} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_{n1} & \cdots & a_{nn} \end{pmatrix}$$

contains an $n \times n$ non-singular submatrix obtained by deleting one of the second n columns.

Proof We see that for $i = 1, \ldots, n$

$$\begin{vmatrix} 1 & a_{11} & \cdots & a_{1\,i-1} & a_{1\,i+1} & \cdots & a_{1n} \\ 1 & a_{21} & \cdots & a_{2\,i-1} & a_{2\,i+1} & \cdots & a_{2n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & a_{n1} & \cdots & a_{n\,i-1} & a_{n\,i+1} & \cdots & a_{2n} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a_{11} & \cdots & a_{1\,i-1} & a_{1\,i+1} & \cdots & a_{1n} \\ 0 & a_{21} - a_{11} & \cdots & a_{2\,i-1} - a_{1\,i-1} & a_{2\,i+1} - a_{1\,i+1} & \cdots & a_{2n} - a_{1n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_{n1} - a_{11} & \cdots & a_{n\,i-1} - a_{1\,i-1} & a_{n\,i+1} - a_{1\,i+1} & \cdots & a_{2n} - a_{1n} \end{vmatrix},$$

thus the matrix $\Omega(A_1, \ldots, A_n)$ contains an $n \times n$ non-singular submatrix obtained by deleting one of the second n columns if and only if

$$\Omega'(A_1, \ldots, A_n) := \begin{pmatrix} 1 & a_{11} & \cdots & a_{1n} \\ 0 & a_{21} - a_{11} & \cdots & a_{2n} - a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n1} - a_{11} & \cdots & a_{nn} - a_{1n} \end{pmatrix}$$

contains a $n \times n$ non-singular submatrix obtained by deleting one of the second n columns. This fact is equivalent to that the submatrix of $\Omega'(A_1, \ldots, A_n)$ obtained by deleting the first row and the first column has maximum rank $n-1$. This last fact means that $\overrightarrow{A_1A_2}, \ldots, \overrightarrow{A_1A_n}$ are linearly independent, i.e. $A_1, \ldots, A_n$ are affinely independent. □

Proposition 3.40 *Let $A_1, \ldots, A_n$ be n affinely independent points of $\mathbb{A}^n(K)$ ($n \geq 2$), where $A_i = (a_{i1}, \ldots, a_{in})$, $i = 1, \ldots, n$, then the hyperplane $\langle A_1, \ldots, A_n \rangle$ has the equation*

$$\begin{vmatrix} 1 & x_1 & \cdots & x_n \\ 1 & a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \vdots \\ 1 & a_{n1} & \cdots & a_{nn} \end{vmatrix} = 0. \tag{3.23}$$

Proof By Corollary 3.17 the subspace $\mathcal{H} := \langle A_1, \ldots, A_n \rangle$ is a hyperplane. On the other hand, by Lemma 3.39, the polynomial

$$\begin{vmatrix} 1 & x_1 & \cdots & x_n \\ 1 & a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \vdots \\ 1 & a_{n1} & \cdots & a_{nn} \end{vmatrix} \tag{3.24}$$

has total degree 1 and represents a hyperplane $\mathcal{H}'$. Since $A_i \in \mathcal{H}'$, $i = 1, \ldots, n$, $\mathcal{H} \subset \mathcal{H}'$ and hence $\mathcal{H} = \mathcal{H}'$ because $\mathcal{H}$ and $\mathcal{H}'$ are two hyperplanes. □

We have seen that a hyperplane of $\mathbb{A}^n(K)$ has an equation of the form (3.20) and this equation is essentially unique as stated in the following

Lemma 3.41 *Let $b_1x_1 + \cdots + b_nx_n = c$ and $b'_1x_1 + \cdots + b'_nx_n = c'$ be two equations that define the same hyperplane, $\mathcal{A}$ of $\mathbb{A}^n(K)$ (with $n \geq 2$ and $(b_1, \ldots, b_n)$, $(b'_1, \ldots, b'_n) \in K^n \setminus \{(0, \ldots, 0)\}$). Then there exists $d \in K \setminus \{0\}$ such that $b'_i = db_i$, $i = 1, \ldots, n$, and $c' = dc$.*

Proof First suppose $c \neq 0$. Then $c' \neq 0$, otherwise $c' = 0$ implies $(0, \ldots, 0) \in \mathcal{A}$ and $c = 0$. Furthermore, for every i such that $b_i \neq 0$ the point $(0, \ldots, 0, cb_i^{-1}, 0, \ldots, 0) \in \mathcal{A}$. Because the n-tuple $(0, \ldots, 0, cb_i^{-1}, 0, \ldots, 0)$ solves the second equation, we get $b'_i cb_i^{-1} = c'$ so that $\dfrac{b'_i}{b_i} = \dfrac{c'}{c}$ for every i with $b_i \neq 0$. In particular, $b_i \neq 0$ implies $b'_i \neq 0$. Interchanging b_i with b'_i we obtain the reciprocal assertion. So our thesis follows if $c \neq 0$.

If $c = 0$ (and consequently $c' = 0$), let i be an index such that $b_i \neq 0$ and let $j \neq i$ be any other index. Then the n-tuple $(\lambda_1, \ldots, \lambda_n)$, where $\lambda_i = -b_j$, $\lambda_j = b_i$ and $\lambda_k = 0$, $\forall k \neq i, j$, solves the equation $b_1x_1 + \cdots + b_nx_n = 0$ and

hence $(\lambda_1, \ldots, \lambda_n) \in \mathcal{A}$. Therefore it solves also $b'_1 x_1 + \cdots + b'_n x_n = 0$ so that $b'_i(-b_j) + b'_j b_i = 0$. Our proof is now complete. □

Proposition 3.42 *Let $\mathcal{A}$ be a k-dimensional affine subspace of $\mathbb{A}^n(K)$ $(0 \leq k \leq n-1,\ n \geq 2)$ whose implicit equations are*

$$F_i(x_1, \ldots, x_n) := \sum_{j=1}^{n} b_{ij} x_j - c_i = 0, \qquad i = k+1, \ldots, n,$$

with $\operatorname{rank}\left(\left(b_{ij}\right)_{\substack{i=k+1,\ldots,n \\ j=1,\ldots,n}}\right) = n-k$. *For every hyperplane $\mathcal{H}$ of $\mathbb{A}^n(K)$ containing $\mathcal{A}$ there exists $(\lambda_{k+1}, \ldots, \lambda_n) \in K^{n-k} \setminus (0, \ldots, 0)$ such that $\mathcal{H}$ has the equation*

$$\lambda_{k+1} F_{k+1}(x_1, \ldots, x_n) + \cdots + \lambda_n F_n(x_1, \ldots, x_n) = 0.$$

Proof Let $\mathcal{A} = A + D(\mathcal{A})$, with $A = (a_1, \ldots, a_n)$. The director subspace $D(\mathcal{A})$ is a K-vector subspace of dimension k of K^n. By Corollary 1.107 we have $\dim_K(D(\mathcal{A})^0) = n-k$ (see also Definition 1.106). Since the linear functionals on K^n

$$f_i(x_1, \ldots, x_n) = \sum_{j=1}^{n} b_{ij} x_j = F_i + c_i, \qquad i = k+1, \ldots, n,$$

with $c_i = f_i(a_1, \ldots, a_n)$, belong to $D(\mathcal{A})^0$ for $i = k+1, \ldots, n$, and $\operatorname{rank}\left(\left(b_{ij}\right)_{\substack{i=k+1,\ldots,n \\ j=1,\ldots,n}}\right) = n-k$ is equivalent to linear independence of $f_{k+1}, \ldots, f_n$, it follows that $f_{k+1}, \ldots, f_n$ is a basis of $D(\mathcal{A})^0$.

On the other hand, every hyperplane $\mathcal{H}$ which contains $\mathcal{A}$ has an equation of the form

$$F(x_1, \ldots, x_n) = f(x_1, \ldots, x_n) - f(a_1, \ldots, a_n) = 0,$$

with $f \in D(\mathcal{A})^0$. Since $f_{k+1}, \ldots, f_n$ is a basis of $D(\mathcal{A})^0$, there are $\lambda_{k+1}, \ldots, \lambda_n \in K$ such that $f = \lambda_{k+1} f_{k+1} + \cdots + \lambda_n f_n$. Because $c_i = f_i(a_1, \ldots, a_n)$, we get $f(a_1, \ldots, a_n) = \lambda_{k+1} c_{k+1} + \cdots + \lambda_n c_n$ which implies $F = \lambda_{k+1} F_{k+1} + \cdots + \lambda_n F_n$. □

Definition 3.43 Under the notation of Proposition 3.42, the set of hyperplanes of $\mathbb{A}^n(K)$ whose intersection is an affine subspace $\mathcal{A}$ is called an *affine linear system* of hyperplanes with base locus $\mathcal{A}$. If $\dim \mathcal{A} = n-2$ we shall use the term *affine pencil of hyperplanes* generated by $\mathcal{A}$; if $\mathcal{A} = P \in \mathbb{A}^2(K)$ we call the set of all lines through P a *pencil* with base locus P.

3.2 Affine Morphisms

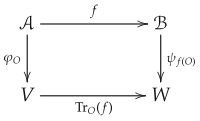
Definition 3.44 Let $(\mathcal{A}, V, \varphi)$ and $(\mathcal{B}, W, \psi)$ be two affine spaces on K, and $f\colon \mathcal{A} \to \mathcal{B}$ a map. Fixing $O \in \mathcal{A}$, from Definition 3.1 (and Proposition 3.2) the maps $\varphi_O\colon \mathcal{A} \to V$ and $\psi_{f(O)}\colon \mathcal{B} \to W$ are bijective. The *trace map with respect to point* O, $\mathrm{Tr}_O(f)\colon V \to W$, is defined by

$$\mathrm{Tr}_O(f) := \psi_{f(O)} \circ f \circ \varphi_O^{-1}.$$

In other words, for every $A \in \mathcal{A}$, we have

$$\mathrm{Tr}_O(f)(\overrightarrow{OA}) = \overrightarrow{f(O)f(A)}, \tag{3.25}$$

i.e. the diagram



is commutative.

If there exists a point $O \in \mathcal{A}$ such that $\mathrm{Tr}_O(f)\colon V \to W$ is a linear map, we call $f\colon \mathcal{A} \to \mathcal{B}$ a *morphism of affine spaces* or *affine map* and we write $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$.

A morphism of affine spaces $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ is called an *affine isomorphism* if there exists a morphism of affine spaces $g\colon (\mathcal{B}, W, \psi) \to (\mathcal{A}, V, \varphi)$ such that $g \circ f = \mathrm{id}_{\mathcal{A}}$ and $f \circ g = \mathrm{id}_{\mathcal{B}}$ (so that $g = f^{-1}$). An isomorphism of affine spaces $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{A}, V, \varphi)$ is called an *affine automorphism* of $(\mathcal{A}, V, \varphi)$. The set of all affine automorphisms of $(\mathcal{A}, V, \varphi)$ will be denoted by $\mathrm{Aut}_K(\mathcal{A}, V, \varphi)$ or simply by $\mathrm{Aut}_K(\mathcal{A})$. We will see that $\mathrm{Aut}_K(\mathcal{A}, V, \varphi)$ is a group which lies at the base of the geometry of the affine space $(\mathcal{A}, V, \varphi)$. If $(\mathcal{A}, V, \varphi) = \mathbb{A}(V)$ we simply write $\mathrm{Aut}_K(\mathbb{A}(V))$.

The following lemma shows that Definition 3.44 does not depend on the choice of $O \in \mathcal{A}$.

Lemma 3.45 *Under the assumptions and notation of* Definition 3.44*, suppose that there is* $O \in \mathcal{A}$ *such that* $\mathrm{Tr}_O(f)\colon V \to W$ *is a linear map. Then for every point* $O' \in \mathcal{A}$ *we have* $\mathrm{Tr}_O(f) = \mathrm{Tr}_{O'}(f)$.

Proof Since $\overrightarrow{OA} = \overrightarrow{OO'} + \overrightarrow{O'A}$ and Tr_O is a linear map, for every $A \in \mathcal{A}$ we have

$$\begin{aligned}\overrightarrow{f(O)f(O')} + \overrightarrow{f(O')f(A)} &= \overrightarrow{f(O)f(A)} \\ &= \mathrm{Tr}_O(f)(\overrightarrow{OA}) \\ &= \mathrm{Tr}_O(f)(\overrightarrow{OO'} + \overrightarrow{O'A}) \\ &= \mathrm{Tr}_O(f)(\overrightarrow{OO'}) + \mathrm{Tr}_O(f)(\overrightarrow{O'A}) \\ &= \overrightarrow{f(O)f(O')} + \mathrm{Tr}_O(f)(\overrightarrow{O'A}),\end{aligned}$$

so that $\mathrm{Tr}_O(f)(\overrightarrow{O'A}) = \overrightarrow{f(O')f(A)}$, $\forall A \in \mathcal{A}$, i.e. $\mathrm{Tr}_O(f) = \mathrm{Tr}_{O'}(f)$. □

Definition 3.46 Let $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ be an affine map and O any point of $\mathcal{A}$. By Lemma 3.45 the linear map $\mathrm{Tr}_O(f)$ does not depend on point O so that we can define

$$\mathrm{Tr}(f)\colon V \to W, \qquad \mathrm{Tr}(f) := \mathrm{Tr}_O(f),$$

and call it the *trace of* f. Therefore we can write, for any fixed $O \in \mathcal{A}$

$$f(A) = f(O) + \mathrm{Tr}(f)(\overrightarrow{OA}), \qquad \forall\, A \in \mathcal{A}.$$

Remark 3.47 Let $(\mathcal{A}, V, \varphi)$ and $(\mathcal{B}, W, \psi)$ two affine spaces, $O \in \mathcal{A}$ and $O' \in \mathcal{B}$ two fixed points. Every linear map $f\colon V \to W$ induces an affine map

$$\widetilde{f}_{O,O'}\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi), \qquad \widetilde{f}_{O,O'}(A) = B \quad \text{such that} \quad f(\overrightarrow{OA}) = \overrightarrow{O'B}.$$

It is immediately seen that $\mathrm{Tr}_O(\widetilde{f}_{O,O'}) = f$.

Proposition 3.48 *Let* $(\mathcal{A}, V, \varphi)$, $(\mathcal{B}, W, \psi)$ *and* $(\mathcal{C}, U, \chi)$ *be affine spaces.*

(iv) *The identity map* $\mathrm{id}_{\mathcal{A}}\colon (\mathcal{A}, V, \varphi) \to (\mathcal{A}, V, \varphi)$ *is an affine automorphism of* $(\mathcal{A}, V, \varphi)$).

(ii) *If* $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *and* $g\colon (\mathcal{B}, W, \psi) \to (\mathcal{C}, U, \chi)$ *are affine maps, then we have* $\mathrm{Tr}(g \circ f) = \mathrm{Tr}(g) \circ \mathrm{Tr}(f)$. *In particular,* $g \circ f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{C}, U, \chi)$ *is an affine map.*

(iii) *An affine map* $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *is*

(a) *injective if and only if* $\mathrm{Tr}(f)\colon V \to W$ *is injective;*
(b) *surjective if and only if* $\mathrm{Tr}(f)\colon V \to W$ *is surjective;*
(c) *bijective if and only if* $\mathrm{Tr}(f)\colon V \to W$ *is bijective.*

In particular, an affine map $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *is an affine isomorphism if and only if* $\mathrm{Tr}(f)$ *is isomorphism of* K*-vector spaces.*

(iv) $\mathrm{Aut}_K(\mathcal{A}, V, \varphi)$ *is a group with respect to composition of automorphisms. Furthermore, the trace map* $\mathrm{Tr}\colon \mathrm{Aut}_K(\mathcal{A}, V, \varphi) \to \mathrm{GL}_K(V)$ *is a surjective homomorphism of groups.*

Proof (i) Obvious. (ii) and (iii) left to the reader as an esercise. (iv) It follows from (ii), (iii) and Remark 3.47. □

Lemma 3.49 *Let* $(\mathcal{A}, V, \varphi)$ *and* $(\mathcal{B}, W, \psi)$ *be an affine spaces. A map* $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *is an affine morphism if and only if for every* A, B, C, $D \in \mathcal{A}$ *and* $k \in K$

$$\overrightarrow{AB} = k\,\overrightarrow{CD} \quad \Longrightarrow \quad \overrightarrow{f(A)f(B)} = k\,\overrightarrow{f(C)f(D)}. \tag{3.26}$$

Proof If f is an affine morphism, then

$$\overrightarrow{f(A)f(B)} = \mathrm{Tr}(f)(\overrightarrow{AB}) = \mathrm{Tr}(f)(k\,\overrightarrow{CD}) = k\,\mathrm{Tr}(f)(\overrightarrow{CD}) = k\,\overrightarrow{f(C)f(D)}.$$

Conversely, if $v \in V$ we can write $v = \overrightarrow{AB}$ so that $\mathrm{Tr}(f)(v) = \mathrm{Tr}(f)(\overrightarrow{AB}) := \overrightarrow{f(A)f(B)}$. If $v = \overrightarrow{A'B'}$, then $\mathrm{Tr}(f)(v) = \overrightarrow{f(A')f(B')} = \overrightarrow{f(A)f(B)}$ by (3.26), choosing $k = 1$, because $\overrightarrow{AB} = v = \overrightarrow{A'B'}$. Therefore $\mathrm{Tr}(f)$ is well-defined. It remains to prove that $\mathrm{Tr}(f)$ is linear. First we note that $\mathrm{Tr}(f)(0_V) = \mathrm{Tr}(f)(\overrightarrow{AA}) = \overrightarrow{f(A)f(A)} = 0_W$. Next, given v, $v' \in V$, let us write $v = \overrightarrow{AB}$ and $v' = \overrightarrow{BC}$. Then

$$\begin{aligned}\mathrm{Tr}(f)(v + v') &= \mathrm{Tr}(f)(\overrightarrow{AB} + \overrightarrow{BC}) = \mathrm{Tr}(f)(\overrightarrow{AC}) = \overrightarrow{f(A)f(C)} \\ &= \overrightarrow{f(A)f(B)} + \overrightarrow{f(B)f(C)} = \mathrm{Tr}(f)(v) + \mathrm{Tr}(f)(v').\end{aligned}$$

Finally, given $v \in V$ and $k \in K$, we again write $v = \overrightarrow{AD}$ and put $B = A + k\overrightarrow{AD}$. Thus $\overrightarrow{AB} = k\overrightarrow{AD}$ and we have

$$\begin{aligned}\mathrm{Tr}(f)(kv) &= \mathrm{Tr}(f)(k\overrightarrow{AD}) = \mathrm{Tr}(f)(\overrightarrow{AB}) = \overrightarrow{f(A)f(B)} \\ &= k\,\overrightarrow{f(A)f(D)} = k\,\mathrm{Tr}(f)(\overrightarrow{AD}) = k\,\mathrm{Tr}(f)(v).\end{aligned}$$

□

Proposition 3.50 *Let* $(\mathcal{A}, V, \varphi)$ *and* $(\mathcal{B}, W, \psi)$ *be two affine spaces and* $f\colon \mathcal{A} \to \mathcal{B}$ *a map. Then* f *is an affine map if and only if for every finite set of points* $\{A_1, \ldots, A_m\} \subset \mathcal{A}$ *and for every finite set of scalars* $\{a_1, \ldots, a_m\} \subset K$ *such that* $a_1 + \cdots + a_m = 1$, *we have*

$$f(a_1A_1 + \cdots + a_mA_m) = a_1 f(A_1) + \cdots + a_m f(A_m).$$

Proof Let f be an affine map. Take $A := a_1A_1 + \cdots + a_mA_m$ and $B := a_1 f(A_1) + \cdots + a_m f(A_m)$. We have to prove that $f(A) = B$, i.e. $\overrightarrow{f(O)f(A)} = \overrightarrow{f(O)B}$. Since $\overrightarrow{OA} = a_1\overrightarrow{OA_1} + \cdots + a_m\overrightarrow{OA_m}$ we have (because $\mathrm{Tr}_O(f)$ is a linear map)

$$\begin{aligned}\overrightarrow{f(O)f(A)} = \mathrm{Tr}_O(f)(\overrightarrow{OA}) &= a_1\,\mathrm{Tr}_O(f)(\overrightarrow{OA_1}) + \cdots + a_m\,\mathrm{Tr}_O(f)(\overrightarrow{OA_m})\\ &= a_1\overrightarrow{f(O)f(A_1)} + \cdots + a_m\overrightarrow{f(O)f(A_m)}\\ &= \overrightarrow{f(O)B}.\end{aligned}$$

The reverse implication is proved in a completely similar manner. □

Proposition 3.51 *Under the assumptions and notation of* Definition 3.44, *if* $\mathrm{char}(K) \neq 2$, $f : \mathcal{A} \to \mathcal{B}$ *is an affine map if and only if*

$$f((1-t)A + tB) = (1-t)f(A) + tf(B), \qquad \forall\, A, B \in \mathcal{A},\ \forall\, t \in K. \tag{3.27}$$

Proof One implication is a special case of Proposition 3.50. Suppose now that (3.27) holds. Fixing $O \in \mathcal{A}$ we have to prove that $\mathrm{Tr}_O(f)$ is linear. Let $v \in V$ and $t \in K$, then $v = \overrightarrow{OM}$ with $M \in \mathcal{A}$. Taking $N = (1-t)O + tM$ we have $\overrightarrow{ON} = (1-t)\overrightarrow{OO} + t\overrightarrow{OM} = tv$ so that $\mathrm{Tr}_O(f)(tv) = \mathrm{Tr}_O(f)(\overrightarrow{ON})$. Since $f(N) = (1-t)f(O) + tf(M)$, $\overrightarrow{f(O)f(N)} = t\overrightarrow{f(O)f(M)}$, hence

$$\begin{aligned}\mathrm{Tr}_O(f)(tv) = \mathrm{Tr}_O(f)(\overrightarrow{ON}) &= \overrightarrow{f(O)f(N)}\\ &= t\overrightarrow{f(O)f(M)} = t\,\mathrm{Tr}_O(f)(\overrightarrow{OM}) = t\,\mathrm{Tr}_O(f)(v).\end{aligned}$$

Now let $v = \overrightarrow{OM}$ and $v' = \overrightarrow{OM'}$. Taking $N = \frac{1}{2}M + \frac{1}{2}M'$ we have $\overrightarrow{ON} = \frac{1}{2}\overrightarrow{OM} + \frac{1}{2}\overrightarrow{OM'}$ so that $v + v' = \overrightarrow{OM} + \overrightarrow{OM'} = 2\overrightarrow{ON}$. Since $f(N) = \frac{1}{2}f(M) + \frac{1}{2}f(M')$ we get $\overrightarrow{f(O)f(N)} = \frac{1}{2}\overrightarrow{f(O)f(M)} + \frac{1}{2}\overrightarrow{f(O)f(M')}$. Hence

$$\begin{aligned}\mathrm{Tr}_O(f)(v+v') = \mathrm{Tr}_O(f)(2\overrightarrow{ON}) &= 2\,\mathrm{Tr}_O(f)(\overrightarrow{ON}) = 2\overrightarrow{f(O)f(N)}\\ = 2\left[\frac{1}{2}\overrightarrow{f(O)f(M)} + \frac{1}{2}\overrightarrow{f(O)f(M')}\right] &= \overrightarrow{f(O)f(M)} + \overrightarrow{f(O)f(M')}\\ &= \mathrm{Tr}_O(f)(v) + \mathrm{Tr}_O(f)(v').\end{aligned}$$

Therefore $\mathrm{Tr}_O(f)$ is linear. □

Corollary 3.52 *Let* $(\mathcal{A}, V, \varphi)$ *and* $(\mathcal{B}, W, \psi)$ *be two affine spaces and* $\{A_0, A_1, \ldots, A_n\}$ *an affine frame of* $(\mathcal{A}, V, \varphi)$.

(i) *If* $\sigma : (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *is an affine isomorphism, then* $\{\sigma(A_0), \sigma(A_1), \ldots, \sigma(A_n)\}$ *is an affine frame of* $(\mathcal{B}, W, \psi)$.

(ii) *Let* $\{B_0, B_1, \ldots, B_n\}$ *be an affine frame of* $(\mathcal{B}, W, \psi)$. *Then there exists a unique affine isomorphism* $\sigma : (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *such that* $\sigma(A_i) = B_i$, $i = 0, 1, \ldots, n$.

(iii) *If* $\sigma : (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ *is an affine isomorphism and* A, B *and* C *are three distinct collinear points of* $\mathcal{A}$, *then* $\sigma(A)$, $\sigma(B)$ *and* $\sigma(C)$ *are three distinct collinear points of* $\mathcal{B}$ *and*

$$\mathrm{ar}(A, B, C) = \mathrm{ar}(\sigma(A), \sigma(B), \sigma(AC)).$$

Proof

(i) It suffices to observe that $\{\overrightarrow{\sigma(A_0)\sigma(A_1)}, \ldots, \overrightarrow{\sigma(A_0)\sigma(A_1)}\}$ is a basis of W since $\{\overrightarrow{A_0A_1}, \ldots, \overrightarrow{A_0A_1}\}$ is a basis of V.

(ii) Since $\{A_0, A_1, \ldots, A_n\}$ is an affine frame every point $A \in \mathcal{A}$ is uniquely written in the form $A = a_0A_0 + a_1A_1 + \cdots + a_nA_n$, with $a_0, a_1, \ldots, a_n \in K$ and $a_0 + a_1 + \cdots + a_n = 1$ (Corollary 3.25). If $\sigma : \mathcal{A} \to \mathcal{A}$ is an affine isomorphism such that $\sigma(A_i) = B_i$, $i = 0, 1, \ldots, n$, by Proposition 3.50 we have the equality

$$\sigma(A) = a_0B_0 + a_1B_1 + \cdots + a_nB_n,$$

which uniquely determines σ. Furthermore, σ is an affine isomorphism by Proposition 3.50.

(iii) If $C = (1-t)A + tB$ then $\sigma(C) = (1-t)\sigma(A) + t\sigma(B)$ by Proposition 3.50. □

Examples 3.53

1. If $\mathcal{A}'$ is a non-empty affine subspace of an affine space $(\mathcal{A}, V, \varphi)$, then the inclusion $i : \mathcal{A}' \hookrightarrow \mathcal{A}$ is an affine map.
2. An affine automorphism $\alpha \in \mathrm{Aut}_K(\mathcal{A}, V, \varphi)$ is called an *affine homothety* if there exists a point $O \in \mathcal{A}$ and a scalar $a \in K^*$ such that $\alpha(O) = O$ and $\overrightarrow{O\alpha(A)} = a\overrightarrow{OA}$, $\forall\, A \in \mathcal{A}$. Hence we have

$$\mathrm{Tr}_O(\alpha)(\overrightarrow{OA}) = \overrightarrow{\alpha(O)\alpha(A)} = \overrightarrow{O\alpha(A)} = a\overrightarrow{OA}.$$

The point O, necessarily unique if $a \neq 1$, is called the *centre* of the homothety α. We shall denote the set of all homotheties of centre O by $\mathcal{H}_O(\mathcal{A}, V, \varphi)$ which is easily seen to be a subgroup of $\mathrm{Aut}_K(\mathcal{A}, V, \varphi)$ isomorphic to the multiplicative group K^*.

3. Let $v \in V$. An affine automorphism $\tau_v \in \mathrm{Aut}_K(\mathcal{A}, V, \varphi)$ is called an *affine translation by the vector* v if $\overrightarrow{A\tau_v(A)} = v, \forall A \in \mathcal{A}$. Hence

$$\mathrm{Tr}_O(\tau_v)(\overrightarrow{OA}) = \overrightarrow{\tau_v(O)\tau(A)} = \overrightarrow{\tau_v(O)O} + \overrightarrow{OA} + \overrightarrow{A\tau_v(A)} = -v + \overrightarrow{OA} + v = \overrightarrow{OA},$$

i.e. $\mathrm{Tr}(\tau_v) = \mathrm{id}_V$. The set of affine translations will be denoted by $\mathrm{T}_K(\mathcal{A}, V, \varphi)$ or simply by $\mathrm{T}_K(\mathcal{A})$. It is immediate to see that $\mathrm{T}_K(\mathcal{A}, V, \varphi) = \mathrm{Ker}(\mathrm{Tr})$ so that $\mathrm{T}_K(\mathcal{A}, V, \varphi)$ is a normal subgroup of $\mathrm{Aut}_K(\mathcal{A}, V, \varphi)$.
4. Let $(\mathcal{A}, V, \varphi)$ be an affine space and $O \in \mathcal{A}$ a fixed point. We can see $\varphi_O : \mathcal{A} \to V$ as an affine isomorphism $\varphi_O : (\mathcal{A}, V, \varphi) \to \mathbb{A}(V)$ where $\mathbb{A}(V)$ is the affine space associated with V (see Example 3.4). In this case we have $\varphi_O(O) = 0_V$ and $\mathrm{Tr}_O(\varphi_O) = \mathrm{id}_V$. For this reason we can always refer to that standard example.

Proposition 3.54 *Let $f : (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ be an affine map. If $\mathcal{A}_1$ and $\mathcal{B}_1$ are two affine subspaces of $\mathcal{A}$ and $\mathcal{B}$, respectively, then $f(\mathcal{A}_1)$ and $f^{-1}(\mathcal{B}_1)$ are affine subspaces of $\mathcal{B}$ and $\mathcal{A}$, respectively. Furthermore $f_{|\mathcal{A}_1} : \mathcal{A}_1 \to f(\mathcal{A}_1)$ is an affine map.*

Proof One can verify the following equalities:

$$f(\mathcal{A}_1) = f(A) + \mathrm{Tr}(f)(D(\mathcal{A}_1)), \quad \text{if } \mathcal{A}_1 = A + D(\mathcal{A}_1),$$
$$f^{-1}((\mathcal{B}_1) = A + \mathrm{Tr}(f)^{-1}(D(\mathcal{B}_1)) \quad \text{if } \mathcal{B}_1 = B + D(\mathcal{B}_1),\ f(A) = B.$$

Since $\mathrm{Tr}(f)(D(\mathcal{A}_1))$ and $\mathrm{Tr}(f)^{-1}(D(\mathcal{B}_1))$ are vector subspaces of W and V, respectively, both assertions are proved. The last statement is now obvious. □

Definition 3.55 Let $O \in (\mathcal{A}, V, \varphi)$. If $\alpha \in \mathrm{Aut}_K(\mathcal{A})$ satisfies the equality $\alpha(O) = O$ we shall say that O is a *fixed point* of α. It is clear that

$$\mathrm{Aut}_K(\mathcal{A}, O) := \{\alpha \in \mathrm{Aut}_K(\mathcal{A}) \mid \alpha(O) = O\}$$

is a subgroup of $\mathrm{Aut}_K(\mathcal{A})$ such that

$$\mathrm{Tr}_{|\mathrm{Aut}_K(\mathcal{A}, O)} : \mathrm{Aut}_K(\mathcal{A}, O) \to \mathrm{GL}_K(V)$$

is an isomorphism of groups.

We recall the following definition from the theory of groups.

Definition 3.56 Let G be a (multiplicative) group, $T \subset G$ a normal subgroup of G and $H \subset G$ a subgroup of G. If $T \cap H = \{1_G\}$ and $G = TH := \{th \mid t \in T,\ h \in H\}$, we shall say that G is a *semidirect product of T by H*, denoted by $G = T \rtimes H$. We have the following equivalences:

(a) $G = T \rtimes H$.
(b) Every $g \in G$ is written in a unique way as $g = th$, with $t \in T$ and $h \in H$.

(c) Every $g \in G$ is written in a unique way as $g = h't'$, with $t' \in T$ and $h' \in H$.
(d) $G = H \rtimes T$.

We note that the equalities $g = th = h't'$ imply $h = h'$ but $t \neq t'$, in general (find an example using Proposition 3.57).

Proposition 3.57 *For every point $O \in (\mathcal{A}, V, \varphi)$, we have*

$$\mathrm{Aut}_K(\mathcal{A}) = \mathrm{T}_K(\mathcal{A}) \rtimes \mathrm{Aut}_K(\mathcal{A}, O).$$

In particular, for every $\alpha \in \mathrm{Aut}_K(\mathcal{A})$, we have the unique decomposition $\alpha = \beta \circ \gamma$, with $\beta \in \mathrm{T}_K(\mathcal{A})$ and $\gamma \in \mathrm{Aut}_K(\mathcal{A}, O)$.

Proof For every point $O \in (\mathcal{A}, V, \varphi)$ we have:

$$\mathrm{Aut}_K(\mathcal{A}, O) \cap \mathrm{T}_K(\mathcal{A}) = \{\mathrm{id}_{\mathcal{A}}\}, \qquad \mathrm{Aut}_K(\mathcal{A}) = \mathrm{T}_K(\mathcal{A})\,\mathrm{Aut}_K(\mathcal{A}, O).$$

The first equality is obvious. If $\alpha \in \mathrm{Aut}_K(\mathcal{A})$ is any automorphism there exists one and only one translation $\beta \in \mathrm{T}_K(\mathcal{A})$ such that $\beta(O) = \alpha(O)$ (i.e. $\overrightarrow{A\beta(A)} := \overrightarrow{O\alpha(O)}$, for all $A \in \mathcal{A}$). Then O is a fixed point of $\gamma := \beta^{-1} \circ \alpha$, hence $\gamma \in \mathrm{Aut}_K(\mathcal{A}, O)$. Therefore $\alpha = \beta \circ \gamma$, with $\beta \in \mathrm{T}_K(\mathcal{A})$ and $\gamma \in \mathrm{Aut}_K(\mathcal{A}, O)$. □

Remark 3.58 An affine map $f\colon (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ induces an affine map $f'\colon \mathbb{A}(V) \to \mathbb{A}(W)$ with the property that the diagram, having fixed the points $O \in \mathcal{A}$ and $O' \in \mathcal{B}$,

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{f} & \mathcal{B} \\ {\scriptstyle \varphi_O}\downarrow & & \downarrow{\scriptstyle \psi_{O'}} \\ V & \xrightarrow[f']{} & W \end{array}$$

is commutative. Thus

$$f'(v) = f'(\overrightarrow{OA}) := \overrightarrow{O'f(A)}, \qquad \forall\, A \in \mathcal{A}.$$

We now have

$$\begin{aligned} \mathrm{Tr}_{0_V}(f')(v) = \mathrm{Tr}_{0_V}(f')(\overrightarrow{OA}) &= \overrightarrow{f'(0_V)f'(\overrightarrow{OA})} = f'(\overrightarrow{OA}) - f'(0_V) \\ &= \overrightarrow{O'f(A)} - \overrightarrow{O'f(O)} = \overrightarrow{f(O)O'} + \overrightarrow{O'f(A)} \\ &= \overrightarrow{f(O)f(A)} = \mathrm{Tr}_O(f)(\overrightarrow{OA}) = \mathrm{Tr}_O(f)(v), \end{aligned}$$

hence $\mathrm{Tr}(f) = \mathrm{Tr}(f')$. It follows that, without loss of generality, we can assume that $(\mathcal{A}, V, \varphi) = \mathbb{A}(V)$, $(\mathcal{B}, W, \psi) = \mathbb{A}(W)$ and $f = f'$. Hence $\varphi(v, v') = v' - v$ and $\psi(w, w') = w' - w$, $\forall\, v, v' \in V$ and $\forall\, w, w' \in W$. If we take $O = 0_V$ and

$O' = f(O) = f(0_V)$, from Definition 3.44 we have $\mathrm{Tr}(f)(A) = f(A) - f(0_V)$, i.e. $f = f(0_V) + \mathrm{Tr}(f)$. Hence f is completely determined by $f(0_V) \in \mathbb{A}(W)$ and by its trace $\mathrm{Tr}(f)\colon V \to W$.

Proposition 3.59 *Let $f\colon \mathbb{A}(V) \to \mathbb{A}(W)$ be an affine map where $\dim_K(V) = n$ and $\dim_K(W) = m$. Let $(x_1, \ldots, x_n)$ be the coordinates of $A \in \mathbb{A}(V)$ relative to a basis $\mathcal{B}_V = \{v_1, \ldots, v_m\}$ of A and $(y_1, \ldots, y_m)$ the coordinates of $f(A) \in \mathbb{A}(W)$ relative to a basis $\mathcal{B}_W = \{w_1, \ldots, w_n\}$ of W, i.e.*

$$A = \sum_{j=1}^{n} x_j v_j, \qquad f(A) = \sum_{i=1}^{m} y_i w_i.$$

Furthermore let $f(0_V) = b_1 w_1 + \cdots + b_m w_m$ and let

$$M_{\mathrm{Tr}(f)}^{\mathcal{B}_V, \mathcal{B}_W} = \big(a_{ij}\big)_{\substack{i=1,\ldots,m \\ j=1,\ldots,n}}$$

be the matrix associated with $\mathrm{Tr}(f)$ with respect to $\mathcal{B}_V$ and $\mathcal{B}_W$. Then we have

$$y_i = b_i + \sum_{j=1}^{m} a_{ij} x_j, \qquad i = 1, \ldots, m, \tag{3.28}$$

Conversely, every map $f\colon \mathbb{A}(V) \to \mathbb{A}(W)$ defined by (3.28) is an affine map whose trace map $\mathrm{Tr}(f)$ is given by

$$\mathrm{Tr}(f)(x_1, \ldots, x_n) = (z_1, \ldots, z_m), \qquad z_i = \sum_{j=1}^{n} a_{ij} x_j, \qquad i = 1, \ldots, m.$$

Proof Indeed, $\mathrm{Tr}(f)(A) = \mathrm{Tr}(f)(\overrightarrow{0_V A}) = \overrightarrow{f(0_V) f(A)}$ for all $A \in \mathbb{A}(V)$. The converse statement is immediate. □

An immediate consequence of Propositions 3.59, 3.48 and Corollary 1.90 is the following result.

Theorem 3.60 *Every affine automorphism $\sigma \in \mathrm{Aut}_K(\mathbb{A}^n(K))$ of the standard affine space $\mathbb{A}^n(K)$ is given by formulas*

$$\sigma(x_1, \ldots, x_n) = (y_1, \ldots, y_n), \quad \textit{with} \quad y_i = \sum_{j=1}^{n} a_{ij} x_j + b_i, \quad i = 1, \ldots, n, \tag{3.29}$$

where $\left(a_{ij}\right)_{i,j=1,\dots,n} \in \mathrm{GL}_n(K)$ *is a non-singular matrix and* $(b_1, \dots, b_n) = \sigma(O)$, *with* $O = (0, \dots, 0)$. *The trace map* $\mathrm{Tr}(\sigma)$ *is a linear automorphism of* K^n *given by*

$$\mathrm{Tr}(\sigma)(x_1, \dots, x_n) = (z_1, \dots, z_n), \quad \textit{with} \quad z_i = \sum_{j=1}^{n} a_{ij} x_j, \quad i = 1, \dots, n. \tag{3.30}$$

Conversely, for every non-singular matrix $\left(a_{ij}\right)_{i,j=1,\dots,n} \in \mathrm{GL}_n(K)$ *and for all* $(b_1, \dots, b_n) \in K^n$, *the map* (3.29) *is an affine automorphism of* $\mathbb{A}^n(K)$.

Examples 3.61 The affine translations of $\mathrm{Aut}_K(\mathbb{A}^n(K))$ have equations of the form

$$y_i = x_i + b_i, \qquad i = 1, \dots, n,$$

where $(b_1, \dots, b_n) \in K^n$ is a fixed n-tuple.

The affine homotheties of $\mathrm{Aut}_K(\mathbb{A}^n(K))$ of centre $O = (0, \dots, 0)$ are given by the formulas

$$y_i = a x_i, \qquad i = 1, \dots, n,$$

where $a \in K \setminus \{0\}$ is a fixed scalar.

Definition 3.62 We will denote the group of all affine automorphisms $\mathrm{Aut}_K(\mathbb{A}^n(K))$ by $\mathrm{AGL}_n(K)$, which we will call the *affine general linear group over* K. The normal subgroup of translations will be denoted by $\mathrm{T}_n(K)$ and the one of homotheties of centre O by $\mathcal{H}^n_O(K)$. From Proposition 3.57 we have the decomposition

$$\mathrm{AGL}_n(K) = \mathrm{T}_n(K) \rtimes \mathrm{AGL}_n(K, O),$$

where $\mathrm{AGL}_n(K, O) := \{\alpha \in \mathrm{AGL}_n(K) \mid \alpha(O) = O\}$.

3.2.1 Dimension Theorem

Let $\mathcal{A}$ and $\mathcal{B}$ be two non-empty affine subspaces of an affine space $(\mathcal{U}, V, \varphi)$ of dimension n over K. We can write $\mathcal{A} = A + D(\mathcal{A})$ and $\mathcal{B} = B + D(\mathcal{B})$ with A and B arbitrary points of $\mathcal{A}$ and $\mathcal{B}$, respectively (see Definition 3.8). We denote by $\langle \mathcal{A} \cup \mathcal{B} \rangle$ the affine subspace generated by $\mathcal{A} \cup \mathcal{B}$, i.e. the smallest affine subspace of $\mathcal{U}$ containing $\mathcal{A}$ and $\mathcal{B}$.

Lemma 3.63 *Under the above assumptions and notation we have*

$$\langle \mathcal{A} \cup \mathcal{B} \rangle = A + (K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B})), \tag{3.31}$$

where $K(\overrightarrow{AB})$ *is the vector subspace of* V *generated by* $\overrightarrow{AB}$.

Proof The right– hand side of (3.31) contains both subspaces $\mathcal{A}$ and $\mathcal{B}$ so that

$$\langle \mathcal{A} \cup \mathcal{B} \rangle \subset A + (K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B})) \quad \text{which implies}$$

$$D(\langle \mathcal{A} \cup \mathcal{B} \rangle) \subset K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B}).$$

Conversely, since $A, B \in \mathcal{A} \cup \mathcal{B}$, it follows that $K(\overrightarrow{AB}) \subset D(\langle \mathcal{A} \cup \mathcal{B} \rangle)$. Furthermore we have the obvious inclusions $D(\mathcal{A}) \subset D(\langle \mathcal{A} \cup \mathcal{B} \rangle)$ and $D(\mathcal{B}) \subset D(\langle \mathcal{A} \cup \mathcal{B} \rangle)$ so that

$$K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B}) \subset D(\langle \mathcal{A} \cup \mathcal{B} \rangle).$$

Hence $D(\langle \mathcal{A} \cup \mathcal{B} \rangle) = K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B})$ which implies our thesis. □

Lemma 3.64 *Under the assumptions and notation of* Lemma 3.63, $\mathcal{A} \cap \mathcal{B} \neq \varnothing$ *if and only if* $K(\overrightarrow{AB}) \subset D(\mathcal{A}) + D(\mathcal{B})$.

Proof First suppose that there exists a point $C \in \mathcal{A} \cap \mathcal{B}$. Then $C = A + u = B + v$, with $u \in D(\mathcal{A})$ and $v \in D(\mathcal{B})$, so that $\overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BC} = u - v \in D(\mathcal{A}) + D(\mathcal{B})$, i.e. $K(\overrightarrow{AB}) \subset D(\mathcal{A}) + D(\mathcal{B})$. Conversely, suppose that $K(\overrightarrow{AB}) \subset D(\mathcal{A}) + D(\mathcal{B})$. Then we have $\overrightarrow{AB} = u + v$ with $u \in D(\mathcal{A})$ and $v \in D(\mathcal{B})$, hence $A + u \in \mathcal{A}$, $B - v \in \mathcal{B}$ and $B = A + u + v$ so that $B - v = A + u$ and $A + u = B - v \in \mathcal{A} \cap \mathcal{B}$. □

Corollary 3.65 *If* $A \in \mathcal{A} \cap \mathcal{B}$, *then* $\mathcal{A} \cap \mathcal{B} = A + D(\mathcal{A}) \cap D(\mathcal{B})$ *and* $\langle \mathcal{A} \cup \mathcal{B} \rangle = A + (D(\mathcal{A}) + D(\mathcal{B}))$.

Proof The inclusion $A + D(\mathcal{A}) \cap D(\mathcal{B}) \subset \mathcal{A} \cap \mathcal{B}$ is immediate. Conversely, if $C \in \mathcal{A} \cap \mathcal{B}$ then $C = A + u$ with $u \in D(\mathcal{A})$ and $C = A + v$ with $v \in D(\mathcal{B})$, hence $u = v \in D(\mathcal{A}) \cap D(\mathcal{B})$. The second equality follows from Lemmas 3.63 and 3.64. □

Corollary 3.66 *Let* $\mathcal{A}$ *and* $\mathcal{B}$ *be two affine subspaces of an affine space* $(\mathcal{U}, V, \varphi)$, *then* $\mathcal{A} \cap \mathcal{B}$ *is an affine subspace.*

Now we are able to prove the following fundamental result.

Theorem 3.67 (Dimension Theorem for Affine Subspaces) *Let* $\mathcal{A}$ *and* $\mathcal{B}$ *be two non-empty affine subspaces of an affine space* $(\mathcal{U}, V, \varphi)$ *of dimension* $n \geq 2$ *over* K.

(i) *If* $\mathcal{A} \cap \mathcal{B} \neq \varnothing$, *then* $\dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle = \dim_K \mathcal{A} + \dim_K \mathcal{B} - \dim_K (\mathcal{A} \cap \mathcal{B})$.

(ii) *If* $\mathcal{A} \cap \mathcal{B} = \varnothing$, *then* $\dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle = \dim_K \mathcal{A} + \dim_K \mathcal{B} - \dim_K (D(\mathcal{A}) \cap D(\mathcal{B})) + 1$.

Proof

(i) By Corollary 3.65 we have $D(\langle \mathcal{A} \cup \mathcal{B} \rangle) = D(\mathcal{A}) + D(\mathcal{B})$ and $D(\mathcal{A} \cap \mathcal{B}) = D(\mathcal{A}) \cap D(\mathcal{B})$. By Grassmann's formula (see Corollary 1.76) we have

$$\dim_K (D(\mathcal{A}) + D(\mathcal{B})) = \dim_K (D(\mathcal{A})) + \dim_K (D(\mathcal{B})) - \dim_K (D(\mathcal{A}) \cap D(\mathcal{B})),$$

i.e. $\dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle = \dim_K \mathcal{A} + \dim_K \mathcal{B} - \dim_K (\mathcal{A} \cap \mathcal{B})$.

(ii) Let $A \in \mathcal{A}$ and $B \in \mathcal{B}$. By Lemmas 3.63 and 3.64 we have $D(\langle \mathcal{A} \cup \mathcal{B} \rangle) = K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B})$ and $K(\overrightarrow{AB}) \not\subset D(\mathcal{A}) + D(\mathcal{B})$. Again by Grassmann's formula we have

$$\begin{aligned} \dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle &= \dim_K (K(\overrightarrow{AB}) + D(\mathcal{A}) + D(\mathcal{B})) = 1 + \dim_K (D(\mathcal{A}) + D(\mathcal{B})) \\ &= 1 + \dim_K (D(\mathcal{A})) + \dim_K (D(\mathcal{B})) - \dim_K (D(\mathcal{A}) \cap D(\mathcal{B})), \end{aligned}$$

i.e. $\dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle = \dim_K \mathcal{A} + \dim_K \mathcal{B} - \dim_K (D(\mathcal{A}) \cap D(\mathcal{B})) + 1$.

□

Definition 3.68 Let $\mathcal{A}$ and $\mathcal{B}$ be two non-empty affine subspaces of an affine space $(\mathcal{U}, V, \varphi)$ of dimension $n \geq 2$ over K. We say that $\mathcal{A}$ is *parallel* to $\mathcal{B}$, written $\mathcal{A} \parallel \mathcal{B}$, if $D(\mathcal{A}) \subset D(\mathcal{B})$ or $D(\mathcal{B}) \subset D(\mathcal{A})$. Therefore, two non-empty affine subspaces $\mathcal{A}$ and $\mathcal{B}$ having the same dimension are parallel if and only if $D(\mathcal{A}) = D(\mathcal{B})$ (see Corollary 1.61). The parallelism is clearly an equivalence relation on the set of non-empty affine subspaces of an affine space.

The subspaces $\mathcal{A}$ and $\mathcal{B}$ are *skew* if $\mathcal{A} \cap \mathcal{B} = \varnothing$ and $\mathcal{A} \nparallel \mathcal{B}$, i.e. $\dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle = \dim_K \mathcal{A} + \dim_K \mathcal{B} + 1$.

Proposition 3.69 *Under the notation of* Definition 3.68 *if* $\mathcal{A} \parallel \mathcal{B}$, *then we have* $\mathcal{A} \subseteq \mathcal{B}$, *or* $\mathcal{B} \subseteq \mathcal{A}$, *otherwise* $\mathcal{A} \cap \mathcal{B} = \varnothing$.

Proof Suppose that there exists a point $P \in \mathcal{A} \cap \mathcal{B}$. Since $\mathcal{A} \parallel \mathcal{B}$ we have (for instance) that $D(\mathcal{A}) \subseteq D(\mathcal{B})$. Then $\mathcal{A} = P + D(\mathcal{A})$ and $\mathcal{B} = P + D(\mathcal{B})$, so that $\mathcal{A} \subseteq \mathcal{B}$. □

Proposition 3.70 *Let* $\mathcal{A}$ *be a non-empty affine subspace and* $\mathcal{B}$ *a hyperplane of an affine space* $(\mathcal{U}, V, \varphi)$ *of dimension* $n \geq 2$ *over* K. *If* $\mathcal{A} \cap \mathcal{B} = \varnothing$, *then* $\mathcal{A} \parallel \mathcal{B}$.

Proof Since $\mathcal{B}$ is a hyperplane and $\mathcal{A} \cap \mathcal{B} = \varnothing$, we have $\dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle = n$ by Theorem 3.67. Again by Theorem 3.67 and Grassmann's formula (Corollary 1.76) we have

$$\dim_K (D(\mathcal{A}) + D(\mathcal{B})) = \dim_K D(\mathcal{A}) + \dim_K D(\mathcal{B}) - \dim_K (D(\mathcal{A}) \cap D(\mathcal{B})) =$$

$$\dim_K \mathcal{A} + \dim_K \mathcal{B} - \dim_K (D(\mathcal{A}) \cap D(\mathcal{B})) = \dim_K \langle \mathcal{A} \cup \mathcal{B} \rangle - 1 = n - 1.$$

Since $D(\mathcal{B}) \subseteq D(\mathcal{A}) + D(\mathcal{B})$, we have $D(\mathcal{B}) = D(\mathcal{A}) + D(\mathcal{B})$ so that $D(\mathcal{A}) \subseteq D(\mathcal{B})$ and $\mathcal{A} \parallel \mathcal{B}$. □

Corollary 3.71 *If $\mathcal{A}$ and $\mathcal{A}'$ are two hyperplanes of an affine space $(\mathcal{U}, V, \varphi)$ of dimension $n \geq 2$ over K, then $\mathcal{A} \parallel \mathcal{A}'$ if and only if $\mathcal{A} = \mathcal{A}'$ or $\mathcal{A} \cap \mathcal{A}' = \varnothing$.*

Proof If $\mathcal{A} \parallel \mathcal{A}'$ we have $\mathcal{A} = \mathcal{A}'$ or $\mathcal{A} \cap \mathcal{A}' = \varnothing$, by Proposition 3.69. The converse follows from Proposition 3.70. □

Corollary 3.72 *Let d be a line and $\mathcal{A}$ a hyperplane of an affine space $(\mathcal{U}, V, \varphi)$ of dimension $n \geq 2$ over K such that $d \cap \mathcal{A} \neq \varnothing$ and $d \not\subset \mathcal{A}$. If d' is any line of $(\mathcal{U}, V, \varphi)$ which is parallel to d, then the intersection $d' \cap \mathcal{A}$ is a point.*

Proof By Corollary 3.66 $d \cap \mathcal{A}$ is an affine subspace and by Theorem 3.67 it is a point. Suppose that $d' \cap \mathcal{A}$ is not a point, then we have $d' \cap \mathcal{A} = \varnothing$ or $d' \subseteq \mathcal{A}$. In the first case, by Proposition 3.70, we get $d' \parallel \mathcal{A}$. If $d' \subseteq \mathcal{A}$, then $D(d') \subseteq D(\mathcal{A})$, i.e. $d' \parallel \mathcal{A}$. Hence in both cases $d \parallel \mathcal{A}$, contradicting Proposition 3.69. □

Proposition 3.73 *Let $\mathcal{A}$ and $\mathcal{A}'$ be two hyperplanes of $\mathbb{A}^n(K)$ $(n \geq 2)$ of equations $b_1x_1 + \cdots + b_nx_n = c$, $b'_1x_1 + \cdots + b'_nx_n = c'$, respectively. Then $\mathcal{A} \parallel \mathcal{A}'$ if and only if there exists $d \in K \setminus \{0\}$ such that $b'_i = db_i$, $i = 1, \ldots, n$. In particular, $c' \neq dc$ if $\mathcal{A}$ and $\mathcal{A}'$ are distinct.*

Proof Since $\mathcal{A}$ and $\mathcal{A}'$ are hyperplanes, $\mathcal{A} \parallel \mathcal{A}'$ if and only if $D(\mathcal{A}) = D(\mathcal{A}')$. The equations of $D(\mathcal{A})$ and $D(\mathcal{A}')$ are $b_1x_1 + \cdots + b_nx_n = 0$ and $b'_1x_1 + \cdots + b'_nx_n = 0$, respectively, so that our thesis follows from Lemma 3.41. □

Euclid's fifth postulate assures that given a line d and a point P there is one and only one line through P parallel to d. The next result shows that any affine plane over a field satisfies Euclid's fifth postulate.

Proposition 3.74 *Let d be a line of an affine plane $(\mathcal{A}, V, \varphi)$ and $A \in \mathcal{A}$ a point. Then there exists one and only one line d' such that $d' \parallel d$ and $A \in d'$.*

Proof Let $d = B + D(d)$, where B is a point of $\mathcal{A}$ and $D(d)$ is a vector subspace of V of dimension 1. A line through A and parallel to d is $d' = A + D(d)$. Let $d'' = C + D(d)$ be another line through A parallel to d. Then $d' = A + D(d) \subset d''$, so that $d' = d''$ since d' and d'' have the same dimension. □

The following result shows that parallelism is an affine concept.

Proposition 3.75 *Let $f : (\mathcal{A}, V, \varphi) \to (\mathcal{B}, W, \psi)$ be an affine map. If $\mathcal{A}_1$ and $\mathcal{A}_2$ are two affine subspaces of $\mathcal{A}$ such that $D(\mathcal{A}_1) \subset D(\mathcal{A}_2)$, then $\mathrm{Tr}(f)(D(\mathcal{A}_1)) \subset D(\mathrm{Tr}(f)(\mathcal{A}_2))$. In particular, if $\mathcal{A}_1 \parallel \mathcal{A}_2$, then $f(\mathcal{A}_1) \parallel f(\mathcal{A}_2)$ and, if f is an affine isomorphism, $\mathcal{A}_1 \parallel \mathcal{A}_2$, if and only if $f(\mathcal{A}_1) \parallel f(\mathcal{A}_2)$.*

Proof Easy exercise (see Proposition 3.54). □

Definition 3.76 Let A, B, C and D be four coplanar points of an affine space $(\mathcal{A}, V, \varphi)$ of dimension ≥ 2 no three of which are collinear. We call the quadruple $\{A, B, C, D\}$ *quadrilateral* and we denote it by $\square ABCD$. The points A, B, C and

D are called the *vertices* of quadrilateral $\{A, B, C, D\}$, while the six lines AB, AC, BC, AD, BD and CD are the *sides* of $\{A, B, C, D\}$.

A quadrilateral $\{A, B, C, D\}$ is a *parallelogram* (denoted by $\square ABCD$) if $\overrightarrow{AB} = \overrightarrow{CD}$. Then we have $\overrightarrow{AC} = \overrightarrow{BD}$, indeed

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}, \qquad \overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD} \quad \Longrightarrow \overrightarrow{AC} - \overrightarrow{BD} = \overrightarrow{AB} - \overrightarrow{CD} = 0_V.$$

As an immediate consequence we have $AB \parallel CD$ and $AC \parallel BD$. Conversely, if $AB \parallel CD$ and $AC \parallel BD$, then $\overrightarrow{AB} = \overrightarrow{CD}$, i.e. $\{A, B, C, D\}$ is a parallelogram. Indeed, from the relations $\overrightarrow{AB} = k\overrightarrow{CD}$ and $\overrightarrow{AC} = h\overrightarrow{BD}$ with $k, h \in K^*$, we get

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = k\overrightarrow{CD} + \overrightarrow{BC} \qquad \text{and} \qquad \overrightarrow{AC} = h\overrightarrow{BD} = h\overrightarrow{BC} + h\overrightarrow{CD},$$

so that $(k-h)\overrightarrow{CD} - (h-1)\overrightarrow{BC} = 0_V$ which implies $h = k = 1$, otherwise B, C and D would be collinear.

Furthermore we can show that BD is not parallel to BC. Indeed, if $\overrightarrow{BD} = k\overrightarrow{AC}$ with $k \in K^*$, then $\overrightarrow{BD} = k(\overrightarrow{AB} + \overrightarrow{BC}) = k(\overrightarrow{AB} + \overrightarrow{AD})$. On the other hand, $\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$ so that $k(\overrightarrow{AB} + \overrightarrow{AD}) = \overrightarrow{BA} + \overrightarrow{AD}$, i.e. $(k+1)\overrightarrow{AB} + (k-1)\overrightarrow{AD} = 0_V$ and A, B and D would be collinear.

Example 3.77 (Skew Lines of $\mathbb{A}^3(K)$) Let

$$r: \begin{cases} a_1x_1 + a_2x_2 + a_3x_3 &= a_4 \\ b_1x_1 + b_2x_2 + b_3x_3 &= b_4 \end{cases} \qquad \text{and} \qquad r': \begin{cases} a'_1x_1 + a'_2x_2 + a'_3x_3 &= a'_4 \\ b'_1x_1 + b'_2x_2 + b'_3x_3 &= b'_4 \end{cases}$$

be two lines of $\mathbb{A}^3(K)$. Then

$$r \quad \text{is coplanar to} \quad r' \iff \begin{vmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ a'_1 & a'_2 & a'_3 & a'_4 \\ b'_1 & b'_2 & b'_3 & b'_4 \end{vmatrix} = 0.$$

We leave the implication $\Longrightarrow$ to the reader as an exercise. Consider the matrices

$$M = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ a'_1 & a'_2 & a'_3 & a'_4 \\ b'_1 & b'_2 & b'_3 & b'_4 \end{pmatrix}, \qquad M' = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ a'_1 & a'_2 & a'_3 \\ b'_1 & b'_2 & b'_3 \end{pmatrix}.$$

Suppose $\text{rank}(M) \leq 3$, then we have $2 \leq \text{rank}(M') \leq \text{rank}(M) \leq 3$. If $\text{rank}(M') = \text{rank}(M)$, the linear system

$$\begin{cases} a_1x_1 + a_2x_2 + a_3x_3 &= a_4 \\ b_1x_1 + b_2x_2 + b_3x_3 &= b_4 \\ a'_1x_1 + a'_2x_2 + a'_3x_3 &= a'_4 \\ b'_1x_1 + b'_2x_2 + b'_3x_3 &= b'_4 \end{cases} \tag{3.32}$$

is consistent so that r intersects r' and they are coplanar. If $\text{rank}(M') \neq \text{rank}(M)'$, i.e. $\varrho = 3$ and $\varrho' = 2$, the system (3.32) is inconsistent, i.e. $r \cap r' = \varnothing$. Since the first two rows of M' are independent, the other two rows are a linear combination of the first, then $D(r) = D(r')$, i.e. $r \parallel r'$ so that r and r' are coplanar.

Thus the four planes $a_1x_1 + a_2x_2 + a_3x_3 = a_4$, $b_1x_1 + b_2x_2 + b_3x_3 = b_4$, $a'_1x_1 + a'_2x_2 + a'_3x_3 = a'_4$ and $b'_1x_1 + b'_2x_2 + b'_3x_3 = b'_4$ have non-empty intersections if and only if r intersects r'. Furthermore, $r \neq r'$ if and only if $\text{rank}(M') = \text{rank}(M) = 3$, while $r = r'$ if and only if $\text{rank}(M') = \text{rank}(M) = 2$.

3.2.2 Projections and Symmetries

We suppose $\text{char}(K) \neq 2$ throughout this subsection.

Definition 3.78 Let $(\mathcal{A}, V, \varphi)$ be an affine space and $\mathcal{B}$ a subspace of $\mathcal{A}$. Let W be a complement of $D(\mathcal{B})$. For every $A \in \mathcal{A}$ there exists one and only one affine subspace $\mathcal{B}_A$ with $D(\mathcal{B}_A) = W$ such that $\mathcal{B}_A \cap \mathcal{B}$ has a single point, which we denote by $p_{\mathcal{B},W}(A)$. We thus obtain a map $p_{\mathcal{B},W} \colon \mathcal{A} \to \mathcal{A}$ (or $p_{\mathcal{B},W} \colon \mathcal{A} \to \mathcal{B}$) which we call *the projection from $\mathcal{A}$ to $\mathcal{B}$ parallel to W*.

Proposition 3.79 *Under the notation of* Definition 3.78*, the projection $p_{\mathcal{B},W}$ is an affine map. Choose any point $O \in \mathcal{B}$, then we have, for all $v \in V$, $v = v_1 + v_2$, $v_1 \in D(\mathcal{B})$, $v_2 \in W$,*

$$p_{\mathcal{B},W}(P) = p_{\mathcal{B},W}(O + v) = O + v_1 = P - v_2. \tag{3.33}$$

Proof We write p for $p_{\mathcal{B},W}$. We apply Lemma 3.49. Given $\overrightarrow{AB} = k\overrightarrow{CD}$, we must prove $\overrightarrow{p(A)p(B)} = k\overrightarrow{p(C)p(D)}$. Since $D(\mathcal{B})$ and W are supplementary, this reduces to proving that

$$\overrightarrow{p(A)p(B)} - k\overrightarrow{p(C)p(D)} \in D(\mathcal{B}) \cap W = \{0_V\}.$$

But by definition of the projection, $p(A)$, $p(B)$, $p(C)$ and $p(D)$ are in $\mathcal{B}$, thus $\overrightarrow{p(A)p(B)}$ and $\overrightarrow{p(C)p(D)}$ are in $D(\mathcal{B})$. So the linear combination $\overrightarrow{p(A)p(B)} -$

$k\overrightarrow{p(C)p(D)} \in D(\mathcal{B})$ as well. It remains to prove that $\overrightarrow{p(A)p(B)} - k\overrightarrow{p(C)p(D)} \in W$. Still by definition of the projection, A and $p(A)$ are both in $\mathcal{B}_A$, so that $\overrightarrow{Ap(A)} \in W$. An analogous argument holds for B, C and D. Then we have

$$\begin{aligned}
&\overrightarrow{p(A)p(B)} - k\,\overrightarrow{p(C)p(D)} \\
&\quad = \overrightarrow{p(A)A} + \overrightarrow{AB} + \overrightarrow{Bp(B)} - k\left(\overrightarrow{p(C)C} + \overrightarrow{CD} + \overrightarrow{Dp(D)}\right) \\
&\quad = \overrightarrow{p(A)A} + k\overrightarrow{CD} + \overrightarrow{Bp(B)} - k\overrightarrow{p(C)C} - k\overrightarrow{CD} - k\overrightarrow{Dp(D)} \\
&\quad = \overrightarrow{p(A)A} + \overrightarrow{Bp(B)} - k\overrightarrow{p(C)C} - k\overrightarrow{Dp(D)},
\end{aligned}$$

and this last row is a vector $\in W$ because $\overrightarrow{p(A)A}$, $\overrightarrow{Bp(B)}$, $\overrightarrow{p(C)C}$ and $\overrightarrow{Dp(D)}$ are in W. □

If $\mathcal{C}$ is a subspace of $\mathcal{A}$, then $p_{\mathcal{B},W}(\mathcal{C})$ is an affine subspace called the *projection* of $\mathcal{C}$ on $\mathcal{B}$ parallel to W.

Remark 3.80 Note that $p^2_{\mathcal{B},W} = p_{\mathcal{B},W}$ and that $\mathrm{Tr}(p_{\mathcal{B},W})$ is the projection on $D(\mathcal{B})$ parallel to W according to Exercise 1.14. Indeed, putting $p_{\mathcal{B},W} := p$, we have $\mathrm{Tr}(p)^2 = \mathrm{Tr}(p)$. Let $O \in \mathcal{B}$, $A \in \mathcal{A}$, then we have the decomposition $V = D(\mathcal{B}) \oplus W$, i.e. $\overrightarrow{OA} = \overrightarrow{Op(A)} + \overrightarrow{p(A)A}$ where $\overrightarrow{Op(A)} \in D(\mathcal{B})$ and $\overrightarrow{p(A)A} \in D(\mathcal{B}_A) = W$. Hence

$$\mathrm{Tr}(p)(\overrightarrow{OA}) = \mathrm{Tr}(p)(\overrightarrow{Op(A)}) + \mathrm{Tr}(p)(\overrightarrow{p(A)A}),$$

with $\mathrm{Tr}(p)(\overrightarrow{Op(A)}) = \overrightarrow{p(O)p^2(A)} = \overrightarrow{Op(A)}$ and $\mathrm{Tr}(p)(\overrightarrow{p(A)A}) = \overrightarrow{p^2(A)p(A)} = \overrightarrow{p(A)p(A)} = 0_V$.

We have also the converse.

Proposition 3.81 *If $p\colon (\mathcal{A}, V, \varphi) \to (\mathcal{A}, V, \varphi)$ is an affine map such that $p^2 = p$, then p is the projection on the affine subspace $\mathcal{B} = p(\mathcal{A})$ parallel to $W = \ker(\mathrm{Tr}(p))$ which is a complement of $D(\mathcal{B}) = \mathrm{Im}(\mathrm{Tr}(p))$.*

Proof First we observe that $\mathrm{Tr}(p)^2 = \mathrm{Tr}(p)$ by Proposition 3.48. We take $W = \ker(\mathrm{Tr}(p))$ and we prove that $V = \mathrm{Im}(\mathrm{Tr}(p)) \oplus \ker(\mathrm{Tr}(p))$. If $v \in \mathrm{Im}(\mathrm{Tr}(p)) \cap \ker(\mathrm{Tr}(p))$, then $v = \mathrm{Tr}(p)(v')$ for some $v' \in V$. Therefore

$$v = \mathrm{Tr}(p)(v') = \mathrm{Tr}(p)\left(\mathrm{Tr}(p)(v')\right) = \mathrm{Tr}(p)(v) = 0.$$

We certainly have $v = \mathrm{Tr}(p)(v) + (v - \mathrm{Tr}(p)(v))$ for every $v \in V$. Since $\mathrm{Tr}(p)(v) \in \mathrm{Im}(\mathrm{Tr}(p))$ we have to show that $(v - \mathrm{Tr}(p)(v)) \in \ker(\mathrm{Tr}(p))$ which is the case because

$$\mathrm{Tr}(p)(v - \mathrm{Tr}(p)(v)) = \mathrm{Tr}(p)(v) - \mathrm{Tr}(p)^2(v) = 0_V.$$

It remains to show that $p(A)$ is the (only) point lying in $\mathcal{B}_A \cap \mathcal{B}$. Certainly $p(A) \in p(\mathcal{A}) = \mathcal{B}$. On the other hand, we have $p(A) = A + \overrightarrow{Ap(A)}$. Since $A \in \mathcal{B}_A$ it suffices to show that $\overrightarrow{Ap(A)} \in W$ to get $p(A) \in \mathcal{B}_A$. Indeed,

$$\mathrm{Tr}(p)(\overrightarrow{Ap(A)}) = \overrightarrow{p(A)p^2(A)} = \overrightarrow{p(A)p(A)} = 0_V.$$

□

Definition 3.82 Let $(\mathcal{A}, V, \varphi)$ be an affine space and $\mathcal{B}$ a subspace of $\mathcal{A}$. Let W be a complement of $D(\mathcal{B})$. The *reflection* $s_{\mathcal{B},W}$ *through* $\mathcal{B}$ *parallel to* W is the map

$$s_{\mathcal{B},W} : \mathcal{A} \to \mathcal{A}, \qquad s_{\mathcal{B},W}(A) = p_{\mathcal{B},W}(A) + \overrightarrow{A\, p_{\mathcal{B},W}(A)}, \qquad \forall\, A \in \mathcal{A}. \tag{3.34}$$

Proposition 3.83 *Under the notation of* Definition 3.82*, the symmetry* $s_{\mathcal{B},W}$ *is an affine map. Furthermore* $s^2_{\mathcal{B},W} = \mathrm{id}_{\mathcal{A}}$ *so that* $s_{\mathcal{B},W}$ *is an affine automorphism. Choose any point* $O \in \mathcal{B}$ *then we have, for all* $v \in V$, $v = v_1 + v_2$, $v_1 \in D(\mathcal{B})$, $v_2 \in W$,

$$s_{\mathcal{B},W}(P) = s_{\mathcal{B},W}(O + v) = O + v_1 - v_2 = P - 2v_2. \tag{3.35}$$

Proof We write s for $s_{\mathcal{B},W}$ and p for $p_{\mathcal{B},W}$. Let $A, B \in \mathcal{A}$. Then we observe

$$\begin{aligned}
\overrightarrow{s(A)s(B)} &= \overrightarrow{s(A)p(A)} + \overrightarrow{p(A)p(B)} + \overrightarrow{p(B)s(B)} \\
&= \overrightarrow{p(A)A} + \overrightarrow{p(A)p(B)} + \overrightarrow{Bp(B)} \\
&= \overrightarrow{p(A)A} + \overrightarrow{AB} + \overrightarrow{p(A)p(B)} + \overrightarrow{Bp(B)} - \overrightarrow{AB} \\
&= \overrightarrow{p(A)p(B)} + \overrightarrow{p(A)p(B)} - \overrightarrow{AB} \\
&= 2\overrightarrow{p(A)p(B)} - \overrightarrow{AB}.
\end{aligned}$$

We apply Lemma 3.49. If $\overrightarrow{AB} = k\,\overrightarrow{CD}$, then we have (by Proposition 3.79)

$$\begin{aligned}
\overrightarrow{s(A)s(B)} &= 2\overrightarrow{p(A)p(B)} - \overrightarrow{AB} \\
&= 2k\overrightarrow{p(C)p(D)} - k\overrightarrow{CD} \\
&= k\left(2\overrightarrow{p(C)p(D)} - \overrightarrow{CD}\right) \\
&= ks(C)s(D).
\end{aligned}$$

By definition $\overrightarrow{s(A)p(A)} = -\overrightarrow{p(A)s(A)} = -\overrightarrow{Ap(A)} \in W$ so that $p(s(A)) = p(A)$. Therefore

$$s(s(A)) = p(A) + \overrightarrow{s(A)p(A)} = p(A) + \overrightarrow{p(A)A} = A.$$

□

If $\mathcal{C}$ is a subspace of $\mathcal{A}$, then $s_{\mathcal{B},W}(\mathcal{C})$ is an affine subspace called the *symmetric* of $\mathcal{C}$ through $\mathcal{B}$ parallel to W.

Remark 3.84 We can say that $s_{\mathcal{B},W}(P)$ is defined by the property that $p_{\mathcal{B},W}(P)$ is the midpoint of P and $s_{\mathcal{B},W}(P)$. Note that $\mathrm{Tr}(s_{\mathcal{B},W})$ is the symmetry through $D(\mathcal{B})$ parallel to W according to Exercise 1.14. Indeed, put $s_{\mathcal{B},W} := s$ and $p_{\mathcal{B},W} := p$, we have $\mathrm{Tr}(s)^2 = \mathrm{id}_V$. Then, if $O \in \mathcal{B}$, $A \in \mathcal{A}$, we have the decomposition $V = D(\mathcal{B}) \oplus W$, i.e. $\overrightarrow{OA} = \overrightarrow{Op(A)} + \overrightarrow{p(A)A}$, where $\overrightarrow{Op(A)} \in D(\mathcal{B})$ and $\overrightarrow{p(A)A} \in D(\mathcal{B}_A) = W$. Hence

$$\begin{aligned}
\mathrm{Tr}(s)(\overrightarrow{OA}) &= \mathrm{Tr}(s)(\overrightarrow{Op(A)}) + \mathrm{Tr}(s)(\overrightarrow{p(A)A}) \\
&= \overrightarrow{s(O)s(p(A))} + \overrightarrow{s(p(A))s(A)} \\
&= \overrightarrow{Op(A)} + \overrightarrow{p(A)s(A)} \\
&= \overrightarrow{Op(A)} + \overrightarrow{Ap(A)} \\
&= \overrightarrow{Op(A)} - \overrightarrow{p(A)A}.
\end{aligned}$$

Example 3.85 Let $(\mathcal{A}, V, \varphi)$ be an affine space and P a point (i.e. a subspace of dimension 0) of $\mathcal{A}$ (in this case $W = V$). We put $s_P := s_{P,V}$ and we call s_P *the symmetry of centre* P. In particular we have

(i) $s_P(P) = P$.
(ii) For every point $Q \in \mathcal{A}$, $Q \neq P$, $s_P(Q) \in PQ \setminus \{Q\}$, and P is the midpoint of $\{Q, s_P(Q)\}$ (i.e. $P = \frac{1}{2}Q + \frac{1}{2}s_P(Q)$).

The point $s_P(Q)$ is called *the symmetric point* of Q with respect to P. If $\mathcal{A} = \mathbb{A}^n(K)$, $P = (p_1, \ldots, p_n)$, $Q = (x_1, \ldots, x_n)$ and $s_P(Q) = (y_1, \ldots, y_n)$, the equations of s_P are

$$y_i = 2p_i - x_i, \qquad i = 1, \ldots, n.$$

We have the converse proposition.

Proposition 3.86 *If* $s\colon (\mathcal{A}, V, \varphi) \to (\mathcal{A}, V, \varphi)$ *is an affine map such that* $s^2 = \mathrm{id}_{\mathcal{A}}$, *then* s *is the affine symmetry through the subspace* $\mathcal{B} = \mathrm{Fix}(s)$, *the set of fixed points of* s, *parallel to* $W = \ker(\mathrm{Tr}(s) + \mathrm{id}_V)$, *which is a complement of* $D(\mathcal{B}) = \mathrm{Ker}(\mathrm{Tr}(s) - \mathrm{id}_V)$.

Proof Since $\mathrm{Tr}(s)^2 = \mathrm{id}_V$ then $V = \mathrm{Ker}(\mathrm{Tr}(s) - \mathrm{id}_V) \oplus \mathrm{Ker}(\mathrm{Tr}(s) + \mathrm{id}_V)$ (Exercise 1.14). If $O \in \mathcal{B}$, then

$$s(O + v) = O + \mathrm{Tr}(s)(v) = O + v \iff v \in \mathrm{Ker}(\mathrm{Tr}(s) - \mathrm{id}_V),$$

i.e. $D(\mathcal{B}) = \mathrm{Ker}(\mathrm{Tr}(s) - \mathrm{id}_V)$. For every $v = v_1 + v_2 \in \mathrm{Ker}(\mathrm{Tr}(s) - \mathrm{id}_V) \oplus \mathrm{Ker}(\mathrm{Tr}(s) + \mathrm{id}_V)$,

$$s(P) = s(O + v) = O + \mathrm{Tr}(s)(v) = O + \mathrm{Tr}(s)(v_1 + v_2) = O + v_1 - v_2$$

so that $s = s_{\mathcal{B},W}$. Note that $\mathcal{B} \neq \varnothing$. Indeed, if $s \neq \mathrm{id}_{\mathcal{A}}$, then there exists $O \in \mathcal{A}$ such that $\overrightarrow{Os(O)} \neq 0_V$ so that the midpoint $M = \frac{1}{2}O + \frac{1}{2}s(O)$ is fixed by s. □

Lemma 3.87 *Let $(\mathcal{A}, V, \varphi)$ be an affine space and $\mathcal{B}$ a subspace of $\mathcal{A}$. Let W be a complement of $D(\mathcal{B})$. If σ is an affine automorphism of $\mathcal{A}$ then*

$$\sigma \circ s_{\mathcal{B},W} = s_{\sigma(\mathcal{B}),\mathrm{Tr}(\sigma)(W)} \circ \sigma.$$

Proof By Proposition 3.83, for every $P \in \mathcal{A}$ we have $s_{\mathcal{B},W}(P) = P - 2v_2$, where $P = O + v$ and $v = v_1 + v_2$, $O \in \mathcal{A}$, $v_1 \in D(\mathcal{B})$, $v_2 \in W$. Hence $\sigma(s_{\mathcal{B},W}(P)) = \sigma(P) - 2\,\mathrm{Tr}(\sigma)(v_2)$. On the other hand, we have $V = \mathrm{Tr}(\sigma)(D(\mathcal{B})) \oplus \mathrm{Tr}(\sigma)(W)$ so that $\sigma(P) = \sigma(O) + \mathrm{Tr}(\sigma)(v) = \sigma(O) + \mathrm{Tr}(\sigma)(v_1) + \mathrm{Tr}(\sigma)(v_2)$ and

$$s_{\sigma(\mathcal{B}),\mathrm{Tr}(\sigma)(W)}(\sigma(P)) = \sigma(P) - 2\,\mathrm{Tr}(\sigma)(v_2).$$ □

3.2.3 Thales' and Ceva's Theorems

Now we show two other applications of barycentric coordinates: Thales' and Ceva's theorems.

Proposition 3.88 (Thales' Theorem) *Let $\triangle ABC$ be a triangle of an affine space $(\mathcal{A}, V, \varphi)$ of dimension $n \geq 2$. Let $M \in AB \setminus \{A, B\}$ and $N \in AC \setminus \{A, C\}$ (see Fig. 3.3). Then the line MN is parallel to the line BC if and only if*

$$\frac{\overrightarrow{MA}}{\overrightarrow{MB}} = \frac{\overrightarrow{NA}}{\overrightarrow{NC}}. \tag{3.36}$$

Proof We can suppose $n = 2$ since $\langle A, B, C\rangle$ is a plane. Write

$$M = (1 - t)A + tB, \qquad N = (1 - s)A + sC, \qquad t, s \in K \setminus \{0, 1\}. \tag{3.37}$$

Fig. 3.3 Thales' theorem

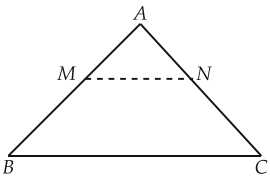
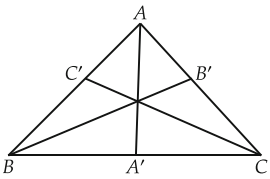


Fig. 3.4 Ceva's theorem



The equality (3.36) is equivalent to $t = s$. Let $X = (1-u)B + uC$ be an arbitrary point of BC and $Y = (1-v)M + vN$ an arbitrary point of MN, with $u, v \in K$. From (3.37) we get $Y = (1-t+v(t-s))A + t(1-v)B + vsC$. Then MN intersects BC in one point at least if and only if $1 - t + v(t-s) = 0$ for one $v \in K$. By Corollary 3.71, since the lines BC and MN are distinct by assumption, we have that $BC \parallel MN$ if and only if $1 - t + v(t-s) \neq 0$ for all $v \in K$, which is equivalent to $t = s$. □

Proposition 3.89 (Ceva's Theorem) *Let $\triangle ABC$ be a triangle of an affine space $(\mathcal{A}, V, \varphi)$ of dimension $n \geq 2$. Let $A' \in BC \setminus \{B, C\}$, $B' \in CA \setminus \{C, A\}$ and $C' \in AB \setminus \{A, B\}$ (see Fig. 3.4). Then the lines AA', BB' and CC' are concurrent or pairwise parallel if and only if we have*

$$\frac{\overrightarrow{A'B}}{\overrightarrow{A'C}} \cdot \frac{\overrightarrow{B'C}}{\overrightarrow{B'A}} \cdot \frac{\overrightarrow{C'A}}{\overrightarrow{C'B}} = -1. \tag{3.38}$$

Proof As in the proof of Thales' theorem we can suppose $n = 2$. There exist t_A, t_B and $t_C \in K \setminus \{0, 1\}$ such that

$$A' = (1-t_A)B + t_A C, \qquad B' = (1-t_B)C + t_B A, \qquad C' = (1-t_C)A + t_C B. \tag{3.39}$$

Therefore the equality (3.38) is equivalent to

$$\frac{t_A t_B t_C}{(t_A - 1)(t_B - 1)(t_C - 1)} = -1. \tag{3.40}$$

First suppose that AA', BB', CC' are concurrent lines and $P := AA' \cap BB' \cap CC'$. Hence there are s_A, s_B, $s_C \in K \setminus \{0, 1\}$ such that $P = (1 - s_A)A + s_A A' = (1 - s_B)B + s_B B' = (1 - s_C)C + s_C C'$. From (3.39) we obtain

$$\begin{aligned} P &= (1 - s_A)A + s_A(1 - t_A)B + s_A t_A C \\ &= s_B t_B A + (1 - s_A)B + s_B(1 - t_B)C \\ &= s_C(1 - t_C)A + s_C t_C B + (1 - s_C)C. \end{aligned}$$

Since the barycentric coordinates with respect to the affine frame $\{A, B, C\}$ are uniquely determined we have the equalities:

$$\begin{aligned} 1 - s_A = s_B t_B = s_C(1 - t_C), \quad & s_A(1 - t_A) \\ = 1 - s_B = s_C t_C, \quad & s_A t_A = s_B(1 - t_B) = 1 - s_C, \end{aligned}$$

so that

$$\begin{aligned} \frac{t_A t_B t_C}{(t_A - 1)(t_B - 1)(t_C - 1)} &= \frac{t_B}{t_C - 1} \cdot \frac{t_C}{t_A - 1} \cdot \frac{t_A}{t_B - 1} \\ &= \left(-\frac{s_C}{s_B}\right)\left(-\frac{s_A}{s_C}\right)\left(-\frac{s_B}{s_A}\right) = -1, \end{aligned}$$

hence (3.40) is verified.

Suppose now that $AA' \parallel BB' \parallel CC'$. Applying Proposition 3.88 (Thales' theorem) to triangles $\triangle BCB'$ and $\triangle BCC'$ we get the equalities:

$$\frac{\overrightarrow{B'C}}{\overrightarrow{B'A}} = \frac{\overrightarrow{BC}}{\overrightarrow{BA'}}, \quad \frac{\overrightarrow{C'A}}{\overrightarrow{C'B}} = \frac{\overrightarrow{CA'}}{\overrightarrow{CB}},$$

which imply $\dfrac{\overrightarrow{B'C}}{\overrightarrow{B'A}} \cdot \dfrac{\overrightarrow{C'A}}{\overrightarrow{C'B}} = \dfrac{\overrightarrow{BC}}{\overrightarrow{BA'}} \cdot \dfrac{\overrightarrow{CA'}}{\overrightarrow{CB}} = -\dfrac{\overrightarrow{CA'}}{\overrightarrow{BA'}} = -\dfrac{\overrightarrow{A'C}}{\overrightarrow{A'B}}$, i.e. the equality (3.38).

Conversely, suppose that (3.38) (or its equivalent form (3.40)) holds. We have to prove that AA', BB' and CC' are concurrent or pairwise parallel lines. We have from (3.39)

$$\begin{aligned} \forall\, P \in BB' \quad P &= (1 - s)B + sB' \\ &= st_B A + (1 - s)B + s(1 - t_B)C, \quad \text{for some } s \in K, \\ \forall\, Q \in CC' \quad Q &= (1 - u)C + uC' \\ &= u(1 - t_C)A + ut_C B + (1 - u)C, \quad \text{for some } u \in K. \end{aligned}$$

To find the intersection $BB' \cap CC'$ we have to solve the system (in the unknowns u and s)

$$\begin{cases} u(1-t_C) = st_B \\ ut_C = 1-s \\ 1-u = s(1-t_B). \end{cases}$$

If this system is consistent, then the solution is

$$s = \frac{1-t_C}{t_Bt_C+1-t_C}, \qquad u = \frac{t_B}{t_Bt_C+1-t_C},$$

therefore the intersection $BB' \cap CC'$ consists of the point

$$P = \frac{t_B(1-t_C)}{t_Bt_C+1-t_C}A + \frac{t_Bt_C}{t_Bt_C+1-t_C}B + \frac{(1-t_B)(1-t_C)}{t_Bt_C+1-t_C}C.$$

Now we have to verify that A, P, A' are collinear. We observe that P can be written in the form

$$P = \frac{t_B(1-t_C)}{t_Bt_C+1-t_C}A + \frac{1}{t_Bt_C+1-t_C}\left(t_Bt_CB + (1-t_B)(1-t_C)C\right). \tag{3.41}$$

Taking account of (3.40), we can write

$$t_Bt_CB + (1-t_B)(1-t_C)C = t_Bt_CB + \frac{t_At_Bt_C}{1-t_A}C = \frac{t_Bt_C}{1-t_A}((1-t_A)B + t_AC)$$
$$= \frac{t_Bt_C}{1-t_A}A',$$

so that we have

$$P = \frac{t_B(1-t_C)}{t_Bt_C+1-t_C}A + \frac{1}{t_Bt_C+1-t_C}\cdot\frac{t_Bt_C}{1-t_A}A'.$$

The sum of coefficients of A and A' is equal to 1 taking account of (3.40). In other words P lies on AA'. If the equation $t_Bt_C + 1 - t_C = 0$ is inconsistent, the we can immediately verify that AA', BB', CC' are pairwise parallel lines. Thus Ceva's theorem is completely proved. □

3.2.4 Real Affine Spaces and Convex Sets

Throughtout this subsection we are concerned only with real affine space. Let $(\mathcal{A}, V, \varphi)$ be a real affine space of dimension n.

Definition 3.90 Let $A \neq B \in \mathcal{A}$ be two distinct points and $C = (1-t)A + tB$, with $t \in \mathbb{R}$, an arbitrary point lying on the line AB. We shall say that C *lies between* (or *strictly lies between*) A and B if $t \in [0, 1]$ (respectively, if $t \in (0, 1)$). The above definition does not depend on the order of A and B; indeed, changing the order of the points, the barycentric coordinate t is substituted by $1 - t$. Using the affine ratio we can also say that C strictly lies between A and B if $\mathrm{ar}(A, B, C) < 0$.

We shall define the *closed affine segment with ends A and B* as the set $[AB]$ of all points of AB lying between A and B and the *open affine segment with ends A and B* as the set (AB) of all points of AB strictly lying between A and B, i.e.

$$[AB] := \{(1-t)A + tB \,|\, 0 \leq t \leq 1\}, \quad (AB) := \{(1-t)A + tB \,|\, 0 < t < 1\}. \tag{3.42}$$

Definition 3.91 Let $M \subset \mathcal{A}$. We shall say that M is *convex* if for every pair of distinct points $A, B \in M$ we have $[AB] \subseteq M$. The empty set and every singleton (i.e. a set consisting of only one point) are conventionally considered convex. The whole space $\mathcal{A}$ is obviously convex.

Lemma 3.92 *Let $\{M_i\}_{i \in I}$ be a non-empty family of convex subsets of $\mathcal{A}$. Then $\bigcap_{i \in I} M_i$ is convex.*

Proof It is an immediate consequence of Definition 3.91. □

Definition 3.93 Let $M \subset \mathcal{A}$ be a non-empty subset of $\mathcal{A}$. The *convex hull* of M, denoted by $\mathrm{conv}(M)$, is the intersection of all convex subsets of $\mathcal{A}$ containing M, i.e. the smallest convex subset of $\mathcal{A}$ containing M. This definition is justified by Lemma 3.92. A non-empty subset $M \subset \mathcal{A}$ is convex if and only if $M = \mathrm{conv}(M)$. A *polytope* of $\mathcal{A}$ is the convex hull of a finite set of points of $\mathcal{A}$.

Theorem 3.94 *Let $M = \{A_1, \ldots, A_m\}$ be a non-empty finite subset of $\mathcal{A}$. Then the convex hull* $\mathrm{conv}(M)$ *of M is the set of all barycentres of points of M with non-negative masses, i.e.*

$$\mathrm{conv}(M) = \left\{ \sum_{i=1}^{m} \lambda_i A_i \,\middle|\, \lambda_i \geq 0,\ \sum_{i=1}^{m} \lambda_i = 1 \right\}.$$

Proof Let

$$M' = \left\{ \sum_{i=1}^{m} \lambda_i A_i \,\middle|\, \lambda_i \geq 0,\ \sum_{i=1}^{m} \lambda_i = 1 \right\}.$$

First we prove $\text{conv}(M) \subseteq M'$. It will suffice to show that $M \subset M'$ (obvious) and that M' is convex. Let $A = \sum_{i=1}^{m} \lambda_i A_i$ and $B = \sum_{i=1}^{m} \mu_i A_i$, with $\lambda_i, \mu_i \geq 0$ and $\sum_{i=1}^{m} \lambda_i = \sum_{i=1}^{m} \mu_i = 1$ be two arbitrary distinct points of M'. We have to prove that $(1-t)A + tB \in M'$ for all $t \in [0, 1]$. Indeed,

$$(1-t)A + tB = (1-t)\sum_{i=1}^{m} \lambda_i A_i + t\sum_{i=1}^{m} \mu_i A_i$$

$$= \sum_{i=1}^{m} \alpha_i A_i, \quad \text{with} \quad \alpha_i = (1-t)\lambda_i + t\mu_i \geq 0, \quad i = 1, \ldots, m,$$

since $t \in [0, 1]$ and $\lambda_i, \mu_i \geq 0$; furthermore $\sum_{i=1}^{m} \alpha_i = (1-t)\sum_{i=1}^{m} \lambda_i + t\sum_{i=1}^{m} \mu_i = (1-t) + t = 1$.

Now we have to prove that $M' \subset \text{conv}(M)$, i.e.

$$\sum_{i=1}^{m} \lambda_i A_i \in \text{conv}(M), \qquad \forall\, \lambda_i \geq 0, \quad \sum_{i=1}^{m} \lambda_i = 1. \tag{3.43}$$

We use the induction with respect to the number $s := \#(\{i \mid \lambda_i \neq 0\})$. If $s \leq 2$ there is nothing to prove because $\text{conv}(M)$ is convex. Hence suppose $s \geq 3$. Renumbering the terms of the sum (3.43) if necessary, we can write $\sum_{i=1}^{m} \lambda_i A_i = \sum_{i=1}^{s} \lambda_i A_i$. If we put $t := \lambda_1 + \cdots + \lambda_{s-1}$, then $t \in [0, 1]$ (because $(\lambda_1 + \cdots + \lambda_{s-1}) + \lambda_s = 1$ and $\lambda_i \geq 0$). By our inductive hypothesis

$$B = \frac{\lambda_1}{t} A_1 + \cdots + \frac{\lambda_{s-1}}{t} A_{s-1} \in \text{conv}(M).$$

Then $\lambda_1 A_1 + \cdots + \lambda_s A_s = tB + \lambda_s A_s \in \text{conv}(M)$, because $t + \lambda_s = 1, t \geq 0, \lambda_s \geq 0$ and $\text{conv}(M)$ is convex. □

Remark 3.95 For sake of completeness we wish to recall the following result (see [4], p. 339):

Theorem 3.96 (Carathéodory) *If $(\mathcal{A}, V, \varphi)$ is a real affine space of dimension n, the convex hull $\text{conv}(M)$ of an arbitrary non-empty subset of $\mathcal{A}$ is given by*

$$\text{conv}(M) = \left\{ \sum_{i=1}^{n+1} \lambda_i A_i \;\middle|\; A_i \in M,\ \lambda_i \geq 0,\ \sum_{i=1}^{n+1} \lambda_i = 1 \right\}.$$

Definition 3.97 We have defined the notions of triangle and parallelogram (see Definitions 3.32 and 3.76) in an affine space over an arbitrary field K. If $K = \mathbb{R}$ we can come back to the more usual definitions as follows. We use symbols different to those of Definitions 3.32 and 3.76 for triangles and parallelograms in a real affine space. If O, A and B are affine independent points of a real affine space $\mathcal{A}$ a *triangle* $\overset{\triangle}{OAB}$ of vertices O, A and B is the convex hull of $\{O, A, B\}$. Its *sides* are the segments $[OA]$, $[OB]$ and $[AB]$ which are intersections of $\overset{\triangle}{OAB}$ with the lines OA, OB and AB (see Exercise 3.21).

If $\square ABCD$ is a parallelogram in the sense of Definition 3.76, where $AB \parallel CD$, $BC \parallel AD$, we shall also call the convex hull of $\{A, B, C, D\}$ a *parallelogram* of vertices A, B, C and D, and denote it by $\overset{\square}{ABCD}$. Its four *sides* are the segments $[AB]$, $[BC]$, $[CD]$ and $[DA]$, which are intersections of $\overset{\square}{ABCD}$ with the lines AB, BC, CD and AD.

3.3 Exercises

Let $\mathcal{A} = (\mathcal{A}, V, \varphi)$ be an affine space over K with $\dim_K(\mathcal{A}) = n \geq 1$.

Exercise 3.1 Let A, B, C, D be four affine independent points of $\mathcal{A}$ (with $n \geq 3$ and $\mathrm{char}(K) \neq 2$). Show that the lines joining the midpoints of pairs $\{A, B\}$ and $\{C, D\}$, $\{A, C\}$ and $\{B, D\}$, $\{A, D\}$ and $\{B, C\}$ are concurrent. (Hint: prove that the common point of the three lines is the barycentre of A, B, C and D).

Exercise 3.2 Prove that in a quadrilateral $ABCD$ of $\mathcal{A}$ (with $n = 2$ and $\mathrm{char}(K) \neq 2$) the lines joining the midpoints of pairs $\{A, B\}$ and $\{C, D\}$, $\{B, C\}$ and $\{A, D\}$, $\{A, C\}$ and $\{B, D\}$, are concurrent.

Exercise 3.3 Let $\{A_1, \ldots, A_m, B_1, \ldots, B_p\}$ be $m + p$ distinct points of $\mathcal{A}$ (with $n \geq 2$) Show that the barycentres

$$\sum_{i=1}^{m} \frac{1}{m} A_i, \qquad \sum_{j=1}^{p} \frac{1}{p} B_j, \qquad \sum_{i=1}^{m} \frac{1}{m+p} A_i + \sum_{j=1}^{p} \frac{1}{m+p} B_j$$

are collinear.

Exercise 3.4 Let $ABCD$ be a quadrilateral in $\mathcal{A}$ (with $n = 2$ and $\mathrm{char}(K) \neq 2$) and $E = AC \cap BD$. Show that $ABCD$ is a parallelogram if and only if E is the midpoint of both $\{A, C\}$ and $\{B, D\}$. (Hint: the reader can prove it directly or by using Thales' theorem.)

Exercise 3.5 Let r and s be two distinct lines of $\mathcal{A}$ (with $n = 2$ and $\mathrm{char}(K) \neq 2$). Suppose that r and s are concurrent at a point V and $P \notin r \cup s$. Prove that there is a

line through P which meets r at A and s at B such that P is the midpoint of $\{A, B\}$. (Hint: use Thales' theorem.)

Exercise 3.6 Let A, B, C, D be four coplanar points no three of which are collinear in the affine space $\mathcal{A}$ (with $n \geq 3$ and $\text{char}(K) \neq 2$). Let O be a point such that $\dim_K \langle O, A, B, C, D\rangle = 3$. Show that there are four points $A' \in OA$, $B' \in OB$, $C' \in OC$ and $D' \in OD$ such that $A'B'C'D'$ is a parallelogram. (Hint: Use Exercises 3.4 and 3.5.)

Exercise 3.7 Let $ABCD$ and $A'B'C'D'$ be two parallelograms of $\mathcal{A}$ (with $n \geq 2$). Fix $t \in K$ and consider the points $M = (1-t)A + tA'$, $N = (1-t)B + tB'$, $P = (1-t)C + tC'$ and $Q = (1-t)D + tD'$. Prove that $MNPQ$ is a parallelogram.

Exercise 3.8 Let $\{A_0, A_1, A_2\}$ be an affine frame of $\mathcal{A}$ (with $n = 2$) and $\sigma : \mathcal{A} \to \mathcal{A}$ the unique affine automorphism such that $\sigma(A_0) = A_1$, $\sigma(A_1) = A_2$ and $\sigma(A_2) = A_0$ (see Corollary 3.52). Find the fixed points of σ (i.e. the points $A \in \mathcal{A}$ such that $\sigma(A) = A$).

Exercise 3.9 Let $f \in \text{Aut}_K(\mathcal{A})$ be an affine automorphism of $\mathcal{A}$. Show that the set of fixed points of f is an affine subspace (possibly empty) of $\mathcal{A}$.

Exercise 3.10 Let $O, O' \in \mathcal{A}$ be two arbitrary points of $\mathcal{A}$. Prove that the subgroups $\mathcal{H}_O$ and $\mathcal{H}_{O'}$ of all homotheties of centres O and O' respectively (see Example 3.53-2), are conjugate in $\text{Aut}_K(\mathcal{A})$, i.e. there exists $g \in \text{Aut}_K(\mathcal{A})$ such that $\mathcal{H}_{O'} = g\mathcal{H}_O g^{-1}$. (Hint: take a suitable affine translation g. Prove that $\mathcal{H}_O \cap \mathcal{H}_{O'} = \{\text{id}_{\text{Aut}_K(\mathcal{A})}\}$ if $O \neq O'$.)

Exercise 3.11 Let $O \in \mathcal{A}$. Prove that every affine homothety $\alpha \in \mathcal{H}_O$ commutes with every affine automorphism of $\text{Aut}_K(\mathcal{A}, O)$. In particular, $\mathcal{H}_O$ is a normal subgroup of $\text{Aut}_K(\mathcal{A}, O)$. Does α commute with every affine automorphism of $\text{Aut}_K(\mathcal{A})$?

Exercise 3.12 Let $\alpha \in \mathcal{H}_O(\mathcal{A}, V, \varphi) \cup \text{T}_K(\mathcal{A}, V, \varphi)$. Find the fixed lines of α, i.e. the lines d such that $\alpha(d) = d$.

Exercise 3.13 Parallel lines r and s lie in two planes of an affine space $\mathcal{A}$ of dimension $n \geq 3$ that intersect along a line l. Prove that $r \parallel l$.

Exercise 3.14 Given three pairwise skew lines, prove that there exist infinitely many lines, each of which intersects all the three of these lines.

Exercise 3.15 Let $O_1, O_2 \in \mathcal{A}$ be two arbitrary distinct points and $\alpha_i \in \mathcal{H}_{O_i}$ such that $\overrightarrow{O_i\alpha_i(A)} = a_i \overrightarrow{O_iA}$, $\forall A \in \mathcal{A}$, with $a_i \in K \setminus \{0\}$, $i = 1, 2$. If $a_1a_2 \neq 1$ show that $\alpha_2 \circ \alpha_1$ is an homothety of centre

$$O = \frac{a_2(1-a_1)}{1-a_1a_2}O_1 + \frac{1-a_2}{1-a_1a_2}O_2$$

such that $\overrightarrow{O\alpha_2 \circ \alpha_1(A)} = a_1 a_2 \overrightarrow{OA}$, $\forall\, A \in \mathcal{A}$. If $a_1 a_2 = 1$ prove that $\alpha_2 \circ \alpha_1$ is an affine translation. (In particular, if α_1, α_2 and α_3 are three affine homotheties $\neq$ id, of centres O_1, O_2 and O_3 respectively, such that $\alpha_1 \circ \alpha_2 = \alpha_3$ then O_1, O_2 and O_3 are collinear.)

Exercise 3.16 Let A, B and C be three non-collinear points of $\mathcal{A}$ with $\mathrm{char}(K) \neq 2$. Show that the medians of $\triangle ABC$ are concurrent if $\mathrm{char}(K) \neq 3$ and parallel if $\mathrm{char}(K) = 3$.

Exercise 3.17 Let $\mathcal{A}$ be an affine space over $\mathbb{R}$ of dimension n. Show that one and only one of three distinct collinear points of $\mathcal{A}$ strictly lies between the remaining two.

Exercise 3.18 Let $\mathcal{A}$ and $\mathcal{A}'$ be two affine spaces over $\mathbb{R}$ of dimension n and m respectively. Let $M \subset \mathcal{A}$ and $N \subset \mathcal{A}'$ be two convex subsets. Let $f : \mathcal{A} \to \mathcal{A}'$ be an affine map. Prove that $f(M)$ is a convex subset of $\mathcal{A}'$ and $f^{-1}(N)$ is a convex subset of $\mathcal{A}$.

Exercise 3.19 Prove Theorem 3.96, adapting the proof of Theorem 3.94.

Exercise 3.20 Prove that the intersection of a polytope of an affine space over $\mathbb{R}$ with a line is a segment, a point or the empty set.

Exercise 3.21 Let M and N be two convex subsets of a real affine space $\mathcal{A}$. Show that

$$\bigcup_{A \in M,\, B \in N} [AB]$$

is convex.

Exercise 3.22 Let d_1 and d_2 be two non-coplanar lines of a real affine space $\mathcal{A}$. Prove that the subset of $\{(1-t)A_1 + tA_2 \mid A_1 \in d_1, A_2 \in d_2, t \in \mathbb{R}\}$ is not convex.

Exercise 3.23 Let $\mathcal{H}$ be a hyperplane of $\mathbb{A}^n(\mathbb{R})$ of equation $a_1 x_1 + \cdots + a_n x_n = b$. Then $\mathbb{A}^n(\mathbb{R}) \setminus \mathcal{H}$ is a disjoint union of two sets

$$\mathcal{H}_+ = \{(x_1, \ldots, x_n) \in \mathbb{A}^n(\mathbb{R}) \mid a_1 x_1 + \cdots + a_n x_n > b\},$$
$$\mathcal{H}_- = \{(x_1, \ldots, x_n) \in \mathbb{A}^n(\mathbb{R}) \mid a_1 x_1 + \cdots + a_n x_n < b\}.$$

Prove that $\mathcal{H}_+$ and $\mathcal{H}_-$ are convex.

Exercise 3.24 Let r and s be two skew lines and $P \notin r \cup s$ a point of $\mathcal{A}$. Under what condition does there exist a line through P which intersects both r and s?

Exercise 3.25 Let S and S' be two affine subspaces of $\mathcal{A}$ with $\dim_K(S) = \dim_K(S') = d < n$. Find all implications between the following assertions:

(i) $S \parallel S'$.
(ii) There exists an affine subspace T with $\dim T = d + 1$ such that $S \subset T$ and $S' \subset T$.
(iii) $S = S'$ or $S \cap S' = \varnothing$.
(iv) Both the conditions (ii) and (iii).

Exercise 3.26 Let $\mathcal{A}$ be an affine plane over a finite field K having k elements. Prove:

(a) $\mathcal{A}$ has k^2 points in all.
(b) There are k points on every line.
(c) There are $k + 1$ lines through every point.
(d) There are $k^2 + k$ lines in all.
(e) There are $k - 1$ lines parallel to a given line.

Exercise 3.27 Let $ABCD$ be a parallelogram. Prove that the lines joining A to the midpoints of the opposite sides trisect the diagonal not through A.

Exercise 3.28 Let $\mathrm{char}(K) \neq 2$. Let AA', BB', CC' be lines concurring in G, as in Ceva's theorem. Prove the following identity:

$$\frac{\overrightarrow{AG}}{\overrightarrow{AA'}} + \frac{\overrightarrow{BG}}{\overrightarrow{BB'}} + \frac{\overrightarrow{CG}}{\overrightarrow{CC'}} = 2.$$

Exercise 3.29 Let $\mathrm{char}(K) \neq 2, 3$. Let $\square ABCD$ be a quadrilateral. Consider the points

$$E = \frac{1}{3}(A + 2B), \quad F = \frac{1}{3}(2B + C), \quad G = \frac{1}{3}(C + 2D), \quad H = \frac{1}{3}(2D + A)$$

and prove that $\square EFGH$ is a parallelogram.

Exercise 3.30 Let $\mathrm{char}(K) \neq 2$. Consider a triangle $\triangle ABC$ and midpoints A', B', C' of the sides $[BC]$, $[CA]$, and $[AB]$ respectively. Show that $A'B' \parallel AB$, $B'C' \parallel BC$, $C'A' \parallel CA$, and that

$$\overrightarrow{B'A'} = \overrightarrow{AC'} = \frac{1}{2}\overrightarrow{AB}.$$

Exercise 3.31 Let $\mathrm{char}(K) \neq 3$. Through each vertex of a triangle, two lines dividing the opposite side into three equal parts are drawn. Let P be the hexagon bounded by these six lines. Prove that the diagonals joining the opposite vertices of P are concurrent.

Chapter 4
Euclidean Spaces



In order to introduce concepts like "distance", "angle" or "orthogonality" in a real affine space $(\mathcal{A}, V, \varphi)$ the vector space V needs to be equipped with an inner product, i.e. V must be a euclidean vector space. Throughout this chapter we are concerned only with real affine spaces of finite dimension.

4.1 Euclidean Affine Spaces

Definition 4.1 A real affine space $(\mathcal{A}, V, \varphi)$ is called a *euclidean affine space* or simply a *euclidean space* if V is equipped with an inner product $\langle \cdot, \cdot \rangle$, i.e. if V is a euclidean vector space; it will be denoted by $(\mathcal{A}, (V, \langle \cdot, \cdot \rangle), \varphi)$.

The standard real affine space $\mathbb{A}^n(\mathbb{R})$ whose orienting space $\mathbb{R}^n$ is equipped with the inner product $\langle \cdot, \cdot \rangle_s$ (see Example 1.140, 1)) is called the *standard euclidean space* and denoted by $\mathbb{E}^n(\mathbb{R})$, i.e.

$$\mathbb{E}^n(\mathbb{R}) = (\mathbb{A}^n(\mathbb{R}), (\mathbb{R}^n, \langle \cdot, \cdot \rangle_s), \varphi),$$

where $\varphi(v, w) = w - v, \forall\, v, w \in \mathbb{R}^n$.

If $\mathcal{B}$ is an affine subspace of a euclidean space $(\mathcal{A}, (V, \langle \cdot, \cdot \rangle), \varphi)$, then $D(\mathcal{B})$ is equipped with the inner product $\langle \cdot, \cdot \rangle_{\mathcal{B}} := \langle \cdot, \cdot \rangle_{\big| D(\mathcal{B}) \times D(\mathcal{B})}$ (see Proposition 1.139) so that $(\mathcal{B}, D(\mathcal{B}), \langle \cdot, \cdot \rangle_{\mathcal{B}}, \varphi)$ is a euclidean space.

Definition 4.2 Let $(\mathcal{A}, (V, \langle \cdot, \cdot \rangle), \varphi)$ be a euclidean space. For every pair of points $A, B \in \mathcal{A}$ the *distance* $d(A, B)$ is defined as being

$$d(A, B) := d(\overrightarrow{OA}, \overrightarrow{OB}), \tag{4.1}$$

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where $O \in \mathcal{A}$ is a fixed point according to Definition 1.151. The scalar $d(A, B)$ is independent of the choice of $O \in \mathcal{A}$. Indeed, if $O' \in \mathcal{A}$ is another point of $\mathcal{A}$, then $\overrightarrow{O'A} = \overrightarrow{OA} + v$ and $\overrightarrow{O'B} = \overrightarrow{OB} + v$, with $v = \overrightarrow{O'O}$, so that $d(\overrightarrow{OA}, \overrightarrow{OB}) = d(\overrightarrow{O'A}, \overrightarrow{O'B})$ by Proposition 1.152, (iv). Hence

$$d(A, B) = \|\overrightarrow{AB}\|, \quad \forall\, A, B \in \mathcal{A}. \tag{4.2}$$

If $A = (a_1, \ldots, a_n)$ and $B = (b_1, \ldots, b_n) \in \mathbb{E}^n(\mathbb{R})$, then

$$d(A, B) = \sqrt{\sum_{i=1}^{n}(a_i - b_i)^2}. \tag{4.3}$$

From Proposition 1.152 one immediately obtains the relations

$$d(A, B) \geq 0, \qquad d(A, B) = 0 \iff A = B, \quad \forall A, B \in \mathcal{A}. \tag{4.4}$$

$$d(A, B) = d(B, A), \quad \forall A, B \in \mathcal{A}. \tag{4.5}$$

$$d(A, C) \leq d(A, B) + d(B, C), \quad \forall A, B, C \in \mathcal{A}, \tag{4.6}$$

so that $\mathcal{A}$ equipped with the map $\mathcal{A} \times \mathcal{A} \to \mathbb{R}, \quad (A, B) \to d(A, B)$, is a metric space.

We define the *length* $\ell([AB])$ of the segment $[AB]$ as being $d(A, B) = \|\overrightarrow{AB}\|$.

Definition 4.3 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space. An affine frame $A_0, A_1, \ldots, A_n$ of $\mathcal{A}$ such that the vectors $\overrightarrow{A_0A_1}, \ldots, \overrightarrow{A_0A_n}$ constitute an on-basis of V is called a *euclidean frame*. Since $\{\overrightarrow{A_0A_1}, \ldots, \overrightarrow{A_0A_n}\}$ is a basis of V there is a unique n-tuple $(x_1(A), \ldots, x_n(A)) \in K^n$ such that $\overrightarrow{A_0A} = x_1(A)\overrightarrow{A_0A_1} + \cdots + x_n(A)\overrightarrow{A_0A_n}$. We call $x_1(A), \ldots, x_n(A) \in K$ *orthogonal coordinates with respect to the euclidean frame* $A_0, A_1, \ldots, A_n$ or also a *system of orthogonal coordinates*.

The standard affine frame $O, E_1, \ldots, E_n$ of $\mathbb{E}^n(\mathbb{R})$ (see Definition 3.23) is a euclidean frame called the *standard euclidean frame*. Let $A_0, A_1, \ldots, A_n$ be an affine frame of $\mathbb{E}^n(\mathbb{R})$ where $\overrightarrow{A_0A_i} = (a_{i1}, \ldots, a_{in})$, $i = 1, \ldots, n$. Then $A_0, A_1, \ldots, A_n$ is a euclidean frame if and only if the matrix $\left(a_{ij}\right)_{i,j=1,\ldots,n}$ is orthogonal. In particular, $\det\left(\left(a_{ij}\right)_{i,j=1,\ldots,n}\right) = \pm 1$. We say that $A_0, A_1, \ldots, A_n$ is *right-handed (left-handed)* if the basis $\{\overrightarrow{A_0A_i}\}_{i=1,\ldots,n}$ is right-handed (left-handed) (see Definition 2.29), i.e. if $\det\left(\left(a_{ij}\right)_{i,j=1,\ldots,n}\right) = 1$ ($\det\left(\left(a_{ij}\right)_{i,j=1,\ldots,n}\right) = -1$). The standard euclidean frame is obviously right-handed.

Angle Between Two Lines

Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space. The *angle* $\measuredangle(d, e)$ between two lines d and e of $\mathcal{A}$ is defined by

$$\measuredangle(d, e) := \arccos\left(\frac{|\langle v, w\rangle|}{\|v\| \cdot \|w\|}\right), \tag{4.7}$$

where $0 \neq v \in D(d)$ and $0 \neq w \in D(e)$. Note that (4.7) does not depend on the choice of v and w and $\measuredangle(d, e) \in [0, \frac{\pi}{2}]$. Furthermore d and e (not necessarily concurrent) are *orthogonal* or *perpendicular* (written $d \perp e$) if and only if $\langle v, w\rangle = 0$, i.e. $D(d) \perp D(e)$.

If d and e are two lines of $\mathbb{E}^n(\mathbb{R})$ having director parameters $(m_1, \ldots, m_n) \neq (0, \ldots, 0)$ and $(k_1, \ldots, k_n) \neq (0, \ldots, 0)$ respectively (see Example 3.38-(2)), we have

$$\measuredangle(d, e) = \arccos\left(\frac{\left|\sum_{i=1}^{n} m_i k_i\right|}{\sqrt{(\sum_{i=1}^{n} m_i^2)(\sum_{i=1}^{n} k_i^2)}}\right). \tag{4.8}$$

Hence d and e are perpendicular if and only if

$$\sum_{i=1}^{n} m_i k_i = 0. \tag{4.9}$$

Definition 4.4 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space and $\overset{\triangle}{OAB}$ a triangle of vertices O, A and B. The *angles of vertex* O, A *and* B of $\overset{\triangle}{OAB}$ are $\measuredangle(\overrightarrow{OA}, \overrightarrow{OB})$, $\measuredangle(\overrightarrow{AO}, \overrightarrow{AB})$ and $\measuredangle(\overrightarrow{BO}, \overrightarrow{BA})$ respectively. We call a triangle $\overset{\triangle}{OAB}$ *right-angled* if one of its angles is $\frac{\pi}{2}$.

Proposition 4.5 (Pythagoras' Theorem) *Let $\overset{\triangle}{OAB}$ be a triangle in a euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$. Then $\overset{\triangle}{OAB}$ is a right-angled triangle (in the vertex O i.e. $\measuredangle(\overrightarrow{OA}, \overrightarrow{OB}) = \frac{\pi}{2}$) if and only if*

$$\ell([AB])^2 = \ell([OA])^2 + \ell([OB])^2.$$

Proof It is an immediate consequence of Proposition 1.150. □

Lines Orthogonal to a Hyperplane

Definition 4.6 Let d be a line and $\mathcal{H}$ a hyperplane of a euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$. We say that d is *orthogonal* (or *the perpendicular*) to $\mathcal{H}$ (written $d \perp \mathcal{H}$) if $D(d) \perp D(\mathcal{H})$, i.e. $D(d) = D(\mathcal{H})^{\perp}$. Hence d is orthogonal to $\mathcal{H}$ if and only if it is orthogonal to every line e contained in $\mathcal{H}$. A line orthogonal to $\mathcal{H}$ is also called the *normal line to* $\mathcal{H}$. If d and d' are perpendicular to $\mathcal{H}$, then $D(d) = D(\mathcal{H})^{\perp} = D(d')$, i.e. d and d' are parallel. Therefore, given a point $P \in \mathcal{A}$ and a hyperplane $\mathcal{H}$, there exists one and only one line d through P perpendicular to $\mathcal{H}$. This line d is called the *perpendicular to* $\mathcal{H}$ *passing through* P, and the point $d \cap \mathcal{H}$ is called the *foot of the perpendicular* to $\mathcal{H}$ passing through P.

If a line d of $\mathbb{E}^n(\mathbb{R})$ has director parameters $(m_1, \ldots, m_n)$ and a hyperplane $\mathcal{H}$ has the equation $a_1x_1 + \cdots + a_nx_n = b$, then the condition $d \perp \mathcal{H}$ is equivalent to the existence of a scalar $\lambda \in \mathbb{R} \setminus \{0\}$ such that

$$a_i = \lambda m_i, \qquad i = 1, \ldots, n. \tag{4.10}$$

Indeed, a line e having parametric equations $x_i = b_i + tk_i$, $i = 1, \ldots, n$, lies on the hyperplane $\mathcal{H}$ if and only if $a_1b_1 + \cdots + a_nb_n = b$ and $a_1k_1 + \cdots + a_nk_n = 0$. Then $d \perp e$ for every line e lying on $\mathcal{H}$ if and only if the equations

$$a_1x_1 + \cdots + a_nx_n = 0 \qquad \text{and} \qquad m_1x_1 + \cdots + m_nx_n = 0$$

have the same solutions $(k_1, \ldots, k_n)$. Let i be an index such that $a_i \neq 0$ and let $j \neq i$ be any other index. Then the n-tuple $(k_1, \ldots, k_n)$, where $k_i = -a_j$, $k_j = a_i$ and $k_l = 0$, $\forall l \neq i, j$, solves the equation $a_1x_1 + \cdots + a_nx_n = 0$ and hence it solves also $m_1x_1 + \cdots + m_nx_n = 0$ so that $m_i(-a_j) + m_ja_i = 0$ and (4.10) follows. Conversely, every line having director parameters $(a_1, \ldots, a_n)$ is obviously perpendicular to $\mathcal{H}$.

Distance Between Points and Hyperplanes

Definition 4.7 Let $\mathcal{H}$ be a hyperplane and P a point of a euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$. If r is the perpendicular to $\mathcal{H}$ through P, by Pythagoras' theorem we have $d(P, Q) \leq d(P, R)$, where $Q := r \cap \mathcal{H}$ and R is every other point of $\mathcal{H}$. Hence

$$d(P, Q) = \inf_{R \in \mathcal{H}} d(P, R),$$

and the scalar $d(P, Q)$ is called the *distance from* P *to* $\mathcal{H}$, and denoted by $d(P, \mathcal{H})$.

Let $\mathcal{H}$ be a hyperplane of $\mathbb{E}^n(\mathbb{R})$ having the equation $a_1x_1 + \cdots + a_nx_n = b$ and $P = (p_1, \ldots, p_n) \in \mathbb{E}^n(\mathbb{R})$. The perpendicular r to $\mathcal{A}$ passing through P has parametric equations $x_i = p_i + ta_i$, $i = 1, \ldots, n$ (see Definition 4.6). If $Q = r \cap \mathcal{H}$,

then $Q = (p_1 + t_0 a_1, \ldots, p_n + t_0 a_n)$, with

$$t_0 = -\frac{a_1 p_1 + \cdots + a_n p_n - b}{a_1^2 + \cdots + a_n^2}.$$

Hence we have

$$d(P, \mathcal{H}) = \frac{|a_1 p_1 + \cdots + a_n p_n - b|}{\sqrt{a_1^2 + \cdots + a_n^2}}. \tag{4.11}$$

We call Q the *foot of the perpendicular* from the point P to the hyperplane $\mathcal{H}$.

Let $\mathcal{H}$ and $\mathcal{K}$ be two parallel hyperplanes of an euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$. We define the *distance* $d(\mathcal{H}, \mathcal{K})$ of $\mathcal{H}$ and $\mathcal{K}$ as follows:

$$d(\mathcal{H}, \mathcal{K}) := d(P, \mathcal{K}) = d(\mathcal{H}, Q),$$

where P is an arbitrary point of $\mathcal{H}$ and Q is an arbitrary point of $\mathcal{K}$. First note that $d(P, \mathcal{K})$ is independent of the choice of P as well as $d(\mathcal{H}, Q)$ is independent of the choice of Q. Indeed, let H be the foot of the perpendicular from the point P to $\mathcal{K}$, P' be any point of $\mathcal{H}$ and H' the foot of the perpendicular from the point P' to $\mathcal{K}$, then $PH \parallel P'H'$. Furthermore, the plane containing the line PP' perpendicular to $\mathcal{K}$ intersects $\mathcal{K}$ in a line parallel to PP' (indeed PP' is coplanar with HH' and cannot intersect HH' since $\mathcal{H} \cap \mathcal{K} = \varnothing$). Hence the quadrilateral $\square PP'HH'$ is a parallelogram so that in particular, we have (see Definition 3.76)

$$d(P, \mathcal{K}) = d(P, H) = \ell([PH]) = \ell(P'H']) = d(P', H') = d(P', \mathcal{K}).$$

We end with a trivial remark: If $A \in \mathcal{H}$ and $B \in \mathcal{K}$ are arbitrary points, then $d(\mathcal{H}, \mathcal{K}) \leq d(A, B)$.

Area of a Triangle

We wish to express the area of a triangle of $\mathbb{E}^2(\mathbb{R})$ as a function of coordinates of its vertices. Let $\triangle ABC$ be a triangle of $\mathbb{E}^2(\mathbb{R})$ whose vertices are $A = (a_1, a_2)$, $B = (b_1, b_2)$ and $C = (c_1, c_2)$. From elementary geometry we know that the area S of $\triangle ABC$ is given by the formula

$$S = \frac{1}{2}\ell([BC])\,\ell([AH]),$$

with $H = BC \cap h$ where h is the perpendicular to BC through A. By Proposition 3.40 the line BC has the equation

$$\begin{vmatrix} x_1 & x_2 & 1 \\ b_1 & b_2 & 1 \\ c_1 & c_2 & 1 \end{vmatrix} = 0.$$

Since $\ell([AH])$ is the distance between the point A and the line BC, by formula (4.11) we have

$$\ell([AH]) = \frac{\left|\det \begin{pmatrix} a_1 & a_2 & 1 \\ b_1 & b_2 & 1 \\ c_1 & c_2 & 1 \end{pmatrix}\right|}{\sqrt{(b_1 - c_1)^2 + (b_2 - c_2)^2}}.$$

We have also $\ell([BC]) = \sqrt{(b_1 - c_1)^2 + (b_2 - c_2)^2}$, so that we get

$$S = \frac{1}{2}\left|\det \begin{pmatrix} a_1 & a_2 & 1 \\ b_1 & b_2 & 1 \\ c_1 & c_2 & 1 \end{pmatrix}\right|. \tag{4.12}$$

4.1.1 Orthogonality

Definition 4.8 We can generalize Definition 4.6 (the perpendicular to a hyperplane passing through an assigned point) to any subspace $\mathcal{S}$ of a euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. Let $\mathcal{S}$ be an affine subspace of $\mathcal{A}$ and $P \in \mathcal{A} \setminus \mathcal{S}$. Let $\mathcal{S}' = \langle \mathcal{S} \cup \{P\} \rangle$ be the subspace generated by $\mathcal{S}$ and P, then $\dim_K \mathcal{S}' = \dim_K \mathcal{S} + 1$ and $\mathcal{S}$ is a hyperplane of $\mathcal{S}'$. Hence we can define *the perpendicular to* $\mathcal{S}$ *passing through* $P \notin \mathcal{S}$ as the perpendicular to $\mathcal{S}$ passing through $P \in \mathcal{S}'$ according to Definition 4.6. In particular it is defined the *foot of the perpendicular* to $\mathcal{S}$ passing through $P \notin \mathcal{S}$. Hence Q is the foot of the perpendicular to $\mathcal{S}$ passing through P if and only if $P \notin \mathcal{S}$, $Q \in \mathcal{S}$ and $\overrightarrow{PQ} \in D(\mathcal{S})^{\perp}$. Note that if $P \in \mathcal{S}$ and $\dim_K(\mathcal{A}) - \dim_K(\mathcal{S}) \geq 2$ there are infinitely many lines through P orthogonal to $\mathcal{S}$.

Sic rebus stantibus, we can state the following:

Proposition 4.9 (Three Perpendiculars Theorem) *Let $\mathcal{A}$ and $\mathcal{B}$ be two affine subspaces of a euclidean space $(\mathcal{S}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ such that $\mathcal{B} \subset \mathcal{A}$. Let $P \in \mathcal{S} \setminus \mathcal{A}$, $A \in \mathcal{A}$ and $B \in \mathcal{B}$.*

(i) *If A is the foot of the perpendicular to $\mathcal{A}$ through P and B is the foot of the perpendicular to $\mathcal{B}$ through A, then B is the foot of the perpendicular to $\mathcal{B}$ through P.*

(ii) *If A is the foot of the perpendicular to $\mathcal{A}$ through P and B is the foot of the perpendicular to $\mathcal{B}$ through P, then B is the foot of the perpendicular to $\mathcal{B}$ through A.*

Proof Since $\mathcal{B} \subseteq \mathcal{A}$ we have $D(\mathcal{B}) \subseteq D(\mathcal{A})$ and by Proposition 1.159 we get

$$D(\mathcal{B})^{\perp} = D(\mathcal{B})^{\perp}_{D(\mathcal{A})} \perp D(\mathcal{A})^{\perp}. \tag{4.13}$$

(i) If A is the foot of the perpendicular to $\mathcal{A}$ through P, then $A \in \mathcal{A}$ and $\overrightarrow{PA} \in D(\mathcal{A})^{\perp}$. If B is the foot of the perpendicular to $\mathcal{B}$ through A, then $B \in \mathcal{B}$, $A \notin \mathcal{B}$ and $\overrightarrow{AB} \in D(\mathcal{B})^{\perp}_{D(\mathcal{A})}$ (if $A \in \mathcal{B}$ then $A = B$ is the foot of the perpendicular to $\mathcal{B}$ through P). Therefore

$$\overrightarrow{PB} = \overrightarrow{PA} + \overrightarrow{AB} \in D(\mathcal{A})^{\perp} + D(\mathcal{B})^{\perp}_{D(\mathcal{A})} = D(\mathcal{B})^{\perp}$$

by virtue of (4.13). Since $B \in \mathcal{B}$, then B is the foot of the perpendicular to $\mathcal{B}$ through P.

(ii) We assume that $A \notin \mathcal{B}$, otherwise $A = B$. By our assumptions we have $\overrightarrow{PA} \in D(\mathcal{A})^{\perp}$ and $\overrightarrow{PB} \in D(\mathcal{B})^{\perp}$. It follows that $\overrightarrow{AB} = \overrightarrow{PA} - \overrightarrow{PB} \in D(\mathcal{B})^{\perp}$ (indeed $D(\mathcal{B}) \subset D(\mathcal{A})$ implies $D(\mathcal{A})^{\perp} \subset D(\mathcal{B})^{\perp}$). Since $A, B \in \mathcal{A}$, $\overrightarrow{AB} \in D(\mathcal{A})$, so that $\overrightarrow{AB} \in D(\mathcal{B})^{\perp}_{D(\mathcal{A})}$ by (4.13), then (ii) follows because $B \in \mathcal{B}$.

□

Definition 4.10 We can generalize the concept of orthogonality between two lines or a line and a hyperplane to a pair of arbitrary subspaces of an euclidean space $(\mathcal{S}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$.

Let $\mathcal{A}$ and $\mathcal{B}$ be two subspaces of a euclidean space $(\mathcal{S}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ of dimension n. We say that $\mathcal{A}$ is *perpendicular to* $\mathcal{B}$, written $\mathcal{A} \perp \mathcal{B}$, if one of the following conditions is satisfied:

(i) $D(\mathcal{A}) \subset D(\mathcal{B})^{\perp}$, if $\dim \mathcal{A} + \dim \mathcal{B} \leq n$,
(ii) $D(\mathcal{B})^{\perp} \subset D(\mathcal{A})$, if $\dim \mathcal{A} + \dim \mathcal{B} \geq n$.

Note that if $\dim \mathcal{A} + \dim \mathcal{B} = n$, the conditions (i) and (ii) of Definition 4.10 are equivalent. Indeed, by Proposition 1.156 we have $\dim D(\mathcal{B})^{\perp}) = n - \dim D(\mathcal{B}) = \dim D(\mathcal{A})$, then the inclusions of (i) and (ii) are equalities.

Proposition 4.11 *Let $(\mathcal{S}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space of dimension n.*

(i) *For every point $P \in \mathcal{S}$ and for every affine subspace $\mathcal{A}$ of $\mathcal{S}$ there exists one and only one subspace $\mathcal{B}$ containing P and perpendicular to $\mathcal{A}$ whose dimension is $n - \dim \mathcal{A}$.*
(ii) *Let $\mathcal{A}$ and $\mathcal{B}$ be two affine subspaces of $\mathcal{S}$.*
   (a) *If $\mathcal{A} \perp \mathcal{B}$, then $\mathcal{B} \perp \mathcal{A}$.*
   (b) *If $\dim \mathcal{A} + \dim \mathcal{B} \leq n$, $\mathcal{A} \perp \mathcal{B}$ and $\mathcal{A}' \subset \mathcal{A}$ is an affine subspaces of $\mathcal{S}$, then $\mathcal{A}' \perp \mathcal{B}$.*
   (c) *If $\dim \mathcal{A} + \dim \mathcal{B} \geq n$, $\mathcal{A} \perp \mathcal{B}$ and $\mathcal{A}' \supset \mathcal{A}$ is an affine subspaces of $\mathcal{S}$, then $\mathcal{A}' \perp \mathcal{B}$.*
   (d) *If $\mathcal{A} \perp \mathcal{B}$ and $\dim \mathcal{A} + \dim \mathcal{B} = n$, then $\mathcal{A} \cap \mathcal{B}$ is a point.*

(e) *Let* $\mathcal{C}$ *be an affine subspace of* $\mathcal{S}$ *such that* $\dim \mathcal{B} = \dim \mathcal{C} = n - \dim \mathcal{A}$. *If* $\mathcal{B} \perp \mathcal{A}$ *and* $\mathcal{C} \perp \mathcal{A}$, *then* $\mathcal{B} \parallel \mathcal{C}$.

Proof

(i) The director subspace of such an affine subspace $\mathcal{B}$ must be $D(\mathcal{A})^\perp$ so that $\mathcal{B} = P + D(\mathcal{A})^\perp$ which proves existence and uniqueness.

(ii)-(a) If $D(\mathcal{A}) \subset D(\mathcal{B})^\perp$, then $D(\mathcal{B}) = D(\mathcal{B})^{\perp\perp} \subset D(\mathcal{A})^\perp$ (by Proposition 1.156). If $D(\mathcal{B})^\perp \subset D(\mathcal{A})$ in analogous manner we have $D(\mathcal{A})^\perp \subset D(\mathcal{B})^{\perp\perp} = D(\mathcal{B})$.

(ii)-(b) and (ii)-(c) follow immediately from Definition 4.10 and from $\mathcal{A}' \subset \mathcal{A} \Rightarrow D(\mathcal{A}') \subset D(\mathcal{A})$.

(ii)-(d) Take two points $A \in \mathcal{A}$ and $B \in \mathcal{B}$. By assumptions we have $D(\mathcal{A}) = D(\mathcal{B})^\perp$, then by Proposition 1.156 we get $V = D(\mathcal{A}) \perp D(\mathcal{B})$. In particular, $\overrightarrow{AB} = u + v$, with $u \in D(\mathcal{A})$ and $v \in D(\mathcal{B})$. Since $A \in \mathcal{A}$ and $u \in D(\mathcal{A})$, there exists a unique point $C \in \mathcal{A}$ such that $\overrightarrow{AC} = u$, therefore $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB} = u - \overrightarrow{AB} = -v \in D(\mathcal{B})$ so that $C \in \mathcal{B}$ and $C \in \mathcal{A} \cap \mathcal{B}$. We get immediately the formula $D(\mathcal{A} \cap \mathcal{B}) = D(\mathcal{A}) \cap D(\mathcal{B})$. Hence

$$\dim_K(\mathcal{A} \cap \mathcal{B}) = \dim_K(D(\mathcal{A} \cap \mathcal{B})) = \dim_K(D(\mathcal{A}) \cap D(\mathcal{B})) = 0,$$

because $D(\mathcal{A})$ and $D(\mathcal{B})$ are perpendicular, so that $\mathcal{A} \cap \mathcal{B} = C$.

(ii)-(e) By assumptions we have $D(\mathcal{B}) = D(\mathcal{A})^\perp = D(\mathcal{C})$.

□

Remark 4.12 If $\dim_K(\mathcal{S}) = 3$, special cases of (ii)-(b) and (ii)-(c) of Proposition 4.11 are the following:

1. If a line d is perpendicular to a plane $\mathcal{P}$, then d is perpendiculat to every line contained in $\mathcal{P}$ (see Definition 4.6).
2. If a plane $\mathcal{P}$ is perpendicular to a plane $\mathcal{Q}$ containing a line d, then $\mathcal{P}$ is perpendicular to d.

Definition 4.13 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space and $\mathcal{S}$ a subspace of $\mathcal{A}$. For every $P \in \mathcal{A}$ consider the affine subspace $\mathcal{T} := P + D(\mathcal{T})$ with $D(\mathcal{T}) = D(\mathcal{S})^\perp$. By Theorem 3.67 the intersection $\mathcal{S} \cap \mathcal{T}$ is a point which will be denoted by $p_\mathcal{S}(P)$ and called *the orthogonal projection of* P *on* $\mathcal{S}$. If $P \notin \mathcal{S}$, the point $p_\mathcal{S}(P)$ is nothing else but the foot of the perpendicular from P to $\mathcal{S}$ according to Definition 4.8. Hence $p_\mathcal{S} : \mathcal{A} \to \mathcal{A}$ is the projection on $\mathcal{S}$ along $D(\mathcal{S})^\perp$ according to Definition 3.78. Let $\mathcal{B}$ be an affine subspace of $\mathcal{A}$. The *orthogonal projection* of $\mathcal{B}$ on $\mathcal{S}$ is the affine subspace $p_\mathcal{S}(\mathcal{B})$. In particular, if $\mathcal{H}$ and d are a hyperplane and a line of $\mathcal{A}$ respectively, the orthogonal projection of d over $\mathcal{H}$ is

$$\begin{cases} \mathcal{H} \cap d & \text{if } d \perp \mathcal{H} \\ \mathcal{H} \cap \pi & \text{otherwise,} \end{cases}$$

where π is the plane containing d and perpendicular to $\mathcal{H}$, i.e. $\pi = P + W$, where P is any point of d, e is any line orthogonal to $\mathcal{H}$ and $W = D(e) \oplus D(d)$.

By Pythagoras' theorem (Theorem 12.66) we have

$$d(P, Q) \geq d(P, p_{\mathcal{S}}(P)), \qquad \forall\, Q \in \mathcal{S},$$

so that it is natural to define the *distance from P to $\mathcal{S}$* by means of

$$d(P, \mathcal{S}) = \inf\{d(P, Q) \mid Q \in \mathcal{S}\} = d(P, p_{\mathcal{S}}(P)). \tag{4.14}$$

More generally, if $\mathcal{S}$ and $\mathcal{S}'$ are affine subspaces the *distance $d(\mathcal{S}, \mathcal{S}')$ between $\mathcal{S}$ and $\mathcal{S}'$* is given by

$$d(\mathcal{S}, \mathcal{S}') = \inf\{d(P, Q) : P \in \mathcal{S},\ Q \in \mathcal{S}'\} = \inf\{d(P, \mathcal{S}') : P \in \mathcal{S}\}. \tag{4.15}$$

The notion of Gram determinant given in Section 2.2 of Chapter 2 is now used to calculate the distance from a point to an affine subspace (see [4]).

Proposition 4.14 *Let $\mathcal{S}$ be a subspace of a euclidean space $(\mathcal{A}, (V, \langle\cdot\cdot\rangle), \varphi)$. Let $\{P_0, \ldots, P_k\}$ be an affine frame of $\mathcal{S}$. Then for every point $P \in \mathcal{A}$ we have*

$$d^2(P, \mathcal{S}) = \frac{\mathrm{Gram}(v, v_1, \ldots, v_k)}{\mathrm{Gram}(v_1, \ldots, v_k)}, \tag{4.16}$$

where

$$v = \overrightarrow{P_0 P}, \qquad v_i = \overrightarrow{P_0 P_i}, \quad i = 1, \ldots, k.$$

Proof Let $Q = p_{\mathcal{S}}(P)$. By our assumptions there are $a_i \in \mathbb{R}$, $i = 1, \ldots, k$, such that $\overrightarrow{P_0 Q} = \sum_{i=1}^{k} a_i v_i$. Since $\overrightarrow{QP} = \overrightarrow{P_0 Q} - \overrightarrow{P_0 P}$, put $w = \overrightarrow{QP}$ so that

$$w = v - \sum_{i=1}^{k} a_i v_i.$$

Then by (2.21) we have

$$\mathrm{Gram}(v, v_1, \ldots, v_k) = \mathrm{Gram}(w, v_1, \ldots, v_k).$$

Since $\langle w, v_i\rangle = 0, i = 1, \ldots, k$, we get

$$\mathrm{Gram}(w, v_1, \ldots, v_k) = \begin{vmatrix} \|w\|^2 & 0 & \ldots & 0 \\ 0 & \|v_1\|^2 & \ldots & \langle v_1, v_p\rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \langle v_k, v_1\rangle & \ldots & \|v_k\|^2 \end{vmatrix} = \|w\|^2 \mathrm{Gram}(v_1, \ldots, v_k).$$

Then (4.16) follows from $\|w\|^2 = d^2(P, \mathcal{S})$. □

Remark 4.15 The reader can deduce (4.11) from (4.16).

Corollary 4.16 *Let r be a line of $\mathbb{E}^3(\mathbb{R})$ having parametric equations $x_i = a_i + \alpha_i t$, $i = 1, 2, 3$; let $P = (p_1, p_2, p_3)$ be a point of $\mathbb{E}^3(\mathbb{R})$ not lying on r. Then we have the formula*

$$d(P, r) = \frac{\sqrt{\begin{vmatrix} p_2 - a_2 & p_3 - a_3 \\ \alpha_2 & \alpha_3 \end{vmatrix}^2 + \begin{vmatrix} p_1 - a_1 & p_3 - a_3 \\ \alpha_1 & \alpha_3 \end{vmatrix}^2 + \begin{vmatrix} p_1 - a_1 & p_2 - a_2 \\ \alpha_1 & \alpha_2 \end{vmatrix}^2}}{\sqrt{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}}. \tag{4.17}$$

Proof One can deduce (4.17) from (4.16) or using the cross-product (see Exercises 3.10– 3.12). Let $Q' = p_r(P)$ and Q any point of r, $Q \neq Q'$. Consider the triangle $\triangle PQQ'$. By Exercise 3.12 we have

$$d(P, r) = d(P, Q') = \|\overrightarrow{PQ'}\| = \|\overrightarrow{PQ}\| \sin\theta = \frac{\|\overrightarrow{PQ} \times \overrightarrow{Q'Q}\|}{\|\overrightarrow{Q'Q}\|},$$

with $\theta = \measuredangle(\overrightarrow{QP}, \overrightarrow{QQ'})$. Note that $\dfrac{\|\overrightarrow{PQ} \times \overrightarrow{Q'Q}\|}{\|\overrightarrow{Q'Q}\|}$ is independent of the choice of $Q \neq Q'$. □

Angles Between Two Hyperplanes and Between a Line and a Hyperplane

Definition 4.17 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space of dimension n, $\mathcal{H}$ and $\mathcal{H}'$ be two hyperplanes of $\mathcal{A}$. Let d and d' be two normal lines to $\mathcal{H}$ and $\mathcal{H}'$ respectively. The *angle* $\measuredangle(H, H')$ between $\mathcal{H}$ and $\mathcal{H}'$ is defined as being $\measuredangle(d, d')$. By Definition 4.6, $\measuredangle(H, H')$ does not depend on the choice of d and d'.

Let $\pi : a_1x_1 + \cdots + a_nx_n = b$ and $\pi' : a'_1x_1 + \cdots + a'_nx_n = b'$ be two hyperplanes of $\mathbb{E}^n(\mathbb{R})$. Let d and d' be two normal lines to π and π' respectively. Hence d and d' have parametric equations of the form

$$d : x_i = p_i + a_i t, \quad i = 1, \ldots, n,$$
$$d' : x_i = p'_i + a'_i t, \quad i = 1, \ldots, n.$$

Therefore we have

$$\alpha := \measuredangle(\pi, \pi') = \arccos\left(\frac{\left|\sum_{i=1}^{n} a_i a_i'\right|}{\sqrt{(\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} a_i'^2)}}\right). \tag{4.18}$$

In particular, $\pi \perp \pi'$ if and only if $\sum_{i=1}^{n} a_i a_i' = 0$.

Let $\mathcal{H}$ and d be a hyperplane and a line of $\mathcal{A}$ respectively. We define the *angle* $\measuredangle(d, \mathcal{H})$ between d and $\mathcal{H}$ as follows:

$$\measuredangle(d, \mathcal{H}) = \begin{cases} 0 & \text{if } d \parallel \mathcal{H}, \\ \dfrac{\pi}{2} - \measuredangle(d, e) & \text{if } d \parallel \mathcal{H}, \end{cases}$$

where e is any normal line to $\mathcal{H}$.

Let d be a line and $\mathcal{H}$ a hyperplane of $\mathbb{E}^n(\mathbb{R})$:

$$D(d) = \langle (m_1, \ldots, m_n) \rangle, \qquad \mathcal{H} : \ a_1 x_1 + \cdots + a_n x_n = b.$$

If e is a normal line to $\mathcal{H}$ then $D(e) = \langle (a_1, \ldots, a_n) \rangle$, hence we have

$$\measuredangle(d, e) = \arccos\left(\frac{\left|\sum_{i=1}^{n} a_i m_i\right|}{\sqrt{(\sum_{i=1}^{n} a_i^2)\left(\sum_{i=1}^{n} m_i^2\right)}}\right),$$

so that (since $\measuredangle(d, \mathcal{H}) = \dfrac{\pi}{2} - \measuredangle(d, e)$) we get

$$\measuredangle(d, \mathcal{H}) = \arcsin\left(\frac{\left|\sum_{i=1}^{n} a_i m_i\right|}{\sqrt{(\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} m_i^2)}}\right). \tag{4.19}$$

In Fig. 4.1, e is chosen to contain the point $A = \pi \cap d$, d' is the orthogonal projection of d over $\mathcal{H}$ (see Definition 4.13) and $\alpha = \measuredangle(d, \mathcal{H})$.

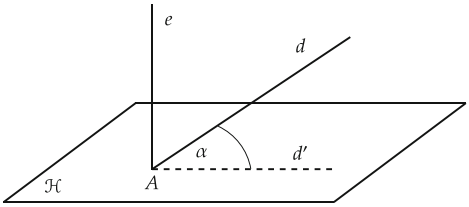


Fig. 4.1 Angle between a line and a hyperplane

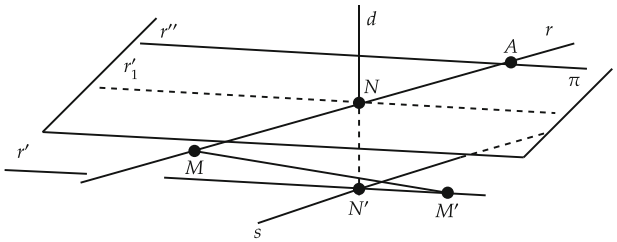


Fig. 4.2 The common perpendicular to two skew lines

Common Perpendicular and Distance Between Two Skew Lines

Proposition 4.18 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space of dimension n. Let r and r' be two skew lines of $\mathcal{A}$.*

(i) *There exists one and only one line d of $\mathcal{A}$ intersecting both r and r' and orthogonal to both r and r' (*see Fig. 4.2*).*

(ii) *If $N = d \cap r$ and $N' = d \cap r'$, then $d(N, N') \leq d(M, M')$ for every $M \in r$ and $M' \in r'$.*

(iii) *Let A, $B \in r$ and B, $B' \in r'$, all points different from N and N'. Then we have*

$$d(N, N')^2 = \frac{\mathrm{Gram}(\overrightarrow{AA'}, \overrightarrow{AB}, \overrightarrow{A'B'})}{\mathrm{Gram}(\overrightarrow{AB}, \overrightarrow{A'B'})}. \tag{4.20}$$

Proof We can assume that $\dim_{\mathbb{R}}(\mathcal{A}) = 3$ since all lines and planes involved in this proof are contained in the affine subspace generated by r and r' which is of dimension 3.

(i) Let A be an arbitrary point of r and r'' the line through A parallel to r'. Since r and r' are skew, the lines r and r'' are distinct (indeed they are not parallel) so that they generate a plane π. We observe that r' is not orthogonal to π, otherwise $r' \nparallel r''$ so that the orthogonal projection of r' over π is a line r'_1. We put $D(r) = \langle v_r\rangle$, $D(r') = \langle v_{r'}\rangle$, $D(r'') = \langle v_{r''}\rangle$ and $D(e) = \langle v_e\rangle$, where e is any line perpendicular to π whose director subspace is $D(\pi) = \langle v_r, v_{r''}\rangle$. Note that from Definition 4.13 we know that r'_1 is the intersection between π and the plane π' containing r' and ortogonal to π, i.e. $D(\pi') = \langle v_{r'}, v_e\rangle$. We claim that

$r_1' \not\parallel r$. indeed, if $r_1' \parallel r$ then $v_{r_1'} = \lambda v_r$ which implies, since $v_{r_1'} = \alpha v_{r'} + \beta v_e$ and $k v_{r''} = v_{r'}$,

$$\lambda v_r = \alpha v_{r'} + \beta v_e = \gamma v_{r''} + \beta v_e,$$

so that v_e, v_r and $v_{r''}$ would be linearly dependent, which is false since e is perpendicular to π. Let $N = r \cap r_1'$ and $N' \in r'$ be the unique point whose projection over π is N. Then the line $d := NN'$ is obviously perpendicular to r in N and to r' in N'. So the existence of d is proved. The uniqueness of d follows from the proof of (ii).

(ii) Consider the plane σ parallel to π and passing through N'. The plane σ is generated by r' and by the line s parallel to r and passing through N'. The line $d = NN'$ is orthogonal to both planes π and σ. Hence $d(N, N')$ is just the distance between π and σ. Let $M \in r$ and $M' \in r'$ be two arbitrary points. Then $M \in \pi$ and $M' \in \sigma$ so that $d(N, N') \leq d(M, M')$, and $d(N, N') = d(M, M')$ if and only if $M = N$ and $M' = N'$ (see Definition 4.7). Thus we have also proved the uniqueness of point (i).

(iii) Since

$$\overrightarrow{AA'} = \overrightarrow{AN} + \overrightarrow{NN'} + \overrightarrow{N'A'}, \quad \overrightarrow{AN} = \lambda \overrightarrow{AB}, \quad \overrightarrow{N'A'} = \mu \overrightarrow{A'B'}, \quad \lambda, \mu \in \mathbb{R}^*,$$

we have, by (2.21),

$$\begin{aligned} \operatorname{Gram}(\overrightarrow{AA'}, \overrightarrow{AB}, \overrightarrow{A'B'}) &= \operatorname{Gram}(\overrightarrow{AN} + \overrightarrow{NN'} + \overrightarrow{N'A'}, \overrightarrow{AB}, \overrightarrow{A'B'}) \\ &= \operatorname{Gram}(\lambda \overrightarrow{AB} + \overrightarrow{NN'} + \mu \overrightarrow{A'B'}, \overrightarrow{AB}, \overrightarrow{A'B'}) \\ &= \operatorname{Gram}(\overrightarrow{NN'}, \overrightarrow{AB}, \overrightarrow{A'B'}) \\ &= \|\overrightarrow{NN'}\|^2 \operatorname{Gram}(\overrightarrow{AB}, \overrightarrow{A'B'}). \end{aligned}$$

Since $\|\overrightarrow{NN'}\| = d(N, N')$ we get (4.20).

□

Definition 4.19 The line d of Proposition 4.18 is called the *common perpendicular* to the skew lines r and r' and the distance $d(N, N')$ of (4.20) is called the *distance* between r and r' and denoted by $d(r, r')$.

Definition 4.20 (Axis of a Segment and Bisector Hyperplanes) Let $(\mathcal{A}, (V, \langle \cdot, \cdot \rangle), \varphi)$ be a euclidean space, $\mathcal{H}$ and $\mathcal{H}'$ two non-parallel hyperplanes of $\mathcal{A}$. The geometric locus of equidistant points of $\mathcal{A}$ from $\mathcal{H}$ and $\mathcal{H}'$ consists of two mutually orthogonal hyperplanes $\mathcal{K}$ and $\mathcal{K}'$ as is easily seen. The hyperplanes $\mathcal{K}$ and $\mathcal{K}'$ are called *bisector hyperplanes* of $\mathcal{H}$ and $\mathcal{H}'$.

Let P and $Q \in \mathcal{A}$ be two distinct points. The geometric locus of equidistant points of $\mathcal{A}$ from P and Q is the hyperplane $\mathcal{H}$ orthogonal to the line PQ passing through the midpoint of the segment $[PQ]$, called the *axis of segment* $[PQ]$.

Definition 4.21 (Spheres) Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be an euclidean space of dimension $n \geq 1$ and C a point of $\mathcal{A}$. The *sphere* $S^{n-1}(C, r)$ of $\mathcal{A}$ of *centre* C and radius $r > 0$ is defined as being the locus of the points $P \in \mathcal{A}$ such that $d(P, C) = r$. If $n = 1$ we have a degenerate sphere $S^0(C, r) = \{P_1, P_2\}$, where $d(C, P_1) = D(C, P_2) = r$. If $C = (p_1, \ldots, p_n) \in \mathbb{E}^n(\mathbb{R})$, we have

$$S^{n-1}(C, r) = \{(x_1, \ldots, x_n) \in \mathbb{E}^n(\mathbb{R}) \mid (x_1 - p_1)^2 + \cdots + (x_n - p_n)^2 = r^2\}.$$

A hyperplane $\mathcal{H}$ is *tangent* to $S^{n-1}(C, r)$ if $d(\mathcal{H}.C) = r$. If $Q \in S^{n-1}(C, r)$, the *tangent hyperplane* to $S^{n-1}(C, r)$ at Q, denoted by $\mathrm{T}_Q(S^{n-1}(C, r))$, is the hyperplane perpendicular to CQ passing through Q. Every line contained in $\mathrm{T}_Q(S^{n-1}(C, r))$ and passing through Q is called a *tangent* to $S^{n-1}(C, r)$ at Q.

In $\mathbb{E}^n(\mathbb{R})$ consider the sphere $S^{n-1}(O, 1)$ of centre the origin $O = (0, \ldots, 0)$ and radius 1. The line joining the "north pole" $N = (0, \ldots, 0, 1)$ and an arbitrary point $P = (u_1, \ldots, u_{n-1}, 0)$ of the hyperplane $H : x_n = 0$ has parametric equations

$$\begin{cases} x_1 & = u_1 t \\ \vdots & \vdots \\ x_{n-1} & = u_{n-1} t \\ x_n & = 1 - t, \end{cases}$$

and meets $S^{n-1}(O, 1) \setminus \{N\}$ at a unique point

$$\begin{cases} x_1 & = \dfrac{2u_1}{u_1^2 + \cdots + u_{n-1}^2 + 1} \\ \vdots & \quad \vdots \quad \vdots \\ x_{n-1} & = \dfrac{2u_{n-1}}{u_1^2 + \cdots + u_{n-1}^2 + 1} \\ x_n & = \dfrac{u_1^2 + \cdots + u_{n-1}^2 - 1}{u_1^2 + \cdots + u_{n-1}^2 + 1}. \end{cases} \tag{4.21}$$

Thus (4.21) represents a bijective map $H \to S^{n-1}(O, 1) \setminus \{N\}$ (a homeomorphism with respect to euclidean topology) whose inverse is called the *stereographic projection from the north pole*. In the same way we define the *stereographic projection from the "south pole"* $S = (0, \ldots, 0, -1)$.

4.2 Orthogonal Affine Morphisms

Definition 4.22 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ and $(\mathcal{B}, (W, \langle\cdot,\cdot\rangle_W), \psi)$ be two euclidean spaces of dimensions n and m respectively. An affine morphism $f : (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) \to (\mathcal{B}, (W, \langle\cdot,\cdot\rangle_W), \psi)$ is called an *orthogonal affine morphism* or simply *orthogonal morphism* if its trace $\mathrm{Tr}(f) : V \to W$ is an orthogonal linear map, i.e. if there exists a point $O \in \mathcal{A}$ such that

$$\langle \mathrm{Tr}(f)(\overrightarrow{OA}), \mathrm{Tr}(f)(\overrightarrow{OB})\rangle_W = \langle \overrightarrow{OA}, \overrightarrow{OB}\rangle_V, \qquad \forall\, A, B \in \mathcal{A}, \tag{4.22}$$

namely if

$$\|\mathrm{Tr}(f)(\overrightarrow{OA})\|_W^2 = \|\overrightarrow{OA}\|_V^2, \qquad \forall\, A \in \mathcal{A}. \tag{4.23}$$

We have just seen (Lemma 3.45) that (4.22) does not depend on the choice of O. If $d_{\mathcal{A}}$ and $d_{\mathcal{B}}$ are the associated distances (see Definition 4.2), (4.23) is equivalent to

$$d_{\mathcal{B}}(f(P), f(Q)) = d_{\mathcal{A}}(P, Q), \qquad \forall\, P, Q \in \mathcal{A}, \tag{4.24}$$

so that an orthogonal morphism preserves the distance. Therefore by (4.24) f is injective. If f is also surjective (i.e. $\mathrm{Tr}(f)$ is surjective) we say that f is an *orthogonal isomorphism* or *isometry*. If $n \leq m$ are two positive integers, the map

$$f : \mathbb{E}^n \to \mathbb{E}^m, \qquad f(x_1, \ldots, x_n) = (x_1, \ldots, x_n, 0, \ldots, 0)$$

is an orthogonal morphism which is not surjective.

If $\dim_K(\mathcal{A}) = \dim_K(\mathcal{B})$ an orthogonal morphism is necessarily an orthogonal isomorphism. In particular, every orthogonal morphism $f : (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) \to (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ is an orthogonal isomorphism called an *orthogonal automorphism*. We denote the set of all orthogonal automorphisms of $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ by $\mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ and the group $\mathrm{O}(\mathbb{E}^n(\mathbb{R}), \langle\cdot,\cdot\rangle_s)$ simply by $\mathrm{O}(\mathbb{E}^n(\mathbb{R}))$. By Proposition 3.48 it is immediate to show that $\mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ is a group. Since $\sigma \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ if and only if $\mathrm{Tr}(\sigma)$ is an orthogonal isomorphism, then $\det(\mathrm{Tr}(\sigma)) = \pm 1$. If $\det(\mathrm{Tr}(\sigma)) = 1$ we call σ *proper motion* or *special isometry*; if $\det(\mathrm{Tr}(\sigma)) = -1$ we call σ *improper motion* or *non-special isometry*. We will denote the set of all proper motions of $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ by $\mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ (and $\mathrm{SO}(\mathbb{E}^n(\mathbb{R}), \langle\cdot,\cdot\rangle_s)$ by $\mathrm{SO}(\mathbb{E}^n(\mathbb{R}))$). As a matter of fact, $\mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ is a normal subgroup of index 2 of $\mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. Indeed, $\mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ is the kernel of the surjective group-homomorphism

$$\chi : \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) \to \{-1, 1\},$$
$$\chi(\sigma) = \det(\mathrm{Tr}(\sigma)), \qquad \forall\, \sigma \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi).$$

It is immediate that a translation τ_v is a proper motion (see Example 3.53-(3) so that the group $\mathrm{T}_{\mathbb{R}}(\mathcal{A})$ of translations is a subgroup of $\mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. The orthogonal symmetries through hyperplanes (see Definitions 4.25) are improper motions, as we will prove in Proposition 4.30.

Foe every $O \in \mathcal{A}$ we introduce the sets

$$\mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi; O) = \{\sigma \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) : \sigma(O) = O\},$$
$$\mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi; O) = \{\sigma \in \mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) : \sigma(O) = O\},$$

which are subgroups of $\mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ and $\mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ respectively. The group isoomorphism (see Definition 3.55)

$$\mathrm{Tr}_{|\mathrm{Aut}_{\mathbb{R}}(\mathcal{A},O)} : \mathrm{Aut}_{\mathbb{R}}(\mathcal{A}, O) \to \mathrm{GL}_K(V)$$

induces two isomorphisms

$$\mathrm{Tr}_{|\mathrm{O}(\mathcal{A},(V,\langle\cdot,\cdot\rangle_V),\varphi;O)} : \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi; O) \to \mathrm{O}(V, \langle\cdot,\cdot\rangle_V), \tag{4.25}$$
$$\mathrm{Tr}_{|\mathrm{SO}(\mathcal{A},(V,\langle\cdot,\cdot\rangle_V),\varphi;O)} : \mathrm{SO}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi; O) \to \mathrm{SO}(V, \langle\cdot,\cdot\rangle_V). \tag{4.26}$$

We have also called the elements of $\mathrm{SO}(V, \langle\cdot,\cdot\rangle_V)$ rotations (see Definition 1.168) so that we can name the special isometries of $\mathcal{A}$ which have fixed point *rotations*.

Proposition 4.23 *For every euclidean frame $\mathcal{F} = \{A_0, A_1, \ldots, A_n\}$ of a euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ of dimension n there exists one and only one orthogonal isomorphism*

$$f : (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) \to \mathbb{E}^n(\mathbb{R})$$

such that $f(A_i) = E_i$, $i = 1, \ldots, n$ and $f(A_0) = O$, where $E_0, \ldots, E_n$ is the standard euclidean frame of $\mathbb{E}^n(\mathbb{R})$.

Proof If $A \in \mathcal{A}$ has coordinates $x_1(A), \ldots, x_n(A)$ with respect to $\mathcal{F}$ the required orthogonal isomorphism is given by

$$f(A) := (x_1(A), \ldots, x_m(A)) \in \mathbb{E}^n(\mathbb{R}).$$

□

Proposition 4.24 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ and $(\mathcal{B}, (W, \langle\cdot,\cdot\rangle_W), \psi)$ be two euclidean spaces. A map*

$$f : (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi) \to (\mathcal{B}, (W, \langle\cdot,\cdot\rangle_W), \psi)$$

is an isometry (i.e. $\|\overrightarrow{f(A)f(B)}\|_W = \|\overrightarrow{AB}\|_V$, $\forall\, A, B \in \mathcal{A}$) if and only if f is an orthogonal isomorphism.

Proof We have only to prove that $\mathrm{Tr}(f)$ is a linear orthogonal isomorphism. Fixing a point $O \in \mathcal{A}$ we have

$$\begin{aligned}\|\mathrm{Tr}(f)(\overrightarrow{OA}) - \mathrm{Tr}(f)(\overrightarrow{OB})\|_W &= \|\overrightarrow{f(O)f(A)} - \overrightarrow{f(O)f(B)}\|_W \\ &= \|-(\overrightarrow{f(A)f(O)} + \overrightarrow{f(O)f(B)})\|_W \\ = \|-\overrightarrow{f(A)f(B)}\|_W = \|\overrightarrow{f(A)f(B)}\|_W &= \|\overrightarrow{AB}\|_V = \|-\overrightarrow{AB}\|_V \\ &= \|-\overrightarrow{AO} - \overrightarrow{OB}\|_V = \|\overrightarrow{OA} - \overrightarrow{OB}\|_V\end{aligned}$$

so that $\mathrm{Tr}(f)$ is an isometry. Furthermore $\mathrm{Tr}(f)(0_V) = 0_W$ so that, by Proposition 1.164 $\mathrm{Tr}(f)$ (with $u_0 = 0_V$), preserves the inner product and it is linear, i.e. it is an orthogonal isomorphism. □

4.2.1 Structure of Isometries

Definition 4.25 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space and $\mathcal{S}$ a subspace of $\mathcal{A}$. Let W be the orthogonal complement of $D(\mathcal{S})$. The *orthogonal symmetry or reflection through* $\mathcal{S}$ is $s_{\mathcal{S}} := s_{\mathcal{S},W}$ as defined in (3.34)

$$s_{\mathcal{S}}(A) = p_{\mathcal{S}}(A) + \overrightarrow{A\, p_{\mathcal{S}}(A)} \qquad \forall\, P \in \mathcal{A}, \tag{4.27}$$

where $p_{\mathcal{S}}$ is the orthogonal projection on $\mathcal{S}$ (see Definition 4.13).

Proposition 4.26 *Under the above notation and assumptions $s_{\mathcal{S}}$ is an orthogonal affine isomorphism.*

Proof We have seen that $\mathrm{Tr}(s_{\mathcal{S}}) = 2\,\mathrm{Tr}(p_{\mathcal{S}}) - \mathrm{id}_V$ (see Remarks 3.80 and 3.84), then by Example 1.161-(2) we conclude that $\mathrm{Tr}(s_{\mathcal{S}})$ is a linear orthogonal isomorphism, i.e. $s_{\mathcal{S}}$ is an orthogonal affine isomorphism. □

Corollary 4.27 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space and $\mathcal{S}$ a subspace of $\mathcal{A}$. If $\sigma \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ then*

$$\sigma \circ s_{\mathcal{S}} = s_{\sigma(\mathcal{S})} \circ \sigma.$$

Proof It follows from Lemma 3.87 since $(\mathrm{Tr}(\sigma)(D(\mathcal{S})))^{\perp} = \mathrm{Tr}(\sigma)(D(\mathcal{S}))^{\perp})$. □

Example 4.28 Let $\mathcal{H}$ be a hyperplane of $\mathbb{E}^n(\mathbb{R})$ having the equation $a_1x_1 + \cdots + a_nx_n = b$. Let $P = (p_1, \ldots, p_n) \in \mathbb{E}^n(\mathbb{R})$ be an arbitrary point. We have seen that the foot $Q = (q_1, \ldots, q_n)$ of the perpendicular to $\mathcal{H}$ through P has coordinates $q_1, \ldots, q_n$ where

$$q_i = p_i + t_0 a_i, \quad i = 1, \ldots, n, \qquad t_0 = -\frac{a_1 p_1 + \cdots + a_n p_n - b}{a_1^2 + \cdots + a_n^2}.$$

Let $P' = (p'_1, \ldots, p'_n)$ be the point of $\mathbb{E}^n(\mathbb{R})$ such that Q is the midpoint of $[PP']$, i.e.

$$p'_i = p_i - 2\,\frac{a_1 p_1 + \cdots + a_n p_n - b}{a_1^2 + \cdots + a_n^2}\,a_i.$$

Then the map $\sigma : \mathbb{E}^n(\mathbb{R}) \to \mathbb{E}^n(\mathbb{R})$ defined by

$$\sigma(x_1, \ldots, x_n) = (y_1, \ldots, y_n),$$
$$y_i = x_i - 2\,\frac{a_1 x_1 + \cdots + a_n x_n - b}{a_1^2 + \cdots + a_n^2}\,a_i, \quad i = 1, \ldots, n \tag{4.28}$$

is the orthogonal symmetry with respect to $\mathcal{H}$. If $n = 2$ the equations (4.28) take the form

$$y_1 = \frac{-a_1^2 + a_2^2}{a_1^2 + a_2^2}\,x_1 - \frac{2a_1a_2}{a_1^2 + a_2^2}\,x_2 + \frac{2a_1 b}{a_1^2 + a_2^2},$$
$$y_2 = -\frac{2a_1a_2}{a_1^2 + a_2^2}\,x_1 + \frac{a_1^2 - a_2^2}{a_1^2 + a_2^2}\,x_2 + \frac{2a_2 b}{a_1^2 + a_2^2}.$$

The matrix

$$\begin{pmatrix} \dfrac{-a_1^2 + a_2^2}{a_1^2 + a_2^2} & -\dfrac{2a_1a_2}{a_1^2 + a_2^2} \\ -\dfrac{2a_1a_2}{a_1^2 + a_2^2} & \dfrac{a_1^2 - a_2^2}{a_1^2 + a_2^2} \end{pmatrix}$$

is obviously orthogonal (of determinant -1).

Remark 4.29 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle), \varphi)$ be a euclidean space and P a point of $\mathcal{A}$. Then the symmetry s_P through P (see Example 3.85) is obviously an orthogonal symmetry.

Proposition 4.30 *The orthogonal symmetries through a hyperplane of a euclidean space $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ are improper motions.*

Proof Let $\dim_{\mathbb{R}}(\mathcal{A}) = n$. Let σ be the symmetry with respect to the hyperplane $\mathcal{H}$ and P a point of $\mathcal{H}$. Consider the vector space

$$\overrightarrow{P\mathcal{A}} = \{\overrightarrow{PA} : A \in \mathcal{A}\}.$$

By Theorem 1.155 there exists an on-basis $\overrightarrow{PA_1}, \ldots, \overrightarrow{PA_{n-1}}$ of the vector subspace $\overrightarrow{P\mathcal{H}}$. Take a point A_n on the perpendicular to $\mathcal{H}$ through P such that $\|\overrightarrow{PA_n}\| = 1$ so that $\overrightarrow{PA_1}, \ldots, \overrightarrow{PA_{n-1}}, \overrightarrow{PA_n}$ is an on-basis of $\overrightarrow{P\mathcal{A}}$. In this way we have built a

euclidean frame $\{P, A_1, \ldots, A_n\}$ of $\mathcal{A}$. If $\{O, E_1, \ldots, E_n\}$ is the standard frame of $\mathbb{E}^n(\mathbb{R})$, by Proposition 4.23, there exists a unique isometry $\tau : \mathcal{A} \to \mathbb{E}^n(\mathbb{R})$ such that $\tau(O) = P$ and $\tau(E_i) = A_i$, $i = 1, \ldots, n$. If $\mathcal{H}'$ is the hyperplane of $\mathbb{E}^n(\mathbb{R})$ having the equation $x_n = 0$, then $\tau(\mathcal{H}) = \mathcal{H}'$. Therefore $\sigma' := \tau^{-1} \circ \sigma \circ \tau$ is the symmetry with respect to $\mathcal{H}'$ whose equations are (see (4.28))

$$y_i = x_i, \quad i = 1, \ldots, n-1, \qquad y_n = -x_n,$$

which is an improper motion Hence $\sigma = \tau \circ \sigma' \circ \tau^{-1}$ is an improper motion. □

Theorem 4.31 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space and $f \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$, $f \neq \mathrm{id}_{\mathcal{A}}$. Let $\mathcal{S}$ be an affine subspace of $\mathcal{A}$, with $\dim(\mathcal{S}) = n - r$, $r \geq 1$, such that $f_{|\mathcal{S}} = \mathrm{id}_{\mathcal{S}}$. Then f is a composition of at most r orthogonal reflections through hyperplanes contaning $\mathcal{S}$.*

Proof We proceed by induction on r, $1 \leq r \leq n$. If $r = 1$ then $\mathcal{S}$ is a hyperplane and an isometry $f \neq \mathrm{id}_{\mathcal{A}}$ having an hyperplane of fixed points is the orthogonal reflection through that hyperplane. Indeed, Let $P \in \mathcal{A}$ such that $f(P) \neq P$ and $Q = p_{\mathcal{S}}(P)$ (the orthogonal projection on $\mathcal{S}$). The line r_{PQ} is orthogonal to $\mathcal{S}$ by definition so that its image $f(r_{PQ}) = r_{f(P)f(Q)} = r_{f(P)Q}$ is orthogonal to $\mathcal{S}$, hence from Q we have two lines orthogonal to $\mathcal{S}$ which implies $r_{PQ} = r_{f(P)Q}$ i.e. P, Q and $f(P)$ are collinear. Furthermore,

$$\|\overrightarrow{Qf(P)}\| = \|\overrightarrow{f(Q)f(P)}\| = \|\overrightarrow{QP}\|$$

so that Q is the midpoint of the segment $[Pf(P)]$ and $f = s_{\mathcal{S}}$ is the orthogonal reflection through the hyperplane $\mathcal{S}$.

Suppose the assertion is true for $r - 1 \leq n - 1$. Let $A \notin \mathcal{S}$ such that $f(A) = B \neq A$. Such a point exists since $\dim(\mathcal{S}) < n$ and $f \neq \mathrm{id}_{\mathcal{A}}$. Let $\mathcal{H}$ be the bisector hyperplane of $[AB]$, then $\mathcal{S} \subset \mathcal{H}$; indeed, if $P \in \mathcal{S}$, then $d(P, A) = d(f(P), f(A)) = d(f(P), B) = d(P, B)$, i.e $P \in \mathcal{H}$. Let $\mathcal{S}'$ be the subspace joining $\mathcal{S}$ and A. If $P \in \mathcal{S}'$ we have

$$(s_{\mathcal{H}} \circ f)(P) = s_{\mathcal{H}}(f(P)) = s_{\mathcal{H}}(P) = P \text{ if } P \in \mathcal{S},$$
$$(s_{\mathcal{H}} \circ f)(A) = s_{\mathcal{H}}(f(A)) = s_{\mathcal{H}}(B) = A,$$

hence $(s_{\mathcal{H}} \circ f)(P) = P$ for every $P \in \mathcal{S}'$. Since $\dim(\mathcal{S}') = n - r + 1 = n - (r - 1)$, by inductive hypothesis we have

$$s_{\mathcal{H}} \circ f = s_{\mathcal{H}_1} \circ s_{\mathcal{H}_2} \circ \cdots \circ s_{\mathcal{H}_k},$$

with $k \leq r - 1$ and $\mathcal{H}_1, \ldots, \mathcal{H}_k$ hyperplanes containing $\mathcal{S}'$ and $\mathcal{S}$ a fortiori. Therefore

$$f = s_{\mathcal{H}} \circ (s_{\mathcal{H}} \circ f) = s_{\mathcal{H}} \circ s_{\mathcal{H}_1} \circ s_{\mathcal{H}_2} \circ \cdots \circ s_{\mathcal{H}_k}$$

where $k \leq r - 1$ and $\mathcal{H}, \mathcal{H}_1, \ldots, \mathcal{H}_k$ are hyperplanes containing $\mathcal{S}$. □

Corollary 4.32 *An isometry $f \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ is the compostion of at most $n+1$ hyperplane reflections.*

Proof If $f = \mathrm{id}_{\mathcal{A}}$ then $f = s_{\mathcal{H}} \circ s_{\mathcal{H}}$, where $\mathcal{H}$ is any hyperplane. Let $f \neq \mathrm{id}_{\mathcal{A}}$, $A \in \mathcal{A}$ with $f(A) \neq A$ and $\mathcal{H}$ the bisector hyperplane of the segment $[Af(A)]$ so that $(s_{\mathcal{H}} \circ f)(A) = A$. By Theorem 4.31 applied to $s_{\mathcal{H}} \circ f$ with $\mathcal{S} = \{A\}$ $(r = n)$, $s_{\mathcal{H}} \circ f$ is a composition of at most n hyperplane reflections. The thesis follows from the obvious relation $f = s_{\mathcal{H}} \circ s_{\mathcal{H}} \circ f$. □

Remark 4.33 The reader can deduce Corollary 4.32 from Corollary 1.174.

Example 4.34 A translation τ_v is a composition of a orthogonal reflection through a hyperplane $\mathcal{H}$ perpendicular to v and a reflection through a hyperplane $\mathcal{H}'$ parallel to $\mathcal{H}$ such that $d(\mathcal{H}, \mathcal{H}') = \dfrac{\|v\|}{2}$.

The following structure theorem leads to classification of the isometries of a euclidean affine space (see Exercises 4.6 and 4.12).

Theorem 4.35 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space and $f \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. There exists a unique decomposition $f = \tau_v \circ g$ where $g \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ whose set $\mathcal{G}$ of fixed points is non-empty, τ_v is a translation with $v \in D(\mathcal{G})$. Under these conditions we also have $D(\mathcal{G}) = \ker(\mathrm{Tr}(f) - \mathrm{id}_V)$ and $\tau_v \circ g = g \circ \tau_v$.*

Proof Note that $V = \ker(s - \mathrm{id}_V) \perp \mathrm{Im}(s - \mathrm{id}_V)$ for every $s \in \mathrm{O}(V, \langle\cdot,\cdot,\rangle)$. Indeed, if $u \in \ker(s - \mathrm{id}_V)$, for every $s \in \mathrm{O}(V, \langle\cdot,\cdot,\rangle)$ we have

$$\langle u, s(v) - v\rangle = \langle u, s(v)\rangle - \langle u, v\rangle = \langle s(u), s(v)\rangle - \langle u, v\rangle = 0.$$

Take $P \in \mathcal{A}$ and write $\overrightarrow{Pf(P)} = v + w$, with $w = \mathrm{Tr}(f)(u) - u$ and $\mathrm{Tr}(f)(v) = v$. Put $Q = P - u$, then

$$\overrightarrow{Qf(Q)} = \overrightarrow{QP} + \overrightarrow{Pf(P)} + \overrightarrow{f(P)f(Q)} = u + v + \mathrm{Tr}(f)(u) - u - \mathrm{Tr}(f)(u) = v.$$

Now we take $g = \tau_v^{-1} \circ f$ so that $g(Q) = f(Q) - v = Q$. From the equivalences

$$g(R) = R \iff f(R) - v = R \iff \overrightarrow{Rf(R)} = v,$$

we conclude that $\mathcal{G} = \{Q \in \mathcal{A} : \overrightarrow{Qf(Q)} = v\}$ with $v \in \ker(f - \mathrm{id}_V)$. If $Q, R \in \mathcal{G}$, then

$$\mathrm{Tr}(f)(\overrightarrow{QR}) = \overrightarrow{f(Q)f(R)} = \overrightarrow{f(Q)Q} + \overrightarrow{QR} + \overrightarrow{Rf(R)} = -v + \overrightarrow{QR} + v = \overrightarrow{QR}$$

so that $\overrightarrow{QR} \in \ker(\mathrm{Tr}(f) - \mathrm{id}_V)$ i.e. $D(\mathcal{G}) \subset \ker(\mathrm{Tr}(f) - \mathrm{id}_V)$. Conversely, if $t \in \ker(\mathrm{Tr}(f) - \mathrm{id}_V)$, then $t = \overrightarrow{QR}$ with $Q \in \mathcal{G}$. The equality

$$\overrightarrow{Rf(R)} = \overrightarrow{RQ} + \overrightarrow{Qf(Q)} + \overrightarrow{f(Q)f(R)} = t + v - t = v$$

implies $R \in \mathcal{G}$ so that we have the opposite inclusion $\ker(\mathrm{Tr}(f) - \mathrm{id}_V) \subset D(\mathcal{G})$. Finally, for every $P \in \mathcal{A}$,

$$\tau_v(g(P)) = g(P) + v = f(P) - v + v = f(P);$$
$$g(\tau_v(P)) = f(P + v) - v = f(P) + \mathrm{Tr}(f)(v) - v = f(P)$$

since $v \in \ker(f - \mathrm{id}_V)$, hence $\tau_v \circ g = g \circ \tau_v$. □

Rotations in Two and Three Dimensional Euclidean Affine Spaces

Definition 4.36 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean plane. Let $C \in \mathcal{A}$ and $\theta = [0, 2\pi)$. A *rotation of angle θ with centre C* is a map $f : \mathcal{A} \to \mathcal{A}$ satysfying the following conditions:

(a) $f(C) = C$;
(b) for every point $P \neq C$,

$$\|\overrightarrow{CP}\| = \|\overrightarrow{Cf(P)}\| \qquad \text{and} \qquad \measuredangle(\overrightarrow{CP}, \overrightarrow{Cf(P)}) = \min\{\theta, 2\pi - \theta\}.$$

A rotation of angle $\theta = 0$ is $\mathrm{id}_{\mathcal{A}}$; if $\theta = \pi$, f is the symmetry with respect to C.

Proposition 4.37 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean plane. Let $f : \mathcal{A} \to \mathcal{A}$ be an arbitrary mapping. The following two conditions are equivalent:*

(a) *f is a proper isometry with a fixed point.*
(b) *f is a rotation (of angle θ with centre C).*

Proof

(a) $\Longrightarrow$ (b). Suppose that $f(C) = C$ for a point $C \in \mathcal{A}$. Then we have for all $P \in \mathcal{A}$, $P \neq C$,

$$\|\overrightarrow{Cf(P)}\| = \|\overrightarrow{f(C)f(P)}\| = \|\mathrm{Tr}(f)(\overrightarrow{CP})\| = \|\overrightarrow{CP}\|$$

because $\mathrm{Tr}(f) \in \mathrm{O}(V, \langle\cdot,\cdot\rangle_V)$ by definition. Since f is proper, $\det(\mathrm{Tr}(f)) = 1$ so that, by Proposition 1.170, $\mathrm{Tr}(f)$ is represented by the matrix (1.65) with respect to any on-basis B. If (x_1, x_2) are the coordinates of $\overrightarrow{CP}$ with

respect to B, then the coordinates of $\mathrm{Tr}(f)(\overrightarrow{CP}) = \overrightarrow{Cf(P)}$ are $(x_1 \cos\theta - x_2 \sin\theta, x_1 \sin\theta + x_2 \cos\theta)$ so that

$$\cos\left(\measuredangle(\overrightarrow{CP}, \overrightarrow{Cf(P)})\right) = \frac{\langle \overrightarrow{CP}, \overrightarrow{Cf(P)}\rangle}{\|\overrightarrow{CP}\|\,\|\overrightarrow{Cf(P)}\|} = \cos\theta.$$

(b) $\Longrightarrow$ (a). Put $g(\overrightarrow{CP}) := \overrightarrow{Cf(P)} = \mathrm{Tr}(f)(\overrightarrow{CP})$. We have to show that $g\colon V \to V$ is an orthogonal linear isomorphism. Let $P, Q \in \mathcal{A}$, $P \neq C$ and $Q \neq C$. By (1.61) we have

$$\langle \overrightarrow{CP}, \overrightarrow{CQ}\rangle_V = \|\overrightarrow{CP}\|\,\|\overrightarrow{CQ}\|\cos\alpha, \qquad \alpha = \measuredangle(\overrightarrow{CP}, \overrightarrow{CQ}),$$

and by hypothesis

$$\measuredangle(\overrightarrow{CP}, \overrightarrow{Cf(P)}) = \measuredangle(\overrightarrow{CQ}, \overrightarrow{Cf(Q)}) = \min\{\theta, 2\pi - \theta\},$$
$$\|\overrightarrow{CP}\| = \|\overrightarrow{Cf(P)}\|, \quad \|\overrightarrow{CQ}\| = \|\overrightarrow{Cf(Q)}\|,$$

so that $\measuredangle(g(\overrightarrow{CP}), g(\overrightarrow{CQ})) = \alpha$ and

$$\langle g(\overrightarrow{CP}), g(\overrightarrow{CQ})\rangle_V = \|\overrightarrow{Cf(P)}\|\,\|\overrightarrow{Cf(Q)}\|\cos\alpha = \langle \overrightarrow{CP}, \overrightarrow{CQ}\rangle_V.$$

If $\det(g) = -1$ then by Proposition 1.170 we should have a line of fixed points so that $\theta = 0$.

□

Definition 4.38 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space of dimension 3. Let r be a line and $\theta \in [0, 2\pi)$. A *rotation of axis r and angle θ* is a map $f\colon \mathcal{A} \to \mathcal{A}$ satisfying the following conditions:

(a) $f(O) = O$ for every $O \in r$;
(b) for every point $O \in r$, f restricted to the affine plane orthogonal to r and containing O is a rotation of angle θ and centre O.

A rotation of angle $\theta = 0$ is $\mathrm{id}_{\mathcal{A}}$; if $\theta = \pi$, f is the orthogonal symmetry with respect to r.

Proposition 4.39 *Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space of dimension 3. Let $f\colon \mathcal{A} \to \mathcal{A}$ be an arbitrary mapping. The following two conditions are equivalent:*

(a) *f is a proper isometry with a fixed point.*
(b) *f is a rotation about an axis r and angle θ.*

Proof

(b) $\Longrightarrow$ (a). Let $O \in r$. Put $g(\overrightarrow{OP}) := \overrightarrow{Of(P)} = \mathrm{Tr}(f)(\overrightarrow{OP})$. Choose an euclidean frame O, A_1, A_2, A_3 such that $\overrightarrow{OA_1}, \overrightarrow{OA_2}, \overrightarrow{OA_3}$ is an on-basis of V and $\overrightarrow{OA_3} \in D(r)$. Then g is represented by the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so that g is an orthogonal linear isomorphism with $\det(g) = 1$.

(a) $\Longrightarrow$ (b). Let O be a fixed point of f and $g := \mathrm{Tr}(f)$. By hypothesis we have $\det(g) = 1$ so that g has an eigenvalue 1 (see Exercise 1.39). Choose an eigenvector e_1 of 1 with $\|e_1\| = 1$. Consider the line

$$r = O + \langle e_1 \rangle.$$

Every point of r is fixed by f, indeed if $P = O + ke_1 \in r$ then

$$f(P) = f(O) + kg(e_1) = O + ke_1 = P.$$

If we choose a euclidean frame O, A_1, A_2, A_3, such that $\overrightarrow{OA_1} = e_1$ and O, A_2, A_3 is an euclidean frame for the plane $O + \langle e_1 \rangle^{\perp}$, the matrix associated with g with respect to on-basis $\{\overrightarrow{OA_1}, \overrightarrow{OA_2}, \overrightarrow{OA_3}\}$ is

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & b_{11} & b_{12} \\ 0 & b_{21} & b_{22} \end{pmatrix},$$

where $B' = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$ is orthogonal with $\det B' = 1$. By Proposition 1.170, $B' = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

is the matrix of a rotation of angle θ about the axis r.

□

4.3 Exercises

Exercise 4.1 Let $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be a euclidean space. Let $A\overset{\square}{B}CD$ be a parallelogram of $\mathcal{A}$. Show that $\ell([AB])^2 + \ell([BC])^2 + \ell([CD])^2 + \ell([AD])^2 = \ell([AC])^2 + \ell([BD])^2$ (see Definition 4.2).

Exercise 4.2 Let $A, B, C, D \in (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. Prove the equality

$$\langle \overrightarrow{AB}, \overrightarrow{CD} \rangle + \langle \overrightarrow{AC}, \overrightarrow{DB} \rangle + \langle \overrightarrow{AD}, \overrightarrow{BC} \rangle = 0.$$

Exercise 4.3 Let $A, B \in (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$. Describe the set of isometries that fix the segment $[AB]$ (not necessarily pointwise).

Exercise 4.4 Let $\mathcal{B}$ and $\mathcal{C}$ be two affine subspaces of $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ such that $\mathcal{C} \subseteq \mathcal{B}$ and $P \in (\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ such that $P \notin \mathcal{B}$. Let $B \in \mathcal{C}$ be the foot of the perpendicular to $\mathcal{C}$ through P. Let $\mathcal{D}$ be the unique affine subspace of $\mathcal{B}$ of dimension $\dim_K \mathcal{B} - \dim_K \mathcal{C}$ containing B and perpendicular to $\mathcal{C}$ (see Proposizione 4.11). Prove that the foot of the perpendicular to $\mathcal{D}$ passing through P coincides with the foot of the perpendicular to $\mathcal{B}$ through P. (Hint: adapt the proof of Proposition 4.9.)

Exercise 4.5 Let $\mathcal{B}$ and $\mathcal{C}$ be two affine subspaces of $(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ such that $\mathcal{B} \perp \mathcal{C}$ and $\dim_K \mathcal{B} + \dim_K \mathcal{C}) \geq \dim_K \mathcal{A}$. Prove the formula $\dim_K \mathcal{B} \cap \mathcal{C}) = \dim_K \mathcal{B} + \dim_K \mathcal{C} - \dim_K \mathcal{A}$. (Hint: adapt the proof of Proposition 4.11.)

Exercise 4.6 Prove that any proper motion of the Euclidean plane $\mathbb{E}^2(\mathbb{R})$ is one of the following:

(a) the identity,
(b) a translation,
(c) a rotation about some point $C \in \mathbb{E}^2(\mathbb{R})$,

and that any improper motion of the Euclidean plane $\mathbb{E}^2(\mathbb{R})$ is one of the following

(d) an orthogonal reflection.
(e) a *glide reflection*, that is, a reflection through a line r followed by a translation τ_v with v parallel to r.

Exercise 4.7 Show that the lines of $\mathbb{E}^3(\mathbb{R})$

$$r: \begin{cases} 2x - y = 0 \\ z - 1 = 0 \end{cases} \qquad s: \begin{cases} 2x + y = 0 \\ z + 1 = 0 \end{cases}$$

are skew and determine the common perpendicular to r and s.

Exercise 4.8 Given $P(1,0,0) \in \mathbb{E}^3(\mathbb{R})$, the line r of parametric equations $x = 1+t$, $y = t$, $z = t$ and the plane π of equation $x - 2y + 3z = 7$,

(i) find the line s such that $P \in s$, $s \perp r$ and $s' \perp r'$, where r' and s' are the orthogonal projections of r and s on π respectively;
(ii) show that r and s' are skew and find their distance.

Exercise 4.9 Write $\varphi\colon \mathbb{E}^3(\mathbb{R}) \to \mathbb{E}^3(\mathbb{R})$, $\varphi(x,y,z) = (x',y',z')$, defined by

$$\begin{cases} x' &= \dfrac{3}{\sqrt{14}}x - \dfrac{5}{\sqrt{70}}z + 1 \\ y' &= -\dfrac{1}{\sqrt{14}}x + \dfrac{2}{\sqrt{5}}y - \dfrac{3}{\sqrt{70}}z - 2 \\ z' &= \dfrac{2}{\sqrt{14}}x + \dfrac{1}{\sqrt{5}}y + \dfrac{6}{\sqrt{70}}z, \end{cases}$$

as a composition of orthogonal symmetries.

Exercise 4.10 Let $f, g \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be two isometries such that $f \circ g = g \circ f$. Suppose that both f and g admit fixed points. Prove that f and g have at least one common fixed point.

Exercise 4.11 Let $f \in \mathrm{O}(\mathcal{A}, (V, \langle\cdot,\cdot\rangle_V), \varphi)$ be an isometry and $\mathcal{H}$ a hyperplane such that $f_{|\mathcal{H}} = \mathrm{id}_{\mathcal{H}}$. Prove that f is the orthogonal reflection through $\mathcal{H}$.

Exercise 4.12 Prove that any proper motion of the Euclidean space $\mathbb{E}^3(\mathbb{R})$ is one of the following:

(a) the identity,
(b) a translation,
(c) a rotation about some line r,
(d) a *screw rotation*, that is, a rotation about some line r followed by a translation τ_v where v is parallel to r,

and that any improper motion of the Euclidean space $\mathbb{E}^3(\mathbb{R})$ is one of the following

(e) an orthogonal reflection through some plane,
(f) a *glide reflection*, that is, a reflection through some plane π followed by a translation τ_v with v parallel to π,
(g) a *rotatory reflection*, that is, a rotation about some line r followed by a reflection in a plane perpendicular to r. (Hint: use Corollary 1.175 and Proposition 4.23.)

Exercise 4.13 Find the lines lying on the plane $\pi : x + y + z = 0$, parallel to the plane $z = 0$ having distance $d = \sqrt{2}$ from $P(0,0,1)$.

Exercise 4.14 Let

$$r: \begin{cases} x - y = 0 \\ 2x - z + 5 = 0 \end{cases} \qquad s: \begin{cases} x - y = 6 \\ x - 2y + z = 6 \end{cases} \qquad t: \begin{cases} 3x - 2z + 2 = 0 \\ 3y + z - 1 = 0 \end{cases}$$

be three lines of $\mathbb{E}^3(\mathbb{R})$.

(i) Find the equations of the line parallel to t which intersects r and s.
(ii) Find the sphere Σ tangent to t at $P(0, 0, 1)$ and passing through $A(1, 1, 2)$ and $B(2, 1, 1)$.
(iii) Determine the circles on Σ of radius $\sqrt{14/3}$ tangent to t.

Exercise 4.15 Let τ_v be the translation of $\mathbb{A}^2(\mathbb{R})$, where $v = (1, 1)$ and σ the orthogonal reflection through the line $r : 2x + y - 1 = 0$.

(i) Classify $\varphi = \tau \circ \sigma$.
(ii) Find two non-parallel lines s and t such that $\varphi(t) \parallel t$ and $\varphi(s) \parallel s$,

Exercise 4.16 Consider the following lines of $\mathbb{E}^2(\mathbb{R})$:

$$d_1: \ y = 0, \quad d_2: \ x = 0, \quad d_3: \ x + y = 1;$$

$$d_1': \ y = 3, \quad d_2': \ x = 4, \quad d_3': \ x - y = 2.$$

Find an isometry f of $\mathbb{E}^2(\mathbb{R})$ such that $f(d_i') = d_i$, $i = 1, 2, 3$.

Exercise 4.17 Let

$$r: \begin{cases} x - y = 0 \\ 2x - z + 5 = 0 \end{cases} \qquad s: \begin{cases} x - y = 6 \\ x - 2y + z = 6 \end{cases}$$

be two lines of $\mathbb{E}^3(\mathbb{R})$ and $S^2(\rho)$ a sphere of radius ρ. Prove that the set

$$\mathcal{R} = \{\rho > 0 \mid S^2(\rho) \text{ is tangent to } r \text{ and } s\},$$

is non-empty and find $\min\{\mathcal{R}\}$ and $\max\{\mathcal{R}\}$.

Exercise 4.18 Generalize the formula (4.12) showing the following result:

Let P be a polygon having vertices $A_1, \ldots, A_n$, where $A_i = (x_i, y_i)$, $i = 1, \ldots, n$ and $n \geq 3$. Then the area A_P of P is given by the formula

$$A_P = \frac{1}{2} \left| \det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix} + \det \begin{pmatrix} x_2 & y_2 \\ x_3 & y_3 \end{pmatrix} + \cdots + \det \begin{pmatrix} x_n & y_n \\ x_1 & y_1 \end{pmatrix} \right|.$$

(Hint: induction on n.)

Chapter 5
Affine Hyperquadrics

5.1 Affine Hypersurfaces

Let $n \geq 1$ be a fixed natural number. For every integer $d \geq 1$, each polynomial $F(X_1, \ldots, X_n) \in K[X_1, \ldots, X_n]_d$ is represented as

$$F = F_d + F_{d-1} + \cdots + F_1 + F_0, \tag{5.1}$$

where F_i is a homogeneous polynomial of degree i, $i = 1, 2, \ldots, d$, $F_0 \in K$ and $F_d \neq 0$. The non-zero homogeneous polynomial F_d is called the *dominant homogeneous polynomial* of $F(X_1, \ldots, X_n) \in K[X_1, \ldots, X_n]_d$. To denote the indeterminates we use both capital letters $X_1, \ldots, X_n$ and small letters $x_1, \ldots, x_n$.

Definition 5.1 We put on $K[X_1, \ldots, X_n]_d$ the equivalence relation

$$\begin{aligned} F \sim G \quad &\Longleftrightarrow \quad \exists\, \lambda \in K \setminus \{0\} \quad \text{such that} \\ &G = \lambda F, \quad F, G \in K[X_1, \ldots, X_n]_d. \end{aligned} \tag{5.2}$$

An *affine hypersurface of degree* d of $\mathbb{A}^n(K)$ is an equivalence class $\widehat{F}$ of $F \in K[X_1, \ldots, X_n]_d$ with respect to $\sim$ defined by (5.2). The polynomial F (or $F(x_1, \ldots, x_n) = 0$) is called an *equation* of $\widehat{F}$. Every other equation of $\widehat{F}$ is of the form $G = \lambda F$, with $\lambda \in K \setminus \{0\}$. A hypersurface $\widehat{F}$ of degree d of $\mathbb{A}^2(K)$ is called an *affine curve* of degree d of $\mathbb{A}^2(K)$; if $d = 2$, $\widehat{F}$ is called an *affine conic*. A hypersurface of degree d of $\mathbb{A}^3(K)$ is called an *affine surface* of degree d of $\mathbb{A}^3(K)$.

We denote the set of all hypersurfaces of degree d of $\mathbb{A}^n(K)$ by $\mathcal{H}_d(n, K)$. In other words $\mathcal{H}_d(n, K)$ is the quotient set of $K[X_1, \ldots, X_n]_d$ modulo the equivalence relation (5.2). With every $\widehat{F} \in \mathcal{H}_d(n, K))$ we associate the subset (possibly empty)

$$\mathcal{V}(\widehat{F}) := \{(x_1, \ldots, x_n) \in \mathbb{A}^n(K) \mid F(x_1, \ldots, x_n) = 0\}$$

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called the *geometric locus* (or *locus of zeros*) of $\widehat{F}$. We indicate $\mathcal{V}(\widehat{F})$ also by $\mathcal{V}(F)$ and $P \in \mathcal{V}(\widehat{F})$ by $P \in \widehat{F}$ or $P \in F$.

If $n = 1$ then $0 \leq \#(\mathcal{V}(\widehat{F})) \leq d$, i.e. $\mathcal{V}(\widehat{F})$ consists of a finite number ($\leq d$) of points or it is empty.

If $\widehat{F}$ and $\widehat{G}$ are two hypersurfaces, we denote the hypersurface $\widehat{FG}$ by $\widehat{F} + \widehat{G}$; moreover, for every positive integer m, we put $m\widehat{F} := \widehat{F^m}$. We say that $\widehat{F} \in \mathcal{H}_d(n, K)$ is *irreducible* if $F \in [X_1, \ldots, X_n]_d$ is a power of an irreducible polynomial. Let $F = kF_1^{m_1} \cdots F_r^{m_r}$ with k a non-zero constant and the F_i be distinct irreducible polynomials, then we have a decomposition (unique up to the order):

$$\widehat{F} = m_1\widehat{F_1} + \cdots + m_r\widehat{F_r} \tag{5.3}$$

and the hypersurfaces $\widehat{F_i}$ are called the *irreducible components of* $\widehat{F}$ and the integer m_i is called the *multiplicity* of the component $\widehat{F_i}$. Every irreducible component of multiplicity $m_i > 1$ is called a *multiple component*. Hypersurfaces without multiple components are said to be *reduced*. If $\widehat{F}$ is irreducible and reduced then F is a irreducible polynomial. We can immediately observe that

$$\mathcal{V}(\widehat{F}) = \mathcal{V}(\widehat{F_1}) \cup \cdots \cup \mathcal{V}(\widehat{F_r}).$$

Lemma 5.2 *Let $\widehat{F} \in \mathcal{H}_d(n, K)$. Suppose that K is a field with at least $d + 1$ elements* (*e.g. K be an infinite field*). *Then:*

(i) $\mathcal{V}(\widehat{F}) \subsetneqq \mathbb{A}^n(K)$.

(ii) *If K is algebraically closed then $\mathcal{V}(\widehat{F}) \neq \varnothing$ and, if $n \geq 2$, $V(\widehat{F})$ is an infinite set.*

Proof

(i) We proceed by induction on n. Let $n = 1$. A polynomial $F(X) \in K[X]$ of degree d can't have more than d roots in K, hence $\mathcal{V}(\widehat{F}) \subsetneqq K = \mathbb{A}^1(K)$. Let $n \geq 2$. Let us write

$$F = F(X_1, \ldots, X_n) = \sum_{i=0}^{m} F_i(X_1, \ldots, X_{n-1})X_n^i, \tag{5.4}$$

where $m \leq d$, $F_i(X_1, \ldots, X_{n-1}) \in K[X_1, \ldots, X_{n-1}]$ and $F_m(X_1, \ldots, X_{n-1}) \neq 0$. By inductive hypothesis there exists $(a_1, \ldots, a_{n-1})$ such that $F_m(a_1, \ldots, a_{n-1}) \neq 0$. If $\mathcal{V}(\widehat{F}) = \mathbb{A}^n(K)$, i.e. $F(a_1, \ldots, a_{n-1}, a_n) = 0$ for all $(a_1, \ldots, a_{n-1}, a_n) \in \mathbb{A}^n(K)$, then, putting

$$G(X_n) = F(a_1, \ldots, a_{n-1}, X_n) = \sum_{i=0}^{m} F_i(a_1, \ldots, a_{n-1})X_n^i,$$

we have $G(a_n) = F(a_1, \ldots, a_{n-1}, a_n) = 0$ for all $a_n \in K$, so that $G(X_n)$ is the zero-polynomial; in particular, $F_m(a_1, \ldots, a_{n-1}) = 0$ contradicting our hypothesis.

(ii) The case $n = 1$ is obvious. Let $n \geq 2$. By (i) there exists a point $P = (a_1, \ldots, a_n) \notin \mathcal{V}(\widehat{F})$. Every line passing through P has parametric equations

$$x_i = a_i + l_i t, \quad i = 1, \ldots, n, \quad (l_1, \ldots, l_n) \in K^n \setminus \{(0, \ldots, 0)\},$$

so that $f(t) := F(a_1 + l_1 t, \ldots, a_n + l_n t)$ is not identically zero and it has at least a root, i.e. there is $Q_{(l_1,\ldots,l_n)} \in V(\widehat{F})$. Since the non-proportional n-tuples $(l_1, \ldots, l_n)$ are infinitely many, so are the points $Q_{(l_1,\ldots,l_n)}$.

□

Remark 5.3 The non-zero polynomial $f(X) = X^p - X \in \mathbb{Z}_p[X]$ vanishes on $\mathbb{A}^1(\mathbb{Z}_p)$.

Definition 5.4 Let $\widehat{F} \in \mathcal{H}_d(n, K)$ and let $\mathcal{A}$ be an affine subspace of dimension h of $\mathbb{A}^n(K)$ not contained in $\widehat{F}$. By a suitable change of coordinates we can suppose that $\mathcal{A}$ has equations $x_{h+1} = x_{h+2} = \cdots = x_n = 0$. It is obvious that

$$\pi_h : \mathcal{A} \to \mathbb{A}^h(K), \qquad \pi_h(x_1, \ldots, x_n) = (x_1, \ldots, x_h),$$

is an affine isomorphism. By means of π_h we can identify $\mathcal{V}(\widehat{F}) \cap \mathcal{A}$ with the geometric locus $\mathcal{V}(\widehat{F_{|\mathcal{A}}})$ of the hypersurface $\widehat{F_{|\mathcal{A}}}$ of $\mathbb{A}^h(K)$, where

$$F_{|\mathcal{A}}(x_1, \ldots, \ldots, x_h) := F(x_1, \ldots, x_h, 0, \ldots, 0).$$

The hypersurface $\widehat{F_{|\mathcal{A}}}$, called the *restriction* of $\widehat{F}$ to $\mathcal{A}$, has degree $\leq d$ and its geometric locus $\mathcal{V}(\widehat{F_{|\mathcal{A}}})$ can be empty.

5.1.1 Tangent Hyperplanes and Multiple Points

Definition 5.5 For any $k = 1, \ldots, n$ the *formal derivative* $\dfrac{\partial F}{\partial X_k}$ of a polynomial

$$F = \sum_{i_1 \geq 0, \ldots, i_n \geq 0} a_{i_1,\ldots,i_n} X_1^{i_1} \cdots X_n^{i_n} \in K[X_1, \ldots, X_n]$$

with respect the indeterminate X_k is defined by

$$\frac{\partial F}{\partial x_k} = \sum_{i_1 \geq 0, \ldots, i_n \geq 0} i_k a_{i_1,\ldots,i_n} X_1^{i_1} \cdots X_{k-1}^{i_{k-1}} X_k^{i_k - 1} X_{k+1}^{i_{k+1}} \cdots X_n^{i_n},$$

where the term $a_{i_1,\dots,i_n} X_1^{i_1} \cdots X_{k-1}^{i_{k-1}} X_k^{i_k-1} X_{k+1}^{i_{k+1}} \cdots X_n^{i_n}$ does not occur if $i_k = 0$. We shall also use the usual *gradient*

$$\nabla F = \left(\frac{\partial F}{\partial X_1}, \dots, \frac{\partial F}{\partial X_n} \right).$$

We recall this elementary result.

Lemma 5.6 *Let $F, F_0, \dots, F_n \in K[X_1, \dots, X_n]$. If $F(F_0, \dots, F_n)$ is the polynomial obtained replacing X_k with F_k, $k = 0, 1, \dots, n$ then*

$$\frac{\partial F(F_0, \dots, F_n)}{\partial X_i} = \sum_{k=0}^{n} \frac{\partial F}{\partial F_k}(F_0, \dots, F_n) \cdot \frac{\partial F_k}{\partial X_i}.$$

Definition 5.7 Let $\widehat{F} \in \mathcal{H}_d(n, K)$. A point $P = (a_1, \dots, a_n) \in \mathcal{V}(\widehat{F})$ is called a *singular point* of $\widehat{F}$ if

$$\frac{\partial F}{\partial x_1}(a_1, \dots, a_n) = \cdots = \frac{\partial F}{\partial x_n}(a_1, \dots, a_n) = 0.$$

A point $P \in \mathcal{V}(\widehat{F})$ is *non-singular* or *smooth* if it is not singular. We will denote the sets of singular points and of non-singular points of $\widehat{F}$ by $\mathrm{Sing}(\widehat{F})$ and $\mathrm{Reg}(\widehat{F})$ respectively. It is obvious that

$$\mathcal{V}(\widehat{F}) = \mathrm{Sing}(\widehat{F}) \cup \mathrm{Reg}(\widehat{F}) \qquad \text{and} \qquad \mathrm{Sing}(\widehat{F}) \cap \mathrm{Reg}(\widehat{F}) = \varnothing.$$

A hypersurface $\widehat{F} \in \mathcal{H}_d(n, K)$ is *non-singular* or *smooth* if $\mathrm{Reg}(\widehat{F}) = \mathcal{V}(\widehat{F})$, i.e. all points of $\widehat{F}$ are non-singular.

Definition 5.8 Let $n \geq 2$. If $P = (a_1, \dots, a_n) \in \mathcal{V}(\widehat{F})$ is a non-singular point of $(\widehat{F})$, then $\dfrac{\partial F}{\partial x_i}(P) \neq 0$ for some $i \in \{1, \dots, n\}$ so that we can associate the hyperplane of $\mathbb{A}^n(K)$ having the equation

$$\frac{\partial F}{\partial x_1}(P)(x_1 - a_1) + \cdots + \frac{\partial F}{\partial x_n}(P)(x_n - a_n) = 0 \tag{5.5}$$

with P. We call the hyperplane (5.5) *the tangent hyperplane* to $\widehat{F}$ at P and denote it by $\mathrm{T}_{\widehat{F},P}$. We can put (5.5) in the form

$$\nabla F(P)(X - P)^t = 0.$$

Definition 5.9 Let $\widehat{F} \in \mathcal{H}_d(n, K)$ with $n \geq 2$. Let $x_i = a_i + l_i t$, $i = 1, \dots, n$, with $(l_1, \dots, l_n) \in K^n \setminus \{(0, \dots, 0)\}$ be parametric equations of a line r passing through

a point $P = (a_1, \ldots, a_n) \in \mathcal{V}(\widehat{F})$. If $r \not\subset \mathcal{V}(\widehat{F})$, the points of $r \cap \mathcal{V}(\widehat{F})$ are found by solving the equation

$$f(t) = 0, \quad \text{where} \quad f(t) := F(a_1 + l_1 t, \ldots, a_n + l_n t).$$

Since $f(0) = F(a_1, \ldots, a_n) = 0$ we say that the line r meets the affine hypersurface $\widehat{F}$ at P with *intersection multiplicity*

$$\mathfrak{m}_P(\widehat{F}, r) := \text{ the order of 0 as a root of } f(t). \tag{5.6}$$

If $r \subset V(\widehat{F})$ we put $\mathfrak{m}_P(\widehat{F}, r) = \infty$. It is easy to see that $\mathfrak{m}_P(\widehat{F}, r)$ does not depend on the linear parametrization of r.

We say that r is *tangent* to $\widehat{F}$ at P if $\mathfrak{m}_P(\widehat{F}, r) \geq 2$ (including $\mathfrak{m}_P(\widehat{F}, r) = \infty$).

Remark 5.10 Let $f(X) \in K[X]$ be a polynomial of degree d and $\mathrm{char}(K) \nmid d!$. Then α is a root of f of order m if and only if

$$f(\alpha) = f'(\alpha) = \cdots f^{(m-1)}(\alpha) = 0 \qquad \text{and} \qquad f^{(m)}(\alpha) \neq 0.$$

Proposition 5.11 *A line r passing through $P = (a_1, \ldots, a_n) \in \mathrm{Reg}(\widehat{F})$ is tangent to $\widehat{F}$ if and only if $r \subset \mathrm{T}_{\widehat{F},P}$. Furthermore, $\mathrm{T}_{\widehat{F},P}$ is the union of all tangent lines to $\widehat{F}$ passing through P.*

Proof We have the following immediate equivalences:

$$0 \text{ is root of } f(t) \text{ of order } \geq 2 \iff \frac{\mathrm{d}f}{\mathrm{d}t}(0) = 0 \iff \sum_{i=1}^{n} l_i \frac{\partial F}{\partial x_i}(P)$$
$$= 0 \iff r \subseteq \mathrm{T}_{\widehat{F},P}.$$

□

Definition 5.12 Let $\widehat{F} \in \mathcal{H}_d(n, K)$ and $P \in \mathbb{A}^n(K)$ with $n \geq 2$. We define the *multiplicity of $\widehat{F}$ at P* or *multiplicity of P on $\widehat{F}$* as the integer

$$\mathfrak{m}_P(\widehat{F}) = \min_{\substack{r \not\subset \mathcal{V}(\widehat{F}) \\ P \in r}} \mathfrak{m}_P(\widehat{F}, r).$$

We also say that P is an $\mathfrak{m}_P(\widehat{F})$*-fold point*. If K has at least $d + 1$ elements there exists a line not contained in $\widehat{F}$ so that

$$0 \leq \mathfrak{m}_P(\widehat{F}) \leq d,$$

where $\mathfrak{m}_P(\widehat{F}) = 0$ if and only if $P \notin \widehat{F}$. If $n = 1$ the multiplicity of $\widehat{F} \in \mathcal{H}_d(1, K)$ at $P \in \mathcal{V}(\widehat{F})$ is the order of P as a zero of the polynomial F. A point $P \in \mathcal{V}(\widehat{F})$ is said to be *simple* if $\mathfrak{m}_P(\widehat{F}) = 1$ and *multiple* if $\mathfrak{m}_P(\widehat{F}) > 1$ (*double* if $\mathfrak{m}_P(\widehat{F}) = 2$

and so on). A point P is simple if and only if it is non-singular. Indeed, if P is non-singular, $Q \notin \mathrm{T}_{\widehat{F},P}$ and r is the line joining Q and P, then $\mathfrak{m}_P(\widehat{F}, r) = 1$. Conversely, if $\mathfrak{m}_P(\widehat{F}) = 1$, then $\mathfrak{m}_P(\widehat{F}, r) = 1$ for some line r, hence (according to the notation of Definition 5.9)

$$0 \neq \frac{\mathrm{d}f}{\mathrm{d}t}(0) = \sum_{i=1}^{n} l_i \frac{\partial F}{\partial x_i}(P), \qquad \text{which implies} \quad \frac{\partial F}{\partial x_i}(P) \neq 0$$

$$\text{for some } i \in \{1, \ldots, n\}.$$

Proposition 5.13 *Let $\widehat{F} \in \mathcal{H}_d(n, K)$. Suppose that* $\mathrm{char}(K) \nmid d!$. *Then $P \in \mathbb{A}^n(K)$ is a point of multiplicity $m \geq 1$ on $\widehat{F}$ if and only if $F(P) = 0$ and, if $m > 1$,*

(i) *all partial derivatives of F up to the order $m - 1$ vanish at P,*
(ii) *there exists at least a partial derivative of F of order m which does not vanish at P.*

Proof Let $P = (a_1, \ldots, a_n)$. By Taylor's expansion of F at P

$$F(X_1, \ldots, X_n) = \sum_{j=m}^{d} \sum_{\substack{e_i \geq 0 \\ e_1 + \cdots + e_n = j}} \left(\frac{1}{e_1! \cdots e_n!}\right) \frac{\partial^j F}{\partial X_1^{e_1} \cdots \partial X_n^{e_n}}(P)$$

$$(X_1 - a_1)^{e_1} \cdots (X_n - a_n)^{e_n} = F_m + F_{m+1} + \cdots + F_d,$$

where $1 \leq m \leq d$, F_i are homogeneous polynomials in $X_1 - a_1, \ldots, X_n - a_n$ of degree i and $F_d \neq 0$, we get

$$F(a_1 + l_1 t, \ldots, a_n + l_n t) = \sum_{j=m}^{d} \sum_{\substack{e_i \geq 0 \\ e_1 + \cdots + e_n = j}} \left(\frac{1}{e_1! \cdots e_n!}\right) \frac{\partial^j F}{\partial X_1^{e_1} \cdots \partial X_n^{e_n}}(P)$$

$$(l_1)^{e_1} \cdots (l_n)^{e_n} t^j.$$

Thus if r is a line passing through P then $\mathfrak{m}_P(\widehat{F}, r) \geq m$, so that $m = \mathfrak{m}_P(\widehat{F})$. □

Definition 5.14 A hypersurface $\widehat{F} \in \mathcal{H}_d(n, K)$ is called an *affine cone*, or simply *cone*, if there exists a point $P \in \mathcal{V}(\widehat{F})$, called a *vertex*, such that for all points $Q \in \mathcal{V}(\widehat{F})$, $P \neq Q$, the line r_{PQ} joining P and Q is contained in $\mathcal{V}(\widehat{F})$. If F is a homogeneous polynomial in the variables $x_1 - p_1, \ldots, x_n - p_n$, then $\widehat{F}$ is a cone with vertex $P = (p_1, \ldots, p_n)$.

An *affine cylinder* is a hypersurface $\widehat{F} \in \mathcal{H}_d(n, K)$ such that $\mathcal{V}(\widehat{F})$ is the union of lines which are all parallel to each other, i.e. have the same director subspace.

Proposition 5.15 *If a point P of a hypersurface $\widehat{F} \in \mathcal{H}_d(n, K)$ is a d-fold point (i.e. $\mathfrak{m}_P(\widehat{F}) = d$), then $\widehat{F}$ is a cone with vertex P. Conversely, if a hypersurface $\widehat{F} \in \mathcal{H}_d(n, K)$ is a cone then each vertex of $\widehat{F}$ is a d-fold point.*

Proof If $Q \in \mathcal{V}(\widehat{F})$, $P \neq Q$, the line r_{PQ} joining P and Q has at least $d+1$ points in common with $\widehat{F}$ (at least d at P plus the point Q) and thus r_{PQ} is contained in $\mathcal{V}(\widehat{F})$. Conversely, let $\widehat{F}$ be a cone and P is any vertex of $\widehat{F}$. Then if r is a line passing through P and not contained in $\mathcal{V}(\widehat{F})$ we have $\mathfrak{m}_P(\widehat{F}, r) = d$ since $r \cap \mathcal{V}(\widehat{F}) = P$. □

We call *the vertex of a cone* $\widehat{F}$ the set of its vertices. The vertex of a cone can be an infinite set: if $F(x, y, z) = xy$ then $\widehat{F}$ is a cone of $\mathbb{A}^3(\mathbb{R})$ whose vertices are the points of the line $x = y = 0$.

Assume the notation and hypothesis of Proposition 5.13 (and of its proof). The *affine tangent cone* to a hypersurface $\widehat{F}$ at a m-fold point P is defined by the equation $F_m = 0$, i.e.

$$\sum_{\substack{e_i \geq 0 \\ e_1+\cdots+e_n=m}} \left(\frac{1}{e_1! \cdots e_n!}\right) \frac{\partial^j F}{\partial X_1^{e_1} \cdots \partial X_n^{e_n}}(P)(X_1 - a_1)^{e_1} \cdots (X_n - a_n)^{e_n} = 0, \tag{5.7}$$

and denoted by $\mathrm{C}_{\widehat{F},P}$. The tangent cone $\mathrm{C}_{\widehat{F},P}$ is the locus of the lines r passing through P such that $\mathfrak{m}_P(\widehat{F}, r) > m$. Indeed, if a line $r\colon x(t) = (a_1 + l_1 t, \ldots, a_n + l_n t)$ lies on $\widehat{F_m}$, then for every $t \in K$ one has

$$F(x(t)) = \sum_{j=m+1}^{d} \sum_{\substack{e_i \geq 0 \\ e_1+\cdots+e_n=j}} \left(\frac{1}{e_1! \cdots e_n!}\right) \frac{\partial^j F}{\partial X_1^{e_1} \cdots \partial X_n^{e_n}}(P)(l_1)^{e_1} \cdots (l_n)^{e_n} t^j$$

so that $\mathfrak{m}_P(\widehat{F}, r) > m$. Conversely, if $\mathfrak{m}_P(\widehat{F}, r) > m$, then

$$F(x(t)) = t^{m+1} G(t) \qquad \Longrightarrow \qquad F_m(x(t)) = 0, \quad \forall\, t \in K.$$

If P is simple then $\mathrm{C}_{\widehat{F},P} = \mathrm{T}_{\widehat{F},P}$.

Example 5.16 Consider the following curves of $\mathbb{A}^2(\mathbb{R})$ and their tangent cones at the origini O: $a \in \mathbb{R}$, $a \neq 0$:

"Bernoulli's lemniscate": $F(X, Y) = (X^2 + Y^2)^2 - a^2(X^2 - Y^2)$: $\mathrm{C}_{\widehat{F},O} : X^2 - Y^2 = 0$;
"trifolium": $F(X, Y) = (X^2 + Y^2)^2 - a(Y^3 - 3X^2 Y)$: $\mathrm{C}_{\widehat{F},O} : Y^3 - 3X^2 Y = 0$;
"quadrifolium": $F(X, Y) = (X^2 + Y^2)^3 - 4a^2 X^2 Y^2$: $\mathrm{C}_{\widehat{F},O} : X^2 Y^2 = 0$.

Proposition 5.17 *Let $\widehat{F} \in \mathcal{H}_d(n, K)$ with $n \geq 2$. Let*

$$\widehat{F} = m_1 \widehat{F}_1 + \cdots + m_k \widehat{F}_r$$

be the decomposition of $\widehat{F}$ into irreducible components. A point $P \in \mathcal{V}(\widehat{F})$ is singular if and only if at least one of the following conditions holds:

(i) *there exists $j \in \{1, \ldots, r\}$ such that $P \in \widehat{F_j}$ with $m_j \geq 2$;*
(ii) *there exists $j \in \{1, \ldots, r\}$ such that P is a singular point of $\widehat{F_j}$;*
(iii) *there exist $h, k \in \{1, \ldots, r\}$, $h \neq k$, such that $P \in \widehat{F_h} \cap \widehat{F_k}$.*

If P is a non-singular point of $\widehat{F}$ then there exists one and only one j, $1 \leq j \leq r$, such that $P \in \widehat{F_j}$ with $m_j = 1$ and P is non-singular for $\widehat{F_j}$; in this case we have

$$\mathrm{T}_{\widehat{F},P} = \mathrm{T}_{\widehat{F_j},P}.$$

Proof Let $F = F_1^{m_1} \cdots F_r^{m_r}$ be the factorization of F into irreducible polynomials F_i. By Leibniz's rule we have

$$\nabla F(P) = \sum_{i=1}^{r} m_i F_1^{m_1}(P) \cdots F_i^{m_i - 1}(P) \cdots F_r^{m_r}(P) \nabla f_i(P). \tag{5.8}$$

If P belongs to two distinct components or to a multiple component $\widehat{F_j}$ all summands on the right– hand side of (5.8) are zero so that P is singular for $\widehat{F}$. If P lies on a simple component $\widehat{F_j}$, from (5.8) we have

$$\nabla F(P) = c \nabla F_j(P), \quad \text{where} \quad c = \prod_{i \neq j} F_i^{m_i}(P) \neq 0. \tag{5.9}$$

Hence $\nabla F(P) = 0$ if and only if $\nabla F_j(P) = 0$. The last assertion follows immediately from (5.9). □

The Action of $\mathrm{AGL}_n(K)$ on $\widehat{F} \in \mathcal{H}_d(n, K)$

Definition 5.18 Let $\sigma \in \mathrm{AGL}_n(K)$ be an affine automorphism of $\mathbb{A}^n(K)$ associated with a matrix $M = \left(a_{ij}\right)_{i,j=1,\ldots,n} \in \mathrm{GL}_n(K)$

$$\sigma(x_1, \ldots, x_n) = (y_1, \ldots, y_n), \qquad y_i = \sum_{j=1}^{n} a_{ij} x_j + b_i, \qquad i = 1, \ldots, n. \tag{5.10}$$

Then

$$\sigma^{-1}(y_1,\dots,y_n)=(x_1,\dots,x_n),\qquad x_i=\sum_{j=1}^{n}a^*_{ij}y_j+b^*_i,\qquad i=1,\dots,n, \tag{5.11}$$

where $(a^*_{ij})_{i,j=1,\dots,n}=M^{-1}$ and

$$\begin{pmatrix}b^*_1\\ \vdots\\ b^*_n\end{pmatrix}=-\begin{pmatrix}\cdots\cdots\cdots\\ \vdots\ \ a^*_{ij}\ \ \vdots\\ \cdots\cdots\cdots\end{pmatrix}\begin{pmatrix}b_1\\ \vdots\\ b_n\end{pmatrix}.$$

Let $F(x_1,\dots,x_n)\in K[x_1,\dots,x_n]_d$ be a non-zero polynomial of degree d. We want define the polynomial $\sigma\cdot F\in K[x_1,\dots,x_n]_d$ in order to have

$$\sigma(\mathcal{V}(\widehat{F}))=\mathcal{V}(\sigma\cdot\widehat{F}). \tag{5.12}$$

Thus we put

$$\begin{aligned}(\sigma\cdot F)(y_1,\dots,y_n)&:=F(\sigma^{-1}(y_1,\dots,y_n))\\ &=F\left(\sum_{j=1}^{n}a^*_{1j}y_j+b^*_1,\dots,\sum_{j=1}^{n}a^*_{nj}y_j+b^*_n\right).\end{aligned} \tag{5.13}$$

It is easy to see that we obtain an action on the left

$$\mathrm{AGL}_n(K)\times K[x_1,\dots,x_n]_d\to K[x_1,\dots,x_n]_d$$

of the group $\mathrm{AGL}_n(K)$ on $K[x_1,\dots,x_n]_d$ (Definition 1.6), i.e.

$$\begin{aligned}&(\sigma\circ\tau)\cdot F=\sigma\cdot(\tau\cdot F),\quad &&\forall\,\sigma,\tau\in\mathrm{AGL}_n(K),\ \forall\,F\in K[x_1,\dots,x_n]_d,\\ &\mathrm{id}_{\mathbb{A}^n(K)}\cdot F=F,\quad &&\forall\,F\in K[x_1,\dots,x_n]_d.\end{aligned}$$

Furthermore, if $G=\lambda F$, with $F,G\in K[x_1,\dots,x_n]_d$, and $\lambda\in K\setminus\{0\}$, then we have $\sigma\cdot G=\sigma\cdot\lambda F$. Hence (5.13) induces a well-defined action

$$\mathrm{AGL}_n(K)\times\mathcal{H}_d(n,K)\to\mathcal{H}_d(n,K),\qquad \sigma\cdot\widehat{F}:=\widehat{\sigma\cdot F}.$$

Let $\widehat{F},\widehat{G}\in\mathcal{H}_d(n,K)$ be two hypersurfaces of $\mathbb{A}^n(K)$. We say that $\widehat{F}$ and $\widehat{G}$ are *affinely equivalent*, written $\widehat{F}\sim\widehat{G}$, if there exists $\sigma\in\mathrm{AGL}_n(K)$ such that $\sigma\cdot\widehat{F}=\widehat{G}$. In other words, $\widehat{F}$ and $\widehat{G}$ are affinely equivalent if $\widehat{F}$ and $\widehat{G}$ belong to

the same orbit of $\mathcal{H}_d(n, K)$ with respect the action (5.13) of $\mathrm{AGL}_n(K)$. According to Definition 1.6 we get an equivalence relation on $\mathcal{H}_d(n, K)$. It is an important task to describe the set of orbits of $\mathcal{H}_d(n, K)$ but in general it is very difficult. Since the group $\mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ of the isometries of $\mathbb{E}^n(\mathbb{R})$ is a subgroup of $\mathrm{AGL}_n(\mathbb{R})$ (5.13) defines an action of $\mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ on $\mathcal{H}_d(n, \mathbb{R})$. Thus we say that $\widehat{F}$ and $\widehat{G}$ are *euclidean equivalent* if there exists $\sigma \in \mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ such that $\sigma \cdot \widehat{F} = \widehat{G}$.

The following result, easy to prove, shows the importance of the action (5.13).

Proposition 5.19 *Let $\sigma \in \mathrm{AGL}_n(K)$ be an automorphism as in (5.10). Let $\widehat{F} = m_1\widehat{F_1} + \cdots + m_k\widehat{F_r}$ be the decomposition of a hypersurface $\widehat{F} \in \mathcal{H}_d(n, K)$ into irreducible components. Then*

$$\widehat{\sigma \cdot F} = m_1\widehat{\sigma \cdot F_1} + \cdots + m_r\widehat{\sigma \cdot F_r} \tag{5.14}$$

is the decomposition of $\widehat{\sigma \cdot F}$ into irreducible components. Furthermore

$$\sigma(\mathrm{Reg}(\widehat{F})) = \mathrm{Reg}(\sigma \cdot \widehat{F}), \quad \sigma(\mathrm{Sing}(\widehat{F})) = \mathrm{Sing}(\sigma \cdot \widehat{F}). \tag{5.15}$$

If $R \in \mathrm{Reg}(\widehat{F})$ is a non-singular point of $\widehat{F}$ then $\sigma(\mathrm{T}_{\widehat{F},P}) = \mathrm{T}_{\sigma\cdot\widehat{F},\sigma(P)}$.

Let r be a line passing through $P \in \mathbb{A}^n(K)$. Then

$$\mathfrak{m}_P(\widehat{F}, r) = \mathfrak{m}_{\sigma(P)}(\sigma \cdot F, \sigma(r)). \tag{5.16}$$

In particular we have

$$\mathfrak{m}_P(F) = \mathfrak{m}_{\sigma(P)}(\sigma \cdot F). \tag{5.17}$$

Proof The relation (5.14) easily follows from (5.13). If we take $F_k = x_k = \sum_{i=1}^{n} a^*_{ki} y_i + b^*_k$, $k = 1, \ldots, n$ in Lemma 5.6 we get for $i = 1, \ldots, n$,

$$\frac{\partial(\sigma \cdot F)}{\partial y_i} = \sum_{k=1}^{n} \frac{\partial F}{\partial x_k}(x_1, \ldots, x_n)\frac{\partial x_k}{\partial y_i} = \sum_{k=1}^{n} \frac{\partial F}{\partial x_k}(x_1, \ldots, x_n)a^*_{ki},$$

so that $\nabla(\sigma \cdot F) = \nabla(F)M^{-1}$ which implies (5.15).

Put $b = (b_1, \ldots, b_n)$. Let r have parametric equations $x(u) = (x_1(u), \ldots, x_n(u))$, then the line $\sigma(r)$ passing through $\sigma(P)$ has parametric equations $y(u)^t = Mx(u)^t + b^t$, where $y(u) = (y_1(u), \ldots, y_n(u))$. The relation (5.16) follows immediately from the equalities

$$\sigma \cdot F(y(u)) = F\left(M^{-1}(y(u) - b)^t)\right) = F\left(M^{-1}(Mx(u)^t\right) = F(x(u))$$

and (5.17) is immediate from (5.16). □

Lemma 5.20 *Let $\widehat{F} \in \mathcal{H}_d(n, K)$ and let $\widehat{H}$ be a hyperplane contained in $\mathcal{V}(\widehat{F})$. If K has at least $d+1$ elements then $\widehat{H}$ is an irreducible component of $\widehat{F}$, i.e. $H \mid F$.*

Proof Up to a suitable change of coordinates we can suppose that H has the equation $x_n = 0$. Then we write (by Theorem 1.10)

$$F(X_1, X_2, \ldots, X_n) = X_n G(X_1, X_2, \ldots, X_n) + R(X_1, X_2, \ldots, X_{n-1}),$$

where G and R are polynomials and R does not contain the indeterminate X_n. Since every point $(x_1, \ldots, x_{n-1}, 0)$ lies on $\mathcal{V}(\widehat{F})$, R is identically zero, otherwise, by Lemma 5.2, we can find an $(n-1)$-tuple $(b_1, \ldots, , b_{n-1})$ such that $R(b_1, \ldots, , b_{n-1}) \neq 0$. Hence the point $(b_1, \ldots, , b_{n-1}, 0)$ would belong to $H \setminus \widehat{F}$, contradicting the hypothesis. □

Definition 5.21 Let $\widehat{F}$ and $\widehat{G}$ be two distinct hypersurfaces of the same degree. We define the *pencil of hypersurfaces* generated by $\widehat{F}$ and $\widehat{G}$ as the set of the hypersurfaces $\widehat{H} = \widehat{H}_{\lambda,\mu}$ such that

$$H_{\lambda,\mu} = \lambda F + \mu G, \qquad (\lambda, \mu) \in \mathbb{R}^2 \setminus \{(0,0)\}.$$

In particular we have $\mathcal{V}(\widehat{H_{\lambda,\mu}}) \supset \mathcal{V}(\widehat{F}) \cap \mathcal{V}(\widehat{G})$.

5.2 Affine Hyperquadrics

Henceforth we assume $\mathrm{char}(K) \neq 2$.

Definition 5.22 The most general form of a polynomial $F \in K[X_1, \ldots, X_N]_2$ is

$$F = F(X_1, \ldots, X_n) = \sum_{i,j=1}^{n} c_{ij} X_i X_j + 2 \sum_{i=1}^{n} c_i X_i + c, \tag{5.18}$$

where $c_{ij} \in K$, $c_{ij} = c_{ji}$ for $i, j = 1, \ldots, n$, $c_{ij} \neq 0$ for some i, j, and $c_i, c \in K$, $i = 1, \ldots, n$. The associated hypersurface $\widehat{F}$ of $\mathbb{A}^n(K)$ is called *affine hyperquadric*. A hyperquadric is called *conic* if $n = 2$, *quadric* if $n = 3$.

The following symmetric non-zero matrices with entries in K are associated with $\widehat{F}$:

$$C_F = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix}, \qquad \overline{C}_F = \begin{pmatrix} c & c_1 & c_2 & \cdots & c_n \\ c_1 & c_{11} & c_{12} & \cdots & c_{1n} \\ c_2 & c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_n & c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix}.$$

We call C_F the *incomplete matrix* and $\overline{C}_F$ the *complete matrix* associated with $\widehat{F}$. If we put

$$X^t := (x_1, \ldots, x_n), \qquad \overline{X}^t = (1, x_1, \ldots, x_n) \qquad C_1 := (c_1, \ldots, c_n),$$

then the equation of $\widehat{F}$ can be written in the following matrix form:

$$X^t C_F X + 2C_1 X + c = 0, \tag{5.19}$$

$$\overline{X}^t \, \overline{C}_F \, \overline{X} = 0. \tag{5.20}$$

Let $\widehat{F}$ be a hyperquadric of $\mathbb{A}^n(K)$ having the equation

$$F = \sum_{i,j=1}^{n} c_{ij} x_i x_j + 2 \sum_{i=1}^{n} c_i x_i + c = 0. \tag{5.21}$$

Let $\sigma \in \mathrm{AGL}_n(K)$ be an affine automorphism of $\mathbb{A}^n(K)$ defined by

$$y_i = \sum_{j=1}^{n} a_{ij} x_j + b_i, \qquad i = 1, \ldots, n, \tag{5.22}$$

where $A := \left(a_{ij}\right)_{i,j=1,\ldots,n}$ is a non-singular matrix. It is convenient to write (5.22) in the matrix form

$$Y = AX + B, \tag{5.23}$$

where $B = (b_1, \ldots, b_n)^t$ and $Y = (y_1, \ldots, y_n)^t$. Then the matrix form of σ^{-1} is

$$X = A^{-1}Y - A^{-1}B. \tag{5.24}$$

Hence $\sigma \cdot \widehat{F}$ is the hyperquadric of the equation

$$\begin{aligned} \sigma \cdot F &= (A^{-1}Y - A^{-1}B)^t C (A^{-1}Y - A^{-1}B) + 2C_1(A^{-1}Y - A^{-1}B) + c \\ &= \left(Y^t (A^{-1})^t - B^t (A^{-1})^t\right) C (A^{-1}Y - A^{-1}B) + 2C_1(A^{-1}Y - A^{-1}B) + c \\ &= Y^t (A^{-1})^t C A^{-1} Y + \cdots \end{aligned} \tag{5.25}$$

Hence $C_{\sigma\cdot F} = (A^{-1})^t C_F A^{-1}$ so that by (1.13) we get

$$\operatorname{rank}(C_{\sigma\cdot F}) = \operatorname{rank}(C_F). \tag{5.26}$$

Furthermore,

$$\det(C_{\sigma\cdot F}) = \det(C_F)\,[\det(A^{-1})]^2 = \frac{\det(C_F)}{[\det(A)]^2}. \tag{5.27}$$

In particular, if $K = \mathbb{R}$, the sign of $\det(C_F)$ is invariant under affine automorphisms.

If we put

$$\overline{Y}^t = (1, y_1, \ldots, y_n), \qquad \overline{A} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ b_1 & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

from (5.23) we have $\overline{Y} = \overline{A}\,\overline{X}$ so that

$$\overline{X}^t\,\overline{C}_F\,\overline{X} = \left((\overline{A})^{-1}\overline{Y}\right)^t\,\overline{C}_F\,\left((\overline{A})^{-1}\overline{Y}\right) = \overline{Y}^t\,((\overline{A})^{-1})^t\,\overline{C}_F\,(\overline{A})^{-1}\,\overline{Y}$$

and

$$\overline{C}_{\sigma\cdot F} = ((\overline{A})^{-1})^t\,\overline{C}_F\,(\overline{A})^{-1}. \tag{5.28}$$

Therefore by (1.13) we have

$$\operatorname{rank}(\overline{C}_{\sigma\cdot F}) = \operatorname{rank}(\overline{C}), \tag{5.29}$$

and

$$\det(\overline{C}_{\sigma\cdot F}) = \det(\overline{A})^{-2}\det(\overline{C}_F) = \det(A)^{-2}\det(\overline{C}_F) = \frac{\det(\overline{C}_F)}{\det(A)^2}. \tag{5.30}$$

In particular, if $K = \mathbb{R}$, the sign of $\det(\overline{C})$ is invariant under affine automorphisms.

Definition 5.23 A hyperquadric $\widehat{F} \in \mathcal{H}_2(n, K)$ (with F as in (5.18)) is called *non-degenerate* if the matrix $\overline{C}_F$ is non-singular, i.e. $\det(\overline{C}_F) \neq 0$ (or also $\operatorname{rank}(\overline{C}_F) = n + 1$). If $\det(\overline{C}_F) = 0$, $\widehat{F}$ is called *degenerate*. We call *simply degenerate* a hyperquadric $\widehat{F}$ such that $\operatorname{rank}(\overline{C}_F) = n$. These definitions are well-posed (i.e. independent of affine coordinates) by (5.29).

Proposition 5.24 *Let $A = (a_1, \ldots, a_n)^t$ be a point lying on the hyperquadric $\widehat{F}$ of $\mathbb{A}^n(K)$ of the equation*

$$\sum_{i,j=1}^{n} c_{ij} x_i x_j + 2 \sum_{i=1}^{n} c_i x_i + c = 0. \tag{5.31}$$

Then A is singular for $\widehat{F}$ if and only if

$$\sum_{j=1}^{n} c_{ij} a_j + c_i = 0, \qquad i = 1, \ldots, n, \tag{5.32}$$

i.e. $\overline{A}^t \overline{C}_F = 0$ where $\overline{A}^t = (1, a_1, \ldots, a_n)$. In particular a non-degenerate hyperquadric is non-singular.
The tangent hyperplane to $\widehat{F}$ at a non-singular point A is given by

$$\sum_{i,j=1}^{n} c_{ij} a_j x_i + \sum_{i=1}^{n} c_i (x_i + a_i) + c = 0, \tag{5.33}$$

i.e. $\overline{A}^t \overline{C}_F \overline{X} = 0$ in matrix form.

Proof From (5.31) we obtain

$$\frac{\partial F}{\partial x_i} = 2 \sum_{j=1}^{n} c_{ij} x_j + 2c_i, \qquad i = 1, \ldots, n, \tag{5.34}$$

from which (5.32) follows immediately. From (5.32) and the identity

$$0 = \sum_{i,j=1}^{n} c_{ij} a_i a_j + 2 \sum_{i=1}^{n} c_i a_i + c = \sum_{i=1}^{n} \left[\left(\sum_{j=1}^{n} c_{ij} a_j \right) a_i + 2 c_i a_i \right] + c$$

we get

$$\sum_{i=1}^{n} a_i c_i + c = 0. \tag{5.35}$$

The relations (5.32) and (5.35) are equivalent to $\overline{A}^t \overline{C}_F = 0$.

Let $A = (a_1, \ldots, a_n)$ be a non-singular point of $\widehat{F}$. From the relations

$$\begin{aligned}\sum_{i=1}^{n} \frac{\partial F}{\partial x_i}(a_1, \ldots, a_n)(x_i - a_i) &= 2 \sum_{i,j=1}^{n} c_{ij}(x_i - a_i)a_j + 2\sum_{i=1}^{n} c_i(x_i - a_i) \\ &= 2 \sum_{i,j=1}^{n} c_{ij}a_j x_i + 2\sum_{i=1}^{n} c_i x_i - 2 \sum_{i,j=1}^{n} c_{ij}a_i a_j \\ &\quad - 2\sum_{i=1}^{n} c_i a_i \\ &= 2 \sum_{i,j=1}^{n} c_{ij}x_i a_j + 2\sum_{i=1}^{n} c_i x_i + 4\sum_{i=1}^{n} c_i a_i \\ &\quad + 2c - 2\sum_{i=1}^{n} c_i a_i \\ &= 2 \sum_{i,j=1}^{n} c_{ij}a_j x_i + 2\sum_{i=1}^{n} c_i(x_i + a_i) + 2c\end{aligned}$$

we get (5.33). □

Remark 5.25 By Proposition 5.24 a non-degenerate hyperquadric is non-singular and irreducible. However, the quadric of $\mathbb{A}^3(\mathbb{R})$ of equation $x^2 + y^2 - 1 = 0$ is degenerate but smooth (in fact simply degenerate) and the quadric cone $x^2 + y^2 - z^2 = 0$ is degenerate but irreducible.

Remark 5.26 In a pencil generated by two hyperquadrics $\widehat{F}$ and $\widehat{G}$ of $\mathbb{A}^n(K)$ there are at most $n + 1$ degenerate hyperquadrics or all hyperquadrics of the pencil are degenerate, indeed the equation (in homogeneous unknowns λ and μ) $\det(\overline{C}_{H_{\lambda,\mu}}) = 0$ vanishes identically or has degree $n + 1$.

Proposition 5.27 *Let $P_1, \ldots, P_m$ be m points of $\mathbb{A}^n(K)$. If $m \leq \dfrac{(n+1)(n+2)}{2} - 1$, then there exists at least one hyperquadric $\widehat{F}$ of $\mathbb{A}^n(K)$ passing through $P_1, \ldots, P_m$.*

Proof Let $F = \sum_{i,j=1}^{n} c_{ij}x_i x_j + 2\sum_{i=1}^{n} c_i x_i + c$. If $\widehat{F}$ passes through $P_1, \ldots, P_m$, its coefficients must satisfy the homogeneous linear system in $\frac{(n+1)(n+2)}{2}$ unknowns

$$\sum_{i,j=1}^{n} c_{ij}\, p_{k,i}\, p_{k,j} + 2\sum_{i=1}^{n} c_i\, p_{k,i} + c = 0, \qquad P_k = (p_{k1}, \ldots, p_{kn}), \quad k = 1, \ldots m. \tag{5.36}$$

By Corollary 1.65 the system (5.36) has non-trivial solutions if

$$\frac{(n+1)(n+2)}{2} - m \geq 1.$$

□

Remark 5.28 Let $\widehat{F}$ be the sphere $S^{n-1}(P, r)$ with centre $P = (p_1, \ldots, p_n)$ and radius r of equation

$$(x_1 - p_1)^2 + \cdots + (x_n - p_n)^2 = r^2.$$

The sphere $S^{n-1}(P, r)$ is smooth with tangent hyperplane at $A = (a_1, \ldots, a_n) \in S^{n-1}(P, r)$ (according to (5.33))

$$\sum_{i=1}^{n}(a_i - p_i)x_i + \sum_{i=1}^{n} p_i(p_i - a_i) - r^2 = 0. \tag{5.37}$$

Since the line AP has $(a_1 - p_1, \ldots, a_n - p_n)$ as director parameters, the tangent hyperplane to $S^{n-1}(P, r)$ at A is orthogonal to AP, in accordance with Definition 4.21. Also the concept of a tangent line given by Definition 4.21 is a special case of the concept of a tangent line to a hypersurface given by Definition 5.9.

Intersection of a Hyperquadric with an Affine Subspace
Let $\widehat{F} \in \mathcal{H}_2(n, K)$ be a hyperquadric and $\mathcal{A}$ an affine subspace of dimension h of $\mathbb{A}^n(K)$ not contained in $\mathcal{V}(\widehat{F})$. By a suitable change of coordinates we can suppose that $\mathcal{A}$ has equations $x_{h+1} = x_{h+2} = \cdots = x_n = 0$. It is obvious that

$$\pi_h : \mathcal{A} \to \mathbb{A}^h(K), \qquad \pi_h(x_1, \ldots, x_n) = (x_1, \ldots, x_h),$$

is an affine isomorphism. By means of π_h we can identify $\mathcal{V}(\widehat{F}) \cap \mathcal{A}$ with the geometric locus $\mathcal{V}(\widehat{G})$ of the hypersurface $\widehat{G}$ of $\mathbb{A}^h(K)$, where

$$G(x_1, \ldots, \ldots, x_h) = F(x_1, \ldots, x_h, 0, \ldots, 0).$$

The hypersurface $\widehat{G}$ has degree ≤ 2 and its geometric locus $\mathcal{V}(\widehat{G})$ can be empty.

Proposition 5.29 *Let $\widehat{F} \in \mathcal{H}_2(n, K)$ be an irreducible hyperquadric.*

(i) *If P is a non-singular point of $\widehat{F}$, then $\widehat{G} := \mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$ is a degenerate hyperquadric of $\mathrm{T}_{P,\widehat{F}}$. Furthermore, if $\widehat{F}$ is non-degenerate, then $\mathrm{rank}(C_G) = n - 1$ and P is the only singular point of $\widehat{G}$.*

(ii) *Conversely, if $\widehat{F}$ is non-degenerate and H is a hyperplane such that $H \cap \widehat{F}$ is a degenerate hyperquadric of H, then $H = \mathrm{T}_{P,\widehat{F}}$ for some $P \in \widehat{F}$.*

Proof We may suppose that $P = (0, \dots, 0)$ and $\mathrm{T}_{P,\widehat{F}} : x_n = 0$. By (5.33) we get $c_1 = \cdots = c_{n-1} = 0$ so that

$$\overline{C}_F = \begin{pmatrix} 0 & 0 & \dots & c_n \\ 0 & c_{11} & \dots & c_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_n & c_{n1} & \dots & c_{nn} \end{pmatrix}.$$

Then $G(x_1, \dots, x_{n-1}) = \sum_{i,j=1}^{n-1} c_{ij} x_i x_j$ is a degenerate hyperquadric of $\mathrm{T}_{P,\widehat{F}}$. Since

$$\overline{C}_F = \begin{pmatrix} 0 & 0 & 0 & \dots & c_n \\ 0 & & & & c_{1n} \\ \vdots & & C_G & & \vdots \\ c_n & c_{n1} & c_{n2} & \dots & c_{nn} \end{pmatrix},$$

we have $\det(\overline{C}_F) = -c_n^2 \det(C_G)$ so that $\det(C_G) \neq 0$ and $P = (0, \dots, 0)$ is the only singular point of $\widehat{G}$.

Conversely, without loss of generality we can suppose that H has the equation $x_n = 0$. If

$$F = \sum_{i,j=1}^{n} c_{ij} x_i x_j + 2 \sum_{i=1}^{n} c_i x_i + c = 0$$

then $H \cap \widehat{F} = \widehat{G}$ where $G = \sum_{i,j=1}^{n-1} c_{ij} x_i x_j + 2 \sum_{i=1}^{n-1} c_i x_i + c = 0$ with

$$\begin{vmatrix} c & c_1 & \dots & c_{n-1} \\ c_1 & c_{11} & \dots & c_{1\,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n-1} & c_{n-1\,1} & \dots & c_{n-1\,n-1} \end{vmatrix} = 0.$$

The vanishing of the above determinant implies the existence of $\lambda_1, \dots, \lambda_{n-1} \in K$ (not all zero) such that

$$c_i = \sum_{j=1}^{n-1} \lambda_j c_{ij}, \ i = 1, \dots, n-1, \qquad c = \sum_{i=1}^{n-1} \lambda_i c_i = \sum_{i,j=1}^{n-1} \lambda_i \lambda_j c_{ij}. \tag{5.38}$$

The point $P = (-\lambda_1, \ldots, -\lambda_{n-1}, 0)$ belongs to $\mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$, indeed by (5.38), we have

$$\sum_{i,j=1}^{n} c_{ij}\lambda_i\lambda_j - 2\sum_{i=1}^{n}\lambda_i c_i + c = \sum_{i,j=1}^{n-1} c_{ij}\lambda_i\lambda_j - \sum_{i=1}^{n-1}\lambda_i c_i - \sum_{i=1}^{n-1}\lambda_i c_i + \sum_{i,j=1}^{n-1}\lambda_i\lambda_j c_{ij} = 0.$$

The tangent hyperplane to $\widehat{F}$ at P has the equation $-\sum_{i,j=1}^{n} c_{ij}\lambda_j x_i + \sum_{i=1}^{n} c_i(x_i - \lambda_i) + c = 0$, i.e.

$$-\left(\sum_{j=1}^{n} c_{nj}\lambda_j\right)x_n - \sum_{i,j=1}^{n-1} c_{ij}\lambda_j x_i + \sum_{i=1}^{n} c_i x_i - \sum_{i=1}^{n-1} c_i\lambda_i + c$$

$$= \left(-\sum_{j=1}^{n} c_{nj}\lambda_j + c_n\right)x_n + \sum_{i=1}^{n-1}\left(-\sum_{j=1}^{n-1} c_{ij}\lambda_j + c_i\right)x_i = \left(-\sum_{j=1}^{n} c_{nj}\lambda_j + c_n\right)x_n.$$

Since $\widehat{F}$ is non-degenerate $-\sum_{j=1}^{n} c_{nj}\lambda_j + c_n \neq 0$, otherwise $\det(\overline{C}_F) = 0$ (by (5.38)), hence $H = \mathrm{T}_{P,\widehat{F}}$. □

Centres of Hyperquadrics

Definition 5.30 Let $\widehat{F}$ be a hyperquadric of $\mathbb{A}^n(K)$. A point $P \in \mathbb{A}^n(K)$ is called a *centre* of $\widehat{F}$ if $s_P \cdot \widehat{F} = \widehat{F}$, s_P being the symmetry of centre P. If P is a centre of $\widehat{F}$ by Lemma 5.19 we have the relation

$$s_P(\mathcal{V}(\widehat{F})) = \mathcal{V}(\widehat{F}), \tag{5.39}$$

which reveals the geometric meaning of centre. It is worthwhile to observe that the notion of a centre is an affine concept.

Proposition 5.31 *Let $\sigma \in \mathrm{AGL}_n(K)$). A point P is a centre of $\widehat{F}$ if and only if $\sigma(P)$ is a centre of $\sigma \cdot \widehat{F}$.*

Proof $\Rightarrow$) For all $Q \in \mathbb{A}^n(K)$

$$P = \frac{1}{2}\sigma^{-1}(Q) + \frac{1}{2}s_P(\sigma^{-1}(Q)) = \frac{1}{2}\sigma^{-1}(Q) + \frac{1}{2}s_P^{-1}(\sigma^{-1}(Q))$$

since $s_P = s_P^{-1}$. We have also

$$\sigma(P) = \frac{1}{2}Q + \frac{1}{2}s_{\sigma(P)}(Q), \quad \text{which implies} \quad P = \frac{1}{2}\sigma^{-1}(Q) + \frac{1}{2}\sigma^{-1}(s_{\sigma(P)}(Q)),$$

hence $\sigma^{-1}(s_{\sigma(P)}(Q)) = s_P^{-1}(\sigma^{-1}(Q))$ and $F(\sigma^{-1}(s_{\sigma(P)}(Q))) = F(s_P^{-1}(\sigma^{-1}(Q)))$, i.e.

$$\begin{aligned}(s_{\sigma(P)} \cdot \sigma \cdot F)(Q) &:= F((s_{\sigma(P)} \circ \sigma)^{-1}(Q)) = F(s_P^{-1}(\sigma^{-1}(Q))) \\ &:= s_P \cdot F(\sigma^{-1}(Q)) = F(\sigma^{-1}(Q)) := \sigma \cdot F(Q).\end{aligned}$$

$\Leftarrow$) Apply σ^{-1}. □

Proposition 5.32 *Let $P \in \mathbb{A}^n(K)$ be an arbitrary point and $\widehat{F} \in \mathcal{H}_d(n, K)$ be a hyperquadric, where F is given by (5.18). Then $P = (p_1, \ldots, p_n)$ is a centre of $\widehat{F}$ if and only if*

$$\sum_{j=1}^{n} c_{ij} p_j + c_i = 0, \qquad i = 1, \ldots, n. \tag{5.40}$$

Proof By Definition 3.85 we get

$$\begin{aligned}s_P \cdot F &= \sum_{i,j=1}^{n} c_{ij}(2p_i - x_i)(2p_j - x_j) + 2\sum_{i=1}^{n} c_i(2p_i - x_i) + c \\ &= 4\sum_{i,j=1}^{n} c_{ij} p_i p_j - 4\sum_{i,j=1}^{n} c_{ij} x_i p_j + \sum_{i,j=1}^{n} c_{ij} x_i x_j + 4\sum_{i=1}^{n} c_i p_i - 2\sum_{i=1}^{n} c_i x_i + c.\end{aligned}$$

Then the condition $\sigma_P \cdot \widehat{F} = \widehat{F}$ is equivalent to

$$\begin{aligned}4\sum_{i,j=1}^{n} c_{ij} p_i p_j - 4\sum_{i,j=1}^{n} c_{ij} x_i p_j + \sum_{i,j=1}^{n} c_{ij} x_i x_j + 4\sum_{i=1}^{n} c_i p_i \\ - 2\sum_{i=1}^{n} c_i x_i + c = \sum_{i,j=1}^{n} c_{ij} x_i x_j + 2\sum_{i=1}^{n} c_i x_i + c,\end{aligned}$$

which implies the polynomial identity

$$\sum_{i=1}^{n}\Big(\sum_{j=1}^{n} c_{ij} p_j + c_i\Big)(x_i - p_i) = 0$$

so that (5.40) follows. □

Corollary 5.33 *Let $\widehat{F} \in \mathcal{H}_2(n, K)$ be a hyperquadric. If $P \in \mathrm{Sing}(\widehat{F})$ then P is a centre of $\widehat{F}$. Conversely, if $P \in \mathcal{V}(\widehat{F})$ is a centre of $\widehat{F}$ then P is a singular point of $\widehat{F}$.*

Proof It is an immediate consequence of Propositions 5.24 and 5.32. □

Corollary 5.34 *Let $\widehat{F} \in \mathcal{H}_2(n, K)$ be a hyperquadric, where F is given by (5.18). If $C_F = \left(c_{ij}\right)_{i,j=1,\dots,n}$ is non-singular, then $\widehat{F}$ has one and only one centre.*

Proof By Proposition 5.32 the coordinates of the centres of $\widehat{F}$ are the solution of the linear system

$$\sum_{j=1}^{n} c_{ij}x_j + c_i = 0, \qquad i = 1, \dots, n,$$

so that the thesis follows from Proposition 1.26. □

Remark 5.35 The parabola $y = x^2$ of $\mathbb{A}^2(\mathbb{R})$ does not have any centre, while the circle $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$ admit $(0, 0)$ as a unique centre. The cone $x^2 + y^2 - z^2 = 0$ of $\mathbb{A}^3(\mathbb{R})$ has the origin $(0, 0, 0)$ as a unique centre. Every point $(0, 0, z)$, $z \in \mathbb{R}$, is a centre of the cylinder $x^2 + y^2 = 1$ of $\mathbb{A}^3(\mathbb{R})$.

5.2.1 The Reduced Equation of a Hyperquadric of $\mathbb{A}^n(K)$

Lemma 5.36 *Let $\widehat{F} \in \mathcal{H}_2(n, K)$ be a hyperquadric, where F is given by (5.18) and $\operatorname{rank}(C_F) = r$. There exists a system of coordinates of $\mathbb{A}^n(K)$ in which $\widehat{F}$ has an equation of the form*

$$\sum_{i=1}^{r} \alpha_i x_i^2 + 2 \sum_{i=r+1}^{n} \beta_i x_i + \gamma = 0. \tag{5.41}$$

Proof Put the equation $F = 0$ in the matrix form $X^t C_F X + 2C_1 X + c = 0$ (see (5.19)). By Theorem 2.14 there is a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ of K^n in which the matrix M of the quadratic form $q(X) = X^t C_F X$ is diagonal, say

$$M = M_F^{\mathcal{B}} = \begin{pmatrix} \alpha_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \alpha_2 & \dots & 0 & \vdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \alpha_r & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}, \tag{5.42}$$

where $\alpha_i \neq 0$ for $i = 1, \ldots, r$. If $\Lambda^{\mathcal{E},\mathcal{B}}$ is the matrix of transition from the canonical basis $\mathcal{E}$ to the basis $\mathcal{B}$ we have

$$M = (\Lambda^{\mathcal{E},\mathcal{B}})^t \, C_F \Lambda^{\mathcal{E},\mathcal{B}}.$$

If we put $Y = (\Lambda^{\mathcal{E},\mathcal{B}})^{-1} X$, $X^t C_F X + 2C_1 X + c = 0$ is changed into

$$Y^t M Y + 2C_1 \Lambda^{\mathcal{E},\mathcal{B}} Y + c = 0$$

so that, if $Y = (y_1, \ldots, y_n)^t$, $\widehat{F}$ has the equation

$$\sum_{i=1}^{r} \alpha_i y_i^2 + 2 \sum_{i=1}^{n} \beta_i y_i + c = 0.$$

By the change of coordinates

$$z_i = y_i + \frac{\beta_i}{\alpha_i}, \ i = 1, \ldots, r, \qquad z_i = y_i, \ i = r+1, \ldots, n,$$

we obtain

$$\sum_{i=1}^{r} \alpha_i z_i^2 + 2 \sum_{i=r+1}^{n} \beta_i z_i + \gamma = 0,$$

i.e. (5.41) after renaming the coordinates. □

Theorem 5.37 *Let $\widehat{F} \in \mathcal{H}_2(n, K)$ be a hyperquadric, where F is given by* (5.18) *and* $\operatorname{rank}(C_F) = r$. *Assume* (5.41), *then there exists a system of coordinates of $\mathbb{A}^n(K)$ with respect to which the equation of the hyperquadric takes one of the following forms:*

$$\Gamma_r^1(K): \ \sum_{i=1}^{r} a_i x_i^2 = 1, \qquad 1 \leq r \leq n, \ \gamma \neq 0, \qquad a_i = -\frac{\alpha_i}{\gamma}, \ i = 1, \ldots, r,$$

$$\Gamma_r^0(K): \ \sum_{i=1}^{r} \alpha_i x_i^2 = 0, \qquad 1 \leq r \leq n,$$

$$\Gamma_r^2(K): \ \sum_{i=1}^{r} \alpha_i x_i^2 = 2x_n, \qquad 1 \leq r \leq n-1.$$

Such an equation is called a reduced equation *of the hyperquadric. Furthermore, hyperquadrics of different types cannot be affinely equivalent.*

Proof If $r = n$ or if $\beta_i = 0$ for $i = r+1, \ldots, n$ from Lemma 5.36, we get $\Gamma_r^1(K)$ if $\gamma \neq 0$ and $\Gamma_r^0(K)$ if $\gamma = 0$. If $\beta_i \neq 0$ for some $i > r$, we can choose $a_{i\,r+1}, \ldots, a_{in} \in K$, $i = r+1, \ldots, n-1$ such that

$$\begin{vmatrix} a_{r+1\,r+1} & \cdots & a_{r+1\,n} \\ \vdots & \ddots & \vdots \\ a_{n-1\,r+1} & \cdots & a_{n-1\,n} \\ \beta_{r+1} & \cdots & \beta_n \end{vmatrix} \neq 0,$$

i.e. we can find a basis of $\mathbb{R}^{n-r}$ containing $(\beta_{r+1}, \cdots, \beta_n)$. Giving rise to a change of coordinates

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \\ y_{r+1} \\ \vdots \\ y_n \end{pmatrix} = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & \begin{array}{ccc} a_{r+1\,r+1} & \cdots & a_{r+1\,n} \\ \vdots & \ddots & \vdots \\ a_{n-1\,r+1} & \cdots & a_{n-1\,n} \\ -\beta_{r+1} & \cdots & -\beta_n \end{array} \end{array}\right) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_r \\ x_{r+1} \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ -\frac{\gamma}{2} \end{pmatrix}$$

we obtain from (5.41)

$$\sum_{i=1}^{r} \alpha_i y_i^2 = 2y_n,$$

i.e. $\Gamma_r^2(K)$ after renaming the coordinates. The last assertion follows from the table

	$\operatorname{rank}(C_F)$	$\operatorname{rank}(\overline{C}_F)$
$\Gamma_r^1(K)$	r	$r+1$
$\Gamma_r^0(K)$	r	r
$\Gamma_r^2(K)$	r	$r+2$

by (5.29). □

Remark 5.38 Since we can choose several diagonal forms (5.42) for C_F, the reduced equations of hyperquadrics are not " canonical equations".

Affine Classification of Complex and Real Hyperquadrics

Definition 5.39 Let $K = \mathbb{C}, \mathbb{R}$. According to Definition 5.18 we say that two hyperquadrics $\widehat{F}$ and $\widehat{G} \in \mathcal{H}_2(n, K)$ are *affinely equivalent*, written $\widehat{F} \sim \widehat{G}$, if there exists $\sigma \in \mathrm{AGL}_n(K)$ such that $\sigma \cdot \widehat{F} = \widehat{G}$. In other words, two hyperquadrics

$\widehat{F}$ and $\widehat{G} \in \mathcal{H}_2(n, K)$ are affinely equivalent if they belong to the same orbit of $\mathcal{H}_2(n, K)$ with respect to the canonical action (see Definition 5.18)

$$\mathrm{AGL}_n(K) \times \mathcal{H}_2(n, K) \to \mathcal{H}_2(n, K), \qquad (\sigma, \widehat{F}) \to \sigma \cdot \widehat{F}, \tag{5.43}$$

of the group $\mathrm{AGL}_n(K)$ on the set of hyperquadrics $\mathcal{H}_2^{\mathrm{a}}(n, K)$.

To classify the real hyperquadrics up to affine isomorphism of $\mathbb{A}^n(K)$ we have to determine a canonical representative in every orbit of $\mathcal{H}_2(n, K)$ with respect to (5.43).

Theorem 5.40 *Let $\widehat{F} \in \mathcal{H}_2(n, \mathbb{C})$ be a complex hyperquadric where F is given by (5.18) and $\mathrm{rank}(C_F) = r$. Assume (5.41), then there exists a system of coordinates of $\mathbb{A}^n(\mathbb{C})$ with respect to which the equation of the hyperquadric takes one of the following forms:*

$$\Gamma_r^1(\mathbb{C}): \sum_{i=1}^{r} x_i^2 = 1, \qquad 1 \le r \le n,$$

$$\Gamma_r^0(\mathbb{C}): \sum_{i=1}^{r} x_i^2 = 0, \qquad 1 \le r \le n,$$

$$\Gamma_r^2(\mathbb{C}): \sum_{i=1}^{r} x_i^2 = 2x_n, \qquad 1 \le r \le n-1.$$

Such an equation is called a canonical equation *of the hyperquadric. Furthermore, hyperquadrics of different types cannot be affinely equivalent.*

Proof By Corollary 2.17 the matrix (5.42) takes the form $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ so we get $\sum_{i=1}^{r} x_i^2 = 1$ from $\sum_{i=1}^{r} -\frac{1}{\gamma} x_i^2 = 1$ of Theorem 5.37 by the change of coordinates

$$y_i = \delta x_i, \quad \delta^2 = -\frac{1}{\gamma}, \ i = 1, \ldots, r, \qquad y_i = x_i, \ i = r+1, \ldots, n.$$

No changes are needed for $\Gamma_r^0(\mathbb{C})$ and $\Gamma_r^2(\mathbb{C})$. □

Assumption 5.41 If A is a real symmetric matrix $\mathrm{Sign}(\lambda A) = \mathrm{Sign}(A)$ if $\lambda > 0$ and $\mathrm{Sign}(\lambda A) = (i_-(A), i_+(A))$ if $\lambda < 0$, thus multiplying the equation of a real hyperquadric $\widehat{F} \in \mathcal{H}_2^{\mathrm{a}}(n, \mathbb{R})$ by -1 we can assume from now on

$$i_+(C_F)) \ge i_-(C_F)), \qquad i_+(\overline{C}_F)) \ge i_-(\overline{C}_F)). \tag{5.44}$$

In this way the signatures $\mathrm{Sign}(C_F)$ and $\mathrm{Sign}(\overline{C}_F)$ depend on the hyperquadric $\widehat{F}$ but not on F.

Theorem 5.42 *Let $\widehat{F} \in \mathcal{H}_2(n, \mathbb{R})$ be a real hyperquadric where F is given by (5.18). Let $\mathrm{rank}(C_F) = r$, $p = i_+(C_F)$, $r - p = i_-(C_F)$, $1 \leq r \leq n$, $p \leq r$. By* Assumption 5.41 *we have $p \geq r - p$. Assume (5.41), then there exists a system of coordinates of $\mathbb{A}^n(\mathbb{R})$ with respect to which the equation of the hyperquadric takes one of the following forms:*

$$\Gamma^1_{p,r}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{r} x_i^2 = 1, \qquad p \geq r - p + 1$$

$$\Gamma^{-1}_{p,r}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{r} x_i^2 = -1, \qquad 2p \neq r.$$

$$\Gamma^0_{p,r}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{r} x_i^2 = 0,$$

$$\Gamma^2_{p,r}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{r} x_i^2 = 2x_n, \qquad 1 \leq r \leq n-1.$$

Such an equation is called a canonical equation *of the hyperquadric. Furthermore, hyperquadrics of different types cannot be affinely equivalent.*

Proof The matrix (5.42) takes the form (by Corollary 2.20)

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_{r-p} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{5.45}$$

where p is the positivity index of C_F. Hence we have

$$\sum_{i=1}^{r} a_i x_i^2 - 1 = \sum_{i=1}^{p} -\frac{1}{\gamma} x_i^2 + \sum_{i=p+1}^{r} \frac{1}{\gamma} x_i^2 - 1. \tag{5.46}$$

If $\gamma < 0$, by the change of coordinates

$$y_i = \delta x_i, \ \delta^2 = -\frac{1}{\gamma} \ i = 1, \ldots, n, \tag{5.47}$$

we obtain $\Gamma^1_{p,r}(\mathbb{R})$. If $\gamma > 0$, multiplying the right– hand side of (5.46) by -1 we get

$$\sum_{i=1}^{p} \frac{1}{\gamma} x_i^2 - \sum_{i=p+1}^{r} \frac{1}{\gamma} x_i^2 + 1.$$

By the change of coordinates

$$y_i = \delta x_i,\ \delta^2 = \frac{1}{\gamma}\ i = 1, \ldots, n, \tag{5.48}$$

we get $\Gamma^{-1}_{p,r}(\mathbb{R})$. If $r = 2p$ the hyperquadric $\Gamma^{-1}_{p,r}(\mathbb{R})$: $x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_{2p}^2 = -1$, i.e. $-x_1^2 + \cdots - x_p^2 + x_{p+1}^2 + \cdots + x_{2p}^2 = 1$, is affinely equivalent to $x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_{2p}^2 = 1$, i.e. $\Gamma^1_{p,r}(\mathbb{R})$, interchanging x_i and x_{p+i}, $i = 1, \ldots, p$. From $\sum_{i=1}^{r} \alpha_i x_i^2 = 0$ and $\sum_{i=1}^{r} \alpha_i x_i^2 = 2x_n$ we get $\Gamma^0_{p,r}(\mathbb{R})$ and $\Gamma^2_{p,r}(\mathbb{R})$ immediately. The last assertion follows from the Table 5.1 by (5.29).

For $F = \Gamma^2_{p,r}(\mathbb{R})$ (here the positive eigenvalues of C_F are $\lambda_1 = \cdots = \lambda_p = 1$ and the negative ones $\lambda_{p+1} = \cdots \lambda_r = -1$) from the equalities (obtained developing the deteminants along their second columns)

$$\begin{vmatrix} -\lambda & & & & & & -1 \\ & \lambda_1 - \lambda & & & & & \\ & & \ddots & & & & \\ & & & \lambda_r - \lambda & & & \\ & & & & -\lambda & & \\ & & & & & \ddots & \\ -1 & & & & & & -\lambda \end{vmatrix} = (\lambda_1 - \lambda) \begin{vmatrix} -\lambda & & & & & & -1 \\ & \lambda_2 - \lambda & & & & & \\ & & \ddots & & & & \\ & & & \lambda_r - \lambda & & & \\ & & & & -\lambda & & \\ & & & & & \ddots & \\ -1 & & & & & & -\lambda \end{vmatrix}$$

Table 5.1 Real hyperquadrics

F	rank(C_F)	rank$(\overline{C}_F)$	Sign(C_F)	Sign$(\overline{C}_F)$
$\Gamma^1_{p,r}(\mathbb{R})$	r	$r+1$	$(p, r-p)$	$(p, r-p+1)$
$\Gamma^{-1}_{p,r}(\mathbb{R})$	r	$r+1$	$(p, r-p)$	$(p+1, r-p)$
$\Gamma^0_{p,r}(\mathbb{R})$	r	r	$(p, r-p)$	$(p, r-p)$
$\Gamma^2_{p,r}(\mathbb{R})$	r	$r+2$	$(p, r-p)$	$(p+1, r-p+1)$

$$= \cdots = (\lambda_1 - \lambda)\cdots(\lambda_r - \lambda)\begin{vmatrix} -\lambda & & & -1 \\ & -\lambda & & \\ & & \ddots & \\ -1 & & & -\lambda \end{vmatrix}$$

$$= \cdots = (\lambda_1 - \lambda)\cdots(\lambda_r - \lambda)(-\lambda)^{n-r-1}\begin{vmatrix} -\lambda & -1 \\ -1 & -\lambda \end{vmatrix}$$

$$= (\lambda_1 - \lambda)\cdots(\lambda_r - \lambda)(-\lambda)^{n-r-1}(\lambda - 1)(\lambda + 1)$$

we deduce immediately that $\mathrm{Sign}(\overline{C}_F) = (p+1, r-p+1)$. □

From Theorem 5.40 we have:

Corollary 5.43 *Let $\widehat{F}$ and $\widehat{G}$ be two real hyperquadrics of equations $\overline{X}^t\,\overline{C}_F\,\overline{X} = 0$ and $\overline{X}^t\,\overline{C}_G\,\overline{X} = 0$ respectively. Then $\widehat{F}$ and $\widehat{G}$ are affinely equivalent if and only if*

$$(\mathrm{Sign}(C_F), \mathrm{Sign}(\overline{C}_F)) = (\mathrm{Sign}(C_G), \mathrm{Sign}(\overline{C}_G)).$$

Corollary 5.44 *Let $\widehat{F} \in \mathcal{H}_2(n, \mathbb{C})$ be a complex hyperquadric. $\widehat{F}$ has a unique centre if and only if it is affinely equivalent to*

$$\Gamma_n^0(\mathbb{C}) : \sum_{i=1}^{n} x_i^2 = 0 \qquad \textit{or to} \qquad \Gamma_n^1(\mathbb{C}) : \sum_{i=1}^{n} x_i^2 = 1. \tag{5.49}$$

$\widehat{F}$ has no centre if and only if it is affinely equivalent to

$$\Gamma_r^0(\mathbb{C}) : \sum_{i=1}^{r} x_i^2 = 2x_n, \qquad 1 \le r \le n-1. \tag{5.50}$$

Proof By Proposition 5.31 we can consider only the hyperquadrics $\Gamma_r^1(\mathbb{C})$, $\Gamma_r^0(\mathbb{C})$ and $\Gamma_r^2(\mathbb{C})$. The origin $O = (0, \ldots, 0)$ is the unique centre of hyperquadrics (5.49). If $r < n$ the linear system $\sum_{j=1}^{r} c_{ij}p_j + c_i = 0$, $i = 1, \ldots, n$, (i.e. the equation $\sum_{j=1}^{r} p_j = 0$ repeated n times) has infinite solutions.

In the case of (5.50) among the equations of the linear system $\sum_{j=1}^{r} c_{ij}p_j + c_i = 0$, $i = 1, \ldots, n$, there is $\sum_{j=1}^{r} c_{nj}p_j - 1 = 0$, i.e. $-1 = 0$. □

Corollary 5.45 *Let $\widehat{F} \in \mathcal{H}_2(n, \mathbb{R})$ be a real hyperquadric. $\widehat{F}$ has a unique centre if and only if it is affinely equivalent to*

$$\Gamma^0_{p,n}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{n} x_i^2 = 0, \qquad p \geq n - p,$$

or to

$$\Gamma^{\pm 1}_{p,n}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{n} x_i^2 = \pm 1, \qquad p \geq n - p.$$

$\widehat{F}$ has no centre if and only if it is affinely equivalent to

$$\Gamma^2_{p,r}(\mathbb{R}) : \sum_{i=1}^{p} x_i^2 - \sum_{i=p+1}^{r} x_i^2 = 2x_n, \qquad 1 \leq r \leq n-1,\ p \geq r - p.$$

Proof Repeat the proof of Proposition 5.44. □

Definition 5.46 A non-degenerate real hyperquadric $\widehat{F} \in \mathcal{H}_2(n, \mathbb{R})$, with $n \geq 3$, without a centre (i.e. if it is affinely equivalent to $\Gamma^2_{p,n}(\mathbb{R})$) is called a *paraboloid*. The remaining non-degenerate hyperquadrics are called *hyperboloids* and *ellipsoids* whose general definitions will be given in Chap. 10.

5.2.2 Euclidean Classification of the Real Hyperquadrics

We want to classify the real hyperquadrics up to isometries of $\mathbb{E}^n(\mathbb{R})$. Assume the notation to be as in Definition 5.22. First we observe that if $\widehat{F} \in \mathcal{H}_2(n, \mathbb{R})$ and $\sigma \in \mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ then, from (5.26) and (5.29), we get

$$\det(C_{\sigma \cdot F}) = \det(C_F), \qquad \det(\overline{C}_{\sigma \cdot F}) = \det(\overline{C}_F), \tag{5.51}$$

since A is orthogonal so that $\det(A) = \pm 1$.

Definition 5.47 We say that two hyperquadrics $\widehat{F}$ and $\widehat{G} \in \mathcal{H}_2(n, \mathbb{R})$ are *isometric*, written $\widehat{F} \cong \widehat{G}$, if there exists $\sigma \in \mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ such that $\sigma \cdot \widehat{F} = \widehat{G}$. In other words, two hyperquadrics $\widehat{F}$ and $\widehat{G} \in \mathcal{H}_2(n, \mathbb{R})$ are isometric if they belong to the same orbit of $\mathcal{H}_2(n, \mathbb{R})$ with respect to the canonical action (see Definition 5.18)

$$\mathrm{O}(\mathbb{E}^n(\mathbb{R})) \times \mathcal{H}_2(n, \mathbb{R}) \to \mathcal{H}_2(n, \mathbb{R}), \qquad (\sigma, \widehat{F}) \to \sigma \cdot \widehat{F}, \tag{5.52}$$

of the group $\mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ on the set of hyperquadrics $\mathcal{H}_2(n, \mathbb{R})$.

To classify the real hyperquadrics up to isometries of $\mathbb{E}^n(\mathbb{R})$ we have to determine a canonical representative in every orbit of $\mathcal{H}_2(n,\mathbb{R})$ with respect to (5.52).

We can now state the fundamental theorem of the euclidean classification of the real hyperquadrics.

Theorem 5.48 *Assume the notation as in* Definition 5.22. *Let* $\widehat{F}$ *be the real hyperquadric of equation*

$$X^t C_F X + 2C_1 X + c = 0$$

with $\operatorname{rank}(C_F) = r$, $i_+(C_F) = p$ *satisfying* Assumption 5.41. *Then* $\widehat{F}$ *is isometric to one and only one of the following* canonical hyperquadrics*:*

$$\begin{aligned}
&(1)\quad \Gamma^0_{p,r;d_1,\dots,d_r}: && d_1x_1^2+\cdots+d_px_p^2-d_{p+1}x_{p+1}^2\\
&&& -\cdots-d_rx_r^2=0, && d_1=1,\\
&(2)\quad \Gamma^{-1}_{p,r;d_1,\dots,d_r}: && d_1x_1^2+\cdots+d_px_p^2-d_{p+1}x_{p+1}^2\\
&&& -\cdots-d_rx_r^2=-1, && r\neq 2p,\\
&(3)\quad \Gamma^1_{p,r;d_1,\dots,d_r}: && d_1x_1^2+\cdots+d_px_p^2-d_{p+1}x_{p+1}^2\\
&&& -\cdots-d_rx_r^2=1, && p\geq r-p+1,\\
&(4)\quad \Gamma^2_{p,r;d_1,\dots,d_r}: && d_1x_1^2+\cdots+d_px_p^2-d_{p+1}x_{p+1}^2\\
&&& -\cdots-d_rx_r^2=2x_n, && 1\leq p\leq r\leq n-1,
\end{aligned}$$

where $0<d_1\leq\cdots\leq d_p$, $0<d_{p+1}\leq\cdots\leq d_r$.

In cases (2), (3) *and* (4) *the reals* $d_1,\dots,d_r$ *are proportional to the non-zero eigenvalues of* C_F *by a positive factor so that they are uniquely determined by* $\widehat{F}$. *In case* (1) *the numbers* $d_1,\dots,d_r$ *are not uniquely determined.*

Proof By Theorem 1.178 there exists an orthogonal matrix $A=\left(a_{ij}\right)_{i,j=1,\dots,n}\in \mathrm{O}(n)$ such that

$$AC_FA^t = AC_FA^{-1} = \begin{pmatrix}
\lambda_1 & 0 & \dots & 0 & 0 & \dots & 0\\
0 & \lambda_2 & \dots & 0 & \vdots & \vdots & \vdots\\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots\\
0 & 0 & \dots & \lambda_r & 0 & \dots & 0\\
0 & 0 & \dots & 0 & 0 & \dots & 0\\
\vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots\\
0 & 0 & \dots & 0 & 0 & \dots & 0
\end{pmatrix}$$

where $\lambda_1, \ldots, \lambda_r$ are the non-zero eigenvalues of C_F and with $i_+(C_F) = p \geq r - p = i_-(C_F)$ (multiplying the equation $F = 0$ by -1 if $p < r - p$). Therefore the affine automorphism σ

$$Y = AX, \qquad y_i = \sum_{j=1}^{n} a_{ij} x_j, \ i = 1, \ldots, n, \quad Y = (y_1, \ldots, y_n)^t,$$

is an isometry. Hence $\widehat{F}$ is isometric to hyperquadric $\sigma \cdot \widehat{F}$ of equation

$$Y^t \, ACA^t \, Y + 2C_1 \, A^t Y + c = 0,$$

i.e.

$$\lambda_1 y_1^2 + \cdots + \lambda_r y_r^2 + 2\beta_1 y_1 + \cdots + 2\beta_n y_n + c = 0. \tag{5.53}$$

Let σ_1 be the isometry

$$x_i = y_i + \beta_i/\lambda_i, \quad i = 1, \cdots, r; \qquad x_j = y_j, \quad j = r+1, \ldots, n.$$

Then $\widehat{F}$ is isometric to $\sigma_1 \circ \sigma \cdot \widehat{F}$ which has the equation

$$\lambda_1 x_1^2 + \cdots + \lambda_r x_r^2 + 2\beta_{r+1} x_{r+1} + \cdots + 2\beta_n x_n = -c'. \tag{5.54}$$

Permuting the variables if necessary (if α is a permutation of $\{1, \ldots, n\}$ then the affine automorphism $\sigma_2 : \ y_i = x_{\alpha(i)}, i = 1, \ldots, n$ is an isometry), we can suppose that $0 < \lambda_1 \leq \cdots \leq \lambda_p$, $0 < \lambda_{p+1} \leq \cdots \leq \lambda_r$ so that $\sigma_2 \circ \sigma_1 \circ \sigma \cdot \widehat{F}$ has the equation

$$\lambda_1 x_1^2 + \cdots + \lambda_p x_p^2 - \lambda_{p+1} x_{p+1}^2 - \cdots - \lambda_r x_r^2 + 2\beta_{r+1} x_{r+1} + \cdots + 2\beta_n x_n = -c', \tag{5.55}$$

with $p \geq r - p$.

(1) If $\beta_{r+1} = \cdots = \beta_n = c' = 0$ we have $\lambda_1 x_1^2 + \cdots + \lambda_p x_p^2 - \lambda_{p+1} x_{p+1}^2 - \cdots - \lambda_r x_r^2 = 0$, where $0 < \lambda_1 \leq \cdots \leq \lambda_p$, $0 < \lambda_{p+1} \leq \cdots \leq \lambda_r$. Dividing by λ_1 we get

$$x_1^2 + d_2 x_2^2 + \cdots + d_p x_p^2 - d_{p+1} x_{p+1}^2 - \cdots - d_r x_r^2 = 0, \qquad 1 \leq r \leq n, \tag{5.56}$$

where $1 < d_2 \leq \cdots \leq d_p$, $0 < d_{p+1} \leq \cdots \leq d_r$.

((2)–(3)) Suppose that $\beta_{r+1} = \cdots = \beta_n = 0$ and $c' \neq 0$. Dividing (5.55) by $|c'|$ we obtain

$$d_1 x_1^2 + \cdots + d_p x_p^2 - d_{p+1} x_{p+1}^2 - \cdots - d_r x_r^2 = -1, \quad \text{if } c' > 0, \tag{5.57}$$

$$d_1 x_1^2 + \cdots + d_p x_p^2 - d_{p+1} x_{p+1}^2 - \cdots - d_r x_r^2 = 1, \quad \text{if } c' < 0, \tag{5.58}$$

where $d_i = \dfrac{\lambda_i}{|c'|}$, $i = 1, \ldots, r \leq n$, and $0 < d_1 \leq \cdots \leq d_p$, $0 < d_{p+1} \leq \cdots \leq d_r$. If $r = 2p$ we have $d_1 x_1^2 + \cdots + d_p x_p^2 - d_{p+1} x_{p+1}^2 - \cdots - d_{2p} x_{2p}^2 = -1$, i.e. $-d_1 x_1^2 - \cdots - d_p x_p^2 + d_{p+1} x_{p+1}^2 + \cdots + d_{2p} x_{2p}^2 = 1$, so that interchanging x_i and x_{p+i}, $i = 1, \ldots, p$, we get $d_{p+1} x_1^2 + \cdots + d_{2p} x_p^2 - d_1 x_{p+1}^2 - \cdots - d_p x_{2p}^2 = 1$ with $0 < d_{p+1} \leq \cdots \leq d_{2p}$, $0 < d_1 \leq \cdots \leq d_p$ i.e. $\Gamma^1_{p,2p;d_1,\ldots,d_{2p}}$ after renumbering the d_i, $i = 1, \ldots, 2p$.

(4) If $\beta_j \neq 0$ for some $j \in \{r+1, \ldots, n\}$ there exists $a > 0$ such that $a^2(\beta_{r+1}^2 + \cdot + \beta_n^2) = 1$. Hence we can find $b_{ij} \in \mathbb{R}$, $i = r+1, \ldots, n-1$, $j = r+1, \ldots, n$, such that

$$\begin{pmatrix} b_{r+1,r+1} & \ldots & b_{r+1,n} \\ \vdots & \ddots & \vdots \\ b_{n-1,r+1} & \ldots & b_{n-1,n} \\ -a\beta_{r+1} & \ldots & -a\beta_n \end{pmatrix}$$

is orthogonal. Indeed, since $a^2(\beta_{r+1}^2 + \cdot + \beta_n^2) = 1$, $(-a\beta_{r+1}, \cdots, -a\beta_n)$ is a unit vector of $\mathbb{R}^{n-r}$, so we can determine an orthonormal basis of $\mathbb{R}^{n-r}$ containing $(-a\beta_{r+1}, \cdots, -a\beta_n)$ via the Gram– Schmidt algorithm (Theorem 1.155). Let σ_3 be the isometry of equations

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \\ y_{r+1} \\ \vdots \\ y_n \end{pmatrix} = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & \begin{matrix} b_{r+1\,r+1} & \ldots & b_{r+1\,n} \\ \vdots & \ddots & \vdots \\ b_{n-1\,r+1} & \ldots & b_{n-1\,n} \\ -a\beta_{r+1} & \ldots & -a\beta_n \end{matrix} \end{array}\right) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_r \\ x_{r+1} \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ -\frac{ac'}{2} \end{pmatrix}.$$

Therefore $\sigma_3 \circ \sigma_2 \circ \sigma_1 \circ \sigma \cdot \widehat{F}$ has the equation (after renaming the variables)

$$d_1 x_1^2 + \cdots + d_p x_p^2 - d_{p+1} x_{p+1}^2 - \cdots - d_r x_r^2 = 2x_n, \qquad 1 \leq r \leq n-1, \tag{5.59}$$

where $d_i = a\lambda_i$, $i = 1, \ldots, r$, and $0 < d_1 \leq \cdots \leq d_p$, $0 < d_{p+1} \leq \cdots \leq d_r$.

□

Uniqueness We have just proved that every real hyperquadric is isometric to one of the $\Gamma^a_{p,r;d_1,\dots,d_r}$, with $a = 0, \pm 1, 2$. First we have to point out the following facts:

(a) If $\widehat{\Gamma}^a_{p,r;d_1,\dots,d_r} \simeq \widehat{\Gamma}^b_{p',r';t_1,\dots,t_{r'}}$, since $\widehat{\Gamma}^a_{p,r;d_1,\dots,d_r}$ is affinely equivalent to $\widehat{\Gamma}^a_{p,r}$, from Theorem 5.40 we get immediately

$$a = b, \quad p = p', \quad r = r'.$$

(b) For two isometric hyperquadrics $\Gamma^0_{p,r;d_1,\dots,d_r}$ and $\Gamma^0_{p',r',t_1,\dots,t_{r'}}$ we can have $d_i \neq t_i$, for some $i = 1, \dots, r$; for instance $\Gamma^0_{1,2;1,2} : x_1^2 - 2x_2^2 = 0$ and $\Gamma^0_{1,2;1,\frac{1}{2}} : x_1^2 - \frac{1}{2}x_2^2 = 0$ are isometric (if $\alpha(x_i, x_2) = (x_2, x_1)$ then $\alpha \cdot \Gamma^0_{1,2;1,2}$ has the equation $-2x_1^2 + x_2^2 = 0$, i.e. $x_1^2 - \frac{1}{2}x_2^2 = 0$).

Lemma 5.49 *If the hyperquadrics $\Gamma^a_{p,r;d_1,\dots,d_r}$ and $\Gamma^a_{p,r;t_1,\dots,t_r}$, with $a = \pm 1$, are isometric, then $d_i = t_i$, $i = 1, \dots, r$.*

Proof Put $G_d = \Gamma^a_{p,r;d_1,\dots,d_r}$ and $G_t = \Gamma^a_{p,r;t_1,\dots,t_r}$. If $\alpha : \widehat{G_d} \to \widehat{G_t}$ is an isometry and $\alpha(X) = AX + B$ with $A = (a_{ij})_{i,j} \in \mathrm{O}(n)$ and $B = (b_1, \dots, b_b)^t$, then from (5.28) we have $\overline{C}_{G_d} = h\,(\overline{A})^t\, \overline{C}_{G_t} \overline{A}$ some $h \in \mathbb{R}^*$, i.e.

$$\begin{pmatrix} -a & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & C_{G_d} & \\ 0 & & & \end{pmatrix} = h \begin{pmatrix} 1 & b_1 & \dots & b_n \\ 0 & & & \\ \vdots & & A^t & \\ 0 & & & \end{pmatrix} \begin{pmatrix} -a & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & C_{G_t} & \\ 0 & & & \end{pmatrix} \begin{pmatrix} 1 & 0 & \dots & 0 \\ b_1 & & & \\ \vdots & & A & \\ b_n & & & \end{pmatrix}. \tag{5.60}$$

It is convenient to write (5.60) in a more compact form (we put $\mathbf{0} = (0, \dots, 0)^t$)

$$\begin{aligned} \begin{pmatrix} -a & \mathbf{0}^t \\ \mathbf{0} & C_{G_d} \end{pmatrix} &= h \begin{pmatrix} 1 & B^t \\ \mathbf{0} & A^t \end{pmatrix} \begin{pmatrix} -a & \mathbf{0}^t \\ \mathbf{0} & C_{G_t} \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}^t \\ B & A \end{pmatrix} \\ &= h \begin{pmatrix} 1 & B^t \\ \mathbf{0} & A^t \end{pmatrix} \begin{pmatrix} -a & \mathbf{0}^t \\ C_{G_t}B & C_{G_t}A \end{pmatrix} \\ &= h \begin{pmatrix} -a + B^t C_{G_t} B & B^t C_{G_t} A \\ A^t C_{G_t} B & A^t C_{G_t} A \end{pmatrix}. \end{aligned} \tag{5.61}$$

From (5.61) we get $B^t C_{G_t} A = \mathbf{0}^t$ (and $A^t C_{G_t} B = \mathbf{0}$) so that $B^t C_{G_t} = \mathbf{0}^t$, $C_{G_t} B = \mathbf{0}$ (since A is invertible). Hence $-ah = -a$ which implies $h = 1$. Therefore $C_{G_d} = A^t C_{G_t} A$. The non-zero eigenvalues of C_{G_d} are $d_1, \dots, d_p, -d_{p+1}, \dots, -d_r$ and those of $A^t C_{G_t} A$ are $t_1, \dots, t_p, -t_{p+1}, \dots, -t_r$ so that we must have

$$d_i = t_i, \qquad i = 1, \dots, r.$$

□

Lemma 5.50 *If the hyperquadrics* $\Gamma^2_{p,r;d_1,\dots,d_r}$ *and* $\Gamma^2_{p,r;t_1,\dots,t_r}$ *are isometric, then* $d_i = t_i$, $i = 1, \dots, r$.

Proof Put $G_d = \Gamma^2_{p,r;d_1,\dots,d_r}$ and $G_t = \Gamma^2_{p,r;t_1,\dots,t_r}$. If $\alpha\colon \widehat{G_d} \to \widehat{G_t}$ is an isometry and $\alpha(X) = AX + B$ with $A = (a_{ij})_{i,j} \in \mathrm{O}(n)$ and $B = (b_1, \dots, b_b)^t$, then from (5.28) we have $\overline{C}_{G_d} = h\,(\overline{A})^t\,\overline{C}_{G_t}\overline{A}$ some $h \in \mathbb{R}^*$, i.e.

$$\begin{pmatrix} 0 & 0 & \dots & -1 \\ 0 & & & \\ \vdots & & C_{G_d} & \\ -1 & & & \end{pmatrix} = h \begin{pmatrix} 1 & b_1 & \dots & b_n \\ 0 & & & \\ \vdots & & A^t & \\ 0 & & & \end{pmatrix} \begin{pmatrix} 0 & 0 & \dots & -1 \\ 0 & & & \\ \vdots & & C_{G_t} & \\ -1 & & & \end{pmatrix} \begin{pmatrix} 1 & 0 & \dots & 0 \\ b_1 & & & \\ \vdots & & A & \\ b_n & & & \end{pmatrix}. \tag{5.62}$$

If $u = (0, \dots, 0, -1)^t$, we have in a more compact form

$$\begin{aligned} \begin{pmatrix} 0 & u^t \\ u & C_{G_d} \end{pmatrix} &= h \begin{pmatrix} 1 & B^t \\ \mathbf{0} & A^t \end{pmatrix} \begin{pmatrix} 0 & u^t \\ u & C_{G_t} \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}^t \\ B & A \end{pmatrix} \\ &= h \begin{pmatrix} 1 & B^t \\ \mathbf{0} & A^t \end{pmatrix} \begin{pmatrix} u^t B & u^t A \\ u + C_{G_t} B & C_{G_t} A \end{pmatrix} \\ &= h \begin{pmatrix} 2B^t u + B^t C_{G_t} B & u^t A + B^t C_{G_t} A \\ A^t u + A^t C_{G_t} B & A^t C_{G_t} A \end{pmatrix}. \end{aligned} \tag{5.63}$$

We write

$$C_{G_t} = \begin{pmatrix} D_t & \mathbf{0}_{r,n-r} \\ \mathbf{0}_{n-r,r} & \mathbf{0}_{n-r,n-r} \end{pmatrix}, \; C_{G_d} = \begin{pmatrix} D_d & \mathbf{0}_{r,n-r} \\ \mathbf{0}_{n-r,r} & \mathbf{0}_{n-r,n-r} \end{pmatrix},$$

$$D_t = \begin{pmatrix} t_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & -t_r \end{pmatrix}, \; D_d = \begin{pmatrix} d_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & -d_r \end{pmatrix}.$$

From (5.62) we get $h\,C_{G_t}A = AC_{G_d}$, i.e.

$$h \begin{pmatrix} D_t P_{r,r} & D_t Q_{r,n-r} \\ \mathbf{0}_{n-r,r} & \mathbf{0}_{n-r,n-r} \end{pmatrix} = \begin{pmatrix} P_{r,r} D_d & \mathbf{0}_{r,n-r} \\ R_{n-r,r} D_d & \mathbf{0}_{n-r,n-r} \end{pmatrix}, \quad \text{where}$$

$$A = \begin{pmatrix} P_{r,r} & Q_{r,n-r} \\ R_{n-r,r} & S_{n-r,n-r} \end{pmatrix},$$

which implies $Q_{r,n-r} = \mathbf{0}_{r,n-r}$ and $R_{n-r,r} = \mathbf{0}_{n-r,r}$ since D_t and D_d are invertible. Hence

$$A = \begin{pmatrix} P_{r,r} & \mathbf{0}_{r,n-r} \\ \mathbf{0}_{n-r,r} & S_{n-r,n-r} \end{pmatrix} = \left(\begin{array}{ccc|ccc} a_{11} & \dots & a_{1r} & & & \\ \vdots & \ddots & \vdots & & \mathbf{0}_{r,n-r} & \\ a_{r1} & \dots & a_{rr} & & & \\ \hline & & & a_{r+1\,r+1} & \dots & a_{n\,r+1} \\ & \mathbf{0}_{n-r,r} & & \vdots & \ddots & \vdots \\ & & & a_{n\,r+1} & \dots & a_{nn} \end{array}\right).$$

Therefore

$$A^t u = \begin{pmatrix} 0 \\ \vdots \\ -a_{n\,r+1} \\ \vdots \\ -a_{nn} \end{pmatrix} \qquad A^t C_{G_t} B = \begin{pmatrix} P_{r,r}^t D_t & \mathbf{0}_{r,n-r} \\ \mathbf{0}_{n-r,r} & \mathbf{0}_{n-r,n-r} \end{pmatrix} \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_r \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Since $h\,A^t u + A^t C_{G_t} B = u$ we obtain

$$h \begin{pmatrix} 0 \\ \vdots \\ -a_{n\,r+1} \\ \vdots \\ -a_{nn} \end{pmatrix} + \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_r \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ -1 \end{pmatrix}$$

so that $a_{n\,r+1} = \cdots = a_{n\,n-1} = 0$ and $h a_{nn} = 1$. Since $a_{n1} = \cdots = a_{nr} = 0$ and $a_{n1}^2 + \cdots + a_{nn}^2 = 1$ we deduce $a_{nn} = \pm 1$ and, because $h > 0$, $a_{nn} = 1$ and $h = 1$ so that

$$d_i = t_i, \qquad i = 1, \dots, r. \qquad \square$$

Remark 5.51 From Theorem 5.48 every hyperquadric of $\mathbb{E}^1(\mathbb{R})$ is isometric to one and only one of the following types:

(1) $x_1^2 = 0$: a real double point.
(2) $d_1 x_1^2 = -1$, with $d_1 > 0$: two complex-conjugate points.
(3) $d_1 x_1^2 = 1$, with $d_1 > 0$: two distinct real points.

Principal Diametral Hyperplanes and Axes of a Hyperquadric

Definition 5.52 Let $\widehat{F} \in \mathcal{H}_2(n, \mathbb{R})$ be a non-degenerate hyperquadric of $\mathbb{E}^n(\mathbb{R})$ ($n \geq 2$). A hyperplane H of $\mathbb{E}^n(\mathbb{R})$ is called a *principal diametral hyperplane* of $\widehat{F}$ if

$$s_H \cdot \widehat{F} = \widehat{F},$$

where s_H is the orthogonal symmetry with respect to H. An *axis* of $\widehat{F}$ is a line intersection of principal diametral hyperplanes. If $n = 2$ principal diametral hyperplanes coincide obviouvsly with axes.

Remark 5.53 Under the assumptions and notation of Lemma 3.87 if $s_{\mathcal{B},W} \cdot \widehat{F} = \widehat{F}$ then

$$s_{\sigma(\mathcal{B}),\mathrm{Tr}(\sigma)(W)} \circ \sigma \cdot \widehat{F} = \sigma \circ s_{\mathcal{B},W} \cdot \widehat{F} = \sigma \cdot \widehat{F}. \tag{5.64}$$

In particular, let $\widehat{F}$ be orthogonally symmetric with respect to a euclidean subspace $\mathcal{B}$, i.e. $s_{\mathcal{B}} \cdot \widehat{F} = \widehat{F}$ where $s_{\mathcal{B}}$ is the orthogonal symmetry with respect to $\mathcal{B}$ (Definition 4.25). If $\sigma \in \mathrm{O}(\mathbb{E}^n(\mathbb{R}))$, then $\sigma \circ s_{\mathcal{B}} = s_{\sigma(\mathcal{B})} \circ \sigma$ by Corollary 4.27 so that $\sigma \cdot \widehat{F}$ is orthogonally symmetric with respect to the euclidean subspace $\sigma(\mathcal{B})$

Now we give some obvious statements concerning principal diametral hyperplanes and axes of the canonical non-degenerate hyperquadrics of Theorem 5.48. These assertions are valid in general thanks to Remark 5.53. In Chap. 10 we will come back to this subject.

Corollary 5.54 *Principal diametral hyperplanes and axes of the canonical non-degenerate hyperquadrics of* Theorem 5.48 *satisfy the following properties:*

1. *The non-degenerate hyperquadrics with centre* $\Gamma^{-1}_{p,n;d_1,\ldots,d_n}$ *and* $\Gamma^{1}_{p,n;d_1,\ldots,d_n}$ *have* n *pairwise orthogonal principal diametral hyperplanes* $x_i = 0$, $i = 1, \ldots, n$ *and* n *axes, each of them being the intersection of* $n-1$ *principal diametral hyperplanes. Furthermore, their centres are the intersections of their axes, i.e. the intersection of* n *pairwise orthogonal principal hyperplanes.*
2. *Every hyperplane or line passing through the centre* $(0, \ldots, 0)$ *of the sphere* $\Gamma^{1}_{n,n;d_1} : d_1x_1^2 + \cdots + d_1x_n^2 = 1$ *is a principal diametral hyperplane or an axis respectively.*
3. *The paraboloid* $\Gamma^{2}_{p,n-1;d_1,\ldots,d_{n-1}}$ *has* $n-1$ *pairwise orthogonal principal diametral hyperplanes* $x_i = 0$, $i = 1, \ldots, n-1$ *and only one axis, which is the intersection of* $n-1$ *pairwise orthogonal principal diametral hyperplanes.*
4. *The non-degenerate hyperquadrics are symmetric with respect to their axes.*

Table 5.2 Real affine conics: see Table 5.1

	rank(C_F)	rank($\overline{C}_F$)	Sign(C_F)	Sign($\overline{C}_F$)
Ellipse: $\Gamma^1_{2,2}$	2	3	(2, 0)	(2, 1)
Imaginary ellipse: $\Gamma^{-1}_{2,2}$	2	3	(2, 0)	(3, 0)
Hyperbola: $\Gamma^{-1}_{1,2}$	2	3	(1, 1)	(2, 1)
Parabola: $\Gamma^2_{1,1}$	1	3	(1, 0)	(2, 1)
Pair of complex-conjugate lines: $\Gamma^0_{2,2}$	2	2	(2, 0)	(2, 0)
Pair of incident real lines: $\Gamma^0_{1,2}$	2	2	(1, 1)	(1, 1)
Pair of parallel real lines: $\Gamma^1_{1,1}$	1	2	(1, 0)	(1, 1)
Pair of parallel complex-conjugate lines: $\Gamma^{-1}_{1,1}$	1	2	(1, 0)	(2, 0)
Double line: $\Gamma^0_{1,1}$	1	1	(1, 0)	(1, 0)

5.3 Real Conics

Ellipse, Hyperbola and Parabola
See Table 5.2 (classification of affine real conics).

Definition 5.55 The ellipse $\Gamma^1_{2,2;d_1,d_2}$ of equation $d_1x_1^2 + d_2x_2^2 = 1$ $(0 < d_1 \leq d_2)$ is usually put in the form

$$\Gamma_{a,b}:\ \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad \text{where } a = \frac{1}{\sqrt{d_1}} \geq b = \frac{1}{\sqrt{d_2}}, \quad x = x_1,\ y = x_2. \tag{5.65}$$

If $a = b$ the ellipse $\Gamma_{a,b}$ is the circle $S^1(O, \sqrt{a})$: $x^2 + y^2 = a$ of centre the origin O and radius $\sqrt{a}$. To the ellipse $\Gamma_{a,b}$ with $a > b$ we associate:

1. The centre (of symmetry) O and the axes (of symmetry) $y = 0$ and $x = 0$.
2. The *foci* $F_1 = (c, 0)$ and $F_2 = (-c, 0)$ with $c = \sqrt{a^2 - b^2}$.
3. The *directrices* $r_2:\ x = -\dfrac{a^2}{c}$ and $r_1:\ x = \dfrac{a^2}{c}$.

Foci and directrices of $\Gamma_{a,b}$ have the following geometrical characterizations whose easy proofs are left to the reader.

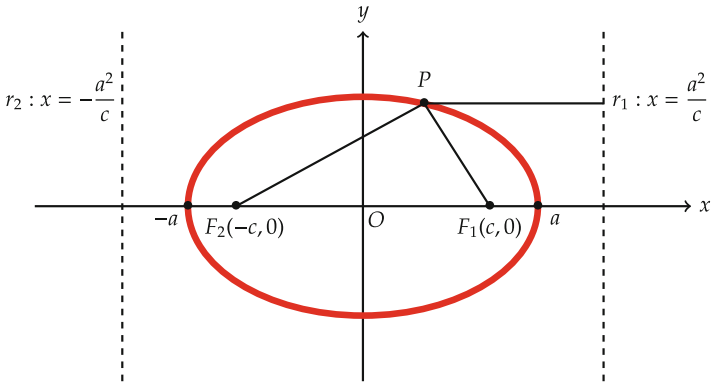


Fig. 5.1 Ellipse: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

Proposition 5.56 *The geometric locus of* $\Gamma_{a,b}$ *can be characterized by each of the following relations:*

$$\begin{aligned}\mathcal{V}(\widehat{\Gamma_{a,b}}) &= \{P \in \mathbb{E}^2(\mathbb{R}) : d(P, F_1) + d(P, F_2) = 2a\},\\ &= \left\{P \in \mathbb{E}^2(\mathbb{R}) : \frac{d(P, F_1)}{d(P, r_1)} = \frac{c}{a}\right\},\\ &= \left\{P \in \mathbb{E}^2(\mathbb{R}) : \frac{d(P, F_2)}{d(P, r_2)} = \frac{c}{a}\right\},\end{aligned}$$

where $c = \sqrt{a^2 - b^2}$ *(Fig. 5.1).*

Definition 5.57 We put the hyperbola $\Gamma^{-1}_{1,2;d_1,d_2}$ of equation $d_1x_1^2 - d_2x_2^2 = -1$ in the more usual form (interchanging the coordinates x_1 and x_2)

$$\Delta_{a,b} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad \text{where } a = \frac{1}{\sqrt{d_1}},\ b = \frac{1}{\sqrt{d_2}},\ x = x_1,\ y = x_2. \tag{5.66}$$

To the hyperbola $\Delta_{a,b}$ we associate:

1. The centre (of symmetry) O and the axes (of symmetry) $y = 0$ and $x = 0$.
2. The *foci* $F_1 = (c, 0)$ and $F_2 = (-c, 0)$ with $c = \sqrt{a^2 + b^2}$.
3. The *directrices* $r_2 : x = -\dfrac{a^2}{c}$ and $r_1 : x = \dfrac{a^2}{c}$.
4. The *asymptotes* $l_2 : \dfrac{x}{a} + \dfrac{y}{b} = 0$ and $l_1 : \dfrac{x}{a} - \dfrac{y}{b} = 0$.

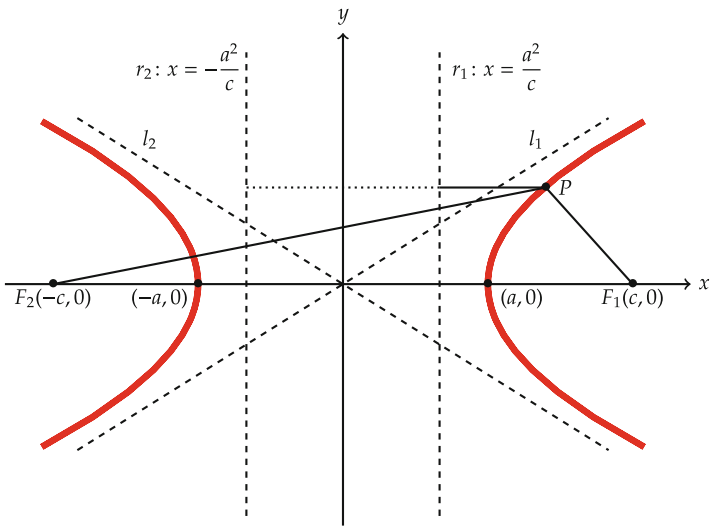


Fig. 5.2 Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

We let the reader prove the following geometrical characterizations:

Proposition 5.58 *The geometric locus of $\Delta_{a,b}$ can be characterized by each of the following relations:*

$$\begin{aligned}\mathcal{V}(\widehat{\Delta_{a,b}}) &= \{P \in \mathbb{E}^2(\mathbb{R}) : |d(P, F_1) - d(P, F_2)| = 2a\},\\ &= \left\{P \in \mathbb{E}^2(\mathbb{R}) : \frac{d(P, F_1)}{d(P, r_1)} = \frac{c}{a}\right\},\\ &= \left\{P \in \mathbb{E}^2(\mathbb{R}) : \frac{d(P, F_2)}{d(P, r_2)} = \frac{c}{a}\right\},\end{aligned}$$

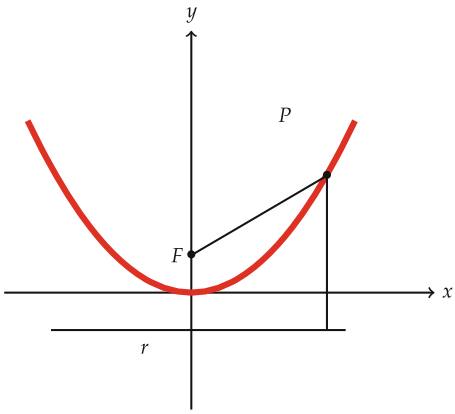
where $c = \sqrt{a^2 + b^2}$ (Fig. 5.2).

Definition 5.59 The parabola $\Gamma^2_{2,1:d_1} : d_1 x_1^2 = 2x_2$, usually written in the form

$$\Lambda_p : \ y = 2px^2, \tag{5.67}$$

where $p = \frac{d_1}{4}$, $x = x_1$ and $y = x_2$, has no centre. The point $F = \left(0, \frac{1}{8p}\right)$ is called *the focus* of Λ_p. and the line $r : \ y = -\frac{1}{8p}$ the *directix* of Λ_p.

Fig. 5.3 Parabola $y = 2px^2$



Proposition 5.60 *The geometric locus of Λ_p is the set of the points $P \in \mathbb{E}^2(\mathbb{R})$ such that*

$$d(P, F) = d(P, d).$$

Remark 5.61 We have defined foci, directrices, asymptotes of a conic referring to the canonical forms (5.65), (5.66) and (5.67). We will see their intrinsic definitions by aid of projective geometry in Chap. 10 (Fig. 5.3).

Intersections Between a Non-degenerate Conic and an Algebraic Plane Curve In order to prove Theorem 5.63 we need a rational parametrization of the non-degenerate conics. We need a homogeneous parametrization for $\Gamma_{a,b}$ and $\Delta_{a,b}$

$$\Gamma_{a,b}: \begin{cases} x = \dfrac{a(u^2 - v^2)}{u^2 + v^2} \\ y = \dfrac{2buv}{u^2 + v^2}, \end{cases} \quad (u, v) \in \mathbb{R}^2 \setminus \{(0, 0)\}, \tag{5.68}$$

$$\Delta_{a,b}: \begin{cases} x = \dfrac{a(u^2 + v^2)}{u^2 - v^2} \\ y = \dfrac{2buv}{u^2 - v^2}, \end{cases} \quad (u, v) \in \mathbb{R}^2 \setminus \{(0, 0)\},\ u \neq \pm v, \tag{5.69}$$

while for Λ_p we simply have $t \to (t, 2pt^2)$, $t \in \mathbb{R}$.

The following theorem is a very simple case of the affine Bézout's theorem and it is an interesting consequence of the euclidean (or affine) classification of conics. We need the following preliminary remark.

Remark 5.62 Let $f(X, Y) = a_0X^n + a_1X^{n-1}Y + \cdots a_{n-1}XY^{n-1} + a_nY^n \in K[X, Y]$ be a homogeneous polynomial of degree n with coefficients in a field K.

Let

$$\mathcal{Z}(f) := \{(a,b) \in K \times K \setminus \{(0,0)\} : f(a,b) = 0\}.$$

We put on $\mathcal{Z}(f)$ the following equivalence relation: $(a,b) \sim (c,d)$ iff $a = kc$, $b = kd$ for some $k \in K^*$. We call the equivalence classes $\widehat{(a,b)} \in \mathcal{Z}(f)/\sim$ *zeros* of f.

The zeros of (x,y) of $f(X,Y)$ with $x \neq 0$ are in bijection with the zeros of the polynomial

$$g(Z) = a_0 + a_1 Z + \cdots + a_{n-1} Z^{n-1} + a_n Z^n \in K[Z].$$

If $f(0,1) = 0$ (i.e. $a_n = 0$), then $f(X,Y) = X(a_0 X^{n-1} + a_1 X^{n-2} Y + \cdots + a_{n-1} Y^{n-1})$. Therefore we can conclude that $f(X,Y)$ has at most n zeros.

Theorem 5.63 *Let $\widehat{F} \in \mathcal{H}_2(2,\mathbb{R})$ be a non-degenerate conic of $\mathbb{A}^2(\mathbb{R})$ and $\widehat{G} \in \mathcal{H}_d(2,\mathbb{R})$ an algebraic curve of degree d of $\mathbb{A}^2(\mathbb{R})$. Suppose that F does not divide G in $\mathbb{R}[X,Y]$, then*

$$\#\left(\mathcal{V}(\widehat{F}) \cap \mathcal{V}(\widehat{G})\right) \leq 2d.$$

In other words the number of solution of the algebraic system $\begin{cases} F(x,y) = 0 \\ G(x,y) = 0 \end{cases}$ *is* $\leq 2d$.

Proof If σ is an affine automorphism of $\mathbb{A}^2(\mathbb{R})$ by (5.12) we have

$$\mathcal{V}(\sigma \cdot \widehat{F}) \cap \mathcal{V}(\sigma \cdot \widehat{G}) = \sigma(\mathcal{V}(\widehat{F})) \cap \sigma(\mathcal{V}(\widehat{G})) = \sigma(\mathcal{V}(\widehat{F}) \cap \mathcal{V}(\widehat{G}))$$

so that we can suppose that $\widehat{F}$ is $\Gamma_{a,b}$, $\Delta_{a,b}$ or Λ_p (if $\widehat{F}$ is an imaginary ellipse then $\mathcal{V}(\widehat{F}) = \varnothing$ so that we have nothing to prove). We can restrict ourselves to the case of the ellipse $F = \Gamma_{a,b}$, the other cases being completely analogous (much easier if $\widehat{F}$ is Λ_p). We have to solve the algebraic system $F(x,y) = G(x,y) = 0$ in order to determine $\mathcal{V}(\widehat{F}) \cap \mathcal{V}(\widehat{G})$. Thanks to parametrization (5.68) we have to find the solutions $(u,v) \neq (0,0)$ of

$$G\left(\frac{a(u^2 - v^2)}{u^2 + v^2}, \frac{2buv}{u^2 + v^2}\right) = 0. \tag{5.70}$$

First we prove that the function $G\left(\frac{a(u^2 - v^2)}{u^2 + v^2}, \frac{2buv}{u^2 + v^2}\right)$ is not identically zero, otherwise F would divide G. Indeed, by Theorem 1.10 in $(\mathbb{R}[Y])[X]$ we write

$$G(X,Y) = F(X,Y)q(X,Y) + A(Y)X + B(Y), \tag{5.71}$$

where $A(Y)$, $B(Y) \in \mathbb{R}[Y]$. We have to prove that $A(Y)$ and $B(Y)$ are null polynomials. Let

$$A(Y) = a_0 + a_1 Y + \cdots + a_n Y^n, \qquad \text{and} \qquad B(Y) = b_0 + b_1 Y + \cdots + b_n Y^n, \tag{5.72}$$

with $n = \max\{\deg(A(Y)), \deg(B(Y))\}$. If $G\left(\frac{a(u^2 - v^2)}{u^2 + v^2}, \frac{2buv}{u^2 + v^2}\right)$ is identically zero, since $F\left(\frac{a(u^2 - v^2)}{u^2 + v^2}, \frac{2buv}{u^2 + v^2}\right)$ is identically zero, it follows that

$$A\left(\frac{2buv}{u^2 + v^2}\right)\frac{a(u^2 - v^2)}{u^2 + v^2} + B\left(\frac{2buv}{u^2 + v^2}\right) = 0, \qquad \forall\, (u, v) \in \mathbb{R}^2 \setminus \{(0, 0)\}. \tag{5.73}$$

Taking $u = 0$ we have $-aa_0 + b_0 = 0$. Furthermore, from (5.72) and (5.73) we get

$$\begin{aligned} aa_0(u^2 + v^2)^n(u^2 - v^2) + 2aba_1uv(u^2 + v^2)^{n-1}(u^2 - v^2) \\ + \cdots + 2^n a_n a b^n u^n v^n (u^2 - v^2) \\ + b_0(u^2 + v^2)^{n+1} + 2bb_1uv(u^2 + v^2)^n \\ + \cdots + 2b_n b^n u^n v^n (u^2 + v^2) = 0 \quad \forall\, (u, v) \in \mathbb{R}^2 \setminus \{(0, 0)\} \end{aligned}$$

so that the coefficient of u^{2n+2} is $aa_0 + b_0 = 0$. Then the equalities $aa_0 + b_0 = aa_0 - b_0 = 0$ imply $a_0 = b_0 = 0$, i.e. $A(Y) = YA'(Y)$ and $B(Y) = YB'(Y)$, with $A'(Y)$, $B'(Y) \in \mathbb{R}[Y]$. Hence the remainder $a(Y)X + b(Y)$ of (5.71) can be rewritten as $Y(a'(Y)X + b'(Y))$. Since $Y = \frac{2buv}{u^2 + v^2}$ is not a null function we have

$$A'\left(\frac{2buv}{u^2 + v^2}\right)\frac{a(v^2 - u^2)}{u^2 + v^2} + B'\left(\frac{2buv}{u^2 + v^2}\right) = 0, \qquad \forall\, (u, v) \in \mathbb{R}^2 \setminus \{(0, 0)\}.$$

Iterating our argument we get $A(Y) = B(Y) = 0$, i.e. $F \mid G$.

Let $G(X, Y) = F_d(X, Y) + F_{d-1}(X, Y) + \cdots + F_0(X, Y)$, where $F_i(X, Y)$ is a homogeneous polynomial of degree i, $i = 0, \ldots, d$, and $F_d(X, Y) \neq 0$. Hence we can write (5.70) in the form

$$\begin{aligned} \frac{1}{(u^2 + v^2)^d} F_d(a(u^2 - v^2), 2buv) + \frac{1}{(u^2 + v^2)^{d-1}} F_{d-1}(a(u^2 - v^2), 2buv) \\ + \cdots + F_0 = 0, \end{aligned}$$

or, equivalently,

$$F_d(a(u^2-v^2),2buv)+(u^2+v^2)F_{d-1}(a(u^2-v^2),2buv)+\cdots+(u^2+v^2)^dF_0=0.$$

The left– hand side is a homogeneous polynomial of $\mathbb{R}[u,v]$ whose degree is $\leq 2d$ so that it has at most $2d$ roots since $G\left(\dfrac{a(u^2-v^2)}{u^2+v^2},\dfrac{2buv}{u^2+v^2}\right)$ is not the null rational function. □

Now we wish to prove the following classical result by means of Theorem 5.63.

Theorem 5.64 (Affine Pascal's Theorem) *Let $\widehat{F}\in\mathcal{H}_2(2,\mathbb{R})$ be a non-degenerate conic of $\mathbb{A}^2(\mathbb{R})$. Let $P_1\dots P_6$ be six points of $\mathcal{V}(\widehat{F})$* (an hexagon inscribed in the conic). *Assume that*

$$P_1P_2\not\parallel P_4P_5,\quad P_3P_4\not\parallel P_1P_6,\quad P_2P_3\not\parallel P_5P_6,$$

then the points $A=P_1P_2\cap P_4P_5$, $B=P_3P_4\cap P_1P_6$ and $C=P_2P_3\cap P_5P_6$ are collinear (*see Fig. 5.4*).

Proof *(Plücker)* Let $\widehat{L_{ij}}$ be the line passing through P_i and P_j. Consider the polynomial

$$F_\lambda:=L_{12}L_{34}L_{56}+\lambda L_{16}L_{23}L_{45},\qquad \lambda\in\mathbb{R}.$$

For every $\lambda\in\mathbb{R}$, F_λ is a non-zero-polynomial of degree ≤ 3. Indeed, the ring $\mathbb{R}[X,Y]$ is factorial so that $F_\lambda\neq 0$ by our hypothesis on the lines $\widehat{L_{ij}}$. We observe that $A,B,C\in\mathcal{V}(\widehat{F_\lambda})$ for all $\lambda\in\mathbb{R}$ but $A,B,C\notin\mathcal{V}(\widehat{F})$, indeed by Theorem 5.63, a line and a non-degenerate conic have two points in common at most. Suppose that F_λ has degree ≤ 2. Since $\{P_1,\dots,P_6\}\subseteq\mathcal{V}(\widehat{F})\cap\mathcal{V}(\widehat{F_\lambda})$ for all λ, by Theorem 5.63, $F_\lambda\mid F$ contradicting what was said above. Hence F_λ has degree 3 for all $\lambda\in\mathbb{R}$.

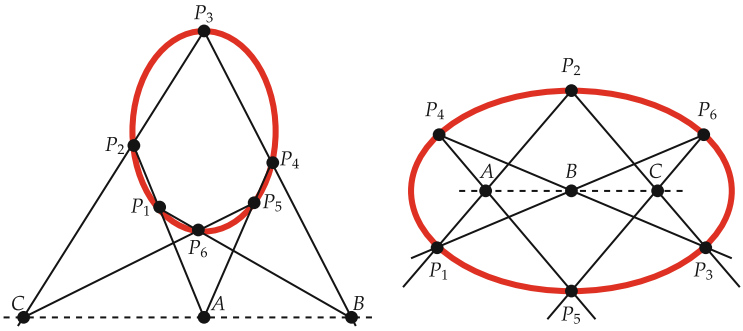


Fig. 5.4 Affine Pascal's theorem

Table 5.3 Canonical forms of real conics: Theorems 5.40 and 5.48

Non-degenerate conics	*Affine equation*	*Euclidean equations*
Ellipse	$x_1^2 + x_2^2 = 1$	$d_1x_1^2 + d_2x_2^2 = 1, \ \ 0 < d_1 \leq d_2$
Imaginary ellipse	$x_1^2 + x_2^2 = -1$	$d_1x_1^2 + d_2x_2^2 = -1, \ \ 0 < d_1 \leq d_2$
Hyperbola	$x_1^2 - x_2^2 = -1$	$d_1x_1^2 - d_2x_2^2 = -1, \ \ d_1, d_2 > 0$
Parabola	$x_1^2 = 2x_2$	$d_1x_1^2 = 2x_2, \ \ 0 < d_1$
Degenerate conics	*Affine equation*	*Euclidean equations*
Pair of complex-conjugate lines	$x_1^2 + x_2^2 = 0$	$x_1^2 + d_2x_2^2 = 0, \ \ 0 < 1 \leq d_2$
Pair of incident real lines	$x_1^2 - x_2^2 = 0$	$x_1^2 - d_2x_2^2 = 0, \ \ 0 < d_2$
Pair of parallel real lines	$x_1^2 = 1$	$d_1x_1^2 = 1, \ \ 0 < d_1$
Pair of parallel complex-conjugate lines	$x_1^2 = -1$	$d_1x_1^2 = -1, \ \ 0 < d_1$
Double line	$x_1^2 = 0$	$x_1^2 = 0$

Choose a point $P \in \mathcal{V}(\widehat{F})$ with $P \neq P_i$, $i = 1, \ldots, 6$. Determine λ such that $F_\lambda(P) = 0$. Since

$$\mathcal{V}(\widehat{F}) \cap \mathcal{V}(\widehat{F_\lambda}) \supseteq \{P_1, \ldots, P_6, P\},$$

by Theorem 5.63 we have $F \mid F_\lambda$, i.e. $F_\lambda = F \cdot L$, where L is a polynomial of degree 1. As noted above

$$A, B, C \in \mathcal{V}(\widehat{F_\lambda}) \setminus \mathcal{V}(\widehat{F}),$$

so that A, B and C belong to $\mathcal{V}(\widehat{L})$. □

Corollary 5.65 *There is at least one conic passing through five points $P_1, \ldots, P_5$ of $\mathbb{A}^2(\mathbb{R})$. Furthermore, if no four of $P_1, \ldots, P_5$ lie on a line, then the conic which contains them is unique and if no three lie on a line the conic is necessarily non-degenerate.*

Proof It follows from Table 5.3 and from Bézout's theorem. □

Corollary 5.66 *There exists one and only one pencil of conics passing through four non-collinear points of $\mathbb{A}^2(\mathbb{R})$.*

Proof It follows from Corollary 5.65. □

5.4 Real Quadrics

Ellipsoids, Hyperboloids and Paraboloids
We need a preliminary definition (Table 5.4).

Table 5.4 Real affine quadrics: see Table 5.1

Non-degenerate quadrics	$\text{rank}(C_F)$	$\text{rank}(\overline{C}_F)$	$\text{Sign}(C_F)$	$\text{Sign}(\overline{C}_F)$
Ellipsoid: $\Gamma^1_{3,3}$	3	4	(3, 0)	(3, 1)
Imaginary ellipsoid: $\Gamma^{-1}_{3,3}$	3	4	(3, 0)	(4, 0)
One-sheet hyperboloid: $\Gamma^1_{2,3}$	3	4	(2, 1)	(2, 2)
Two-sheet hyperboloid: $\Gamma^{-1}_{2,3}$	3	4	(2, 1)	(3, 1)
Hyperbolic paraboloid: $\Gamma^2_{1,3}$	2	4	(1, 1)	(2, 2)
Elliptic paraboloid: $\Gamma^2_{2,3}$	2	4	(2, 0)	(3, 1)
Simply degenerate quadrics	$\text{rank}(C_F)$	$\text{rank}(\overline{C}_F)$	$\text{Sign}(C_F)$	$\text{Sign}(\overline{C}_F)$
Cone: $\Gamma^0_{2,3}$	3	3	(2, 1)	(2, 1)
Imaginary cone: $\Gamma^0_{3,3}$	3	3	(3, 0)	(3, 0)
Elliptic cylinder: $\Gamma^1_{2,2}$	2	3	(2, 0)	(2.1)
Hyperbolic cylinder: $\Gamma^{-1}_{1,2}$	2	3	(1, 1)	(2, 1)
Parabolic cylinder: $\Gamma^2_{1,1}$	1	3	(1, 0)	(2, 1)
Imaginary cylinder: $\Gamma^{-1}_{2,2}$	2	3	(2, 0)	(3, 0)
Reducible quadrics	$\text{rank}(C_F)$	$\text{rank}(\overline{C}_F)$	$\text{Sign}(C_F)$	$\text{Sign}(\overline{C}_F)$
2 distinct incident planes: $\Gamma^0_{1,2}$	2	2	(1, 1)	(1, 1)
2 complex-conjugate planes: $\Gamma^0_{2,2}$	2	2	(2, 0)	(2, 0)
2 distinct parallel planes: $\Gamma^1_{1,1}$	1	2	(1, 0)	(1, 1)
2 parallel complex-conjugate planes: $\Gamma^{-1}_{1,1}$	1	2	(1, 0)	(2, 0)
Double plane:	1	1	(1, 0	(1, 0)

Definition 5.67 A *ruled surface* of $\mathbb{A}^3(K)$ is defined by the property that through every point in the surface, there is at least one straight line which also lies on the surface. The lines are called the *generators* of the surface. Examples of ruled surfaces are the cones and the cylinders (Definition 5.14).

We obtain for non-degenerate quadrics their classical forms taking $a = \dfrac{1}{\sqrt{d_1}}$, $b = \dfrac{1}{\sqrt{d_2}}$, $c = \dfrac{1}{\sqrt{d_3}}$, ($x = x_1$, $y = x_2$ and $z = x_3$). For real ellipsoid $\widehat{F_{re}}$ and imaginary ellipsoid $\widehat{F_{ie}}$ we have

$$\widehat{F_{re}} : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \qquad \widehat{F_{ie}} : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1 \tag{5.74}$$

The principal diametral planes of $\widehat{F_{re}}$ are $x = 0$, $y = 0$, $z = 0$ and its axes are $x = y = 0$, $y = z = 0$, $x = z = 0$ while the origin is the centre.

The canonical equations of one-sheet hyperboloid $\widehat{F_{\mathrm{i}1}}$ and two-sheet hyperboloid $\widehat{F_{\mathrm{i}2}}$ are

$$\widehat{F_{\mathrm{i}1}} : \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1, \qquad \widehat{F_{\mathrm{i}2}} : \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1. \tag{5.75}$$

They have the same principal diametral planes, axes and centres as $\widehat{F_{\mathrm{re}}}$. Furthermore, one-sheet hyperboloids $\widehat{F_{\mathrm{i}1}}$ is a ruled surface covered by the lines:

$$r_t : \begin{cases} \dfrac{x}{a} - \dfrac{z}{c} & = t\left(1 - \dfrac{y}{b}\right) \\ t\left(\dfrac{x}{a} + \dfrac{z}{c}\right) & = 1 + \dfrac{y}{b}, \end{cases} \quad t \in \mathbb{R},$$

$$r_u : \begin{cases} \dfrac{x}{a} + \dfrac{z}{c} & = u\left(1 - \dfrac{y}{b}\right) \\ u\left(\dfrac{x}{a} - \dfrac{z}{c}\right) & = 1 + \dfrac{y}{b}, \end{cases} \quad u \in \mathbb{R}, \tag{5.76}$$

and

$$r_{t=\infty} : \begin{cases} \dfrac{x}{a} + \dfrac{z}{c} & = 0 \\ 1 - \dfrac{y}{b} & = 0, \end{cases} \qquad r_{u=\infty} : \begin{cases} \dfrac{x}{a} - \dfrac{z}{c} & = 0 \\ 1 - \dfrac{y}{b} & = 0. \end{cases} \tag{5.77}$$

The sets $\{r_t \mid t \in \mathbb{R} \cup \{\infty\}\}$ and $\{r_u \mid u \in \mathbb{R} \cup \{\infty\}\}$ are called the *rulings* of $\widehat{F_{\mathrm{i}1}}$.

Elliptic and hyperbolic parabolids take the forms

$$\widehat{F_{\mathrm{e}p}} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 2z, \qquad \widehat{F_{\mathrm{h}p}} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z. \tag{5.78}$$

Their principal diametral planes are $x = 0$, $y = 0$ and $x = y = 0$ is the axis of symmetry. They don't have any centre.

The hyperbolic paraboloid $\widehat{F_{\mathrm{h}p}}$ is a ruled surface covered by the lines

$$r'_t : \begin{cases} \dfrac{x}{a} - \dfrac{y}{b} & = t \\ t\left(\dfrac{x}{a} + \dfrac{y}{b}\right) & = 2z, \end{cases} \quad t \in \mathbb{R}, \qquad r'_u : \begin{cases} \dfrac{x}{a} + \dfrac{y}{b} & = u \\ u\left(\dfrac{x}{a} - \dfrac{y}{b}\right) & = 2z, \end{cases} \quad u \in \mathbb{R}. \tag{5.79}$$

The sets $\{r_t \mid t \in \mathbb{R}\}$ and $\{r_u \mid u \in \mathbb{R}\}$ are the rulings of $\widehat{F_{\mathrm{h}p}}$.

The only ruled non-degenerate quadrics are the one-sheet hyperboloid and the hyperbolic paraboloid. All the degenerate quadrics (with non-empty geometric

locus) are obviously ruled. The reader can easily prove the following properties of the rulings of a ruled non-degenerate quadric $\widehat{F}$:

(i) Two lines belonging to the same ruling are skew.
(ii) Every line of a ruling meets every line of the other ruling.
(iii) For every point $P \in \mathcal{V}(\widehat{F})$ there is one and only one line of both the rulings passing through P.
(iv) For every point $P \in \mathcal{V}(\widehat{F})$ the tangent plane to $\widehat{F}$ at P cuts $\mathcal{V}(\widehat{F})$ along the lines of the two rulings passing through P.

Thanks to Proposition 5.29 and Table 5.2 we have the following

Definition 5.68 Let $\widehat{F}$ be an irreducible quadric of $\mathbb{A}^3(\mathbb{R})$. A non-singular point $P \in \mathcal{V}(\widehat{F})$ is:

1. *elliptic* if $\mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$ is a pair of complex-conjugate lines,
2. *hyperbolic* if $\mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$ is a pair of distinct real lines,
3. *parabolic* if $\mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$ is a double line.

Proposition 5.69 *Let $\widehat{F}$ be an irreducible quadric of $\mathbb{A}^3(\mathbb{R})$. The regular points of $\widehat{F}$ are all:*

(i) *elliptic if and only if $\widehat{F}$ is an ellipsoid, a two-sheet hyperboloid or an elliptic paraboloid,*
(ii) *hyperbolic if and only if $\widehat{F}$ is a one-sheet hyperboloid or a hyperbolic paraboloid,*
(iii) *parabolic if and only if $\widehat{F}$ is a cone or a cylinder.*

Proof If $\widehat{F}$ is non-degenerate, $\gamma := \mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$ is a degenerate conic of rank 2 so that it is a pair of distinct real lines or two complex-conjugate lines by Proposition 5.29. If $\widehat{F}$ does not contain lines, then every point is elliptic. Suppose that $\widehat{F}$ contains a line r, then for every point $P \in r$, the line r is contained in $\mathrm{T}_{P,\widehat{F}}$ and hence also in γ, which is consequently the union of two lines through P. If instead $P \in \widehat{F} \setminus r$, the plane joining r and P intersects $\widehat{F}$ in another line s passing through P so that P is hyperbolic.

If $\widehat{F}$ is a cone or a cylinder, one and only one line passes through a non-singular points so that $\widehat{F}$ has only parabolic points. Conversely, if $\widehat{F}$ has a parabolic point P, then $r = \mathrm{T}_{P,\widehat{F}} \cap \widehat{F}$ is a double line passing through P, then $\widehat{F}$ must be simply degenerate so that it is a cone or a cylinder. □

Quadrics of Revolution

Definition 5.70 Let $\widehat{F}$ be a non-degenerate or simply degenerate quadric of $\mathbb{E}^3(\mathbb{R})$ such that $\#(\mathcal{V}(\widehat{F})) > 1$. If there exists a line r such that $\varphi(\mathcal{V}(\widehat{F})) = \mathcal{V}(\widehat{F})$ for all rotations φ about r we say that $\widehat{F}$ is a *quadric of revolution about r* and we call r an *axis of revolution*. Every plane orthogonal to an axis of revolution of $\widehat{F}$ cuts $\widehat{F}$ in a circle (possibly reduced to a point) or in the empty set. From Table 5.5 we see that the only possible quadrics of revolution are the ellipsoids, the hyperboloids (one-sheet and two-sheet), the ellipitic paraboloids, the cones and the cylinders. Indeed a

Table 5.5 Real euclidean quadrics: see Theorem 5.48

Non-degenerate quadrics	*Affine equation*	*Euclidean equations*
Ellipsoid	$x_1^2 + x_2^2 + x_3^2 = 1$	$d_1x_1^2 + d_2x_2^2 + d_3x_3^2 = 1,\ 0 < d_1 \leq d_2 \leq d_3$
Imaginary ellipsoid	$x_1^2 + x_2^2 + x_3^2 = -1$	$d_1x_1^2 + d_2x_2^2 + d_3x_3^2 = -1,\ 0 < d_1 \leq d_2 \leq d_3$
One-sheet hyperboloid	$x_1^2 + x_2^2 - x_3^2 = 1$	$d_1x_1^2 + d_2x_2^2 - d_3x_3^2 = 1,\ 0 < d_1 \leq d_2, 0 < d_3$
Two-sheet hyperboloid	$x_1^2 + x_2^2 - x_3^2 = -1$	$d_1x_1^2 + d_2x_2^2 - d_3x_3^2 = -1,\ 0 < d_1 \leq d_2, 0 < d_3$
Hyperbolic paraboloid	$x_1^2 - x_2^2 = 2x_3$	$d_1x_1^2 - d_2x_2^2 = 2x_3,\ 0 < d_1, 0 < d_2$
Elliptic paraboloid	$x_1^2 + x_2^2 = 2x_3$	$d_1x_1^2 + d_2x_2^2 = 2x_3,\ 0 < d_1 \leq d_2$
Simply degenerate quadrics	*Affine equation*	*Euclidean equations*
Cone	$x_1^2 + x_2^2 - x_3^2 = 0$	$x_1^2 + d_2x_2^2 - d_3x_3^2 = 0,\ 1 \leq d_2, d_3 > 0$
Imaginary cone	$x_1^2 + x_2^2 + x_3^2 = 0$	$x_1^2 + d_2x_2^2 + d_3x_3^2 = 0,\ 1 \leq d_2 \leq d_3$
Elliptic cylinder	$x_1^2 + x_2^2 = 1$	$d_1x_1^2 + d_2x_2^2 = 1,\ 0 < d_1 \leq d_2$
Hyperbolic cylinder	$x_1^2 - x_2^2 = -1$	$d_1x_1^2 - d_2x_2^2 = -1,\ 0 < d_1, 0 < d_2$
Parabolic cylinder	$x_1^2 = 2x_3$	$d_1x_1^2 = 2x_3,\ d_1 > 0$
Imaginary cylinder	$x_1^2 + x_2^2 = -1$	$d_1x_1^2 + d_2x_2^2 = -1,\ 0 < d_1 \leq d_2$
Reducible quadrics	*Affine equation*	*Euclidean equations*
2 distinct incident planes	$x_1^2 - x_2^2 = 0$	$x_1^2 - d_2x_2^2 = 0,\ d_2 > 0$
2 complex-conjugate planes	$x_1^2 + x_2^2 = 0$	$x_1^2 + d_2x_2^2 = 0,\ 1 \leq d_2$
2 parallel planes	$x_1^2 = 1$	$d_1x_1^2 = 1,\ d_1 > 0$
2 parallel complex-conjugate planes	$x_1^2 = -1$	$d_1x_1^2 = -1,\ d_1 > 0$
Double plane	$x_1^2 = 0$	$x_1^2 = 0$

non-degenerate or simply degenerate real quadric $\widehat{F}$ of $\mathbb{E}^3(\mathbb{R})$ such that $\#(\mathcal{V}(\widehat{F})) > 1$ is of revolution if and only if its matrix C_F has a double non-zero eigenvalue. If the eigenvalues of C_F are triple then $\widehat{F}$ is a sphere, which has every line through its centre as an axis of revolution. In the remaining cases there is only one axis of revolution (*the axis of revolution*) whose director subspace is generated by an eigenvector of the simple eigenvalue. Every axis of revolution is an axis of symmetry of $\widehat{F}$ and every plane containing the axis of revolution is a principal diametral plane of $\widehat{F}$. Canonical forms of quadrics of revolution are:

- $d_1x_1^2 + d_2x_2^2 + d_3x_3^2 = 1$, with $0 < d_1 = d_2 < d_3$: ellipsoid of revolution with axis the line $x_1 = x_2 = 0$;
- $d_1x_1^2 + d_2x_2^2 - d_3x_3^2 = \pm 1$, with $d_1 = d_2 > 0$, $d_1 > 0$: hyperboloid (one-sheet and two-sheet), with axis the line $x_1 = x_2 = 0$;
- $d_1x_1^2 + d_2x_2^2 = 2x_3$, with $d_1 = d_2 > 0$: elliptic paraboloid with axis the line $x_1 = x_2 = 0$;
- $d_1x_1^2 + d_2x_2^2 = 1$, with $d_1 = d_2 > 0$: cylinder with axis the line $x_1 = x_2 = 0$;
- $x_1^2 + d_2x_2^2 - d_3x_3^2 = 0$, with $d_2 = 1$: cone with axis the line $x_1 = x_2 = 0$.

5.5 Exercises

Exercise 5.1 Find the multiplicity of $O = (0, 0)$ and the tangent cone for the following real plane curves:

$$
\begin{array}{lll}
x + 2x^2 + y^3 = 0; & 2x + 2x^2 + y^4 = 0; & x - y = y^3; \\
3xy = x^3 + y^3; & xy - y^2 = x^3; & xy = x^4 + xy^3 + y^3; \\
2xy + x^4 + y^4 = 0; & 8xy = (x^2 + y^2)^2; & x^2 - y^3 = 0; \\
x^2 - yx^2 + y^4 = 0; & xy^2 = x^4 + y^4. &
\end{array}
$$

Exercise 5.2 Find the multiplicity of $O = (0, 0, 0)$ and the tangent cone for the following urfaces of $\mathbb{A}^3(\mathbb{R})$:

$$
\begin{array}{lll}
z = xy + x^3; & xy - z^2 + x^3 - y^3 = 0; & xy + z^3 = 0; \\
y^2 + x^3 + z^2 = 0; & xyz + x^3 + y^3 = 0; & xyz = x^2y^2 + y^2z^2 + z^2x^2; \\
x^3 + xy^2 + z^3 = 0; & x^2 + xy^2 + z^3 = 0; & z(x^2 + y^2) = xy; \\
z^2 = x(x^2 + y^2); & x + y^2 + z^3 = 0; & xyz = (x + y + z)^3; \\
xyz = 1. & &
\end{array}
$$

Exercise 5.3 ([12]) Let $\widehat{F}$ be a hypersurface of $\mathbb{A}^n(K)$, $P \in \widehat{F}$ and $\widehat{H}$ a hyperplane passing through P and not contained in $\widehat{F}$. Prove the following assertions:

(a) $\mathfrak{m}_P(\widehat{F} \cap \widehat{H}) \geq \mathfrak{m}_P(\widehat{F})$; in particular, if P is a singular point of $\widehat{F}$, then it is a singular point for $\widehat{F} \cap \widehat{H}$.

(b) The hypersurface $\widehat{F} \cap \widehat{H}$ of $\widehat{H}$ is singular at P if and only if $\widehat{H} \subset T_P(\widehat{F})$.
(c) There exists a hyperplane $\widehat{H}$ such that $\mathfrak{m}_P(\widehat{F} \cap \widehat{H}) = \mathfrak{m}_P(\widehat{F})$.

Exercise 5.4 Let $F = 0$ and $G = 0$ be the equations of two non-parallel planes of $\mathbb{A}^3(K)$ and $\varphi \in K[u, v]$ be a polynomial in the indeterminates u and v. Show that $\varphi(F, G) = 0$ represents a cylinder whose generators are parallel to the line $A = B = 0$.

Exercise 5.5 A *conoid* is a ruled surface whose generators stay parallel to a plane called the *director plane* of the conoid, while intersecting a line L, called the *axis* of the conoid. Prove that a surface $\widehat{F}$ of $\mathbb{A}^3(K)$ is a conoid whose director plane is $A = 0$ and its axis is the line $B = C = 0$ if it can be written in the form $\varphi\left(A, \dfrac{B}{C}\right) = 0$, where $\varphi \in K[u, v]$ is a polynomial in the indeterminates u and v, and $A = 0$, $B = 0$ and $C = 0$ are the equations of three planes of $\mathbb{A}^3(K)$ not belonging to the same pencil.

Exercise 5.6 Consider the following surfaces:

$x - 2y + 3z - 2(3x + 2y - 1)^3 = 0$	$(x - 2y + z) - 3(x + y + 2z)^3 = (2x - y + 3z)^2$
$2x^3 - 3y^3 + z^3 + x^2z = 0$	$(x - 1)(y - 2)^2 + 3(z - 1)^3 = 0$
$(x - y)^2 + 2(y - z)^3 - 3(z - x)^5 = 0$	$y = 3x(z - x)^3$
$(x - 2y + z)^2 = (x^3 - x + 1)(x + y + z)^2$	$x^2z = y^2$ (Whitney's umbrella).

Determine which are cones, cylinders or conoids and find their vertices, director planes and axes.

Exercise 5.7 Let $F(x, y, z) = xy^2 - z^2 \in \mathbb{R}[x, y, z]$.

(a) Find the singular points of $\widehat{F}$.
(b) Determine the lines contained in $\mathcal{V}(\widehat{F})$.
(c) Say whether $\widehat{F}$ is irreducible.

Exercise 5.8 Consider the non-degenerate hyperquadric $\Gamma^1_{p,n;d_1,\ldots,d_n}$ of $\mathbb{E}^n(\mathbb{R})$ (see Theorem 5.48). Prove that

$$\min\left\{d(O, P)^2 \mid P \in \mathcal{V}(\Gamma^1_{p,n;d_1,\ldots,d_n})\right\} = \frac{1}{d_p},$$

where $d(O, P)$ is the euclidean distance of P from the origin $O = (0, \ldots, 0)$, and determine the point of minimum.

Exercise 5.9 Let $S^{n-1}(A, R)$ be the $(n-1)$-dimensional sphere of $\mathbb{E}^n(\mathbb{R})$ having centre $A = (a_1, \ldots, a_n)$ and radius R whose equation is $F = \sum_{i=0}^{n}(x_i - a_i)^2 - R^2$ (see Definition 4.21). Let d be a line passing through a fixed point $B = (b_1, \ldots, b_n)$ and intersecting $S^{n-1}(A, R)$ at B_1 and B_2 (where we may have $B_1 = B_2$). The real number $F(b_1, \ldots, b_n)$ is called the *power of* B with respect to $S^{n-1}(A, R)$ and denoted by $P_B(S^{n-1}(A, R))$. Prove that:

1. $P_B(S^{n-1}(A, R)) = \langle \overrightarrow{BB_1}, \overrightarrow{BB_2} \rangle$,
2. $d(B, B_1) \cdot d(B, B_2) = |P_B(S^{n-1}(A, R))|$.

In particular, $P_B(S^{n-1}(A, R))$ depends only on B and not on the line d.

Exercise 5.10 Let $S^{n-1}(A, R)$ and $S^{n-1}(A', R')$ be two eccentric spheres (i.e. $A \neq A'$). Show that the set

$$\{B \in \mathbb{E}^n(\mathbb{R}) : \ P_B(S^{n-1}(A, R)) = P_B(S^{n-1}(A', R'))\}$$

is a hyperplane orthogonal to the line AA', called the *radical hyperplane* (or *radical axis* if $n = 2$) of $S^{n-1}(A, R)$ and $S^{n-1}(A', R')$. If the two spheres are secant, the radical hyperplane is merely the hyperplane containing their intersection.

Prove that the geometric locus of the points of $\mathbb{E}^3(\mathbb{R})$ having the same power with respect three spheres, whose centres are not collinear, is a line perpendicular to the plane determined by the centres of the spheres.

Exercise 5.11 Determine the affine type of the quadrics of $\mathbb{E}^3(\mathbb{R})$

$$\begin{array}{ll}
xy - yz = 1, & 3z^2 + 2xy = 1, \\
x^2 + 4y^2 - 6z^2 - 2x + 8y - 2 = 0, & x^2 - 6xy = 2z + 3, \\
x^2 + y^2 + z^2 + z(3x - y - 1) = 0, & 3x^2 - xy + y^2 = z + 3, \\
xy + yz + xz = 1, & xy + yz + xz = -1, \\
2x^2 + y^2 - 2xy + 2x - z^2 = 0, & (x + y - 6z)^2 = 4z, \\
x^2 + y^2 + 3yz - 2x + 4y = 0, & z^2 - y^2 = 3, \\
x^2 - xy + y^2 - 2x = 1, & (2x - 1)^2 + z^2 - 3z = 0, \\
xy - xz + z^2 = 1, & 2x^2 - xy + xz - yz + z^2 - 3x + z = 0.
\end{array}$$

(a) Say which of the above quadrics are of revolution and find their axes of revolution.
(b) Say which of the above quadrics are ruled and describe their rulings.
(c) Find the euclidean canonical forms for some of the non-degenerate quadrics.
(d) Determine their possible centres, principal diametral planes and axes of symmetry.

Exercise 5.12 Determine the planes passing through the line $r : x = y = z + 2$ of $\mathbb{E}^3(\mathbb{R})$ and tangent to the sphere $x^2 + y^2 + z^2 - 2x = 0$.

Exercise 5.13 Determine the cone of vertex V *circumscribed* around a quadric $\widehat{F}$ of $\mathbb{E}^3(\mathbb{R})$, i.e. the locus of all lines passing through V and tangent to $\widehat{F}$. By a more geometric method determine the cone of vertex the origin $O = (0, 0, 0)$ circumscribed around the sphere of centre $C = (2, 3, 4)$ and radius 1.

Exercise 5.14 Prove that the asymptotes and the parallel lines to the asymptotes of a hyperbola Γ passing through a point $P \in \mathcal{V}(\Gamma)$ form a parallelogram of constant area.

Exercise 5.15 Let C_1 and C_2 be two tangent circles. Find the geometric locus of the centres of all circles tangent to C_1 and C_2.

Exercise 5.16 Find all parabolas whose axis is the line $y = 2x$ and passing through the point $A(1, 0)$.

Exercise 5.17 Determine the equilateral hyperbolas (i.e. with orthogonal asymptotes) having the line $y = 3x$ as an asymptote and passing through the point $(2, 0)$.

Exercise 5.18 Find the conic whose axes are the lines $x = y$ and $x = -y$, and having the line $y = 3x - 5$ as tangent at the point $(2, 1)$.

Exercise 5.19 Determine all the conics with centre at the point $(0, 1)$, tangent to the line $2x - y = 1$ at the point $(1, 1)$ and tangent to the line $x = 2y$.

Exercise 5.20 Let r_1 and r_2 be two perpendicular lines. Determine the locus

$$\{P \in \mathbb{E}^2(\mathbb{R}) : d(P, r_1)d(P, r_2) = k\},$$

where $k \in \mathbb{R}$ is fixed.

Exercise 5.21 Let Γ be a hyperbola with asymptotes the line r_1 and r_2. Show that the product $d(P, r_1)d(P, r_2)$ does not depend on $P \in \Gamma$.

Exercise 5.22 Let Γ_λ be the conic of $\mathbb{E}^2(\mathbb{R})$ having equation

$$(2 - \lambda)x^2 - 2\lambda xy + (2 - \lambda)y^2 - 8x - 8y + 8 = 0,$$

where $\lambda > 0$.

(i) Show that all conics Γ_λ have a common symmetry axis.
(ii) Determine the affine type of Γ_λ.
(iii) Find the euclidean canonical forms for Γ_λ with $\lambda = 1$.

Exercise 5.23 Let r and s be two skew lines of $\mathbb{E}^3(\mathbb{R})$. Show that the locus

$$\left\{P \in \mathbb{E}^3(\mathbb{R}) : \frac{d(P, r)}{d(P, s)} = \lambda > 0\right\}$$

is a hyperbolic paraboloid if $\lambda = 1$ and a one-sheet hyperboloid if $\lambda \neq 1$.

Exercise 5.24 Prove that all quadrics of the pencil generated by a quadric of revolution and a sphere are of revolution having all their axes parallel and lying on the same plane.

Exercise 5.25 Determine the planes passing through the line $r : x - z = y - 2z - 1 = 0$ of $\mathbb{E}^3(\mathbb{R})$ and tangent to the quadric $z = xy$.

Exercise 5.26 Find the affine type of the following conics:

$$\begin{cases} x - y = 0 \\ x^2 + y^2 - yz = 1, \end{cases} \qquad \begin{cases} (x - 2y)^2 = z \\ 3x + y + 4z = 8, \end{cases} \qquad \begin{cases} xy + z^2 = 1 \\ 2x = y + 3. \end{cases}$$

Exercise 5.27 (Group Law on a Non-degenerate Affine Conic) Let $\widehat{f}$ be a non-degenerate conic of $\mathbb{A}^2(K)$, and let N be any fixed point on it. We define an operation as follows:

1. Let $P, Q \in \mathcal{V}(\widehat{f})$, $P \neq Q$. Through N we draw a line r_N parallel to PQ. This line must intersect $\widehat{f}$ at another point, say R. We define
$$P + Q := R.$$
2. If $P = Q$, then the line PQ is tangent to $\widehat{f}$ at P. So to find $P + P$, which we denote by $2P$, we draw a line through N parallel to the tangent at P to $\widehat{f}$; then its second point of intersection with $\widehat{f}$ is $2P$.
3. If the line through N parallel to PQ is tangent to $\widehat{f}$ at N, then we take $R = N$, i.e. $P + Q = N$.

Prove that $\mathcal{V}(\widehat{f})$ is an abelian group equipped with the operation above defined. (Hint: Use Pascal's Theorem 5.64 to check associativity.)

Chapter 6
Projective Spaces



6.1 Some Elementary Synthetic Projective Geometry

We start by giving some notions on projective geometry from a synthetic point of view. However, the main topic of this chapter is the projective space associated to a K-vector space, where K is a field. Good references for the synthetic approach are [1, 5, 16] and [23].

Projective Planes

Definition 6.1 Let $\mathcal{X}$ be a non-empty set and $\mathcal{D}$ a non-empty subset of $\mathcal{P}(\mathcal{X})$, the set of all subsets of $\mathcal{X}$. The elements of $\mathcal{X}$ will be called *points* and those of $\mathcal{D}$, *lines*. If a point A belongs to the line d we say that A lies on d or also that d passes through A.

A pair $(\mathcal{X}, \mathcal{D})$ is an *incidence structure* if it satisfies the following axiom:

(P_1) For every two distinct points $A, B \in \mathcal{X}$ there is one and only one line $d \in \mathcal{D}$ such that d *joins* A and B, i.e. $A \in d$ and $B \in d$. The line d will be denoted by AB.

It is immediately seen that the intersection of two lines d and d' of $(\mathcal{X}, \mathcal{D})$ is either d if $d = d'$, one point, or $\varnothing$.

Example 6.2 Let $(\mathcal{A}, V, \varphi)$ be an affine space and $\mathcal{D}$ the set of all affine subspaces of dimension 1 of $\mathcal{A}$. Then the pair $(\mathcal{A}, \mathcal{D})$ is an incidence structure by Corollary 3.28.

Definition 6.3 Let $\mathcal{Y} \subset \mathcal{X}$ be a non-empty subset of an incidence structure $(\mathcal{X}, \mathcal{D})$. We say that $\mathcal{Y}$ is a *linear subvariety* of $(\mathcal{X}, \mathcal{D})$ if for every two distinct points A, $B \in \mathcal{Y}$ the line $AB \in \mathcal{D}$ is contained in $\mathcal{Y}$, i.e. if the pair $(\mathcal{Y}, \mathcal{D}')$ is an incidence structure, where $\mathcal{D}'$ is the set of the lines of $\mathcal{D}$ contained in $\mathcal{Y}$. The points and the lines of $(\mathcal{X}, \mathcal{D})$ are examples of linear subvarieties. We also consider the empty set $\varnothing$ as a linear subvariety of $\mathcal{X}$.

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Definition 6.4 Let $(\mathcal{X}, \mathcal{D})$ be an incidence structure and $A, B, C \in \mathcal{X}$ three points. We say that A, B, C are *collinear* if there exists a line $d \in \mathcal{D}$ such that $A, B, C \in d$. In other words, A, B, C are collinear if and only if $C \in AB$. Since $AC = AB$ and $BC = AC$ this definition does not depend on the order of the points.

Definition 6.5 An incidence structure $(\mathcal{X}, \mathcal{D})$ is called a *projective plane* if it satisfies the following axioms:

(P_2) For every two lines $d, d' \in \mathcal{D}$, $d \cap d' \neq \varnothing$.
(P_3) There exist four points $A, B, C, D \in \mathcal{X}$ and no three of them are collinear.

The elements of $\mathcal{D}$ will be called *projective lines*.

Definition 6.6 An *affine plane* is an incidence structure $(\mathcal{X}, \mathcal{D})$ such that:

(A_2) given any line $d \in \mathcal{D}$ and any point P not on d, there is exactly one line $d' \in \mathcal{D}$ through P that does not meet d;
(A_3) there exist four points $A, B, C, D \in \mathcal{X}$ and no three of them are collinear.

We say that two lines are *parallel* if they are equal, or if they have no points in common.

An affine space $(\mathcal{A}, V, \varphi)$ of dimension 2 is an affine plane according to the above definition (for (A_2) see Proposition 3.74 and in $\mathbb{A}^2(K)$ the points $(0, 0), (1, 0), (0, 1), (1, 1)$ satisfy axiom (A_3)).

Example 6.7 Consider the sets $\mathcal{X} = \{1, 2, 3, 4, 5, 6, 7\}$ and

$$\mathcal{D} = \{\{1, 2, 6\}, \{2, 3, 4\}, \{1, 3, 5\}, \{1, 4, 7\}, \{2, 5, 7\}, \{3, 6, 7\}, \{4, 5, 6\}\}.$$

Then $(\mathcal{X}, \mathcal{D})$ is a projective plane called *Fano's plane* (see Fig. 6.1). It is the "smallest" projective plane.

Example 6.8 Let $\mathcal{D}'$ be the set of all affine lines of the affine plane $\mathbb{A}^2(K)$. We put on $\mathcal{D}'$ the following equivalence relation:

$$d \sim d' \iff d \parallel d'.$$

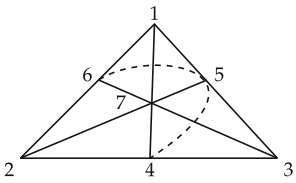


Fig. 6.1 Fano's projective plane

If we denote the equivalence class of d by $[d]$ we can put

$$[d] = \begin{cases} m & \text{if } d \text{ has the equation } y = mx + n, \\ \infty & \text{if } d \text{ has the equation } x = a. \end{cases}$$

Thus the quotient set $\omega := \mathcal{D}'/\sim$ can be identified with $K \cup \{\infty\}$. Let $\mathcal{X} := \mathbb{A}^2(K) \cup \omega$ (disjoint union). The points of $\mathbb{A}^2(K)$ will be called the *proper points* of $\mathcal{X}$; the points of ω the *improper points* or the *points at infinity* of $\mathcal{X}$. The set ω is called the *line at infinity* of $\mathcal{X}$. To every line $d \in \mathcal{D}'$ we associate

$$\widetilde{d} := d \cup \{[d]\}.$$

Then $\widetilde{d} \subset \mathcal{X}$. Finally, we define the set of projective lines

$$\mathcal{D} := \{\widetilde{d} \mid d \in \mathcal{D}'\} \cup \{\omega\}.$$

Lemma 6.9 *The pair* $(\mathcal{X}, \mathcal{D})$ *of* Example 6.8 *is a projective plane.*

Proof We have to check axiom (P_1). Let $A, B \in \mathcal{X}$, $A \neq B$.

- If A and B are proper points there exists only one affine line $d \in \mathcal{D}'$ passing through A and B. Then $\widetilde{d}$ is the unique line of $\mathcal{D}$ passing through A and B.
- If A is a proper point and $B \in \omega$, then $B = [d]$, with $d \in \mathcal{D}'$. By Proposition 3.74 there is a unique affine line d' parallel to d through A; hence $\widetilde{d'}$ is the unique line of $\mathcal{D}$ passing through A and $B = [d] = [d']$.
- If $A, B \in \omega$, then ω is the unique line of $\mathcal{D}$ through A and B.

Now we check axiom (P_2). Let $\widetilde{d}$ and $\widetilde{d'}$ be two distinct lines of $\mathcal{D}$, with d, $d' \in \mathcal{D}'$ (hence $d \neq d'$). If d and d' are not parallel, i.e. $[d] \neq [d']$ and $d \cap d' \neq \varnothing$, then *a fortiori* $\widetilde{d} \cap \widetilde{d'} \neq \varnothing$. If $d \parallel d'$ (i.e. $[d] = [d'] =: A \in \omega$) we have $\widetilde{d} \cap \widetilde{d'} = \{A\}$. Finally, every line $\widetilde{d}$, with $d \in \mathcal{D}'$, meets the line at infinity ω in the point at infinity $[d]$.

Axiom (P_3) is obviously satisfied by the points $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$ of $\mathbb{A}^2(K)$. □

Lemma 6.10 *Let* $(\mathcal{X}, \mathcal{D})$ *be a projective plane and* $A, B, C \in \mathcal{X}$ *three non-collinear points. Then we have*

$$\mathcal{X} = \bigcup_{F \in BC} AF.$$

Proof We have only to prove the inclusion $\subset$. Let $D \in \mathcal{X}$, $D \neq A$. By axiom (P_2), $AD \cap BC \neq \varnothing$, hence there exists a point $F \in AD \cap BC$, so that $D \in AF$, with $F \in BC$. □

Lemma 6.11 *Let* $(\mathcal{X}, \mathcal{D})$ *be a projective plane and* d, $d' \in \mathcal{D}$ *two arbitrary lines. Then there exists a bijection* $f : d \to d'$.

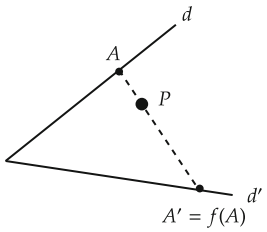


Fig. 6.2 Bijection between two projective lines

Proof We can obviously suppose that $d \neq d'$. We claim that there exists a point $P \in \mathcal{X} \setminus (d \cup d')$. By axiom (P_3) let A, B, C, $D \in \mathcal{X}$ be four points such that no line contains all three of them. If all these points belong to $d \cup d'$, we should have A, $B \in d \setminus d'$ and C, $D \in d' \setminus d$ for instance. We claim that $AD \cap BC = P \in \mathcal{X} \setminus (d \cup d')$. Indeed, if $P \in d$ (or $P \in d'$) then A, B and D (A, D and C respectively) would be collinear.

Fix a point $P \in \mathcal{X} \setminus (d \cup d')$. We can define the map $f : d \to d'$ such that $f(A) := AP \cap d'$, for all $A \in d$ (see Fig. 6.2). Let $g : d' \to d$ be the map defined by $g(B') := B'P \cap d$ for all $B' \in d'$. By axiom (P_1), g is the inverse of f so that f is bijective. □

We introduce two more axioms concerning an incidence structure $(\mathcal{X}, \mathcal{D})$:

(P_3') Every line $d \in \mathcal{D}$ contains at least three points,
(P_3'') There exist three non-collinear points in $\mathcal{X}$.

Proposition 6.12 *Let $(\mathcal{X}, \mathcal{D})$ be an incidence structure. The following conditions are equivalent:*

(i) *$(\mathcal{X}, \mathcal{D})$ is a projective plane.*
(ii) *$(\mathcal{X}, \mathcal{D})$ satisfies axioms (P_2), (P_3') and (P_3'').*

Proof

(i)$\Longrightarrow$ (ii). It is obvious that (P_3) implies (P_3''). Let $A, B, C, D \in \mathcal{X}$ be four points no three collinear. then the intersection $AD \cap BC$ is a point E distinct from B and from C. Hence the line BC has at least three distinct points. By Lemma 6.11 every other line of $\mathcal{X}$ has at least three distinct points, so that (P_3) implies (P_3').

(ii)$\Longrightarrow$ (i). By axiom (P_3'') there exist three non-collinear points $A, B, C \in \mathcal{X}$. By axiom (P_3') the line BC contains a point E with $E \neq B$ and $E \neq C$. By the same axiom AE contains a point D different from A and E. Hence A, B, C, D are four points such that no three of them lie on a line, so that axiom (P_3) is verified.

□

Proposition 6.13 *A projective plane $(\mathcal{X}, \mathcal{D})$ satisfies the following axiom:*

$(P_3^\star)$ *Through every point $P \in \mathcal{X}$ there pass at least three lines of $\mathcal{D}$.*

Proof Let P be an arbitrary point of $\mathcal{X}$ and r be a line which does not contain P. Take three distinct points Q, R and S on r, then through P there pass the distinct

lines PQ, PR and PS. For each P a line r not containing P can be found by taking two distinct lines s and t through P, a point $A \in s$, A point $B \in t$, $A \neq P$ and $B \neq P$, so that line AB does not contain P. □

Definition 6.14 Let $(\mathcal{X}, \mathcal{D})$ and $(\mathcal{X}', \mathcal{D}')$ two projective planes A map $f : \mathcal{X} \to \mathcal{X}'$ is called an *isomorphism of projective planes* if it is bijective and for every three collinear points $A, B, C \in \mathcal{X}$, $f(A)$, $f(B)$ and $f(C)$ are collinear points of $\mathcal{X}'$. A projective plane $(\mathcal{X}, \mathcal{D})$ is isomorphic to the projective plane $(\mathcal{X}', \mathcal{D}')$ if there is a isomorphism of projective planes $f : \mathcal{X} \to \mathcal{X}'$. If $f : \mathcal{X} \to \mathcal{X}'$ and $g : \mathcal{X}' \to \mathcal{X}''$ are isomorphisms, then $g \circ f : \mathcal{X} \to \mathcal{X}''$ is clearly an isomorphism. An isomorphism $f : \mathcal{X} \to \mathcal{X}$ is called an *automorphism* of $\mathcal{X}$. The *isomorphism of affine planes* is defined in the same manner.

The following proposition states that the relation of isomorphism of projective planes is symmetric.

Proposition 6.15 *If $f : \mathcal{X} \to \mathcal{X}'$ is an isomorphism of projective (affine) planes, then also $f^{-1} : \mathcal{X}' \to \mathcal{X}$ is an isomorphism of projective (affine) planes.*

Proof Let $A', B', C' \in \mathcal{X}'$ be three points lying on a line d'. We have to show that $A := f^{-1}(A')$, $B := f^{-1}(B')$, $C := f^{-1}(C')$ are collinear in $\mathcal{X}$. We prove it by contradiction. Suppose that A, B, C are not collinear. Let $P \in \mathcal{X}$ be an arbitrary point. We shall prove that $f(P) \in d'$, where d' is the line through $A' = f(A)$, $B' = f(B)$ and $C' = f(C)$ which contradicts the surjectivity of f ($d' \neq \mathcal{X}'$). If $P = A$ then it obvious that $f(P) \in d'$. Let $P \neq A$ and $D := AP \cap BC$. Since f is an isomorphism $f(D) \in B'C' = d'$ and then $f(P) \in f(A)f(D) = d'$. In the affine case, if $AP \parallel BC$ then $f(A)f(P) \parallel f(B)f(C) = B'C'$ so that $A' \notin B'C'$, contradicting the hypothesis. □

Corollary 6.16 *The set* $\mathrm{Aut}(\mathcal{X})$ *of all automorphisms of a projective (or affine) plane $\mathcal{X}$ is a group with respect to composition of automorphisms.*

Every line of Fano's plane (Example 6.7) contains exactly three points. We also have the converse.

Proposition 6.17 *Let $(\mathcal{X}', \mathcal{D}')$ be a projective plane whose all lines contain exactly three points. Then $(\mathcal{X}', \mathcal{D}')$ is isomorphic to Fano's plane (see* Exercise 1.3).

Proof Since every line contains exactly three points, by Lemma 6.10 we see immediately that $\#(\mathcal{X}') = 7$. Let $A, B, C \in \mathcal{X}'$ be three non-collinear points. The remaining points are as in Fig. 6.3. It is easy to see that the map $f : \{1, 2, 3, 4, 5, 6, 7\} \to$

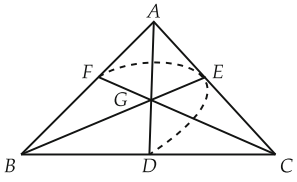


Fig. 6.3 Fano's projective plane

$\{A, B, C, D, E, F, G\}$, where $f(1) = A$, $f(2) = B$, $f(3) = C$, $f(4) = D$, $f(5) = E$, $f(6) = F$ and $f(7) = G$, is an isomorphism of projective planes (compare it with Fig. 6.1). □

Proposition 6.18 *Let $(\mathcal{X}, \mathcal{D})$ be a projective plane, $r \in \mathcal{D}$ a fixed line, $\mathcal{X}_r = \mathcal{X} \setminus r$ and $\mathcal{D}_r = \{d_r := d \cap \mathcal{X}_r : d \in \mathcal{D},\ d \neq r\}$. Then $(\mathcal{X}_r, \mathcal{D}_r)$ is an affine plane.*

Conversely, let $(\mathcal{X}, \mathcal{D})$ be an affine plane. Consider the equivalence relation on $\mathcal{D}$ defined by $d \sim d'$ if and only if $d \parallel d'$ and denote the equivalence class of a line d by $\widehat{d}$. Put

$$\overline{\mathcal{X}} := \mathcal{X} \cup \{\widehat{d} : d \in \mathcal{D}\}, \quad d_\infty := \{\widehat{d} : d \in \mathcal{D}\}, \quad \overline{\mathcal{D}} := \{\overline{d} := d \cup \widehat{d} : d \in \mathcal{D}\} \cup \{d_\infty\}.$$

Then $(\overline{\mathcal{X}}, \overline{\mathcal{D}})$ is a projective plane called the projective closure *of $(\mathcal{X}, \mathcal{D})$ (see* Example 6.8).

Finally, if $(\mathcal{X}, \mathcal{D})$ is a projective plane, then $\overline{\mathcal{X}_r} = \mathcal{X}$ and $\overline{\mathcal{D}_r} = \mathcal{D}$, where $d_\infty = r$. Analogously, if $(\mathcal{X}, \mathcal{D})$ is an affine plane, then $\overline{\mathcal{X}}_{d_\infty} = \mathcal{X}$ and $\overline{\mathcal{D}}_{d_\infty} = \mathcal{D}$.

Proof Two lines d_r and e_r of $\mathcal{X}_r$ are parallel if and only if $d \cap e \in r$. The axioms (A_2) and (A_3) of Definition 6.6 are easily proved. First we immediately realize that $(\mathcal{X}_r, \mathcal{D}_r)$ is an incidence structure. Let P be a point, d_r be a line of $\mathcal{X}_r$ not containing P and $e = PQ$ where $Q = r \cap d$, then the line $e_r := e \cap \mathcal{X}_r$ is the only line of $\mathcal{X}_r$ passing through P and parallel to d_r. Finally, let A, B, C, D be four points of $\mathcal{X}$ such that no three of them lie on a line. We can check axioms (A_2) and (A_3) on the Fano's plane and then suppose that every projective line contains four points at least. If $A \in r$ but B, C and D don't lie on r we can choose a point E on AB, $E \neq A$, B and $E \notin CD$ so that E, B, C, D satisfy axiom (A_3). Indeed E, B, C are not collinear, otherwise A, B, C would be collinear; thus also E, B, D are not collinear. The points B, C and D are not collinear by hypothesis as well as E, C and D. By a similar argument we see that if A, $B \in r$ but C and D don't lie on r we can find two points E, $F \in \mathcal{X}_r$ such that C, D, E and F satisfy axiom (A_3). We let the reader prove the converse assertion as an exercise. □

Finally, we observe that an isomorphism of projective planes $f : (\mathcal{X}, \mathcal{D}) \to (\mathcal{Y}, \mathcal{E})$ induces an isomorphism of affine planes $f_r : (\mathcal{X}_r, \mathcal{D}_r) \to (\mathcal{Y}_{f(r)}, \mathcal{E}_{f(r)})$. Analogously an isomorphism of affine planes $f : (\mathcal{X}, \mathcal{D}) \to (\mathcal{Y}, \mathcal{E})$ extends to an isomorphism of projective planes $\overline{f} : (\overline{\mathcal{X}}, \overline{\mathcal{D}}) \to (\overline{\mathcal{Y}}, \overline{\mathcal{E}})$ defined by

$$\overline{f}(P) = f(P), \quad \forall\, P \in \mathcal{X}, \qquad \overline{f}(\widehat{d}) = \widehat{f(d)}, \quad \forall\, \widehat{d} \in d_\infty.$$

Remark 6.19 From Lemma 6.11 and from Propositions 6.12 and 6.18 we deduce immediately:

1. Let d, $d' \in \mathcal{D}$ be two arbitrary lines in an affine plane. Then there exists a bijection $f : d \to d'$.
2. Every line in an affine space contains at least two points.

6.1.1 General Projective Spaces

We are going to extend Definitions 6.5 and 6.14.

Definition 6.20 An incidence structure $(\mathcal{X}, \mathcal{D})$, satisfying the following axioms:

(P_3'') there exist three non-collinear points in $\mathcal{X}$,
(P_4) for every three non-collinear points $A, B, C \in \mathcal{X}$ there exists a linear subvariety $\mathcal{Y}$ of $(\mathcal{X}, \mathcal{D})$, containing A, B, C, which is a projective plane,

is called a *general projective space*. In particular a projective plane according to Definition 6.5 is a general projective space.

If $(\mathcal{X}, \mathcal{D})$ is a projective space and if $A, B, C \in \mathcal{X}$ are three non-collinear points, by Lemma 6.10 it follows that every projective plane $\mathcal{Y}$ containing A, B, C coincides (as a set) with $\bigcup_{P \in BC} AP$. In particular, $\mathcal{Y}$ is uniquely determined by every three non-collinear points $A, B, C \in \mathcal{Y}$; it is called the *projective plane passing through* A, B, C; $\mathcal{Y}$ will be denoted by ABC. Hence every projective space satisfies axiom (P_3'), i.e. every line contains three distinct points at least. On the other hand, axiom (P_2) does not hold in a general projective space. We can assert that a projective space $(\mathcal{X}, \mathcal{D})$ satisfies axiom (P_2) if and only if it is a projective plane. Implication $\Longleftarrow$ is obvious. Let $A, B, C \in \mathcal{X}$ be three non-collinear points and $\mathcal{Y}$ the plane passing through them. Let $r = AB$, $s = AC$ so that $A = r \cap s$. Let D be any point of $\mathcal{X} \setminus (r \cup s)$. If t is the line joining A and D and $u = BC$, by axiom (P_2) we have $t \cap u = E$ so that both A and E belong to $\mathcal{Y}$. Then we get $t \subset \mathcal{Y}$ and $D \in \mathcal{Y}$.

A projective space $(\mathcal{X}, \mathcal{D})$ contains four points A, B, C, D such that A, B, C are not collinear and $D \notin ABC$, if $(\mathcal{X}, \mathcal{D})$ is not a projective plane. Four points of a projective space $(\mathcal{X}, \mathcal{D})$ are called *non-coplanar* if they are not contained in a plane of $\mathcal{X}$. In particular, if four points are not coplanar then no three of them are collinear.

The concept of projective isomorphism between projective spaces is the same as for projective planes.

Definition 6.21 Let $(\mathcal{X}, \mathcal{D})$ and $(\mathcal{X}', \mathcal{D}')$ be two projective spaces. A bijective map $f : \mathcal{X} \to \mathcal{X}'$ is called an *isomorphism of projective spaces* if for every $A, B, C \in \mathcal{X}$

$$A, B, C \quad \text{are collinear in } \mathcal{X} \iff f(A), f(B), f(C) \quad \text{are collinear in } \mathcal{X}'.$$

In other words, an isomorphism $f : \mathcal{X} \to \mathcal{X}'$ is characterized by the following property:

- for every subset $\mathcal{Y} \subset \mathcal{X}$, $\mathcal{Y}$ is a line of $\mathcal{X}$ (i.e. $\mathcal{Y} \in \mathcal{D}$) if and only if $f(\mathcal{Y})$ is a line of $\mathcal{X}'$ (i.e. $f(\mathcal{Y}) \in \mathcal{D}'$).

The reader can immediately show that a map $f : \mathcal{X} \to \mathcal{X}'$ is an isomorphism of projective spaces if and only if the following conditions are satisfied:

(i) For every three collinear points $A, B, C \in \mathcal{X}$, then $f(A)$, $f(B)$ and $f(C)$ are collinear in $\mathcal{X}'$.

(ii) There exists a map $g\colon \mathcal{X}' \to \mathcal{X}$ such that $g \circ f = \mathrm{id}_{\mathcal{X}}$, $f \circ g = \mathrm{id}_{\mathcal{X}'}$, and for every three collinear points A', B', $C' \in \mathcal{X}'$, then $g(A')$, $g(B')$ and $g(C')$ are collinear in $\mathcal{X}$.

Definition 6.22 Let $(\mathcal{X}, \mathcal{D})$ be a projective space. An isomorphism of projective spaces $f\colon \mathcal{X} \to \mathcal{X}$ is called an *automorphism* of $(\mathcal{X}, \mathcal{D})$, (or simply of $\mathcal{X}$. The set of all automorphisms $\mathcal{X}$ will be denoted by $\mathrm{Aut}(\mathcal{X})$. It is immediately seen that $\mathrm{Aut}(\mathcal{X})$ is a group (in general non commutative) with respect to composition of automorphisms.

Projective Subspaces

Definition 6.23 Let $(\mathcal{X}, \mathcal{D})$ be a projective space. A subset $\mathcal{Y} \subset \mathcal{X}$ is called a *projective subspace*, or simply a *subspace* of $\mathcal{X}$, if $\mathcal{Y}$ is a linear subvariety of the incidence structure $(\mathcal{X}, \mathcal{D})$. In other words, $\mathcal{Y}$ is a subspace of $\mathcal{X}$ if for every two distinct points A and $B \in \mathcal{Y}$, the line AB is contained in $\mathcal{Y}$. We agree that the points and the lines of $\mathcal{X}$ as well as the empty set $\varnothing$ are projective subspaces of $\mathcal{X}$. If $\mathcal{Y}$ is the empty set, a point or a line, we can define the *dimension* $\dim \mathcal{Y}$ as follows:

$$\dim \mathcal{Y} = \begin{cases} -1 & \text{if } \mathcal{Y} = \varnothing, \\ 0 & \text{if } \mathcal{Y} \text{ is a point,} \\ 1 & \text{if } \mathcal{Y} \text{ is a line.} \end{cases}$$

If $\mathcal{Y}$ is a subspace of $\mathcal{X}$ and $\mathcal{D}_{\mathcal{Y}} := \{d \in \mathcal{D} \mid d \subset \mathcal{Y}\}$, then $(\mathcal{Y}, \mathcal{D}_{\mathcal{Y}})$ is a projective space according to Definition 7.1 provided that $\mathcal{Y}$ contains at least three non-collinear point. A *hyperplane* $\mathcal{Y}$ is a subspace of $\mathcal{X}$ such that if $\mathcal{Z}$ is a subspace of $\mathcal{X}$ such that $\mathcal{Y} \subset \mathcal{Z} \subset \mathcal{X}$ then $\mathcal{Z} = \mathcal{Y}$ or $\mathcal{Z} = \mathcal{X}$.

If $f\colon \mathcal{X} \to \mathcal{X}'$ is an isomorphism of projective spaces and $\mathcal{Y}$ is a subspace of $\mathcal{X}$, it is immediately seen that $\mathcal{Y}' = f(\mathcal{Y})$ is a subspace of $\mathcal{X}'$ and $f|_{\mathcal{Y}}\colon \mathcal{Y} \to \mathcal{Y}'$ is an isomorphism of projective spaces.

Lemma 6.24 *Let $\{\mathcal{Y}_i\}_{i\in I}$ be an arbitrary family of projective subspaces of a projective space $\mathcal{X}$. Then $\bigcap_{i\in I} \mathcal{Y}_i$ is a projective subspace of $\mathcal{X}$.*

Proof It is an immediate consequence of the above definition. □

Thanks to 6.24 we can give the following:

Definition 6.25 Let M be a subset of a projective space $\mathcal{X}$. The intersection of all subspaces of $\mathcal{X}$ containing M is a subspace of $\mathcal{X}$ called the *subspace generated by* M; it will be denoted by $\langle M \rangle$. It is the smallest subspace containing M.

Let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ be two subspaces of $\mathcal{X}$. We shall denote the subspace $\langle \mathcal{Y}_1 \cup \mathcal{Y}_2 \rangle$ by $\mathcal{Y}_1 + \mathcal{Y}_2$ and we call it the *sum* of $\mathcal{Y}_1$ and $\mathcal{Y}_2$. If $M = \{A_1, \ldots, A_m\}$, where $A_i \in \mathcal{X}$, $i = 1, \ldots, m$, then $\langle M \rangle$ will be denoted by $A_1 + \cdots + A_m$. For instance, if A and B are two distinct points of $\mathcal{X}$, the subspace $A + B$ is the line AB. If $\mathcal{X}$ is a plane and d_1, d_2 are two distinct lines of $\mathcal{X}$, then $d_1 + d_2 = \mathcal{X}$.

The following theorem, which is a generalization of Lemma 6.10, gives an explicit description of the sum of two subspaces.

Theorem 6.26 *Let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ be two non-empty subspaces of a projective space $\mathcal{X}$ such that $\mathcal{Y}_1 \cup \mathcal{Y}_2$ contains at least two distinct points. Then we have*

$$\mathcal{Y}_1 + \mathcal{Y}_2 = \bigcup_{A_1 \in \mathcal{Y}_1, A_2 \in \mathcal{Y}_2, A_1 \neq A_2} A_1 A_2. \tag{6.1}$$

Proof Let $\mathcal{X}'$ be the right– hand side of (6.1). Both the inclusions $\mathcal{X}' \subseteq \mathcal{Y}_1 + \mathcal{Y}_2$ and $\mathcal{Y}_1 \cup \mathcal{Y}_2 \subseteq \mathcal{X}'$ are obvious, so that we have only to show that $\mathcal{X}'$ is a subspace of $\mathcal{X}$ to prove (6.1).

Consider two arbitrary points P_1, $P_2 \in \mathcal{X}'$, $P_1 \neq P_2$. There exist A_i, $B_i \in \mathcal{Y}_i$, $i = 1, 2$, such that $A_1 \neq A_2$, $B_1 \neq B_2$, $P_1 \in A_1 A_2$ and $P_2 \in B_1 B_2$. Two cases are possible:

(a) The points A_1, A_2, B_1 and B_2 are coplanar, i.e. the subspace $A_1 + A_2 + B_1 + B_2$ is contained in a plane $\mathcal{Z}$. If A_1, A_2, B_1 and B_2 lie on the same line r there is nothing to prove since $P_1 P_2$ coincides with r (which is contained in both the subspaces $\mathcal{Y}_1$ and $\mathcal{Y}_2$). Otherwise we have the equality $A_1 + A_2 + B_1 + B_2 = \mathcal{Z}$. By axiom (P_2) (applied to $\mathcal{Z}$) the lines $A_1 A_2$ and $B_1 B_2$ intersect in a point $R \in \mathcal{Z}$. If P is an arbitrary point of $P_1 P_2$, by axiom (P_2) the line RP meets $A_1 B_1$ and $A_2 B_2$ in the points C_1 and C_2 respectively (see Fig. 6.4).

Since $\mathcal{Y}_i$, $i = 1, 2$, are subspaces, then $C_i \in \mathcal{Y}_i$, $i = 1, 2$, and $P \in C_1 C_2$. Therefore in case a) we checked that the line $P_1 P_2$ lies on $\mathcal{X}'$.

(b) The points A_1, A_2, B_1 and B_2 are not coplanar. As in the previous case, let $P \in P_1 P_2$. By axiom (P_4) we have the plane $A_1 A_2 P$. Since P_1, $P \in A_1 A_2 P$, then $P_1 P = P_1 P_2 \subset A_1 A_2 P$, so that $P_2 \in A_1 A_2 P$. By axiom (P_2) (applied to $A_1 A_2 P$) there exists a point Q such that $A_1 P \cap A_2 P_2 = Q$ (see Fig. 6.5).

Furthermore, $Q \in A_2 P_2 \subset A_2 B_1 B_2$ so that, by axiom (P_2) applied to the plane $A_2 B_1 B_2$, there exists a point C_2 such that $B_1 Q \cap A_2 B_2 = C_2$. Finally, since $Q \in B_1 C_2 \subset A_1 B_1 C_2$ and $P \in A_1 Q \subset A_1 B_1 C_2$, by axiom (P_2) applied to the plane $A_1 B_1 C_2$ there exists a point C_1 such that $C_2 P \cap A_1 B_1 = C_1$. Thus we have obtained the points $C_1 \in A_1 B_1$ and $C_2 \in A_2 B_2$ (hence $C_1 \in \mathcal{Y}_1$ and $C_2 \in \mathcal{Y}_2$ because $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are subspaces) such that $P \in C_1 C_2$. This ends the proof that $\mathcal{X}'$ is a subspace of $\mathcal{X}$.

□

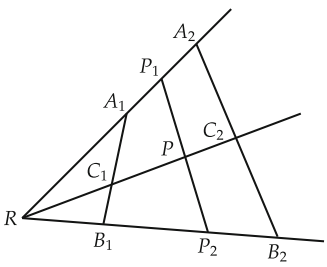


Fig. 6.4 Three coplanar lines

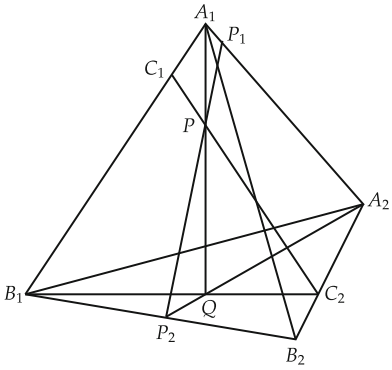


Fig. 6.5 Three non-coplanar lines

Remark 6.27 Let $(\mathcal{X}, \mathcal{D})$ be a projective space. If $\mathcal{Y}$ is a hyperplane of $\mathcal{X}$ and d a line, then $d \cap \mathcal{Y} = d$ or $d \cap \mathcal{Y}$ is only one point. Indeed, if $d \cap \mathcal{Y} = \varnothing$, take a point $P \in d$. Consider the subspace $P + \mathcal{Y}$. By Theorem 6.26 we have

$$\mathcal{Y} \subsetneq P + \mathcal{Y} \subsetneq \mathcal{X},$$

contradicting the assumption that $\mathcal{Y}$ is a hyperplane.

6.2 The Projective Space Associated with a K-Vector Space

Definition 6.28 Let V be a K-vector space of dimension $n + 1$ (with $n \geq 2$). Consider the set $\mathcal{X}$ of all vector subspaces U of V with $\dim_K(U) = 1$. Let W be a vector subspace of dimension 2 and d_W be the set of all subspaces of dimension 1 contained in W (i.e. $d_W := \{\langle v \rangle \mid v \in W,\ v \neq 0_V\}$). We define the set of lines

$$\mathcal{D} := \{d_W \mid W \text{ vector subspace of } V,\ \dim_K W = 2\}.$$

We call $\mathbb{P}(V) := (\mathcal{X}, \mathcal{D})$ the *projective space associated with the K-vector space V*. The above definition is meaningful also if $\dim V = 2$; in that case $\mathbb{P}(V) := (\mathcal{X}, \mathcal{D})$ is called the *projective line associated with the K-vector space V* since $\mathcal{X} = d_V$ and $\mathcal{D} = \{d_V\}$. If $V = K^{n+1}$ the projective space $\mathbb{P}^n(K) := \mathbb{P}(K^{n+1})$ is called the *standard projective space of dimension n over K*. There is a canonical identification

$$(V \setminus \{0_V\})/\sim \longrightarrow \mathbb{P}(V), \qquad \widehat{v} \to \langle v \rangle, \tag{6.2}$$

where $v \sim w$ if and only if $v = kw$ for some $k \in K^*$. We will denote the equivalence class of $(x_0, \ldots, x_n) \in K^{n+1}$ by $[x_0, \ldots, x_n]$.

Theorem 6.29 *The projective space $\mathbb{P}(V) = (\mathcal{X}, \mathcal{D})$ associated with the K-vector space V with* ($\dim V \geq 3$) *is a general projective space according to* Definition 7.1.

Proof First we have to show that $\mathbb{P}(V)$ is an incidence structure. Let $A, B \in \mathcal{X}$ be two distinct points, i.e. $A = \langle v \rangle$ and $B = \langle w \rangle$, with $v, w \in V$, $v, w \neq 0_V$ such that v, w are linearly independent vectors. Let $W = \langle v, w \rangle$, then $\dim_K W = 2$ and the line $d_W \in \mathcal{D}$ contains A and B. Let W' be a vector subspace of dimension 2 such that $A, B \in d_{W'}$, i.e. $v, w \in W'$. Since $\dim_K W' = 2$, $v, w \in W'$ and v, w are linearly independent we get $W' = \langle v, w \rangle$ so that $W = W'$ and $d_W = d_{W'}$. Therefore axiom (P_1) Definition 6.1 is verified and $\mathbb{P}(V)$ is an incidence structure. □

We need the following result.

Lemma 6.30 *Suppose* $\dim_K V = 3$, *then* $\mathbb{P}(V) = (\mathcal{X}, \mathcal{D})$ *is a projective plane.*

Proof

Axiom (P_2). Let d_W and $d_{W'}$ be two distinct lines of $\mathbb{P}(V)$, i.e. $W \neq W'$. Since $\dim_K V = 3$ and $\dim_K W = \dim_K W' = 2$, then we have $W + W' = V$. By Grassmann's formula (Corollary 1.76) we get

$$\dim_K(W \cap W') = \dim_K W + \dim_K W' - \dim_K(W + W') = 2 + 2 - 3 = 1,$$

so that $W \cap W'$ is a point of $\mathcal{X}$ common to d_W and $d_{W'}$.

Axiom (P_3'). Let $d_W \in \mathcal{D}$ be a line and $\{v_1, v_2\}$ be a K-basis of W. Then $\langle v_1 \rangle$, $\langle v_2 \rangle$ and $\langle v_1 + v_2 \rangle$ are three distinct subspaces of W of dimension 1 so that they represent three distinct points of d_W.

Axiom (P_3''). Let $\{v_1, v_2, v_3\}$ be a basis of V. It is clear that the vector subspaces $W_i := K v_i$, $i = 1, 2, 3$ define three non-collinear points of $\mathcal{X}$.

□

By Lemma 6.30 we can suppose $n \geq 3$, i.e. $\dim_K V \geq 4$. The proof of axiom (P_3'') is exactly the same as for Lemma 6.30.

Axiom (P_4). Let A, B, C be three non-collinear points of $\mathcal{X}$, i.e. $A = K v_1$, $B = K v_2$ and $C = K v_3$ such that v_1, v_2 and v_3 are linearly independent. Let $W = \langle v_1, v_2, v_3 \rangle$. Since $\dim_K W = 3$, by Lemma 6.29 $\mathbb{P}(W)$ (the set of all 1-dimensional vector subspaces of V contained in W) is a projective plane. Thus the projective plane $\mathbb{P}(W)$ is a linear subvariety of the incidence structure $\mathbb{P}(V)$ containing A, B and C.

Theorem 6.31 *Let* V *be a* K*-vector space of dimension* ≥ 3. *Consider the canonical map*

$$\pi : V \setminus \{0_V\} \to \mathbb{P}(V), \quad \pi(v) = K v.$$

The correspondence

$$\mathcal{Y} \to V_{\mathcal{Y}} := \pi^{-1}(\mathcal{Y}) \cup \{0_V\} \tag{6.3}$$

between the set of all non-empty subspaces of $\mathbb{P}(V)$ *and the set of all non-null vector subspaces of* V *is bijective and preserves the inclusions. The correspondence* (6.3) *can also be seen as*

$$W \to \mathbb{P}(W) \subset \mathbb{P}(V), \qquad W \text{ vector subspace of } V,\ W \neq 0_V. \tag{6.4}$$

Thus if W *is a vector subspace of* V *we also can identify* $\mathbb{P}(W)$ *with* $(W \setminus \{0_V\})/\sim$.

Proof Let $\mathcal{Y}$ be a non-empty projective subspace of $\mathbb{P}(V)$. First we have to check that $V_{\mathcal{Y}}$ is a vector subspace of V. Let $v_1, v_2 \in V_{\mathcal{Y}}$. If either v_1 or v_2 is 0_V then it is obvious that $v_1 + v_2 \in V_{\mathcal{Y}}$. Suppose $v_i \neq 0_V$, $i = 1, 2$. Then $\pi(v_i) = Kv_i$, $i = 1, 2$ determine two points P_1 and P_2 of $\mathcal{Y}$. If $P_1 = P_2$ i.e. $Kv_1 = Kv_2$, then v_1, $v_2 \in Kv_1$, hence $v_1 + v_2 \in V_{\mathcal{Y}}$. Suppose now that $Kv_1 \neq Kv_2$ i.e. $P_1 \neq P_2$, then $P_1P_2 \subset \mathcal{Y}$ since $\mathcal{Y}$ is a subspace. According to Definition 6.28, P_1P_2 is the set of all 1-dimensional vector subspaces of $Kv_1 + Kv_2$, hence $v_1 + v_2 \in V_{\mathcal{Y}}$. Let $v \in V_{\mathcal{Y}}$ and $\lambda \in K$. If $\lambda = 0_K$, then $\lambda v = 0_V \in V_{\mathcal{Y}}$. If $\lambda \neq 0_K$, since $Kv = K(\lambda v)$ we have $\lambda v \in V_{\mathcal{Y}}$.

The correspondence $\mathcal{Y} \to V_{\mathcal{Y}}$ clearly preserves the inclusions. Now we prove that it is bijective. Let W be a non-null vector subspace of V. If $\dim_K W = 1$, then W represents a uniquely determined point $A \in \mathbb{P}(V)$ such that $V_A = W$. If $\dim_K W = 2$, then W is a uniquely determined line d of $\mathbb{P}(V)$ such that $V_d = W$. Hence suppose $\dim_K W \geq 3$. The set $\mathcal{Y}_W$ of all 1-dimensional vector subspaces of V contained in W is a projective subspace of $\mathbb{P}(V)$ such that $V_{\mathcal{Y}_W} = W$. Hence the map $W \to \mathcal{Y}_W$ is the inverse of $\mathcal{Y} \to V_{\mathcal{Y}}$. □

Definition 6.32 Under the hypothesis and notation of Theorem 6.31, we define the *dimension* of a non-empty projective subspace $\mathcal{Y}$ of $\mathbb{P}(V)$ as the integer

$$\dim_K \mathcal{Y} := \dim_K V_{\mathcal{Y}} - 1,$$

where $\dim_K V_{\mathcal{Y}}$ is the dimension of $V_{\mathcal{Y}}$ as a vector subspace of V. In particular we have $\dim \mathbb{P}(V) := \dim_K V - 1$ and $\dim_K \mathcal{Y} = 0$ if and only if $\mathcal{Y}$ is a point. Finally, we put

$$\dim_K \mathcal{Y} = -1, \text{ if } \mathcal{Y} = \varnothing.$$

From now on we use the notation $\dim \mathcal{Y}$ (instead of $\dim_K \mathcal{Y}$) if the field K is fixed.

If $\dim \mathbb{P}(V) = n$ and $\mathcal{Y}$ is a projective subspace of $\mathbb{P}(V)$, then $\dim \mathcal{Y} \leq n$. More generally, if $\mathcal{Z}$ is another projective subspace of $\mathbb{P}(V)$ such that $\mathcal{Y} \subset \mathcal{Z}$, then $\dim \mathcal{Y} \leq \dim \mathcal{Z}$, and $\dim \mathcal{Y} = \dim \mathcal{Z}$ if and only if $\mathcal{Y} = \mathcal{Z}$. If $\dim \mathbb{P}(V) = n \geq 2$, a projective subspace $\mathcal{Y}$ of $\mathbb{P}(V)$ is called a *projective line* if $\dim \mathcal{Y} = 1$, *projective plane* if $\dim \mathcal{Y} = 2$, and, when $n \geq 4$, *projective hyperplane* if $\dim \mathcal{Y} = n - 1$.

The concept of dimension for projective subspaces of Definition 6.23 can be found in Chapter 8 of [1].

The analogue of Grassmann's formula (1.26) is given by the following:

Corollary 6.33 *Let $\mathcal{Y}$ and $\mathcal{Z}$ be two projective subspaces of $\mathbb{P}(V)$. We have the following identity:*

$$\dim \mathcal{Y} + \dim \mathcal{Z} = \dim(\mathcal{Y} \cap \mathcal{Z}) + \dim(\mathcal{Y} + \mathcal{Z}), \tag{6.5}$$

called the projective Grassmann formula.

Proof We obtain (6.5) by Grassmann's formula of Corollary 1.76 and Definition 6.32, taking into account the following relations:

$$V_{\mathcal{Y}+\mathcal{Z}} = V_{\mathcal{Y}} + V_{\mathcal{Z}}, \qquad V_{\mathcal{Y}\cap\mathcal{Z}} = V_{\mathcal{Y}} \cap V_{\mathcal{Z}}.$$

□

There are some immediate consequences of the projective Grassmann formula.

Corollary 6.34 *Let $\mathcal{Y}$ and $\mathcal{Z}$ be two projective subspaces of $\mathbb{P}(V)$, where* $\dim \mathbb{P}(V) = n$.

(i) *If $\mathcal{Y}$ is a line and $\mathcal{Z}$ is a hyperplane, then $\mathcal{Y} \cap \mathcal{Z} \neq \varnothing$. If $\mathcal{Y} \not\subset \mathcal{Z}$ then $\mathcal{Y} \cap \mathcal{Z}$ is a point.*
(ii) *If $\mathcal{Y}$ is a plane and $\mathcal{Z}$ is a hyperplane, then* $\dim(\mathcal{Y} \cap \mathcal{Z}) \geq 1$. *If $\mathcal{Y} \not\subset \mathcal{Z}$ then $\mathcal{Y} \cap \mathcal{Z}$ is a line.*
(iii) *If $\mathcal{Y}$ and $\mathcal{Z}$ are distinct hyperplanes, then* $\dim(\mathcal{Y} \cap \mathcal{Z}) = n - 2$.

Proof (i). It is obvious that

$$\mathcal{Y} \subset \mathcal{Z} \implies \mathcal{Y} \cap \mathcal{Z} = \mathcal{Y} \neq \varnothing, \qquad \mathcal{Y} \not\subset \mathcal{Z} \implies \mathcal{Z} \subsetneqq \mathcal{Y} + \mathcal{Z}.$$

Since $\dim \mathcal{Z} = n - 1$ we have $\dim(\mathcal{Y} + \mathcal{Z}) > n - 1$, which implies $\dim(\mathcal{Y} + \mathcal{Z}) = n$ (i.e. $\mathcal{Y} + \mathcal{Z} = \mathbb{P}(V)$). By the projective Grassmann formula we get

$$\dim(\mathcal{Y} \cap \mathcal{Z}) = \dim \mathcal{Y} + \dim \mathcal{Z} - \dim(\mathcal{Y} + \mathcal{Z}) = 1 + (n - 1) - n = 0,$$

so that $\mathcal{Y} \cap \mathcal{Z}$ is a point. We let the reader prove the remaining points (ii) and (iii) as an exercise. □

Definition 6.35 Let $B_0, B_1, \ldots, B_m$ be $m + 1$ points ($m \geq 0$) of the projective space $\mathbb{P}(V)$ of dimension n. Then we have

$$\dim(B_0 + B_1 + \cdots + B_m) \leq m. \tag{6.6}$$

Indeed, if B_i corresponds to the vector subspace $K v_i$ of dimension 1, $i = 0, 1, \ldots, m$, then

$$V_{B_0+B_1+\cdots+B_m} = K v_0 + K v_1 + \cdots + K v_m,$$

so that $\dim(V_{B_0+B_1+\cdots+B_m}) \leq m + 1$ and (6.6).

We say that the set $\{B_0, B_1, \ldots, B_m\}$ is:

- *projectively independent*, or simply *independent*, if $\dim(B_0+B_1+\cdots+B_m) = m$;
- *projectively dependent* or simply *dependent* if $\dim(B_0 + B_1 + \cdots + B_m) < m$;
- a *system of generators* of a subspace $\mathcal{Y}$ of $\mathbb{P}(V)$ if $\mathcal{Y} = B_0 + B_1 + \cdots + B_m$; We also say that $B_0, B_1, \ldots, B_m$ generate $\mathcal{Y}$;
- a *basis* of $\mathbb{P}(V)$ if it is a system of generators of $\mathbb{P}(V)$ which is projectively independent.

If $B_i = K v_i$, $i = 0, 1, \ldots, m$, we easily get the following equivalences:

1. $\{B_0, B_1, \ldots, B_m\}$ is a projectively independent subset of $\mathbb{P}(V)$ if and only if $v_0, v_1, \ldots, v_m$ are linearly independent vectors of V. In particular $m \leq n + 1$.
2. $\{B_0, B_1, \ldots, B_m\}$ is a system of generators of $\mathbb{P}(V)$ if and only if $v_0, v_1, \ldots, v_m$ generate the vector space V. In particular $m \geq n + 1$.
3. $\{B_0, B_1, \ldots, B_m\}$ is a basis of $\mathbb{P}(V)$ if and only if $\{v_0, v_1, \ldots, v_m\}$ is a basis of V. In particular $m = n + 1$.

Every subset of a projectively independent set is obviously projectively independent and every (finite) subset containing a systems of generators of $\mathbb{P}(V)$ generates $\mathbb{P}(V)$.

More generally, we shall say that r projective subspaces $\mathcal{Y}_1, \ldots, \mathcal{Y}_r$ of $\mathbb{P}(V)$ are *projectively independent* if

$$\dim(\mathcal{Y}_1 + \cdots + \mathcal{Y}_r) = \dim(\mathcal{Y}_1) + \cdots + \dim(\mathcal{Y}_r) + r - 1.$$

Example 6.36 Two points $B_0, B_1 \in \mathbb{P}(V)$ are projectively independent if and only if $B_0 \neq B_1$; three points $B_0, B_1, B_2 \in \mathbb{P}(V)$ are projectively independent if and only if they are not collinear. Four points $B_0, B_1, B_2, B_3 \in \mathbb{P}(V)$ are projectively independent if and only if they are not coplanar.

We say that $m + 1$ points $B_0, B_1, \ldots, B_m$ are *in general position* if:

- $m \leq n$ and $B_0, B_1, \ldots, B_m$ are projectively independent;
- $m > n$ and every subset of $n+1$ points contained in $\{B_0, \ldots, B_m\}$ is projectively independent.

For instance, if $n = 2$, $m + 1 \geq 3$ distinct points $B_0, B_1, \ldots, B_m \in \mathbb{P}(V)$ are in general position if no three of them are collinear. Any ordered set $\{B_0, \ldots, B_n, B_{n+1}\}$ consisting of $n + 2$ points in general position is called a *projective frame* of $\mathbb{P}(V)$; the points $B_0, B_1, \ldots, B_n$ are called *the fundamental points* of the frame, while B_{n+1} is called *the unit point* of the frame. In $\mathbb{P}^n(K)$ the frame {[1, 0, ..., 0], [0, 1, ..., 0], ..., [0, 0, ..., 1], [1, 1, ..., 1]} is called *the standard projective frame*; a corresponding normalized basis is the canonical basis of K^{n+1}

Theorem 6.37 *Let $\mathbb{P}(V)$ be a projective space of dimension $n \geq 2$.*

(i) *Every projectively independent subset of $\mathbb{P}(V)$ is contained in a basis of $\mathbb{P}(V)$.*
(ii) *Every system of generators of $\mathbb{P}(V)$ contains a basis of $\mathbb{P}(V)$.*

Proof It is immediate from the above definitions and Theorem 1.49 and Corollary 1.53. □

Corollary 6.38 *Let $\mathbb{P}(V)$ be a projective space of dimension n. Let $\mathcal{B}$ be a finite subset of $\mathbb{P}(V)$. The following assertions are equivalent:*

(i) *$\mathcal{B}$ is a maximal projectively independent subset (with respect to inclusion) of $\mathbb{P}(V)$.*
(ii) *$\mathcal{B}$ is a minimal system of generators of $\mathbb{P}(V)$.*
(iii) *$\mathcal{B}$ is a basis of $\mathbb{P}(V)$.*

Proof It follows immediately from Theorem 6.37. □

Definition 6.39 Let $\mathcal{F} = \{P_0, P_1, \ldots, P_n, P_{n+1}\}$ be a projective frame of $\mathbb{P}(V)$. For any $u \in V$, $u \neq 0_V$, there exists a unique basis $\mathcal{B}_u = \{v_0, \ldots, v_n\}$ of $\mathbb{P}(V)$ such that

$$P_i = \langle v_i \rangle,\ i = 0, \ldots, n, \qquad P_{n+1} = \langle u \rangle, \qquad u = v_0 + \cdots + v_n. \tag{6.7}$$

Any basis $\mathcal{B}_u$ satisfying (6.7) is called a *normalized basis* of V. If $P = \langle v \rangle$ and if $v = x_0 v_0 + \cdots x_n v_n$ (i.e. $(x_0, \ldots, x_n)$ are the coordinates of v with respect to $\mathcal{B}_u$), we say that $(x_0, \ldots, x_n)$ are the *homogeneous coordinates* of P with respect to the frame $\mathcal{F}$. These coordinates are uniquely determined up to a non-zero scalar (indeed, for any $k \in K^*$ we have $\mathcal{B}_{ku} = \{kv_0, \ldots, kv_n\}$), hence by homogeneous coordinates of P we mean a homogeneous $(n+1)$-tuple $[P]_{\mathcal{F}} = [x_0, \ldots, x_n] \in \mathbb{P}^n(K)$. $[x_0, \ldots, x_n]$. In particular, we have

$$\begin{aligned} [P_0]_{\mathcal{F}} &= [1, 0, \ldots, 0], \quad [P_1]_{\mathcal{F}} = [0, 1, \ldots, 0], \quad \ldots, [P_n]_{\mathcal{F}} \\ &= [0, 0, \ldots, 1], \quad [P_{n+1}]_{\mathcal{F}} = [1, 1, \ldots, 1]. \end{aligned}$$

Once a projective frame of $\mathbb{P}(V)$ has been chosen, we can simply write $P = [x_0, \ldots, x_n]$.

Choosing a projective frame $\mathcal{F}$ is equivalent to choosing a projective isomorphism

$$\mathbb{P}(V) \to \mathbb{P}^n(K), \qquad P \to [P]_{\mathcal{F}},$$

induced by the linear isomorphism which sends $v \in V$ to its coordinates with respect to the basis $\mathcal{B}$, where $\mathcal{B}$ is any normalized basis associated with $\mathcal{F}$.

Definition 6.40 Let V and V' be two K-vector spaces with $\dim_K(V), \dim_K(V') \geq 2$. A map

$$T : \mathbb{P}(V) \to \mathbb{P}(V')$$

is called a *linear projective transformation* if there exists an injective linear map $\phi\colon V \to V'$ such that

$$T(\langle v\rangle) = \langle \phi(v)\rangle, \qquad \forall\, v \in V \setminus \{0_V\}. \tag{6.8}$$

Lemma 6.41 *The linear map which induces a linear projective transformation is determined only up to multiplication by a non-zero scalar.*

$$T_\phi = T_\psi \quad \Longleftrightarrow \quad \exists\, k \in K^* \quad \textit{such that } \phi = k\psi.$$

Proof The implication $\Longleftarrow$ is immediate. Suppose $T_\phi = T_\psi$, i.e. $\phi(v) = k_v\psi(v)$, $k_v \in K^*$ for every $v \in V$. Take a basis $\{e_i,\ i = 0, \ldots, n\}$ of V, then $\phi(e_i) = k_{e_i}\psi(e_i)$. $i = 0, \ldots, n$. Hence

$$\phi\left(\sum_{i=0}^{n} e_i\right) = \sum_{i=0}^{n} \phi(e_i) = \sum_{i=0}^{n} k_{e_i}\psi(e_i)$$

and

$$\phi\left(\sum_{i=0}^{n} e_i\right) = k_{e_0+\cdots+e_n}\psi\left(\sum_{i=0}^{n} e_i\right) = \sum_{i=0}^{n} k_{e_0+\cdots+e_n}\psi(e_i).$$

By linear independence of $\psi(e_i)$ it follows that

$$k_{e_0} = \cdots = k_{e_n} = k_{e_0+\cdots+e_n} = k.$$

Therefore for every $v = \sum_{i=0}^{n} \lambda_i e_i \in V$, we have

$$\phi(v) = \phi\left(\sum_{i=0}^{n} \lambda_i e_i\right) = \sum_{i=0}^{n} \lambda_i \phi(e_i) = \sum_{i=0}^{n} \lambda_i k\psi(e_i) = k\psi(v).$$

□

We call any such linear map ϕ *associated with* T and write $T = T_\phi$.

If ϕ is an isomorphism of vector spaces, we say that T_ϕ is a *linear projective isomorphism* of projective spaces. Two projective spaces over a field K are said to be *linearly isomorphic* if there exists a linear projective isomorphism between them. Two projective spaces over a field K are linearly isomorphic if and only if they have the same dimension. A linear projective isomorphism $T\colon \mathbb{P}(V) \to \mathbb{P}(V)$ is also called a *linear projectivity* (or a *linear projective automorphism*) of $\mathbb{P}(V)$. We denote the set of all linear automorphism of $\mathbb{P}(V)$ by $\mathrm{PGL}_K(V)$; if $V = K^{n+1}$

we use the notation $\mathrm{PGL}_n(K)$. Two subsets $\mathcal{Y}$ and $\mathcal{Z}$ of $\mathbb{P}(V)$ are said to be *linearly equivalent* if there exists a linear projectivity T of $\mathbb{P}(V)$ such that $T(\mathcal{Y}) = \mathcal{Z}$.

Proposition 6.42 *Let $T: \mathbb{P}(V) \to \mathbb{P}(V')$ and $T': \mathbb{P}(V') \to \mathbb{P}(V'')$ be two linear projective transformations. Then $T' \circ T: \mathbb{P}(V) \to \mathbb{P}(V'')$ is a linear projective transformation. If $T: \mathbb{P}(V) \to \mathbb{P}(V')$ is a linear projective isomorphism then $T^{-1}: \mathbb{P}(V') \to \mathbb{P}(V)$ is a linear projective isomorphism.*

Proof Let $T = T_\phi$ and $T' = T_{\phi'}$, then we have $T' \circ T = T_{(\phi' \circ \phi)}$. Furthermore $T^{-1} = T_{\phi^{-1}}$. □

Corollary 6.43 *The above– defined set $\mathrm{PGL}_K(V)$ is a group with respect to composition of automorphisms called the* general linear projective group *of $\mathbb{P}(V)$.*

Proposition 6.44 *Let $T: \mathbb{P}(V) \to \mathbb{P}(V')$ be a linear projective transformation. Then:*

(i) *T is an injective map.*
(ii) *If P_1, P_2 and P_3 are collinear points of $\mathbb{P}(V)$, then $T(P_1)$, $T(P_2)$ and $T(P_3)$ are collinear points of $\mathbb{P}(V')$.*
(iii) *If T is an isomorphism according to* Definition 6.40 *and $\dim \mathbb{P}(V) \geq 2$, then T is an isomorphism according to* Definition 6.21.
(iv) *If $\mathcal{Y}$ is a subspace of $\mathbb{P}(V)$, then $T(\mathcal{Y})$ is a subspace of $\mathbb{P}(V')$ having the same dimension as $\mathcal{Y}$ and $T_{|\mathcal{Y}}: \mathcal{Y} \to f(\mathcal{Y})$ is a linear projective isomorphism.*

Proof

(i) Suppose $T(P) = T(Q)$ with $P = \langle v \rangle$ and $Q = \langle w \rangle$. Then, if $T = T_\phi$, we have $\langle \phi(v) \rangle = \langle \phi(w) \rangle$ so that $\phi(v) = k\phi(w) = \phi(kw)$ for some $k \in K^*$ so that $v = kw$ and $P = Q$.
(ii) If $P_1 = \langle v_1 \rangle$, $P_2 = \langle v_2 \rangle$ and $P_3 = \langle v_3 \rangle$, then v_1, v_2, v_3 are linearly dependent vectors of V. Let $T = T_\phi$. Since ϕ is an injective linear map, $\phi(v_1)$, $\phi(v_2)$, $\phi(v_3)$ are linearly dependent, i.e. $T(P_1)$, $T(P_2)$ and $T(P_3)$ are three collinear points of $\mathbb{P}(V')$.
(iii) We have only to show that if $T(P_1)$, $T(P_2)$ and $T(P_3)$ are collinear points of $\mathbb{P}(V')$, then P_1, P_2 and P_3 are collinear points of $\mathbb{P}(V)$. Since T^{-1} is a linear projective isomorphism, by (ii) $T^{-1}(T(P_1)) = P_1$, $T^{-1}(T(P_2)) = P_2$ and $T^{-1}(T(P_3)) = P_3$ are collinear.
(iv) If $\mathcal{Y} = \mathbb{P}(W)$ and $T = T_\phi$, then $T(\mathcal{Y}) = \mathbb{P}(\phi(W))$.

□

Definition 6.45 Let V be a K-vector space of dimension $n + 1$ ($n \geq 2$) and $\mathbb{P}(V)$ the associated projective space. By Proposition 6.42 the map $T: \mathrm{GL}_K(V) \to \mathrm{Aut}(\mathbb{P}(V))$, defined by $T(\phi) = T_\phi$ is a homomorphism of groups with $\ker(T) = \{k\mathrm{id}_V \mid k \in K^*\}$ (by Lemma 6.41). Since $T(\mathrm{GL}_K(V)) = \mathrm{PGL}_K(V)$ by definition, T induces a group isomorphism $GL_K(V)/\ker(T) \cong \mathrm{PGL}_K(V)$ by Theorem 1.5.

Example 6.46 If $V = K^{n+1}$, identifying $\mathrm{GL}_K(V)$ with $\mathrm{GL}_{n+1}(K)$ (the invertible $(n+1) \times (n+1)$ matrices with entries in K), we associate to every matrix $A = (a_{ij})_{i,j=0,\dots,n}$ the linear projective isomorphism

$$T_A \colon \mathbb{P}^n(K) \to \mathbb{P}^n(K), \qquad P \to AP, \qquad \forall\, P = (x_0, \dots, x_n)^t \tag{6.9}$$

i.e.

$$T_A([x_0, x_1, \dots, x_n]) := [y_0, y_1, \dots, y_n], \quad \text{with}$$
$$y_i = \sum_{j=0}^{n} a_{ij} x_j, \quad i = 0, 1, \dots, n.$$

The relations $y_i = \sum_{j=0}^{n} a_{ij} x_j, \quad i = 0, 1, \dots, n$ are also called the *equations of* T_A. They can be written in matricial form $\varrho y^t = Ax^t$, where $x = (x_0, \dots, x_n)$, $y = (y_0, \dots, y_n)$ and $\varrho \in K^*$. In this case we denote the above groups homomorphism T by T_n; so $T_n \colon \mathrm{GL}_{n+1}(K) \to \mathrm{PGL}_n(K)$. Thus we have the isomorphism $\mathrm{GL}_{n+1}(K)/\ker(T_n) \cong \mathrm{PGL}_n(K)$.

The existence of a linear projective transformation $T \colon \mathbb{P}(V) \to \mathbb{P}(V')$ is guaranteed by the fundamental theorem of projective transformations:

Theorem 6.47 (Fundamental Theorem of Projective Transformations) *Let $\mathbb{P}(V)$ and $\mathbb{P}(V')$ be projective spaces over the field K such that $\dim \mathbb{P}(V) = n \leq \dim \mathbb{P}(V')$. Assume that $\{P_0, \dots, P_n, P_{n+1}\}$ is a projective frame of $\mathbb{P}(V)$ and that $\{Q_0, \dots, Q_n, Q_{n+1}\}$ is a projective frame of an n-dimensional subspace $\mathbb{P}(W)$ of $\mathbb{P}(V')$. Then there exists a unique linear projective transformation $T \colon \mathbb{P}(V) \to \mathbb{P}(V')$ such that $T(P_i) = Q_i$ for each $i = 0, 1, \dots, n+1$. If $\mathbb{P}(W) = \mathbb{P}(V')$, then T is a linear projective isomorphism and vice versa.*

Proof Let $P_i = \langle v_i \rangle$, $i = 0, \dots, n+1$, and $Q_i = \langle w_i \rangle$, $i = 0, \dots, n+1$, where $\{v_0, \dots, v_{n+1}\}$ is a basis of V and $\{w_0, \dots, w_{n+1}\}$ is a basis of W. By the conditions $T(P_i) = Q_i$ we have to define a linear map $\varphi \colon V \to V'$ such that $\varphi(v_i) = \lambda_i w_i$, with $\lambda_i \neq 0$, $i = 0, \dots, n+1$. Dividing $\lambda_0, \lambda_1, \dots, \lambda_n, \lambda_{n+1}$ by λ_{n+1}^{-1} (that does not change the projective transformation T we are looking for), we can suppose $\lambda_{n+1} = 1$. By a suitable choice of λ_i we can get also $T(P_{n+1}) = Q_{n+1}$. Indeed, since $\{v_0, \dots, v_n\}$ is a basis of V and $\{w_0, \dots, w_n\}$ is a basis of W we have

$$v_{n+1} = \sum_{i=0}^{n} \mu_i v_i \qquad \text{and} \qquad w_{n+1} = \sum_{i=0}^{n} \mu_i' w_i, \tag{6.10}$$

with $\mu_i, \mu_i' \neq 0$ for each $i = 0, 1, \ldots, n$ (since all $n+1$ vectors of $\{v_0, \ldots, v_{n+1}\}$ or of $\{w_0, \ldots, w_{n+1}\}$ are linearly independent). Then

$$\varphi(v_{n+1}) = \sum_{i=0}^{n} \mu_i \varphi(v_i) = \sum_{i=0}^{n} \mu_i \lambda_i w_i. = w_{n+1} \tag{6.11}$$

if we have $\mu_i' = \lambda_i \mu_i$, $i = 0, 1, \ldots, n$. We satisfy $T(P_{n+1}) = Q_{n+1}$ by choosing $\lambda_i = \dfrac{\mu_i'}{\mu_i}$, $i = 0, 1, \ldots, n$. □

Remark 6.48 In general we have a strict inclusion $\mathrm{PGL}_K(V) \subsetneq \mathrm{Aut}(\mathbb{P}(V))$, i.e. not all projective automorphism of $\mathbb{P}(V)$ (according to Definition 6.21) are linear projective automorphisms (see Chap. 8).

Definition 6.49 Let $\mathcal{S}, \mathcal{T}$ be projective subspaces of the projective space $\mathbb{P}(V)$ of dimension n such that $\mathcal{S} \cap \mathcal{T} = \varnothing$ and $\mathbb{P}(V) = \mathcal{S} + \mathcal{T}$. If $\mathcal{S} = \mathbb{P}(W)$ and $\mathcal{T} = \mathbb{P}(U)$, then $V = W \oplus U$. The projection $p_{U,W} : V \to U$ onto U along W (see Exercise 1.14) induces a map

$$\pi_{\mathcal{S}} : \mathbb{P}(V) \setminus \mathcal{S} \to \mathcal{T}, \qquad \pi_{\mathcal{S}}(\langle v \rangle) = \langle p_{U,W}(v) \rangle, \qquad \forall\, v \in V \setminus W,$$

($\langle p_{U,W}(v) \rangle = \langle W \cup \{v\} \rangle \cap U$) called *the projection onto $\mathcal{T}$ centred at $\mathcal{S}$*.

Let $\mathcal{T}_1, \mathcal{T}_2$ be two projective subspaces of the projective space $\mathbb{P}(V)$ of dimension n with $\dim(\mathcal{T}_1) = \dim(\mathcal{T}_2) = m$ and let $\mathcal{S}$ be a subspace such that

$$\mathcal{S} \cap \mathcal{T}_1 = \mathcal{S} \cap \mathcal{T}_2 = \varnothing, \qquad \dim(\mathcal{S}) = n - m - 1.$$

Then the restriction to $\mathcal{T}_1$ of the projection onto $\mathcal{T}_2$ centred at $\mathcal{S}$ is a projective isomorphism $f : \mathcal{T}_1 \to \mathcal{T}_2$, called *perspectivity centred at* $\mathcal{S}$. It is induced by the linear isomorphism $p_1 : U_1 \to U_2$, where $\mathcal{T}_2 = \mathbb{P}(U_2)$, $\mathcal{S} = \mathbb{P}(W)$, $\mathcal{T}_1 = \mathbb{P}(U_1)$ and p_1 is the restiction to U_1 of $p_{U_2,W}$. The inverse isomorphism $f^{-1} : \mathcal{T}_2 \to \mathcal{T}_1$ is a perspectivity centred at $\mathcal{S}$ and the restriction of f to $\mathcal{T}_1 \cap \mathcal{T}_2$ is the identity map.

6.2.1 *Projective Subspaces of the Standard Projective Space*

Definition 6.50 Let $\mathcal{Y}$ be a projective subspace of $\mathbb{P}^n(K)$ with $\dim \mathcal{Y} = m$. We choose a base $\{v_0, \ldots, v_m\}$ of the associated vector subspace $V_{\mathcal{Y}}$ of K^{n+1} of dimension $m+1$. Let $v_j = (a_{0j}, \ldots, a_{nj})$, $j = 0, \ldots, m$ and $\big(a_{ij}\big)_{\substack{i=0,1,\ldots,n \\ j=0,1,\ldots,m}}$, a $(n+1) \times (m+1)$-matrix of maximum rank $m+1$. Consider the linear map $\phi : K^{m+1} \to K^{n+1}$ which sends the canonical basis $\{e_0, \ldots, e_n\}$ of K^{m+1} (see

Example 1.62-(1)) to $\{v_0, \ldots, v_m\}$, i.e. $\phi(e_i) = v_i$, $i = 0, \ldots, m$. Thus $\phi(W) = V_{\mathcal{Y}}$, where $W = \langle e_0, \ldots, e_m \rangle$. Then we have

$$\phi(t_0, t_1, \ldots, t_m) = \left(\sum_{j=0}^{m} a_{0j} t_j, \sum_{j=0}^{m} a_{1j} t_j, \ldots, \sum_{j=0}^{m} a_{nj} t_j \right).$$

The linear map ϕ induces a projective transformation $f : \mathbb{P}^m(K) \to \mathbb{P}^n(K)$ defined by

$$f([t_0, t_1, \ldots, t_m]) = \left[\sum_{j=0}^{m} a_{0j} t_j, \sum_{j=0}^{m} a_{1j} t_j, \ldots, \sum_{j=0}^{m} a_{nj} t_j \right]$$

such that $f(\mathbb{P}^m(K)) = \mathcal{Y}$. The components of the analytic representation of f

$$x_i = \sum_{j=0}^{m} a_{ij} t_j, \quad i = 0, 1, \ldots, n, \quad \forall [t_0, t_1, \ldots, t_m] \in \mathbb{P}^m(K), \tag{6.12}$$

are called *parametric equations* of $\mathcal{Y}$. As in the affine case, the parametric equations of a projective subspace $\mathcal{Y}$ are not uniquely determined, indeed they depend on the choice of a base for $V_{\mathcal{Y}}$.

Example 6.51 Let $A_0 = [a_{00}, a_{10}, \ldots, a_{n0}]$ and $A_1 = [a_{01}, a_{11}, \ldots, a_{n1}]$ be two distinct points of $\mathbb{P}^n(K)$ and $\mathcal{Y}$ be the line $A_0 A_1$. Then its parametric equations are given by

$$x_i = a_{i0} t_0 + a_{i1} t_1, \quad i = 0, 1, \ldots, n, \quad \forall [t_0, t_1] \in \mathbb{P}^1(K), \tag{6.13}$$

We can choose $\{(a_{00}, a_{10}, \ldots, a_{n0}), (a_{01}, a_{11}, \ldots, a_{n1})\}$ as a basis of $V_{\mathcal{Y}}$.

Definition 6.52 Let $m + 1 \geq 2$ projectively independent points $P_i = [p_{i0}, p_{i1}, \ldots, p_{in}] \in \mathbb{P}^n(K)$, $i = 0, 1, \ldots, m$. Consider the subspace $\mathcal{Y} = P_0 + P_1 + \cdots + P_m$ and the matrix

$$M = \begin{pmatrix} x_0 & x_1 & \cdots & x_n \\ p_{00} & p_{01} & \cdots & p_{0n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m0} & p_{m1} & \cdots & p_{mn} \end{pmatrix}.$$

We have the following equivalences:

$$[x_0, \ldots, x_n] \in \mathcal{Y} \iff (x_0, \ldots, x_n) \in V_{\mathcal{Y}} \iff \operatorname{rank} M = m + 1.$$

By Proposition 1.25 (Kronecker's rule) $\operatorname{rank} M = m + 1$ if and only if M contains an invertible $(m+1) \times (m+1)$ submatrix M' such that the determinant of every $(m+2) \times (m+2)$ submatrix of M containing M' vanishes; since there are $n - m$ such matrices, this provides a *Cartesian representation* (or *implicit representation*) of $\mathcal{Y}$ with $n - m$ homogeneous linear equations.

If $m = n - 1$, the hyperplane $\mathcal{Y} = P_1 + \cdots + P_{n-1}$ has the equation

$$\begin{vmatrix} x_0 & \cdots & x_n \\ p_{00} & \cdots & p_{0n} \\ \vdots & \vdots & \vdots \\ p_{n-1\,0} & \cdots & p_{n-1\,n} \end{vmatrix} = 0, \qquad \text{i.e.} \qquad \alpha_0 x_0 + \cdots + \alpha_n x_n = 0,$$

where

$$\alpha_j = (-1)^j \begin{vmatrix} p_{00} & \cdots & \widehat{p_{0j}} & \cdots & p_{0n} \\ \vdots & & \vdots & & \vdots \\ p_{n-1\,0} & \cdots & \widehat{p_{n-1\,j}} & \cdots & p_{n-1\,n} \end{vmatrix}, \qquad j = 0, \ldots, n,$$

where $\widehat{\ }$ means that the j-th column is deleted.

We also have the converse result.

Proposition 6.53 *For every matrix* $(c_{ij})_{\substack{i=1,\ldots,n-m \\ j=0,1,\ldots,n}} \in \mathcal{M}_{n-m,n+1}(K)$ *of rank* $n - m$, *the system of homogeneous linear equations*

$$\sum_{j=0}^{n} c_{ij} x_j = 0, \qquad i = 1, \ldots, n - m$$

represents a projective subspace of $\mathbb{P}^n(K)$ *of dimension* m.

Proof Use Corollary 1.65. □

Remark 6.54 It is obvious that Cartesian equations of a projective subspace are not uniquely determined. Indeed, different choices of projectively independent points (to generate a subspace) produce different homogeneous linear systems. However, if $F = a_0 x_0 + a_1 x_1 + \cdots + a_n x_n$ and $G = b_0 x_0 + b_1 x_1 + \cdots + a_n x_n$ are two homogeneous polynomials of degree 1 in the indeterminates $x_0, x_1, \ldots, x_n$ over a field K ($n \geq 2$), then

$$\left\{ [x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K) \;\middle|\; \sum_{i=0}^{n} a_i x_i = 0 \right\}$$

$$= \left\{ [y_0, y_1, \ldots, y_n] \in \mathbb{P}^n(K) \;\middle|\; \sum_{i=0}^{n} b_i y_i = 0 \right\} \tag{6.14}$$

if and only if there exists $c \in K \setminus \{0\}$ such that $F = cG$ (i.e. $a_i = cb_i$, $i = 0, 1, \ldots, n$).

Suppose that (6.14) is true and let $0 \leq i < j \leq n$ be two arbitrary indices such that $a_i \neq 0$ or $a_j \neq 0$. Then the $(0, \ldots, 0, a_j, 0, \ldots, 0, -a_i, 0, \ldots, 0)$ (with a_j at the place i and $-a_i$ at the place j) is a zero of F. Hence it also is a zero of G, i.e. $b_i a_j - b_j a_i = 0$. Since i and j are arbitrary, it follows that $a_0, a_1, \ldots, a_n$ and $b_0, b_1, \ldots, b_n$ are proportional, i.e. $a_i = cb_i$, $i = 0, 1, \ldots, n$ with $c \neq 0$. □

Hence if $\mathcal{Y}$ is a hyperplane of $\mathbb{P}^n(K)$ defined by the equation $c_0x_0 + c_1x_1 + \cdots + c_nx_n = 0$, any other equation of $\mathcal{Y}$ is of the form $kc_0x_0 + kc_1x_1 + \cdots + kc_nx_n = 0$ for $k \in K^*$. Thus we are allowed to say that $c_0x_0 + c_1x_1 + \cdots + c_nx_n = 0$ is *the equation* of $\mathcal{Y}$. Therefore there is a bijective correspondence between the set of all hyperplanes of $\mathbb{P}^n(K)$ and ${}_hK[X_0, \ldots, X_n]_1/\sim$ (see Example 1.33-(2)), where $F \sim G$ if and only if $F = kG$ for some $k \in K^*$.

6.3 Dual Projective Space and Projective Duality

Definition 6.55 Let V be a K-vector space of dimension $n + 1 \geq 3$ and denote by V^* its dual space. Since $\dim_K(V) = \dim_K(V^*)$, V is isomorphic to V^* but there is no canonical isomorphism $V \to V^*$. However, if we fix a basis $\mathcal{B} = \{v_0, \ldots, v_n\}$ of V and take the dual basis $\mathcal{B}^* = \{v_0^*, \ldots, v_n^*\}$ we have the isomorphism (depending only on the choice of $\mathcal{B}$ of V) we might call "semi-canonical"

$$\phi_{\mathcal{B}} : V \to V^*, \qquad \phi_{\mathcal{B}}\left(\sum_{i=0}^{n} a_i v_i\right) = \sum_{i=0}^{n} a_i v_i^*, \qquad a_i \in K,\ i = 0, \ldots, n. \tag{6.15}$$

The projective space $\mathbb{P}(V^*)$ is called the *dual projective space*; if $V = K^{n+1}$ we will denote $\mathbb{P}(V^*)$ by $\mathbb{P}^n(K)^*$. The isomorphism $\phi_{\mathcal{B}}$ induces an isomorphism of projective spaces $T_{\mathcal{B}} : \mathbb{P}(V) \to \mathbb{P}(V^*)$ according to Definition 6.40.

Notation Let V be a K-vector space of dimension $n + 1 \geq 3$. We denote the set of m-dimensional projective subspaces of $\mathbb{P}(V)$ by $\mathcal{S}_m(\mathbb{P}(V))$, where $0 \leq m \leq n$. For instance $\mathcal{S}_0(\mathbb{P}(V))$ coincides with $\mathbb{P}(V)$ as a set; $\mathcal{S}_{n-1}(\mathbb{P}(V))$ is the set of hyperplanes of $\mathbb{P}(V)$, while $\mathcal{S}_n(\mathbb{P}(V)) = \{\mathbb{P}(V)\}$. The set

$$\mathcal{S}(\mathbb{P}(V)) = \bigcup_{m=0}^{n} \mathcal{S}_m(\mathbb{P}(V))$$

is clearly a lattice partially ordered by inclusion.

Theorem 6.56 *Assume the above hypothesis and notation. There exists a bijective correspondence*

$$\alpha : \mathbb{P}(V^*) \to \mathcal{S}_{n-1}(\mathbb{P}(V)) \tag{6.16}$$

such that A_1, A_2 and A_3 are collinear points of $\mathbb{P}(V^*)$ if and only if $\dim(\alpha(A_1)) \cap \alpha(A_2)) \cap \alpha(A_3)) \geq n-2$.

Proof By the projective Grassmann formula (Corollary 6.33) for three hyperplanes H_1, H_2 and H_3 of $\mathbb{P}(V)$ we have

$$\dim(H_1 \cap H_2 \cap H_3) \geq n-3$$

(we have $=$ if H_1, H_2 and H_3 are pairwise distinct and $H_1 \cap H_2 \not\subset H_3$).

Let $f \in V^* \setminus \{0\}$, then $W_f := \mathrm{Ker}(f)$ is a vector subspace of V of dimension n. By Theorem 6.31 there exists a unique hyperplane $\mathcal{Y}_f$ $(= \mathbb{P}(W_f))$ of $\mathbb{P}(V)$ such that $V_{\mathcal{Y}_f} = W_f$. Note that if $g = \lambda \cdot f$, with $\lambda \in K \setminus \{0\}$, then $\mathrm{Ker}(f) = \mathrm{Ker}(g)$, i.e. $W_f = W_g$ or also $\mathcal{Y}_f = \mathcal{Y}_g$. Thus we obtain a map

$$\alpha \colon \mathbb{P}(V^*) \to \mathcal{S}_{n-1}(\mathbb{P}(V)), \quad \alpha(\langle f \rangle) := \mathcal{Y}_f \tag{6.17}$$

(as usual, $\langle f \rangle$ is the subspace of V^* generated by f).

Conversely, let $\mathcal{Y}$ be a hyperplane of $\mathbb{P}(V)$ and $V_{\mathcal{Y}}$ be the n-dimensional vector subspace of V given by Theorem 6.31. The annihilator $V_{\mathcal{Y}}^\circ$ of $V_{\mathcal{Y}}$ has dimension 1 so that there exists a linear functional $f_{\mathcal{Y}} \neq 0_{V*}$ such that $V_{\mathcal{Y}}^\circ = \langle f_{\mathcal{Y}} \rangle$. Thus we are able to define the map

$$\beta \colon \mathcal{S}_{n-1}(\mathbb{P}(V)) \to \mathbb{P}(V^*), \qquad \beta(\mathcal{Y}) := \langle f_{\mathcal{Y}} \rangle.$$

It follows immediately that $\alpha \circ \beta = \mathrm{id}$ and $\beta \circ \alpha = \mathrm{id}$, so that α is bijective.

We have to prove the last assertion. Suppose $A_i = \langle f_i \rangle$, with $f_i \in V^*$, $f_i \neq 0_{V*}$, $i = 1, 2, 3$. Then A_1, A_2 and A_3 are collinear if and only if f_1, f_2 and f_3 are linearly dependent. By Theorem 6.31 we have

$$\begin{aligned}
\dim(\alpha(A_1)) \cap \alpha(A_2)) \cap \alpha(A_3)) &= \dim(\mathcal{Y}_{f_1} \cap \mathcal{Y}_{f_2} \cap \mathcal{Y}_{f_3}) \\
&= \dim(\mathbb{P}(\ker f_1) \cap \mathbb{P}(\ker f_2) \cap \mathbb{P}(\ker f_3)) \\
&= \dim(\mathbb{P}((\ker f_1) \cap (\ker f_2) \cap (\ker f_3)).
\end{aligned}$$

Therefore $\dim(\alpha(A_1)) \cap \alpha(A_2)) \cap \alpha(A_3)) \geq n-2$ if and only if $\dim((\ker f_1) \cap (\ker f_2) \cap (\ker f_3)) \geq n-1$. This last inequality holds if and only if f_1, f_2 and f_3 are linearly dependent (we let the reader show this last equivalence). □

Remark 6.57 We have just seen (Theorem 6.56) that the dual projective space $\mathbb{P}(V^*)$ induces a structure of projective space on the set $\mathcal{S}_{n-1}(\mathbb{P}(V))$ of all hyperplanes of $\mathbb{P}(V)$. Thus the map $\alpha \colon \mathbb{P}(V^*) \to \mathcal{S}_{n-1}(\mathbb{P}(V))$ is a canonic isomorphism of projective spaces so that $\mathbb{P}(V^*)$ and $\mathcal{S}_{n-1}(\mathbb{P}(V))$ can be identified.

The composition of maps $\alpha_{\mathcal{B}} = \alpha \circ T_{\mathcal{B}}$ gives the isomorphism of projective spaces

$$\alpha_{\mathcal{B}} \colon \mathbb{P}(V) \to \mathcal{S}_{n-1}(\mathbb{P}(V)), \qquad \alpha_{\mathcal{B}}([a_0, \ldots, a_n]) = \mathcal{Y}_{[a_0, a_1, \ldots, a_n]}, \tag{6.18}$$

where $\mathcal{Y}_{[a_0,a_1,\ldots,a_n]}$ is the hyperplane of $\mathbb{P}(V)$ of equation $a_0x_0+a_1x_1+\cdots+a_nx_n = 0$ (here $x_0,\ldots,x_n$ are the projective coordinates on $\mathbb{P}(V)$ associated with the base $\mathcal{B}$).

Definition 6.58 Let $\dim \mathbb{P}(V) = n$ and $\mathcal{Y}$ be a subspace of $\mathbb{P}(V)$ of dimension m, i.e. $\mathcal{Y} = \mathbb{P}(W)$, where $\dim_K(W) = m+1$ (Theorem 6.31). The annihilator W° of W (Definition 1.106) has dimension $n-m$ (Corollary 1.107) as a subspace of $\mathbb{P}(V^*)$. Thus we have the natural map, called *duality correspondence*,

$$\delta\colon \mathcal{S}(\mathbb{P}(V)) \to \mathcal{S}(\mathbb{P}(V^*)), \qquad \delta(\mathcal{Y}) = \mathbb{P}(W^\circ), \tag{6.19}$$

which reverses inclusions. From Proposition 1.107 we get immediately the following relations:

$$\begin{aligned} &\dim(\delta(\mathcal{Y})) = n-m-1, \qquad \delta(\mathcal{Y}_1\cap\mathcal{Y}_2) = \delta(\mathcal{Y}_1)+\delta(\mathcal{Y}_2),\\ &\delta(\mathcal{Y}_1+\mathcal{Y}_2) = \delta(\mathcal{Y}_1)\cap\delta(\mathcal{Y}_2). \end{aligned} \tag{6.20}$$

In particular we have $\delta(\varnothing) = \mathbb{P}(V^*)$ and $\delta(\mathbb{P}(V)) = \varnothing$. When $m = n-1$ we recover the bijective correspondence between hyperplanes of $\mathbb{P}(V))$ and points of $\mathbb{P}(V^*)$ and when $m = 0$ between points of $\mathbb{P}(V)$ and hyperplanes of $\mathbb{P}(V^*)$. We have a bijective correspondence between the set of hyperplanes containing a subspace $\mathcal{Y}$ of $\mathbb{P}(V)$ and the points of $\delta(\mathcal{Y})$.

The isomorphism $T_{\mathcal{B}}^{-1}\colon \mathbb{P}(V^*) \to \mathbb{P}(V)$ induces a bijective map

$$(T_{\mathcal{B}}^{-1})_{\mathcal{S}}\colon \mathcal{S}(\mathbb{P}(V^*)) \to \mathcal{S}(\mathbb{P}(V)), \qquad (T_{\mathcal{B}}^{-1})_{\mathcal{S}}(\mathcal{Y}) = T_{\mathcal{B}}^{-1}(\mathcal{Y}), \qquad \forall\, \mathcal{Y}\in\mathcal{S}(\mathbb{P}(V^*)),$$

so that we have the bijection

$$\delta_{\mathcal{B}}\colon \mathcal{S}(\mathbb{P}(V)) \to \mathcal{S}(\mathbb{P}(V)), \quad \delta_{\mathcal{B}} = (T_{\mathcal{B}}^{-1})_{\mathcal{S}}\circ\delta, \tag{6.21}$$

which transforms subspaces of dimension m of $\mathbb{P}(V)$ into subspaces of *dual dimension* $n-m-1$ of $\mathbb{P}(V)$. More explicitly, if $\mathcal{Y} = P_0+\cdots+P_m$, where $P_0,\ldots,P_m$ are $m+1$ projectively independent points, and $P_i = [p_{i0},\ldots,p_{in}]$, $i = 0,\ldots,m$ with respect to the base $\mathcal{B}$, then, by (6.20), we have

$$\delta_{\mathcal{B}}(\mathcal{Y}) = \delta_{\mathcal{B}}(P_0)\cap\cdots\cap\delta_{\mathcal{B}}(P_m)$$

so that $\delta_{\mathcal{B}}(\mathcal{Y})$ is described by the system of homogeneous linear equations

$$\sum_{j=0}^{n} p_{ij}x_j = 0, \qquad i = 0,\ldots,m. \tag{6.22}$$

The *duality principle*, we are going to explain, was discovered by geometers of École Polytechnique of Paris about two centuries ago. This principle, which does

not hold in affine geometry, is very important in projective geometry since one can deduce new results from known ones immediately.

Theorem 6.59 (Duality Principle) *Let $\mathbb{P}(V)$ be a projective space of dimension n. Let $\mathcal{P}$ be a statement about subspaces of $\mathbb{P}(V)$, their intersections, joins, and dimensions. If $\mathcal{P}^*$ is the* dual statement *obtained from $\mathcal{P}$ by replacing the words* intersection, join, contained, containing, dimension *by* join, intersection, containing, contained, dual dimension *respectively, then $\mathcal{P}$ holds if and only if $\mathcal{P}^*$ holds.*

Proof It follows immediately from the properties of δ ((6.20)). □

Examples 6.60 Here are some assertions together with their dual assertion.

$\mathcal{P}$: Every line of $\mathbb{P}(V)$ (with $\dim \mathbb{P}(V) = 2$) contains at least three distinct points.
$\mathcal{P}^*$: For every point P of $\mathbb{P}(V)$ there are at least three distinct lines passing through P.
$\mathcal{P}_1$: The lines d and e of $\mathbb{P}(V)$ (with $\dim \mathbb{P}(V) = 3$) are concurrent at a point P.
$\mathcal{P}_1^*$: The lines $\delta(d)$ and $\delta(e)$ are contained in the plane $\delta(P)$.
$\mathcal{P}_2$: If the line d of $\mathbb{P}(V)$ (with $\dim \mathbb{P}(V) = 3$) does not pass through the point P there exists a unique plane π containing $d \cup \{P\}$.
$\mathcal{P}_2^*$: If the line $\delta(d)$ is not contained in the plane $\delta(P)$, then $\delta(d) \cap \delta(P) = \delta(\pi)$.

Let $\dim \mathbb{P}(V) = n \geq 2$.

$\mathcal{P}_3$: Two distinct points of $\mathbb{P}(V)$ generate a line.
$\mathcal{P}_3^*$: Two distinct hyperplanes meet at a subspace of dimension $n - 2$.
$\mathcal{P}_4$: n points of $\mathbb{P}(V)$ generate a subspace of dimension $\leq n - 1$.
$\mathcal{P}_4^*$: n hyperplanes meet at least at a point.
$\mathcal{P}_5$: n projectively independent points generate a hyperplane.
$\mathcal{P}_5^*$: n projectively independent hyperplanes (as points of the projective space $\mathcal{S}_{n-1}(\mathbb{P}(V))$) meet exactly at one point.

The duality principle allows us to apply the concepts of projective geometry of $\mathbb{P}(V)$ to the sets whose elements are subspaces (of fixed dimension) of $\mathbb{P}(V)$ that is very useful in projective geometry of higher dimensions. For instance, we shall see in Chap. 9 how the cross-ratio of 4 collinear points is immediately defined for 4 coplanar lines or for 4 hyperplanes passing through a subspace of codimension 2. The reader can find further developments in the beautiful book by Enriques [11].

Remark 6.61 The concepts of dual projective space and duality can be defined also for the general projective spaces of Definition 7.1. The reader can find a thorough exposition in [1] and in [11].

Remark 6.62 The canonical isomorphism $\alpha_V : V \to V^{**}$ (see Definition 1.98) induces a canonical isomorphism of projective spaces

$$\overline{\alpha}_V : \mathbb{P}(V) \to \mathbb{P}(V^{**}), \qquad \overline{\alpha}_V(\langle v \rangle) = \langle \alpha(v) \rangle, \qquad \forall\, v \in V, \tag{6.23}$$

where $\mathbb{P}(V^{**})$ is the dual projective space of $\mathbb{P}(V^*)$.

Proposition 6.63 *Let* $\dim \mathbb{P}(V) = n$ *and* $\mathcal{Y}$ *be a subspace of* $\mathbb{P}(V)$ *of dimension* m, $0 \leq m \leq n-1$. *If the equations of* $\mathcal{Y}$ *with respect to a projective coordinate system* $\mathcal{B}$ *of* $\mathbb{P}(V)$ *are*

$$F_i := \sum_{j=0}^{n} c_{ij} x_j = 0, \qquad i = 1, 2, \ldots, n-m.$$

and $\mathcal{Z}$ *is a hyperplane of* $\mathbb{P}(V)$ *containing* $\mathcal{Y}$, *then there exists* $(\lambda_1, \ldots, \lambda_{n-m}) \in K^{n-m} \setminus (0, \ldots, 0)$ *such that* $\mathcal{Z}$ *has an equation of the form*

$$\lambda_1 F_1 + \cdots + \lambda_{n-m} F_{n-m} = 0. \tag{6.24}$$

Proof We have just seen that $\delta(\mathcal{Y})$ is a subspace of $\mathbb{P}(V^*)$ of dimension $n-m-1$ and that $\{\mathcal{Z} \in \mathcal{S}_{n-1}(\mathbb{P}(V)) \mid \mathcal{Y} \subset \mathcal{Z}\}$ is in a bijective correspondence with $\delta(\mathcal{Y})$. On the other hand, the map

$$f : \mathbb{P}^{n-m-1}(K) \to \mathbb{P}(V^*) \qquad f([\lambda_1, \ldots, \lambda_{n-m}]) = \lambda_1 F_1 + \cdots + \lambda_{n-m} F_{n-m}$$

is a projective transformation such that $f(\mathbb{P}^{n-m-1}(K)) = \mathcal{V} \subset \delta(\mathcal{Y})$. Since $\dim \mathcal{V} = \dim(\delta(\mathcal{Y}))$, we get $\mathcal{V} = \delta(\mathcal{Y})$. □

Definition 6.64 Let S_r be a r-dimensional subspace of $\mathbb{P}(V)$, $r \leq n-1$. Let s be an integer such that $r \leq s \leq n-1$. The set $\Phi_s(S_r) = \{S_s : S_s \supset S_r\}$ of all s-dimensional subspaces of $\mathbb{P}(V)$ containing S_r is called *the star* of s-dimensional subspaces with *base locus* S_r. The union $\Phi(S_r) = \bigcup_{s=r+1}^{n-1} \Phi_s(S_r)$ is the *star* with *base locus* S_r. The star $\Phi_{n-1}(S_r)$ of all hyperplanes passing through S_r (which coincides with $\delta(S_r)$) is called a (*projective*) *linear system* with *base locus* S_r. If $r = n-2$, $\Phi_{n-1}(S_r)$ is also called the *pencil* of hyperplanes passing through S_r.

Definition 6.65 Let V be a K-vector space of dimension $n+1 \geq 3$. The $\mathbb{G}(r, V)$ of r-dimensional vector subspaces M of L is called a *Grassmann variety*. In particular we have $\mathbb{G}(1, V) = \mathbb{P}(V)$ and the bijective canonical map

$$\mathbb{G}(n, V) \longleftrightarrow \mathbb{G}(1, V^*) = \mathbb{P}(V^*), \quad M \longleftrightarrow M^0. \tag{6.25}$$

Let $S_r = \mathbb{P}(W)$ be a fixed subspace of $\mathbb{P}(V)$ of dimension $0 \leq r < n-1$. Let s be an integer with $r < s \leq n-1$. If $S_s \supset S_r$, then $S_s = \mathbb{P}(U)$ with $U \supset W$ (so that $U^0 \subset W^0$). Hence, by duality, $\Phi_s(S_r)$ can be seen as the set of subspaces $\delta(S_s) = \mathbb{P}(U^0)$, $\delta(S_s) \subset \delta(S_r) = \mathbb{P}(W^0)$. Therefore there is a bijective correspondence

$$\Phi_s(S_r) \longleftrightarrow \mathbb{G}(n-s, W^0) \tag{6.26}$$

between $\Phi_s(S_r)$ and the set of $(n-s)$-dimensional vector subspaces of V^* contained in W^0 (which has dimension $n-r$). From (6.26) we get

$$\Phi_{n-1}(S_r) \longleftrightarrow \mathbb{G}(1, W^0) \longleftrightarrow \mathbb{P}(W^0) \longleftrightarrow \mathbb{P}^{n-r-1}(K). \tag{6.27}$$

Now let K be an infinite field and V be a K-vector space of dimension $n+1$. We say that $\mathbb{P}(V)$ has ∞^n *points* since every point is uniquely determined by a homogeneous $(n+1)$-tuple of parameters. In particular the projective line $\mathbb{P}^1(K)$ has ∞^1 points, the projective plane $\mathbb{P}^2(K)$ has ∞^2 points and so on. We can also say that in $\mathbb{P}(V)$ there are ∞^{n-r-1} hyperplanes passing through a subspace S_r. Thus we shall call $\Phi_{n-1}(S_r)$ (and every set in bijective correspondence with $\mathbb{P}^{n-r-1}(K)$) *form of* $(n-r-1)^{th}$ *kind*. The following sets are forms of the first kind:

1. A projective line.
2. A pencil of hyperplane passing through a subspace S_{n-2} of $\mathbb{P}(V)$. In particular a pencil of lines in $\mathbb{P}^2(K)$ passing through a point and a pencil of planes in $\mathbb{P}^3(K)$ containing a line.

Forms of the second kind are:

1. A projective plane.
2. A linear system of planes in $\mathbb{P}^3(K)$ passing through a point.

Let $1 \leq m < n$ be an integer and $S_{m-1} = \mathbb{P}(W)$ a subspace of $\mathbb{P}(V)$ having dimension $m-1$. From (6.26) we have a bijective correspondence

$$\Phi_m(S_{m-1}) \longleftrightarrow \mathbb{G}(n-m, W^0),$$

where $\dim W^0 = n+1-m$. Hence by (6.25) we get

$$\mathbb{G}(n-m, W^0) = \mathbb{G}(n+1-m-1, W^0) \longleftrightarrow \mathbb{P}((W^0)^*).$$

Therefore there are ∞^{n-m} subspaces S_m containing a fixed S_{m-1} in $\mathbb{P}(V)$. In particular, a star $\Phi_1(S_0)$ of lines of $\mathbb{P}^3(K)$ with base locus a point S_0 is a form of the second kind.

6.4 Exercises

Exercise 6.1 Show that a projective plane $\mathcal{X}$ contains m^2-m+1 exactly if there exists in $\mathcal{X}$ a line consisting of m points. Furthermore, prove that exactly m distinct lines pass through any point $P \in \mathcal{X}$.

Exercise 6.2 Under the assumptions of Exercise 1.1 calculate the number of elements of the following set: $\{(A, B, C) \in \mathcal{X}\times\mathcal{X}\times\mathcal{X} \mid \text{with } A,\ B,\ C \text{ non-collinear}\}$.

Exercise 6.3 Let K be a finite field with p^k elements, where p is a prime number and $k \geq 1$.

(a) Prove that $\mathbb{P}^n(K)$ (with $n \geq 2$) consists of $\dfrac{p^{k(n+1)} - 1}{p - 1}$ points.

(b) Calculate the number of the lines and the planes of $\mathbb{P}^3(K)$.

Exercise 6.4 Show that $\mathbb{P}^2(\mathbb{Z}_2)$ is isomorphic to Fano's plane.

Exercise 6.5 Prove that $\mathbb{P}^2(\mathbb{R})$ is isomorphic to the projective plane of Example 6.8.

Exercise 6.6 Let A and B be two points of a projective space $\mathbb{P}(V)$ of dimension ≥ 2. Prove that there exists a hyperplane of $\mathbb{P}(V)$ passing through A not containing B.

Exercise 6.7 Let $\{E_0, E_1 \ldots, E_n\}$ be a basis of a projective space $\mathbb{P}(V)$ of dimension $n \geq 2$. For every $i = 0, 1, \ldots, n$ denote the hyperplane $E_0 + \cdots + \widehat{E_i} + \cdots + E_n$ (where $\widehat{E_i}$ means that E_i is omitted) by $\mathcal{X}_i$. Prove that $\bigcap_{i=0}^{n} \mathcal{X}_i = \varnothing$. (Hint: induction on n.)

Exercise 6.8 Let $\mathbb{P}(V)$ be a projective space of dimension $n \geq 2$. Let $\mathcal{Y}_0, \mathcal{Y}_1, \ldots, \mathcal{Y}_n$ be an $n+1$ hyperplane such that $\mathcal{Y}_0 \cap \mathcal{Y}_1 \cap \ldots \cap \mathcal{Y}_n = \varnothing$. Show that $\mathcal{X} \neq \mathcal{Y}_0 \cup \mathcal{Y}_1 \cup \cdots \cup \mathcal{Y}_n$. (Hint: induction on n.)

Exercise 6.9 Let P, Q be two distinct points of a projective space $\mathbb{P}(V)$ of dimension ≥ 2. Prove that there exists a hyperplane $\mathcal{Z}$ of $\mathbb{P}(V)$ such that $P, Q \notin \mathcal{Z}$.

Exercise 6.10 Let $\mathcal{Z}$ be a subspace of $\mathbb{P}(V)$ such that $\dim \mathcal{Z} \leq \dim \mathbb{P}(V) - 2$, and $P, Q \in \mathbb{P}(V) \setminus \mathcal{Z}$. Prove that there exists a hyperplane $\mathcal{Y}$ of $\mathbb{P}(V)$ containing $\mathcal{Z}$ and such that $P, Q \notin \mathcal{Y}$.

Exercise 6.11 Let $\mathbb{P}(V)$ be a projective space and $\mathcal{Y}$ be a subspace of dimension $k \geq 0$.

(i) Show that there exists a subspace $\mathcal{Z}$ of $\mathbb{P}(V)$ with $\dim \mathcal{Z} = n - k - 1$ such that $\mathcal{Y} \cap \mathcal{Z} = \varnothing$.

(ii) Prove that for every i, $0 \leq i \leq k$, and for every subspace $\mathcal{Z}$ of dimension $n - k + i$ one has $\dim(\mathcal{Y} \cap \mathcal{Z}) \geq i$. Furthermore, there exists a subspace $\mathcal{Z}$ of dimension $n - k + i$ such that $\dim(\mathcal{Y} \cap \mathcal{Z}) = i$.

In both cases (i) and (ii) we have $\mathcal{Y} + \mathcal{Z} = \mathbb{P}(V)$.

Exercise 6.12 Let $\mathcal{Y}$ be a subspace of $\mathbb{P}(V)$ whose of dimension $\dim(\mathbb{P}(V)) - 2$. Prove that there are at least three distinct hyperplanes of $\mathbb{P}(V)$ containing $\mathcal{Y}$.

Exercise 6.13 Let d and d' be two arbitrary lines of $\mathbb{P}(V)$ ($\dim(\mathbb{P}(V)) \geq 2$). Prove that there exists a projective isomorphism $f : \mathbb{P}(V) \to \mathbb{P}(V)$ such that $f(d) = d'$.

Exercise 6.14 Let $\mathbb{P}(V)$ be a projective space of dimension $n \geq 2$ containing a line d consisting exactly of m points ($m \geq 3$). Prove that

$$\#(\mathbb{P}(V)) = \frac{(m-1)^{n+1} - 1}{m-2}.$$

(Hint: Theorem 6.26.)

Exercise 6.15 Let $\mathbb{P}(V)$ be a projective space of dimension $n \geq 2$. Suppose that there exists a line d in $\mathbb{P}(V)$ consisting of an infinite number of points. Prove that $\mathbb{P}(V)$ is not a finite union of hyperplanes.

Exercise 6.16 Let $\dim(\mathbb{P}(V)) = 3$. Let d and d' be two skew lines of $\mathbb{P}(V)$ and $A \in \mathbb{P}(V) \setminus (d \cup d')$. Show that there exists a unique line e through A which meets both d and d' (see Exercise 3.24). Give the dual statement of the above assertion.

Exercise 6.17 Let d_1, d_2 and d_3 be three lines in a projective space $\mathbb{P}(V)$ of dimension 4, no two of them being coplanar. Prove that there exists a line of $\mathbb{P}(V)$ which meets d_1, d_2 and d_3. Is such a line unique?

Exercise 6.18 Prove that in a projective space $\mathbb{P}(V)$ of dimension 5 there exist three lines d_1, d_2 and d_3 such that no other line of $\mathbb{P}(V)$ meets d_1, d_2 and d_3.

Exercise 6.19 Give the dual statement of the following assertions:

(i) Two arbitrary planes of $\mathbb{P}(V)$ with $\dim(\mathbb{P}(V)) = 4$ have a non-empty intersection.
(ii) Two distinct planes of $\mathbb{P}(V)$ with $\dim(\mathbb{P}(V)) = 4$ have a line in common.

Exercise 6.20 Let K be an infinite field and $\mathbb{P}(V)$ be a projective space of dimension $n \geq 2$. For every finite subset A of $\mathbb{P}(V)$ there exists a hyperplane H such that $H \cap A = \varnothing$. What is the dual statement?.

Exercise 6.21 Let r and s be two lines of $\mathbb{P}^2(K)$. Prove the following assertions:

(i) If $r \neq s$ and $P = r \cap s$, then a projectivity $\varphi\colon r \to s$ is a perspectivity if and only if $\varphi(P) = P$.
(ii) Every projectivity $\varphi\colon r \to s$ is a composition of at least two perspectivities if $r \neq s$ and of at least three perspectivities if $r = s$.

Chapter 7
Desargues' Axiom

Desargues' axiom (and Pappus' axiom) are extremely important configuration properties as will come clear in the next chapter (Remark 8.15). This chapter is devoted to Desargues' axiom, while Pappus' axiom will be treated in Chap. 8.

Let $(\mathcal{X}, \mathcal{D})$ be a projective space according to Definition 7.1. *Desargues' axiom* asserts:

(P_5) For every three lines d_1, d_2, d_3 of $(\mathcal{X}, \mathcal{D})$ concurrent at a point O and for every pair of distinct points $A_i, B_i \in d_i \setminus \{O\}$, $i = 1, 2, 3$, the points $D_{12} := A_1A_2 \cap B_1B_2$, $D_{13} := A_1A_3 \cap B_1B_3$, $D_{23} := A_2A_3 \cap B_2B_3$ are collinear (see Fig. 7.1).

In other words, if the lines joining corresponding vertices of two triangles are concurrent (in Fig. 7.1 $\triangle A_1A_2A_3$ and $\triangle B_1B_2B_3$, then the intersections of corresponding pairs of sides are collinear.

Theorem 7.1 (Desargues) *The projective space $\mathcal{X} = \mathbb{P}(V)$ satisfies Desargues' axiom (P_5).*

Proof

Case 1: Suppose that d_1, d_2 and d_3 are not coplanar, hence $\dim(d_1 + d_2 + d_3) = 3$. We can observe that $D_{12} \in A_1A_2A_3 \cap B_1B_2B_3$. Analogously, $D_{13}, D_{23} \in A_1A_2A_3 \cap B_1B_2B_3$. Since A_i and B_i are distinct for $i = 1, 2, 3$, and the subspace $A_1A_2A_3 + B_1B_2B_3 = d_1 + d_2 + d_3$ has dimension 3, by Grassmann's formula (Corollary 6.33) the intersection between the planes $A_1A_2A_3$ and $B_1B_2B_3$ is a line d which passes through D_{12}, D_{13} and D_{23} so our assertion is proved.

Case 2: Now suppose that d_1, d_2 and d_3 are coplanar, contained in the plane $\mathcal{P} := d_1 + d_2 + d_3$. If A_1, A_2, A_3 (or B_1, B_2, B_3) are collinear, then $D_{12} = D_{13}$ so that D_{12}, D_{13} and D_{23} are obviously collinear. Therefore we can suppose that A_1, A_2, A_3 and B_1, B_2, B_3 are not collinear. Since $n \geq 3$, there exists a line d' in $\mathbb{P}(V)$ passing through O and not lying on the plane $\mathcal{P}$. Take two points $O' \neq O''$ distinct from O (i.e. $O', O'' \notin \mathcal{P}$) on d' (every line contains at least three distinct

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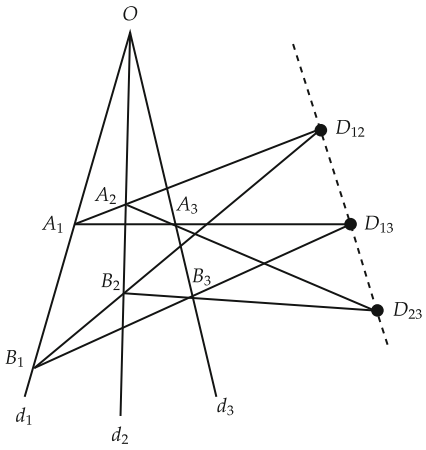


Fig. 7.1 Desargues' axiom

points). Put $C_i := O'A_i \cap O''B_i \in d_i + d'$, $i = 1, 2, 3$. Since A_1, A_2, A_3 are not collinear, the lines $O'A_1$, $O'A_2$, $O'A_3$ are not coplanar, so that we are under Case 1 and we can deduce that $P_{12} := A_1A_2 \cap C_1C_2$, $P_{13} := A_1A_3 \cap C_1C_3$ and $P_{23} := A_2A_3 \cap C_2C_3$ are collinear. Analogously, since the lines $O''B_1$, $O''B_2$ and $O''B_3$ are not coplanar, by Case 1, $Q_{12} := B_1B_2 \cap C_1C_2$, $Q_{13} := B_1B_3 \cap C_1C_3$ and $Q_{23} := B_2B_3 \cap C_2C_3$ are collinear points. On the other side the line C_iC_j does not lie on $\mathcal{P}$, so that the intersection (in the subspace $d' + \mathcal{P}$ of dimension 3) $C_iC_j \cap \mathcal{P}$ $(i < j)$ is a point. Hence we have $P_{ij} = C_iC_j \cap \mathcal{P} = Q_{ij}$ $(i < j)$, thus we deduce that $P_{ij} = Q_{ij} = D_{ij}$, for every $i < j$, so that D_{12}, D_{13} and D_{23} are collinear.

□

The converse of Desargues' axiom is the following statement:

(P_5') Under the notation of Fig. 7.1, given three lines d_1, d_2, d_3 and three pairs of distinct points $A_i \neq B_i$ on every line d_i, $i = 1, 2, 3$, such that the intersections D_{12}, D_{13} and D_{23} $(D_{ij} = A_iA_j \cap B_iB_j)$, are collinear, then the lines d_1, d_2 and d_3 are concurent.

The dual statement of Desargues' axiom (P_5) in a projective plane $\mathbb{P}(V)$ (according to Theorem 6.59) coincides with the converse of Desargues' axiom (P_5') (Exercise 7.1). Thus from Theorem 7.1, and Theorem 6.59 we get:

Corollary 7.2 *A projective plane $\mathbb{P}(V)$ satisfies the converse of Desargues' axiom (P_5').*

A Projective Plane which Does Not Satisfy Desargues' Axiom

Definition 7.3 A *configuration* is a pair of sets $(\mathcal{X}, \mathcal{D})$ where the elements of $\mathcal{X}$ are called *points* and $\mathcal{D}$ is a collection of subsets of $\mathcal{X}$, called *lines* of $\mathcal{X}$, which satisfies the following axiom:

(C_1) Two distinct points of $\mathcal{X}$ lie on at most one line of $\mathcal{X}$.

It follows that two distinct lines have at most one point in common. If $\mathcal{Y} \subset \mathcal{X}$, $\mathcal{E}$ is a collection of subsets of $\mathcal{Y}$ and $\mathcal{E} \subset \mathcal{D}$, then $(\mathcal{Y}, \mathcal{E})$ is a configuration, called a *subconfiguration* of $(\mathcal{X}, \mathcal{D})$.

Examples 7.4

(i) Every non-empty set $\mathcal{X}$ with $\mathcal{D} = \varnothing$ (i.e. there is no line) is a configuration. If $\mathcal{X}$ has exactly four distinct points A, B, C, D, then no three of them are collinear (since $\mathcal{D} = \varnothing$).

(ii) Every incidence structure is a configuration.

(iii) The set of points and lines which occurs in Fig. 7.1,

$$\mathcal{X} = \{O, A_1, A_2, A_3, B_1, B_2, B_3, D_{12}, D_{13}, D_{23}\},$$

$$\mathcal{D} = \big\{\{O, A_1, B_1\},\ \{O, A_2, B_2\},\ \{O, A_3, B_3\},\ \{A_1, A_2, D_{12}\},$$

$$\{B_1, B_2, D_{12}\},\ \{A_1, A_3, D_{13}\},$$

$$\{B_1, B_3, D_{13}\},\ \{A_2, A_3, D_{23}\},\ \{B_2, B_3, D_{23}\},\ \{D_{12}, D_{13}, D_{23}\}\big\},$$

is a configuration which has ten points and ten lines and every line consists of exactly three distinct points, while through every point there pass exactly three distinct lines.

This configuration is called *Desargues' configuration*.

Definition 7.5 (The Free Projective Plane Generated by a Configuration) Let $(\mathcal{X}_0, \mathcal{D}_0)$ be a configuration. We will construct a new configuration $(\mathcal{X}_1, \mathcal{D}_1)$ such that $\mathcal{X}_1 := \mathcal{X}_0$, and $\mathcal{D}_1$ consists of all lines of $\mathcal{D}_0$ and of all subsets of $\mathcal{X}_1 = \mathcal{X}_0$ of the form $\{A, B\}$, with $A, B \in \mathcal{X}_0$ not on a line of $\mathcal{D}_0$. We emphasize that the new lines (i.e. those belonging to $\mathcal{D}_1 \setminus \mathcal{D}_0$) contain exactly two distinct points. Therefore $(\mathcal{X}_1, \mathcal{D}_1)$ satisfies the following property:

(a) Every two distinct points of $\mathcal{X}_1$ lie on one and only one line of $\mathcal{D}_1$ (i.e. $(\mathcal{X}_1, \mathcal{D}_1)$ is an incidence structure).

Construct $(\mathcal{X}_2, \mathcal{D}_2)$ from $(\mathcal{X}_1, \mathcal{D}_1)$ as follows. The points of $\mathcal{X}_2$ are the points of $\mathcal{X}_1$, plus, for each pair of lines d and d' of $\mathcal{D}_1$ which do not meet, a new point denoted by $d \cdot d'$. If e, e' are two lines which do not meet, then $d \cdot d' = e \cdot e'$ if and only if $\{d, d'\} = \{e, e'\}$ by definition. Each line $\widehat{d}$ of $\mathcal{D}_2$ is a line $d \in \mathcal{D}_1$ extended with all possible points of the form $d \cdot d'$, with $d' \in \mathcal{D}_1$ and $d \cap d' = \varnothing$. In particular, the map $f : \mathcal{D}_1 \to \mathcal{D}_2$ defined by $f(d) = \widehat{d}$ is bijective. By construction through a point of $\mathcal{X}_2$ of the form $d \cdot d'$ (hence not belonging to $\mathcal{X}_1$) they pass only two lines: $\widehat{d}$ and $\widehat{d'}$. Therefore $(\mathcal{X}_2, \mathcal{D}_2)$ satisfies the following property:

(b) Every pair of distinct lines of $\mathcal{D}_2$ meets in one (and only one) point.

We have to observe that $(\mathcal{X}_2, \mathcal{D}_2)$ no longer has the property (a), indeed a pair of points $d \cdot d'$ and $e \cdot e'$, with $d \cdot d' \neq e \cdot e'$, does not lie on any line of $\mathcal{D}_2$.

Suppose by induction that the configuration $(\mathcal{X}_n, \mathcal{D}_n)$ has been constructed. We have to consider two cases:

(i) If n is even, then $\mathcal{X}_{n+1} := \mathcal{X}_n$. A line of $\mathcal{D}_{n+1}$ is a line of $\mathcal{D}_n$ or a subset of the form $\{A, B\}$, where A and B are distinct points of $\mathcal{X}_n$ not joined by any line of $\mathcal{D}_n$. As said above, the new lines (i.e. those belonging to $\mathcal{D}_{n+1} \setminus \mathcal{D}_n$) contain exactly two distinct points. The configuration $(\mathcal{X}_{n+1}, \mathcal{D}_{n+1})$ has the property (a).

(ii) If n is odd, $\mathcal{X}_{n+1}$ contains $\mathcal{X}_n$ and the new points $d \cdot d'$, where d, d' are two lines of $\mathcal{D}_n$ which do not meet. The lines of $\mathcal{D}_{n+1}$ are the lines of $\mathcal{D}_n$ extended as described above for $n = 1$. Through a point of $\mathcal{X}_{n+1}$ of the form $d \cdot d'$ (hence not belonging to $\mathcal{X}_n$) they pass only two lines: $\widehat{d}$ and $\widehat{d'}$. Thus the configuration $(\mathcal{X}_{n+1}, \mathcal{D}_{n+1})$ satisfies the property (b) (but no longer (a)).

Now we define

$$\mathcal{X} := \bigcup_{n=0}^{\infty} \mathcal{X}_n, \qquad \mathcal{D} = \{L \subset \mathcal{X} \mid L \cap \mathcal{X}_n \in \mathcal{D}_n \ \text{ for all large enough } n\}.$$

In other words, a subset L of $\mathcal{X}$ is a line of $(\mathcal{X}, \mathcal{D})$ if and only if there exists a natural number n_L (which depends on L) such that $L \cap \mathcal{X}_n$ is a line of $\mathcal{D}_n$ for every $n \geq n_L$. The *level* of $P \in \mathcal{X}$ is the smallest integer $n \geq 0$ such that $P \in \mathcal{X}_n$. Analogously, the *level* of a line $d \in \mathcal{D}$ is the smallest integer $n \geq 0$ such that $d \cap \mathcal{X}_n$ is a line of $(\mathcal{X}_n, \mathcal{D}_n)$.

From the above construction we see that three points $A, B, C \in \mathcal{X}_n$ are collinear in $(\mathcal{X}_n, \mathcal{D}_n)$ if and only if they are collinear in $(\mathcal{X}_{n+1}, \mathcal{D}_{n+1})$ for every $n \geq 0$. Therefore if A, B and C are collinear in $(\mathcal{X}_n, \mathcal{D}_n)$ they are collinear in $(\mathcal{X}, \mathcal{D})$. Conversely, if they are collinear in $(\mathcal{X}, \mathcal{D})$, then they are collinear in $(\mathcal{X}_n, \mathcal{D}_n)$ for every $n \geq 0$.

Proposition 7.6 *Under the above notation, suppose that $\mathcal{X}_0$ contains at least four points no three of which lie on a line in $(\mathcal{X}_0, \mathcal{D}_0)$, then $(\mathcal{X}, \mathcal{D})$ is a projective plane.*

Proof Since $(\mathcal{X}_n, \mathcal{D}_n)$ satisfies the condition (a) for every odd integer n and the condition (b) for every even integer $n \geq 2$, then $(\mathcal{X}, \mathcal{D})$ has both the properties (a) and (b). In other words, $(\mathcal{X}, \mathcal{D})$ satisfies axioms (P_1) and (P_2) of the projective plane (Definitions 6.1 and 6.5). Moreover, if A, B, C, $D \in \mathcal{X}_0$ are four points no three of which lie on a line in $(\mathcal{X}_0, \mathcal{D}_0)$, then the same property holds for A, B, C, D in $(\mathcal{X}, \mathcal{D})$. Thus $(\mathcal{X}, \mathcal{D})$ satisfies axiom (P_3). □

Definition 7.7 The pair $(\mathcal{X}, \mathcal{D})$ constructed above is called the *free projective plane* generated by the configuration $(\mathcal{X}_0, \mathcal{D}_0)$.

Definition 7.8 A configuration $(\mathcal{X}, \mathcal{D})$ is called *confined* if each point lies on at least three lines, and each line contains at least three points. Desargues' configuration of Example 7.4-(iii) is confined. A projective plane is confined by Propositions 6.12 and 6.13.

Proposition 7.9 *Any finite, confined subconfiguration of the free projective plane $(\mathcal{X}, \mathcal{D})$ generated by $(\mathcal{X}_0, \mathcal{D}_0)$, is contained in $(\mathcal{X}_0, \mathcal{D}_0)$.*

Proof Let $(\mathcal{Y}, \mathcal{D}')$ be a finite, confined subconfiguration of $(\mathcal{X}, \mathcal{D})$. Let n be the maximum of all levels of the points and the lines of $(\mathcal{Y}, \mathcal{D}')$ which is finite. We have to prove that $n = 0$. By *reductio ad absurdum* suppose that $n \geq 1$.

Case 1. If the maximum level n is reached by a line $d \in \mathcal{D}'$, then it follows that $d \cap \mathcal{X}_n \in \mathcal{D}_n$ and $d \cap \mathcal{X}_{n-1} \notin \mathcal{D}_{n-1}$. Hence $d \cap \mathcal{X}_n$ is a line of $\mathcal{X}_n$ having the form (A, B), where A and B are two distinct points of $\mathcal{X}_{n-1}$ which do not lie on a line of $\mathcal{D}_{n-1}$. Since the level of each point $P \in \mathcal{Y}$ is $\leq n$, then $P \in \mathcal{X}_n$, so that $\mathcal{D}' \subset \mathcal{D}_n$ and $(\mathcal{Y}, \mathcal{D}')$ is a subconfiguration of $(\mathcal{X}_n, \mathcal{D}_n)$ such that the line d consists only of the points A and B, contradicting the hypothesis that $(\mathcal{Y}, \mathcal{D})$ is confined.

Case 2. Suppose now that n is the level of a point P of $\mathcal{Y}$, i.e. $\mathcal{Y} \subset \mathcal{X}_n$. We can also suppose that the level of each line of $\mathcal{D}'$ is $\leq n-1$ (otherwise we are in Case 1). Then the point $P \in \mathcal{X}_n \setminus \mathcal{X}_{n-1}$ so that it is of the form $d \cdot d'$, where $\{d, d'\}$ is a uniquely determined pair of lines of $\mathcal{D}_{n-1}$ such that $d \cap d' = \varnothing$. It follows that through the point $P \in \mathcal{Y}$ pass exactly two distinct lines of $\mathcal{D}_n$ (i.e. $\widehat{d}$ and $\widehat{d'}$), hence exactly two distinct lines of $\mathcal{D}'$, contradicting the hypothesis that $(\mathcal{Y}, \mathcal{D})$ is confined.

□

Corollary 7.10 *The free projective plane $(\mathcal{X}, \mathcal{D})$ generated by a configuration $(\mathcal{X}_0, \mathcal{D}_0)$, where $\mathcal{X}_0$ is a set consisting of four distinct points and $\mathcal{D}_0 = \varnothing$ (see* Example 7.4-i)) *is infinite. In particular, every line of $\mathcal{D}$ is an infinite set.*

Proof If $\mathcal{X}$ is a finite set, then $(\mathcal{X}, \mathcal{D})$ is a finite, confined subconfiguration of itself, hence, by Proposition 7.9 $(\mathcal{X}, \mathcal{D})$ is contained in $(\mathcal{X}_0, \mathcal{D}_0)$, which is absurd. □

Example 7.11 (A Non-Desarguesian Projective Plane) Let $(\mathcal{X}_0, \mathcal{D}_0)$ be the configuration with $\mathcal{X}_0 = \{O, A_1, A_2, A_3\}$ and $\mathcal{D}_0 = \varnothing$ (see Example 7.4-i)). We claim that the free projective plane $(\mathcal{X}, \mathcal{D})$ generated by $(\mathcal{X}_0, \mathcal{D}_0)$ does not satisfy Desargues' axiom (P_5). Since the line OA_i $(\in \mathcal{D})$ is infinite $(i = 1, 2, 3)$ we can find on OA_i a point B_i $(i = 1, 2, 3)$ such that B_1, B_2 and B_3 are not collinear. Since $\mathcal{D}_0 = \varnothing$ the points A_1, A_2 and A_3 are not collinear in $(\mathcal{X}, \mathcal{D})$. Suppose that $(\mathcal{X}, \mathcal{D})$ satisfies Desargues' axiom. Then $D_{12} := A_1A_2 \cap B_1B_2$, $D_{13} := A_1A_3 \cap B_1B_3$ and $D_{23} := A_2A_3 \cap B_2B_3$ are collinear. The pair $(\mathcal{Y}, \mathcal{E})$ where $\mathcal{Y} = \{O, A_1, A_2, A_3, B_1, B_2, B_3, D_{12}, D_{13}, D_{23}\}$ and

$$\begin{aligned}\mathcal{E} = \big\{&\{O, A_1, B_1\},\ \{O, A_2, B_2\},\ \{O, A_3, B_3\},\\ &\{A_1, A_2, D_{12}\},\ \{B_1, B_2, D_{12}\},\ \{A_1, A_3, D_{13}\},\\ &\{B_1, B_3, D_{13}\},\ \{A_2, A_3, D_{23}\},\ \{B_2, B_3, D_{23}\},\ \{D_{12}, D_{13}, D_{23}\}\big\},\end{aligned}$$

is a finite, confined subconfiguration of $(\mathcal{X}, \mathcal{D})$ (it is Desargues' configuration of Example 7.4, (iii)). Then by Proposition 7.9 we get $\mathcal{Y} \subseteq \mathcal{X}_0$, which is impossible because $\mathcal{Y}$ consists of ten points while $\mathcal{X}_0$ has only four points. Therefore the

projective plane $(\mathcal{X}, \mathcal{D})$ is non-Desarguesian, hence $(\mathcal{X}, \mathcal{D})$ is not isomorphic to any projective plane associated with a K-vector space.

Remark 7.12 The non-Desarguesian projective plane of Example 7.11 is infinite. The reader can find examples of finite non-Desarguesian projective planes in [16], Exercises 26, 27, 28 and 29.

7.1 Exercises

Exercise 7.1 Prove that the dual assertion of Desargues' axiom in a projective plane $\mathbb{P}(V)$ coincides with the converse of Desargues' axiom.

Exercise 7.2 (The Moulton Plane) Let $\mathcal{X} = \mathbb{R}^2$. The set of lines $\mathcal{D}$ is as follows:

$$\mathcal{D} = \{x = a \,|\, a \in \mathbb{R}\} \cup \{y = mx + n \,|\, m \geq 0\} \cup \{\ell_{m,n} \,|\, m \leq 0, n \in \mathbb{R}\},$$

where

$$\ell_{m,n} = \begin{cases} mx + n & x \leq 0 \\ 2mx + n & x \geq 0. \end{cases}$$

Prove that $(\mathcal{X}, \mathcal{D})$ is a projective plane which does not satisfy Desargues' axiom.

Chapter 8
General Linear Projective Automorphisms



8.1 Projective Homotheties and Translations

Let $(\mathcal{X}, \mathcal{D})$ be a general projective space and $\mathrm{Aut}(\mathcal{X})$ be the groups of all its projective automorphisms.

Definition 8.1 Let $\sigma \in \mathrm{Aut}(\mathcal{X})$. A point $A \in \mathcal{X}$ is called a *fixed point* of σ if $\sigma(A) = A$. A subspace $\mathcal{Y}$ of $\mathcal{X}$ is called a *fixed subspace* of σ if $\sigma(\mathcal{Y}) = \mathcal{Y}$. If $\sigma(P) = P$ for all $P \in \mathcal{Y}$ we say that $\mathcal{Y}$ is a *subspace of fixed points* of σ. A point $O \in \mathcal{X}$ is called the *centre of* σ if every line through O is a fixed line of σ. Since there are two distinct lines d and d' passing through O, we have $O = d \cap d'$ so that $\sigma(O) = \sigma(d \cap d') = \sigma(d) \cap \sigma(d') = d \cap d' = O$. Hence the centre of an automorphism σ is a fixed point of σ. A hyperplane $\mathcal{Y}$ of $\mathcal{X}$ is called an *axis of* σ if $\mathcal{Y}$ consists of fixed points of σ.

Proposition 8.2 *Let $\sigma \in \mathrm{Aut}(\mathcal{X})$. If σ has two distinct centres (or two distinct axes), then $\sigma = \mathrm{id}_{\mathcal{X}}$, the identity map of $\mathcal{X}$.*

Proof Suppose that σ has two centres $O, O' \in \mathcal{X}$, with $O \neq O'$. Since O and O' are fixed points of σ, we have to prove that $\sigma(A) = A$ for each $A \in \mathcal{X} \setminus \{O, O'\}$. Let $A \notin OO'$, then $A = AO \cap AO'$, so that $\sigma(A) = \sigma(AO \cap AO') = \sigma(AO) \cap \sigma(AO') = AO \cap AO' = A$. Let $B \in OO' \setminus \{O, O'\}$. The line AB contains a third point C (which does not belong to OO' so that $\sigma(C) = C$). Hence $B = AC \cap OO'$ which implies $\sigma(B) = \sigma(AC \cap OO') = \sigma(AC) \cap \sigma(OO') = AC \cap OO' = B$. Therefore $\sigma = \mathrm{id}_{\mathcal{X}}$.

Let $\mathcal{Y}$ and $\mathcal{Z}$ be two distinct axes of σ. Since $\mathcal{Y} \cup \mathcal{Z}$ consists of fixed points of σ we have only to examine the points $A \in \mathcal{X} \setminus (\mathcal{Y} \cup \mathcal{Z})$. Let B and C be two distinct points of hyperplane $\mathcal{Y}$. By Remark 6.27 $BA \cap \mathcal{Z}$ and $CA \cap \mathcal{Z}$ are two points B' and C'. Then $A = BB' \cap CC'$ so that $\sigma(A) = \sigma(BB' \cap CC') = \sigma(BB') \cap \sigma(CC') = BB' \cap CC' = A$, since B, B', C, C' are fixed points of σ. □

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Theorem 8.3 *Let $\sigma \in \mathrm{Aut}(\mathcal{X}) \setminus \{\mathrm{id}_{\mathcal{X}}\}$ be an automorphism of a projective space $(\mathcal{X}, \mathcal{D})$. Then σ has a centre if and only if it has an axis.*

Proof First suppose that σ has the hyperplane $\mathcal{Y}$ as its axis.

Case 1: *There exists a fixed point $O \in \mathcal{X} \setminus \mathcal{Y}$ of σ*. We wish to prove that O is the centre of σ. Indeed, let d be a line passing through O and $A = d \cap \mathcal{Y}$ (see Remark 6.27). Then $\sigma(A) = A$ and $\sigma(d) = \sigma(OA) = \sigma(O)\sigma(A) = OA = d$, since $\sigma(O) = O$.

Case 2: *For every $P \in \mathcal{X} \setminus \mathcal{Y}$ we have $\sigma(P) \neq P$*. Since $\mathcal{Y}$ is an axis, $P \notin \mathcal{Y}$ so that $\sigma(P) \notin \mathcal{Y}$ (otherwise $P = \sigma^{-1}(\sigma(P)) \in \mathcal{Y}$). Fix a point P and put $O := P\sigma(P) \cap \mathcal{Y}$. First observe that $\sigma(OP) = \sigma(O)\sigma(P) = O\sigma(P) = OP$ (since $O \in P\sigma(P)$), so OP is a fixed line of σ. Let d be another line passing through O. If $d \subset \mathcal{Y}$, then d is obviously fixed. Let $d \not\subset \mathcal{Y}$. Take a point $Q \in d \setminus \mathcal{Y}$ and put $U := PQ \cap \mathcal{Y}$. Thus $Q \in UP$ and $\sigma(Q) \in \sigma(U)\sigma(P) = U\sigma(P)$. In particular P, Q, $\sigma(P)$ and $\sigma(Q)$ are coplanar. On the plane determined by P, Q, $\sigma(P)$ and $\sigma(Q)$ consider $O' := P\sigma(P) \cap Q\sigma(Q)$. Since the line $Q\sigma(Q)$ is fixed (let $Q\sigma(Q) \cap \mathcal{Y} = E$, then $\sigma(EQ) = \sigma(E)\sigma(Q) = E\sigma(Q) = Q\sigma(Q)$) if we prove that $O = O'$, then $d = Q\sigma(Q)$ so that d is fixed and O is the centre of σ. The point O' is fixed because it is the intersection of the fixed lines $P\sigma(P)$ and $Q\sigma(Q)$. By the above assumption (Case 2) O' must belong to $\mathcal{Y}$. If $O' \neq O$ we should have $P\sigma(P) = OO' \subset \mathcal{Y}$, contradicting the hypothesis $P \notin \mathcal{Y}$, hence $O' = O$.

Now suppose that σ has a centre O. We have to consider two cases.

Case 1′: *There exists a hyperplane $\mathcal{Y}$ of $\mathcal{X}$ such that $\sigma(\mathcal{Y}) = \mathcal{Y}$ and $O \notin \mathcal{Y}$*. Then $\mathcal{Y}$ is an axis of σ. Indeed, if $P \in \mathcal{Y}$ the line OP is fixed (since O is a centre) and $\sigma(P) \in OP \cap \mathcal{Y}$. Since $O \notin \mathcal{Y}$ the intersection $OP \cap \mathcal{Y}$ consists of the point P only so that $\sigma(P) = P$.

Case 2′: *For each hyperplane $\mathcal{Z}$ which does not contain O one has $\sigma(\mathcal{Z}) \neq \mathcal{Z}$*. Since we have not introduced the concept of dimension of a general projective subspace (Definition 7.1) we have to suppose that $\mathcal{X} = \mathbb{P}(V)$ (with $\dim(\mathbb{P}(V) = n \geq 2$). However, the proof of the general case is the same as the one we are giving below (see [1]). Fix such a hyperplane $\mathcal{Z}$. Since $\sigma(\mathcal{Z}) \neq \mathcal{Z}$, by projective Grassmann formula (Corollary 6.33) we have $\dim(\sigma\mathcal{Z} \cap \mathcal{Z}) = n - 2$. Let $\mathcal{U} := \sigma(\mathcal{Z}) \cap \mathcal{Z}$. By Corollary 6.33 the subspace $\mathcal{Y} := O + \mathcal{U}$ is a hyperplane. We shall show that $\mathcal{Y}$ is an axis of σ. First we prove that every point of $\mathcal{U}$ is fixed. Indeed, if $P \in \mathcal{U}$, then $\sigma(P) \in OP$ (OP is fixed) and $\sigma(P) = P$ since $OP \cap \sigma(\mathcal{Z}) = \sigma(P)$ and $P \in OP \cap \sigma(\mathcal{Z})$. By Theorem 6.26 we have

$$\mathcal{Y} = \bigcup_{P \in \mathcal{U}} OP.$$

To finish our proof it remains to prove that every point $R \in OP$ is fixed (for arbitrary $P \in \mathcal{U}$). By *reductio ad absurdum* suppose there is $R \in OP$ such that $\sigma(R) \neq R$. Take a hyperplane $\mathcal{V}$ of $\mathbb{P}(V)$) passing through R and not containing

O. By Corollary 6.33, $\mathcal{V} \cap \sigma(\mathcal{V})$ is a subspace of dimension $n-2$ and every point of $\mathcal{V} \cap \sigma(\mathcal{V})$ is a fixed point. Indeed, if $P \in \mathcal{V} \cap \sigma(\mathcal{V})$, then $\sigma(P) \in OP$ (OP is fixed) and $\sigma(P) = P$ since $OP \cap \sigma(\mathcal{V}) = \sigma(P)$ and $P \in OP \cap \sigma(\mathcal{V})$. Furthermore, there exists $S \in \mathcal{V} \cap \sigma(\mathcal{V}) \setminus \mathcal{U}$, otherwise $\mathcal{V} \cap \sigma(\mathcal{V}) = \mathcal{U}$ (both the subspaces are of dimension $n-2$), hence, in particular, $P \in \mathcal{V} \cap \sigma(\mathcal{V}) \subset \mathcal{V}$. Since $R \in \mathcal{V}$, we have $RP \subset \mathcal{V}$, so that $O \in \mathcal{V}$ (since $O \in RP$) contradicting the assumptions on the hyperplane $\mathcal{V}$. Take a point $S \in \mathcal{V} \cap \sigma(\mathcal{V}) \setminus \mathcal{U}$ and consider the hyperplane

$$S + \mathcal{U} = \bigcup_{T \in \mathcal{U}} ST.$$

It is fixed since S and T are fixed points; moreover it does not contain O. But this contradicts the assumption of Case 2′.

□

Definition 8.4 Let $\sigma \in \mathrm{Aut}(\mathcal{X} \setminus \{\mathrm{id}_{\mathcal{X}}\}$ be an automorphism of a general projective space $(\mathcal{X}, \mathcal{D})$.

- σ is a *projective translation* if σ has an axis $\mathcal{Y}$ and a centre O such that $O \in \mathcal{Y}$.
- σ is a *projective homothety* if σ has an axis $\mathcal{Y}$ and a centre O such that $O \notin \mathcal{Y}$.

Let $\sigma \in \mathrm{Aut}(\mathcal{X})$ be an automorphism of axis $\mathcal{Y}$ and centre O (i.e. a projective translation or homothety). If $P \in \mathcal{X} \setminus (\mathcal{Y} \cup \{O\})$, then $\sigma(P) \in PO$ and $\sigma(P) \notin (\mathcal{Y} \cup \{O\}))$.

The identity $\mathrm{id}_{\mathcal{X}}$ is a projective translation as well as a projective homothety by definition. Every point of $\mathcal{X}$ is a centre and every hyperplane of $\mathcal{X}$ is an axis.

Notation Let $\mathcal{Y}$ be a hyperplane of $(\mathcal{X}, \mathcal{D})$. We introduce the following notation:

- $\mathcal{S}_{\mathcal{Y}}$: the set of all projective automorphisms di $\mathrm{Aut}(\mathcal{X})$ of axis $\mathcal{Y}$.
- $\mathcal{T}_{\mathcal{Y}}$: the set of all projective translations of axis $\mathcal{Y}$.
- $\mathcal{T}_{\mathcal{Y},U}$: the set of all projective translations of axis $\mathcal{Y}$ and centre U.
- $\mathcal{H}_{\mathcal{Y},O}$: the set of all projective homotheties of axis $\mathcal{Y}$ and centre O.

The following inclusions are obvious: $\mathcal{T}_{\mathcal{Y},U} \subseteq \mathcal{T}_{\mathcal{Y}} \subseteq \mathcal{S}_{\mathcal{Y}}$ and $\mathcal{H}_{\mathcal{Y},O} \subseteq \mathcal{S}_{\mathcal{Y}}$.

Proposition 8.5 *Under the above notation we have:*

(i) $\mathcal{S}_{\mathcal{Y}}$ *is a subgroup of* $\mathrm{Aut}(\mathcal{X})$.
(ii) $\mathcal{T}_{\mathcal{Y}}$ *is a normal subgroup of* $\mathcal{S}_{\mathcal{Y}}$.
(iii) *For every* $U, V \in \mathcal{Y}$, $\mathcal{T}_{\mathcal{Y},U}$ *is a subgroup of* $\mathcal{T}_{\mathcal{Y}}$ *and* $\mathcal{T}_{\mathcal{Y},U} \cap \mathcal{T}_{\mathcal{Y},V} = \{\mathrm{id}_{\mathcal{X}}\}$ *if* $U \neq V$.
(iv) *For every* $O \in \mathcal{X} \setminus \mathcal{Y}$, $\mathcal{H}_{\mathcal{Y},O}$ *is a subgroup of* $\mathcal{S}_{\mathcal{Y}}$ *and* $\mathcal{H}_{\mathcal{Y},O} \cap \mathcal{T}_{\mathcal{Y}} = \{\mathrm{id}_{\mathcal{X}}\}$.
(v) *An automorphism* $\alpha \in \mathrm{Aut}(\mathcal{X})$ *such that* $\alpha(\mathcal{Y}) = \mathcal{Y}'$ *and* $\alpha(O) = O'$ *induces two group isomorphisms*

$$\mathcal{H}_{\mathcal{Y},O} \to \mathcal{H}_{\mathcal{Y}',O'}, \quad \mathcal{T}_{\mathcal{Y},O} \to \mathcal{T}_{\mathcal{Y}',O'}, \qquad \sigma \to \alpha \circ \sigma \circ \alpha^{-1}.$$

Proof The assertions (i), (iii), (iv) and (v) easily follow from Definition 8.1 and from Proposition 8.2 (exercise left to the reader).
(ii) First we prove that $\mathcal{T}_\mathcal{Y}$ is a subgroup $\mathcal{S}_\mathcal{Y}$. It is obvious that if $\sigma \in \mathcal{T}_{\mathcal{Y},U}$ then $\sigma^{-1} \in \mathcal{T}_{\mathcal{Y},U}$. Let $\sigma, \tau \in \mathcal{T}_\mathcal{Y}$ of centres $U, V \in \mathcal{Y}$ respectively. The axis of $\sigma \circ \tau$ is clearly $\mathcal{Y}$. If $U = V$ the centre of $\sigma \circ \tau$ is obviously U. Then suppose that both the automorphisms are distinct from $\mathrm{id}_\mathcal{X}$ (otherwise there is nothing to prove) and $U \neq V$. If $P \in \mathcal{X} \setminus \mathcal{Y}$, then $\sigma(P) \in UP$, $\sigma(P) \neq P$ (otherwise every line through P, since it meets $\mathcal{Y}$, would be fixed and P would be a second centre of σ, which is impossible by Proposition 8.2 since $\sigma \neq \mathrm{id}_\mathcal{X}$). Analogously, $\tau(P) \in VP$ and $\tau(P) \neq P$. It follows that $\sigma(\tau(P)) \in U\tau(P)$, hence $P \neq \sigma(\tau(P))$ for each $P \in \mathcal{X} \setminus \mathcal{Y}$. Therefore the centre of $\sigma \circ \tau \in \mathcal{S}_\mathcal{Y}$ (which exists by Theorem 8.3) belongs to $\mathcal{Y}$, i.e. $\sigma \circ \tau \in \mathcal{T}_\mathcal{Y}$.

Now we show that $\mathcal{T}_\mathcal{Y}$ is a normal subgroup of $\mathcal{S}_\mathcal{Y}$. Let $\sigma \in \mathcal{T}_\mathcal{Y}$ and $\alpha \in \mathcal{S}_\mathcal{Y}$. We have to prove that $\alpha^{-1} \circ \sigma \circ \alpha \in \mathcal{T}_\mathcal{Y}$. Let $U \in \mathcal{Y}$ be the centre of σ. We shall prove that U is also the centre of $\alpha^{-1} \circ \sigma \circ \alpha$. Let d be a line of $\mathcal{X}$ through U. Since $\alpha(U) = U$ the line $\alpha(d)$ also passes through U, and because U is the centre of σ, we have $\sigma(\alpha(d)) = \alpha(d)$, therefore $\alpha^{-1}(\sigma(\alpha(d))) = \alpha^{-1}(\alpha(d)) = d$, i.e. d is a fixed line of $\alpha^{-1} \circ \sigma \circ \alpha$. □

Throughout this chapter K will be a fixed field and $n \geq 1$ a fixed integer. Most results of this chapter also hold if K is a skew field (e.g see [2] or [1]).

Example 8.6 Let $A = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a \end{pmatrix}$, with $a \in K \setminus \{0\}$. Since $\det(A) = a^n \neq 0$, $A \in \mathrm{GL}_{n+1}(K)$. The general linear projective automorphism $\sigma_a := T_A \in \mathrm{PGL}_n(K)$:

$$\sigma_a([x_0, x_1, \ldots, x_n]) = [x_0, ax_1, \ldots, ax_n],$$

is a homothety with centre $O = [1, 0, \ldots, 0]$ and axis the hyperplane $\mathcal{Y} : x_0 = 0$. The set $\{\sigma_a \mid a \in K^*\}$ is a commutative subgroup of $\mathrm{PGL}_n(K)$, being isomorphic to the multiplicative group K^*.

Example 8.7 For every $(a_1, \ldots, a_n) \in K^n$ consider the following matrix:

$$A_{(a_1,\ldots,a_n)} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ a_1 & 1 & 0 & \cdots & 0 \\ a_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_n & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Since $\det(A_{(a_1,\dots,a_n)}) = 1$, $A_{(a_1,\dots,a_n)} \in \mathrm{GL}_{n+1}(K)$, the linear projective automorphism

$$\tau_{(a_1,\dots,a_n)}([x_0, x_1, \dots, x_n]) = [x_0, a_1x_0 + x_1, \dots, a_nx_0 + x_n].$$

is a projective translation of centre $U := [0, a_1, \dots, a_n]$ and axis the hyperplane $\mathcal{Y} : x_0 = 0$. The set $\{\tau_{(a_1,\dots,a_n)} \mid (a_1, \dots, a_n) \in K^n\}$ (with $\tau_{(0,\dots,0)} = \mathrm{id}_{\mathbb{P}^n(K)}$) is a commutative subgroup of $\mathrm{PGL}_n(K)$ isomorphic to the additive group of K^n (note that $A_{(a_1,\dots,a_n)}A_{(b_1,\dots,b_n)} = A_{(a_1+b_1,\dots,a_n+b_n)}$).

The following result shows the importance of Desargues' axiom.

Proposition 8.8 *Let $(\mathcal{X}, \mathcal{D})$ be a projective space which satisfies Desargues' axiom. Let $\mathcal{Y}$ be a hyperplane of $(\mathcal{X}, \mathcal{D})$ and O be a point of $\mathcal{X}$. For every pair of points P, $P' \in \mathcal{X} \setminus (\mathcal{Y} \cup \{O\})$ such that P, P' and O are collinear, there exists one and only one automorphism $\sigma \in \mathrm{Aut}(\mathcal{X})$ of axis $\mathcal{Y}$ and with centre O such that $\sigma(P) = P'$.*

Proof First we prove uniqueness, which does not require Desargues' axiom. Let $\sigma, \tau \in \mathrm{Aut}(\mathcal{X})$ be two automorphisms which satisfy the assertion of the proposition. Then $\alpha := \tau^{-1} \circ \sigma$ is an automorphism of axis $\mathcal{Y}$ and with centre O such that $\alpha(P) = P$. Every line d through P is fixed for α since d joins P and $Q = d \cap \mathcal{Y}$. Hence α has two distinct centres that implies $\alpha = \mathrm{id}$ by Proposition 8.2.

For a synthetic proof of existence in the general case see [1]. We suppose that $X = \mathbb{P}^n(K)$ with K field.

Case of projective homothety. Let $\mathcal{Y} : x_0 = 0$ and $O = [1, 0, \dots, 0]$. Suppose $OP \cap \mathcal{Y} = Q = [0, a_1, \dots, a_n]$. Then $P = [1, \lambda a_1, \dots, \lambda a_n]$ and $P' = [1, \lambda' a_1, \dots, \lambda' a_n]$, with $\lambda, \lambda' \in K \setminus \{0\}$. Put $a = \lambda' \cdot \lambda^{-1}$ and $\sigma_a = T_A$, where

$$A = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a \end{pmatrix}.$$

Then σ_a is a projective homothety with axis $\mathcal{Y}$ and centre O such that $\sigma_a(P) = P'$.

Case of projective translation. Let $\mathcal{Y} : x_0 = 0$, $O = [0, a_1, \dots, a_n] \in \mathcal{Y}$ and $P = [1, c_1, \dots, c_n]$. Then $P' = [1, c_1 + \lambda a_1, \dots, c_n + \lambda a_n]$, with $\lambda \in K$. Put $\tau_{(\lambda a_1,\dots,\lambda a_n)} = T_A$, where

$$A = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ \lambda a_1 & 1 & 0 & \cdots & 0 \\ \lambda a_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda a_n & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Then $\tau_{(\lambda a_1,\ldots,\lambda a_n)}$ is a projective translation with axis $\mathcal{Y}$ and centre O such that $\tau_{(\lambda a_1,\ldots,\lambda a_n)}(P) = P'$. Observe that the projective homothety and the projective translation above determined are in $\mathrm{PGL}_n(K)$.
Let $\mathcal{Y}'$ be an arbitrary hyperplane of $\mathbb{P}^n(K)$ and $O' \in \mathbb{P}^n(K)$ an arbitrary point. There exists $\sigma \in \mathrm{PGL}_n(K)$ such that $\sigma(\mathcal{Y}) = \mathcal{Y}'$ and $\sigma(O) = O'$ (Theorem 6.47). If $O' \notin \mathcal{Y}'$, then $\sigma \circ \sigma_a \circ \sigma^{-1} \in H_{\mathcal{Y}',O'}$ sends $\sigma(P)$ to $\sigma(P')$. If $O' \in \mathcal{Y}'$, then $\sigma \circ \tau_{\lambda a_1,\ldots,\lambda a_n} \circ \sigma^{-1} \in \mathcal{T}_{\mathcal{Y}',O'}$ sends $\sigma(P)$ to $\sigma(P')$. □

Corollary 8.9 *Every projective translation (resp. projective homothety) of $\mathbb{P}^n(K)$ is an element of $\mathrm{PGL}_n(K)$. Furthermore, for every hyperplane $\mathcal{Y}$ of $\mathbb{P}^n(K)$ and for every $O \notin \mathcal{Y}$ and $U \in \mathcal{Y}$, we have the following group isomorphisms:*

$$\mathcal{H}_{\mathcal{Y},O} \cong K^*, \qquad \mathcal{T}_{\mathcal{Y},U} \cong K.$$

In particular $\mathcal{H}_{\mathcal{Y},O}$ and $\mathcal{T}_{\mathcal{Y},U}$ are commutative subgroups of $\mathrm{PGL}_n(K)$.

Proof The first statement follows from the proof of Proposition 8.8 (existence and uniqueness). The isomorphism $\mathcal{H}_{\mathcal{Y},O} \cong K^*$ follows from Example 8.6. From the identity

$$\tau_{(\lambda a_1,\ldots,\lambda a_n)} \circ \tau_{(\mu a_1,\ldots,\mu a_n)} = \tau_{((\lambda+\mu)a_1,\ldots,(\lambda+\mu)a_n)}, \qquad \lambda, \mu \in K,$$

we deduce the isomorphism $T_{\mathcal{Y},U} \cong K$. □

Theorem 8.10 *Let $\mathcal{Y}$ be a hyperplane of a general projective space $(\mathcal{X}, \mathcal{D})$. If $(\mathcal{X}, \mathcal{D})$ is a plane we require that $\mathcal{X}$ satisfies Desargues' axiom (P_5). Then:*

(i) *The group $\mathcal{T}_{\mathcal{Y}}$ of translation with axis $\mathcal{Y}$ is commutative.*
(ii) *Let $O \in \mathcal{X} \setminus \mathcal{Y}$. For each $\sigma \in \mathcal{S}_{\mathcal{Y}}$ there exists $\tau \in \mathcal{T}_{\mathcal{Y}}$ and $\alpha \in \mathcal{H}_{\mathcal{Y},O}$ such that $\sigma = \tau \circ \alpha$.*

Proof From Theorem 7.1 and from the hypothesis it follows that $\mathcal{X}$ satisfies Desargues' axiom (P_5).

(i) Let $\tau_1, \tau_2 \in \mathcal{T}_{\mathcal{Y}}$ be two translations $\neq \mathrm{id}_{\mathcal{X}}$ of centres U_1 and U_2 respectively.

Case 1: $U_1 \neq U_2$. Let $P \in \mathcal{X}$ an arbitrary point. Put $P_1 := \tau_1(P)$ and $P_2 := \tau_2(P)$. We can assume $P \notin \mathcal{Y}$ (otherwise $\tau_1(\tau_2(P)) = P = \tau_2(\tau_1(P))$). Since $\tau_i \neq \mathrm{id}_{\mathcal{X}}$, $P_i \neq P$, $i = 1, 2$ (otherwise P would be another centre of τ_1 and τ_2). The points P_2, $\tau_2(P_1)$ and $\tau_2(U_1) = U_1$ are collinear since $P_1 \in U_1P$. It follows that $\tau_2(P_1) \in U_1P_2$. Furthermore $\tau_2(P_1) \in U_2P_1$. Hence $\tau_2(P_1) = U_1P_2 \cap U_2P_1$ $(U_1 \neq U_2)$. By symmetry $\tau_1(P_2) = U_1P_2 \cap U_2P_1$, hence $\tau_1 \circ \tau_2 = \tau_2 \circ \tau_1$ if $U_1 \neq U_2$. Thus we have proved the equality $\tau_1 \circ \tau_2 = \tau_2 \circ \tau_1$ if $U_1 \neq U_2$ without assuming Desargues' axiom.

Case 2: $U_1 = U_2 =: U$. Choose a point $V \in \mathcal{Y}$, $V \neq U$, and two distinct points A, $A' \notin \mathcal{Y}$, such that A, A' and V are collinear. Since $\mathcal{X}$ satisfies Desargues' axiom, by Proposition 8.8 there exists a unique projective translation τ of axis

$\mathcal{Y}$ and centre V such that $\tau(A) = A'$. If W is the centre of $\tau_2 \circ \tau$, then $W \neq U$ (otherwise $A, U, \tau_2(\tau(A))$ would be collinear as well as $A', U, \tau_2(\tau(A))$ so that $U \in AA'$ but $AA' \cap \mathcal{Y} = V \neq U$). Since $V \neq U$, by Case 1 we have

$$\tau_1 \circ \tau_2 = \tau_1 \circ \tau_2 \circ \tau \circ \tau^{-1} = \tau_1 \circ (\tau_2 \circ \tau) \circ \tau^{-1} = (\tau_2 \circ \tau) \circ \tau_1 \circ \tau^{-1}$$
$$= \tau_2 \circ (\tau \circ \tau_1) \circ \tau^{-1} = \tau_2 \circ (\tau_1 \circ \tau) \circ \tau^{-1} = \tau_2 \circ \tau_1$$

and (i) is proved.

(ii) Let $\sigma \in \mathcal{S}_{\mathcal{Y}}$. If $\sigma \in \mathcal{H}_{\mathcal{Y},O}$ we take $\tau = \mathrm{id}$ and $\alpha = \sigma$. If we suppose $\sigma \notin \mathcal{H}_{\mathcal{Y},O}$, then we have $O' := \sigma(O) \neq O$. By Proposition 8.8 there exists a unique projective translation $\tau \in \mathcal{T}_{\mathcal{Y}}$ of centre $U = OO' \cap \mathcal{Y}$ such that $\tau(O) = O'$. Therefore $\tau^{-1} \circ \sigma(O) = O$, thus if we put $\alpha := \tau^{-1} \circ \sigma$ we get $\alpha \in \mathcal{H}_{\mathcal{Y},O}$. □

Corollary 8.11 *Assume the hypothesis of* Proposition 8.8. *Let $A, B, C \in \mathcal{X} \setminus \mathcal{Y}$.*

(a) *If τ_B^A is the unique translation $\tau \in \mathcal{T}_{\mathcal{Y}}$ such that $\tau(A) = B$ then $\tau_A^B = (\tau_B^A)^{-1}$.*
(b) $\tau_C^A = \tau_C^B \circ \tau_B^A$.
(c) $\tau_B^A(C) = \tau_C^A(B)$.
(d) *If σ_B^A is the unique homothety $\sigma \in \mathcal{H}_{\mathcal{Y},O}$ such that $\sigma(A) = B$ then $\sigma_A^B = (\sigma_B^A)^{-1}$.*
(e) $\sigma_C^A = \sigma_C^B \circ \sigma_B^A$, *where O, A, B and C are necessarily collinear.*
(f) $\sigma_B^A(C) = \sigma_C^A(B)$, *where O, A, B and C are necessarily collinear.*

Proof (c) It is clear that $\tau_B^A(C) = \tau_C^A(B)$ if and only if $\tau_A^B(\tau_B^A(C)) = \tau_A^B(\tau_C^A(B))$. Since $\tau_A^B(\tau_B^A(C)) = C$ by (a) and $\tau_A^B(\tau_C^A(B)) = \tau_C^A(\tau_A^B(B)) = \tau_C^B(B) = C$ (because $\mathcal{T}_{\mathcal{Y}}$ is commutative and by (b)), (c) follows.

As above, $\sigma_B^A(C) = \sigma_C^A(B)$ if and only if $\sigma_A^B(\sigma_B^A(C)) = \sigma_A^B(\sigma_C^A(B))$. By (d) $\sigma_A^B(\sigma_B^A(C)) = C$ and by commutativity of $\mathcal{H}_{\mathcal{Y},O}$, $\sigma_A^B(\sigma_C^A(B)) = \sigma_C^A(\sigma_A^B(B)) = \sigma_C^B(B) = C$. □

8.1.1 *Desargues' Axiom, Pappus' Axiom and the Division Ring of the Coordinates*

Definition 8.12 Let $(\mathcal{X}, \mathcal{D})$ be a general projective space (according to Definition 7.1). *Pappus' axiom* holds if:

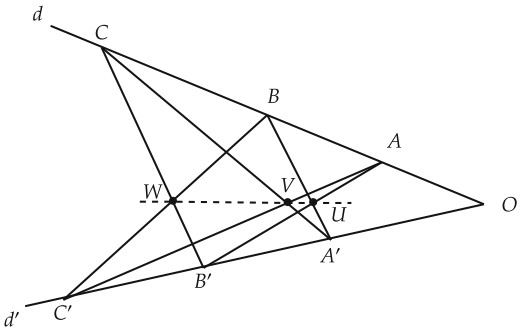


Fig. 8.1 Pappus' theorem

(P_6) For every two lines d and d' of $(\mathfrak{X}, \mathfrak{D})$ concurrent at a point O and for every six distinct points $A, B, C \in d \setminus \{O\}$ and $A', B', C' \in d' \setminus \{O\}$ the points of intersection

$$U := AB' \cap A'B, \quad V := AC' \cap A'C, \quad W := BC' \cap B'C$$

lie on a common line in the plane $d + d'$ (see Fig. 8.1).

Now we give an application of projective homotheties and translations by proving an important classical theorem of projective geometry.

Theorem 8.13 (Pappus' Theorem) *Pappus' axiom holds in a projective space* $\mathbb{P}(V)$ *associated with a* K*-vector space* V *of dimension* ≥ 3.

Proof Assume the notation of Definition 8.12.

Case 1. Suppose $O \in UW$ and put $\mathcal{Y} = UW$. Let $P, Q \in d \setminus \{O\}$ and $P', Q' \in d' \setminus \{O\}$. By Proposition 8.8 there is only one projective translation $\tau_Q^P \in \mathcal{T}_{\mathcal{Y},O}$ (resp. $\tau_{Q'}^{P'} \in \mathcal{T}_{\mathcal{Y},O}$) such that $\tau_Q^P(P) = Q$ (resp. $\tau_{Q'}^{P'}(P') = Q'$). We claim that there are the following equalities:

$$\tau_B^A = \tau_{A'}^{B'}, \qquad \tau_C^B = \tau_{B'}^{C'}.$$

Indeed to prove the first equality (the second is analogous) it is enough to observe that $\tau_B^A(B') \in AU$ since B, U, A' are collinear. Furthermore, $\tau_B^A(B') \in d'$, hence $\tau_B^A(B') = AU \cap d' = A'$, i.e. $\tau_B^A = \tau_{A'}^{B'}$. Therefore

$$\tau_C^A = \tau_C^B \circ \tau_B^A = \tau_{B'}^{C'} \circ \tau_{A'}^{B'} = \tau_{A'}^{B'} \circ \tau_{B'}^{C'} = \tau_{A'}^{C'}$$

($\tau_{B'}^{C'} \circ \tau_{A'}^{B'} = \tau_{A'}^{B'} \circ \tau_{B'}^{C'}$ follows from Corollary 8.9 which implies that $\mathcal{T}_{\mathcal{Y},O}$ is a commutative group). In other words $\tau_C^A(AC') = CA'$. Furthermore $\tau_C^A(V) \in OV$ which implies $\tau_C^A(V) = \tau_C^A(AC') \cap OV = CA' \cap OV = V$, so that $V \in \mathcal{Y}$.

Case 2. Now suppose $O \notin UW$ and put $\mathcal{Y} = UW$. In this case the subgroup $\mathcal{T}_{\mathcal{Y},O}$ will be replaced by the subgroup $\mathcal{H}_{\mathcal{Y},O}$ which is commutative too (Corollary 8.9).

Let $P, Q \in d \setminus (\mathcal{Y} \cup \{O\})$ (resp. if $P', Q' \in d' \setminus (\mathcal{Y} \cup \{O\})$). By Proposition 8.8 there is only one projective homothety $\sigma_Q^P \in \mathcal{H}_{\mathcal{Y},O}$ (resp. $\sigma_{Q'}^{P'} \in \mathcal{H}_{\mathcal{Y},O}$) such that $\sigma_Q^P(P) = Q$ (resp. $\sigma_{Q'}^{P'}(P') = Q'$. As above we get $\sigma_B^A = \sigma_{A'}^{B'}$ and $\sigma_C^B = \sigma_{B'}^{C'}$. Hence

$$\sigma_C^A = \sigma_C^B \circ \sigma_B^A = \sigma_{B'}^{C'} \circ \sigma_{A'}^{B'} = \sigma_{A'}^{B'} \circ \sigma_{B'}^{C'} = \sigma_{A'}^{C'}$$

(the equality $\sigma_{B'}^{C'} \circ \sigma_{A'}^{B'} = \sigma_{A'}^{B'} \circ \sigma_{B'}^{C'}$ holds because the subgroup $\mathcal{H}_{\mathcal{Y},O}$ is commutative). In other words $\sigma_C^A(AC') = CA'$. Furthermore $\sigma_C^A(V) \in OV$ which implies $\sigma_C^A(V) = \sigma_C^A(AC') \cap OV = CA' \cap OV = V$, so that $V \in \mathcal{Y}$. □

Theorem 8.14 (Hessenberg's Theorem) *Let* $(\mathcal{X}, \mathcal{D})$ *be a general projective space. If Pappus' axiom holds in* $(\mathcal{X}, \mathcal{D})$ *then Desargues' axiom is also true in* $(\mathcal{X}, \mathcal{D})$.

Proof See [1] pp. 147–148. □

Remark 8.15 Let $(\mathcal{X}, \mathcal{D})$ be a general projective space satisfying Desargues' axiom. Fix a hyperplane $\mathcal{Y}$ and two points $O, E \in \mathcal{X} \setminus \mathcal{Y}$ and put $K_{O,E,\mathcal{Y}} := OE \setminus \mathcal{Y}$. Then by means of projective homotheties and translations we can define a structure of division ring (in general non-commutative) on $K_{O,E,\mathcal{Y}}$ (see Sect. 8.2.1 when $\mathcal{X} = \mathbb{P}^n(K)$). We point out some basic properties which underline the relevance of Desargues' and Pappus' axioms.

1. If we fix another triple $(O', E', \mathcal{Y}')$ as above, then $K_{O,E,\mathcal{Y}}$ is isomorphic to $K_{O',E',\mathcal{Y}'}$, thus we can define $K(\mathcal{X}) := K_{O,E,\mathcal{Y}}$ the *division ring of the coordinates* of $\mathcal{X}$ which is unique up to division ring isomorphisms.
2. We can extend Definition 6.28 also to the vector spaces over a non-commutative division ring K without any change. In particular we can introduce the left standard projective space $\mathbb{P}_l^n(K)$ as well as the right standard projective space $\mathbb{P}_r^n(K)$. Every projective plane $\mathbb{P}_l^2(K)$ (or $\mathbb{P}_r^2(K)$) satisfies Desargues' axiom. The projective spaces over the skew field of quaternions play an important role in algebraic topology (see [17]).
3. There exists a projective isomorphism $\mathcal{X} \to \mathbb{P}_l^n(K)$ (or $\mathcal{X} \to \mathbb{P}_r^n(K)$), where $K \cong K(\mathcal{X})$.
4. The division ring of the coordinates of $\mathcal{X}$ is commutative if and only if $\mathcal{X}$ satisfies Pappus' axiom. In this case $K_{O,E,\mathcal{Y}}$ will be called *the field of coordinates*.

In Chapter 11 of [1] one can find a very complete account of this subject.

8.2 The Group of Projective Automorphisms of $\mathbb{P}^n(K)$

Definition 8.16 Let $\mathrm{Aut}(K)$ be the group (in general non-commutative) of all field automorphisms of K with respect to composition of maps. Every $\alpha \in \mathrm{Aut}(K)$ induces a well-defined map $S_\alpha : \mathbb{P}^n(K) \to \mathbb{P}^n(K)$, where $n \geq 2$, given by

$$S_\alpha([x_0, x_1, \ldots, x_n]) = [\alpha(x_0), \ldots, \alpha(x_n)].$$

It is immediately seen that S_α is bijective. Since

$$\begin{aligned} S_\alpha([\lambda a_0 + \mu b_0, \ldots, \alpha a_n + \mu b_n]) = \alpha(\lambda) S_\alpha([a_0, a_1, \ldots, a_n]) \\ + \alpha(\mu) S_\alpha([b_0, b_1, \ldots, b_n]) \end{aligned}$$

S_α maps lines into lines and so it is a projective automorphism of $\mathbb{P}^n(K)$. Furthermore, since $S_{\alpha\beta} = S_\alpha \circ S_\beta$, $\forall\, \alpha, \beta \in \mathrm{Aut}(K)$, the map

$$S_n : \mathrm{Aut}(K) \to \mathrm{Aut}(\mathbb{P}^n(K)), \qquad S_n(\alpha) = S_\alpha,$$

is an injective group homomorphism. We shall denote the subgroup $S_n(\mathrm{Aut}(K))$ of $\mathrm{Aut}(\mathbb{P}^n(K))$ by $\mathrm{PAut}_n(K)$. Since S_n is an injective group homomorphism $\mathrm{PAut}_n(K)$ is isomorphic to $\mathrm{Aut}(K)$.

Proposition 8.17 $\mathrm{Aut}(\mathbb{R}) = \{\mathrm{id}_\mathbb{R}\}$, *i.e.* $\mathrm{id}_\mathbb{R}$ *is the unique automorphism of* $\mathbb{R}$.

Proof Let $f \in \mathrm{Aut}(\mathbb{R})$. Since $f(1) = 1$, we have $f(2) = f(1+1) = f(1) + f(1) = 1 + 1 = 2$. Suppose by induction that $f(n) = n$, then $f(n+1) = f(n) + f(1) = n + 1$. Hence $f(n) = n$ for all $n \in \mathbb{N}$. Furthermore $f(0) = 0$ and $f(-n) = -f(n) = -n$, $\forall\, n \in \mathbb{N}$, so that $f(n) = n$, $\forall\, n \in \mathbb{Z}$. Let $n > 0$ be a positive integer. Then

$$1 = \underbrace{\frac{1}{n} + \ldots + \frac{1}{n}}_{n \text{ times}} \quad \Longrightarrow \quad 1 = f(1) = f\underbrace{\left(\frac{1}{n} + \cdots + \frac{1}{n}\right)}_{n \text{ times}}$$

$$= \underbrace{f\left(\frac{1}{n}\right) + \cdots + f\left(\frac{1}{n}\right)}_{n \text{ times}} = nf\left(\frac{1}{n}\right),$$

i.e. $f\left(\frac{1}{n}\right) = \frac{1}{n}$. If $m > 0$ is another positive integer, we have

$$f\left(\frac{m}{n}\right) = f\underbrace{\left(\frac{1}{n} + \cdots + \frac{1}{n}\right)}_{m \text{ times}} = \underbrace{f\left(\frac{1}{n}\right) + \cdots + f\left(\frac{1}{n}\right)}_{m \text{ times}} = mf\left(\frac{1}{n}\right) = \frac{m}{n}.$$

Hence $f(q) = q$ for every $q \in \mathbb{Q}$, $q > 0$. Therefore $f(-q) = -q$, $\forall\, q > 0$ so that $f(q) = q$ for all $q \in \mathbb{Q}$.

If $x \in \mathbb{R}$ and $x > 0$, then $x = a^2$ for some $a \in \mathbb{R}$, so that $f(x) = f(a^2) = f(a)^2 > 0$. If $x, y \in \mathbb{R}$ with $x < y$, then $f(y) - f(x) = f(y - x) > 0$ since $y - x > 0$, hence f is strictly increasing.

Let $x \in \mathbb{R}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exist two sequences of rational numbers $\{p_n\}_{n\in\mathbb{N}}$ and $\{q_n\}_{n\in\mathbb{N}}$ such that

$$\lim_{n\to\infty} p_n = x, \qquad \lim_{n\to\infty} q_n = x, \qquad p_n < x < q_n, \forall\, n \in \mathbb{N},$$

such that $\{p_n\}_{n\in\mathbb{N}}$ is strictly increasing and $\{q_n\}_{n\in\mathbb{N}}$ strictly decreasing. Since f s strictly increasing we have $f(p_n) = p_n < f(x) < f(q_n) = q_n, \forall\, n \in \mathbb{N}$. Therefore $f(x) = \lim_{n\to\infty} p_n = \lim_{n\to\infty} q_n$, so that $f(x) = x$ by uniqueness of limits. □

Corollary 8.18 $\mathrm{PAut}_n(\mathbb{R}) = \{\mathrm{id}_{\mathbb{R}}\}$.

Remark 8.19 From the proof of Proposition 8.17) we have $\mathrm{PAut}_n(\mathbb{Q}) = \{\mathrm{id}_{\mathbb{Q}}\}$. Also $\mathrm{PAut}_n(\mathbb{Z}_p) = \{\mathrm{id}_{\mathbb{Z}_p}\}$ (Exercise 12.2). On the contrary, $\mathrm{PAut}_n(\mathbb{C})$ is uncountable (see [38]). However, only the identity $\mathrm{id}_{\mathbb{C}}$ and the conjugation $z \to \bar{z}$ are continuous with respect to euclidean topology.

8.2.1 Geometric Characterization of the Field K

In order to determine the structure of $\mathrm{Aut}(\mathbb{P}^n(K))$ we need to give a geometric characterization of the field K.

Definition 8.20 Let

$$\begin{aligned}\big\{O' = [1, 0, \ldots, 0],\, E' = [1, 1, \ldots, 1],\, E'_1 \\ = [0, 1, 0, \ldots, 0], \ldots, E'_n = [0, 0, \ldots, 0, 1]\big\}\end{aligned} \tag{8.1}$$

be the standard projective frame of $\mathbb{P}^n(K)$ and $\mathcal{Y}_0$ be the hyperplane generated by $E'_1, \ldots, E'_n$ (whose equation is $x_0 = 0$). The line $O'E'$ (of equations $x_1 = x_2 = \cdots = x_n$) meets $\mathcal{Y}_0$ at the point $U' = [0, 1, \ldots, 1]$. Hence the triple $\{O', E', U'\}$ is a projective frame of $O'E'$. Put

$$K_{O',E',\mathcal{Y}_0} := O'E' \setminus \{U'\} = \{[1, a, \ldots, a] \,|\, a \in K\}.$$

Then the natural map

$$\varphi_{O',E',\mathcal{Y}_0} \colon K \to K_{O',E',\mathcal{Y}_0}, \qquad \varphi_{O',E',\mathcal{Y}_0}(a) = [1, a, \ldots, a], \quad \forall\, a \in K, \tag{8.2}$$

is bijective. Therefore by means of $\varphi_{O',E',\mathcal{Y}_0}$ we can carry the field structure of K to $K_{O',E',\mathcal{Y}_0}$. More precisely, the triple $(K_{O',E',\mathcal{Y}_0}, +, \cdot)$ becomes a field canonically isomorphic to K with respect to the operations defined by

$$\begin{aligned} +[1, b, \dots, b] &= [1, a+b, \dots, a+b], \\ [1, a, \dots, a] \cdot [1, b, \dots, b] &= [1, ab, \dots, ab], \quad \forall\, a, b \in K. \end{aligned} \tag{8.3}$$

If $[1, a, \dots, a], [1, b, \dots, b] \in K_{O'.E',\mathcal{Y}_0}$ are two elements of $K_{O',E',\mathcal{Y}_0}$ the addition of $K_{O',E',\mathcal{Y}_0}$ defined by (8.3) can be seen geometrically as

$$\tau^{O'}_{A'+B'} = \tau^{O'}_{A'} \circ \tau^{O'}_{B'} \qquad \forall\, A' = [1, a, \dots, a],\ B' = [1, b, \dots, b], \tag{8.4}$$

where $\tau^{O'}_{A'}$ is the unique projective translation of axis $\mathcal{Y}_0$ and centre $U' = [0, 1, \dots, 1]$ which sends $O' = [1, 0, \dots, 0]$ to $A' = [1, a, \dots, a]$. Indeed, from the proof of Proposition 8.8, $\tau^{O'}_{A'}$, $\tau^{O'}_{B'}$ and $\tau^{O'}_{A'+B'}$ correspond to the matrices

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ a & 1 & 0 & \cdots & 0 \\ a & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a & 0 & 0 & \cdots & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ b & 1 & 0 & \cdots & 0 \\ b & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & 0 & 0 & \cdots & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ a+b & 1 & 0 & \cdots & 0 \\ a+b & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a+b & 0 & 0 & \cdots & 1 \end{pmatrix}$$

respectively. Analogously the multiplication of $K_{O',E',\mathcal{Y}_0}$ defined by (8.3) is geometrically given by

$$\sigma^{E'}_{A'.B'} = \sigma^{E'}_{A'} \circ \sigma^{E'}_{B'}, \qquad \forall\, A' = [1, a, \dots, a],\ B' = [1, b, \dots, b],\ a, b \in K \setminus \{0\}, \tag{8.5}$$

where $\sigma^{E'}_{A'}$ is the unique projective homothety of axis $\mathcal{Y}_0$ and centre O' which sends E' to $A' = [1, a, \dots, a]$. Indeed, from the proof of Proposition 8.8, $\sigma^{E'}_{A'}$, $\sigma^{E'}_{B'}$ and $\sigma^{E'}_{A'.B'}$ correspond to the matrices

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & b & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & ab & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & ab \end{pmatrix}$$

respectively. In particular we have

$$E' \cdot A' = A' \qquad \forall\, A' = [1, a, \dots, a] \in K_{O',E',\mathcal{Y}_0}. \tag{8.6}$$

If $ab = 0$ then $A' = O'$ (or $B' = O'$) and we put by definition $O' \cdot A' = O'$ for every $A' = [1, a, \ldots, a] \in K_{O',E',\mathcal{Y}_0}$.

Proposition 8.21 *Assume the notation of* Definition 8.20. *Let $\mathcal{Y}$ be a hyperplane of $\mathbb{P}^n(K)$ and O, $E \in \mathbb{P}^n(K) \setminus \mathcal{Y}$ be two distinct points. Let*

$$K_{O,E,\mathcal{Y}} := OE \setminus \mathcal{Y}, \qquad U := OE \cap \mathcal{Y}.$$

Then there exists a natural field structure on $K_{O,E,\mathcal{Y}}$ whose algebraic operations are given by

$$\tau^O_{A+B} = \tau^O_A \circ \tau^O_B, \quad \textit{i.e. } A + B = \tau^O_A \circ \tau^O_B(O) = \tau^O_A(B), \quad \forall\, A, B \in K_{O,E,\mathcal{Y}}, \tag{8.7}$$

$$\sigma^E_{A\cdot B} = \sigma^E_A \circ \sigma^E_B \quad \textit{i.e. } A \cdot B = \sigma^E_A \circ \sigma^E_B(E) = \sigma^E_A(B), \quad \forall\, A, B \in K_{O,E,\mathcal{Y}} \setminus \{O\}, \tag{8.8}$$

$$O \cdot A = O \qquad \forall\, A \in K_{O,E,\mathcal{Y}}, \tag{8.9}$$

$$E \cdot A = A \qquad \forall\, A \in K_{O,E,\mathcal{Y}}, \tag{8.10}$$

where τ^O_A is the projective translation of axis $\mathcal{Y}$ and centre U such that $\tau^O_A(O) = A$; σ^E_A is the projective homothety of axis $\mathcal{Y}$ and centre O such that $\sigma^E_A(E) = A$, if $A \neq O$.

Furthermore, there exists a canonical field isomorphism $\varphi_{O,E,\mathcal{Y}} \colon K \to K_{O,E,\mathcal{Y}}$ such that $\varphi_{O,E,\mathcal{Y}}(0) = O$ and $\varphi_{O,E,\mathcal{Y}}(1) = E$ given by (8.13).

Proof Choose n points $E_1, \ldots, E_n \in \mathcal{Y}$ which generate $\mathcal{Y}$ (hence $E_1, \ldots, E_n \in \mathcal{Y}$ are projectively independent). Therefore $\{O, E, E_1, \ldots, E_n\}$ is a projective frame of $\mathbb{P}^n(K)$ and the set $\{O, E, U\}$ is a projective frame of OE. By Theorem 6.47 there exists a unique $\sigma \in \mathrm{PGL}_n(K)$ such that

$$\sigma(O') = O,\ \sigma(E') = E,\ \sigma(E'_1) = E_1,\ \ldots,\ \sigma(E'_n) = E_n.$$

In particular, $\sigma(\mathcal{Y}_0) = \mathcal{Y}$. Since $U' = O'E' \cap \mathcal{Y}_0$ and $U = OE \cap \mathcal{Y}$, we also have $\sigma(U') = U$. Furthermore, if $\mathcal{Y}^*$ is another hyperplane of $\mathbb{P}^n(K)$ passing through U such that $O,\ E \in \mathbb{P}^n(K) \setminus \mathcal{Y}^*$ and if $E^*_1, \ldots, E^*_n \in \mathcal{Y}^*$ are n points generating $\mathcal{Y}^*$, then the set $\{O, E, E^*_1, \ldots, E^*_n\}$ is another projective frame of $\mathbb{P}^n(K)$. By Theorem 6.47 there exists a unique automorphism $\tau \in \mathrm{PGL}_n(K)$ such that

$$\tau(O') = O,\ \tau(E') = E,\ \tau(E'_1) = E^*_1,\ \ldots,\ \tau(E'_n) = E^*_n.$$

Since the hyperplanes $\mathcal{Y}$ and $\mathcal{Y}^*$ contain U, it follows that $\sigma(U') = \tau(U') = U$. Therefore the restrictions $\sigma|_{O'E'} \colon O'E' \to OE$ and $\tau|_{O'E'} \colon O'E' \to OE$ are the same projective linear isomorphism $O'E' \to OE$. In particular, the projective frame $\{O, E, U\}$ of the line OE does not depend on the choice of the hyperplane $\mathcal{Y}$

passing through U and such that $O, E \in \mathbb{P}^n(K) \setminus \mathcal{Y}$. If we put

$$\widetilde{\sigma}_{O,E}\colon K_{O',E',\mathcal{Y}_0} \to K_{O,E,\mathcal{Y}}, \qquad \widetilde{\sigma}_{O,E} := \sigma|_{O'E' \smallsetminus \{U'\}} = \tau|_{O'E' \smallsetminus \{U'\}} \tag{8.11}$$

($\widetilde{\sigma}_{O,E}\colon O'E' \smallsetminus \{U'\} \to OE \smallsetminus \{U\}$ since $\sigma(U') = U$), then $\widetilde{\sigma}_{O,E}(O') = O$ and $\widetilde{\sigma}_{O,E}(E') = E$.

Now let $A, B \in K_{O,E,\mathcal{Y}}$. By that seen above there exist uniquely determined $a, b \in K$ such that $A' = [1, a, \ldots, a]$ and $B' = [1, b, \ldots, b]$ with $A = \sigma(A') = \widetilde{\sigma}_{O,E}(A')$ and $B = \sigma(B') = \widetilde{\sigma}_{O,E}(B')$. Since $\sigma \circ \tau_{A'}^{O'} \circ \sigma^{-1}$ is a projective translation with axis $\mathcal{Y}$ which sends O into A and $\sigma \circ \sigma_{A'}^{E'} \circ \sigma^{-1}$ is a projective homothety with axis $\mathcal{Y}$ and centre O which sends E into A, by Proposition 8.8 we have the following formulas:

$$\tau_A^O = \sigma \circ \tau_{A'}^{O'} \circ \sigma^{-1} \qquad \text{and} \qquad \sigma_A^E = \sigma \circ \sigma_{A'}^{E'} \circ \sigma^{-1}. \tag{8.12}$$

By means of (8.4), (8.5), (8.6) and (8.12) the operations of $K_{O,E,\mathcal{Y}}$ are naturally defined by the formulas from (8.7) to (8.10). Therefore $\widetilde{\sigma}_{O,E}$ is an isomorphisms of fields:

$$\begin{aligned}
\widetilde{\sigma}_{O,E}(A' \cdot B') &= \widetilde{\sigma}_{O,E}(\sigma_{A'}^{E'}(B')) = \widetilde{\sigma}_{O,E}(\sigma^{-1} \circ \sigma_A^E \circ \sigma(B')) \\
&= \sigma_A^E(B) = A \cdot B = \widetilde{\sigma}_{O,E}(A') \cdot \widetilde{\sigma}_{O,E}(B'); \\
\widetilde{\sigma}_{O,E}(A' + B') &= \widetilde{\sigma}_{O,E}(\tau_{A'}^{O'}(B')) = \widetilde{\sigma}_{O,E}(\sigma^{-1} \circ \tau_A^O \circ \sigma(B')) \\
&= \tau_A^O(B) = A + B = \widetilde{\sigma}_{O,E}(A') + \widetilde{\sigma}_{O,E}(B').
\end{aligned}$$

Thus the composition

$$\varphi_{O,E,\mathcal{Y}}\colon K \to K_{O,E,\mathcal{Y}}, \qquad \varphi_{O,E,\mathcal{Y}} = \widetilde{\sigma}_{O,E} \circ \varphi_{O',E',\mathcal{Y}_0} \tag{8.13}$$

is a canonical isomorphism of fields. □

Corollary 8.22 *Let $f \in \mathrm{Aut}(\mathbb{P}^n(K))$. Let $\mathcal{Y}$ be a hyperplane and $O, E \in \mathbb{P}^n(K) \smallsetminus \mathcal{Y}$ be two distinct points. Then the restriction $f|_{K_{O,E,\mathcal{Y}}}\colon K_{O,E,\mathcal{Y}} \to K_{f(O),f(E),f(\mathcal{Y})}$ is a field isomorphism.*

Proof Let:

(a) τ_A^O be the unique projective translation of axis $\mathcal{Y}$ and centre $U = OE \cap \mathcal{Y}$ such that $\tau_A^O(O) = A$,
(b) σ_B^E be the unique projective homothety of axis $\mathcal{Y}$ and centre O such that $\sigma_B^E(E) = B \neq O$.

Then by Proposition 8.8

$$\begin{aligned}\tau_{f(A)}^{f(O)} &= f \circ \tau_A^O \circ f^{-1}, \quad \forall\, A \in K_{O,E,\mathcal{Y}};\\ \sigma_{f(A)}^{f(E)} &= f \circ \sigma_A^E \circ f^{-1}, \quad \forall\, A \in K_{O,E,\mathcal{Y}} \setminus \{O\},\end{aligned} \tag{8.14}$$

where $\tau_{f(A)}^{f(O)}$ has axis $f(\mathcal{Y})$ and centre $f(U)$, while $\sigma_{f(A)}^{f(E)}$ has axis $f(\mathcal{Y})$ and centre $f(O)$. By the group structure on $K_{f(O),f(E),f(\mathcal{Y})}$ and by (8.14) we have the relations

$$\begin{aligned}\tau_{f(A)+f(B)}^{f(O)} &= \tau_{f(A)}^{f(O)} \circ \tau_{f(B)}^{f(O)} = (f \circ \tau_A^O \circ f^{-1}) \circ (f \circ \tau_B^O \circ f^{-1})\\ &= f \circ (\tau_A^O \circ \tau_B^O) \circ f^{-1} = f \circ \tau_{A+B}^O \circ f^{-1} = \tau_{f(A+B)}^{f(O)}\end{aligned}$$

and

$$\begin{aligned}\sigma_{f(A)\cdot f(B)}^{f(E)} &= \sigma_{f(A)}^{f(E)} \circ \sigma_{f(B)}^{f(E)} = (f \circ \sigma_A^E \circ f^{-1}) \circ (f \circ \sigma_B^E \circ f^{-1})\\ &= f \circ (\sigma_A^E \circ \sigma_B^E) \circ f^{-1} = f \circ \sigma_{A\cdot B}^E \circ f^{-1} = \sigma_{f(A\cdot B)}^{f(E)}.\end{aligned}$$

This concludes the proof. □

Definition 8.23 The field $K_{O,E,\mathcal{Y}}$ is called the *coordinate field* of $\mathbb{P}^n(K)$ (according to Remark 8.15), unique up to field isomorphisms.

Proposition 8.24 *Let*

$$\begin{aligned}\Big\{O' = [1, 0, \ldots, 0],\ E' = [1, 1, \ldots, 1],\ E_1' = [0, 1, 0, \ldots, 0], \ldots, E_n'\\ = [0, 0, \ldots, 0, 1]\Big\}\end{aligned} \tag{8.15}$$

be the standard projective frame of $\mathbb{P}^n(K)$ with $n \geq 2$. If $\alpha \in \mathrm{Aut}(\mathbb{P}^n(K))$ is such that $\alpha(O') = O'$, $\alpha(E_i') = E_i'$, $i = 1, 2, \ldots, n$, $\alpha(E') = E'$, then $\alpha \in \mathrm{PAut}_n(K)$.

Proof We assume the notation of the discussion coming before Proposition 8.21 to (8.3) and put $K' := K_{O',E',\mathcal{Y}_0}$. By Lemma 8.22 $\alpha|_{K'}\colon K' \to K'$ is a group isomorphism. By means of the canonical isomorphism $\varphi' := \varphi_{O',E',\mathcal{Y}_0}\colon K \to K'$ defined by (8.2), we have the automorphism $\beta\colon K \to K$ given by $\beta(a) = b$, where $\alpha|_{K'}([1, a, \ldots, a]) = [1, b, \ldots, b]$ for every $a \in K$. We have to show that $\alpha([a_0, a_1, \ldots, a_n]) = [\beta(a_0), \beta(a_1), \ldots, \beta(a_n)]$ for each $P = [a_0, a_1, \ldots, a_n] \in \mathbb{P}^n(K)$.

First suppose $P = [a_0, a_1, \ldots, a_n] \notin \mathcal{Y}$. Assume that $a_0 = 1$. Put $\alpha(P) = [1, b_1, \ldots, b_n]$ and fix $i \in \{1, \ldots, n\}$. Consider the point $O'E' \cap (\alpha(P) + E_1' + \cdots + \widehat{E_i'} + \cdots E_n')$, where $\widehat{E_i'}$ means that E_i' is omitted. Then

$$\begin{aligned}O'E' \cap (\alpha(P) + E_1' + \cdots + \widehat{E_i'} + \cdots E_n') &= \alpha(O'E') + \alpha(P + E_1' + \cdots\\ &+ \widehat{E_i'} + \cdots E_n') = \alpha(O'E' \cap (P + E_1' + \cdots + \widehat{E_i'} + \cdots E_n'))\end{aligned}$$

since $\alpha(O') = O'$, $\alpha(E'_i) = E'_i$, $i = 1, 2, \ldots, n$, $\alpha(E') = E'$. The hyperplane $(\alpha(P) + E'_1 + \cdots + \widehat{E'_i} + \cdots E'_n)$ has the equation $b_i x_0 - x_i = 0$ so that

$$O'E' \cap (\alpha(P) + E'_1 + \cdots + \widehat{E'_i} + \cdots E'_n) = [1, b_i, \ldots, b_i], \quad i = 1, 2, \ldots, n.$$

The hyperplane $P + E'_1 + \cdots + \widehat{E'_i} + \cdots E'_n$ has the equation $a_i x_0 - x_i = 0$ so that

$$O'E' \cap (P + E'_1 + \cdots + \widehat{E'_i} + \cdots E'_n) = [1, a_i, \ldots, a_i] \quad i = 1, 2, \ldots, n.$$

Therefore $\alpha([1, a_i, \ldots, a_i]) = [1, b_i, \ldots, b_i]$ i.e. $\beta(a_i) = b_i$, $i = 1, 2, \ldots, n$.

Now suppose $P = [0, a_1, \ldots, a_n] \in \mathcal{Y}_0$. Take $Q = [1, a_1, \ldots, a_n]$ so that $P = O'Q \cap \mathcal{Y}_0$. By the above case $\alpha(Q) = [1, \beta(a_1), \ldots, \beta(a_n)]$. Then $\alpha(P) = \alpha(O')\alpha(Q) \cap \mathcal{Y}_0 = [0, b_1, \ldots, b_n]$ so that $b_i = \beta(a_i)$, $i = 1, \ldots, n$. □

The fundamental result concerning the structure of $\mathrm{Aut}(\mathbb{P}^n(K))$ is the following:

Theorem 8.25 *Let $\alpha \in \mathrm{Aut}(\mathbb{P}^n(K))$ $(n \geq 2)$ be any automorphism. Then there exist $\beta \in \mathrm{PGL}_n(K)$ and $\gamma \in \mathrm{PAut}_n(K)$ such that α has a unique expression of the form $\alpha = \beta \circ \gamma$. Since $\mathrm{PGL}_n(K)$ is a normal subgroup of $\mathrm{Aut}(\mathbb{P}^n(K))$ and $\mathrm{PGL}_n(K) \cap \mathrm{PAut}_n(K) = \{\mathrm{id}\}$, then $\mathrm{Aut}(\mathbb{P}^n(K))$ is the semi-direct product $\mathrm{PGL}_n(K) \rtimes \mathrm{PAut}_n(K)$ according to* Definition 3.56.

Proof Let $\{O, E_1, \ldots, E_n, E\}$ be the projective frame as in Proposition 8.24. By Theorem 6.47 there exists $\beta \in \mathrm{PGL}_n(K)$ such that $\beta(O) = \alpha(O)$, $\beta(E_i) = \alpha(E_i)$, $i = 1, 2, \ldots, n$, $\beta(E) = \alpha(E)$. If we put $\gamma := \beta^{-1} \circ \alpha$ then we get $\gamma(O) = O$, $\gamma(E_i) = E_i$, $i = 1, 2, \ldots, n$ and $\gamma(E) = E$, so that by Proposition 8.24 it follows that $\gamma \in \mathrm{PAut}_n(K)$. Hence $\alpha = \beta \circ \gamma$, with $\beta \in \mathrm{PGL}_n(K)$ and $\gamma \in \mathrm{PAut}_n(K)$.

Now let $\alpha \in \mathrm{PGL}_n(K) \cap \mathrm{PAut}_n(K)$. Then $\alpha = T_A$ with $A = (a_{ij})_{i,i=0,\ldots,n}$ a non-singular matrix (Definition 6.45). Since $O, E_1, \ldots, E_n$ are fixed points of α, then A is diagonal, i.e. $a_{ij} = 0, \forall\, i \neq j$. Since also E is a fixed point of α, then $a_{00} = a_{11} = \ldots = a_{nn} = a \neq 0$, so that $\alpha = \mathrm{id}_{\mathbb{P}^n(K)}$.

If $\alpha = \beta \circ \gamma = \beta' \circ \gamma'$ are two expressions of α with $\beta, \beta' \in \mathrm{PGL}_n(K)$ and $\gamma, \gamma' \in \mathrm{PAut}_n(K)$, then $\beta \circ \beta'^{-1} = \gamma' \circ \gamma^{-1}$. Since $\beta \circ \beta'^{-1} \in \mathrm{PGL}_n(K)$ and $\gamma' \circ \gamma^{-1} \in \mathrm{PAut}_n(K)$ we get immediately $\beta = \beta'$ and $\gamma = \gamma'$.

We have now to show that $\mathrm{PGL}_n(K)$ is a normal subgroup of $\mathrm{Aut}(\mathbb{P}^n(K))$. Let $\sigma \in \mathrm{PGL}_n(K)$ and $\alpha \in \mathrm{Aut}(\mathbb{P}^n(K))$. We have just proved that there are $\beta \in \mathrm{PGL}_n(K)$ and $\gamma \in \mathrm{PAut}_n(K)$ such that $\alpha = \beta \circ \gamma$. Then $\alpha \circ \sigma \circ \alpha^{-1} = \beta \circ \gamma \circ \sigma \circ \gamma^{-1} \circ \beta^{-1}$. Let $\sigma = T_A$ with $A = (a_{ij})_{i,i=0,\ldots,n}$. Since

$$\sigma(\gamma^{-1}([x_0, \ldots, x_n])) = \sigma([\gamma^{-1}(x_0), \ldots, \gamma^{-1}(x_n)] = [y_0, \ldots, y_n], \quad \text{with}$$

$$y_i = \sum_{j=0}^{n} a_{ij}\gamma^{-1}(x_j), \qquad i = 0, \ldots, n$$

we have $\gamma(\sigma(\gamma^{-1}([x_0,\dots,x_n]))) = \gamma([y_0,\dots,y_n]) = [\gamma(y_0),\dots,\gamma(y_n)]$, where

$$\gamma(y_i) = \sum_{j=0}^{n} \gamma(a_{ij})\gamma(\gamma^{-1}(x_j)) = \sum_{j=0}^{n} \gamma(a_{ij})x_j, \qquad i = 0,\dots,n.$$

Hence $\gamma \circ \sigma \circ \gamma^{-1} = T_B \in \mathrm{PGL}_n(K)$, where $B = \big(\gamma(a_{ij})\big)_{i,i=0,\dots,n}$. □

Corollary 8.26 *If $K = \mathbb{R}$, $K = \mathbb{Q}$ or $K = \mathbb{Z}_p$, then for every $n \geq 2$ one has $\mathrm{Aut}(\mathbb{P}^n(K)) = \mathrm{PGL}_n(K)$. Furthermore, if $P_0, P_1, \dots, P_{n+1}$ and $Q_0, Q_1, \dots, Q_{n+1}$ are two projective frames of $\mathbb{P}^n(K)$, then there exists one and only one linear automorphism $\alpha \in \mathrm{PGL}_n(K)$ such that $\alpha(P_i) = Q_i$, $i = 0, 1, \dots, n+1$.*

Proof The first assertion follows from Theorem 8.25, Proposition 8.17 and Remark 8.19. The second assertion is a consequence of Theorem 6.47, Theorem 8.25 and Proposition 8.17. □

Automorphisms of the Projective Line

Definition 8.27 Let $\mathcal{P}(\mathbb{P}^1(K))$ be the group of all bijective maps $\mathbb{P}^1(K) \to \mathbb{P}^1(K)$ (i.e. the permutation group of $\mathbb{P}^1(K)$). We introduced $\mathrm{PGL}_1(K)$ (Definition 6.40). We can extend Definition 8.16 to the case $n = 1$:

$$S_1 : \mathrm{Aut}(K) \to \mathcal{P}(\mathbb{P}^1(K)), \quad S_1(\alpha) = S_\alpha, \quad \text{where} \quad S_\alpha([x_0, x_1]) = [\alpha(x_0), \alpha(x_1)], \quad \forall\, \alpha \in \mathrm{Aut}(K).$$

We put $\mathrm{PAut}_1(K) := S_1(\mathrm{Aut}(K))$ and define the group $\mathrm{Aut}(\mathbb{P}^1(K))$ of *projective automorphisms* of the standard projective line $\mathbb{P}^1(K)$ as the subgroup of $\mathcal{P}(\mathbb{P}^1(K))$ generated by the union $\mathrm{PGL}_1(K) \cup \mathrm{PAut}_1(K)$.

Corollary 8.28 *Let $A, B, C \in \mathbb{P}^1(K)$ be three distinct points. For every triple $A', B', C' \in \mathbb{P}^1(K)$ of distinct points there exists one and only one $\alpha \in \mathrm{PGL}_1(K)$ such that $\alpha(A) = A'$, $\alpha(B) = B'$ and $\alpha(C) = C'$.*

Proof It is an immediate consequence of Theorem 6.47 since any three distinct points are a projective frame for a projective line. □

Following the proof of Theorem 8.25, by Corollary 8.28 we get:

Corollary 8.29 *The statement of* Theorem 8.25 *holds true also for $n = 1$.*

Definition 8.30 (Dual Projectivity) Let $A \in \mathrm{GL}_{n+1}(K)$ and $T_A : \mathbb{P}^n(K) \to \mathbb{P}^n(K)$ be the associated linear projective automorphism. Thus if $T_A(x) = y$, where $x = [x_0,\dots,x_n]$ and $y = [y_0,\dots,y_n]$, we have

$$\varrho\, y^t = A\, x^t, \qquad \varrho\, x^t = A^{-1}\, y^t, \qquad \varrho \in K^*.$$

Let $u = [u_0, \ldots, u_1]$ be a hyperplane of $\mathbb{P}^n(K)$. For each point x lying on u we have $u\, x^t = 0$ so that $u A^{-1}\, y^t = 0$. By means the identification (6.16) we define the *dual projectivity* T_A^* of T_A by

$$T_A^* \colon \mathbb{P}^n(K)^* \to \mathbb{P}^n(K)^*, \qquad T_A^* = T_{(A^t)^{-1}} \text{ i.e. } T_A^*(u) = (A^t)^{-1}\, u^t. \tag{8.16}$$

8.3 Exercises

Exercise 8.1 Let σ be a permutation of $\{0, 1, \ldots, n\}$ $(n \geq 1)$. Show that $f \colon \mathbb{P}^n(K) \to \mathbb{P}^n(K)$ given by $f([x_0, x_1, \ldots, x_n]) = [x_{\sigma(0)}, x_{\sigma(1)}, \ldots, x_{\sigma(n)}]$ is a linear projective automorphism, i.e. $f \in \mathrm{PGL}_n(K)$.

Exercise 8.2 Prove that the identity is the unique automorphism of the field $\mathbb{Z}_p$, where $p \geq 2$ is a prime number.

Exercise 8.3 Show that $\mathrm{PGL}_n(K)$ acts on $\mathbb{P}^n(K)$ transitively, i.e. for every two points $A, B \in \mathbb{P}^n(K)$ there exists $\sigma \in \mathrm{PGL}_n(K)$ such that $\sigma(A) = B$.

Exercise 8.4 Show the following structure theorem.

Let $A \in \mathrm{GL}_{n+1}(K)$ and $T_A \colon \mathbb{P}^n(K) \to \mathbb{P}^n(K)$ be the associated projectivity (as in (6.9)). Let $\lambda_1, \ldots, \lambda_t$ be the distinct eigenvalues of A belonging to K and $V_1, \ldots, V_t$ be the corresponding eigenspaces. Then the locus of fixed points of T_A is

$$\bigcup_{i=1}^{t} \mathcal{S}_i, \qquad \mathcal{S}_i = \mathbb{P}(V_i), \quad i = 1, \ldots, t.$$

The subspaces $\mathcal{S}_i$ are called the *fundamental subspaces* of T_A. They are projectively independent and $\dim(\mathcal{S}_i) = h_i := n - k_i$, where $k_i = \mathrm{rank}(A - \lambda_i I_{n+1})$, $i = 1, \ldots, t$.

Exercise 8.5 Assume the hypothesis and notation of Exercise 8.4 and prove the following assertion:

There exist t independent *fundamental linear systems* $\Sigma_1, \ldots, \Sigma_t$ of fixed hyperplanes with respect to T_A whose base loci are the dual spaces of $\mathcal{S}_1, \ldots, \mathcal{S}_t$, respectively.

Exercise 8.6 Determine fixed points and fixed lines of $\sigma \in \mathrm{PGL}_2(\mathbb{R})$ defined by

$$\sigma([x_0, x_1, x_2]) = [x_0 - x_1, x_0 + 3x_1, 2x_2].$$

Exercise 8.7 Find all the automorphisms $\sigma \in \mathrm{PGL}_2(\mathbb{R})$ which fix the points $[1, 1, 1]$, $[2, -1, 1]$ and $[0, 1, 1]$.

Exercise 8.8 Let

$$d : x_0 - x_1 = 0, \quad d' : x_0 + x_1 = 0, \quad e : x_0 + x_1 + x_2 = 0, \quad e' : x_1 + x_2 = 0$$

be 4 lines of $\mathbb{P}^2(\mathbb{R})$. Find all the automorphisms $\sigma \in \mathrm{PGL}_2(\mathbb{R})$ such that $\sigma(d) = d'$, $\sigma(e) = e'$ and $\sigma([1, 2, 1]) = [1, 0, 0]$.

Exercise 8.9 Prove that the order of the group $\mathrm{PGL}_2(\mathbb{Z}_p)$ (with $p \geq 2$ prime number) is $p^3(p^3 - 1)(p^2 - 1)$. Generalize this result by determining the order of $\mathrm{PGL}_n(\mathbb{Z}_p)$, with $n \geq 2$.

Exercise 8.10 Let

$$f : \mathbb{P}^2(\mathbb{R}) \to \mathbb{P}^2(\mathbb{R}), \qquad f([x_0, x_1, x_2]) = [5x_0 + x_2, 5x_1 + x_2, 5x_2].$$

Show that f is a projective translation.

Exercise 8.11 Assume the hypothesis and notation of Example 6.46. Prove that

$$T_A : \mathbb{P}^n(\mathbb{R}) \to \mathbb{P}^n(\mathbb{R})$$

is a projective translation $\neq$ id if and only if the characteristic polynomial of A is of the form $a(T - \lambda)^{n+1}$, with $a = \pm 1$ and $\lambda \in \mathbb{R}$, i.e. A has a unique eigenvalue λ whose eigenspace has dimension n.

Exercise 8.12 Assume the hypothesis and notation of Example 6.46. Prove that

$$T_A : \mathbb{P}^n(\mathbb{R}) \to \mathbb{P}^n(\mathbb{R})$$

is a projective homothety $\neq$ id if and only if A has two distinct eigenvalues λ_1 and λ_2, whose multiplicity is 1 and n respectively, and the eigenspace of λ_2 has dimension n.

Exercise 8.13 Show that $T_A : \mathbb{P}^{2n}(\mathbb{R}) \to \mathbb{P}^{2n}(\mathbb{R})$ as in Exercise 9.1 has at least a fixed point.

Exercise 8.14 Find the fixed points and the fixed lines of

$$f : \mathbb{P}^2(\mathbb{R}) \to \mathbb{P}^2(\mathbb{R}), \qquad f([x_0, x_1, x_2]) = [5x_0 + x_1, 5x_1 + x_2, 5x_2].$$

Exercise 8.15 Prove the affine Desargues' theorem.

> Let $\triangle ABC$, $\triangle A'B'C'$ be triangles in an affine space. Assume they have no vertices in common and that their sides are parallel: $AB \parallel A'B'$, $AC \parallel A'C'$, $BC \parallel B'C'$. Then the three lines AA', BB', CC' are either parallel or concurrent.

(Hint: Look at the proofs of Menelaus' and Ceva's theorems.)

Exercise 8.16 Show that the following affine assertion is a consequence of the projective Desargues' theorem:

> Let d_1, d_2 and d_3 be three distinct parallel lines of $\mathbb{A}^2(K)$. Take two distinct points A_i and B_i on d_i, $i = 1, 2, 3$, such that the pairs of lines (A_1A_2, B_1B_2), (A_1A_3, B_1B_3) and (A_2A_3, B_2B_3) are concurrent at the points D_{12}, D_{13} and D_{23} respectively. Then D_{12}, D_{13} and D_{23} are collinear.
>
> Find also an alternative affine proof.

Chapter 9
Affine Geometry and Projective Geometry

Throughout this chapter we focus on the standard projective space $\mathbb{P}^n(K)$ of dimension $n \geq 2$ over a field K.

9.1 Affine Space Structure on the Complement of a Hyperplane of $\mathbb{P}^n(K)$

Definition 9.1 Let $\mathcal{Y}$ be a fixed hyperplane of $\mathbb{P}^n(K)$. We shall prove that on $\mathcal{A} := \mathbb{P}^n(K) \setminus \mathcal{Y}$ there is a natural structure of affine space over K of dimension n. Fix one point $O \in \mathbb{P}^n(K) \setminus \mathcal{Y}$ and put

$$\overrightarrow{O\mathcal{A}} := \{(O, X) : X \in \mathcal{A}\}.$$

The pair (O, A) will be denoted by $\overrightarrow{OA}$.

For every point $X \in \mathcal{A}$ let $\tau_X^O \in \mathcal{T}_{\mathcal{Y}}$ be the unique projective translation of axis $\mathcal{Y}$ such that $\tau_X^O(O) = X$; if $X \neq O$ the centre of τ_X^O is the point $\mathcal{Y} \cap OX$, while if $X = O$, then $\tau_X^O = \mathrm{id}_{\mathbb{P}^n(K)}$. The *sum* on $\overrightarrow{O\mathcal{A}}$ is defined by

$$\overrightarrow{OX} + \overrightarrow{OY} = \overrightarrow{OZ}, \qquad \text{with} \qquad Z := \tau_X^O \circ \tau_Y^O(O) = \tau_X^O(Y), \qquad \forall\, X, Y \in \mathcal{A}. \tag{9.1}$$

Writing X and Y instead of $\overrightarrow{OX}$ and $\overrightarrow{OY}$ respectively, the sum $\overrightarrow{OX} + \overrightarrow{OY}$ is also given by

$$\tau_{X+Y}^O = \tau_X^O \circ \tau_Y^O. \tag{9.2}$$

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Now fix another point $E \in \mathbb{P}^n(K) \setminus \mathcal{Y}$, $E \neq O$. For each point $A \in K_{O,E,\mathcal{Y}}$, $A \neq O$ (see Proposition 8.21), let $\sigma_A^E \in \mathcal{H}_{\mathcal{Y},O}$ be the unique projective homothety (with centre O and axis $\mathcal{Y}$) such that $\sigma_A^E(E) = A$. The *scalar multiplication* $K_{O,E,\mathcal{Y}} \times \overrightarrow{O\mathcal{A}} \to \overrightarrow{O\mathcal{A}}$ is defined by the formulas

$$A \cdot \overrightarrow{OX} = \overrightarrow{OZ}, \qquad \text{with } Z := \sigma_A^E(X), \qquad \forall\, A \in K_{O,E,\mathcal{Y}} \setminus \{O\}, \quad \forall\, X \in \mathcal{A}. \tag{9.3}$$

$$O \cdot \overrightarrow{OX} = \overrightarrow{OO}, \qquad \forall\, X \in \mathcal{A}. \tag{9.4}$$

Theorem 9.2 *The set $\overrightarrow{O\mathcal{A}}$ equipped with the operations* (9.1), (9.3) *and* (9.4) *is a vector space over $K_{O,E,\mathcal{Y}}$, hence, by* Proposition 8.21 *also a K-vector space* (*in particular $a \cdot \overrightarrow{OX} := \varphi_{O,E,\mathcal{Y}}(a) \cdot \overrightarrow{OX}$, $\forall\, a \in K$*).

Proof First we prove that $(\overrightarrow{O\mathcal{A}}, +)$ is an abelian group. For every $X, Y, Z \in \mathcal{A}$ we have

$$\tau^O_{X+(Y+Z)} = \tau^O_X \circ \tau^O_{Y+Z} = \tau^O_X \circ (\tau^O_Y \circ \tau^O_Z) = (\tau^O_X \circ \tau^O_Y) \circ \tau^O_Z = \tau^O_{X+Y} \circ \tau^O_Z = \tau^O_{(X+Y)+Z}$$

since the composition of translations is associative, hence the sum (defined by (9.1) (or by (9.2)) is associative. Since the group of translations $\mathcal{T}_{\mathcal{Y}}$ of axis $\mathcal{Y}$ is commutative the sum (9.1) is commutative. Furthermore, from (9.2) it follows that

$$\tau^O_{O+X} = \tau^O_O \circ \tau^O_X = \tau^O_X$$

(because $\tau^O_O = \mathrm{id}_{\mathbb{P}^n(K)}$), i.e. $\overrightarrow{OO}$ is the identity of the sum. Finally, for every $X \in \mathcal{A}$ we define $X' := (\tau^O_X)^{-1}(O)$ which is the opposite element of X with respect to the sum. Indeed, one has $\tau^O_{X'}(O) = X'$ and

$$\tau^O_X \circ \tau^O_{X'}(O) = \tau^O_X(X') = \tau^O_X((\tau^O_X)^{-1}(O)) = O \quad \text{i.e.} \quad \overrightarrow{OX} + \overrightarrow{OX'} = \overrightarrow{OO}.$$

Thus we have proved that $(\overrightarrow{O\mathcal{A}}, +)$ is an abelian group.

Now we have to verify the axioms concerning the scalar multiplication defined by (9.3) and (9.4). It is clear that $E \cdot \overrightarrow{OX} = \overrightarrow{OX}$, $\forall X \in \mathcal{A}$ since $\sigma_E^E = \mathrm{id}_{\mathbb{P}^n(K)}$. The next step is to show the equality

$$A \cdot (\overrightarrow{OX} + \overrightarrow{OY}) = A \cdot \overrightarrow{OX} + A \cdot \overrightarrow{OY}, \qquad \forall\, A \in K_{O,E,\mathcal{Y}},\ \forall\, X, Y \in \mathcal{A}, \tag{9.5}$$

which is obvious if $A = E$ or if either X or Y coincides with O. Hence suppose A, $X, Y \neq O$. Since $A, X \neq O$, from $A \cdot \overrightarrow{OX} = \overrightarrow{OZ}$ with $Z = \sigma_A^E(X) \neq O$ we have $A \cdot \overrightarrow{OX} \neq \overrightarrow{OO}$. In order to prove (9.5) we need the following relations:

$$\begin{aligned}
\overrightarrow{OX} + \overrightarrow{OY} &= \overrightarrow{OZ}, \quad \text{with } Z = \tau_X^O(Y), \\
A \cdot (\overrightarrow{OX} + \overrightarrow{OY}) &= A \cdot \overrightarrow{OZ} = \overrightarrow{OZ'} \quad \text{with} \\
Z' &= \sigma_A^E(Z) = \sigma_A^E(\tau_X^O(Y)) = (\sigma_A^E \circ \tau_X^O \circ \tau_Y^O)(O), \\
A \cdot \overrightarrow{OX} &= \overrightarrow{OX'} \quad \text{with } X' = \sigma_A^E(X), \\
A \cdot \overrightarrow{OY} &= \overrightarrow{OY'} \quad \text{with } Y' = \sigma_A^E(Y), \\
\overrightarrow{OX'} + \overrightarrow{OY'} &= \overrightarrow{OZ"} \quad \text{with } Z" = \tau_{X'}^O(Y') = \tau_{X'}^O \circ \tau_{Y'}^O(O).
\end{aligned}$$

We claim that

$$\tau_{X'}^O = \sigma_A^E \circ \tau_X^O \circ \sigma_E^A, \tag{9.6}$$

$$\tau_{Y'}^O = \sigma_A^E \circ \tau_Y^O \circ \sigma_E^A. \tag{9.7}$$

Indeed, $\mathfrak{T}_\mathcal{Y}$ is a normal subgroup of $\mathfrak{S}_\mathcal{Y}$ (Proposition 8.5), so that $\sigma_A^E \circ \tau_X^O \circ \sigma_E^A = \sigma_A^E \circ \tau_X^O \circ (\sigma_A^E)^{-1}$ is a translation as well as $\sigma_A^E \circ \tau_Y^O \circ \sigma_E^A = \sigma_A^E \circ \tau_Y^O \circ (\sigma_A^E)^{-1}$. Thus in order to prove (9.6) and (9.7) it suffices to see that they coincide at O (Proposition 8.8):

$$\begin{aligned}
\tau_{X'}^O(O) = X' \quad \text{and} \quad \sigma_A^E \circ \tau_X^O \circ \sigma_E^A(O) &= \sigma_A^E(\tau_X^O(\sigma_E^A(O))) \\
= \sigma_A^E(\tau_X^O(O)) = \sigma_A^E(X) &= X'.
\end{aligned}$$

In the same way we check that $\tau_{Y'}^O(O) = Y' = \sigma_A^E \circ \tau_Y^O \circ \sigma_E^A(O)$. From the above relations we get

$$\begin{aligned}
Z" &= \tau_{X'}^O \circ \tau_{Y'}^O(O) = (\sigma_A^E \circ \tau_X^O \circ \sigma_E^A) \circ (\sigma_A^E \circ \tau_Y^O \circ \sigma_E^A)(O) \\
&= (\sigma_A^E \circ \tau_X^O \circ \tau_Y^O \circ \sigma_E^A)(O) = (\sigma_A^E \circ \tau_Z^O)(O) = Z',
\end{aligned}$$

i.e. (9.5). Now we have to show the equality

$$(A + B) \cdot \overrightarrow{OX} = A \cdot \overrightarrow{OX} + B \cdot \overrightarrow{OX}, \quad \forall A, B \in K_{O,E,\mathcal{Y}}, \ \forall X \in \mathcal{A}. \tag{9.8}$$

Bypassing the trivial cases we can assume $A, B, X \neq O$. If $X \in OE$ the relation (9.8) comes from the right distributivity of product in $K_{O,E,\mathcal{Y}}$ with respect to sum ($K_{O,E,\mathcal{Y}} = OE \setminus U$, where $U = OE \cap \mathcal{Y}$). Now let $X \notin OE$. Put $C := A + B$ and

$$\begin{aligned} \overrightarrow{OA'} &:= A \cdot \overrightarrow{OX}, \quad \text{with} \quad A' := \sigma_A^E(X), \\ \overrightarrow{OB'} &:= B \cdot \overrightarrow{OX}, \quad \text{with} \quad B' := \sigma_B^E(X), \\ \overrightarrow{OC'} &= C \cdot \overrightarrow{OX}, \quad \text{with} \quad C' := \sigma_C^E(X). \end{aligned} \tag{9.9}$$

Hence A', B', C' lie on OX, while A, B, C lie on the line OE. Furthermore, all the lines AA', BB' and CC' pass through the point $EX \cap \mathcal{Y}$ by (9.9) (indeed, for instance, $EX \cap \mathcal{Y} = \sigma_A^E(EX \cap \mathcal{Y}) = \sigma_A^E(EX) \cap \mathcal{Y} = AA' \cap \mathcal{Y}$). We have to prove the equality $\overrightarrow{OC'} = \overrightarrow{OA'} + \overrightarrow{OB'}$. Starting from the obvious relations

$$\begin{aligned} \tau_{A'}^O &= \tau_{A'}^A \circ \tau_A^O \\ \tau_{B'}^O &= \tau_{B'}^B \circ \tau_B^O \\ \tau_{C'}^O &= \tau_{C'}^C \circ \tau_C^O \end{aligned}$$

and taking into account that $\tau_C^O = \tau_A^O \circ \tau_B^O$, we get (using commutativity of $\mathcal{T}_{\mathcal{Y}}$)

$$\tau_{C'}^O \circ (\tau_{A'}^O)^{-1} \circ (\tau_{B'}^O)^{-1} = \tau_A^{A'} \circ \tau_B^{B'} \circ \tau_{C'}^C. \tag{9.10}$$

The left– hand side of (9.10) is a translation of centre $OX \cap \mathcal{Y}$, while the right– hand side of (9.10) is a translation of centre $EX \cap \mathcal{Y}$. Since we supposed $X \notin OE$, these two centres are distinct. By Proposition 8.2, both sides of (9.10) coincide with $\mathrm{id}_{\mathbb{P}^n(K)}$. In particular, $\tau_{C'}^O = \tau_{A'}^O \circ \tau_{B'}^O$ i.e. $\overrightarrow{OC'} = \overrightarrow{OA'} + \overrightarrow{OB'}$.

It remains to show that

$$(A \cdot B) \cdot \overrightarrow{OX} = A \cdot (B \cdot \overrightarrow{OX}), \qquad \forall\, A, B \in K_{O,E,\mathcal{Y}},\ \forall\, X \in \mathcal{A}. \tag{9.11}$$

Bypassing the trivial cases, we can suppose $A, B, X \neq O$. Hence (9.11) is equivalent to the equality

$$\sigma_{A \cdot B}^E(X) = \sigma_A^E(\sigma_B^E(X)),$$

which follows from $\sigma_{A \cdot B}^E = \sigma_A^E \circ \sigma_B^E$ (see (8.8) which defines the product in $K_{O,E,\mathcal{Y}}$). □

Remark 9.3 If $\mathcal{Y}_0$ is the hyperplane of equation $x_0 = 0$, $O = [1, 0, \ldots, 0]$ and $E = [1, 1, \ldots, 1]$, the structure of a standard K-vector space on

$$\mathbb{U}_0 := \mathbb{P}^n(K) \setminus \mathcal{Y}_0 \cong K^n, \qquad [1, x_1, \ldots, x_n] \longleftrightarrow (x_1, \ldots, x_n)$$

with respect to the sum defined by (9.1) and to the scalar multiplication defined by (9.3) and (9.4) appears clearly.

Theorem 9.4 *Under the above notation we have* $\dim_{K_{O,E,\mathcal{Y}}}(\overrightarrow{O\mathcal{A}}) = n$ *or also* $\dim_K(\overrightarrow{O\mathcal{A}}) = n$.

Proof Assume the notation of Definition 8.20. Choose a projective frame $\mathcal{R} := \{O, E_1, \ldots, E_n, E\}$ of $\mathbb{P}^n(K)$. There exists a unique automorphism $\sigma \in \mathrm{PGL}_n(K)$ such that $\sigma(O') = O$, $\sigma(E') = E$, $\sigma(E') = E$ and $\sigma(E'_i) = E_i$, $i = 1, \ldots, n$. Taking into account the proofs of Proposition 8.21 and Theorem 9.2, the map $\widetilde{\sigma}\colon \overrightarrow{O'\mathbb{U}_0} \to \overrightarrow{O\mathcal{A}}$, defined by $\widetilde{\sigma}(\overrightarrow{O'X}) = \overrightarrow{O\sigma(X)}$ for every $X \in \mathbb{U}_0$, is an isomorphism of K-vector spaces. By Remark 9.3, $\overrightarrow{O'\mathbb{U}_0}$ is canonically identified with K^n. □

Remark 9.5 Assume the above notation. Take another point $E'' \in OE \setminus \mathcal{Y}$ such that $E'' \neq O$. Consider the coordinates field $K_{O,E'',\mathcal{Y}}$ and the map

$$\varphi\colon K_{O,E,\mathcal{Y}} \to K_{O,E'',\mathcal{Y}}, \qquad \varphi(A) := \sigma_A^E(E''), \quad \forall\, A \in K_{O,E,\mathcal{Y}},\ A \neq O.$$

We put $\varphi(O) = O$ so that φ is a (canonical) field isomorphism. Indeed,

$$\sigma_A^E = \sigma_{\varphi(A)}^{E''} \qquad \forall\, A \in K_{O,E,\mathcal{Y}} \setminus \{O\}.$$

($\sigma_A^E = \sigma_{\sigma_A^E(E'')}^{E''}$ by Proposition 8.8 since they have the same axis $\mathcal{Y}$, the same centre O and $\sigma_A^E(E'') = \sigma_{\sigma_A^E(E'')}^{E''}(E'')$). Thus for every $A, B \in K_{O,E,\mathcal{Y}} \setminus \{O\}$ we have the equalities

$$\begin{aligned}\varphi(A \cdot B) &= \sigma_{A\cdot B}^E(E'') = \sigma_A^E(\sigma_B^E(E'')) = \sigma_{\sigma_A^E(E'')}^{E''}(\sigma_B^E(E''))\\ &= \sigma_{\varphi(A)}^{E''}(\varphi(B)) = \varphi(A)\cdot\varphi(B).\end{aligned}$$

We have now to check that $\varphi(A + B) = \varphi(A) + \varphi(B)$: by (f) of Corollary 8.11 we have

$$\varphi(A + B) = \sigma_{A+B}^E(E'') = \sigma_{\tau_A^O(B)}^E(E'') = \sigma_{E''}^E(\tau_A^O(B)).$$

On the other hand,

$$\begin{aligned}\varphi(A) + \varphi(B) &= \sigma_A^E(E'') + \sigma_B^E(E'') = \tau_{\sigma_A^E(E'')}^O(\sigma_B^E(E'')) = \tau_{\sigma_{E''}^E(A)}^O(\sigma_{E''}^E(B))\\ &= \tau_{\sigma_{E''}^E(A)}^{\sigma_{E''}^E(O)}(\sigma_{E''}^E(B)).\end{aligned}$$

By (8.14) with $f = \sigma^E_{E''}$

$$\tau^{\sigma^E_{E''}(O)}_{\sigma^E_{E''}(A)}(\sigma^E_{E''}(B)) = \sigma^E_{E''}(\tau^O_A(B)).$$

In other words, φ is defined in such a way that

$$A \cdot \overrightarrow{OX} = \varphi(A) \cdot \overrightarrow{OX}, \qquad \forall\, A \in K_{O,E,\mathcal{Y}},\ \forall\, X \in \mathcal{A}.$$

Therefore the structure of $K_{O,E,\mathcal{Y}}$-vector space of $\overrightarrow{O\mathcal{A}}$ is essentially independent of the choice of the point $E \in \mathbb{P}^n(K) \setminus (\{O\} \cup \mathcal{Y})$.

Let O and E be as above. Take another point $O'' \in \mathbb{P}^n(K) \setminus \mathcal{Y}$ and put $E'' := \tau^O_{O''}(E)$, where $\tau^O_{O''}$ is the only projective translation with axes $\mathcal{Y}$ sending O to O''. First observe that

$$\tau^O_X = \tau^{O''}_{\tau^O_{O''}(X)}, \qquad \forall\, X \notin \mathcal{Y}. \tag{9.12}$$

It is enough to see that $\tau^O_X(O'') = \tau^{O''}_{\tau^O_{O''}(X)}(O'')$. Indeed, by (c) of Corollary 8.11 $\tau^O_X(O'') = \tau^O_{O''}(X)$, so that we have

$$O'' = \tau^X_O(\tau^O_X(O'')); \qquad \tau^X_O(\tau^{O''}_{\tau^O_{O''}(X)}(O'')) = \tau^X_O(\tau^{O''}_{\tau^O_X(O'')}(O'')) = \tau^X_O(\tau^O_X(O'') = O''.$$

Then the map $\psi : K_{O,E,\mathcal{Y}} \to K_{O'',E'',\mathcal{Y}}$, defined by $\psi(A) = \tau^O_{O''}(A)$, is a canonical field isomorphism. Indeed,

$$\psi(A + B) = \tau^O_{O''}(A + B) = \tau^O_{O''}(\tau^O_A(B))$$

and

$$\psi(A) + \psi(B) = \tau^O_{O''}(A) + \tau^O_{O''}(B) = \tau^O_A(O'') + \tau^O_B(O'') = \tau^O_A(\tau^O_B(O'')) = \tau^O_A(\tau^O_{O''}(B)) = \tau^O_{O''}(\tau^O_A(B)).$$

Furthermore we have

$$\psi(A \cdot B) = \tau^O_{O''}(A \cdot B) = \tau^O_{O''}(\sigma^E_A(B)),$$

and (by (8.14) with $f = \tau^O_{O''}$)

$$\psi(A) \cdot \psi(B) = \sigma^{E''}_{\tau^O_{O''}(A)}(\tau^O_{O''}(B)) = \sigma^{\tau^O_{O''}(E)}_{\tau^O_{O''}(A)}(\tau^O_{O''}(B)) = \tau^O_{O''}(\sigma^E_A(B)).$$

The reader can easily verify that

$$f\colon \overrightarrow{O\mathcal{A}} \to \overrightarrow{O''\mathcal{A}}, \qquad f(\overrightarrow{OX}) := \overrightarrow{O''X'}, \quad \text{where } X' := \tau^{O}_{O''}(X) \tag{9.13}$$

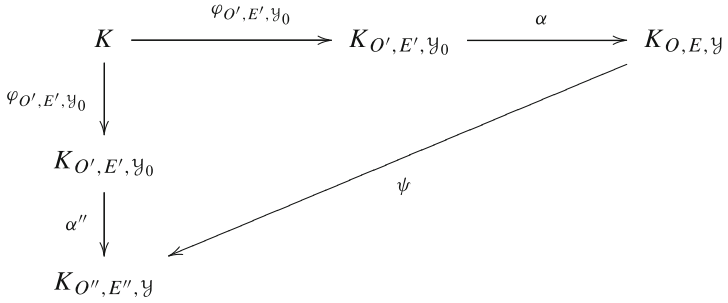
is a bijective map and satisfies the relations

$$\begin{aligned} f(\overrightarrow{OX} + \overrightarrow{OY}) &= f(\overrightarrow{OX}) + f(\overrightarrow{OY}), \qquad \forall\, X, Y \in \mathcal{A}, \\ f(A \cdot \overrightarrow{OX}) &= \psi(A) \cdot f(\overrightarrow{OX}), \qquad \forall\, A \in K_{O,E,\mathcal{Y}},\ \forall\, X \in \mathcal{A}. \end{aligned} \tag{9.14}$$

Let $\alpha = \sigma_{|K_{O',E',\mathcal{Y}_0}}$ and $\alpha'' = \sigma''_{|K_{O',E',\mathcal{Y}_0}}$, where $\sigma, \sigma'' \in \mathrm{PGL}_n(K)$ are the unique projective automorphisms such that

$$\begin{aligned} &\sigma(O') = O,\ \sigma(E') = E,\ \sigma(E'_1) = E_1,\ \ldots,\ \sigma(E'_n) = E_n,\ \sigma''(O') = O'',\ \sigma''(E') \\ &\quad = E'',\ \sigma''(E'_1) = E_1,\ \ldots,\ \sigma''(E'_n) = E_n. \end{aligned}$$

By the commutative diagram (and Proposition 8.21)



(9.15)

we see that $f\colon \overrightarrow{O\mathcal{A}} \to \overrightarrow{O''\mathcal{A}}$ is an isomorphism of K-vector spaces.

The above argument and Theorem 9.2 immediately imply the following:

Theorem 9.6 *Under the above notation the triple* $(\mathcal{A}, V, \varphi)$ *where*

$$\varphi\colon \mathcal{A} \times \mathcal{A} \to V, \qquad \varphi(X, Y) = \overrightarrow{OY} - \overrightarrow{OX},$$

is an affine space of dimension n *over* K.

Remark 9.7 It is easily seen that three points A, B and C of $\mathcal{A}$ are collinear in $\mathbb{P}^n(K)$ if and only if they are collinear in $\mathcal{A}$. This fact means that there is a close link between affine geometry and projective geometry.

Corollary 9.8 *Under the hypothesis of* Remark 9.3 *the triple* $(\mathbb{U}_0, K^n, \varphi)$ *is canonically identified with the standard affine space* $\mathbb{A}^n(K)$.

We can consider other *standard triples* in addition to $(\mathbb{U}_0, K^n, \varphi)$ of Remark 9.3 and Corollary 9.8.

Definition 9.9 Let $\mathcal{Y}_i := \{[x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K) \mid x_i = 0\}$, $i = 0, 1, \ldots, n$, and

$$\mathbb{U}_i := \mathbb{P}^n(K) \setminus \mathcal{Y}_i = \{[x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K) \mid x_i \neq 0\}. \tag{9.16}$$

There are $n+1$ canonical maps $\varphi_i : \mathbb{A}^n(K) \to \mathbb{U}_i$ defined by

$$\begin{aligned} \varphi_i(x_1, \ldots, x_n) &= [y_0, y_1, \ldots, y_n], \quad \text{where } y_j \\ &= \begin{cases} x_{j+1}, & j = 0, \ldots, i-1, \\ y_i = 1, & \\ y_k = x_k, & k = i+1, \ldots, n, \end{cases} \qquad i = 0, \ldots, n. \end{aligned} \tag{9.17}$$

Conversely, if we put

$$\begin{aligned} &\psi_i : \mathbb{U}_i \to \mathbb{A}^n(K), \qquad \psi_i([x_0, x_1, \ldots, x_n]) \\ &= \left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right), \quad i = 0, 1, \ldots, n, \end{aligned}$$

then ψ_i is well-defined for $i = 0, 1, \ldots, n$ and we have the obvious relations $\varphi_i \circ \psi_i = \mathrm{id}_{\mathbb{U}_i}$, $\psi_i \circ \varphi_i = \mathrm{id}_{\mathbb{A}^n(K)}$. Therefore every $\mathbb{U}_i$ can be canonically identified with $\mathbb{A}^n(K)$ by means the bijective map φ_i. The triples $(\mathbb{U}_i, K^n, \varphi_i)$ are called the *standard affine charts* of $\mathbb{P}^n(K)$.

Remark 9.10 The affine line $\mathbb{A}^1(K)$ has two natural inclusions:

$$\mathbb{A}^1(K) \to \mathbb{P}^1(K) \setminus \{[0, 1]\}, \qquad a \to [1, a] \tag{9.18}$$

and

$$\mathbb{A}^1(K) \to \mathbb{P}^1(K) \setminus \{[1, 0]\}, \qquad a \to [a, 1]. \tag{9.19}$$

9.2 Projective Closure of an Affine Subspace

Definition 9.11 Let $\mathcal{B}$ be an affine subspace of $\mathbb{A}^n(K)$ (of dimension $k \geq 0$) whose equations are

$$\sum_{j=1}^{n} b_{ij} x_j = c_i, \qquad i = 1, \ldots, n-k,$$

where the matrix $\left(b_{ij}\right)_{\substack{i=1,\dots,n-k \\ j=1,\dots,n}} \in \mathcal{M}_{n-k,n}(K)$ has rank $n-k$ and $c_{k+1}, \dots, c_n \in K$. The *projective closure of* $\mathcal{B}$ *with respect to canonical immersion* φ_0 (Definition 9.9) is defined as the projective subspace ${}^{x_0}\mathcal{B}$ of $\mathbb{P}^n(K)$ whose equations are

$$\sum_{j=1}^{n} b_{ij} x_j = c_i x_0, \quad i = 1, \dots, n-k, \tag{9.20}$$

so that $\dim({}^{x_0}\mathcal{B}) = k$. Thus ${}^{x_0}\mathcal{B}$ is the smallest projective subspace containing $\mathcal{B}$. We can also define the projective closure ${}^{x_i}\mathcal{B}$ with respect any canonical immersion φ_i, $i = 1, \dots, n$, in a completely analogous way, but we usually use (9.20). Henceforth we denote ${}^{x_0}\mathcal{B}$ by $\overline{\mathcal{B}}$ if there is no misunderstanding.

The hyperplane $\mathcal{Y}_0$ is called the *hyperplane at infinity* (or *improper hyperplane*) and the intersection $\mathcal{B}_\infty := \overline{\mathcal{B}} \cap \mathcal{Y}_0$ is called the *subspace at infinity of* $\mathcal{B}$ (or the *improper part* of $\mathcal{B}$). The points of $\mathcal{B}_\infty$ are called the *points at infinity* of $\mathcal{B}$ (or the *improper points* of $\mathcal{B}$). Hence $\mathcal{B}_\infty$ is a projective subspace of dimension $k-1$ of $\mathbb{P}^n(K)$ which is contained in $\mathcal{Y}_0$; its equations are

$$x_0 = 0, \quad \sum_{j=1}^{n} b_{ij} x_j = 0, \quad i = 1, \dots, n-k. \tag{9.21}$$

We also have the disjoint union $\overline{\mathcal{B}} = \mathcal{B} \cup \mathcal{B}_\infty$.

If $D(\mathcal{B})$ is the orienting subspace of $\mathcal{B}$, from (9.21) we have

$$\mathcal{B}_\infty = \mathbb{P}(D(\mathcal{B})). \tag{9.22}$$

On the other hand, according to the notation of Theorem 6.31, we get

$$D(\mathcal{B}) = V_{\mathcal{B}_\infty}. \tag{9.23}$$

The following simple result gives a projective meaning to parallelism.

Proposition 9.12 *Let* $\mathcal{B}$ *and* $\mathcal{C}$ *be two affine subspaces of* $\mathbb{A}^n(K)$ *(with* $n \geq 2$*). Then* $\mathcal{B}$ *is parallel to* $\mathcal{C}$ *if and only if* $\mathcal{B}_\infty \subset \mathcal{C}_\infty$ *or* $\mathcal{C}_\infty \subset \mathcal{B}_\infty$.

Proof By Definition 3.68, $\mathcal{B} \parallel \mathcal{C}$ if $D(\mathcal{B}) \subset D(\mathcal{C})$ or $D(\mathcal{C}) \subset D(\mathcal{B})$. By Theorem 6.31 and (9.23), the above inclusions are equivalent to $\mathcal{B}_\infty \subset \mathcal{C}_\infty$ or $\mathcal{C}_\infty \subset \mathcal{B}_\infty$ by (9.22). □

Let $\mathcal{D}$ be a projective subspace of $\mathbb{P}^n(K)$ (of dimension $k \geq 0$) whose equations are

$$F_i(x_0, \dots, x_n) = 0 \qquad i = 1, \dots, n-k,$$

where $F_i(x_0, \ldots, x_n)$ are homogeneous polinomials of degree 1, $i = 1, \ldots, n-k$. The *affine part of* $\mathcal{D}$ *with respect to canonical immersion* φ_0 is defined as being the affine subspace ${}_{x_0}\mathcal{D}$ of $\mathbb{A}^n(K)$ whose equations are

$$F_i(1, x_1, \ldots, x_n) \quad i = 1, \ldots, n-k. \tag{9.24}$$

Therefore ${}_{x_0}\mathcal{D} = \mathcal{D} \cap \mathbb{U}_0$. If $F_i(x_0, \ldots, x_n) = cx_0$ for some $i \in \{1, \ldots, n-k$ and $c \in K^*$ then ${}_{x_0}\mathcal{D} = \varnothing$, otherwise $\dim({}_{x_0}\mathcal{D}) = \dim(\mathcal{D})$. A completely analogous definition can be given for the affine part of $\mathcal{D}$ with respect to canonical immersion $\varphi_i, i = 1, \ldots, n$.

9.2.1 Projective Automorphisms and Affine Automorphisms

Proposition 9.13 *Assuming the canonical identification* $\varphi_0 \colon \mathbb{A}^n(K) \to \mathbb{U}_0$, *we have*

$$\mathrm{AGL}_n(K) = \{\sigma \in \mathrm{PGL}_n(K) \mid \sigma(\mathcal{Y}_0) = \mathcal{Y}_0\}.$$

Proof Let $\sigma \in \mathrm{AGL}_n(K)$ be an affine automorphism given by

$$y_i = \sum_{i=1}^{n} a_{ij} x_j + b_i, \qquad i = 1, \ldots, n, \tag{9.25}$$

where $(a_{ij})_{i,j=1,\ldots,n} \in \mathrm{GL}_n(K)$. Since $\varphi_0(x_1, \ldots, x_n) = [z_0, \ldots, z_n]$ with $x_i = z_i/z_0$, $i = 1, \ldots, n$, so that from the Eq. (9.25) we can define the projective automorphism $\overline{\sigma} \in \mathrm{PGL}_n(K)$

$$t_0 = z_0, \quad t_i = b_i z_0 + \sum_{j=0}^{n} a_{ij} z_j, \qquad i = 1, \ldots, n,$$

whose associated matrix is

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ b_1 & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

Therefore the hyperplane $\mathcal{Y}_0$ is fixed by $\overline{\sigma}$ and we have the inclusion $\mathrm{AGL}_n(K) \subset \{\sigma \in \mathrm{PGL}_n(K) \mid \sigma(\mathcal{Y}_0) = \mathcal{Y}_0\}$.

Conversely, let $\sigma \in \mathrm{PGL}_n(K)$ defined by

$$t_i = \sum_{j=0}^{n} b_{ij} z_j, \qquad i = 0, 1, \ldots, n,$$

where the matrix $\left(b_{ij}\right)_{i,j=0,1,\ldots,n}$ is invertible. We observe that

$$\sigma(\mathcal{Y}_0) = \mathcal{Y}_0 \quad \Longleftrightarrow \quad b_{01} = b_{02} = \cdots = b_{0n} = 0.$$

Since $\det\left(\left(b_{ij}\right)_{i,j=0,1,\ldots,n}\right) \neq 0$ we have $b_{00} \neq 0$ and $\det\left(\left(b_{ij}\right)_{i,j=1,\ldots,n}\right) \neq 0$, therefore we can suppose $b_{00} = 1$ (multiplying each b_{ij} by b_{00}^{-1}) without changing the automorphism. Thus σ is given by

$$t_0 = z_0, \quad t_i = \sum_{j=0}^{n} b_{ij} z_j, \qquad i = 1, \ldots, n. \tag{9.26}$$

Since σ is bijective and $\sigma(\mathcal{Y}_0) \subset \mathcal{Y}_0$, we have $\sigma(\mathcal{Y}_0) = \mathcal{Y}_0$ so that $\sigma(\mathbb{U}_0) = \mathbb{U}_0$. Therefore $\sigma_{|\mathbb{U}_0} \colon \mathbb{U}_0 \to \mathbb{U}_0$ is the affine automorphism $\alpha \colon \mathbb{A}^n(K) \to \mathbb{A}^n(K)$ of equations

$$y_i = \sum_{j=1}^{n} b_{ij} x_j + b_i, \qquad i = 1, \ldots, n,$$

where in (9.26) we put $x_i := z_i/z_0$, $y_i := t_i/t_0$, and $b_i := b_{i0}$, $i = 1, \ldots, n$. It follows that $\sigma = \overline{\alpha}$ and we get the inclusion $\mathrm{AGL}_n(K) \supset \{\sigma \in \mathrm{PGL}_n(K) \mid \sigma(\mathcal{Y}_0) = \mathcal{Y}_0\}$. □

Remark 9.14 By Proposition 9.13 we get the map

$$\varphi \colon \mathrm{AGL}_n(K) \to \mathrm{PGL}_{n-1}(K), \qquad \varphi(\alpha) := \overline{\alpha}\,|\mathcal{Y}_0, \tag{9.27}$$

$\mathcal{Y}_0$ being identified with $\mathbb{P}^{n-1}(K)$ by the projective isomorphism $[0, z_1, \ldots, z_n] \to [z_1, \ldots, z_n]$. More explicitly, if $\alpha \in \mathrm{AGL}_n(K)$ is given by the equations

$$y_i = \sum_{j=1}^{n} a_{ij} x_j + b_i, \qquad i = 1, \ldots, n,$$

where the matrix $(a_{ij})_{i,j=1,\ldots,n}$ is non-singular, then $\varphi(\alpha) \in \mathrm{PGL}_{n-1}(K)$ is the projective automorphism of $\mathbb{P}^{n-1}(K)$ (equipped with homogeneous coordinates $[x_1, \ldots, x_n]$) of equations

$$y_i = \sum_{j=1}^{n} a_{ij} x_j, \qquad i = 1, \ldots, n.$$

Corollary 9.15 *The map φ defined by (9.27) is a surjective group homomorphism.*

Proof Let $\alpha \in \mathrm{AGL}_n(K)$, $\alpha(x) = \left(Ax^t + a^t\right)^t = xA^t + a$, where $x = (x_1, \ldots, x_n)$, $a = (a_1, \ldots, a_n)$ and $A = (a_{ij})_{i,j=1,\ldots,n}$; let $\beta \in \mathrm{AGL}_n(K)$, $\beta(x^t) = \left(Bx^t + b^t\right)^t = xB^t + b$, where $x = (x_1, \ldots, x_n)$, $b = (b_1, \ldots, b_n)$ and $B = (b_{ij})_{i,j=1,\ldots,n}$. Then $(\alpha \circ \beta)(x) = (xB^t + b)A^t + a = (ABx^t + Ab^t + a^t)^t$. Hence $\overline{\alpha} \circ \overline{\beta} = \overline{\alpha \circ \beta}$; indeed,

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ a_1 & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_n & a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ b_1 & b_{11} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & b_{n1} & \cdots & b_{nn} \end{pmatrix} = \begin{pmatrix} 1 & & 0 \cdots 0 \\ a_1 + \sum_{j=1}^{n} a_{1j} b_j & & \\ & AB & \\ a_n + \sum_{j=1}^{n} a_{nj} b_j & & \end{pmatrix}. \qquad \square$$

Remark 9.16 Under the notation and hypothesis of Examples 8.6 and 8.7 we observe that the restrictions

$$\sigma_{a\,|\mathbb{U}_0} \colon \mathbb{U}_0 \cong \mathbb{A}^n(K) \to \mathbb{A}^n(K)$$

is the affine homothety

$$y_i = a x_i, \qquad i = 1, \ldots, n.$$

and

$$\tau_{(a_1,\ldots,a_n)\,|\mathbb{U}_0} \colon \mathbb{U}_0 \cong \mathbb{A}^n(K) \to \mathbb{A}^n(K)$$

is the affine translation

$$y_i = x_i + a_i, \qquad i = 1, \ldots, n.$$

9.3 Cross-Ratio (or Anharmonic Ratio) of Four Collinear Points

Definition 9.17 Let A_1, A_2, A_3, $A_4 \in \mathbb{P}^n(K)$ be four collinear and pairwise distinct points. Following the notation of Proposition 8.21 put $O := A_1$, $E := A_4$, $U := A_2$. Let $\mathcal{Y}$ be a hyperplane through U not containing the line $d := OE$ (so $d \cap \mathcal{Y} = \{U\}$). By Proposition 8.21 we know that the coordinate field $K_{O,E,\mathcal{Y}}$ does not depend on the choice of $\mathcal{Y}$ and there exists a canonical isomorphism

$\varphi_{A_1,A_4,A_2} := \varphi_{O,E,\mathcal{Y}} \colon K \to K_{O,E,\mathcal{Y}}$. The *cross-ratio* (or *anharmonic ratio*) $\rho(A_1, A_2, A_3, A_4) \in K$ of A_1, A_2, A_3, A_4 is defined as

$$\rho(A_1, A_2, A_3, A_4) := \varphi^{-1}_{A_1,A_4,A_2}(A_3).$$

Since φ_{A_1,A_4,A_2} is a bijective map between K and $d \setminus \{U\}$, $\rho(A_1, A_2, A_3, A_4) = \rho(O, U, A, E)$ is the coordinate of $A = A_3 \in d$ with respect to the projective frame O, E, U of d. More explicitly

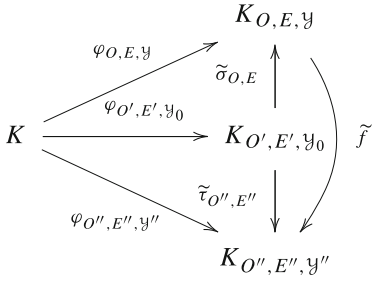
$$O = [1, 0], \quad E = [1, 1], \quad U = [0, 1], \quad A_3 = [1, \rho(A_1, A_2, A_3, A_4)]. \tag{9.28}$$

We observe that $\rho(A_1, A_2, A_3, A_4) \neq 0, 1$.

Lemma 9.18 *Let $f \in \mathrm{PGL}_n(K)$ be a general linear projective automorphism. For every four collinear and pairwise distinct points $A_1, A_2, A_3, A_4 \in \mathbb{P}^n(K)$ one has*

$$\rho(A_1, A_2, A_3, A_4) = \rho(f(A_1), f(A_2), f(A_3), f(A_4)).$$

Proof Assume the notation of Definition 9.17 and Proposition 8.21. Look at the commutative diagram



where $\sigma(O') = O$, $\sigma(E') = E$, $\sigma(\mathcal{Y}_0) = \mathcal{Y}$, $\tau(O') = O''$, $\tau(E') = E''$, $\tau(\mathcal{Y}_0) = \mathcal{Y}''$, $f(O) = O''$, $f(E) = E''$, $f(\mathcal{Y}) = \mathcal{Y}''$, $\widetilde{f} = f_{|K_{O,E,\mathcal{Y}}}$. Hence

$$\begin{aligned}\rho(A_1, A_2, A_3, A_4) &= \rho(O, U, A_3, E) = \varphi^{-1}_{O,E,U}(A_3)\\ &= \varphi^{-1}_{f(A_1),f(A_4),f(A_2)}(f(A_3))\\ &= \rho(f(A_1), f(A_2), f(A_3), f(A_4)).\end{aligned}$$

□

Remark 9.19 By Lemma 9.18 the cross-ratio of four collinear and pairwise distinct points is an invariant of $\mathrm{PGL}_n(K)$ (Fig. 9.1).

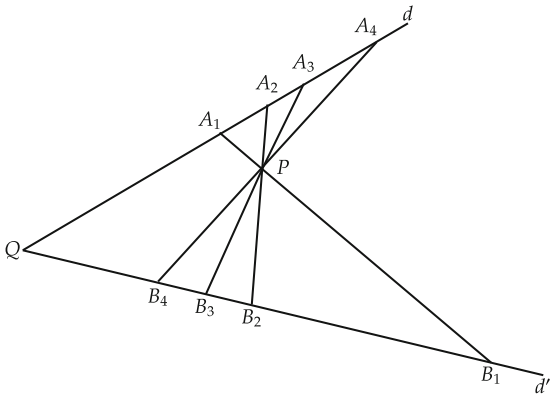


Fig. 9.1 $\rho(A_1, A_2, A_3, A_4) = \rho(B_1, B_2, B_3, B_4)$

An important consequence of Lemma 9.18 is the following property of the cross-ratio.

Corollary 9.20 *Let d and d' be two distinct coplanar lines of $\mathbb{P}^n(K)$ ($n \geq 2$) and $P \in d + d' \setminus (d \cup d')$ be a point. Let $A_1, A_2, A_3, A_4 \in d$ be four pairwise distinct points and $B_i := PA_i \cap d'$, $i = 1, 2, 3, 4$. Then*

$$\rho(A_1, A_2, A_3, A_4) = \rho(B_1, B_2, B_3, B_4).$$

Proof By Lemma 9.18 it will suffice to prove the existence of $\sigma \in \mathrm{PGL}_n(K)$ such that $B_i = \sigma(A_i)$, $i = 1, 2, 3, 4$. Let $\mathcal{Y}$ be a hyperplane passing through PQ, with $Q := d \cap d'$ but not containing the plane $d + d'$. Let σ be the projective translation of axes $\mathcal{Y}$ such that $\sigma(A_1) = B_1$. Hence the centre of $\sigma \in \mathrm{PGL}_n(K)$ is P, so that $\sigma(A_i) = B_i$, $i = 1, 2, 3, 4$. □

Remark 9.21 According to (9.28), if $A_1 = [1, 0] = [\lambda_1, \mu_1]$, $A_1 = [0, 1] = [\lambda_2, \mu_2]$, $A_3 = [1, \rho(A_1, A_2, A_3, A_4)] = [\lambda_3, \mu_3]$ $A_4 = [1, 1] = [\lambda_4, \mu_4]$ we have

$$\rho(A_1, A_2, A_3, A_4) = \frac{\begin{vmatrix}\lambda_1 & \mu_1\\ \lambda_3 & \mu_3\end{vmatrix} \cdot \begin{vmatrix}\lambda_2 & \mu_2\\ \lambda_4 & \mu_4\end{vmatrix}}{\begin{vmatrix}\lambda_1 & \mu_1\\ \lambda_4 & \mu_4\end{vmatrix} \cdot \begin{vmatrix}\lambda_2 & \mu_2\\ \lambda_3 & \mu_3\end{vmatrix}}. \tag{9.29}$$

Lemma 9.22 *Assume the above notation. Let $\{B, C, D\}$ be another projective frame of d with respect to which $A_i = [\alpha_i, \beta_i]$, $i = 1, 2, 3, 4$. Then we have*

$$\frac{\begin{vmatrix}\lambda_1 & \mu_1\\ \lambda_3 & \mu_3\end{vmatrix} \cdot \begin{vmatrix}\lambda_2 & \mu_2\\ \lambda_4 & \mu_4\end{vmatrix}}{\begin{vmatrix}\lambda_1 & \mu_1\\ \lambda_4 & \mu_4\end{vmatrix} \cdot \begin{vmatrix}\lambda_2 & \mu_2\\ \lambda_3 & \mu_3\end{vmatrix}} = \frac{\begin{vmatrix}\alpha_1 & \beta_1\\ \alpha_3 & \beta_3\end{vmatrix} \cdot \begin{vmatrix}\alpha_2 & \beta_2\\ \alpha_4 & \beta_4\end{vmatrix}}{\begin{vmatrix}\alpha_1 & \beta_1\\ \alpha_4 & \beta_4\end{vmatrix} \cdot \begin{vmatrix}\alpha_2 & \beta_2\\ \alpha_3 & \beta_3\end{vmatrix}}.$$

Proof By Theorem 6.47 there is a non-singular 2×2 matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that $(\alpha_i, \beta_i) = (\lambda_i, \mu_i)M$, $i = 1, 2.3, 4$. Starting from the obvious identities

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (a, b) \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot (c, d)^t, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}^t = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

we have

$$\begin{aligned} \begin{vmatrix} \alpha_i & \beta_i \\ \alpha_j & \beta_j \end{vmatrix} &= (\alpha_i, \beta_i) \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot (\alpha_j, \beta_j)^t \\ &= (\lambda_i, \mu_i) \cdot M \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot [(\lambda_j, \mu_j) \cdot M]^t \\ &= (\lambda_i, \mu_i) \cdot M \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot M^t \cdot (\lambda_j, \mu_j)^t \\ &= \det(M) \left[(\lambda_i, \mu_i) \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot (\lambda_j, \mu_j)^t \right] \\ &= \det(M) \begin{vmatrix} \lambda_i & \mu_i \\ \lambda_j & \mu_j \end{vmatrix}, \qquad j = 1, 2, 3, 4. \end{aligned}$$

Hence

$$\frac{\begin{vmatrix} \alpha_1 & \beta_1 \\ \alpha_3 & \beta_3 \end{vmatrix} \cdot \begin{vmatrix} \alpha_2 & \beta_2 \\ \alpha_4 & \beta_4 \end{vmatrix}}{\begin{vmatrix} \alpha_1 & \beta_1 \\ \alpha_4 & \beta_4 \end{vmatrix} \cdot \begin{vmatrix} \alpha_2 & \beta_2 \\ \alpha_3 & \beta_3 \end{vmatrix}} = \frac{\begin{vmatrix} \lambda_1 & \mu_1 \\ \lambda_3 & \mu_3 \end{vmatrix} \cdot \begin{vmatrix} \lambda_2 & \mu_2 \\ \lambda_4 & \mu_4 \end{vmatrix}}{\begin{vmatrix} \lambda_1 & \mu_1 \\ \lambda_4 & \mu_4 \end{vmatrix} \cdot \begin{vmatrix} \lambda_2 & \mu_2 \\ \lambda_3 & \mu_3 \end{vmatrix}}.$$

□

By (9.29) and Lemma 9.22 we get the following explicit formula of the cross ratio of 4 collinear points.

Corollary 9.23 *Let A_1, A_2, A_3, A_4 be four pairwise distinct points lying on a line d of $\mathbb{P}^n(K)$. Let $\{B, C, D\}$ be a projective frame of d with respect to which $A_i = [\alpha_i, \beta_i]$, $i = 1, 2, 3, 4$. Then*

$$\rho(A_1, A_2, A_3, A_4) := \frac{\begin{vmatrix} \alpha_1 & \beta_1 \\ \alpha_3 & \beta_3 \end{vmatrix} \cdot \begin{vmatrix} \alpha_2 & \beta_2 \\ \alpha_4 & \beta_4 \end{vmatrix}}{\begin{vmatrix} \alpha_1 & \beta_1 \\ \alpha_4 & \beta_4 \end{vmatrix} \cdot \begin{vmatrix} \alpha_2 & \beta_2 \\ \alpha_3 & \beta_3 \end{vmatrix}}. \tag{9.30}$$

We have the following proposition:

Proposition 9.24 *Let d be a line and $\mathcal{Y}$ be a hyperplane of $\mathbb{P}^n(K)$ with $n \geq 2$. Let A_1, A_2, A_3, $A_4 \in d \setminus \mathcal{Y}$ be four pairwise distinct points. In the affine space $\mathcal{A} := \mathbb{P}^n(K) \setminus \mathcal{Y}$ (according to* Theorem 9.6*), we have the following relation* (*see* Definition 3.31)*:*

$$\rho(A_1, A_2, A_3, A_4) = \frac{\mathrm{ar}(A_1, A_2, A_3)}{\mathrm{ar}(A_1, A_2, A_4)} = \frac{\dfrac{\overrightarrow{A_3A_1}}{\overrightarrow{A_3A_2}}}{\dfrac{\overrightarrow{A_4A_1}}{\overrightarrow{A_4A_2}}}. \tag{9.31}$$

Proof There exists a projective automorphisms $f \in \mathrm{PGL}_n(K)$ such that $f(\mathcal{Y})$ has the equation $x_0 = 0$ and $f(d)$ is the line of equations $x_2 = \cdots = x_n = 0$ with $B_1 = f(A_1) = [1, 0, \ldots, 0]$ and $B_2 = f(A_2) = [1, 1, 0, \ldots, 0]$. Let $f(A_3) = B_3 = [1, r, 0, \ldots, 0]$ and $f(A_4) = B_4 = [1, s, 0, \ldots, 0]$ with $r \neq 1$, $s \neq 1$, $r \neq s$. Choosing $\{[1, 0, \ldots, 0], [0, 1, 0, \ldots, 0], [1, 1, 0, \ldots, 0]\}$ as the projective frame of $f(d)$, we have

$$B_1 = [1, 0], \quad B_2 = [1, 1], \quad B_3 = [1, r], \quad B_4 = [1, s].$$

By Corollary 9.23 we get

$$\rho(B_1, B_2, B_3, B_4) = \frac{r(s-1)}{s(r-1)}. \tag{9.32}$$

So 0 is the affine coordinate of B_1, 1 of B_2, r of B_3 and s of B_4. From the relation

$$\mathrm{ar}(B_1, B_2, B_3) = \frac{\overrightarrow{B_3B_1}}{\overrightarrow{B_3B_2}} = \frac{t}{t-1} \qquad \text{where} \qquad B_3 = (1-t)B_1 + tB_2$$

we get $t = r$; so

$$\frac{\overrightarrow{B_3B_1}}{\overrightarrow{B_3B_2}} = \frac{r}{r-1}.$$

Analogously, from

$$\mathrm{ar}(B_1, B_2, B_4) = \frac{\overrightarrow{B_4B_1}}{\overrightarrow{B_4B_2}} = \frac{t}{t-1} \qquad \text{where} \qquad B_4 = (1-t)B_1 + tB_2$$

we get $t = s$; so

$$\frac{\overrightarrow{B_4B_1}}{\overrightarrow{B_4B_2}} = \frac{s}{s-1}.$$

By Lemma 9.18 (invariance of cross-ratio with respect to projective automorphism f) and Corollary 3.52-(iii) (invariance of affine ratio with respect to affine isomorphism $f_{|\mathbb{P}^n(K)\setminus\mathcal{Y}} : \mathbb{P}^n(K) \setminus \mathcal{Y} \to \mathbb{P}^n(K) \setminus f(\mathcal{Y})$) we get (9.31). □

Proposition 9.25 (Cross-Ratio Symmetries) *Let $A_1, A_2, A_3, A_4 \in \mathbb{P}^n(K)$ be four pairwise distinct points. If $\rho(A_1, A_2, A_3, A_4) = r$, then we have*

$$\rho(A_1, A_2, A_4, A_3) = \frac{1}{r}, \qquad \rho(A_1, A_3, A_2, A_4) = 1 - r,$$

$$\rho(A_1, A_3, A_4, A_2) = \frac{1}{1-r}, \quad \rho(A_1, A_4, A_2, A_3), = \frac{r-1}{r},$$

$$\rho(A_1, A_4, A_3, A_2) = \frac{r}{r-1}.$$

Hence the 24 *possible cross-ratios of four points on a line reduce to six values.*

If four pairwise distinct collinear points are named in a stated order and if two pairs of points are then interchanged, the value of the cross-ratio is unchanged:

$$\rho(A_1, A_2, A_3, A_4) = \rho(A_4, A_3, A_2, A_1),$$
$$\rho(A_1, A_2, A_3, A_4) = \rho(A_3, A_4, A_1, A_2),$$
$$\rho(A_1, A_2, A_3, A_4) = \rho(A_2, A_1, A_4, A_3).$$

Proof Easy computation using Corollary 9.23. □

9.3.1 Harmonic Ratio

Definition 9.26 Let $A, B, P, Q \in \mathbb{P}^n(\mathbb{R})$ be four pairwise distinct collinear points. We say that P and Q are *separated* (*non-separated*) by A and B if $\rho(A, B, P, Q) < 0$ ($\rho(A, B, P, Q) > 0$ respectively). By Proposition 9.25, the sign of $\rho(A, B, P, Q)$ does not depend on the order of A and B or on the order of P and Q.

Thus two points A and B determine two *segments* of the line $r = AB$ whose endpoints are A and B: P and $Q \in r \setminus \{A, B\}$, *lie on the same segment of endpoints A and B* if they are not separated by A and B, i.e. $\rho(A, B, P, Q) > 0$.

The concept of an interval (or segment) of the affine real line $\mathbb{A}^1(\mathbb{R})$ (see Definition 3.90) is consistent with the separation relation defined above. If $A =$

$[1,-1]$, $B=[1,1]$, $P=[1,p]$ and $Q=[1,q]$ are four pairwise distinct points of $\mathbb{A}^1(\mathbb{R}) \to \mathbb{P}^1(\mathbb{R})$, $x \to [1,x]$, then we have

$$\rho(A,B,P,Q)=\frac{p+1}{q+1}\cdot\frac{q-1}{p-1}.$$

Hence P and Q are both inside the segment $[AB]$ (or *interior points* of $[AB]$), i.e. $-1<p,q<1$ or both outside $[AB]$ (or *exterior points* of $[AB]$), i.e. $p,q>1$ or $p,q<-1$ or $p<-1$ and $q>1$ or $p>1$ and $q<-1$, if $\rho(A,B,P,Q)>0$. On the other hand, P and Q are one inside and one outside the segment $[AB]$ if $\rho(A,B,P,Q)<0$.

Let $\mathrm{char}(K)\neq 2$. A 4-tuple (A,B,C,D) of pairwise distinct collinear points is called a *harmonic quadruple*) if $\rho(A,B,C,D)=-1$. We say also that the pairs (A,B) and (C,D) are *harmonically separate* or that D is the *harmonic conjugate* of C with respect to A and B. If $A=[1,p]$, $B=[1,q]$, $M=[1,\frac{p+q}{2}]$ and $P_\infty=[0,1]$, then

$$\rho(A,B,M,P_\infty)=-1$$

so that (A,B) and (M,P_∞), where M is the midpoint of the segment $[AB]$ and P_∞ is the point at infinity of r harmonically separate. Conversely, given $A=[1,p]$, $B=[1,q]$, $M=[1,t]$ and $P_\infty=[0,1]$ such that $\rho(A,B,M,P_\infty)=-1$, then $M=[1,\frac{p+q}{2}]$, the midpoint of the segment $[AB]$.

The following result by K. G. C. von Staudt (1798– 1867) (see [34]) underlines the important role of the harmonic cross-ratio as a projective invariant.

Theorem 9.27 (von Staudt) *Let K be a field with characteristic $\neq 2$ and $\varphi\colon \mathbb{P}^1(K)\to\mathbb{P}^1(K)$ be a surjective map. Then $\varphi\in \mathrm{Aut}(\mathbb{P}^1(K))$ if and only if φ preserves the harmonic cross-ratio, i.e.*

$$\rho(A,B,C,D)=-1 \qquad \Longrightarrow \qquad \rho(\varphi(A),\varphi(B),\varphi(C),\varphi(D))=-1.$$

In particular, if $K=\mathbb{R}$, φ preserves the harmonic cross-ratio if and only if $\varphi\in \mathrm{PGL}_1(\mathbb{R})$, i.e. φ is a linear projective automorphism of $\mathbb{P}^1(\mathbb{R})$.

Proof We follow [27, Proposizione 7, pp. 53– 54]. Obviously, the condition is necessary. To prove the converse, we put $\mathbb{P}^1(K)=K\cup\{\infty\}$, where $x=[1,x]\in\mathbb{P}^1(K)$, $\forall\, x\in K$ and $\infty=[0,1]$.

Any pair of distinct points of $\mathbb{P}^1(K)$ can always be completed to a harmonic quadruple. Indeed, since $x\neq y\in\mathbb{P}^1(K)$ and the points of a harmonic quadruple

are distinct, we can assume $x, y \in K$. Since $\mathrm{char}(K) \neq 2$, consider the quadruple of distinct points $\left(x, y, \dfrac{x+y}{2}, \infty\right)$. By Corollary 9.23 one has

$$\rho\left(x, y, \frac{x+y}{2}, \infty\right) = \frac{\begin{vmatrix} 1 & x \\ 1 & \frac{x+y}{2} \end{vmatrix} \cdot \begin{vmatrix} 1 & y \\ 0 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & x \\ 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & y \\ 1 & \frac{x+y}{2} \end{vmatrix}} = -1,$$

i.e. $\left(x, y, \dfrac{x+y}{2}, \infty\right)$ is a harmonic quadruple. In particular, φ is injective and therefore it is bijective. The points 0, 1 and ∞ are distinct so that $\varphi(0)$, $\varphi(1)$ and $\varphi(\infty)$ are also distinct. By Proposition 8.28 there exists only one linear projective automorphism $\psi \in \mathrm{PGL}_1(K)$ such that $\psi(\varphi(0)) = 0$, $\psi(\varphi(1)) = 1$, $\psi(\varphi(\infty)) = \infty$. Therefore $\chi := \psi \circ \varphi \colon \mathbb{P}^1(K) \to \mathbb{P}^1(K)$ is also bijective, preserves the harmonic cross-ratio and

$$\chi(0) = 0, \qquad \chi(1) = 1, \qquad \chi(\infty) = \infty.$$

Now the proof is finished once we have shown that $\chi|_K$ is an automorphism of the field K, since then by Corollary 8.29 $\varphi = \psi^{-1} \circ \chi \in \mathrm{Aut}(\mathbb{P}^1(K))$, i.e. φ is a projective automorphism of $\mathbb{P}^1(K)$.

Let $x, y \in K$, $x \neq y$. Since $\left(x, y, \dfrac{x+y}{2}, \infty\right)$ is a harmonic quadruple, so is $\left(\chi(x), \chi(y), \chi(\frac{x+y}{2}), \chi(\infty)\right)$, i.e. (by Corollary 9.23)

$$\rho(\chi(x), \chi(y), \chi(\tfrac{x+y}{2}), \chi(\infty)) = \frac{\begin{vmatrix} 1 & \chi(x) \\ 1 & \chi(\frac{x+y}{2}) \end{vmatrix} \cdot \begin{vmatrix} 1 & \chi(y) \\ 0 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & \chi(x) \\ 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & \chi(y) \\ 1 & \chi(\frac{x+y}{2}) \end{vmatrix}} = -1, \qquad \forall\, x \neq y.$$

Hence

$$\chi\left(\frac{x+y}{2}\right) = \frac{\chi(x) + \chi(y)}{2}, \qquad \forall\, x \neq y. \tag{9.33}$$

Setting $y = 0$ in (9.33) we conclude from $\chi(0) = 0$

$$\chi\left(\frac{x}{2}\right) = \frac{\chi(x)}{2}, \qquad \forall\, x \in K, \tag{9.34}$$

and from $x = 2y$ in (9.34) that

$$\chi(2y) = 2\chi(y), \qquad \forall\, y \in K. \tag{9.35}$$

(9.33) and (9.35) imply

$$\chi(x + y) = \chi(x) + \chi(y), \qquad \forall\, x, y \in K. \tag{9.36}$$

From $\chi(0) = 0$ and (9.36) we conclude that

$$\chi(-x) = -\chi(x), \qquad \forall\, x \in K. \tag{9.37}$$

Since

$$\rho(-x, x, 1, x^2) = \frac{\begin{vmatrix} 1 & -x \\ 1 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & x \\ 1 & x^2 \end{vmatrix}}{\begin{vmatrix} 1 & -x \\ 1 & x^2 \end{vmatrix} \cdot \begin{vmatrix} 1 & x \\ 1 & 1 \end{vmatrix}} = -1, \qquad \forall\, x \in K,$$

we have

$$\rho(\chi(-x), \chi(x), 1, \chi(x^2)) = \frac{\begin{vmatrix} 1 & \chi(-x) \\ 1 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & \chi(x) \\ 1 & \chi(x^2) \end{vmatrix}}{\begin{vmatrix} 1 & \chi(-x) \\ 1 & \chi(x^2) \end{vmatrix} \cdot \begin{vmatrix} 1 & \chi(x) \\ 1 & 1 \end{vmatrix}} = -1, \qquad \forall\, x \in K.$$

This last equation and (9.37) imply

$$\chi(x^2) = \chi(x)^2, \qquad \forall\, x \in K. \tag{9.38}$$

Inserting $x + y$ instead of x into (9.38), and applying (9.35) and (9.36) yields

$$\chi(xy) = \chi(x)\chi(y). \qquad \forall x, y \in K. \tag{9.39}$$

Since $\chi|_K$ is bijective, by (9.36) and (9.39), $\chi|_K$ is an automorphism of K.

By Proposition 8.17 $\mathrm{Aut}(\mathbb{R}) = \{\mathrm{id}_{\mathbb{R}}\}$ so that $\mathrm{Aut}(\mathbb{P}^1(\mathbb{R})) = \mathrm{PGL}_1(\mathbb{R})$ (Corollary 8.18). □

Remark 9.28 For $K = \mathbb{R}$ the assertion can be proved without assuming the surjectivity of φ. As seen above, φ is necessarily iniective. (9.36) implies $\chi(mx) = \chi((m-1)x) + \chi(x)$ and by induction we get

$$\chi(mx) = m\chi(x), \qquad \forall\, m \in \mathbb{N}. \tag{9.40}$$

Since $\chi(1) = 1$, from (9.40) we conclude $\chi(m) = m$ and $\chi(1) = n\chi(1/n)$, i.e. $\chi(1/n) = 1/n$ for every $m, n \in \mathbb{N}$. Therefore

$$\chi\left(\frac{m}{n}\right) = m\chi\left(\frac{1}{n}\right) = \frac{m}{n} \quad \text{and by (9.37)}, \quad \chi\left(-\frac{m}{n}\right) = -\chi\left(\frac{m}{n}\right) = -\frac{m}{n},$$

so that

$$\chi(x) = x, \qquad \forall\, x \in \mathbb{Q}. \tag{9.41}$$

Let $h \in \mathbb{R}$, $h = x^2$ for $x \in \mathbb{R}$. By (9.38)

$$\chi(h) = \chi(x^2) = \chi(x)^2 > 0, \tag{9.42}$$

so that by (9.36), for every fixed $x \in \mathbb{R}$,

$$\chi(x + h) - f(x) = f(h) > 0, \qquad \forall\, h > 0. \tag{9.43}$$

Hence χ is a strictly increasing function. For each real number x there is a strictly increasing sequence of rational numbers $\{a_n\}$ and a strictly decreasing sequence of rational numbers $\{b_n\}$ such that

$$a_n < x < b_n, \quad \forall\, n \geq 1, \qquad \lim_{n\to+\infty} a_n = \lim_{n\to+\infty} b_n = x.$$

By (9.41) and (9.43), we obtain

$$a_n = \chi(a_n) < \chi(x) < \chi(b_n) = b_n, \qquad \forall\, n \geq 1,$$

which implies

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} b_n = x = \chi(x).$$

Hence $\chi = \mathrm{id}_{\mathbb{P}^1(\mathbb{R})}$.

Definition 9.29 Let S_{n-2} be a projective subspace of $\mathbb{P}^n(K)$ of dimension $n-2$ and let $\Phi_{n-1}(S_{n-2})$ be the form of first kind with base locus S_{n-2}. If $\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mathcal{Y}_4$ are four distinct hyperplanes of $\Phi_{n-1}(S_{n-2})$, then the *cross-ratio* of $\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mathcal{Y}_4$ is defined by

$$\rho(\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mathcal{Y}_4) := \rho(P_1, P_2, P_3, P_4),$$

where $P_i = r \cap \mathcal{Y}_i$, $i = 1, 2, 3, 4$, and r is any line skew to S_{n-2}. In particular if $n = 2$, i.e. S_{n-2} is a point P, then the cross-ratio of four lines r_1, r_2, r_3, r_4 through P is the cross-ratio of the four points, where a line r not passing through P meets r_1, r_2, r_3, r_4. Moreover, since two lines not passing through P are perspective with

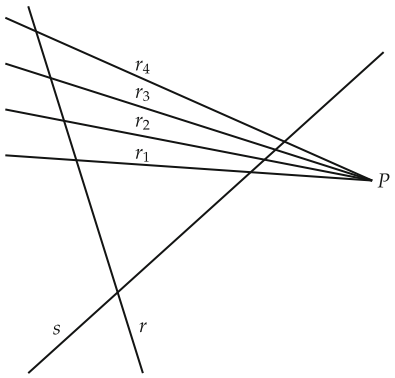


Fig. 9.2 Cross-ratio of four lines

respect to P so that the above definition is independent of the choice of r (see Fig. 9.2).

In the general case, if r and s are skew to S_{n-2}, then they belong to the same plane π which meets S_{n-2} in a single point P. Thus r and s are perspective with respect to P and $\rho(\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mathcal{Y}_4)$ is independent of the choice of r.

9.3.2 *Involutions of* $\mathbb{P}^1(K)$

Definition 9.30 Let $\varphi \in \mathrm{PGL}_1(K)$ be a linear projective automorphism $\neq \mathrm{id}_{\mathbb{P}^1(K)}$ and $M = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix}$ be a matrix associated with φ. Then

$$\begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \lambda \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} \quad \Longleftrightarrow \quad \begin{vmatrix} a_{00} - \lambda & a_{01} \\ a_{10} & a_{11} - \lambda \end{vmatrix} = 0,$$

i.e. φ has a fixed point $[x_0, x_1]$ if and only if M has an eigenvalue $\lambda \in K$. Furthermore,

(a) φ has only one fixed points if and only if M has two coincident eigenvalues; in this case φ is called *parabolic*.
(b) φ has two distinct fixed points if and only if M has two distinct eigenvalues; in this case φ is called *non-parabolic*.

If $\varphi \in \mathrm{PGL}_1(\mathbb{R})$ is non-parabolic, then φ has no fixed points if and only if M has two complex-conjugate eigenvalues; in this case φ is called *elliptic*. We call a non-parabolic projective automorphism φ of $\mathbb{P}^1(\mathbb{R})$ *hyperbolic* if φ has two distinct fixed points.

Definition 9.31 Let $\varphi \in \mathrm{PGL}_1(K)$ be a projective automorphism having two distinct fixed points A and B. Then it can be shown that (Exercise 9.5)

$$\rho(A, B, P, \varphi(P)) = \frac{\lambda_1}{\lambda_2}, \qquad \forall\, P \in \mathbb{P}^1(K) \setminus \{A, B\}, \tag{9.44}$$

where λ_1, λ_2 are the eigenvalues corresponding to A and B respectively. In particular the cross-ratio $\rho(A, B, P, \varphi(P))$ does not depend on the choice of $P \neq A, B$. We call $\rho(A, B, P, \varphi(P))$ *the characteristic* of φ and denote it by $\mathrm{ch}(\varphi)$. Clearly the characteristic depends on the order of the fixed points: if we exchange A and B, then the characteristic takes the value $\dfrac{\lambda_2}{\lambda_1}$.

Definition 9.32 Let $\mathrm{char}(K) \neq 2$. A linear projectivity $\varphi \in \mathrm{PGL}_1(K), \varphi \neq \mathrm{id}_{\mathbb{P}^1(K)}$ such that $\varphi^2 = \varphi \circ \varphi = \mathrm{id}_{\mathbb{P}^1(K)}$ is called an *involution*. Let M be any matrix associated with φ. Then it can be shown that (Exercise 9.6)

1. φ is an involution if and only if $\mathrm{tr}(M) = 0$.
2. φ is an involution if and only if there exist two distinct points $A, B \in \mathbb{P}^1(K)$ such that $\varphi(A) = B$ and $\varphi(B) = A$.
3. If φ is an involution, then φ either has two different fixed points or none.
4. If A, B are two different fixed points of the involution φ, then we have $\mathrm{ch}(\varphi) = -1$.
5. Any projectivity $\varphi \in \mathrm{PGL}_1(K), \varphi \neq \mathrm{id}_{\mathbb{P}^1(K)}$, is the composition of at most two involutions.

If $K = \mathbb{R}$ we say that an involution is *hyperbolic* if it has two fixed points, otherwise *elliptic*.

9.3.3 Homographies of the Complex Affine Line $\mathbb{A}^1(\mathbb{C})$

Definition 9.33 To every matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C})$ we associate the map

$$M_A \colon \mathbb{A}^1(\mathbb{C}) \to \mathbb{A}^1(\mathbb{C}), \qquad M_A(z) = \frac{az + b}{cz + d}, \tag{9.45}$$

which is called the *Möbius transform* or *homography* of the complex affine line $\mathbb{A}^1(\mathbb{C})$. More precisely

$$M_A \colon \mathbb{A}^1(\mathbb{C}) \setminus \left\{-\frac{d}{c}\right\} \to \mathbb{A}^1(\mathbb{C}) \setminus \left\{\frac{a}{c}\right\}, \qquad \text{if} \quad c \neq 0.$$

Since $M_B(z) = M_A(z)$ if and only if $B = aA$ with $a \in \mathbb{C}^*$ and $M_A \circ M_B = M_{AB}$, we can identify the group G of all homographies of $\mathbb{A}^1(\mathbb{C})$ with the group $\mathrm{PSL}_2(\mathbb{C}) := \mathrm{GL}_2(\mathbb{C})/D_2(\mathbb{C})$, where $D_2(\mathbb{C})$ is the subgroup of scalar matrices of $\mathrm{GL}_2(\mathbb{C})$.

Every homography can be extended to a projective automorphism $\overline{M_A}$ of $\mathbb{P}^1(\mathbb{C}) = \mathbb{A}^1(\mathbb{C}) \cup \{\infty\}$, where $\infty := [0, 1]$ by means of the identification

$$\begin{aligned} &j : \mathbb{P}^1(\mathbb{C}) \setminus \{[0,1]\} \to \mathbb{A}^1(\mathbb{C}), \quad j([x_0, x_1]) = x_1/x_0, \\ &i : \mathbb{A}^1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C}) \setminus \{[0,1]\}, \quad i(t) = [1, t], \end{aligned} \tag{9.46}$$

$$\overline{M_A}(z) = \begin{cases} \dfrac{az+b}{cz+d} & z \neq \infty, z \neq -d/c, c \neq 0 \\ \infty & z = \infty, c = 0, \quad \text{or if } z = -d/c, c \neq 0 \\ \dfrac{a}{c} & z = \infty, c \neq 0, \end{cases} \tag{9.47}$$

that in the coordinates $[x_0, x_1]$ takes the form

$$\overline{M_A}([x_0, x_1]) = [dx_0 + cx_1, bx_0 + ax_1].$$

Conversely, a linear projective automorphism $\sigma_A \in \mathrm{PGL}_1(\mathbb{C})$, defined by

$$\sigma([x_0, x_1]) = [ax_0 + bx_1, cx_0 + dx_1], \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}),$$

induces the homography (according to (9.46))

$$M_\sigma(z) = \frac{dz + c}{bz + a}.$$

Proposition 9.34 *The map $\sigma \to M_\sigma$ is a group isomorphism $\mathrm{PGL}_1(\mathbb{C}) \cong \mathrm{PSL}_2(\mathbb{C})$.*

Proof Let

$$\sigma([x_0, x_1]) = [ax_0 + bx_1, cx_0 + dx_1], \qquad \tau([x_0, x_1]) = [ex_0 + fx_1, gx_0 + hx_1].$$

Then

$$M_\sigma(z) = \frac{dz + c}{bz + a}, \qquad M_\tau(z) = \frac{hz + g}{fz + e}$$

and

$$M_\sigma \circ M_\tau(z) = \frac{d\left(\frac{hz+g}{fz+e}\right)+c}{b\left(\frac{hz+g}{fz+e}\right)+a} = \frac{dhz+dg+cfz+ce}{bhz+bg+afz+ae}$$
$$= \frac{(cf+dh)z+ce+dg}{(af+bh)z+ae+by} = M_{\sigma\circ\tau}(z).$$

From the above we see that $\sigma \to M_\sigma$ is bijective and this concludes the proof. □

Definition 9.35 Let z_1, z_2, z_3, z_4 be four distinct elements of $\mathbb{P}^1(\mathbb{C}) = \mathbb{A}^1(\mathbb{C}) \cup \{\infty\}$. By means of the immersion $i : \mathbb{A}^1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C}) \setminus \{[0,1]\}$, $i(z) = [1, z]$, we can rewrite the cross-ratio by inhomogeneous coordinates

$$\rho(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_4} : \frac{z_2 - z_3}{z_2 - z_4}, \qquad z_1, z_2, z_3, z_4 \in \mathbb{A}^1(\mathbb{C}),$$

and

$$\rho(z_1, z_2, z_3, \infty) = \frac{z_1 - z_3}{z_2 - z_3}, \quad \rho(z_1, z_2, \infty, z_4) = \frac{z_2 - z_4}{z_1 - z_4},$$
$$\rho(z_1, \infty, z_3, z_4) = \frac{z_1 - z_3}{z_1 - z_4}, \quad \rho(\infty, z_2, z_3, z_4) = \frac{z_2 - z_4}{z_2 - z_3}.$$

From Lemma 9.18 we have:

Corollary 9.36 *Let $\overline{M_A}$ be as in (9.47) and z_1, z_2, z_3, z_4 be four distinct elements of $\mathbb{P}^1(\mathbb{C})$. Then*

$$\rho(\overline{M_A}(z_1), \overline{M_A}(z_2), \overline{M_A}(z_3), \overline{M_A}(z_4)) = \rho(z_1, z_2, z_3, z_4).$$

Remark 9.37 Let K be a field. Denote the subgroup of $\mathrm{GL}_n(K)$ of the scalar matrices by $D_n(K)$. Consider the groups:

(a) the *special linear group* $\mathrm{SL}_n(K) := \{A \in \mathrm{GL}_n(K) : \det A = 1\}$;
(b) the *projective special linear group* $\mathrm{PSL}_n(K) := \mathrm{SL}_n(K)/(\mathrm{SL}_n(K) \cap D_n(K))$;
(c) the *projective general linear group* $\mathrm{PGL}_n(K) := \mathrm{GL}_n(K)/D_n(K)$.

It is clear that $\mathrm{PSL}_n(K)$ is a subgroup of $\mathrm{PGL}_n(K)$. However, $\mathrm{PSL}_n(K) = \mathrm{PGL}_n(K)$ if and only if every element of K has an n-root in K.

9.4 Exercises

Exercise 9.1 Let A, B, C, D and E be five pairwise distinct collinear points of $\mathbb{P}^n(K)$. Prove the equality $\rho(A, B, C, D)\rho(A, B, D, E) = \rho(A, B, C, E)$.

Exercise 9.2 Let P_1, P_2, P_3, A and B be five pairwise distinct collinear points of $\mathbb{P}^n(K)$. Prove the equality $\rho(P_1, P_2, A, B)\rho(P_2, P_3, A, B)\rho(P_3, P_1, A, B) = 1$.

Exercise 9.3 Let A, B, C, D and A', B', C', D' be two quadruple of pairwise distinct points of $\mathbb{P}^1(K)$. Prove that there exists $\sigma \in \mathrm{PGL}_1(K)$ such that

$$\sigma(A) = A', \quad \sigma(B) = B', \quad \sigma(C) = C', \quad \sigma(D) = D'$$

if and only if $\rho(A, B, C, D) = \rho(A', B', C', D')$. (Hint: Corollary 8.28 and Lemma 9.18.)

Exercise 9.4 Prove all assertions of Definition 9.30 (see Exercise 9.1).

Exercise 9.5 Let $\varphi \in \mathrm{PGL}_1(K)$ be a non-parabolic projective automorphism whose fixed points are denoted by A and B. Show (9.44) of Definition 9.31.

Exercise 9.6 Prove all assertions of Definition 9.32.

Exercise 9.7 Let z_1, z_2, z_3 be three distinct elements of $\mathbb{P}^1(\mathbb{C}) = \mathbb{A}^1(\mathbb{C}) \cup \{\infty\}$. Show that there is a unique $\sigma \in \mathrm{PSL}_2(\mathbb{C})$ such that $\sigma(z_1) = 0$, $\sigma(z_2) = 1$, and $\sigma(z_3) = \infty$.

Exercise 9.8 Let $\sigma(z) = \dfrac{az + b}{cz + d}$ be a homography of $\mathbb{A}^1(\mathbb{C})$. Show that σ transforms lines and circles into lines and circles. (Hint: the projective general linear group $\mathrm{PGL}_1(\mathbb{C})$ is generated by

$$\{\sigma(z) = az + b : a, b \in \mathbb{C},\ a \neq 0\} \cup \{\tau(z) = \frac{1}{z}\}.$$

Moreover, the general equation of a line or a circle in $\mathbb{C}$ is of the form $\alpha(x^2 + y^2) + \beta x + \gamma y + \delta = 0$, where $x = \mathrm{Re}(z)$, $y = \mathrm{Im}(z)$ and $\alpha, \beta, \gamma, \delta \in \mathbb{R}$.)

Exercise 9.9 Let z_1, z_2, z_3, z_4 be 4 distinct elements of $\mathbb{P}^1(\mathbb{C}) = \mathbb{A}^1(\mathbb{C}) \cup \{\infty\}$. Prove that z_1, z_2, z_3 and z_4 are collinear or lie on one and the same circle if and only if $\rho(z_1, z_2, z_3, z_4) \in \mathbb{R}$. (Hint: use Exercise 9.8.)

Exercise 9.10 Let $u(z) = \dfrac{az + b}{cz + d}$ be a homography of $\mathbb{A}^1(\mathbb{C})$. Taking into account that

$$\mathrm{Im}\left(\frac{az + b}{cz + d}\right) = \frac{(ad - bc)\mathrm{Im}(z)}{|cz + d|^2}, \qquad a, b, c, d \in \mathbb{R},$$

show that the upper half plane $\mathbb{H} := \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}$ is preserved by u if and only if $a, b, c, d \in \mathbb{R}$ and $ad - bc > 0$ i.e. $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{R})$. Therefore we can identify the subgroups of homographies of $\mathbb{A}^1(\mathbb{C})$ preserving the upper half plane with $\mathrm{PSL}_2(\mathbb{R})$.

(a) Determine the fixed points of u.
(b) Let v be another homography that fixes $\mathbb{H}$ such that $u \circ v = v \circ u$. Prove that u and v have the same fixed points.

Exercise 9.11 Prove that the subgroup of $\mathrm{PSL}_2(\mathbb{C})$ generated by

$$\left\{z, \frac{1}{z}, 1 - z, \frac{1}{1 - z}, \frac{z - 1}{z}, \frac{z}{z - 1}\right\}$$

is isomorphic to $\mathcal{S}_3$, the group of permutations of a set of three elements (see Definition 1.16).

Exercise 9.12 Let K be a field of characteristic $\neq 2$. Let $A \neq B \in \mathbb{A}^n(K) = \mathbb{P}^n(K) \setminus \mathcal{Y}_0$, where $\mathcal{Y}_0$ has the equation $x_0 = 0$. Let C be the midpoint of $\{A, B\}$ and $D = AB \cap \mathcal{Y}_0$. Prove $\rho(A, B, C, D) = -1$, i.e. A, B, C and D are harmonic conjugates.

Exercise 9.13 Let A, B, C be three non-collinear points of $\mathbb{P}^2(K)$. Let d be a line such that

$$d \cap AB = P \neq A, B, \quad d \cap AC = Q \neq A, C, \quad d \cap BC = N \neq B, C.$$

Consider the points $D := BQ \cap CP$ and $M := BC \cap AD$ and prove that $\rho(B, C, M, N) = -1$. This method to obtain a harmonic quadruple is called *the method of complete quadrilaterals*. (Hint: observe that $\{A, B, C, D\}$ is a projective frame of $\mathbb{P}^2(K)$ and use Theorem 6.47, Lemma 9.18 and Corollary 9.23.)

Exercise 9.14 Using the stereographic projection (see Definition 4.21) prove that:

(a) $\mathbb{P}^1(\mathbb{R})$ can be identified with $S^1(O, r)$, for any $r > 0$.
(b) $\mathbb{P}^1(\mathbb{C})$ can be identified with $S^2(O, r)$, for any $r > 0$.

The above (natural) identifications are actually homeomorphisms.

Chapter 10
Projective Hyperquadrics



10.1 Projective Hypersurfaces

Definition 10.1 Let

$$F(X_0, X_1, \ldots, X_n) = \sum_{i_h \geq 0,\ i_0+\cdots+i_n=d} a_{i_0 i_1 \ldots i_n} X_0^{i_0} X_1^{i_1} \cdots X_n^{i_n} \in$$
$$\in {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d^*.$$

As observed in Example 1.33-(2), ${}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d$ is a K-vector space. Using a double induction on n and d one can prove that

$$\dim_K({}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d) = \binom{n+d}{d}. \tag{10.1}$$

We say that F and $G \in {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d^*$ are *equivalent*, written $F \sim G$, if there exists $\lambda \in K^*$ such that $F = \lambda G$. It obvious that $\sim$ is an equivalence relation. We define the quotient space

$$\mathcal{H}_d^{\mathrm{p}}(n, K) := {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d^* / \sim .$$

In other words, $\mathcal{H}_d^{\mathrm{p}}(n, K)$ is the projective space associated with the K-vector space ${}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d$, hence

$$\dim(\mathcal{H}_d^{\mathrm{p}}(n, K)) = \binom{n+d}{d} - 1.$$

A *projective hypersurface of degree* d of $\mathbb{P}^n(K)$ is any equivalence class $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$. We say that a point $[x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K)$ is a *zero* of

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$F(X_0, X_1, \ldots, X_n)$ if $F(x_0, x_1, \ldots, x_n) = 0$. It is clear that this definition is well-posed since $F(ax_0, ax_1, \ldots, ax_n) = a^d F(x_0, x_1, \ldots, x_n)$ for any $a \in K \setminus \{0\}$. The *geometric locus* (or *locus of the zeros*) of $F \in {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d^*$ is defined by

$$\mathcal{V}_+(F) := \{[x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K) \mid F(x_0, x_1, \ldots, x_n) = 0\}.$$

If $F \sim G$, then clearly one has $\mathcal{V}_+(F) = \mathcal{V}_+(G)$ so that we can define the *geometric locus* or *support* of a hepersurface $\widehat{F}$ as being

$$\mathcal{V}_+(\widehat{F}) := \mathcal{V}_+(F).$$

We say that F is *an equation* of $\widehat{F}$ and we write $P \in \widehat{F}$ or $P \in \mathcal{V}_+(\widehat{F})$ to say that P lies on the support of $\widehat{F}$. The sets $\mathcal{V}_+(F) \subset \mathbb{P}^n(K)$ are fundamental objects of projective geometry. The hypersurfaces of degree 1 are the hyperplanes of $\mathbb{P}^n(K)$. The hypersurfaces of degree d of $\mathbb{P}^2(K)$ are called *plane projective curves* of degree d (*cubic* if $d = 3$). The hypersurfaces of degree d of $\mathbb{P}^3(K)$ are called *projective surfaces* of degree d.

If $\widehat{F}$ and $\widehat{G}$ are two hypersurfaces, we denote the hypersurface $\widehat{FG}$ by $\widehat{F} + \widehat{G}$; moreover, for every positive integer m, we denote $m\widehat{F} := \widehat{F^m}$. A hypersurface $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ is *irreducible* if $F \in {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d^*$ is a power of an irreducible polynomial. If $\widehat{F}$ is not irreducible, it is called *reducible*. It is clear that the condition of irreducibility of $\widehat{F}$ depends only on $\widehat{F}$ and not on F. Let $F = kF_1^{m_1} \cdots F_r^{m_r}$ with k a non-zero constant and the F_i be distinct irreducible polynomials, then

$$\widehat{F} = m_1\widehat{F}_1 + \cdots + m_r\widehat{F}_r$$

and the hypersurfaces $\widehat{F}_i$ are called the *irreducible components of* $\widehat{F}$ and the integer m_i is called the *multiplicity* of the component $\widehat{F}_i$. Every irreducible component of multiplicity $m_i > 1$ is called a *multiple component*. Hypersurfaces without multiple components are said to be *reduced*. If $\widehat{F}$ is irreducible and reduced then F is an irreducible polynomial. It is immediately seen that

$$\mathcal{V}_+(\widehat{F}) = \mathcal{V}_+(\widehat{F}_1) \cup \cdots \cup \mathcal{V}_+(\widehat{F}_r). \tag{10.2}$$

Remark 10.2 A hypersurface of degree 1 is uniquely determined by its support (Exercise 10.9). Distinct hypersurfaces of degree $d > 1$ may have the same support: if $K = \mathbb{R}$, $F = x_0^2 + x_1^2 + \cdots + x_n^2$ and $G = 2x_0^2 + x_1^2 + \cdots + x_n^2$, then $\widehat{F} \neq \widehat{G}$ but $\mathcal{V}_+(\widehat{F}) = \mathcal{V}_+(\widehat{G}) = \varnothing$.

Definition 10.3 Let

$$\mathcal{G}_d(n, K) := \{\mathcal{V}_+(\widehat{F}) \mid F \in {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]_d^*\},$$

be the set of all (possibly empty) geometric loci of $\mathbb{P}^n(K)$ of all hypersurfaces of degree d. The surjective map

$$\varphi\colon \mathcal{H}_d^{\mathrm{p}}(n, K) \to \mathcal{G}_d(n, K), \qquad \varphi(\widehat{F}) := \mathcal{V}_+(\widehat{F}),$$

is not injective in general, but if $d = 1$ then $\varphi\colon \mathcal{H}_1^{\mathrm{p}}(n, K) \to \mathcal{G}_1(n, K)$ is bijective by the previous remark.

Definition 10.4 Let $F = a_0x_0^s + a_1x_0^{s-1}x_1 + \cdots + a_{s-1}x_0x_1^{s-1} + a_sx_1^s \in {}_{\mathrm{h}}K[x_0, x_1]_s^*$ be a homogeneous polynomial of degree $s \geq 1$. Let $[c_0, c_1] \in \mathbb{P}^1(K)$.

(i) If $[c_0, c_1] = [0, 1]$, let $f(x) = F(x, 1) = a_0x^s + a_1x^{s-1} + \cdots + a_{s-1}x + a_s$. We call the multiplicity of f at 0 the *multiplicity* $m_{[0,1]}(F)$ of F at $[0, 1]$: if $t \geq 0$ is the greatest integer such that $a_t \neq 0$, then $f(x) = a_0x^s + a_1x^{s-1} + \cdots + a_tx^{s-t}$ and $m_{[0,1]}(F) = s - t$. If $[0, 1] \notin \mathcal{V}_+(F)$, then $m_{[0,1]}(F) = 0$ since $t = s$.

(ii) If $c_0 \neq 0$, let $g(x) = F(1, x) = a_0 + a_1x + \cdots + a_sx^s$. We call the multiplicity of g at $\dfrac{c_1}{c_0}$ the *multiplicity* $m_{[c_0,c_1]}(F)$ of F at $[c_0, c_1]$. If $[c_0, c_1] \notin \mathcal{V}_+(F)$ we set $m_{[c_0,c_1]}(F) = 0$.

If F is the zero-polynomial then we set $m_P(F) = \infty$ for every $P \in \mathbb{P}^1(K)$.

Proposition 10.5 *Let* $F = a_0x_0^s + a_1x_0^{s-1}x_1 + \cdots + a_{s-1}x_0x_1^{s-1} + a_sx_1^s \in {}_{\mathrm{h}}K[x_0, x_1]_s^*$ *be a homogeneous polynomial of degree* $s \geq 1$. *Then*

$$0 \leq \#(\mathcal{V}_+(F)) \leq s, \qquad \#(\mathcal{V}_+(F)) > 0 \quad \text{if } K \text{ is algebraically closed.}$$

Furthermore

$$\sum_{P \in \mathcal{V}_+(F)} m_P(F) \leq s, \qquad \sum_{P \in \mathcal{V}_+(F)} m_P(F) = s \quad \text{if } K \text{ is algebraically closed.}$$

Proof The set $\{[c_0, c_1] \in \mathcal{V}_+(F) \mid c_0 \neq 0\}$ is in a bijective correspondence with the set of the roots of $g(x) = F(1, x) = a_0 + a_1x + \cdots + a_sx^s$.

$$[c_0, c_1] \to \frac{c_1}{c_0}.$$

If, as above, $t \geq 0$ is the greatest integer such that $a_t \neq 0$ we have $g(x) = a_0 + a_1x + \cdots + a_tx^t$, hence if $b_1, \ldots, b_r$ are the distinct roots of $g(x)$ with multiplicity $m_1, \ldots, m_r$ respectively, we have

$$m_1 + \cdots + m_r \leq t, \qquad m_1 + \cdots + m_r = t \quad \text{if } K \text{ is algebrically closed.}$$

Therefore, if $P_i = [c_0, c_i]$ with $\dfrac{c_i}{c_0} = b_i$, $i = 1, \ldots, r$, we get

$$m_{P_1}(F) + \cdots + m_{P_r}(F) + m_{[0,1]}(F) \leq t + (s - t) = s$$

and $m_{P_1}(F) + \cdots + m_{P_r}(F) + m_{[0,1]}(F) = s$, if K is algebrically closed. □

Corollary 10.6 *Let K be an algebraically closed field and $F \in {}_{\mathrm{h}}K[x_0, x_1]_s^*$ be a homogeneous polynomial of degree $s \geq 1$. Then*

$$F(x_0, x_1) = C \prod_{[c_0,c_1]\in\mathcal{V}_+(F)} (c_1x_0 - c_0x_1)^{m_{[c_0,c_1]}(F)}$$

where $C \in K^*$.

Proof Assume the notation of Definition 10.4. If $[0, 1]$ is a zero of F with multiplicity $m_{[0,1]}(F)$, then

$$F(x_0, x_1) = a_0x_0^s + a_1x_0^{s-1}x_1 + \cdots + a_tx_0^{m_{[0,1]}(F)}x_1^t = x_0^{m_{[0,1]}(F)}(\cdots).$$

If $[c_0, c_1]$ with $c_0 \neq 0$ is a root of F with multiplicity $m_{[c_0,c_1]}(F)$, then

$$g(x) = \left(x - \frac{c_1}{c_0}\right)^{m_{[c_0,c_1]}(F)} (\cdots).$$

Hence

$$\begin{aligned} F(x_0, x_1) = x_0^t g\left(\frac{x_1}{x_0}\right) &= x_0^t \left(\frac{x_1}{x_0} - \frac{c_1}{c_0}\right)^{m_{[c_0,c_1]}(F)} (\cdots) \\ &= (c_1x_0 - c_0x_1)^{m_{[c_0,c_1]}(F)} (\cdots). \end{aligned}$$

In particular, the polynomial

$$\prod_{[c_0,c_1]\in\mathcal{V}_+(F)} (c_1x_0 - c_0x_1)^{m_{[c_0,c_1]}(F)}$$

divides F. By Proposition 10.5 this product has degree s so that it coincides with F (up to a constant factor $C \in K^*$). □

Lemma 10.7 *Let $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of $\mathbb{P}^n(K)$. Assume that $\#(K) \geq d + 1$ (e.g. K infinite). Then:*

(i) $\mathcal{V}_+(\widehat{F}) \subsetneqq \mathbb{P}^n(K)$.
(ii) *If K is algebraically closed, then $\mathcal{V}_+(\widehat{F}) \neq \varnothing$.*

Proof

(i) We proceed by induction on n. Order F according to the increasing powers of X_n

$$F = F(X_0, \ldots, X_n) = \sum_{i=0}^{d} F_i(X_0, \ldots, X_{n-1})X_n^i, \tag{10.3}$$

where $F_d(X_1, \dots, X_{n-1}) \neq 0$ and $\deg(F_i) = d - i$, $i = 0, \dots, d$; in particular for $n = 1$ we have

$$F(X_0, X_1) = F_0(X_0) + F_1(X_0)X_1 + \cdots + F_d(X_0)X_1^d, \qquad F_d(X_0) \neq 0.$$

If $F(x_0, x_1) = 0$ for all $[x_0, x_1] \in \mathbb{P}^1$, then

$$G(X_1) = F_0(x_0) + F_1(x_0)X_1 + \cdots + F_d(x_0)X_1^d$$

vanishes for all $x_1 \in K$. Since K has at least $d+1$ elements $G(X_1)$ must be the zero-polynomial. In particular, $F_d(x_0) = 0$ for all $x_0 \in K$. Since $F_d(X_0) \neq 0$ (and $\#(K) \geq d+1$) there exists $k \in K$ such that $F_d(k) \neq 0$, a contradiction.

Let $n \geq 2$. Assume the thesis true for $n-1$, hence there is $(a_0, \dots, a_{n-1})$ such that $F_d(a_0, \dots, a_{n-1}) \neq 0$. Suppose that $\mathcal{V}(\widehat{F}) = \mathbb{P}^n(K)$, i.e. $F(a_0, \dots, a_{n-1}, a_n) = 0$ for all $[a_1, \dots, a_{n-1}, a_n] \in \mathbb{P}^n(K)$, and consider the polynomial

$$G(X_n) = F(a_0, \dots, a_{n-1}, X_n) = \sum_{i=0}^{d} F_i(a_0, \dots, a_{n-1})X_n^i.$$

We should have $G(a_n) = F(a_0, \dots, a_{n-1}, a_n) = 0$ for all $a_n \in K$, so that $G(X_n)$ should be the zero-polynomial and $F_d(a_0, \dots, a_{n-1}) = 0$ contradicting our inductive hypothesis.

(ii) The case $n = 1$ is Proposition 10.5. Let $n = 2$ (the reader can easily prove the case $n \geq 3$ by induction on n). After renumbering the indeterminates we can always suppose

$$F = a_0(X_0, X_1)X_2^r + a_1(X_0, X_1)X_2^{r-1} + \cdots + a_r(X_0, X_1)$$

arranged according to the increasing powers of X_2, with $a_i(X_0, X_1) \in K[X_0, X_1]$ homogeneous polynomial of degrees $d - i$, $i = 0, 1, \dots, r$ and $a_0(X_0, X_1) \neq 0$. By Proposition 10.5 there exists $[\alpha_0, \alpha_1] \in \mathbb{P}^1(K)$ such that $a_0(\alpha_0, \alpha_1) \neq 0$. Therefore the polynomial

$$G(X_2) := a_0(\alpha_0, \alpha_1)X_2^r + a_1(\alpha_0, \alpha_1)X_2^{r-1} + \cdots + a_r(\alpha_0, \alpha_1) \in K[X_2]$$

is non-constant and it has a root $\alpha_2 \in K$. Hence $F(\alpha_0, \alpha_1, \alpha_2) = 0$, i.e. $[\alpha_0, \alpha_1, \alpha_2] \in \mathcal{V}_+(\widehat{F})$. □

Corollary 10.8 *Let $n \geq 2$, K algebraically closed and $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of $\mathbb{P}^n(K)$. Then $\mathcal{V}_+(\widehat{F})$ is an infinite set.*

Proof Let $P = [p_0, \dots, p_n]$ and $Q = [q_0, \dots, q_n]$ be two distinct points of $\mathbb{P}^n(K)$. The line PQ has parametric equations

$$x_i = p_i t_0 + q_i t_1, \quad i = 0, 1, \dots, n, \quad \forall [t_0, t_1] \in \mathbb{P}^1(K).$$

By (i) we can take $P \notin \mathcal{V}_+(F)$ so that

$$G(t_0, t_1) = F(p_0 t_0 + q_0 t_1, \ldots, p_n t_0 + q_n t_1)$$

is not identically zero and it has a root by Proposition 10.5. Thus we obtain infinitely many points of $\mathcal{V}_+(\widehat{F})$ varying Q in $\mathbb{P}^n(K)$, $Q \neq P$. □

10.1.1 Smooth Points and Tangent Hyperplanes

Let $F \in K[X_0, \ldots, X_n]$ be a homogeneous polynomial of degree $d \geq 0$. Then, for any $k = 0, 1, \ldots, n$, the formal derivative $\dfrac{\partial F}{\partial X_k}$ is either a homogeneous polynomial of degree $d - 1$ or the zero polynomial.

The following proposition is very easy to prove (left to the reader) but it is absolutely crucial.

Proposition 10.9 (Euler) *Let $F \in K[X_0, \ldots, X_n]$ be a homogeneous polynomial of degree $d \geq 0$. Then* Euler's formula of degree 1

$$d \cdot F = \sum_{k=0}^{n} X_k \frac{\partial F}{\partial X_k} \tag{10.4}$$

holds in $K[X_0, \ldots, X_n]$. More generally, we have Euler's formula of degree $m \geq 1$

$$\sum_{i_1=0}^{n} \sum_{i_2=0}^{n} \cdots \sum_{i_m=0}^{n} X_{i_1} X_{i_2} \cdot \cdots \cdot X_{i_m} \cdot \frac{\partial^m F}{\partial X_{i_1} \partial X_{i_2} \cdots \partial X_{i_m}}$$
$$= d(d-1) \cdots (d-m+1) F(X_0, \ldots, X_n). \tag{10.5}$$

We also need the following elementary result.

Corollary 10.10 (Leibniz–Newton) *Let $F \in K[x_0, \ldots, x_n]$ be a homogeneous polynomial of degree d. Let $(a_0, \ldots, a_n)$, $(b_0, \ldots, b_n) \in K^{n+1}$. If $\operatorname{char}(K) \nmid d!$ (in particular if $\operatorname{char}(K) = 0$) we have*

$$F(a_0 + b_0, \ldots, a_n + b_n) = F(a_0, \ldots, a_n)$$
$$+ \sum_{s=1}^{d-1} \frac{1}{s!} \sum_{\substack{i_\alpha \in \{0,\ldots,n\} \\ \alpha = 1,\ldots,s \\ i_1 + \cdots i_s = s}} \frac{\partial^s F}{\partial X_{i_1} \partial X_{i_2} \cdots \partial X_{i_s}} (a_0, \ldots, a_n)\, b_{i_1} b_{i_2} \cdots b_{i_s} + F(b_0, \ldots, b_n). \tag{10.6}$$

Proof In Taylor's formula centred at $(a_0, \dots, a_n)$

$$F(x_0, \dots, x_n) = F(a_0, \dots, a_n) + \sum_{s=1}^{d} \frac{1}{s!} \sum_{\substack{i_\alpha \in \{0,\dots,n\} \\ \alpha=1,\dots,s \\ i_1+\cdots i_s=s}} \frac{\partial^s F}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_s}}(a_0, \dots, a_n)\,(x_{i_1} - a_{i_1}) \cdots (x_{i_s} - a_{i_s})$$

we put $x_i = a_i + b_i$, $i = 0, \dots, n$. Since $\deg F = d$, $\dfrac{\partial^d F}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_d}}$, with $i_1 + \cdots + i_d = d$, is constant, we have

$$\frac{\partial^d F}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_d}}(a_0, \dots, a_n) = \frac{\partial^d F}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_d}}(b_0, \dots, b_n).$$

Therefore

$$\frac{1}{d!} \sum_{\substack{i_\alpha \in \{0,\dots,n\} \\ \alpha=1,\dots,d \\ i_1+\cdots i_d=d}} \frac{\partial^d F}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_d}}(a_0, \dots, a_n)\, b_{i_1} \cdots b_{i_d}$$

$$= \frac{1}{d!} \sum_{\substack{i_\alpha \in \{0,\dots,n\} \\ \alpha=1,\dots,d \\ i_1+\cdots i_d=d}} \frac{\partial^d F}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_d}}(b_0, \dots, b_n)\, b_{i_1} \cdots b_{i_d} = F(b_0, \dots, b_n)$$

by Proposition 10.9. □

Definition 10.11 Let $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of $\mathbb{P}^n(K)$ $(n \geq 2)$ of degree $d \geq 1$. We say that $P = [p_0, p_1, \dots, p_n] \in \mathcal{V}_+(\widehat{F})$ is a *singular point* of $\widehat{F}$ if

$$\frac{\partial F}{\partial x_k}(P) = 0, \qquad k = 0, 1, \dots, n.$$

We call the set of all singular points of $\widehat{F}$ the *singular locus* of $\widehat{F}$ and we denote it by $\mathrm{Sing}(\widehat{F})$. A point $P \in \mathcal{V}_+(\widehat{F})$ is called *smooth* or *non-singular* of $\widehat{F}$ if P is not singular. We call the set of all smooth points of $\widehat{F}$ the *regular locus* and we denote it by $\mathrm{Reg}(\widehat{F})$, A hypersurface $\widehat{F}$ is called *smooth* if $\mathrm{Sing}(\widehat{F}) = \varnothing$, i.e. if $\mathrm{Reg}(F) = \mathcal{V}_+(\widehat{F})$. It is obvious that the above definitions depend only on $\widehat{F}$.

Proposition 10.12 *Let $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of degree d. If* $\mathrm{char}(K) \nmid d$ *(in particular* $\mathrm{char}(K) = 0$*), then for each point $P \in \mathbb{P}^n(K)$*

$$\frac{\partial F}{\partial x_i}(P) = 0, \quad i = 0, 1, \dots, n, \quad \Longrightarrow \quad P \in \mathcal{V}_+(\widehat{F}). \tag{10.7}$$

Proof Euler's formula

$$d \cdot F(P) = \sum_{k=0}^{n} x_k \frac{\partial F}{\partial x_k}(P)$$

and the equalities $\frac{\partial f}{\partial x_i}(P) = 0$, $i = 0, 1, \ldots, n$, imply $d \cdot F(P) = 0$. So if $\mathrm{char}(K) \nmid d$, then $F(P) = 0$. □

Remark 10.13 Let $\mathrm{char}(K) = 2$, $F = x_1^2 - x_0 x_2 \in K[x_0, x_1, x_2]$ and $P = [0, 1, 0]$. Then $\dfrac{\partial F}{\partial x_k}(P) = 0$, $k = 0, 1, 2$, but $F(P) \neq 0$.

The proof of the following lemma is immediate.

Lemma 10.14 *Let $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of degree d. Then*

$$\mathrm{Reg}(\widehat{F}) \cup \mathrm{Sing}(\widehat{F}) = \mathcal{V}_+(\widehat{F}), \qquad \mathrm{Reg}(\widehat{F}) \cap \mathrm{Sing}(\widehat{F}) = \varnothing,$$

$$\mathrm{Sing}(\widehat{F}) = \bigcap_{k=0}^{n} \mathcal{V}_+\left(\widehat{\frac{\partial F}{\partial x_k}}\right).$$

Definition 10.15 Let $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of degree d with $n \geq 2$, and $P = [p_0, p_1, \ldots, p_n] \in \mathcal{V}_+(\widehat{F})$ a smooth point of $\widehat{F}$. The *projective tangent hyperplane* (or simply *tangent hyperplane*) to $\widehat{F}$ at P is defined by

$$\nabla(F)(P)x^t = 0 \quad \text{i.e} \quad \sum_{k=0}^{n} \frac{\partial F}{\partial x_k}(P)x_k = 0. \qquad x = (x_0, \ldots, x_n), \tag{10.8}$$

and denoted by $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$, (or simply $\mathrm{T}_{\widehat{F},P}$). The above definition clearly depends only of $\widehat{F}$. By Euler's formula $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$ passes through P. If $\widehat{F}$ is a hyperplane of $\mathbb{P}^n(K)$, $\mathcal{Y} = \mathcal{V}_+(\widehat{F})$, then every $P \in \mathcal{Y}$ is smooth and $\mathrm{T}^{\mathrm{p}}_{\mathcal{Y},P} = \mathcal{Y}$.

Proposition 10.16 *Let $n \geq 2$ and $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$. Let*

$$\widehat{F} = m_1\widehat{F}_1 + \cdots + m_k\widehat{F}_r$$

be the decomposition of $\widehat{F}$ into irreducible components. Then P is singular for $\widehat{F}$ if and only if at least one of the following conditions holds:

(i) *there exists $j \in \{1, \ldots, r\}$ such that $P \in \widehat{F_j}$ and $m_j \geq 2$;*
(ii) *there exists $j \in \{1, \ldots, r\}$ such that $P \in \mathrm{Sing}(\widehat{F_j})$;*
(iii) *there exist $h, k \in \{1, \ldots, r\}$, $h \neq k$, such that $P \in \widehat{F_h} \cap \widehat{F_k}$.*

If $P \in \mathrm{Reg}(\widehat{F})$, then there exists one and only one j, $1 \le j \le r$, such that $P \in \widehat{F}$ with $m_j = 1$ and $P \in \mathrm{Reg}(\widehat{F_j})$; furthermore

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},P} = \mathrm{T}^{\mathrm{p}}_{\widehat{F_j},P}.$$

Proof Let $F = cF_1^{m_1} \cdots F_r^{m_r}$ be the factorization of F, where $c \in K$, the F_i are mutually coprime irreducible homogeneous polynomials and $m_i > 0$ for every $i = 1, \ldots, r$. By Leibniz's rule

$$\nabla F(P) = \sum_{i=1}^{r} m_i F_1^{m_1}(P) \cdots F_i^{m_i-1}(P) \cdots F_r^{m_r}(P) \nabla f_i(P). \tag{10.9}$$

If P belongs to at least two distinct components of $\widehat{F}$, or it belongs to a component $\widehat{F_j}$ of multiplicity $m_j \ge 2$, then all the terms of the right– hand side of (10.9) vanish and therefore P is singular for $\widehat{F}$. This case corresponds to conditions (i) and (iii).

If P belongs to only one component $\widehat{F_j}$ of multiplicity $m_j = 1$, by (10.9) one has

$$\nabla F(P) = c \nabla F_j(P), \quad \text{where} \quad c = \prod_{i \neq j} F_i^{m_i}(P) \neq 0. \tag{10.10}$$

Hence $\nabla F(P) = 0$ if and only if $\nabla F_j(P) = 0$. The last assertion immediately follows from (10.10). □

Corollary 10.17 *If K is an algebraically closed field then a smooth hypersurface $\widehat{F}$ of $\mathbb{P}^n(K)$ is irreducible.*

Proof Assume by contradiction that $\widehat{F}$ is reducible and that consequently there exist hypersurfaces $\widehat{F_1}$ and $\widehat{F_2}$ of $\mathbb{P}^n(K)$ such that $\widehat{F} = \widehat{F_1} + \widehat{F_2}$. By Remark 10.37 there exists $P \in \mathbb{P}^n(K)$ such that $P \in \mathcal{V}_+(\widehat{F_1}) \cap \mathcal{V}_+(\widehat{F_2})$. By Proposition 10.16 the point P is singular for $\widehat{F}$, contradicting the assumption that $\widehat{F}$ is smooth. □

Definition 10.18 (Tangent Lines) Let $n \ge 2$ and $\widehat{F} \in \mathcal{H}^{\mathrm{p}}_d(n, K)$ be a hypersurface of $\mathbb{P}^n(K)$. Let $A = [a_0, \ldots, a_n]$ and $B = [b_0, \ldots, b_n]$ be two distinct points of $\mathbb{P}^n(K)$. The variable point on the line $r = AB$ is

$$Q_{\lambda,\mu} = [\lambda a_0 + \mu b_0, \ldots, \lambda a_n + \mu b_n], \qquad (\lambda, \mu) \in K \times K, \quad (\lambda, \mu) \neq (0, 0), \tag{10.11}$$

and it belongs to $\mathcal{V}_+(F)$ if and only if

$$g(\lambda, \mu) := F(\lambda a_0 + \mu b_0, \ldots, \lambda a_n + \mu b_n) = 0. \tag{10.12}$$

If $[\lambda_0, \mu_0]$ is a root of multiplicity m of the polynomial $g(\lambda, \mu)$ (Definition 10.4) then we say that $\widehat{F}$ and r have *multiplicity of intersection* $m_P(\widehat{F}, r) := m$ at the

corresponding point $P = \lambda_0 A + \mu_0 B$. Note that the multiplicity of intersection is well defined, because it does not depend on the choice of the two points A and B on the line r. We set by definition $m_P(\widehat{F}, r) = 0$ if $P \notin r \cap \mathcal{V}_+(F)$ and $m_P(\widehat{F}, r) = \infty$ if $r \subset \mathcal{V}_+(F)$. Concerning the notion of multiplicity just introduced, one can easily verify that:

(a) If $\widehat{F} = \widehat{F}_1 + \widehat{F}_2$, then $m_P(\widehat{F}, r) = m_P(\widehat{F}_1, r) + m_P(\widehat{F}_2, r)$.
(b) If $r \not\subset \mathcal{V}_+(F)$ then (Proposition 10.5)

$$\sum_{P \in r} m_P(\widehat{F}, r) \leq d, \qquad \sum_{P \in r} m_P(\widehat{F}, r) = d \quad \text{if } K \text{ is algebrically closed.}$$

A line r is *tangent* at a point P if $m_P(\widehat{F}, r) \geq 2$.

Corollary 10.19 *Under the above hypothesis, let* $\mathrm{char}(K) \nmid d!$, $P = [p_0, \ldots, p_n] \in \mathcal{V}_+(F)$, $Y = [y_0, \ldots, y_n]$ *and* $r = PY \not\subset \mathcal{V}_+(F)$. *If*

$$\Delta_y^s F(P) := \sum_{i_1=0}^{n} \sum_{i_2=0}^{n} \cdots \sum_{i_s=0}^{n} \frac{\partial^s F}{\partial X_{i_1} \partial X_{i_2} \cdots \partial X_{i_s}}(P)\, y_{i_1} y_{i_2} \cdots y_{i_s},$$

then $m_P(\widehat{F}, r) = s$, *where* s *is the integer such that*

$$\Delta_y^1 F(P) = \Delta_y^2 F(P) = \cdots = \Delta_y^{s-1} F(P) = 0, \qquad \Delta_y^s F(P) \neq 0. \tag{10.13}$$

Proof By Corollary 10.10 (with $a_i = \lambda p_i$ and $b_i = \mu y_i$, $i = 0, \ldots, n$) one has

$$g(\lambda, \mu) = \lambda^d F(P) + \lambda^{d-1} \mu \Delta_y^1 F(P) + \frac{1}{2!} \lambda^{d-2} \mu^2 \Delta_y^2 F(P) + \cdots + \frac{1}{s!} \lambda^{d-s} \mu^s \Delta_y^s F(P) + \cdots + \mu^d F(Y) = 0 \tag{10.14}$$

for (λ, μ) such that $Q_{\lambda,\mu} = [\lambda p_0 + \mu y_0, \ldots, \lambda p_n + \mu y_n] \in \mathcal{V}_+(F)$. Since $P \in \mathcal{V}_+(F)$, the pair $(1, 0)$ is a solution of (10.14) so that $m_P(\widehat{F}, r)$ is the multiplicity at 0 of

$$\frac{\mu}{\lambda} \Delta_y^1 F(P) + \frac{1}{2!} \left(\frac{\mu}{\lambda}\right)^2 \Delta_y^2 F(P) + \cdots + \frac{1}{s!} \left(\frac{\mu}{\lambda}\right)^s \Delta_y^s F(P) + \cdots + \left(\frac{\mu}{\lambda}\right)^d F(Y) = 0.$$

□

Proposition 10.20 *The tangent lines to a hypersurface* $\widehat{F}$ *at a smooth point* P *are exactly the lines through* P *lying on* $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$. *If* P *is singular, then every line through* P *is tangent to* $\widehat{F}$.

Proof By Corollary 10.19 we have the equivalences

$$m_P(\widehat{F}, r) \geq 2 \iff \Delta_y^1 F(P) = 0 \text{ i.e. } \sum_{i=0}^{n} \frac{\partial F}{\partial X_i}(p_0, \ldots, p_n) y_i = 0$$
$$\iff Y \in \mathrm{T}^{\mathrm{p}}_{\widehat{F}, P} \iff YP \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F}, P}.$$

If P is singular, we have $\Delta_y^1 F(P) = 0$ for every $Y \in \mathbb{P}^n(K)$, hence $m_P(\widehat{F}, r) \geq 2$. □

Corollary 10.21 *If a line $r \subset \mathcal{V}_+(\widehat{F})$ passes through a smooth point P of $\widehat{F}$, then $r \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F}, P}$.*

Definition 10.22 Let $\widehat{F}$ be a hypersurface of degree d and $P \in \mathbb{P}^n(K)$.The *multiplicity of $\widehat{F}$ at P* is the integer

$$m_P(\widehat{F}) := \min_{\substack{r \not\subset \mathcal{V}(\widehat{F}) \\ P \in r}} m_P(\widehat{F}, r).$$

If K has at least $d+1$ elements there exists at least a line not contained in $\widehat{F}$ so that

$$0 \leq m_P(\widehat{F}) \leq d,$$

where $m_P(\widehat{F}) = 0$ if and only if $P \notin \widehat{F}$.

From Euler's formula (10.5) we have immediately:

Corollary 10.23 *Let* $\mathrm{char}(K) \nmid d!$, *and $\widehat{F}$ be a hypersurface of degree d. Let $P \in \mathbb{P}^n(K)$. Then $m_P(\widehat{F}) = s$ if and only if all partial derivatives of F of order $\leq s-1$ vanish at P and at least one partial derivative of order s does not vanish.*

Definition 10.24 A projective hypersurface $\widehat{F} \in \mathcal{H}^{\mathrm{p}}_d(n, K)$ is called a *projective cone* if there exists a point $P \in \mathcal{V}_+(\widehat{F})$, called a *vertex*, such that, for every $Q \in \mathcal{V}_+(\widehat{F})$, $Q \neq P$, the line joining P and Q is contained in $\widehat{F}$. *The vertex* of a cone is the set of all vertices: the set of vertices of a cone may be infinite (in $\mathbb{P}^3(\mathbb{R})$ the vertex of the cone $x_0 x_1 = 0$ is the line $x_0 = x_1 = 0$).

The *projective tangent cone* to a hypersurface $\widehat{F} \in \mathcal{H}^{\mathrm{p}}_d(n, K)$ is the projective hypersurface $C^{\mathrm{p}}_P(\widehat{F})$ of equation

$$\sum_{i_1=0}^{n} \sum_{i_2=0}^{n} \cdots \sum_{i_m=0}^{n} \frac{\partial^m F}{\partial X_{i_1} \partial X_{i_2} \cdots \partial X_{i_m}}(P) X_{i_1} X_{i_2} \cdots X_{i_m} = 0, \qquad m = m_P(\widehat{F}), \tag{10.15}$$

whose degree is $m_P(\widehat{F})$. If $m_P(\widehat{F}) = 1$, then $C_P^{\mathrm{p}}(\widehat{F}) = \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$. Note that $C_P^{\mathrm{p}}(\widehat{F})$ is the locus of the lines r passing through P such that

$$m(\widehat{F}, r; P) > m_P(\widehat{F}).$$

Definition 10.25 A projective subspace $\mathcal{S}$ is *tangent* to a hypersurface $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ at a point $P \in \mathcal{V}_+(\widehat{F}) \cap \mathcal{S}$ if every line through P lying on $\mathcal{S}$ is tangent to $\widehat{F}$. If P is not singular then $\mathcal{S} \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$.

The Action of $\mathrm{PGL}_n(K)$ on $\mathcal{H}_d^{\mathrm{p}}(n, K)$

Definition 10.26 Let $A = \left(a_{ij}\right)_{i,j=0,\dots,n} \in \mathrm{GL}_{n+1}(K)$ and

$$T_A \in \mathrm{PGL}_n(K), \qquad T_A([x_0, \dots, x_n]) = [y_0, \dots, y_n],$$
$$y_i = \sum_{j=0}^{n} a_{ij} x_j, \qquad i = 0, 1, \dots, n,$$

be the projective automorphism associated with A (see Example 6.45). Then $T_A^{-1} = T_{A^{-1}}$ and

$$T_{A^{-1}}([y_0, \dots, y_n]) = [x_0, \dots, x_n], \qquad x_i = \sum_{j=0}^{n} a^*_{ij} y_j, \quad j = 0, 1, \dots, n,$$

where $A^{-1} = \left(a^*_{ij}\right)_{i,j=0,\dots,n}$. If $F \in K[x_0, \dots, x_n]$ we define

$$\sigma_A(F)([y_0, \dots, y_n]) := F(T_{A^{-1}}([y_0, \dots, y_n])) = F\left(\sum_{j=0}^{n} a^*_{0j} y_j, \dots, \sum_{j=0}^{n} a^*_{nj} y_j\right).$$

Proposition 10.27 *The map $\sigma_A \colon K[x_0, \dots, x_n] \to K[x_0, \dots, x_n]$ is an automorphism of K-algebras for every $A \in \mathrm{GL}_{n+1}(K)$. Furthermore*

$$\sigma_{AB} = \sigma_A \circ \sigma_B, \qquad \forall\, A, B \in \mathrm{GL}_{n+1}(K), \tag{10.16}$$

$$\sigma_A \circ \sigma_{A^{-1}} = \sigma_{A^{-1}} \circ \sigma_A = \mathrm{id}_{K[x_0,\dots,x_n]}, \quad \forall\, A \in \mathrm{GL}_{n+1}(K), \tag{10.17}$$

$$\sigma_A({}_{\mathrm{h}}K[X_0, X_1, \dots, X_n]^*_d) = {}_{\mathrm{h}}K[X_0, X_1, \dots, X_n]^*_d, \qquad \forall\, A \in \mathrm{GL}_{n+1}(K). \tag{10.18}$$

Proof Left to the reader. □

Definition 10.28 The map

$$\mathrm{PGL}_n(K) \times \mathcal{H}^{\mathrm{p}}_d(n, K) \to \mathcal{H}^{\mathrm{p}}_d(n, K), \qquad (\sigma_A, \widehat{F}) \to \widehat{\sigma_A(F)}, \tag{10.19}$$

is well-defined. Indeed, if $B \in \mathrm{GL}_{n+1}(K)$ is another matrix such that $T_A = T_B$ then by Lemma 6.41 we have $B = aA$, with $a \in K^*$. Hence $B^{-1} = a^{-1}A^{-1}$ and $\sigma_B(F) = a^{-d}\sigma_A(F)$ so that $\widehat{\sigma_A(F)} = \widehat{\sigma_B(F)}$. Therefore, by Proposition 10.27, (10.19) defines an action of $\mathrm{PGL}_n(K)$ on $\mathcal{H}^{\mathrm{p}}_d(n, K)$. Thus if $\alpha = T_A$ we denote $\widehat{\sigma_A(F)}$ by $\alpha \cdot \widehat{F}$. Hence we have the following relations:

$$(\alpha \circ \beta) \cdot \widehat{F} = \alpha \cdot (\beta \cdot \widehat{F}), \quad \forall\, \alpha, \beta \in \mathrm{PGL}_n(K), \quad \forall\, \widehat{F} \in \mathcal{H}^{\mathrm{p}}_d(n, K),$$

$$\mathrm{id}_{\mathrm{PGL}_n(K)} \cdot \widehat{F} = \widehat{F}, \quad \forall\, \widehat{F} \in \mathcal{H}^{\mathrm{p}}_d(n, K).$$

Corollary 10.29 *Under assumptions and notation of* Definition 10.28 *we have*

$$\alpha(\mathcal{V}_+(\widehat{F})) = \mathcal{V}_+(\alpha \cdot \widehat{F}), \qquad \forall\, \alpha \in \mathrm{PGL}_n(K).$$

Proof Let $P \in \mathcal{V}_+(\widehat{F})$ and set $\alpha = T_A$. Then $\sigma_A(F)(T_A(P)) = F(T_{A^{-1}}(T_A(P))) = F(P) = 0$, i.e. $\alpha(\mathcal{V}_+(\widehat{F})) \subset \mathcal{V}_+(\alpha \cdot \widehat{F})$. If $Q \in \mathcal{V}_+(\alpha \cdot \widehat{F})$ then $0 = \sigma_A(F)(Q) = F(T_{A^{-1}}(Q))$, hence $P = T_{A^{-1}}(Q) \in \mathcal{V}_+(\widehat{F})$. Therefore $Q = T_A(P) \in \alpha(\mathcal{V}_+(\widehat{F}))$. □

Definition 10.30 Two hypersurfaces $\widehat{F}$ and $\widehat{G} \in \mathcal{H}^{\mathrm{p}}_d(n, K)$ are *projectively equivalent*, written $\widehat{F} \sim \widehat{G}$, if there exists $\alpha \in \mathrm{PGL}_n(K)$ such that $\widehat{G} = \alpha(\widehat{F})$, i.e. if $\widehat{G} \in \mathrm{Orb}(\widehat{F})$ with respect to the action of $\mathrm{PGL}_n(K)$ on $\mathcal{H}^{\mathrm{p}}_d(n, K)$ above defined.

Proposition 10.31 *Let $\widehat{F}$ be a hypersurface of $\mathbb{P}^n(K)$ whose support contains a hyperplane $\widehat{H} \subset \mathbb{P}^n(K)$. Then $\widehat{H}$ is an irreducible component of $\widehat{F}$.*

Proof We may choose homogeneous coordinates such that $\widehat{H}$ is defined by the equation $x_0 = 0$. There exists a homogeneous (possibly zero) polynomial $Q(x_0, \ldots, x_n)$ and a homogeneous (possibly zero) polynomial $R(x_0, \ldots, x_n)$ such that

$$F(x_0, \ldots, x_n) = x_0 Q(x_0, \ldots, x_n) + R(x_1, \ldots, x_n).$$

For every $(a_1, \ldots, a_n) \in K^n$ we have $F(0, a_1, \ldots, a_n) = 0$, so that $R(a_1, \ldots, a_n) = 0$. By the identity principle of polynomials R is the zero polynomial, hence $x_0 \mid F$. □

Corollary 10.32 *Under the notation of* Definition 10.26*, for every $F \in {}_{\mathrm{h}}K[X_0, X_1, \ldots, X_n]^*_d$ and for every $A \in \mathrm{GL}_{n+1}(K)$ we have*

$$\nabla(\sigma_A(F)) = \nabla(F)A^{-1}.$$

Proof If we put $F_k = x_k = \sum_{i=0}^{n} a^*_{ki} x_i$, $k = 0, 1, \ldots, n$, in Lemma 5.6, we get

$$\begin{aligned}\frac{\partial(\sigma_A(F))}{\partial y_i} &= \sum_{k=0}^{n} \frac{\partial F}{\partial x_k}(x_0, \ldots, x_n)\frac{\partial x_k}{\partial y_i} \\ &= \sum_{k=0}^{n} \frac{\partial F}{\partial x_k}(x_0, \ldots, x_n) a^*_{ki}, \qquad i = 0, \ldots, n.\end{aligned}$$

□

Proposition 10.33 *Let $\widehat{F} \in \mathcal{H}^{\mathrm{p}}_d(n, K)$ be a hypersurface of degree d and $\alpha \in \mathrm{PGL}_n(K)$. Then*

$$\alpha(\mathrm{Sing}(\widehat{F})) = \mathrm{Sing}(\alpha \cdot \widehat{F}), \qquad \alpha(\mathrm{Reg}(\widehat{F})) = \mathrm{Reg}(\alpha \cdot \widehat{F}). \tag{10.20}$$

If $P \in \mathrm{Reg}(\widehat{F})$, then

$$\alpha(\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}) = \mathrm{T}^{\mathrm{p}}_{\alpha\cdot\widehat{F},\alpha(P)}. \tag{10.21}$$

Furthermore, if r is a line passing through a point P, then

$$m_P(\widehat{F}, r) = m_{\alpha(P)}(\alpha \cdot \widehat{F}, \alpha(r)). \tag{10.22}$$

In particular

$$m_P(\widehat{F}) = m_{\alpha(P)}(\alpha \cdot \widehat{F}). \tag{10.23}$$

Proof By Corollary 10.32 we get (10.20) and (10.21) immediately. Let $P \in \mathbb{P}^n(K)$ and r be a line through P of parametric equations $x(u, v) = (x_0(u, v), \ldots, x_n(u, v))$, then $\alpha(r)$ passes through $\alpha(P)$ and has parametric equations $y(u, v)^t = Ax(u, v)^t$, where $y(u, v) = (y_0(u, v), \ldots, y_n(u, v))$. Hence the relation

$$\alpha \cdot \widehat{F}(y(u, v)) = F(A^{-1}y(u, v)^t) = F(A^{-1}Ax(u, v)^t) = F(x(u, v))$$

implies (10.22) and (10.23). □

Definition 10.34 Let $\widehat{F}$ and $\widehat{G}$ be two distinct projective hypersurfaces of the same degree. We define the *pencil of hypersurfaces* generated by $\widehat{F}$ and $\widehat{G}$ as the set of the hypersurfaces $\widehat{H} = \widehat{H}_{\lambda,\mu}$ such that

$$H_{\lambda,\mu} = \lambda F + \mu G, \qquad (\lambda, \mu) \in \mathbb{R}^2 \setminus \{(0, 0)\}.$$

In particular we have $\mathcal{V}_+(\widehat{H_{\lambda,\mu}}) \supset \mathcal{V}_+(\widehat{F}) \cap \mathcal{V}(\widehat{G})$.

Restriction of a Hypersurface to a Projective Subspace

Definition 10.35 Let $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be a hypersurface of $\mathbb{P}^n(K)$ and $\mathcal{S} \subset \mathbb{P}^n(K)$ be a projective subspace of dimension m, $1 \leq m \leq n-1$. Suppose that $\mathcal{S}$ is given by parametric equations

$$x_i = \sum_{j=0}^{m} a_{ij} t_j, \quad i = 0, 1, \ldots, m, \text{ with } [t_0, t_1, \ldots, x_m] \in \mathbb{P}^m(K) \text{ and}$$

$$\operatorname{rank} \left(a_{ij}\right)_{i=0,\ldots,n,\, j=0,\ldots,m} = m+1, \tag{10.24}$$

so that

$$\sigma([t_0, t_1, \ldots, t_m]) = \left[\sum_{j=0}^{m} a_{0j} t_j, \sum_{j=0}^{m} a_{1j} t_j, \ldots, \sum_{j=0}^{m} a_{nj} t_j \right] \tag{10.25}$$

is a projective isomorphism $\sigma : \mathbb{P}^m(K) \to \mathcal{S}$ (see Definition 6.50). Let

$$\begin{aligned} F_{|\mathcal{S},\sigma}(t_0, t_1, \ldots, t_m) &:= F(\sigma([t_0, t_1, \ldots, t_m])) \\ &= F\left(\sum_{j=0}^{m} a_{0j} t_j, \sum_{j=0}^{m} a_{1j} t_j, \ldots, \sum_{j=0}^{m} a_{nj} t_j \right), \end{aligned} \tag{10.26}$$

then $F_{|\mathcal{S},\sigma}(t_0, t_1, \ldots, t_m)$ is a homogeneous polynomial of degree d (which obviously depends on the choice of σ, although its degree does not). Note that $F_{|\mathcal{S},\sigma}(t_0, t_1, \ldots, t_m)$ is not the zero-polynomial if and only if $\mathcal{S} \not\subseteq \mathcal{V}_+(\widehat{F})$. Suppose that $\mathcal{S} \not\subseteq \mathcal{V}_+(\widehat{F})$, then the hypersurface $\widehat{F_{|\mathcal{S},\sigma}}$ of $\mathbb{P}^m(K) \cong \mathcal{S}$ is called the *restriction* of $\widehat{F} \in \mathcal{H}(n, d, K)$ to $\mathcal{S}$. If we may understand σ we also denote $\widehat{F_{|\mathcal{S},\sigma}}$ by $\widehat{F} \cap \mathcal{S}$. We have to observe that $\mathcal{V}_+(\widehat{F}) \neq \varnothing$ does not imply that $\mathcal{V}_+(\widehat{F_{|\mathcal{S},\sigma}}) \neq \varnothing$ as the following example in $\mathbb{P}^2(\mathbb{R})$ shows: $F = x_0^2 + x_1^2 - x_2^2$ and $\mathcal{S}$ the line $x_2 = 0$.

If we choose another projective isomorphism $\tau : \mathbb{P}^m(K) \cong \mathcal{S}$, then $\alpha := \tau^{-1} \circ \sigma \in \mathrm{PGL}_m(K)$ and

$$\widehat{F_{|\mathcal{S},\tau}} = \alpha \cdot \widehat{F_{|\mathcal{S},\sigma}},$$

so that

$$\tau(\mathcal{V}_+(\widehat{F_{|\mathcal{S},\tau}})) = \tau(\mathcal{V}_+(\alpha \cdot \widehat{F_{|\mathcal{S},\sigma}})) = \tau(\alpha(\mathcal{V}_+(\widehat{F_{|\mathcal{S},\sigma}}))) = \sigma(\mathcal{V}_+(\widehat{F_{|\mathcal{S},\sigma}})). \tag{10.27}$$

The following is now immediate:

Corollary 10.36 *Assume the above notation. Let* $\mathcal{S} \not\subseteq \mathcal{V}_+(\widehat{F})$*, then*

$$\mathcal{V}_+(\widehat{F}) \cap \mathcal{S} = \sigma(\mathcal{V}_+(\widehat{F_{|\mathcal{S},\sigma}})),$$

where $\sigma : \mathbb{P}^m(K) \cong \mathcal{S}$ *is the isomorphism* (10.25).

If $\mathcal{S}$ has equations $x_{m+1} = \cdots = x_n = 0$ we choose the isomorphism $\sigma([t_0, \ldots, t_m]) = [t_0, \ldots, t_m, 0, \ldots, 0]$ and we denote $F_{|\mathcal{S},\sigma}$ simply by $F_{|\mathcal{S}}$.

Remark 10.37 Let K be an algebraically closed field and $\widehat{F}, \widehat{G} \in \mathcal{H}_d^{\mathrm{p}}(n, K)$ be two hypersurfaces, then

$$\mathcal{V}_+(\widehat{F}) \cap \mathcal{V}_+(\widehat{G}) \neq \varnothing.$$

Let $\mathcal{Y}$ be the plane of equations $x_3 = x_4 = \cdots = x_n = 0$ and suppose that $\mathcal{Y} \not\subset \mathcal{V}_+(\widehat{F})$ and $\mathcal{Y} \not\subset \mathcal{V}_+(\widehat{G})$. Consider the curves of $\mathcal{Y} \cong \mathbb{P}^2(K)$

$$F_{|\mathcal{Y}}(x_0, x_1, x_2) = F(x_0, x_1, x_2, 0, \ldots, 0),$$
$$F_{|\mathcal{Y}}(x_0, x_1, x_2) = G(x_0, x_1, x_2, 0, \ldots, 0).$$

By Bézout's theorem (Theorem 11.1), which is proved in Chapter 11, there exists

$$P[a, b, c] \in \mathcal{V}_+(\widehat{F_{|\mathcal{Y}}}) \cap \mathcal{V}_+(\widehat{G_{|\mathcal{Y}}}),$$

hence $[a, b, c, 0. \ldots, 0] \in \mathcal{V}_+(\widehat{F}) \cap \mathcal{V}_+(\widehat{G})$.

Projective Closure of an Affine Hypersurface

Definition 10.38 Let $\widehat{f}$ be an affine hypersurface of $\mathbb{A}^n(K)$, We set

$$^{x_0}(f)(x_0, x_1, \ldots, x_n) := x_0^d f\left(\frac{x_1}{x_0}, \ldots \frac{x_n}{x_0}\right). \tag{10.28}$$

Since $\deg(f) = d \geq 1$, then $^{x_0}(f)(x_0, x_1, \ldots, x_n)$ is a homogeneous polynomial of degree d in the indeterminates $x_0, x_1, \ldots, x_n$. The following relation is immediate:

$$^{x_0}(fg) = {}^{x_0}(f)^{x_0}(g), \qquad \forall\, f, g \in K[x_1, \ldots, x_n], \quad \deg(f) \geq 1, \deg(g) \geq 1. \tag{10.29}$$

The hypersurface $\widehat{^{x_0}(f)}$ is called the *projective closure of* $\widehat{f} \subset \mathbb{A}^n(K)$ with respect to the canonical identification (Definition 9.9)

$$\mathbb{A}^n(K) \cong \mathbb{U}_0 := \{[x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K) \mid x_0 \neq 0\}, \qquad (x_1, \ldots, x_n)$$
$$\longleftrightarrow [1, x_1, \ldots, x_n]. \tag{10.30}$$

We remark that

$$\mathcal{V}_+(\widehat{{}^{x_0}(f)}) \cap \mathbb{U}_0 = \mathcal{V}(\widehat{f}). \tag{10.31}$$

For instance, if $f = x_1 + x_2^2 + \cdots + x_n^2$ then ${}^{x_0}(f) = x_0 x_1 + x_2^2 + \cdots + x_n^2 = 0$.

The *affine part in the chart* $\mathbb{U}_0$ of a projective hypersurface $\widehat{F}$ of $\mathbb{P}^n(K)$ is the affine hypersurface $\widehat{{}_{x_0}(F)}$, where ${}_{x_0}(F)(x_1, \ldots, x_n) = F(1, x_1, \ldots, x_n)$. The affine part $\widehat{{}_{x_0}F}$ has the same degree as $\widehat{F}$ if and only if $x_0 \nmid F$ We have immediately ${}_{x_0}(FG) = {}_{x_0}(F)_{x_0}(G)$, for all homogeneous polynomials F, G of positive degrees. The support of $\widehat{{}_{x_0}F}$ is given by

$$\mathcal{V}(\widehat{{}_{x_0}F}) = \mathbb{U}_0 \cap \mathcal{V}_+(\widehat{F}). \tag{10.32}$$

Furthermore

$$\begin{aligned} {}_{x_0}\left({}^{x_0}(f)\right) &= f; \qquad {}^{x_0}\left({}_{x_0}(F)\right) \\ &= x_0^{-m} F, \quad \text{where } x_0^m \text{ is the highest power of } x_0 \text{ which divides } F. \end{aligned} \tag{10.33}$$

We could obviously define the projective closure and the affine part with respect to the identifications ($i = 0, \ldots, n$)

$$\begin{aligned} \mathbb{A}^n(K) \cong \mathbb{U}_i &:= \{[x_0, x_1, \ldots, x_n] \in \mathbb{P}^n(K) \mid x_i \neq 0\}, \quad (x_0, \ldots, \widehat{x_i}, \ldots, x_n) \\ &\longleftrightarrow [x_0, x_1, \ldots, \underbrace{1}_{\text{place } i}, \ldots, x_n], \end{aligned}$$

or any other immersion of $\mathbb{A}^n(K)$ into $\mathbb{P}^n(K)$ (see Chap. 9).

We shall denote the hyperplane $\mathcal{Y}_0 : x_0 = 0$ also by $\mathcal{Y}_\infty$. The hypersurface $\widehat{{}^{x_0}(f)} \cap \mathcal{Y}_\infty$ of $\mathcal{Y}_\infty \cong \mathbb{P}^{n-1}(K)$ is called the *hypersurface at infinity* of $\widehat{f}$ and denoted by $\widehat{f}_\infty$.

Proposition 10.39 *Assume the notation of* Definition 10.38. *Let $\widehat{f}$ be a hypersurface of $\mathbb{A}^n(K)$, $P = (a_1, \ldots, a_n) \in \mathcal{V}(f)$, $\widehat{{}^{x_0}(f)}$ be the projective closure of $\widehat{f}$ and $\overline{P} = [1, a_1, \ldots, a_n]$. Then:*

(i) *$P \in \mathrm{Sing}(\widehat{f})$ if and only if $\overline{P} \in \mathrm{Sing}(\widehat{{}^{x_0}(f)})$.*
(ii) *If $\overline{r}$ is the projective closure of an affine line r* (Definition 9.11), *then*

$$\mathfrak{m}_P(\widehat{f}, r) = m_{\overline{P}}(\widehat{{}^{x_0}(f)}, \overline{r}). \tag{10.34}$$

In particular, r is tangent to $\widehat{f}$ at P if and only if $\overline{r}$ is tangent to $\widehat{{}^{x_0}(f)}$ at $\overline{P}$.
(iii) $m_{\overline{P}}(\widehat{{}^{x_0}(f)}) = \mathfrak{m}_P(\widehat{f})$.

(iv) *The tangent hyperplane to* $\widehat{{}^{x_0}(f)}$ *at* $\overline{P}$ *is the projective closure of the tangent hyperplane to* $\widehat{f}$ *at* P *i.e.*

$$\overline{\mathrm{T}_{\widehat{f},P}} = \mathrm{T}^{\mathrm{p}}_{\widehat{{}^{x_0}(f)},\overline{P}}.$$

Proof

(i) From the equality $f(x_1,\dots,x_n) = {}^{x_0}(f)(1,x_1,\dots,x_n)$ we get

$$\frac{\partial f}{\partial x_i}(a_1,\dots,a_n) = \frac{\partial[{}^{x_0}(f)]}{\partial x_i}(1,a_1,\dots,a_n), \quad i=1,\dots,n. \tag{10.35}$$

Euler's formula (10.4) implies

$$0 = d\cdot {}^{x_0}(f)(1,a_1,\dots,a_n) = \frac{\partial[{}^{x_0}(f)]}{\partial x_0}(1,a_1,\dots,a_n) + \sum_{i=1}^{n}\frac{\partial[{}^{x_0}(f)]}{\partial x_i}(1,a_1,\dots,a_n)a_i. \tag{10.36}$$

By (10.35) and (10.36) we conclude that

$$\frac{\partial f}{\partial x_i}(a_1,\dots,a_n)=0, \quad i=1,\dots,n \iff \frac{\partial[{}^{x_0}(f)]}{\partial x_i}(1,a_1,\dots,a_n)=0, \quad i=0,1,\dots,n.$$

(ii) Let $x_i = l_i t + a_i$, $i=1,\dots,n$, be parametric equations of $r \ni P$. If $P \in V(\widehat{f})$ then $m_P(\widehat{f},r)$ is the order of 0 as a root of $f(l_1t+a_1,\dots,l_nt+a_n)$. The projective closure $\overline{r}$ of r passes through $[1,a_1,\dots,a_n]$ and $[0,l_1,\dots,l_n]$ so that its parametric equations are $x_i = a_i u + l_i v$, $i=0,\dots,n$ (with $a_0=1$ and $l_0=0$). Therefore

$$\begin{aligned}{}^{x_0}(f)(a_0u+l_0v,\dots,a_nu+l_nv) &= u^d f\left(\frac{a_1u+l_1v}{u},\dots,\frac{a_nu+l_nv}{u}\right)\\ &= u^d f\left(a_1+l_1\frac{v}{u},\dots,a_n+l_n\frac{v}{u}\right)\end{aligned}$$

from which we deduce (10.34). (iii) and (iv) easily follow.

□

Proposition 10.40 *Assume the notation of* Definition 10.38.

(i) $P=(a_1,\dots,a_n) \in \mathrm{Sing}(\widehat{{}^{x_0}(F)})$ *if and only if* $\overline{P} = [1,a_1,\dots,a_n] \in \mathrm{Sing}(\widehat{F})$.

(ii) *The tangent hyperplane* $\mathrm{T}_{\widehat{x_0(F)},P}$ *at* P *is the affine part of the projective tangent hyperplane* $\mathrm{T}^{\mathrm{p}}_{\widehat{F},\overline{P}}$ *at* $\overline{P}$.

(iii) *Let* s *be a line through* $\overline{P}$, *then* $m_{\overline{P}}(\widehat{F}, s) = \mathfrak{m}_P(\widehat{x_0(F)}, r)$, *where* $r = s \cap \mathbb{U}_0$. *In particular,* r *is tangent to* $\widehat{x_0(F)}$ *at* P *if and only if* s *is tangent to* $\widehat{F}$ *at* $\overline{P}$.

(iv) $m_{\overline{P}}(\widehat{F}) = \mathfrak{m}_P(\widehat{x_0(F)})$.

Proof

(i) Since ${}_{x_0}(F)(x_1, \ldots, x_n) = F(1, x_1, \ldots, x_n)$ we have

$$\frac{\partial_{x_0}(F)}{\partial x_i}(a_1, \ldots, a_n) = \frac{\partial F}{\partial x_i}(1, a_1, \ldots, a_n), \qquad i = 1, \ldots, n.$$

By Euler's formula

$$0 = dF(1, a_1, \ldots, a_n) = \frac{\partial F}{\partial x_0}(1, a_1, \ldots, a_n) + \sum_{i=1}^{n} \frac{\partial F}{\partial x_i}(1, a_1, \ldots, a_n)a_i,$$

so that

$$\frac{\partial_{x_0}(F)}{\partial x_i}(a_1, \ldots, a_n) = 0, \quad i = 1, \ldots, n \iff \frac{\partial F}{\partial x_i}(1, a_1, \ldots, a_n) = 0,$$
$$i = 0, 1, \ldots, n.$$

(iii) Let $x_i = a_i u + b_i v$, with $a_0 = 1, b_0 = 0, i = 0, 1, \ldots, n$, parametric equations of s, then $x_i = a_i + b_i t$, $(t = \dfrac{v}{u}$, $i = 1, \ldots, n$, are parametric equations of r so that

$${}_{x_0}(F)(a_1 + b_1 t, \ldots, a_n + t b_n) = F(1, a_1 u + b_1 v, \ldots, a_n u + b_n v).$$

Thus we deduce (iii). (ii) and (iv) easily follow.

□

Remark 10.41 Let $\widehat{f}$ and $\widehat{g}$ be two affinely equivalent hypersurfaces of $\mathbb{A}^n(K)$ and ϕ an affinity of $\mathbb{A}^n(K)$ such that $\phi(\widehat{f}) = \widehat{g}$. By Proposition 9.13 we may regard ϕ as the restriction to $\mathbb{U}_0$ of a projectivity Φ of $\mathbb{P}^n(K)$ that fixes $\mathcal{Y}_0$ and such that $\Phi(\widehat{x_0 f}) = \widehat{x_0 g}$ and $\Phi_{|\mathcal{Y}_0}(\widehat{x_0 f} \cap \mathcal{Y}_0) = \widehat{x_0 g} \cap \mathcal{Y}_0$ Therefore, the projective closures and the parts at infinity of affinely equivalent hypersurfaces are projectively equivalent.

Analogously, let $\widehat{F}$ and $\widehat{G}$ be two projectively equivalent hypersurfaces of $\mathbb{P}^n(K)$ and α a projectivity of $\mathbb{P}^n(K)$ such that $\alpha(\widehat{F}) = \widehat{G}$ and $\alpha(\mathcal{Y}_0) = \mathcal{Y}_0$. Suppose that $x_0 \nmid F$ (hence $x_0 \nmid G$). Therefore α is represented by a block matrix of the form

$$A = \begin{pmatrix} 1 & 0 \dots 0 \\ b_1 & \\ \vdots & C \\ b_n & \end{pmatrix}$$

where $b_1, \dots, b_n \in K$ and C is an invertible square matrix of order n. The restriction of α to $\mathbb{U}_0$ is the affinity of $\mathbb{A}^n(K)$ given by $x \to Cx^t + b^t$, $x = (x_1, \dots, x_n)$, $b = (b_1, \dots, b_n)$, and transforms $\widehat{F} \cap \mathbb{U}_0$ into $\widehat{G} \cap \mathbb{U}_0$, while the restriction of α to the hyperplane $\mathcal{Y}_0$ is the projectivity of $\mathcal{Y}_0$ represented by the matrix C that transforms $\widehat{F} \cap \mathcal{Y}_0$ into $\widehat{G} \cap \mathcal{Y}_0$.

10.2 Projective Hyperquadrics

Definition 10.42 A projective hypersurface $\widehat{F}$ of degree 2 of $\mathbb{P}^n(K)$ is called *projective hyperquadric*. If $n = 1, 2$, $\widehat{F}$ is called *projective conic* and if $n = 3$, *projective quadric*. Two hyperquadrics of $\mathbb{P}^n(K)$ are *projectively equivalent* if they are projectively equivalent according to Definition 10.30.

Convention In all sections concerning projective hyperquadrics, K is a field of characteristic distinct from 2.

Let $\widehat{F}$ be a hyperquadric where

$$F(x_0, x_1, \dots, x_n) = \sum_{i,j=0}^{n} c_{ij} x_i x_j, \qquad c_{ij} \in K. \tag{10.37}$$

There exists a unique symmetric matrix

$$C := \begin{pmatrix} c_{00} & c_{01} & \cdots & c_{0n} \\ c_{10} & c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n0} & c_{n1} & \cdots & c_{nn} \end{pmatrix}, \qquad c_{ij} = c_{ji},$$

such that

$$F(x_0, x_1, \dots, x_n) = X^t C X, \qquad X = (x_0, x_1, \dots, x_n)^t. \tag{10.38}$$

Definition 10.43 Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a hyperquadric, where $F = X^t C X$. The integer $\operatorname{rank}(C)$ is called the *rank* of $\widehat{F}$, i.e. $\operatorname{rank}(\widehat{F}) := \operatorname{rank}(C)$. If $\operatorname{rank}(\widehat{F}) = n+1$, $\widehat{F}$ is called *non-degenerate*, otherwise it is called *degenerate*.

Remark 10.44 In a pencil generated by two hyperquadrics $\widehat{F}$ and $\widehat{G}$ of $\mathbb{P}^n(K)$ either there are at most $n+1$ degenerate hyperquadrics or all hyperquadrics of the pencil are degenerate.

Proposition 10.45 *Let $P_1, \ldots, P_m$ be m points of $\mathbb{P}^n(K)$. If $m \leq \dfrac{(n+1)(n+2)}{2} - 1$, there exists at least one hyperquadric $\widehat{F}$ of $\mathbb{P}^n(K)$ passing through $P_1, \ldots, P_m$.*

Proof See the proof of Proposition 5.27. □

Lemma 10.46 *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a hyperquadric*

$$F = X^t C X = \sum_{i,j=0}^{n} c_{ij} x_i x_j, \qquad c_{ij} = c_{ji},$$

and $\alpha = T_A \in \mathrm{PGL}_n(K)$. Then the hyperquadric $\alpha \cdot \widehat{F} = \widehat{G}$ has the equation

$$G = X^t D X = \sum_{i,j=0}^{n} d_{ij} x_i x_j, \qquad D = \left(c_{ij}\right)_{i,j=0,\ldots,n}, \quad D = (A^{-1})^t C A^{-1}.$$

Therefore $F(X, X) = G(AX, AX)$. In particular, projectively equivalent hyperquadrics have the same rank.

Proof Let $A = \left(a_{ij}\right)$ and $A^{-1} = \left(a_{ij}^*\right)$, then by Definition 10.26 we have

$$\begin{aligned} G(x_0, x_1, \ldots, x_n) &= \sum_{i,j=0}^{n} c_{ij} \left(\sum_{p=0}^{n} a_{ip}^* x_p \right) \left(\sum_{q=0}^{n} a_{jq}^* x_q \right) \\ &= \sum_{p,q=0}^{n} \left(\sum_{i,j=0}^{n} c_{ij} a_{ip}^* a_{jq}^* \right) x_p x_q. \end{aligned}$$

Hence $d_{pq} = \sum\limits_{i,j=0}^{n} c_{ij} a_{ip}^* a_{jq}^*$, i.e. $D = (A^{-1})^t C A^{-1}$. The last assertion holds true since (see Definition 1.24) $\operatorname{rank}(MN) = \operatorname{rank}(NM) = \operatorname{rank}(N)$, where M is an invertible $n \times n$-matrix and N is any $n \times n$-matrix. □

Remark 10.47 A conic $\widehat{F}$ of $\mathbb{P}^1(K)$, where $F = a_{00}x_0^2 + 2a_{01}x_0x_1 + a_{11}x_1^2$ is non-degenerate if

$$\begin{vmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{vmatrix} \neq 0$$

so that its geometric locus consists of two distinct points if $-\det C = a_{01}^2 - a_{00}a_{11}$ has a square root in K, otherwise it is empty. If $\det C = 0$, then $\mathcal{V}_+(\widehat{F})$ is a single point.

Lemma 10.48 *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a hyperquadric and $\mathcal{Y}$ be a hyperplane such that $\mathcal{Y} \subset \mathcal{V}_+(\widehat{F})$. If $G \in {}_{\mathrm{h}}K[x_0, \ldots, x_n]_1^*$ is an equation of $\mathcal{Y}$ then $F = G \cdot H$, with $H \in {}_{\mathrm{h}}K[x_0, \ldots, x_n]_1^*$. In particular, if $\widehat{F}$ is irreducible, then $\mathcal{V}_+(\widehat{F})$ contains no hyperplane.*

Proof Let $F = \sum_{i,j=0}^{n} c_{ij}x_ix_j$. Since there exists $\sigma \in \mathrm{PGL}_n(K)$ such that $\sigma(\mathcal{Y}) = \mathcal{Y}_0$, by Corollary 10.29 we may suppose that $\mathcal{Y} = \mathcal{Y}_0$ so that $G = x_0$. Since

$$\mathcal{Y}_0 \subset \mathcal{V}_+(\widehat{F}) \iff \sum_{i,j=1}^{n} c_{ij}x_ix_j = 0, \quad \forall\, [0, x_1, \ldots, x_n] \in \mathcal{Y}_0,$$

the homogeneous polynomial $F_0(x_1, \ldots, x_n) := F(0, x_1, \ldots, x_n) \in {}_{\mathrm{h}}K[x_0, \ldots, x_n]_2^*$ vanishes on the whole $\mathbb{P}^{n-1}(K)$. By Lemma 10.7 (since $\mathrm{char}(K) \neq 2$, then K has at least $3 = 2 + 1$ elements), F_0 is the zero-polynomial i.e. $F = c_{00}x_0^2 + 2\sum_{i=1}^{n} c_{0i}x_0x_i$, so that $x_0 \mid F$. □

Restriction of a Hyperquadric to a Projective Subspace

Assume the notation of Definition 10.42. Let $\mathcal{S} \subset \mathbb{P}^n(K)$ be a subspace of dimension m, with $1 \leq m \leq n-1$, of equations $x_i = \sum_{k=0}^{m} a_{ik}t_k, \quad i = 0, \ldots, n$. Then the restriction $\widehat{F_{|\mathcal{S},\sigma}}$ (according to Definition 10.35) is the hyperquadric given by

$$\begin{aligned} F_{|\mathcal{S},\sigma}(t_0, \ldots, t_m) &= F\left(\sum_{j=0}^{m} a_{0j}t_j, \sum_{j=0}^{m} a_{1j}t_j, \ldots, \sum_{j=0}^{m} a_{nj}t_j\right) \\ &= \sum_{i,j=0}^{n} c_{ij} \sum_{k=0}^{m} a_{ik}t_k \sum_{l=0}^{m} a_{jl}t_l \\ &= \sum_{k,l=0}^{m} \left(\sum_{i,j=0}^{n} c_{kl}a_{ik}a_{jl}\right) t_kt_l, \end{aligned} \tag{10.39}$$

which may be equivalently written

$$\left(\sum_{k=0}^{m} a_{0k}t_k, \ldots, \sum_{k=0}^{m} a_{nk}t_k\right) C \left(\sum_{l=0}^{m} a_{0l}t_l, \ldots, \sum_{l=0}^{m} a_{nl}t_l\right)^t.$$

Singular Points; Tangent Lines and Tangent Hyperplanes to a Hyperquadrics

Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a hyperquadric of $\mathbb{P}^n(K)$, where $F = \sum_{i,j=0}^{n} c_{ij}x_i x_j$ and $C = (c_{ij})_{i,j=0,\ldots,n}$. Then

$$\frac{\partial F}{\partial x_i} = 2\sum_{j=0}^{n} c_{ij}x_j =: 2F_i, \qquad i = 0, 1, \ldots, n.$$

Therefore a point $P = [p_0, p_1, \ldots, p_n] \in \mathcal{V}_+(\widehat{F})$ is singular if and only if

$$F_i(P) = \sum_{j=0}^{n} c_{ij}p_j = 0, \quad \forall\, i \in \{0, 1, \ldots, n\},$$
$$\iff \quad P^t C = 0^t_{K^n}, \quad P^t = (p_0, p_1, \ldots, p_n).$$

If P is not singular, then the tangent hyperplane to $\widehat{F}$ at $P = [p_0, p_1, \ldots, p_n]$ has the equation

$$\sum_{i=0}^{n} \frac{\partial F}{\partial x_i}(P)x_i = 0, \quad \text{i.e.} \quad P^t C X = 0. \tag{10.40}$$

Proposition 10.49 *Let $\widehat{F}$ be a hyperquadric of $\mathbb{P}^n(K)$ of equation $X^t C X = 0$.*

(a) *The singular locus* $\mathrm{Sing}(\widehat{F})$ *is a projective subspace of dimension* $n - \mathrm{rank}(C)$.
(b) $\widehat{F}$ *is non-degenerate if and only if it is non-singular.*
(c) $\widehat{F}$ *is a cone whose vertex is* $\mathrm{Sing}(\widehat{F})$.
(d) *If* $P \in \mathrm{Reg}(\widehat{F})$ *then* $\mathrm{Sing}(\widehat{F}) \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$.

Proof

(a) + (b) Since $\mathrm{Sing}(\widehat{F}) = \{P \in \mathcal{V}_+(\widehat{F}) : P^t C = 0\}$, $\mathrm{Sing}(\widehat{F})$ is a projective subspace of dimension $n - \mathrm{rank}(C)$. We have the equivalences

$$\widehat{F} \quad \text{is non-degenerate} \quad \iff \quad \mathrm{rank}(C) = n + 1 \quad \iff \quad n - \mathrm{rank}(C) = -1$$
$$\iff \quad \mathrm{Sing}(\widehat{F}) = \varnothing.$$

(c) Let $P = [p_0, \ldots\ldots, p_n] \in \mathrm{Reg}(\widehat{F})$, $Z = [z_0, \ldots\ldots, z_n] \in \mathrm{Sing}(\widehat{F})$ and $r = PZ$. A point $Q = uZ + vP$ (with $u, v \in K$, $(u, v) \neq (0, 0)$) of r belongs to $\mathcal{V}_+(\widehat{F})$ if and only if

$$(uZ + vP)^t C(uZ + vP) = 0 \iff u^2 Z^t C Z + uv(Z^t C P + P^t C Z) + v^2 P^t C P = 0,$$

i.e. (since $Z^t C P = P^t C Z$)

$$u^2 Z^t C Z + 2uv Z^t C P + v^2 P^t C P = 0. \tag{10.41}$$

The Eq. (10.41) vanishes identically since $Z^t C Z = Z^t C P = 0$ ($Z \in \mathrm{Sing}(\widehat{F})$) and $P^t C P = 0$ ($P \in \widehat{F}$), so that $r \subset \mathcal{V}_+(\widehat{F})$.

(d) If $Q \in \mathrm{Sing}(\widehat{F})$ then $Q^t C = 0_{K^n}$, i.e. $P^t C Q = 0$ so that $Q \in \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$. □

Proposition 10.50 *Let $\widehat{F}$ be a non-degenerate hyperquadric of $\mathbb{P}^n(K)$ with $n > 1$, $P \in \widehat{F}$ and $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$ the tangent hyperplane to $\widehat{F}$ at P. Then $\widehat{F} \cap \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$ is a hyperquadric cone of $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$ whose only vertex is P.*

Conversely, if a non-degenerate hyperquadric $\widehat{F}$ meets a hyperplane $\mathcal{Y}$ along a degenerate hyperquadric $\widehat{F}_0$ of $\mathcal{Y}$, then $\mathcal{Y}$ is a tangent hyperplane to $\widehat{F}$.

Proof Without loss of generality we can suppose that $P = [1, 0, \ldots, 0]$ and that $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$ is the hyperplane $\mathcal{Y}_n : x_n = 0$. Then the matrix C associated with $\widehat{F}$ has the form

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & \varrho \\ 0 & c_{11} & \cdots & c_{1\,n-1} & c_{1n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & c_{n-1\,1} & \cdots & c_{n-1\,n-1} & c_{n-1\,n} \\ \varrho & c_{n1} & \cdots & c_{n\,n-1} & c_{nn} \end{pmatrix}, \qquad \varrho \neq 0.$$

Therefore the hyperquadric $\widehat{F} \cap \mathcal{Y}_n$ (the restriction of $\widehat{F}$ to $\mathcal{Y}_n$ according to Definition 10.35)) has the equation $\sum_{i,j=0}^{n-1} a_{ij} x_i x_j = 0$ in the coordinates $[x_0, \ldots, x_{n-1}]$ of $\mathcal{Y}_n$. Hence the matrix C_0 associated with $\widehat{F} \cap \mathcal{Y}_n$ is

$$\begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & c_{11} & \cdots & c_{1\,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & c_{n-1\,1} & \cdots & c_{n-1\,n-1} \end{pmatrix},$$

which is singular. Since $\widehat{F}$ is non-degenerate, then $\operatorname{rank}(C_0) = n-1$, otherwise if $\operatorname{rank}(C_0) < n-1$, we should have

$$D = \begin{vmatrix} c_{11} & \cdots & c_{1n-1} \\ \vdots & \ddots & \vdots \\ c_{n-1\,1} & \cdots & c_{n-1\,n-1} \end{vmatrix} = 0$$

and $\det(C) = \varrho^2 D = 0$. Therefore $\operatorname{Sing}(\widehat{F} \cap \mathcal{Y}_n) = \{P\}$ and, by Proposition 10.49, $\widehat{F} \cap \mathcal{Y}_n$ is a cone with vertex P.

Conversely, suppose $\mathcal{Y} = \mathcal{Y}_n$ so that the matrix C_0 associated with $\widehat{F}_0$ is

$$\begin{pmatrix} c_{00} & \cdots & c_{0n-1} \\ \vdots & \ddots & \vdots \\ c_{n-1\,0} & \cdots & c_{n-1\,n-1} \end{pmatrix}$$

and $\det(C_0) = 0$. Then there exists $(\lambda_0, \ldots, \lambda_{n-1}) \in K^n \setminus \{(0, \ldots, 0)\}$, such that

$$\lambda_0 \begin{pmatrix} c_{00} \\ \vdots \\ c_{n-1\,0} \end{pmatrix} + \cdots + \lambda_{n-1} \begin{pmatrix} c_{0n-1} \\ \vdots \\ c_{n-1\,n-1} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Hence

$$\begin{pmatrix} c_{00} & \cdots & c_{0n} \\ \vdots & \ddots & \vdots \\ c_{n\,0} & \cdots & c_{n\,n} \end{pmatrix} \begin{pmatrix} \lambda_0 \\ \vdots \\ \lambda_{n-1} \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \varrho \end{pmatrix},$$

where $\varrho \neq 0$ since $\det(C) \neq 0$. Thus $P = [\lambda_0, \ldots, \lambda_{n-1}, 0] \in \mathcal{V}_+(\widehat{F})$, since $P^t C P = 0$. Therefore the tangent hyperplane $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$, having the equation $P^t C X = 0$, i.e. $x_n = 0$, coincides with $\mathcal{Y}_n$. □

Proposition 10.51 *Let $\widehat{F}$ be a hyperquadric $\mathbb{P}^n(K)$ with $n > 1$, $\operatorname{rank}(\widehat{F}) > 1$ and K an algebraically closed field. Let $\operatorname{Sing}(\widehat{F})$ be the singular locus of $\widehat{F}$ so that $\dim(\operatorname{Sing}(\widehat{F})) = n - \operatorname{rank}(\widehat{F})$. Then there exists a projective subspace S skew with $\operatorname{Sing}(\widehat{F})$ of dimension $\operatorname{rank}(\widehat{F}) - 1$ and a non-degenerate hyperquadric $\widehat{G}$ of S such that $\widehat{F}$ is the cone with vertex $\operatorname{Sing}(\widehat{F})$ which projects $\widehat{G}$, i.e. the locus of the lines which join every point of S to every point of $\widehat{G}$.*

Proof Let $r = \operatorname{rank}(\widehat{F})$. We can suppose that $\operatorname{Sing}(\widehat{F}) = E_r + E_{r+1} + \cdots + E_n$ (where E_i are points of the the standard projective frame) so that the equations of

$\mathrm{Sing}(\widehat{F})$ are $x_0 = x_1 = \cdots = x_r = 0$. The equation $F = \sum_{i,j=0}^{n} c_{ij}x_ix_j$ is rewritten in the form $F = \sum_{i,j=0}^{r} c_{ij}x_ix_j$ with $\det((c_{ih})_{i,h=0,\ldots,r}) \neq 0$. The equation

$$G(x_0, \ldots, x_r) = \sum_{i,j=0}^{r} c_{ij}x_ix_j$$

represents a non-degenerate hyperquadric in the space S whose equations are $x_{r+1} = \cdots x_n = 0$ (which is skew with $\mathrm{Sing}(\widehat{F})$).

Let $P = [0, \ldots, 0, \alpha_{r+1}, \ldots, \alpha_n] \in \mathrm{Sing}(\widehat{F})$ and $Q = [y_0, \ldots, y_r, 0, \ldots, 0] \in \mathcal{V}_+(\widehat{G})$. The line joining P with Q is the locus of the points

$$[\mu y_0, \ldots, \mu y_r, \lambda\alpha_{r+1}, \ldots, \lambda\alpha_n], \qquad (\lambda, \mu \in \mathbb{C}, (\lambda, \mu) \neq (0, 0)), \tag{10.42}$$

which is contained in $\mathcal{V}_+(\widehat{F})$. Conversely, every point of $\widehat{F}$ lies on a line of the form (10.42) by Proposition 10.49. □

Remark 10.52 If $\mathrm{rank}(\widehat{F}) = 1$, then $\mathrm{Sing}(\widehat{F}) = \mathcal{V}_+(\widehat{F})$; if $\mathrm{rank}(\widehat{F}) = n + 1$, then $\mathrm{Sing}(\widehat{F}) = \varnothing$.

10.2.1 Reduced Equations of Projective Hyperquadrics

Theorem 10.53 *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a hyperquadric $\mathbb{P}^n(K)$ with $\mathrm{rank}(\widehat{F}) = r + 1$. Then $\widehat{F}$ is projectively equivalent to the hyperquadric $\widehat{\Gamma_{d_{00},\ldots,d_{rr}}}$ given by*

$$\Gamma_{d_{00},\ldots,d_{rr}}(x_0, \ldots, x_n) = d_{00}x_0^2 + d_{11}x_1^2 + \cdots + d_{rr}x_r^2, \quad 0 \leq r \leq n,$$
$$d_{ii} \neq 0, \quad i = 0, 1, \ldots, r. \tag{10.43}$$

(i) *If K is algebraically closed, then $\widehat{F}$ is projectively equivalent to the hyperquadric $\widehat{\Gamma}_r$ defined by*

$$\Gamma_r(x_0, \ldots, x_n) = x_0^2 + x_1^2 + \cdots + x_r^2 = 0, \qquad 0 \leq r \leq n. \tag{10.44}$$

(ii) *If $K = \mathbb{R}$, $\widehat{F}$ is projectively equivalent to the hyperquadric $\widehat{\Gamma_{p,r}}$, where*

$$\Gamma_{p,r}(x_0, \ldots, x_n) = x_0^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2 = 0,$$
$$p \leq r \leq n, \quad p \geq \frac{r-1}{2}. \tag{10.45}$$

Proof Let $F(x_0,\dots,x_n)=\sum_{i,j=0}^{n} c_{ij}x_ix_j$. Since $C=(c_{ij})$ is a symmetric matrix with $\operatorname{rank}(C)=r+1$ by Corollary 2.15 it is congruent to a diagonal matrix, i.e. there exists a non-singular matrix Λ such that

$$\Lambda^t C\Lambda=\begin{pmatrix} d_{00} & 0 & \dots & 0 & 0 & \dots & 0\\ 0 & d_{11} & \dots & 0 & \vdots & \vdots & \vdots\\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots\\ 0 & 0 & \dots & d_{rr} & 0 & \dots & 0\\ 0 & 0 & \dots & 0 & 0 & \dots & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}, \qquad d_{ii}\neq 0, \quad i=0,1,\dots,r.$$

Let $\alpha=T_{\Lambda^{-1}}\in \mathrm{PGL}_n(K)$ Then by Lemma 10.46 the hyperquadric $\alpha\cdot\widehat{F}$ has the equation $Y^t(\Lambda^tC\Lambda)Y=0$, i.e. (10.43).

If $K=\mathbb{C}$, applying Corollary 2.17 by the above argument we get (10.44).

If $K=\mathbb{R}$, by Corollary 2.20 we obtain $\Gamma_{p,r}(x_0,\dots,x_n)=x_0^2+\cdots+x_p^2-x_{p+1}^2-\cdots-x_r^2=0$. Up to multiplying by -1, we assume that the equation already obtained for $\widehat{F}$ has at least as many positive terms as negative, i.e. $p+1\geq r-p$. After a suitable renumbering of the coordinates (see Exercise 12.1) we obtain (10.45). □

Remark 10.54 By Theorem 10.53 we have two reduced forms for a conic of $\mathbb{P}^1(K)$:

$$d_{00}x_0^2+d_{11}x_1^2=0, \qquad \text{(non-degenerate)}, \qquad x_0^2=0, \qquad \text{(degenerate)}.$$

Complex Hyperquadrics

Theorem 10.55 (Projective Classification of Complex Hyperquadrics) *Any hyperquadric of $\mathbb{P}^n(\mathbb{C})$ has a unique reduced equation. For any two quadrics $\widehat{F}$ and $\widehat{G}$ of $\mathbb{P}^n(\mathbb{C})$, the following three conditions are equivalent:*

(i) *$\widehat{F}$ and $\widehat{G}$ are projectively equivalent.*
(ii) *$\widehat{F}$ and $\widehat{G}$ have equal projective reduced equations.*
(iii) *$\widehat{F}$ and $\widehat{G}$ have the same rank.*

Proof The implications (i) $\Longrightarrow$ (iii) follow from Lemma 10.46, (iii) $\Longrightarrow$ (ii) is obvious (since projective reduced equations depend only on rank), (ii) $\Longrightarrow$ (i) since $\widehat{F}\cong\widehat{\Gamma}_r\cong\widehat{G}$. □

Definition 10.56 Thanks to Theorem 10.55 we can call $x_0^2+x_1^2+\cdots+x_r^2=0$ the *canonical equations* of the complex hyperquadrics.

Examples 10.57 Here is the list of canonical equations of conics of $\mathbb{P}^2(\mathbb{C})$:

(i) $x_0^2 + x_1^2 + x_2^2 = 0$ (non-degenerate conic)
(ii) $x_0^2 + x_1^2 = 0$ (two distinct lines)
(iii) $x_0^2 = 0$ (double line)

Here is the list of canonical equations of quadrics of $\mathbb{P}^3(\mathbb{C})$:

(i) $x_0^2 + x_1^2 + x_2^2 + x_3^2 = 0$ (non-degenerate quadric)
(ii) $x_0^2 + x_1^2 + x_2^2 = 0$ (cone)
(iii) $x_0^2 + x_1^2 = 0$ (two distinct planes)
(iv) $x_0^2 = 0$ (double plane)

Proposition 10.58 *The map* $\varphi\colon \mathcal{H}_2^{\mathrm{p}}(n,\mathbb{C}) \to \mathcal{G}_2(n,\mathbb{C})$, $\varphi(\widehat{F}) = \mathcal{V}_+(F)$ *of* Definition 10.3 *is bjective. In other words, a complex hyperquadric is determined by its geometric locus.*

Proof By Theorem 10.53,(i), there exists $\sigma \in \mathrm{PGL}_n(\mathbb{C})$ such that $\sigma \cdot \widehat{F} = \Gamma_r$, with $0 \leq r \leq n$. By Lemma 10.29 the equality $\mathcal{V}_+(\widehat{F}) = \mathcal{V}_+(\widehat{G})$ implies $\mathcal{V}_+(\sigma \cdot \widehat{F}) = \mathcal{V}_+(\sigma \cdot \widehat{G})$. Hence, without loss of generality, we can suppose $F = x_0^2 + \cdots + x_r^2$, $0 \leq r \leq n$, and $G = \sum_{i,j} c_{ij} x_i x_j$. If $r = 0, 1$ we conclude it immediately by Proposition 10.31.

Let $r \geq 2$. Since $[0, \ldots, 0, x_{r+1}, \ldots, x_n] \in \mathcal{V}_+(\widehat{F})$, $\forall\, (x_{r+1}, \ldots, x_n) \in K^{n-r} \setminus \{(0, \ldots, 0)\}$, then

$$G(0, \ldots, 0, x_{r+1}, \ldots, x_n) = \sum_{i,j \geq r+1} c_{ij} x_i x_j = 0, \qquad \forall\, (x_{r+1}, \ldots, x_n) \times \in K^{n-r} \setminus \{(0, \ldots, 0)\}$$

so that $c_{ij} = 0$ for $i, j > r$.

Let d_{ij}, with $0 \leq i \leq r$, $0 \leq j \leq r$, $i \neq j$, be the line of equations $x_k = 0$, $\forall\, k \neq i, j$. Since $\mathcal{V}_+(F) \cap d_{ij} = \mathcal{V}_+(G) \cap d_{ij}$ the equations

$$x_i^2 + x_j^2 = 0, \qquad c_{ii} x_i^2 + 2c_{ij} x_i x_j + c_{jj} x_i^2 = 0$$

having the same solutions must be proportional (Corollary 10.6) so that $c_{ii} = c_{jj} = a \in \mathbb{C}^*$ and $c_{ij} = 0$. If $i > r$ and $j \leq r$ (or if $i \leq r$ and $j > r$) we have

$$x_j^2 = 0, \qquad c_{ii} x_i^2 + 2c_{ij} x_i x_j + c_{jj} x_i^2 = 0$$

so that $c_{ii} = 0$ and $c_{ij} = 0$. Hence $G = aF$. □

Remark 10.59 A famous general result due to Hilbert (Hilbert's Nullstellensatz) states that Proposition 10.58 holds true for any hypersurface of $\mathbb{P}^n(K)$, where K is an algebraically closed field (see [25], Theorem 1, p. 8).

Real Hyperquadrics

Definition 10.60 Let $\widehat{F}$ be a hyperquadric of $\mathbb{P}^n(\mathbb{R})$. The *index* of $\widehat{F}$ is the maximum of the dimensions of the projective subspaces disjoint with $\mathcal{V}_+(\widehat{F})$. It will be denoted by $\mathbf{i}(\widehat{F})$.

Proposition 10.61 *Let $\widehat{F}$ be a hyperquadric of $\mathbb{P}^n(\mathbb{R})$ and $\alpha \in \mathrm{PGL}_n(\mathbb{R})$ be a linear projective automorphism. Then $\mathbf{i}(\alpha \cdot \widehat{F}) = \mathbf{i}(\widehat{F})$. In particular, any two projectively equivalent hyperquadric of $\mathbb{P}^n(\mathbb{R})$ have the same index.*

Proof Every linear projective automorphism transforms a subspace into a subspace of the same dimension. □

Proposition 10.62 *Let*

$$\Gamma_{p,r}(x_0, \ldots, x_n) = x_0^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2 = 0,$$
$$p \le r \le n, \quad p \ge \frac{r-1}{2}. \tag{10.46}$$

Then

$$\mathbf{i}(\widehat{\Gamma_{p,r}}) = p.$$

Furthermore, $n-p-1$ is the maximum of the dimensions of the projective subspaces contained in $\mathcal{V}_+(\widehat{\Gamma_{p,r}})$.

Proof The subspace $\mathcal{S}$, with equations

$$x_{p+1} = \cdots = x_n = 0,$$

has dimension $n-(n-p) = p$. Obviously $\mathcal{S} \cap \mathcal{V}_+(\widehat{\Gamma_{p,r}}) = \varnothing$, since $x_0^2 + \cdots + x_p^2 = 0$ forces $x_0 = \cdots = x_p = 0$.

Since $p + 1 \ge r - p$, the equations

$$\begin{cases} x_0 & = x_{p+1} \\ \vdots & \vdots \\ x_{r-p-1} & = x_r \\ x_{r-p} & = 0 \\ \vdots & \vdots \\ x_p & = 0, \end{cases}$$

define a subspace $\mathcal{T}$ of dimension $n - (p + 1) = n - p - 1$ contained in $\mathcal{V}_+(\widehat{\Gamma_{p,r}})$. If a subspace $\mathcal{S}'$ has a dimension greater than p then $\mathcal{S}' \cap \mathcal{T} \neq \varnothing$ (by the projective Grassmann formula) so that $\mathcal{S}' \cap \mathcal{V}_+(\widehat{\Gamma_{p,r}}) \neq \varnothing$. Similarly, if a subspace $\mathcal{T}'$ has a dimension $> n - p - 1$, then $\mathcal{S} \cap \mathcal{T}' \neq \varnothing$ so that $\mathcal{T}' \not\subset \mathcal{V}_+(\widehat{\Gamma_{p,r}})$. □

Theorem 10.63 (Projective Classification of Real Hyperquadrics) *Any hyperquadric of $\mathbb{P}^n(\mathbb{R})$ has a unique reduced equation. For any two quadrics $\widehat{F}$ and $\widehat{G}$ of $\mathbb{P}^n(\mathbb{R})$, the following three conditions are equivalent:*

(i) *$\widehat{F}$ and $\widehat{G}$ are projectively equivalent.*
(ii) *$\widehat{F}$ and $\widehat{G}$ have equal projective reduced equations.*
(iii) *$\widehat{F}$ and $\widehat{G}$ have the same rank and the same index.*

Proof The implications (i) $\Longrightarrow$ (iii) follow from Lemma 10.46 and Proposition 10.61; (iii) $\Longrightarrow$ (ii) is obvious (since projective reduced equations depend only on rank and index by Proposition 10.62); finally, (ii) $\Longrightarrow$ (i) since $\widehat{F} \cong \widehat{\Gamma_{p,r}} \cong \widehat{G}$. □

Examples 10.64 Here is the list of the canonical equations of conics of $\mathbb{P}^2(\mathbb{R})$:

(i) $x_0^2 + x_1^2 + x_2^2 = 0$ (non-degenerate conic without real points)
(ii) $x_0^2 + x_1^2 - x_2^2 = 0$ (non-degenerate conic)
(iii) $x_0^2 + x_1^2 = 0$ (two complex-conjugate lines)
(iv) $x_0^2 - x_1^2 = 0$ (two distinct lines)
(v) $x_0^2 = 0$ (double real line)

Here is the list of the canonical equations of quadrics of $\mathbb{P}^3(\mathbb{R})$:

(i) $x_0^2 + x_1^2 + x_2^2 + x_3^2 = 0$ (non-degenerate quadric without real points)
(ii) $x_0^2 + x_1^2 + x_2^2 - x_3^2 = 0$ (non-degenerate quadric)
(iii) $x_0^2 + x_1^2 - x_2^2 - x_3^2 = 0$ (non-degenerate ruled quadric)
(iv) $x_0^2 + x_1^2 + x_2^2 = 0$ (cone without real points)
(v) $x_0^2 + x_1^2 - x_2^2 = 0$ (cone)
(vi) $x_0^2 + x_1^2 = 0$ (two complex-conjugate planes)
(vii) $x_0^2 - x_1^2 = 0$ (two distinct real planes)
(viii) $x_0^2 = 0$ (double plane)

Remark 10.65 By Proposition 10.50 a plane $\mathcal{Y}$ is tangent to a non-degenerate quadric of $\widehat{F} \in \mathcal{H}(3, 2, \mathbb{C})$ if and only if $\mathcal{Y} \cap \mathcal{V}_+(\widehat{F})$ is a pair of distinct complex lines. Analogously, a plane $\mathcal{Y}$ is tangent to a non-degenerate quadric $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(3, \mathbb{R})$ if and only if $\mathcal{Y} \cap \mathcal{V}_+(\widehat{F})$ is a pair of distinct complex-conjugate lines or a pair of distinct real lines. In the former case $\widehat{F}$ does not contain real lines, while in the latter $\widehat{F}$ is a ruled surface (see Definition 10.87).

If $K = \mathbb{R}$ we have only a partial result analogous to that of Proposition 10.58:

Proposition 10.66 *Let $\widehat{F}$, $\widehat{G} \in \mathcal{H}(n, 2, \mathbb{R})$ be two hyperquadrics of $\mathbb{P}^n(\mathbb{R})$ such that $\mathcal{V}_+(\widehat{F}) = \mathcal{V}_+(\widehat{G})$. Suppose that either*

$$\mathbf{i}(\widehat{F}) < \operatorname{rank}(\widehat{F}) - 1 \quad \textit{or} \quad \operatorname{rank}(\widehat{F}) = 1$$

then $\widehat{F} = \widehat{G}$.

Proof By Theorem 10.53 we may suppose that F coincides with one of the following polynomials:

(i) $F = x_0^2$, i.e. $\operatorname{rank}(\widehat{F}) = 1$,

(ii) $F = x_0^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2$, with $\dfrac{r-1}{2} \leq p = \mathbf{i}(\widehat{F}) < r = \operatorname{rank}(\widehat{F}) - 1$.

Let $G = \displaystyle\sum_{i,j=0}^{n} d_{ij} x_i x_j$.

Case (i): $\mathcal{V}_+(\widehat{G}) = \mathcal{Y}_0$, the hyperplane of equation $x_0 = 0$. Then by Lemma 10.48 $x_0 \mid G$ so that

$$\begin{aligned} G(x_0, \ldots, x_n) &= x_0(a_0 x_0 + \cdots + a_n x_n) \\ &:= x_0 H(x_0, \ldots, x_n), \quad \text{and} \quad \mathcal{V}_+(\widehat{G}) = \mathcal{Y}_0 \cup \mathcal{V}_+(\widehat{H}). \end{aligned}$$

Therefore $H \mid x_0$ i.e. $H(x_0, \ldots, x_n) = a x_0$ (with $a \in \mathbb{R}^*$) which implies $G(x_0, \ldots, x_n) = a x_0^2$ and $\widehat{F} = \widehat{G}$.

Case (ii): $\mathcal{V}_+(G) = \mathcal{V}_+(x_0^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2)$. We proceed as in the complex case. Let $\ell_{h,k}$ be the line of equations $x_i = 0$, $i = 0, \ldots, n$, $i \neq h, k$. Since

$$\mathcal{V}_+(x_0^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2) \cap \ell_{i,j} = \mathcal{V}_+(G) \cap \ell_{i,j}, \quad 0 \leq i \leq p, \ p+1 \leq j \leq r$$

the equations

$$x_i^2 - x_j^2 = 0, \qquad \text{and} \qquad d_{ii} x_i^2 + 2 d_{ij} x_i x_j + d_{jj} x_j^2 = 0, \quad 0 \leq i \leq p, \quad j = p+1, \ldots, r,$$

have the same solutions, so that they must be proportional, i.e.

$$\begin{aligned} &d_{ii} = \lambda \neq 0, \quad 0 \leq i \leq p, \qquad d_{jj} = -\lambda \neq 0, \quad j = p+1, \ldots, r, \\ &d_{ij} = 0, \qquad 0 \leq i \leq p, \ j = p+1, \ldots, r. \end{aligned}$$

If $r < n$, $0 \leq i \leq p$ and $r+1 \leq j \leq n$, we obtain the equations

$$x_i^2 = 0 \qquad \text{and} \qquad d_{ii} x_i^2 + 2 d_{ij} x_0 x_j + d_{jj} x_j^2 = 0,$$

which have the common double solution [0, 1] so that $d_{ij} = 0$ for $0 \leq i \leq p$ and $r+1 \leq j \leq n$ and $d_{jj} = 0$ for $r+1 \leq j \leq n$. Therefore we have

$$d_{00} = \cdots = d_{pp} = -d_{p+1,p+1} = \cdots = -d_{rr} =: \lambda \neq 0, \ \ d_{ii} = 0, \ \ r < i \leq n,$$

and

$$d_{ij} = 0, \quad i = 0, \ldots, p,\ j = p+1, \ldots, n. \tag{10.47}$$

Since $(d_{ij})_{i,j}$ is symmetric we have only to prove that $d_{ij} = 0$ if $i \neq j \in \{0, \ldots, p\}$ or if $i \neq j \in \{p+1, \ldots, r\}$.

Now fix i and j such that $i \neq j \in \{p+1, \ldots, r\}$. The intersection of $\mathcal{V}_+(G)$ and $\mathcal{V}_+(x_0^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2)$ with the line

$$\begin{cases} x_0 - x_i = 0 \\ x_k = 0,\ \forall k \in \{0, \ldots, n\} \setminus \{0, i, j\}, \end{cases}$$

are given by the equations

$$-x_j^2 = 0 \quad \text{and} \quad d_{00}x_0^2 + 2d_{0i}x_0x_i + d_{ii}x_i^2 + 2d_{0j}x_0x_j + d_{jj}x_j^2 + 2d_{ij}x_ix_j = 0.$$

By (10.47) we have $d_{00}x_0^2 + d_{ii}x_i^2 = 0$ and $d_{0i} = d_{0j} = 0$, which imply $d_{jj}x_j^2 + 2d_{ij}x_ix_j = 0$. Then $d_{ij} = 0$.

Fix now i and j such that $i \neq j \in \{0, \ldots, p\}$. Proceed as above using the line

$$\begin{cases} x_r - x_i = 0 \\ x_k = 0,\ \forall k \in \{0, \ldots, n\} \setminus \{i, j, r\}, \end{cases}$$

to obtain $d_{ij} = 0$. Therefore $\widehat{F} = \widehat{G}$. □

Remark 10.67 For the hyperquadrics $\widehat{\Gamma_{r,r}}$ with $r \geq 1$ Theorem 10.66 is clearly false. Indeed, let $F(x_0, \ldots, x_n) = c_{00}x_0^2 + \cdots + c_{rr}x_r^2$ and $G(x_0, \ldots, x_n) = d_{00}x_0^2 + \cdots + d_{rr}x_r^2$ where $(c_{00}, \ldots, c_{rr})$ is not proportional to $(d_{00}, \ldots, d_{rr})$, i.e. $\widehat{F} \neq \widehat{G}$, nevertheless

$$\mathcal{V}_+(F) = \mathcal{V}_+(G) = \{[x_0, \ldots, x_n] \in \mathbb{P}^n(\mathbb{R}) : x_0 = \ldots = x_r = 0\},$$

of $\mathcal{V}_+(F) = \mathcal{V}_+(G) = \varnothing$ if $r = n$.

Let $\mathcal{A}_r$ $(r \geq 1)$ be the orbit of $\widehat{\Gamma_{r,r}}$ with respect to the action of $\mathrm{PGL}_n(\mathbb{R})$ on the set $\mathcal{H}_2^{\mathrm{p}}(n, \mathbb{R})$ of all hyperquadrics $\mathbb{P}^n(\mathbb{R})$ and let $\mathcal{A} := \mathcal{A}_1 \cup \cdots \cup \mathcal{A}_n$. Then Theorem 10.66 claims that the restriction $\varphi' \colon \mathcal{H}_2^{\mathrm{p}}(n, \mathbb{R}) \setminus \mathcal{A} \to \mathcal{G}_2(n, \mathbb{R}) \setminus \varphi(\mathcal{A})$ of the map $\varphi \colon \mathcal{H}_2^{\mathrm{p}}(n, \mathbb{R}) \to \mathcal{G}_2(n, \mathbb{R})$ (see Definition 10.3) is bjective.

Remark 10.68 (Affine Part of a Projective Hyperquadric and Projective Closure of an Affine Hyperquadric) It is immediate that the projective closure $\widehat{x_0(f)}$ of a non-degenerate affine hyperquadric $\widehat{f}$ of $\mathbb{A}^n(K)$ is a non-degenerate projective hyperquadric of $\mathbb{P}^n(K)$ since the complete matrix associated with f (Definition 5.22) coincides with the matrix associated with $\widehat{x_0(f)}$ (up to a factor

$k \in K^*$). For that reason the affine part $\widehat{x_0(F)}$ of a non-degenerate projective hyperquadric $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ is a non-degenerate affine hyperquadric of $\mathbb{A}^n(K)$. The same assertions obviously hold for a projective closure and for an affine part with respect to any other hyperplane $x_i = 0$, $i = 1, \ldots, n$. It is obvious that two affine parts of a projective hyperquadric may be not affinely equivalent, e.g. if $K = \mathbb{R}$ and $F = x_0^2 + x_1^2 - x_2^2$, then $\widehat{x_2(F)} : x^2 + y^2 = 1$ (circle) and $\widehat{x_0(F)} : x^2 - y^2 = -1$ (hyperbola).

10.3 Polarity with Respect to a Hyperquadric

Definition 10.69 Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a hyperquadric of $\mathbb{P}^n(K)$ with $n \geq 2$ and $F = X^t C X$. Let $P = [p_0, \ldots, p_n]^t$, $Q = [q_0, \ldots, q_n]^t$ be two points of $\mathbb{P}^n(K)$. We say that P and Q are *conjugate to Q with respect to the hyperquadric $\widehat{F}$* if

$$P^t C Q = 0, \quad \text{i.e.} \quad \sum_{i,j=0}^{n} c_{ij} p_i q_j = 0.$$

In particular $\mathcal{V}_+(\widehat{F})$ is the locus of the *self-conjugate points* with respect to $\widehat{F}$. It is clear that the above definition depends only on $\widehat{F}$ and not on the choice of F.

Proposition 10.70 *A point P is conjugate to each point $Q \in \mathbb{P}^n(K)$ with respect to a hyperquadric $\widehat{F}$ if and only if $P \in \mathrm{Sing}(\widehat{F})$.*

Proof If $P \in \mathrm{Sing}(\widehat{F})$ then $P^t C = 0_{K^n}^t$ and $P^t C Q = 0$. Conversely, if $P^t C Q = 0$ for every $Q \in \mathbb{P}^n(K)$, then $P^t C = 0_{K^n}^t$ so that $P \in \mathrm{Sing}(\widehat{F})$. □

Let $P \in \mathbb{P}^n(K) \setminus \mathrm{Sing}(\widehat{F})$. The set of the points $Q \in \mathbb{P}^n(K)$ conjugate to P with respect $\widehat{F}$ is the hyperplane $P^t C X = 0$, i.e.

$$\sum_{j=0}^{n} u_j x_j = 0, \qquad \text{with} \qquad u_j := \sum_{i=0}^{n} c_{ij} p_i. \tag{10.48}$$

The condition $P \notin \mathrm{Sing}(\widehat{F})$ implies that not all coefficients u_j are zero. We denote the hyperplane (10.48) by $\mathcal{P}_P^{\widehat{F}}$ (or simply $\mathcal{P}_P$, if $\widehat{F}$ is understood), and we call it the *polar hyperplane* of P with respect to the hyperquadric $\widehat{F}$; the point P will be called the *pole* of $\mathcal{P}_P^{\widehat{F}}$. If $P \in \mathrm{Reg}(\widehat{F})$ then

$$\mathcal{P}_P^{\widehat{F}} = \mathrm{T}_{\widehat{F},P}^{\mathrm{p}},$$

i.e. polar and tangent hyperplane at P coincide.

By means of the identification (6.16) we can define the map

$$\pi^{\widehat{F}} : \mathbb{P}^n(K) \setminus \mathrm{Sing}(\widehat{F}) \to \mathbb{P}^n(K)^*, \quad \pi^{\widehat{F}}(P) = \mathcal{P}_P^{\widehat{F}} \quad \text{i.e.} \quad \pi^{\widehat{F}}(P) = P^t C, \tag{10.49}$$

which is called the *polarity* of P with respect to $\widehat{F}$, where $\mathbb{P}^n(K)^*$ is the dual projective space of $\mathbb{P}^n(K)$.

Proposition 10.71 *Let $\sigma \in \mathrm{PGL}_n(K)$ be a general linear projective automorphism and $\widehat{F}$ be a hyperquadric. Then P and Q are conjugate with respect to $\widehat{F}$ if and only if $\sigma(P)$ and $\sigma(Q)$ are conjugate with respect to the hyperquadric $\sigma \cdot \widehat{F}$.*

Proof Let $\sigma = T_A$ with $A \in \mathrm{GL}_{n+1}(K)$, $F = X^t C X$ and $\sigma_A(F) = X^t D X$ (see Lemma 10.46). Then

$$P^t C Q = (AP)^t (A^{-1})^t C A^{-1} AQ = \sigma(P)^t D \sigma(Q).$$

□

Lemma 10.72 *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ with $F = X^t C X$.*

(i) *$Q \in \mathcal{P}_P^{\widehat{F}}$ if and only if $P \in \mathcal{P}_Q^{\widehat{F}}$* (reciprocity).

(ii) *$P \in \mathcal{P}_P^{\widehat{F}}$ if and only if $P \in \mathcal{V}_+(\widehat{F})$.*

(iii) *For every $P \in \mathbb{P}^n(K) \setminus \mathrm{Sing}(\widehat{F})$, the polar hyperplane $\mathcal{P}_P^{\widehat{F}}$ contains the singular locus $\mathrm{Sing}(\widehat{F})$.*

(iv) *Conversely, for every hyperplane $\mathcal{Y}$ of $\mathbb{P}^n(K)$ such that $\mathrm{Sing}(\widehat{F}) \subset \mathcal{Y}$, there exists a point $P \in \mathbb{P}^n(K) \setminus \mathrm{Sing}(\widehat{F})$ such that $\mathcal{Y} = \mathcal{P}_P^{\widehat{F}}$.*

(v) *if $P = [p_0, \ldots, p_n] \notin \mathcal{V}_+(\widehat{F})$ and $Q \in \mathcal{V}_+(\widehat{F}) \cap \mathcal{P}_P^{\widehat{F}}$, then the line PQ is tangent to $\widehat{F}$ at Q.*

Proof (i) and (ii) are immediate. (iii) Since every singular point $Q \in \mathrm{Sing}(\widehat{F})$ is conjugate to all points of $\mathbb{P}^n(K)$, in particular we have $\mathrm{Sing}(\widehat{F}) \subseteq \mathcal{P}_P^{\widehat{F}}$.

(iv) By Theorem 10.53 there exists an automorphism $\sigma \in \mathrm{PGL}_n(K)$ such that $\sigma \cdot \widehat{F} = \widehat{G}$, where $G = \Gamma_{d_{00},\ldots,d_{rr}}$. Since $\mathrm{Sing}(\widehat{G})$ is the subspace $\mathcal{S} : x_0 = \cdots = x_r = 0$, a hyperplane $\mathcal{Y}$ which contains $\mathcal{S}$ has an equation of the form $a_0 x_0 + \cdots + a_r x_r = 0$. Then the required point is

$$Q = \left[\frac{a_0}{d_{00}}, \ldots, \frac{a_r}{d_{rr}}, 0, \ldots, 0\right].$$

By Proposition 10.33 $\mathrm{Sing}(\widehat{F}) = \sigma^{-1}(\mathrm{Sing}(\widehat{G}))$ and any hyperplane containing $\mathrm{Sing}(\widehat{F})$ is of the form $\sigma^{-1}(\mathcal{Y})$, where $\mathcal{Y}$ is a hyperplane containing $\mathcal{S}$. Thus $\mathcal{Y} = \mathcal{P}_Q^{\widehat{G}}$. If we take $P := \sigma^{-1}(Q)$ one has $P \notin \mathrm{Sing}(\widehat{F})$, and $\sigma^{-1}(\mathcal{Y}) = \mathcal{P}_P^{\widehat{F}}$ by Proposition 10.71.

(v) Since $Q = [q_0, \ldots, q_n] \in \mathcal{V}_+(\widehat{F}) \cap \mathcal{P}_P^{\widehat{F}}$ then

$$\sum_{i=0}^{n}\sum_{j=0}^{n} c_{ij} q_j p_i = 0, \quad \text{i.e.} \quad \sum_{i=0}^{n} \frac{\partial F}{\partial x_i}(q_0, \ldots, q_n) p_i = 0,$$

so that the line PQ is tangent to $\widehat{F}$ at Q by Corollary 10.19. □

Remark 10.73 The polar hyperplane of a point not lying on the hyperquadric might not intersect the hyperquadric at all. For instance the polar line of $[0, 0, 1]$ with respect to the real conic $x_0^2 + x_1^2 - x_2^2 = 0$ is the line $x_2 = 0$ which does not meet the conic in any real point.

Theorem 10.74 *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a non-singular hyperquadric. Then the polarity $\pi^{\widehat{F}} : \mathbb{P}^n(K) \to \mathbb{P}^n(K)^*$ is a linear projective isomorphism.*

Proof Let $F = X^t C X$, $C = (c_{ij})$. Since $\mathrm{Sing}(\widehat{F}) = \varnothing$, $\pi^{\widehat{F}}$ is defined on the whole $\mathbb{P}^n(K)$. By Lemma 10.72, (ii) $\pi^{\widehat{F}}$ is surjective. Let $P, Q \in \mathbb{P}^n(K)$ such that $\pi^{\widehat{F}}(P) = \pi^{\widehat{F}}(Q)$ which is equivalent to the existence of $\lambda \in K^*$ such that $\sum_{i=0}^{n} c_{ij} q_i = \lambda \sum_{i=0}^{n} c_{ij} p_i$, $j = 0, 1, \ldots, n$, i.e.

$$\sum_{i=0}^{n} c_{ij}(q_i - \lambda p_i) = 0, \qquad j = 0, 1, \ldots, n. \tag{10.50}$$

Since $\det(C) \neq 0$ the equalities (10.50) imply $q_i - \lambda p_i = 0$, $i = 0, 1, \ldots, n$, i.e. $P = Q$, hence $\pi^{\widehat{F}}$ is bjiective. Furthermore, let $\phi : K^{n+1} \to (K^{n+1})^*$ be the linear isomorphism given by $\phi(X) = X^t C$ with respect to the canonical basis on K^{n+1} and to its dual basis (see Definition 1.97) on $(K^{n+1})^*$. Then

$$\pi^{\widehat{F}} = T_\phi$$

according to Definition 6.40. □

Definition 10.75 If $\widehat{F}$ is non-singular, the coefficients u_j, $j = 0, \ldots, n$ of (10.48) are called the *tangential coordinates of the polar hyperplane* $\mathcal{P}_P^{\widehat{F}}$ with respect to $\widehat{F}$. They are uniquely determined up to a constant $\neq 0$.

Example 10.76 If $F = x_0^2 + x_1^2 + \cdots + x_n^2$, then $\mathcal{P}_P^{\widehat{F}}$ is the hyperplane $p_0 x_0 + p_1 x_1 + \cdots + p_n x_n = 0$. In particular, the tangential coordinates of $\mathcal{P}_P^{\widehat{F}}$ coincide with the projective coordinates $[p_0, p_1, \ldots, p_n]$ of P.

The correspondence point $\longleftrightarrow$ polar hyperplane is generalized by the following:

Definition 10.77 Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a non-singular hyperquadric and let $\mathcal{S}$ be a projective subspace of $\mathbb{P}^n(K)$. The *polar subspace* $\mathcal{S}_{\widehat{F}}$ of $\mathcal{S}$ relative to $\widehat{F}$ is the set of points conjugate to all points of $\mathcal{S}$. If $\mathcal{S} = P_1 + \cdots + P_{m+1}$ is the projective subspace of $\mathbb{P}^n(K)$ generated by the projectively independent points $P_1, \ldots, P_{m+1}$, then

$$\pi^{\widehat{F}}(\mathcal{S}) = \{\mathcal{P}_P^{\widehat{F}} : P \in \mathcal{S}\} = \mathcal{P}_{P_1}^{\widehat{F}} + \cdots + \mathcal{P}_{P_m}^{\widehat{F}}$$

is a projective subspace (of hyperplanes) of dimension m so that

$$\mathcal{S}_{\widehat{F}} = \bigcap_{i=1}^{m+1} \mathcal{P}_{P_i}^{\widehat{F}} \tag{10.51}$$

Therefore $\dim(\mathcal{S}_{\widehat{F}}) = n - m - 1$.

The source of the following proposition is Exercise 182 of [12].

Proposition 10.78 *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(3, K)$ be a non-singular quadric and let r be a line of $\mathbb{P}^3(K)$.*

(a) *The polar subspace $r_{\widehat{F}}$ of r is the line base locus of the pencil of the polar planes $\pi^{\widehat{F}}(P)$, P varying on r. We simply call $r_{\widehat{F}}$* the polar *of r.*
(b) *The polar of $r_{\widehat{F}}$ is r.*
(c) *$r = r_{\widehat{F}}$ if and only if $r \subset \mathcal{V}_+(\widehat{F})$.*
(d) *If $r \neq r_{\widehat{F}}$ and $r \cap r_{\widehat{F}} = Q$, then r and $r_{\widehat{F}}$ are tangent to $\widehat{F}$ at Q.*
(e) *r is tangent to $\widehat{F}$ at a point Q if and only if $r_{\widehat{F}}$ is tangent to $\widehat{F}$ at the same point Q.*
(f) *If r is not tangent to $\widehat{F}$, then there exist two distinct planes passing through r and tangent to $\widehat{F}$ if and only if $r_{\widehat{F}}$ intersects $\mathcal{V}_+(\widehat{F})$ exactly at two distinct points, say Q_1 and Q_2. These planes are tangent to $\widehat{F}$ at Q_1 and Q_2.*

Proof

(a) follows immediately from Theorem 10.74.
(b) is an immediate consequence of Lemma 10.72-(i).
(c) If $r = r_{\widehat{F}}$ we have $r \subset \pi^{\widehat{F}}(P)$ for all $P \in r$, hence $P \in \pi^{\widehat{F}}(P)$ so that $P \in \mathcal{V}_+(\widehat{F})$ and $r \subset \mathcal{V}_+(\widehat{F})$. Conversely, if $r \subset \mathcal{V}_+(\widehat{F})$ we have $r \subset \pi^{\widehat{F}}(P)$ for all $P \in r$, hence $r = r_{\widehat{F}}$.
(d) Let $\mathcal{Y}$ be the plane containing r and $r_{\widehat{F}}$. Since $Q \in r$, then $r_{\widehat{F}} \subset \pi^{\widehat{F}}(Q)$. Since $Q \in r_{\widehat{F}}$, then $r \subset \pi^{\widehat{F}}(Q)$ by (b). Therefore $\mathcal{Y} = \pi^{\widehat{F}}(Q)$. In particular, $Q \in \pi^{\widehat{F}}(Q)$, hence $Q \in \mathcal{V}_+(\widehat{F})$ and $\pi^{\widehat{F}}(Q) = \mathrm{T}_{\widehat{F},Q}^{\mathrm{p}}$. It follows that r and $r_{\widehat{F}}$ are tangent to $\widehat{F}$ at Q.

(e) If $r = r_{\widehat{F}}$, the thesis trivially follows. Suppose $r \neq r_{\widehat{F}}$. If r is tangent to $\widehat{F}$ at a point Q, then $r \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q}$, i.e. $P \in \pi^{\widehat{F}}(Q)$ for all $P \in r$. By reciprocity

$$Q \in \bigcap_{P \in r} \pi^{\widehat{F}}(P) = r_{\widehat{F}}.$$

Hence r and $r_{\widehat{F}}$ are distinct and incident at Q. By (d) also $r_{\widehat{F}}$ is tangent to $\widehat{F}$ at Q. The converse follows from (b) and what we proved.

(f) If $r_{\widehat{F}} \cap \mathcal{V}_+(\widehat{F}) = \{Q_1, Q_2\}$, then $r = \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_1} \cap \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_2}$ by (b). Conversely, if there is a plane $\mathcal{Y}$ passing through r and tangent to $\widehat{F}$ at a point Q, then $Q \in r_{\widehat{F}}$, since $r_{\widehat{F}}$ is the locus of poles of the planes through r. Hence $Q \in r_{\widehat{F}} \cap \mathcal{V}_+(\widehat{F})$. Moreover, $r_{\widehat{F}}$ is not tangent to $\widehat{F}$, hence it intersects $\mathcal{V}_+(\widehat{F})$ exactly at two distinct points.

□

Definition 10.79 Assume the notation and hypothesis of Theorem 10.74 and its proof. The inverse isomorphism $\phi^{-1}\colon (K^{n+1})^* \to (K^{n+1})$, $\phi^{-1}(u) = uC^{-1}u^t$ induces the *dual polarity* (by means of the identification (6.23))

$$(\pi^{\widehat{F}})^*\colon \mathbb{P}^n(K)^* \to \mathbb{P}^n(K),$$

whose associated hyperquadric of $\mathbb{P}^n(K)^*$, denoted by $\widehat{F}^*$ and called the *dual hyperquadric* or *hyperplane-hyperquadric*, has the equation

$$uC^{-1}u^t = 0.$$

We observe that $(\pi^{\widehat{F}})^*(\mathcal{P}^{\widehat{F}}_P) = P$. Therefore $(\pi^{\widehat{F}})^*(\mathcal{Y})$ is the *pole* of the hyperplane $\mathcal{Y}$ of $\mathbb{P}^n(K)$. The dual hyperquadric $\widehat{F}^*$ is the locus of polar hyperplanes at the points of $\widehat{F}$, i.e. the locus of tangent hyperplanes to $\widehat{F}$. We say that two hyperplanes $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are *conjugate* with respect to $\widehat{F}$ if $(\pi^{\widehat{F}})^*(\mathcal{Y}_1) \in \mathcal{Y}_2$ ($\Longleftrightarrow (\pi^{\widehat{F}})^*(\mathcal{Y}_2) \in \mathcal{Y}_1$).

A projectivity $\psi\colon \mathbb{P}^n(K) \to \mathbb{P}^n(K)$ (associated with a non-degenerate matrix M with respect to a coordinate system of $\mathbb{P}^n(K)$) preserves $\widehat{F}$ if and only if $CM = (M^t)^{-1}C$. The projectivity ψ induces a dual projectivity $\psi^*\colon \mathbb{P}^n(K)^* \to \mathbb{P}^n(K)^*$ whose associated matrix with respect to the dual coordinate system of $\mathbb{P}^n(K)^*$ is $(M^t)^{-1}$. Since

$$CM = (M^t)^{-1}C \iff M = C^{-1}(M^t)^{-1}C \iff MC^{-1} = C^{-1}(M^t)^{-1}$$

ψ preserves $\widehat{F}$ if and only if ψ^* preserves $\widehat{F}^*$. By abuse of language we say that a projectivity which preserves a hyperquadric $\widehat{F}$ maps also the hyperplane-hyperquadric $\widehat{F}^*$ onto itself.

We also can define the polar subspace $\mathcal{S}_{\widehat{F}^*}$ of a subspace $\mathcal{S}$ of $\mathbb{P}^n(K)^*$ as being the set of all hyperplanes conjugate to all hyperplanes of $\mathcal{S}$, i.e.

$$\mathcal{S}_{\widehat{F}^*} = \Phi_{n-1}((\pi^{\widehat{F}})^*(\mathcal{S})),$$

the linear system with base locus $(\pi^{\widehat{F}})^*(\mathcal{S})$. It is immediately seen that $(\mathcal{S}_{\widehat{F}})_{\widehat{F}^*} = \mathcal{S}$ for every subspace $\mathcal{S}$ of $\mathbb{P}^n(K)$, indeed $\dim((\mathcal{S}_{\widehat{F}})_{\widehat{F}^*}) = \dim(\mathcal{S})$ and every point of $\mathcal{S}$ is conjugate to all points of $\mathcal{S}_{\widehat{F}}$.

A Special Case of a Bertini's Theorem

Proposition 10.80 *Assume the notation and hypothesis of* Definition 10.35. *Let* $\mathcal{S}$ *be a subspace of dimension* > 0 *and* $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ *be a hyperquadric, then:*

(a) $\mathcal{S} \subset \mathcal{V}_+(\widehat{F})$ *if and only if any pair of points* $P, Q \in \mathcal{S}$ *are conjugate with respect to* $\widehat{F}$. *In particular, if* $\mathcal{S}$ *contains a non-singular point* P *of* $\widehat{F}$, *then* $\mathcal{S} \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$.

(b) *If* $\mathcal{S} \not\subset \mathcal{V}_+(\widehat{F})$, $P, Q \in \mathcal{S}$ *are conjugate with respect to* $\widehat{F}$ *if and only if they are conjugate with respect to* $\widehat{F_{|\mathcal{S},\sigma}}$, *the restriction of* $\widehat{F}$ *to* $\mathcal{S}$.

Proof

(a) For any pair of points $P, Q \in \mathcal{S}$ the line $PQ \subset \mathcal{S}$, therefore if $\mathcal{S} \subset \mathcal{V}_+(\widehat{F})$, then $\lambda P + \mu Q \in \mathcal{V}_+(\widehat{F})$ for all $(\lambda, \mu) \in K^2 \setminus \{(0,0)\}$, i.e. the equation

$$\lambda^2 P^t C P + 2\lambda\mu P^t C Q + \mu^2 Q^t C A = 0$$

vanishes identically; this implies $P^t C Q = 0$, i.e. P and Q are conjugate with respect to $\widehat{F}$. Conversely, if any pair of points $P, Q \in \mathcal{S}$ are conjugate with respect to $\widehat{F}$, then in particular $P^t C P = Q^t C Q = 0$ so that $P, Q \in \mathcal{V}_+(\widehat{F})$, i.e. $\mathcal{S} \subset \mathcal{V}_+(\widehat{F})$.

(b) Let $P = \sigma([p_0, \ldots, p_m])$ and $Q = \sigma([q_0, \ldots, q_m])$, i.e.

$$x_k^P = \sum_{j=0}^{m} a_{kj} p_j, \quad k = 0, \ldots, n, \qquad \text{and} \quad x_l^Q = \sum_{h=0}^{m} a_{lh} q_h, \quad l = 0, \ldots, n.$$

The points P and Q are conjugate with respect to $\widehat{F}$ if and only if

$$\sum_{k,l=0}^{n} c_{kl} x_k^P x_l^Q = 0, \tag{10.52}$$

while P and Q are conjugate with respect to $\widehat{F_{|\mathcal{S},\sigma}}$ if and only if

$$\sum_{j,k=0}^{m} \left(\sum_{k,l=0}^{n} c_{kl} a_{kj} a_{lh} \right) p_j q_h = 0. \tag{10.53}$$

The equivalence between (10.52) and (10.53) follows from the equalities (10.39) which define $\widehat{F_{|\mathcal{S},\sigma}}$. □

Corollary 10.81 *Let $P \in \mathrm{Reg}(\widehat{F})$ be a non-singular point of an irreducible hyperquadric $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ and $\mathcal{Y}$ be a hyperplane of $\mathbb{P}^n(K)$ such that $P \in \mathcal{Y}$. Then $\mathcal{Y} = \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$ if and only if P is singular for the restriction of $\widehat{F}$ to $\mathcal{Y}$.*

Proof Since $\widehat{F}$ is irreducible, by Lemma 10.48 $\mathcal{V}_+(\widehat{F})$ does not contain any hyperplane of $\mathbb{P}^n(K)$ passing through P. Let Q any point of $\mathcal{Y}$. If $\mathcal{Y} = \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$, then Q is conjugate to P with respect to $\widehat{F}$, hence by Proposition 10.80, Q is conjugate to P with respect to the restriction of $\widehat{F}$ to $\mathcal{Y}$. Therefore P is singular for the restriction of $\widehat{F}$ to $\mathcal{Y}$ by Proposition 10.70. Conversely, if P is singular for the restriction of $\widehat{F}$ to $\mathcal{Y}$, then P is conjugate to any point Q of $\mathcal{Y}$ with respect to the restriction of $\widehat{F}$ to $\mathcal{Y}$. By Proposition 10.80 P is conjugate to any point Q of $\mathcal{Y}$ with respect to $\widehat{F}$ so that $\mathcal{Y} \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$. □

Remark 10.82 By Proposition 10.49 and Corollary 10.81 the restriction of an irreducible hyperquadric $\widehat{F}$ to a hyperplane $\mathcal{Y}$ which does not meet the vertex of $\widehat{F}$ is non-degenerate.

Corollary 10.83 (Bertini) *Let $\widehat{F} \in \mathcal{H}_2^{\mathrm{p}}(n, K)$ be a non-singular hyperquadric of $\mathbb{P}^n(K)$ with $n \geq 3$. Then there exists a hyperplane $\mathcal{Y}$ of $\mathbb{P}^n(K)$ such that the restriction of $\widehat{F}$ to $\mathcal{Y}$ is a non-singular hyperquadric.*

Proof By Lemma 10.7 there exists a point $P \notin \mathcal{V}_+(\widehat{F})$ so that, by Theorem 10.74, $\pi^{\widehat{F}}(P) \neq \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q}$ for all $Q \in \mathcal{V}_+(\widehat{F})$. Take $\mathcal{Y} = \pi^{\widehat{F}}(P)$. By Corollary 10.81, for every $Q \in \mathcal{V}_+(\widehat{F}) \cap \mathcal{Y}$ is non-singular for the restriction of $\widehat{F}$ to $\mathcal{Y}$, i.e. the restriction of $\widehat{F}$ to $\mathcal{Y}$ is non-singular. □

Definition 10.84 We say that a projective basis $P_0, \ldots, P_n$ of $\mathbb{P}^n(K)$ is a *self-polar* $(n+1)$*-hedron* for a hyperquadric $\widehat{F}$ if

$$\pi^{\widehat{F}}(P_i) = P_0 + \cdots + P_{i-1} + P_{i+1} + \cdots P_n, \qquad i = 0, \ldots, n.$$

If $n = 2$ we speak of a *self-polar triangle* and if $n = 3$ of a *self-polar tetrahedron*.

It is immediately seen that taking a self-polar $(n+1)$-hedron as vertices of a projective frame, $\widehat{F}$ assumes the reduced form of Theorem 10.53 up to a projective isomorphism.

Proposition 10.85 *For any hyperquadric $\widehat{F}$ of $\mathbb{P}^n(K)$ there is a self-polar $(n+1)$-hedron for $\widehat{F}$.*

Proof By Lemma 10.7 there exists a point $P_0 \notin \mathcal{V}_+(\widehat{F})$. Let $\mathcal{Y} = \mathcal{P}^{\widehat{F}}_{P_0}$ its polar hyperplane; since $P_0 \notin \mathcal{V}_+(\widehat{F})$, $P_0 \notin \mathcal{Y}$. If $n = 1$ then $\mathcal{Y} = \{P_1\}$, where $P_1 \neq P_0$ and P_0 and P_1 are conjugate. Therefore $\{P_0, P_1\}$ is a self-polar 2-hedron for $\widehat{F}$.

If $n > 1$, then, using induction, we assume the claim to be true for any hyperquadric of $\mathbb{P}^{n-1}(K)$. If $\mathcal{Y} \subset \mathcal{V}_+(\widehat{F})$, we take any independent points $P_1, \ldots, P_n \in \mathcal{Y}$. Otherwise, $\widehat{F} \cap \mathcal{Y}$ is a hyperquadric of $\mathcal{Y}$ and by the induction hypothesis we have a self-polar n-hedron $P_1, \ldots, P_n$ for $\widehat{F} \cap \mathcal{Y}$. Since $P_0 \notin \mathcal{Y}$, in both cases $P_0, \ldots, P_n$ are independent points and we take $P_0, \ldots, P_n$ as a self-polar $(n+1)$-hedron for $\widehat{F}$. Also in both cases P_0 is conjugate to all P_i , $i = 1, \ldots, n$ because they have been taken in $\mathcal{Y}$. In the case $\mathcal{Y} \subset \mathcal{V}_+(\widehat{F})$ any two points P_i, P_j, $i, j = 1, \ldots, n$, are conjugate with respect to $\widehat{F}$. If $\mathcal{Y} \not\subset \mathcal{V}_+(\widehat{F})$ by the way they have been chosen, any P_i, P_j, $i \neq j$, $i, j = 1, \ldots, n$, are conjugate with respect to $\widehat{F} \cap \mathcal{Y}$ and so they are also conjugate with respect to $\widehat{F}$ by Proposition 10.80. Thus in both cases $P_0, \ldots, P_n$ is a self-polar $(n+1)$-hedron as wanted. □

Non-degenerate Ruled Quadric of $\mathbb{P}^3(\mathbb{C})$ and $\mathbb{P}^3(\mathbb{R})$
First we point out the following elementary fact.

Remark 10.86 Let $\widehat{F}$ be a non-degenerate hyperquadric of $\mathbb{P}^n(K)$ and $\mathcal{S}_h \subset \mathbb{P}^n(K)$ be a projective subspace of dimension h. If $\mathcal{S}_h \subset \mathcal{V}_+(\widehat{F})$, then every point of $\mathcal{S}_h$ belongs to its polar hyperplane so that $\mathcal{S}_h$ is contained in its polar subspace $\mathcal{S}_{n-h-1}$. Hence $h \leq n - h - 1$, i.e.

$$h \leq \left[\frac{n-1}{2}\right],$$

where $[x]$ is the integer part of the real number x. Nevertheless, if K is algebraically closed, then there exist infinitely many projective subspaces of maximal dimension $\left[\frac{n-1}{2}\right]$ (see [3]).

Definition 10.87 By Remark 10.65, through any point P of a quadric $\widehat{F}$ of $\mathbb{P}^3(\mathbb{C})$ there pass two distinct lines a and b which lie on $\widehat{F}$ and on the tangent plane $\mathrm{T}^{\mathrm{p}}_{\widehat{F},P}$. Let $\Phi_{\lambda,\mu}(a)$ be the pencil of planes passing through a and let $\sigma_{\lambda,\mu}(a)$ be a plane of $\Phi_{\lambda,\mu}(a)$, then $\sigma_{\lambda,\mu}(a) \cap \mathcal{V}_+(\widehat{F})$ is a degenerate conic $\sigma_{\lambda,\mu}(a) \cap \mathcal{V}_+(\widehat{F})$ consisting of a and another line $r_{a,\lambda,\mu} \neq a$. The family $R_a := \{r_{a,\lambda,\mu}\}$ is called a *ruling* of $\widehat{F}$. A second ruling R_b is obtained in the same manner.

Let $\widehat{F}$ be a non-degenerate quadric of $\mathbb{P}^3(\mathbb{R})$. Suppose that there exists a point P such that

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},P} \cap \mathcal{V}_+(\widehat{F}) = r \cup s,$$

where r and s are distinct real lines, then if Q is any other point of $\widehat{F}$ we have

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},Q} \cap \mathcal{V}_+(\widehat{F}) = r' \cup s',$$

where r', s' are distinct real lines. Indeed, by contradictory assumption, if for a point Q the lines r', s' are complex conjugate lines, then the plane $r + Q$ should

meet $\mathcal{V}_+(\widehat{F})$ in r and in a real line t passing through Q so that $t \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q}$ which contradicts our hypothesis.

Every non-degenerate quadric of $\mathbb{P}^3(\mathbb{R})$ is projectively isomorphic to $\widehat{\Gamma_{3,3}}$, $\widehat{\Gamma_{2,3}}$ or to $\widehat{\Gamma_{1,3}}$. It is easily seen that $\widehat{\Gamma_{3,3}}$, $\widehat{\Gamma_{2,3}}$ do not contain real lines (for $\widehat{\Gamma_{2,3}}$ take the point $P_0 = [1, 0, 0, -1]$). The quadric $\widehat{\Gamma_{1,3}}$ is ruled quadric, indeed the tangent plane at $P_0 = [1, 0, 1, 0] \in \mathcal{V}_+(\widehat{F})$ has the equation $x_0 - x_2 = 0$ and

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},P_0} \cap \mathcal{V}_+(\widehat{F}) = r \cup s,$$

where $r : x_0 - x_2 = x_1 - x_3 = 0$ and $s : x_0 - x_2 = x_1 + x_3 = 0$. From the equation $x_0^2 - x_2^2 = x_3^2 - x_1^2$ we deduce immediately that all the lines of $\widehat{\Gamma_{1,3}}$ are

$$\{r_\lambda\} : \begin{cases} x_0 - x_2 = \lambda(x_3 + x_1) \\ \lambda(x_0 + x_2) = x_3 - x_1 \end{cases}$$

$$\{s_\mu\} : \begin{cases} x_0 - x_2 = \mu(x_3 - x_1) \\ \mu(x_0 + x_2) = x_3 + x_1 \end{cases} \qquad \begin{cases} x_0 + x_2 = 0 \\ x_3 - x_1 = 0 \end{cases} \qquad \begin{cases} x_0 + x_2 = 0 \\ x_3 + x_1 = 0. \end{cases}$$

Proposition 10.88 *Under the above notation and hypothesis we have*

(i) *For every point $Q \in \mathcal{V}_+(\widehat{F})$ there exists one and only one pair of lines $r_a \in R_a$ and $r_b \in R_b$ such that $Q = r_a \cap r_b$ and*

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},Q} \cap \mathcal{V}_+(\widehat{F}) = r_a \cup r_b.$$

(ii) *Two lines r_1 and r_2 belonging to the same ruling are skew.*
(iii) *Every line of a ruling meets every line of the other ruling.*

Proof

(i) If $Q \notin a \cup b$, then $r_a = (a+Q) \cap \mathcal{V}_+(\widehat{F})$ and $r_b = (b+Q) \cap \mathcal{V}_+(\widehat{F})$; moreover $r_a \neq r_b$, otherwise $P \in r_a \cap r_b \cap a \cap b$. If $Q \in a$ and $Q \notin b$, then

$$r_a = \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q} \cap \mathcal{V}_+(\widehat{F}),$$

which is a line of the pencil $\Phi_{\lambda,\mu}(a)$ (since $a \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q}$) and $r_b = a$; if $Q = a \cap b$, then $r_a = b$ and $r_b = a$.

(ii) The lines r_1 and r_2 lie on two different planes of $\Phi_{\lambda,\mu}(a)$ so that if $r_1 \cap r_2 = Q$, we have $Q \in a$. Hence $\widehat{F}$ would be degenerate because three distinct lines lying on $\mathcal{V}_+(\widehat{F})$ pass through $Q \in \mathcal{V}_+(\widehat{F})$.

(iii) Any line r_a meets any plane $\sigma_{\lambda,\mu}(b) \in \Phi_{\lambda,\mu}(b)$ at a point $Q \in \mathcal{V}_+(\widehat{F})$ which must lie on r_b ($Q \notin b$ since every r_a is skew with b which belongs to R_a). □

Involutions Generated by a Polarity

Definition 10.89 Let $\widehat{F}$ be a non-degenerate hyperquadric of $\mathbb{P}^n(K)$ with $n \geq 2$ and $\pi^{\widehat{F}}$ its polarity. Let r be a line which is not tangent to $\widehat{F}$. The *involution induced by* $\pi^{\widehat{F}}$ *on* r is defined as being the map

$$\varphi_r : r \to r, \qquad \varphi_r(P) = \mathcal{P}_P^{\widehat{F}} \cap r, \quad \forall\ P \in r, \tag{10.54}$$

It is clear that φ_r is an involution according to Definition 9.32. We may extend the above definition to any form of first kind (Definition 6.65). If $\mathcal{S}$ is a subspace of dimension $n-2$ not tangent to $\widehat{F}$ and $\Phi_{n-1}(\mathcal{S})$ is the pencil of the hyperplanes passing through $\mathcal{S}$ (see Definition 6.65) we may define the *involution induced by* $\pi^{\widehat{F}}$ *on* $\Phi_{n-1}(\mathcal{S})$

$$\Phi_{\mathcal{S}} : \Phi_{n-1}(\mathcal{S}) \to \Phi_{n-1}(\mathcal{S}), \qquad \Phi_{\mathcal{S}}(\mathcal{Y}) = \mathcal{S} + (\pi^{\widehat{F}})^*(\mathcal{Y}), \quad \forall\ \mathcal{Y} \in \Phi_{n-1}(\mathcal{S}). \tag{10.55}$$

Proposition 10.90 *The maps* (10.54) *and* (10.55) *are involutions according to* Exercise 9.32. *Furthermore:*

(a) $r \cap \mathcal{V}_+(\widehat{F}) = \{P_1, P_2\}$, *where* P_1 *and* P_2 *are the (distinct) fixed points of* φ_r *(for instance if* K *is algebraically closed) or* $r \cap \mathcal{V}_+(\widehat{F}) = \varnothing$.

(b) *Two distinct points* P *and* Q *of* $\mathbb{P}^n(K)$ *such that the line* $r = PQ$ *meets* $\mathcal{V}_+(\widehat{F})$ *only at two distinct points* P_1 *and* P_2 *are conjugate if and only if*

$$\rho(P_1, P_2, P, Q) = -1. \tag{10.56}$$

(c) *There exist exactly two hyperplanes passing through* $\mathcal{S}$ *which are tangent to* $\widehat{F}$ *and represent the fixed points of* $\Phi_{\mathcal{S}}$ *(for instance if* K *is algebraically closed) or none.*

(d) *If* $\Phi_{\mathcal{S}}$ *has the hyperplanes* $\mathcal{Y}_1$ *and* $\mathcal{Y}_2$ *as fixed points, then two distinct hyperplanes* $\mathcal{Z}_1$ *and* $\mathcal{Z}_2$ *are conjugate if and only if*

$$\rho(\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Z}_1, \mathcal{Z}_2) = -1. \tag{10.57}$$

Proof First we observe that r is not contained in $\mathcal{P}_P^{\widehat{F}}$ for all $P \in r$, otherwise $P \in \mathcal{V}_+(\widehat{F})$ (being self-conjugate) and r would be tangent to $\widehat{F}$ at P. Therefore φ_r is well-defined. Since every hyperplane $\mathcal{Y} \in \Phi_{n-1}(\mathcal{S})$ has its pole $(\pi^{\widehat{F}})^*(\mathcal{Y})$ which does not belong to $\mathcal{S}$ (otherwise $\mathcal{S}$ would be tangent to $\widehat{F}$ at $(\pi^{\widehat{F}})^*(\mathcal{Y})$), also the map $\Phi_{\mathcal{S}}$ is well-defined.

We may suppose that r has equations $x_2 = \cdots = x_n = 0$ so that $[x_0, x_1]$ are projective coordinates on r. Let $F = X^t C X$, where $C = (c_{ij})$. Since r is not tangent to $\widehat{F}$, then $c_{01}^2 - c_{00}c_{11} \neq 0$. The equation of $\mathcal{P}_P^{\widehat{F}}$ at $P = [p_0, p_1, 0, \ldots, 0]$ is

$$(c_{00}p_0 + c_{10}p_1)x_0 + (c_{01}p_0 + c_{11}p_1)x_1 + \cdots + (c_{0n}p_0 + c_{1n}p_1)x_n = 0$$

so that

$$\mathcal{P}_P^{\widehat{F}} \cap r = \{[c_{01}p_0 + c_{11}p_1, -(c_{00}p_0 + c_{10}p_1)]\} \quad \text{i.e}$$

$$\varphi_r([p_0, p_1]) = \begin{pmatrix} c_{01} & c_{11} \\ -c_{00} & -c_{10} \end{pmatrix} \begin{pmatrix} p_0 \\ p_1 \end{pmatrix}.$$

Hence φ_r is an involution. Similarly, we may assume that $\mathcal{S}$ has equations $x_0 = x_1 = 0$ so that any hyperplane $\mathcal{Y} \in \Phi_{n-1}(\mathcal{S})$ has the equation $\lambda x_0 + \mu x_1 = 0$ and $[\lambda, \mu]$ are projective coordinates on $\Phi_{n-1}(\mathcal{S})$. Let $C^{-1} = (d_{ij})$. Since $\mathcal{S}$ is not tangent to $\widehat{F}$ the poles of any $\mathcal{Y} \in \Phi_{n-1}(\mathcal{S})$ do not lie on $\mathcal{S}$. Since

$$(\pi^{\widehat{F}})^*(\mathcal{Y}) = [\lambda, \mu, 0, \ldots, 0]\, C^{-1} = [\lambda d_{00} + \mu d_{10}, \lambda d_{01} + \mu d_{11}, \ldots, \lambda d_{0n} + \mu d_{1n}], \tag{10.58}$$

the condition $(\pi^{\widehat{F}})^*(\mathcal{Y}) \in \mathcal{Y}$ is equivalent to the equation

$$d_{00}\lambda^2 + 2d_{01}\lambda\mu + d_{11}\mu^2 = 0 \tag{10.59}$$

which does not vanish identically (if $d_{00} = d_{01} = d_{11} = 0$, then $(\pi^{\widehat{F}})^*(\mathcal{Y}) \in \mathcal{S}$). If $d_{01}^2 - d_{00}d_{11} = 0$ then $d_{00} \neq 0$ (otherwise $d_{01} = d_{11} = 0$), hence $\lambda = -\frac{d_{01}}{d_{00}}$ and $\mu = 1$. Therefore from (10.58) we get

$$(\pi^{\widehat{F}})^*(\mathcal{Y}) = [0, 0, \lambda d_{02} + \mu d_{12}, \ldots, \lambda d_{0n} + \mu d_{1n}] \in \mathcal{S},$$

so that we have $d_{01}^2 - d_{00}d_{11} \neq 0$ and the Eq. (10.59) has two distinct roots. The hyperplane $\mathcal{S} + (\pi^{\widehat{F}})^*(\mathcal{Y})$ has the equation

$$\begin{vmatrix} x_0 & x_1 & x_2 & x_3 \cdots & x_n \\ \lambda d_{00} + \mu d_{10} & \lambda d_{01} + \mu d_{11} & \lambda d_{02} + \mu d_{12} & \cdots\cdots & \lambda d_{0n} + \mu d_{1n} \\ 0 & 0 & 1 & 0 \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \ddots & \vdots \\ 0 & 0 & 0 & \cdots\cdots & 1 \end{vmatrix} = 0,$$

i.e. $(\lambda d_{01} + \mu d_{11})x_0 - (\lambda d_{00} + \mu d_{10})x_1 = 0$. Therefore we get the relation

$$\Phi_{\mathcal{S}}([\lambda, \mu]) = \begin{pmatrix} d_{01} & d_{11} \\ -d_{00} & -d_{01} \end{pmatrix} \begin{pmatrix} \lambda \\ \mu \end{pmatrix}$$

that shows that $\Phi_{\mathcal{S}}$ is an involution. The assertions (a), (b), (c) and (d) easily follow from Exercise 9.32. □

Remark 10.91 If in Definition 10.89 we take $n = 2$, then $\mathcal{S}$ is a point P which is not self-conjugate, i.e. $P \notin \mathcal{V}_+(\widehat{F})$.

Remark 10.92 We can associate to a non-degenerate conic $F : a_{00}x_0^2 + 2a_{01}x_0x_1 + a_{11}x_1^2$ of $\mathbb{P}^1(K)$ the polarity $\pi^{\widehat{F}} : \mathbb{P}^1(K) \to \mathbb{P}^1(K)^*$ defined by

$$[x_0, x_1] \to [u_0, u_1] = [a_{00}x_0 + a_{01}x_1, a_{01}x_0 + a_{11}x_1].$$

The point coordinates of the "line" $[u_0, u_1]$ are $[u_1, -u_0]$ just because $u_0u_1 - u_1u_0 = 0$, therefore the conjugate point of $[x_0, x_1]$ has coordinates $[a_{01}x_0 + a_{11}x_1, -(a_{00}x_0 + a_{01}x_1)]$ so that the map $\mathbb{P}^1(K) \to \mathbb{P}^1(K)$ defined by

$$[x_0, x_1] \to [a_{01}x_0 + a_{11}x_1, -(a_{00}x_0 + a_{01}x_1)] = \begin{pmatrix} a_{01} & a_{11} \\ -a_{00} & -a_{01} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

is the *involution of conjugation*. Let $\overline{K}$ be an algebraic closure of K, and $j : \mathbb{P}^1(K) \to \mathbb{P}^1(\overline{K})$ be the natural inclusion, then $P, Q \in \mathbb{P}^1(K)$ are conjugate with respect to $\widehat{F}$ if and only if

$$\rho(A, B, P, Q) = -1,$$

where $\{A, B\} = \mathrm{V}_+(\widehat{F}) \subset \mathbb{P}^1(\overline{K})$.

10.3.1 Affine and Euclidean Geometry in a Projective Setting

Several affine (or euclidean) definitions and properties of a hyperquadric $\widehat{f}$ can be rewritten in terms of the projective closure $\widehat{x_0(f)}$ (with respect to the hyperplane at infinity $\mathcal{Y}_0 : x_0 = 0$) as well as of the complexification of $\widehat{f}$ and $\widehat{x_0(f)}$. This is a very fruitful approach as shown, e.g. in Exercise 10.19.

We wish to give a *projective version* of some concepts which occur in the theory of hyperquadrics of $\mathbb{A}^n(\mathbb{R})$ or $\mathbb{E}^n(\mathbb{R})$. We shall follow [12]. Let us fix some notation. Let $\widehat{f}$ be a non-degenerate hyperquadric of $\mathbb{A}^n(\mathbb{R})$, with associated matrices

$$C_f = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix}, \qquad \overline{C}_f = \begin{pmatrix} c & c_1 & c_2 & \cdots & c_n \\ c_1 & c_{11} & c_{12} & \cdots & c_{1n} \\ c_2 & c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_n & c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix}.$$

Then the projective closure $\widehat{x_0(f)}$ (with respect to the hyperplane at infinity $\mathcal{Y}_0$: $x_0 = 0$) is associated to the matrix

$$\overline{C}_f^{\,\mathrm{p}} := \begin{pmatrix} c_{00} & c_{01} & c_{02} & \cdots & c_{0n} \\ c_{10} & c_{11} & c_{12} & \cdots & c_{1n} \\ c_{20} & c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{n0} & c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix}, \qquad c_{00} = c, \quad c_{0i} = c_i,\ i = 1, \ldots n.$$

The *hyperquadric at infinity* of $\widehat{f}$ is $\widehat{f}_\infty := \widehat{x_0(f)} \cap \mathcal{Y}_0$.

Definition 10.93 (Centre of a Hyperquadric) If $\widehat{f}$ is a non-degenerate quadric with centre $P = [p_1, \ldots, p_n]$, the point $\overline{P} = [1, p_1, \ldots, p_n]$ coincides with the pole of $\mathcal{Y}_0$. Indeed,

$$\overline{C}_f^{\,\mathrm{p}}\, \overline{P}^t = (1, \mathbf{0})^t, \qquad \mathbf{0} = (0, \ldots, 0),$$

by Proposition 5.31. The pole $Q = [q_0, q_1, \ldots, q_n]$ of $\mathcal{Y}_0$ is improper, that is, $q_0 = 0$, if and only if $\widehat{f}$ is a paraboloid. Indeed, since

$$(\overline{C}_f^{\,\mathrm{p}})^{-1} = \begin{pmatrix} \dfrac{\det(C_f)}{\det(\overline{C}_f)} & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ \cdots & \cdots & \cdots & \cdots \end{pmatrix},$$

then $(\overline{C}_f^{\,\mathrm{p}})^{-1}(1, \mathbf{0})^t = (0, q_1, \ldots, q_n)^t$. Therefore $\widehat{x_0(f)}$ is tangent to $\mathcal{Y}_0$ at the point Q which may be called the *improper centre* of a paraboloid. We can complete our classification of affine hyperquadrics of $\mathbb{A}^n(\mathbb{R})$ (see Definition 5.46). A hyperquadric $\widehat{f}$ is called:

(a) a *hyperboloid* if non-degenerate, with centre such that $\widehat{f}_\infty$ is a non-degenerate quadric of $\mathcal{Y}_0$ with non-empty support;
(b) an *ellipsoid* if non-degenerate, with centre such that $\widehat{f}_\infty$ is a non-degenerate quadric of $\mathcal{Y}_0$ with empty support.

Suppose that a point $\overline{P} = [1, p_1, \ldots, p_n]$ is the pole of the hyperplane at infinity $\mathcal{Y}_0$, then any line r passing through $P = (p_1, \ldots, p_n)$ meets $\mathcal{V}(\widehat{f})$ in two distinct points P_1, P_2 or in none. Indeed, if $r \cap \mathcal{V}(\widehat{f}) = \{Q\}$, then r being tangent to $\widehat{f}$ would be contained in the polar plane of Q; therefore, by reciprocity, Q would belong to $\mathcal{Y}_0$ which is absurd. Furthermore $\varphi_r(P) = \mathcal{Y}_0 \cap r$ (the improper point of r), where φ_r is the involution induced by $\pi^{\widehat{F}}$ on r, hence

$$\varrho(P_1, P_2, P, \varphi_r(P)) = -1,$$

i.e. P is the midpoint of the segment $\{P_1, P_2\}$ (see Definition 9.26) so that P is the centre according to Definition 5.30.

All what was said above clearly holds true for any field of characteristic $\neq 2$.

Definition 10.94 (Diametral Hyperplanes, Axes and Asymptotes of a Hyperquadric) An affine hyperplane is called a *diametral hyperplane* of a non-degenerate hyperquadric $\widehat{f}$ (*diametral plane* if $n = 3$, *diameter* if $n = 2$) if its projective closure is the polar hyperplane of a point of the hyperplane at infinity $\mathcal{Y}_0$. Every line (not lying on $\mathcal{Y}_0$) passing through the centre (proper or improper) is called a *diameter* of $\widehat{f}$ (if $n = 2$ the notions of diametral hyperplane and diameter coincide). The diameters tangent to $\widehat{x_0(f)}$ are called the *asymptotes* of $\widehat{f}$. If $\widehat{f}$ has a (proper) centre C and d is a diameter tangent to $\widehat{x_0(f)}$ at a point P, then d is contained in the polar hyperplane $\mathcal{P}_P^{\widehat{x_0(f)}}$ so that $P \in \mathcal{P}_C^{\widehat{x_0(f)}} = \mathcal{Y}_0$ by reciprocity. Therefore an asymptote is tangent to $\widehat{x_0(f)}$ in a point of the improper locus $\widehat{f}_\infty := \widehat{x_0(f)} \cap \mathcal{Y}_0$ of $\widehat{f}$. For this reason the set of all lines passing through C and tangent to $\widehat{x_0(f)}$, which is the cone circumscribed to $\widehat{x_0(f)}$ from C (see Exercise 10.7), is called the *asymptote cone* to $\widehat{f}$.

If P is the centre of the hyperquadric, then a hyperplane is diametral if and only if its projective closure passes through the centre P by reciprocity. If $\widehat{f}$ is a paraboloid, all diametral hyperplanes are parallel to one line, because their projective closures pass through the improper centre Q of $\widehat{f}$ (so that all their orienting subspaces contain the 1-dimensional vector subspace V_Q).

In order to define the principal diametral hyperplanes (for hyperquadrics of $\mathbb{E}^n(\mathbb{R})$) we need the following:

Proposition 10.95 *Let K be any field of characteristic $\neq 2$. Let $\widehat{f}$ be a non-degenerate quadric of $\mathbb{A}^n(K)$, let $\mathcal{Y}$ be a diametral hyperplane and let $P = [(0, p_1, \ldots, p_n)] \in \mathbb{P}^n(K)$ be the pole of $\overline{\mathcal{Y}}$ with respect to $\widehat{x_0(f)}$. Then:*

(a) *$P \in \mathcal{Y}$ if and only if the vector $v = (p_1, \ldots, p_n)$ is parallel to $\mathcal{Y}$.*
(b) *Suppose $P \notin \overline{\mathcal{Y}}$, then*

$$s_{\mathcal{Y},\langle v\rangle}(\mathcal{V}(\widehat{f})) = \mathcal{V}(\widehat{f}),$$

where $s_{\mathcal{Y},\langle v\rangle}$ is the symmetry through $\mathcal{Y}$ parallel to the vector subspace $\langle v\rangle$ (Definition 3.82).

Proof

(a) Let $\mathcal{Y} : a_1x_1 + \cdots + a_nx_n + b = 0$, then $\overline{\mathcal{Y}}$ has the equation $bx_0 + a_1x_1 + \cdots + a_nx_n = 0$. Therefore $P \in \overline{\mathcal{Y}}$ if and only if $a_1p_1 + \cdots + a_np_n = 0$, i.e. if and only if v is parallel to $\widehat{f}$.
(b) Since $P \notin \overline{\mathcal{Y}}$ by part (a) we can take affine coordinates $x_1, \ldots, x_n$ such that $v = (0, 0, \ldots, 0, 1)$ and $\mathcal{Y} : x_n = 0$. With respect to this system of coordinates we have

$$s_{\mathcal{Y},\langle v\rangle}(x_1, \ldots, x_{n-1}, x_n) = (x_1, \ldots, x_{n-1}, -x_n).$$

Since the polar hyperplane of $[0, \ldots, 0, 1]$ has the equation $x_n = 0$, i.e.

$$\begin{pmatrix} c_{00} & c_{01} & c_{02} & \cdots & c_{0n} \\ c_{10} & c_{11} & c_{12} & \cdots & c_{1n} \\ c_{20} & c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{n0} & c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix},$$

we get $c_{i0} = 0$ for $i = 0, \ldots, n-1$. Therefore

$$f(x_1, \ldots, x_n) = \sum_{i=0}^{n-1} c_{ij} x_i x_j + c_{nn} x_n^2,$$

which implies $s_{\mathcal{Y},\langle v \rangle} \cdot f = f$. □

If the diametral hyperplane $\mathcal{Y}$ relative to a point $P = [0, p_1, \ldots, p_n] \in \mathcal{Y}_0$ is orthogonal to the vector $(p_1, \ldots, p_n)$, then $\mathcal{Y}$ is called a *principal diametral hyperplane* of $\widehat{f}$ (see Definition 5.52 and Corollary 5.54). Moreover, we define an *axis* of $\widehat{f}$ as every line of $\mathbb{E}^n(\mathbb{R})$ which is the intersection of principal diametral hyperplanes (if $n = 2$ the notions of principal diametral hyperplane and axis coincide), and a *vertex* of a non-degenerate hyperquadric $\widehat{f}$ as every point where $\widehat{f}$ intersects an axis.

The following proposition asserts that the notions of principal diametral hyperplane of Definitions 5.52 and 10.94 coincide.

Proposition 10.96 *Let $\mathcal{Y}$ be a hyperplane of $\mathbb{E}^n(\mathbb{R})$ and $s_{\mathcal{Y}}$ be the orthogonal symmetry through $\mathcal{Y}$.*

(a) *If the projective closure $\overline{\mathcal{Y}}$ is tangent to $\widehat{x_0(f)}$, then $s_{\mathcal{Y}} \cdot \widehat{f} \neq \widehat{f}$.*
(b) *A hyperplane $\mathcal{Y}$ is a principal diametral hyperplane of $\widehat{f}$ if and only if the orthogonal symmetry $s_{\mathcal{Y}}$ through $\mathcal{Y}$ preserves $\widehat{f}$ i.e. $s_{\mathcal{Y}} \cdot \widehat{f} = \widehat{f}$.*

Proof

(a) Let $\overline{\mathcal{Y}}$ be tangent to $\widehat{x_0(f)}$ at a point $\overline{P}$. If $\overline{P}$ is a proper point, we may suppose that $P = (0, \ldots, 0)$ and $\mathcal{Y}$ is defined by the equation $x_1 = 0$ (up to an isometry). With respect to this system of coordinates, we have

$$f(x_1, \ldots, x_n) = x_1 + g(x_1, \ldots, x_n)$$

where g is a homogeneous polynomial of degree 2 (indeed $c = c_2 = \cdots = c_n = 0$ and $c_1 = 1$). The orthogonal reflection through $\mathcal{Y}$ is given by

$$s_{\mathcal{Y}}(x_1, \ldots, x_n) = (-x_1, x_2, \ldots, x_n)$$

so that $s_{\mathcal{Y}} \cdot \widehat{f}$ is defined by

$$(s_{\mathcal{Y}} \cdot f)(x_1, \ldots, x_n) = -x_1 + g(-x_1, \ldots, x_n).$$

Suppose by contradiction that $s_{\mathcal{Y}} \cdot \widehat{f} = \widehat{f}$. Then there exists $\lambda \in \mathbb{R}$ such that $f(-x_1, \ldots, x_n) = \lambda f(x_1, \ldots, x_n)$. Since $f(-x_1, \ldots, x_n) = -x_1 + g(-x_1, \ldots, x_n)$ we must have $\lambda = -1$ and $g(-x_1, \ldots, x_n) = -g(x_1, \ldots, x_n)$. Therefore each monomial appearing in g is of type $x_1 x_j$, for some $2 \leq j \leq n$. Hence g is divisible by x_1, and the same is true for f. Since $\widehat{f}$ is non-degenerate this is impossible.

Let $\overline{P}$ be an improper point. We can always suppose as above that $\mathcal{Y}$ is the hyperplane of equation $x_1 = 0$, and $\overline{P} = [0, \ldots, 0, 1]$. In this system of coordinates, we have

$$f(x_1, \ldots, x_n) = c + g(x_1, \ldots, x_n) + h(x_1, \ldots, x_n),$$

where $c \in \mathbb{R}$, the polynomial g is either zero or homogeneous of degree 1, and $h(x_1, \ldots, x_n) = x_1 x_n + r(x_1, \ldots, x_{n-1})$, where r is a homogeneous polynomial of degree 2 in $x_1, \ldots, x_{n-1}$ (indeed $c_n = c_{2n} = \cdots c_{nn} = 0$ and $c_{1n} = 1$). In this case $s_{\mathcal{Y}} \cdot \widehat{f}$ is defined by

$$f(-x_1, \ldots, x_n) = c - x_1 x_n + g(-x_1, \ldots, x_n) + r(-x_1, \ldots, x_{n-1}).$$

Then, as in the previous case, if $s_{\mathcal{Y}} \cdot \widehat{f} = \widehat{f}$ then x_1 divides f, against the hypothesis that $\widehat{f}$ is non-degenerate.

(b) Let $v = (v_1, \ldots, v_n) \in \mathbb{R}^n \setminus \mathbf{0}$ be a vector orthogonal to $\mathcal{Y}$, and $R = [0, v_1, \ldots, v_n] \in \mathbb{P}^n(\mathbb{R})$ be the point at infinity corresponding to the direction v. Of course we have $R \notin \overline{\mathcal{Y}}$. Moreover, $\mathcal{Y}$ is a principal hyperplane if and only if R is the pole of $\overline{\mathcal{Y}}$ with respect to $\widehat{x_0(f)}$. If $\mathcal{Y}$ is a principal hyperplane, then $s_{\mathcal{Y}} \cdot \widehat{f} = \widehat{f}$ by Proposition 10.95. Conversely, suppose that $s_{\mathcal{Y}} \cdot \widehat{f} = \widehat{f}$ and let P be the pole of $\overline{\mathcal{Y}}$. Let $\overline{s_{\mathcal{Y}}} \colon \mathbb{P}^n(\mathbb{R}) \to \mathbb{P}^n(\mathbb{R})$ the projectivity induced by $s_{\mathcal{Y}}$ (Proposition 9.13). Let $x = (x_1, \ldots, x_n)$, $b = (b_1, \ldots, b_n)$, $A = (a_{ij})_{i,j=1,\ldots,n}$ and $s_{\mathcal{Y}}(x) = Ax^t + b^t$. Since v is orthogonal to $\mathcal{Y}$, then $Av^t = kv$ for $k \in \mathbb{R}^*$, therefore

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ b_1 & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} 0 \\ v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} 0 \\ kv_1 \\ \vdots \\ kv_n \end{pmatrix}$$

so that $\overline{s_{\mathcal{Y}}}(R) = R$, hence the set $\overline{\mathcal{Y}} \cup \{R\}$ is the fixed locus of $\overline{s_{\mathcal{Y}}}$. Since $\overline{s_{\mathcal{Y}}} \cdot \widehat{x_0(f)} = \widehat{x_0(f)}$ we also have $\overline{s_{\mathcal{Y}}}(P) = P$. By (a) the projective hyperplane $\overline{\mathcal{Y}}$ is

not tangent to $\widehat{x_0(f)}$ so that the point P does not belong to $\overline{\mathcal{Y}}$, hence $P = R$ and $\overline{\mathcal{Y}}$ is a principal hyperplane. □

Complexification
Let

$$\sigma_{\mathbb{C}} \colon \mathbb{C}^n \to \mathbb{C}^n, \qquad \sigma_{\mathbb{C}}(z_1, \ldots, z_n) = (\overline{z_1}, \ldots, \overline{z_n}), \tag{10.60}$$

where $\overline{z_j}$ is the complex conjugate of z_j, $j = 1, \ldots, n$. For every polynomial (homogeneous or not homogeneous)

$$f = \sum_{i_1 \geq 0, \ldots, i_n \geq 0} a_{i_1, \ldots, i_n} X_1^{i_1} \cdots X_n^{i_n} \in \mathbb{C}[X_1, \ldots, X_n]$$

we define the *conjugation* $\sigma_{\mathbb{C}}$ as follows:

$$\sigma_{\mathbb{C}}(f) := \sum_{i_1 \geq 0, \ldots, i_n \geq 0} \overline{a_{i_1, \ldots, i_n}} X_1^{i_1} \cdots X_n^{i_n} \in \mathbb{C}[X_1, \ldots, X_n] \tag{10.61}$$

obtained from f by conjugating each coefficient. For any polynomial $f \in \mathbb{C}[X_1, \ldots, X_n]$ the affine hypersurface $\sigma_{\mathbb{C}}(\widehat{f}) := \widehat{\sigma_{\mathbb{C}}(f)}$ is called the *complex conjugate* of the affine hypersurface $\widehat{f}$. If $\widehat{f} \in \mathcal{H}_d(n, \mathbb{C})$, the intersection $\mathcal{V}_{\mathbb{R}}(\widehat{f}) := \mathcal{V}(\widehat{f}) \cap \mathbb{A}^n(\mathbb{R})$ is the set of the *real points* of $\widehat{f}$. Analogously, for any homogeneous polynomial $F \in \mathbb{C}[X_0, \ldots, X_n]$ the projective hypersurface $\sigma_{\mathbb{C}}(\widehat{F}) := \widehat{\sigma_{\mathbb{C}}(F)}$ is called the *complex conjugate* of the projective hypersurface $\widehat{F}$. If $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, \mathbb{C})$, the intersection $V_{\mathbb{R}})_+(\widehat{F}) := \mathcal{V}_+(\widehat{F}) \cap \mathbb{P}^n(\mathbb{R})$ is the set of the *real points* of $\widehat{F}$.

If $f \in \mathbb{R}[X_1, \ldots, X_n]_d$ with $d \geq 1$, then we can consider both the equivalence class $\widehat{f}$ in $\mathcal{H}_d(n, \mathbb{R})$ and the equivalence class $\widehat{f_{\mathbb{C}}}$ in $\mathcal{H}_d(n, \mathbb{C})$. We call the hypersurface $\widehat{f_{\mathbb{C}}}$ the *complexification* of $\widehat{f}$. Since every affine subspace $\mathcal{S}$ of dimension k in $\mathbb{A}^n(\mathbb{R})$ is the intersection of $n-k$ independent hyperplanes $\mathcal{Y}_i = \widehat{f_i}$, $i = 1, \ldots, n-k$, its complexification $\mathcal{S}_{\mathbb{C}}$ is an affine subspace of dimension k which is the intersection of the $n-k$ independent hyperplanes $\widehat{(f_i)_{\mathbb{C}}}$. Any affinity of $\mathbb{A}^n(\mathbb{R})$ trivially extends to an affinity of $\mathbb{A}^n(\mathbb{C})$. Conversely, an affinity $\varphi(X) = MX^t + N^t$ (with $X = (x_1, \ldots, x_n) \in \mathbb{C}^n$, $M \in \mathrm{GL}_n(\mathbb{C})$ and $N = (b_1, \ldots, b_n) \in \mathbb{C}^n$) maps points of $\mathbb{A}^n(\mathbb{R})$ to points of $\mathbb{A}^n(\mathbb{R})$ if and only if $M \in \mathrm{GL}_n(\mathbb{R})$ and $N = (b_1, \ldots, b_n) \in \mathbb{R}^n$. In this case the restriction of φ to $\mathbb{A}^n(\mathbb{R})$ is an affinity of $\mathbb{A}^n(\mathbb{R})$. It is immediately seen that:

1. A hyperquadric $\widehat{f}$ of $\mathbb{A}^n(\mathbb{R})$ is non-degenerate if and only if $\widehat{f_{\mathbb{C}}}$ is a non-degenerate hyperquadric of $\mathbb{A}^n(\mathbb{C})$.
2. A non-degenerate hyperquadric $\widehat{f}$ of $\mathbb{A}^n(\mathbb{R})$ has a centre if and only if $\widehat{f_{\mathbb{C}}}$ has a centre; moreover the two centres coincide.

We note that the curve of equation $x^2 + y^2 = 0$ is reducible when regarded as a complex curve, while it is irreducible when regarded as a real curve. If $\widehat{f} \in \mathcal{H}_d(n, \mathbb{R})$, the subset $\mathcal{V}(\widehat{f_{\mathbb{C}}}) \setminus \mathcal{V}(\widehat{f})$ of $\mathbb{A}^n(\mathbb{C})$ is the set of *complex-conjugate points* of $\widehat{f}$.

Analogous considerations can be made in the projective case. If $F \in {}_{\mathrm{h}}\mathbb{R}[X_0, \ldots, X_n]_d^*$ with $d \geq 1$ (see Definition 10.1), then we can consider both the equivalence class $\widehat{F}$ in $\mathcal{H}_d(n, \mathbb{R})$ and the equivalence class $\widehat{F_{\mathbb{C}}}$ in $\mathcal{H}_d(n, \mathbb{C})$. We call the hypersurface $\widehat{F_{\mathbb{C}}}$ the *complexification* of $\widehat{F}$. Since every projective subspace $\mathcal{S}$ of dimension k in $\mathbb{P}^n(\mathbb{R})$ is intersection of $n-k$ independent hyperplanes $\mathcal{Y}_i = \widehat{F_i}$, $i = 1, \ldots, n-k$, its complexification $\mathcal{S}_{\mathbb{C}}$ is a projective subspace of dimension k which is the intersection of the $n-k$ independent hyperplanes $\widehat{(F_i)_{\mathbb{C}}}$. It is immediate to see that a hyperquadric $\widehat{F}$ of $\mathbb{P}^n(\mathbb{R})$ is non-degenerate if and only if $\widehat{F_{\mathbb{C}}}$ is a non-degenerate hyperquadric of $\mathbb{P}^n(\mathbb{C})$. Every projectivity of $\mathbb{P}^n(\mathbb{R})$ trivially extends to a projectivity of $\mathbb{P}^n(\mathbb{C})$. Conversely, a projectivity ϕ of $\mathbb{P}^n(\mathbb{C})$ represented by a matrix $A \in \mathrm{GL}_{n+1}(\mathbb{C})$ transforms points of $\mathbb{P}^n(\mathbb{R})$ into points of $\mathbb{P}^n(\mathbb{R})$ if and only if there exists $k \in \mathbb{C}^*$ such that $kA \in \mathrm{GL}_{n+1}(\mathbb{R})$ so that the restriction of ϕ to $\mathbb{P}^n(\mathbb{R})$ is a projectivity of $\mathbb{P}^n(\mathbb{R})$. If $\widehat{F} \in \mathcal{H}_d^{\mathrm{p}}(n, \mathbb{R})$, the subset $\mathcal{V}_+(\widehat{F_{\mathbb{C}}}) \setminus \mathcal{V}_+(\widehat{F})$ of $\mathbb{P}^n(\mathbb{C})$ is the set of *complex-conjugate points* of $\widehat{F}$.

Lemma 10.97 *A hypersurface $\widehat{F}$ of $\mathbb{P}^n(\mathbb{C})$ is the complexification $\widehat{G_{\mathbb{C}}}$ of a hypersurface $\widehat{G}$ of $\mathbb{P}^n(\mathbb{R})$ if and only if $\sigma_{\mathbb{C}}(\widehat{F}) = \widehat{F}$.*

Proof If $\widehat{F} = \widehat{G_{\mathbb{C}}}$, then we can take $F = G \in {}_{\mathrm{h}}\mathbb{R}[X_0, \ldots, X_n]_d^*$ so that $\sigma_{\mathbb{C}}(F) = F$ and $\sigma_{\mathbb{C}}(\widehat{F}) = \widehat{F}$.

Conversely, let $\sigma_{\mathbb{C}}(\widehat{F}) = \widehat{F}$ with $F \in \mathbb{C}[z_0, \ldots, z_n]$ homogeneous. The conjugate hypersurface is defined by $\sigma_{\mathbb{C}}(F) = 0$ so there exists $k \in \mathbb{C}^*$ such that $\sigma_{\mathbb{C}}(F) = kF$. Write $z = (z_0, \ldots, z_n)$, $F(z) = G(z) + iH(z)$ with G and $H \in \mathbb{R}[z_0, \ldots, z_n]$ homogeneous. Since $F \neq 0$, up to replacing F by iF we may assume $G \neq 0$ so that G defines a real hypersurface $\widehat{G}$. Therefore we have

$$G(z) = \frac{1}{2}\left(F(z) + \sigma_{\mathbb{C}}(F)(z)\right) = \frac{1+k}{2}F(z),$$

with $k \neq -1$ so that $\widehat{F} = \widehat{G_{\mathbb{C}}}$. □

Lemma 10.98 *Let r be a line of $\mathbb{P}^2(\mathbb{C})$.*

(a) *If $r = \sigma_{\mathbb{C}}(r)$, then r contains infinitely many real points.*
(b) *If $r \neq \sigma_{\mathbb{C}}(r)$, then the point $P = r \cap \sigma_{\mathbb{C}}(r)$ is the only real point of r.*

Proof

(a) If $r = \sigma_{\mathbb{C}}(r)$, by Lemma 10.97 the line r is the complexification of a real line s, so it contains infinitely many real points.
(b) If $r \neq \sigma_{\mathbb{C}}(r)$, then the point $P = r \cap \sigma_{\mathbb{C}}(r)$ is a real point since $\sigma_{\mathbb{C}}(P) \in \sigma_{\mathbb{C}}(r) \cap \sigma_{\mathbb{C}}(\sigma_{\mathbb{C}}(r)) = \sigma_{\mathbb{C}}(r) \cap r$ so that $\sigma_{\mathbb{C}}(P) = P$. If $Q \in r$ is a real point, then $Q = \sigma_{\mathbb{C}}(Q) \in r \cap \sigma_{\mathbb{C}}(r) = P$. □

Remark 10.99 Let $\widehat{F}$ be a non-degenerate quadric of $\mathbb{P}^3(\mathbb{R})$ and let $\widehat{F_{\mathbb{C}}}$ be its complexification. Suppose that $\widehat{F}$ is not ruled and $\mathcal{V}_+(\widehat{F}) \neq \varnothing$. Consider the rulings R_a and R_b of $\widehat{F_{\mathbb{C}}}$. The conjugation $\sigma_{\mathbb{C}}$ preserves incidence relations, so that $\sigma_{\mathbb{C}}$ either transforms each ruling into itself or switches R_a and R_b. Let $P \in \mathcal{V}_+(\widehat{F})$. Then $\sigma_{\mathbb{C}}(\mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},P}) = \mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},P}$, where $\sigma_{\mathbb{C}}$ is the conjugation defined by (10.61). Therefore $\sigma_{\mathbb{C}}$ preserves the degenerate conic $\mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},P} \cap \mathcal{V}_+(\widehat{F_{\mathbb{C}}})$ consisting of two complex-conjugate lines l_1 and l_2. Hence $\sigma_{\mathbb{C}}(l_1) = l_2$ and $\sigma_{\mathbb{C}}(l_2) = l_1$, i.e. $\sigma_{\mathbb{C}}$ switches R_a and R_b.

Definition 10.100 Let $\widehat{F}$ be a non-degenerate quadric of $\mathbb{P}^3(\mathbb{R})$ and let r be a line not contained in $\mathcal{V}_+(\widehat{F})$. We say that r is *secant* to $\widehat{F}$ if $r \cap \mathcal{V}_+(\widehat{F})$ consists of two distinct real points, and that r is *external* to $\widehat{F}$ if $r \cap \mathcal{V}_+(\widehat{F})$ consists of two complex-conjugate points.

The source of the following proposition is Exercise 185 of [12].

Proposition 10.101 *Let $\widehat{F}$ be a non-degenerate quadric of $\mathbb{P}^3(\mathbb{R})$ and let r be a line not contained in $\mathcal{V}_+(\widehat{F})$. Let $r_{\widehat{F}}$ be the polar line of r. If $\widehat{F}$ is ruled, then:*

(a) *r is secant if and only if $r_{\widehat{F}}$ is secant.*
(b) *There exist two planes of $\mathbb{P}^3(\mathbb{R})$ that pass through r and are tangent to $\widehat{F}$ if and only if r is secant.*

If $\widehat{F}$ is not ruled and $\mathcal{V}_+(\widehat{F}) \neq \varnothing$, then:

(c) *r is secant if and only if $r_{\widehat{F}}$ is external.*
(d) *There exist two planes of $\mathbb{P}^3(\mathbb{R})$ that pass through r and are tangent to $\widehat{F}$ if and only if r is external.*

Proof

(a) If $r_{\widehat{F}}$ is secant to $\widehat{F}$ at distinct points Q_1 and Q_2, by Proposition 10.78-(f) there exist two planes $\mathcal{Y}_1$ and $\mathcal{Y}_2$ passing through r and tangent to $\widehat{F}$ at the points Q_1 and Q_2. Let R_a and R_b be the rulings of lines contained in $\widehat{F}$. Hence $\mathcal{Y}_1 \cap \mathcal{V}_+(\widehat{F}) = a_1 \cap b_1$ and $\mathcal{Y}_2 \cap \mathcal{V}_+(\widehat{F}) = a_2 \cap b_2$, with $a_1, a_2 \in R_a$ and $b_1, b_2 \in R_b$. Since r is not contained in $\mathcal{V}_+(\widehat{F})$, then the four lines a_1, a_2, b_1, b_2 are distinct. The lines a_1 and b_2 meet at a point of $\widehat{F}$, and on the other hand, $a_1 \cap b_2 \in \mathcal{Y}_1 \cap \mathcal{Y}_2 = r$, so that the point $a_1 \cap b_2 \in r \cap \mathcal{V}_+(\widehat{F})$. Analogously, $a_2 \cap b_1 \in r \cap \mathcal{V}_+(\widehat{F})$. The points $a_1 \cap b_2$ and $a_2 \cap b_1$ are distinct, otherwise there are three lines contained in $\mathcal{V}_+(\widehat{F})$ passing through a point. The converse immediately follows from the fact that r is the polar of $r_{\widehat{F}}$.
(b) is an immediate consequence of (a) and Proposition 10.78-(f).
(c) Let $r_{\widehat{F}}$ be secant to $\widehat{F}$ and $r_{\widehat{F}} \cap \mathcal{V}_+(\widehat{F}) = \{Q_1, Q_2\}$. Then

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_1} \cap \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_2} = \pi^{\widehat{F}}(Q_1) \cap \pi^{\widehat{F}}(Q_1) = r$$

and

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_1} \cap \mathcal{V}_+(\widehat{F}) = Q_1, \qquad \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_2} \cap \mathcal{V}_+(\widehat{F}) = Q_2,$$

since $\widehat{F}$ is not ruled. The lines r and $r_{\widehat{F}}$ are distinct because r is not contained in $\mathcal{V}_+(\widehat{F})$, and they are not incident, since r is not tangent to $\widehat{F}$ (Proposition 10.78). Therefore, if there exists a point $R \in r \cap \mathcal{V}_+(\widehat{F})$, then we should have $R \neq Q_1$, contradicting the fact that

$$r \cap \mathcal{V}_+(\widehat{F}) \subset \mathrm{T}^{\mathrm{p}}_{\widehat{F},Q_1} \cap \mathcal{V}_+(\widehat{F}) = Q_1.$$

Conversely, suppose that r is external. By Proposition 10.78-(e) $r_{\widehat{F}}$ is not tangent to $\widehat{F}$. Let $r_{\mathbb{C}} \cap \mathcal{V}_+(\widehat{F_{\mathbb{C}}}) = \{Q_1, Q_2\}$, where $r_{\mathbb{C}}$ is the complexification of r. The points Q_1 and Q_2 are complex conjugate, as are the tangent planes $\mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_1}$ and $\mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_2}$. Therefore

$$\mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_1} \cap \mathcal{V}_+(\widehat{F_{\mathbb{C}}}) = a_1 \cup b_1, \qquad \mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_2} \cap \mathcal{V}_+(\widehat{F_{\mathbb{C}}}) = a_2 \cup b_2,$$

where $a_1, a_2 \in R_a$ and $b_1, b_2 \in R_b$ (see Remark 10.99). Hence $\sigma_{\mathbb{C}}(a_1) = b_2$ and $\sigma_{\mathbb{C}}(a_2) = b_1$. Therefore

$$a_1 \cap b_2 = R \in \mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_1} \cap \mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_2} = s$$

and

$$a_2 \cap b_1 = S \in \mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_2} \cap \mathrm{T}^{\mathrm{p}}_{\widehat{F_{\mathbb{C}}},Q_1} = s,$$

where s is the polar of $r_{\mathbb{C}}$. Since R and S are real then $R, S \in r_{\widehat{F}}$.

(d) is an immediate consequence of (c) and Proposition 10.78-(f). □

Definition 10.102 (The Absolute Hyperquadric) We define the *absolute hyperquadric* (*conic* if $n = 2, 3$) or simply *absolute* of the standard euclidean space $\mathbb{E}^n(\mathbb{R})$ as the non-degenerate hyperquadric $\mathrm{Ab}_n := (\mathcal{Y}_0)_{\mathbb{C}} \cap \widehat{(\Gamma_n)_{\mathbb{C}}}$ of the hyperplane at infinity $(\mathcal{Y}_0)_{\mathbb{C}}$, i.e.

$$\mathcal{V}_+(\mathrm{Ab}_n) = \{[x_0, \ldots, x_n] \in \mathbb{P}^n(\mathbb{C}) : x_0^2 + x_1^2 + \cdots + x_n^2 = 0 = x_0 = 0\}.$$

In the case $n = 2$ the absolute is the pair of complex-conjugate points

$$I = [0, 1, i], \qquad J = [0, 1, -i].$$

These points are called the *cyclic* or *circular points* of the plane and a line of $\mathbb{A}^2(\mathbb{C})$ having a cyclic point as improper point is called an *isotropic line*. In general, we call any two complex-conjugate points lying on the absolute Ab_n *circular points* of

$\mathbb{E}^n(\mathbb{R})$. It easy to see that any hyperquadric $\widehat{f}$ of $\mathbb{E}^n(\mathbb{R})$ is a sphere if and only if the hyperquadric at infinity of $\widehat{f_{\mathbb{C}}}$ is Ab_n; in particular, any conic of $\mathbb{E}^2(\mathbb{R})$ is a circle if and only if its complexification has the cyclic points as improper points, and two circles have the same centre if and only if they are tangent at the cyclic points, i.e. they have the same tangent lines at the cyclic points.

Let $v = (x_1, \ldots, x_n)$ and $w = (y_1, \ldots, y_n)$ be two vectors of $\mathbb{R}^n$, then they are orthogonal if $x_1y_1 + \cdots + x_ny_n = 0$, i.e. if the point $P = [0, x_1, \ldots, x_n]$ and $Q = [0, y_1, \ldots, y_n]$ are conjugate with respect to the absolute hyperquadric Ab_n. Thus we are led to the following:

Proposition 10.103 *Let S and T be two subspaces of $\mathbb{E}^n(\mathbb{R})$ of dimension $m > 0$ and $r > 0$ respectively. Then:*

(a) *If $m+r \leq n$, $S \perp T$ if and only if the improper part of either of them is contained in the polar space of the improper part of the other relative to the absolute hyperquadric, i.e. $S_\infty \subset (T_\infty)_{\mathrm{Ab}_n}$ (equivalent to $T_\infty \subset (S_\infty)_{\mathrm{Ab}_n}$, where $S_\infty = \mathbb{P}(D(S))$ and $T_\infty = \mathbb{P}(D(T))$ according to notation of* Definition 10.77.
(b) *If $m + r \geq n$, $S \perp T$ if and only if $(T_\infty)_{\mathrm{Ab}_n} \subset S_\infty$ (equivalent to $(S_\infty)_{\mathrm{Ab}_n} \subset T_\infty$).*

Proof (a) If $m + r \leq n$ and $S \perp T$, then, by Definition 4.10, $D(S) \subset D(T)^\perp$ ($\Longleftrightarrow$ $D(T) \subset D(S)^\perp$) so that $S_\infty \subset (T_\infty)_{\mathrm{Ab}_n}$, which is equivalent to $T_\infty \subset (S_\infty)_{\mathrm{Ab}_n}$). Conversely, if $S_\infty \subset (T_\infty)_{\mathrm{Ab}_n}$ we have $m-1 \leq n-1-(r-1)-1 = n-r-1$, i.e. $m+r \leq n$ and $D(S) \subset D(T)^\perp$, i.e. $S \perp T$.
(b) If $m + r \geq n$ and $S \perp T$, then $D(T)^\perp \subset D(S)$ ($\Longleftrightarrow$ $D(S)^\perp \subset D(T)$) so that $(T_\infty)_{\mathrm{Ab}_n} \subset S_\infty$ which is equivalent to $(S_\infty)_{\mathrm{Ab}_n} \subset T_\infty$. Conversely, if $(T_\infty)_{\mathrm{Ab}_n} \subset S_\infty$ we get $n - 1 - (r - 1) - 1 \leq m - 1$ i.e. $m + r \geq n$ and $D(T)^\perp \subset D(S)$, i.e. $S \perp T$.

□

In particular we have:

(a) Two lines r and s of $\mathbb{E}^n(\mathbb{R})$ are orthogonal if and only if their improper points are conjugate relative to Ab_n. If $n = 2$ and R and S are the improper points of r and s respectively, r and s are orthogonal if and only if (see Remark 10.92)

$$\rho(I, J, R, S) = -1.$$

(b) Two planes σ, σ' of $\mathbb{E}^3(\mathbb{R})$ are orthogonal if and only if their improper lines are conjugate relative to Ab_3.
(c) A line r and a plane σ of $\mathbb{E}^3(\mathbb{R})$ are orthogonal if and only if the improper point of r is the pole of the improper line of σ relative to Ab_3.

Definition 10.104 Let $\widehat{f}$ be a non-degenerate conic, let r_0 be the improper line $x_0 = 0$ and $\widehat{x_0(f)}$ the projective closure of $\widehat{f}$. We recall some definitions (some of them given above for a general hyperquadric):

1. The centre of $\widehat{f}$ is the pole of r_0.
2. A diameter of $\widehat{f}$ is the polar of a point of r_0.
3. An axis of $\widehat{f}$ is a diameter d such that the pole of d and its point at infinity are conjugate with respect to the absolute $x_1^2 + x_2^2 = 0$ of r_0.
4. According to Definition 10.89, if $\widehat{f}$ has a proper centre C it is defined the *involution of conjugate diameters*

$$\Phi_C \colon \Phi_1(C) \to \Phi_1(C), \qquad \Phi_C(d) = (\pi^{\widehat{x_0(f)}})^*(d) + C, \quad \forall\, d \in \Phi_1(C), \tag{10.62}$$

   where $\Phi_1(C)$ is the pencil of lines passing through C. Therefore two diameters d and d' are conjugate if and only if the pole of d is the point at infinity of d' (and conversely).
5. The fixed points of Φ_C are the asymptotes, so that the asymptotes are the self-coniugate diameters. If $\widehat{f}$ is an ellipse, then Φ_C is an elliptic involution (see Definition 9.32) and the asymptotes are complex-conjugate lines. If $\widehat{f}$ is a hyperbola, then Φ_C is an hyperbolic involution and the asymptotes are real lines.
6. If $\widehat{F}$ is a parabola the diameters are real lines parallel to the axis and we may improperly say that all asymptotes coincide with the improper line r_0.
7. Let $P \in \mathbb{P}^2(\mathbb{R})$ be a non-self-conjugate point and $\Phi_P \colon \Phi_1(P) \to \Phi_1(P)$ be the associated involution. We say that Φ_P is *circular* if two conjugate lines are orthogonal, i.e. their poles are conjugate with respect the absolute $x_1^2 + x_2^2 = 0$ of r_0.
8. If $\widehat{f}$ has a proper centre C, the line at infinity $r_0 : x_0 = 0$ is not tangent to $\widehat{x_0(f)_{\mathbb{C}}}$. If $\widehat{f}$ is not a circle, then $\widehat{x_0(f)_{\mathbb{C}}}$ does not contain the cyclic points. By Proposition 10.90-(c) there are two distinct lines s_1 and s_2 of $\mathbb{P}^2(\mathbb{C})$ that pass through I and are tangent to $\widehat{x_0(f)_{\mathbb{C}}}$ at the proper points $Q_1 = s_1 \cap \widehat{x_0(f)_{\mathbb{C}}}$ and $Q_2 = s_2 \cap \widehat{x_0(f)_{\mathbb{C}}}$. The complex-conjugate lines $t_1 = \sigma_{\mathbb{C}}(s_1))$ and $t_2 = \sigma_{\mathbb{C}}(s_2))$ pass through J and they also are tangent to $\widehat{x_0(f)_{\mathbb{C}}}$ since $\sigma_{\mathbb{C}}(\widehat{x_0(f)_{\mathbb{C}}}) = \widehat{x_0(f)_{\mathbb{C}}}$. By Lemma 10.98 $F_1 := s_1 \cap t_1$ and $F_2 := s_2 \cap t_2$ are the only real points of $s_1 \cup s_2 \cup t_1 \cup t_2$. We call F_1 and F_2 the *foci* of $\widehat{f}$. Moreover, F_1 and F_2 are distinct from the centre C (if $F_i = C$ for $i = 1$ or $i = 2$, then $C \in s_i = \pi^{\widehat{x_0(f)_{\mathbb{C}}}}(Q_i)$, so that $Q_i \in r_0$).

   If $\widehat{f}$ is a circle, then I and J belong to $\widehat{x_0(f)_{\mathbb{C}}}$. If r is the line tangent to $\widehat{x_0(f)_{\mathbb{C}}}$ at I, then $r \neq r_0$ since the improper line r_0 intersects $\widehat{x_0(f)_{\mathbb{C}}}$ at I and J. The complex-conjugate line $s = \sigma_{\mathbb{C}}(r)$ is tangent to $\widehat{x_0(f)_{\mathbb{C}}}$ at J, hence it is distinct from r. The lines r and s, being the polars of points at infinity, intersect in the centre C of $\widehat{f}$, which can be also called the *focus* of $\widehat{f}$.

   If $\widehat{f}$ is a parabola, then two of the tangents s_1, s_2, t_1, t_2 from the cyclic points coincide with r_0 and the remaining lines meet at the only real point F, which is called the *focus* of $\widehat{f}$.
9. A *directix* of a conic $\widehat{f}$ which is not a circle is the polar of a focus.

Proposition 10.105 *Let $\widehat{f}$ be a non-degenerate conic.*

(a) *If F is a focus of $\widehat{f}$, then the involution Φ_F is circular.*
(b) *If $\widehat{f}$ has a proper centre but it is not a circle, then the foci F_1 and F_2 lie on an axis called* the focal axis. *Moreover the directrices are orthogonal to the focal axis.*
(c) *The focus of a parabola lies on its axis; moreover the directrix is orthogonal to the axis.*

Proof

(a) Let s and t be two complex-conjugate tangent lines from I and J. By Proposition 10.90-(d), two lines u and v through F are coniugate with respect to $\widehat{f}$ (or equivalently with respect to $\widehat{x_0(f)_{\mathbb{C}}}$) if and only if $\rho(s,t,u,v) = -1$, which is equivalent to $\rho(I,J,U,V) = -1$ (see Definition 9.29), where $U = u \cap r_0$ and $V = v \cap r_0$, i.e. U and V are conjugate with respect the absolute of r_0. Therefore u and v are orthogonal.
(b) Consider now the line $r = F_1F_2$, its improper point P and the conjugate lines $s_1 = \Phi_{F_1}(r)$ and $s_2 = \Phi_{F_2}(r)$ through F_1 and F_2 respectively. Then s_1 and s_2 pass through the pole Q of r by definition. If A_1 and A_2 are the improper points of s_1 and s_2 respectively, then $\rho(I,J,A_1,P) = \rho(I,J,A_2,P) = -1$ (by definition) so that $A_1 = A_2$, hence $A_1 = A_2 = Q$ ($s_1 \neq s_2$) so that $\rho(I,J,Q,P) = -1$, i.e. r is an axis of $\widehat{f}$. Since a directrix d is the polar of a focus then d contains the pole Q of r (since r contains the pole of d) which is an improper point so that Q is the point at infinity of d and we just know that $\rho(I,J,Q,P) = -1$, i.e. $d \perp r$. Since the pole Q of r belongs to r_0 then the centre C belongs to r by reciprocity.
(c) If $\widehat{f}$ is a parabola and r and s are the isotropic lines meeting at the focus F, then (by Proposition 10.90-(d)) $\rho(r,s,u,v) = -1$, where u is the line joining F and the centre C and v its conjugate with respect to $\widehat{x_0(f)_{\mathbb{C}}}$. Therefore $\rho(I,J,C,U) = -1$, where U is the pole of FC (which lies on r_0 since the pole of r_0, i.e. C lies on FC), hence FC is the axis of $\widehat{f}$. Let d be the directrix of $\widehat{f}$ and P its improper point. Since d is the polar of F, then d contains the pole of FC i.e. $U \in d$, therefore $U = P$ so that $d \perp FC$. □

Definition 10.106 Let $\pi_0 : x_0 = 0$ be the plane at infinity of $\mathbb{E}^3(\mathbb{R})$. Also for non-degenerate quadrics $\widehat{f}$ of $\mathbb{E}^3(\mathbb{R})$ we recall the following definitions:

(a) The *centre* C of $\widehat{f}$ if the pole of the improper plane π_0.
(b) A *diametral plane* of $\widehat{f}$ is the polar plane of a point of π_0.
(c) A *diameter* of $\widehat{f}$ is the polar line of a line of π_0.
(d) A *principal diametral plane* π of $\widehat{f}$ is a diametral plane such that its pole O is the pole of the improper line $\pi \cap \pi_0$ with respect to the absolute Ab_3.
(e) An *axis* of $\widehat{F}$ is the line intersection of two principal diametral planes.

Remark 10.107 To find foci, asymptotes or directrices of a conic $\widehat{f}$ in the affine/euclidean space we can avoid having to determine its canonical form thanks to the above definitions which involve the projective closure of $\widehat{f}$ and its complexification.

We end this chapter by introducing a concept which will be crucial in Chap. 13.

Interior and Exterior of a Real Hyperquadric

Definition 10.108 Let $P \in \mathbb{P}^n(\mathbb{R})$ and let $\widehat{F}$ be a non-singular hyperquadric of $\mathbb{P}^n(\mathbb{R})$, $n > 1$. We say that P is *interior* to $\widehat{F}$ if its polar hyperplane $\pi^{\widehat{F}}(P)$ shares no real points with $\widehat{F}$ (see Definition 10.89). Since the points of $\widehat{F}$ lie in their polar hyperplanes, no interior point belongs to $\widehat{F}$. One may equivalently say that the interior points are those where no real line through them is tangent to $\widehat{F}$. A point $P \notin \mathcal{V}_+(\widehat{F})$ is *exterior* to $\widehat{F}$ if $\mathcal{V}_+(\widehat{F}) \cap \pi^{\widehat{F}}(P)$ has some real points, i.e. through P there is some real line tangent to $\widehat{F}$. We denote the sets of interior (exterior) points by $\mathrm{Int}(\widehat{F})$ ($\mathrm{Ext}(\widehat{F})$ respectively). Thus we have a decomposition of $\mathbb{P}^n(\mathbb{R})$ in three pairwise disjoint subsets

$$\mathbb{P}^n(\mathbb{R}) = \mathrm{Int}(\widehat{F}) \cup \mathcal{V}_+(\widehat{F}) \cup \mathrm{Ext}(\widehat{F}). \tag{10.63}$$

It is easy to see that the above splitting (10.63) has a projective character: if $\varphi\colon \mathbb{P}^n(\mathbb{R}) \to \mathbb{P}^n(\mathbb{R})$ is a projectivity, then

$$\varphi(\mathrm{Int}(\widehat{F})) = \mathrm{Int}(\varphi(\widehat{F})), \quad \varphi(\mathrm{Ext}(\widehat{F})) = \mathrm{Ext}(\varphi(\widehat{F})). \tag{10.64}$$

By the definition of an index of a hyperquadric (see Definition 10.60) we immediately have:

Proposition 10.109 *Let $\widehat{F}$ be a non-singular hyperquadric of $\mathbb{P}^n(\mathbb{R})$, $n > 1$.*

(a) *If $\mathbf{i}(\widehat{F}) = n$, then $\mathrm{Int}(\widehat{F}) = \mathbb{P}^n(\mathbb{R})$. This is the case of $F = x_0^2 + x_1^2 + \cdots + x_n^2$.*
(b) *If $\mathbf{i}(\widehat{F}) \leq n-2$, then $\mathrm{Int}(\widehat{F}) = \emptyset$.*

Therefore we have to focus our attention on the non-singular hyperquadrics $\widehat{F}$ of $\mathbb{P}^n(\mathbb{R})$, $n > 1$, with $\mathbf{i}(\widehat{F}) = n-1$, i.e. with reduced equation $x_0^2 + x_1^2 + \cdots + x_{n-1}^2 - x_n^2 = 0$.

Definition 10.110 In order to study the splitting (10.63) concerning the hyperquadrics of index $n-1$ it is useful to introduce the notion of *sign of a point with respect to a hyperquadric*. Let $\widehat{F}$ be a non-singular hyperquadric of $\mathbb{P}^n(\mathbb{R})$, $n > 1$. Once the representative F is fixed, we may assign to each point $P = [x] \notin \mathcal{V}_+(\widehat{F})$ the sign, positive or negative, of $F(x, x)$ which is independent of the choice of x but depends on the choice of F. We say that two points $P = [x]$ and $Q = [y]$, $P, Q \notin \mathcal{V}_+(\widehat{F})$, have *the same sign* or *opposite signs relative to* $\widehat{F}$ if $F(x, x)F(y, y) > 0$ or $F(x, x)F(y, y) < 0$ respectively. Thus comparing signs is independent of the choice of representative of $\widehat{F}$.

If $\alpha = T_A \in \mathrm{PGL}_n(\mathbb{R})$ and $\widehat{G} = \alpha \cdot \widehat{F}$ (see Lemma 10.46), then for each point $P = [x]$ one has $F(x, x) = G(Ax, Ax)$ so that two points P and Q have the same sign relative to $\widehat{F}$ if and only if $\alpha(P)$ and $\alpha(Q)$ have the same sign relative to $\widehat{G}$, i.e. this notion is a projective invariant.

Lemma 10.111 *Let $\widehat{F}$ be a non-singular hyperquadric of $\mathbb{P}^n(\mathbb{R})$ such that $\mathbf{i}(\widehat{F}) = n - 1$, $n > 1$, then any self-polar $(n + 1)$-hedron S has only one vertex interior to $\widehat{F}$ and all the other vertices are exterior. The interior vertex is called the* lone vertex *of S. Moreover the lone vertex and all other vertices have opposite signs.*

Proof Let $S = \{P_0, \ldots\ldots, P_n\}$ be a self-polar $(n + 1)$-hedron relative to $\widehat{F}$, hence

$$\pi^{\widehat{F}}(P_i) = P_0 + \cdots \widehat{P_i} + \cdots P_n, \qquad i = 0, \ldots, n.$$

We may assume $P_0 = [1, 0, \ldots, 0]$, $P_1 = [0, 1, 0, \ldots, 0], \ldots, P_n = [0, 0, \ldots, 1]$ so that $\widehat{F}$ has the equation

$$x_0^2 + x_1^2 + \cdots + x_{n-1}^2 - x_n^2 = 0.$$

Since $\pi^{\widehat{F}}(P_n)$ has the equation $x_n 0 = 0$ we have $\pi^{\widehat{F}}(P_0) \cap \mathcal{V}_+(\widehat{F}) = \varnothing$ so that P_0 is interior to $\widehat{F}$. If $i < n$, $\pi^{\widehat{F}}(P_i)$ has the equation $x_i = 0$ and $\pi^{\widehat{F}}(P_i) \cap \mathcal{V}_+(\widehat{F}) \neq \varnothing$, so that P_i is exterior to $\widehat{F}$. The last assertion is immediately verified. □

Corollary 10.112 *Let $\widehat{F}$ be a non-singular hyperquadric of $\mathbb{P}^n(\mathbb{R})$ such that $\mathbf{i}(\widehat{F}) = n - 1$, $n > 1$, then*

$$\mathrm{Int}(\widehat{F}) \neq \varnothing, \qquad \mathrm{Ext}(\widehat{F}) \neq \varnothing.$$

If $F = x_0^2 + x_1^2 + \cdots + x_{n-1}^2 - x_n^2$ then

$$\mathrm{Int}(\widehat{F}) = \{[x_0, \ldots, x_n] : x_0^2 + x_1^2 + \cdots + x_{n-1}^2 - x_n^2 < 0\}.$$

Proof The point $[a_0, a_1, \ldots, a_n] \in \mathrm{Int}(\widehat{F})$ if and only if

$$\begin{cases} a_0 x_0 + a_1 x_1 + \cdots + a_{n-1} x_{n-1} - a_n x_n = 0 \\ x_0^2 + x_1^2 + \cdots + x_{n-1}^2 - x_n^2 = 0 \end{cases}$$

has no real roots i.e. $a_0^2 + a_1^2 + \cdots + a_{n-1}^2 - a_n^2 < 0$. □

Corollary 10.113 *Assume the notation and hypothesis of* Lemma 10.111. *If P is a point interior to $\widehat{F}$, then P and each point Q conjugate to P have opposite signs with respect to $\widehat{F}$. Therefore two points both interior (exterior) to $\widehat{F}$ have the same sign, while an interior point and an exterior point have opposite signs.*

Proof Let $Q \in \pi^{\widehat{F}}(P)$. Since $\pi^{\widehat{F}}(P) \cap \mathcal{V}_+(\widehat{F}) = \varnothing$ we have that $Q \notin \mathcal{V}_+(\widehat{F})$. Since $P \notin \pi^{\widehat{F}}(P)$ we may find a self-polar $(n+1)$-hedron containing P and Q (see Proposition 10.85). By Lemma 10.111, Q is exterior because P is interior by assumption and they have opposite signs.

If both P and P' are interior points, then each point $Q \in \pi^{\widehat{F}}(P) \cap \pi^{\widehat{F}}(P')$ has the opposite sign with respect to P and P', which necessarily have the same sign.

If Q is exterior there exists a self-polar $(n+1)$-hedron S with vertex Q and the lone vertex P of S is interior and conjugate to Q such that P and Q have opposite signs.

Finally, if Q and Q' are exterior points, then, being conjugate to two interior points P and P', they have opposite signs to P and P' (which have the same sign) and therefore they have the same sign. □

Proposition 10.114 *Assume the notation and hypothesis of* Lemma 10.111. *If P is a point interior to $\widehat{F}$, then each line through P meets $\widehat{F}$ at two distinct real points. Moreover, each line which meets $\widehat{F}$ at two distinct real points P_1 and P_2 contains both interior and exterior points. More precisely, one open segment with endpoints P_1 and P_2 consists of interior points, while the other consists of exterior points.*

Proof Let $P = [x]$ be an interior point and let r be a line through P. Since $P \notin \pi^{\widehat{F}}(P)$, then $r \cap \pi^{\widehat{F}}(P) = Q \neq P$. Therefore P and $Q = [y]$ have opposite signs by Corollary 10.113, i.e. $F(x,x)F(y,y) < 0$. Since P and Q are conjugate, $F(x,y) = 0$ so that the equation $F(x,x)\lambda^2 + F(y,y)\mu^2 = 0$ has two distinct real solutions, i.e. $\mathcal{V}_+(\widehat{F}) \cap r$ consists of two distinct real points.

Let r be a line which meets $\widehat{F}$ at two distinct real points Q_1 and Q_2. Let $P \in r$, $P \neq Q_1, Q_2$. Take P' as the fourth harmonic of Q_1, Q_2 and P. Then P and P' are conjugate (by Proposition 10.90-(b)) so that one is interior and the other is exterior. If $P = [x]$ and $P' = [y]$ are two points belonging to an open segment with endpoints P_1 and P_2, then by (13.15) (in the proof of Lemma 13.4-(b)) $F(x,x)F(y,y) > 0$ so that P and P' have the same sign and the last claim follows from Corollary 10.113. □

10.4 Exercises

Exercise 10.1 Prove the formula $\dim_K({}_{\mathrm{h}}K[x_0,\ldots,x_n]_d) = \binom{n+d}{d}$. (Hint: find the number of monomials $x_0^{i_0}x_1^{i_1}\cdots x_n^{i_n}$, with $i_0 + i_1 + \cdots + i_n = d$, $i_j \geq 0$.)

Exercise 10.2 Let σ be a permutation of $\{0,1,\ldots,n\}$ $(n \geq 1)$, and $f\colon \mathbb{P}^n(K) \to \mathbb{P}^n(K)$ be the map defined by $f([x_0,x_1,\ldots,x_n]) = [x_{\sigma(0)},x_{\sigma(1)},\ldots,x_{\sigma(n)}]$, $\forall\,[x_0,x_1,\ldots,x_n] \in \mathbb{P}^n(K)$. Show that $f \in \mathrm{PGL}_n(K)$.

Exercise 10.3 Let $F = x_0^3 + x_1^3 + x_2^3 + a(x_0+x_1+x_2)^3$. Find all complex numbers a such that the curve $\widehat{F}$ of $\mathbb{P}^2(\mathbb{C})$ has singular points. and determine such points.

Exercise 10.4 Find the singular locus $\mathrm{Sing}(\widehat{F})$ of the Steiner surface $\widehat{F}$ of $\mathbb{P}^3(\mathbb{C})$, where $F = x_0^2x_1^2 + x_1^2x_2^2 + x_0^2x_2^2 - x_0x_1x_2x_3$.

Exercise 10.5 Find the singular locus $\mathrm{Sing}(\widehat{F})$ of the dual Steiner surface $\widehat{F}$ of $\mathbb{P}^3(\mathbb{C})$, where $F = x_1x_2x_3 + x_0x_2x_3 + x_0x_1x_3 + x_0x_1x_2$.

Exercise 10.6 Let $\widehat{F} \in \mathcal{H}_3^{\mathrm{p}}(2, \mathbb{C})$ be a curve of $\mathbb{P}^2(\mathbb{C})$ of degree 3 which has three non-collinear singular points A, B, C. Show that $\mathcal{V}(\widehat{F})$ is the union of the lines AB, BC and CA.

Exercise 10.7 (Cone Circumscribed to a Hyperquadric) Let $\widehat{F}$ be a hyperquadric of $\mathbb{P}^n(K)$, where $F = xCx^t$, $x = (x_0, \ldots, x_n)$ and $C = (c_{ij})_{i,j=1,\ldots,n}$. Fix a point $Y = [y_0, \ldots, y_n] \in \mathbb{P}^n(K) \setminus \mathcal{V}_+(\widehat{F})$. Show that the equation

$$(yC^tx)^2 - (yC^ty)(xC^tx) = 0, \qquad y = (y_0, \ldots, y_n), \tag{10.65}$$

represents the locus of the lines passing through Y and tangent to $\widehat{F}$ called the *cone circumscribed to* $\widehat{F}$ *from* Y.

Exercise 10.8 Prove that a non-degenerate hyperquadric $\widehat{F}$ of $\mathbb{P}^n(\mathbb{C})$ contains infinitely many lines. (Hint: use Corollary 10.83 (Bertini's theorem).)

Exercise 10.9 Prove that a hypersurface of degree 1 is uniquely determined by its support.

Exercise 10.10 The *Fermat cubic* $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$ of $\mathbb{P}^3(\mathbb{C})$ contains exactly 27 lines. (Hint: Up to a permutation of coordinates, every line in $\mathbb{P}^3(\mathbb{C})$ is given by two linear equations of the form

$$\begin{cases} x_0 = a_2x_2 + a_3x_3 \\ x_1 = b_2x_2 + b_3x_3 \end{cases}$$

for suitable $a_2, a_3, b_2, b_3 \in \mathbb{C}$.)

Exercise 10.11 Prove that for any non-degenerate ruled quadric $\widehat{F}$ of $\mathbb{P}^3(\mathbb{C})$ and $\mathbb{P}^3(\mathbb{R})$ there is a projective frame relative to which $\widehat{F}$ has the equation $x_0x_1 - x_2x_3 = 0$.

Exercise 10.12 Let $P_1, \ldots, P_5$ be five points of $\mathbb{P}^2(K)$ in general position.

1. Prove that there is one and only one non-degenerate conic passing through $P_1, \ldots, P_5$. (Hint: choose a convenient projective frame.)
2. Let $\{P_0, \ldots, P_2, U\}$ be a projective frame of $\mathbb{P}^2(K)$. Prove that there is one and only one non-degenerate conic passing through P_1, P_2, U whose tangent lines at P_1 and P_2 meet at P_0.

Exercise 10.13 Let $\widehat{f}$ be an affine non-degenerate conic, $\widehat{x_0(f)}$ its projective closure and $r_0 : x_0 = 0$ the line at infinity. Prove that:

(a) $\widehat{f}$ is a parabola if and only if $\widehat{f}_\infty := \widehat{{}^{x_0}(f)} \cap r_0$ is a double point, i.e. r_0 is tangent to $\widehat{{}^{x_0}(f)}$.
(b) $\widehat{f}$ is an ellipse if and only if $\widehat{f}_\infty$ is a pair of complex-conjugate points.
(c) $\widehat{f}$ is a hyperbola if and only if $\widehat{f}_\infty$ is a pair of distinct real points.

Exercise 10.14 Show that a conic $\widehat{f}$ of $\mathbb{E}^2(\mathbb{R})$ is an equilateral hyperbola (i.e. with orthogonal asymptotes) if and only if the cyclic points I and J are conjugate with respect to $\widehat{{}^{x_0}(f)_{\mathbb{C}}}$.

Exercise 10.15 Prove that the axes of a hyperbola are the bisector lines of the asymptotes. The converse is true for equilateral hyperbolas.

Exercise 10.16 Find centres, axes, asymptotes, foci, directrices of the following real conics:

$$\begin{array}{ll} x^2+2y^2-3xy+8=0, & x^2+xy+y^2=1, \\ x^2+xy-y^2=1, & x^2+2xy+y^2-4x=0, \\ x^2+y^2-(x+y-1)^2=0, & x^2+4xy+4y^2-24x+32y=0. \end{array}$$

Exercise 10.17 Let $\widehat{f}$ be an affine quadric, let $\widehat{{}^{x_0}f}$ be its projective closure and $\widehat{f}_\infty := \widehat{{}^{x_0}f} \cap \pi_0$ be its conic at infinity. Prove that the classification of non-degenerate quadrics of $\mathbb{A}^3(\mathbb{R})$ can be rephrased as follows (compare with the one given in Chapter 5):

(a) $\widehat{f}$ is a elliptic paraboloid if $\widehat{f}_\infty$ is a pair of complex-conjugate lines.
(b) $\widehat{f}$ is a hyperbolic paraboloid if $\widehat{f}_\infty$ is a pair of real lines.
(c) $\widehat{f}$ is a real ellipsoid if $\widehat{f}_\infty$ is a non-degenerate conic such that $\mathcal{V}(\widehat{f}_\infty) = \varnothing$ and $\mathcal{V}(\widehat{f}) \neq \varnothing$.
(d) $\widehat{f}$ is an imaginary ellipsoid if $\widehat{f}_\infty$ is a non-degenerate conic such that $\mathcal{V}(\widehat{f}_\infty) = \varnothing$ and $\mathcal{V}(\widehat{f}) = \varnothing$.
(e) $\widehat{f}$ is one-sheeted hyperboloid if $\widehat{f}_\infty$ is a non-degenerate conic and $\widehat{f}$ is ruled.
(f) $\widehat{f}$ is a two-sheeted hyperboloid if $\widehat{f}_\infty$ is a non-degenerate conic and $\widehat{f}$ is not ruled.

Exercise 10.18 Determine the possible centres, principal diametral planes and axes of symmetry of the non-degenerate quadrics of Exercise 5.11.

Exercise 10.19 (Circles on a Real Non-degenerate Quadric) Let $\widehat{f}$ be a non-degenerate quadric of $\mathbb{E}^3(\mathbb{R})$ which is not a sphere. Let $\widehat{f}_\infty$ its conic at infinity, and let

$$\widehat{f}_\infty \cap \mathrm{Ab}_3 = \{A, \overline{A}, B, \overline{B}\}.$$

First suppose $A \neq B$. Then we have two pencils of parallel planes passing through the real line $A\overline{A}$ and the real line $B\overline{B}$. These planes cut $\mathcal{V}(\widehat{f})$ into circles (why?) and these are the only circles contained in $\mathcal{V}(\widehat{f})$. If $A = B$, i.e. $\widehat{f}_\infty$ and Ab_3 are tangent at $A = B$ and at $\overline{A} = \overline{B}$, then there is only one pencil $\Phi_2(r)$ of parallel planes

passing through the real line $r := A\overline{A}$ that cut $\mathcal{V}(\widehat{f})$ into circles (the only circles on $\mathcal{V}(\widehat{f})$). In this case $\widehat{f}$ has a proper centre (indeed $\widehat{f}_\infty$ is a non-degenerate conic) and it is of revolution around the line passing through the centre and orthogonal to the planes of $\Phi_2(r)$.

Find all circles lying on the quadric $z^2 + xy + 1 = 0$.

Exercise 10.20 Here is a procedure to determine a self-polar tetrahedron for a non-degenerate quadric $\widehat{F}$ of $\mathbb{P}^3(K)$.

1. Let $A \notin \mathcal{V}_+(\widehat{F})$ and $\pi_A := \pi^{\widehat{F}}(A)$ its polar plane.
2. Take $B \in \pi_A \setminus \mathcal{V}_+(\widehat{F})$. The polar plane π_B passes through A and $\pi_B \cap \pi_A$ is a line $r \not\subset \mathcal{V}_+(\widehat{F})$, otherwise π_B would be tangent to $\widehat{F}$ and $B \in \mathcal{V}_+(\widehat{F})$.
3. Take $C \in r \setminus \mathcal{V}_+(\widehat{F})$. The polar plane π_C passes through AB and $\pi_C \cap \pi_A$ is a line s not lying on $\widehat{F}$ (otherwise π_C would be tangent to $\widehat{F}$ and $C \in \mathcal{V}_+(\widehat{F})$).
4. Let $D = r \cap s$ and π_D its polar plane. Then $D \notin \mathcal{V}_+(\widehat{F})$, otherwise (since $\pi_D \supset \{A, B, C\}$) π_D would be tangent to $\widehat{F}$ and we should have $D \in \pi_D$ and r, $s \subset \pi_D$ which implies $\pi_D = \pi_A$, which is absurd.

The quadruple $\{A, B, C, D\}$ is a self-polar tetrahedron for $\widehat{F}$, indeed

$$\{B, C, D\} \subset \pi_A,\ \{A, C, D\} \subset \pi_B,\ \{B, A, D\} \subset \pi_C,\ \{A, B, C\} \subset \pi_D.$$

Determine a self-polar tetrahedron for the quadric $3x_0^2 + x_1^2 + 3x_2^2 + 2x_0x_2 + 2x_0x_3 - 2x_1x_3 + 6x_2x_3 + 3x_3^2 = 0$.

Chapter 11
Bézout's Theorem for Curves of $\mathbb{P}^2(K)$



In this chapter we shall prove Bézout's theorem for curves of $\mathbb{P}^2(K)$, where K is a fixed algebraically closed field of arbitrary characteristic. We begin stating the weak form of Bézout's theorem (see [1, 13, 26], or [30]):

Theorem 11.1 (Weak Form of Bézout's Theorem) *If two projective plane curves $\mathcal{C}$ and $\mathcal{D}$ of degrees $m \geq 1$ and $n \geq 1$ have no common component, then*

$$0 < \#(\mathcal{C} \cap \mathcal{D}) \leq mn. \tag{11.1}$$

Before giving a general proof of Theorem 11.1, we prove its very special case by using projective classification of conics.

11.1 Proof of a Simple Case of Weak Bézout's Theorem

We suppose that $\mathcal{C} = \widehat{F}$ is a curve of degree ≤ 2, $\mathcal{D} = \widehat{G}$ is a curve of degree n and $\mathrm{char}(K) \neq 2$. By Proposition 10.27 we may suppose that F is of the following type:

(i) $F = x_0$ ($\mathcal{C}$ is a line).
(ii) $F = x_0^2 + x_1^2 + x_2^2 = 0$ ($\mathcal{C}$ is a non-degenerate conic).
(iii) $F = x_0^2 + x_1^2$ ($\mathcal{C}$ two distinct lines).
(iv) $F = x_0^2$ ($\mathcal{C}$ is a double line).

(i) We can take $x_0 = 0$, $x_1 = s$, $x_2 = t$, with $[s, t] \in \mathbb{P}^1(K)$ as parametric equations of $\mathcal{C}$. Therefore finding the points of $\mathcal{C} \cap \mathcal{D}$ is equivalent to determining the roots (in $\mathbb{P}^1(K)$) of the equation $G(0, s, t) = 0$. Since x_0 does not divide $G(x_0, x_1, x_2)$, the homogeneous polynomial $H(s, t) := G(0, s, t)$ of degree n in s and t is not identically zero. Therefore by Proposition 10.5, if p

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is the number of the roots of $H(s,t)$, then $1 \leq p \leq n$ so that $\#(\mathcal{C} \cap \mathcal{D}) \leq n$. The above argument works also if $\mathrm{char}(K) = 2$. We can also say that x_0 divides $G(x_0, x_1, x_2)$ if and only if $\mathcal{C} \subseteq \mathcal{D}$.

(ii) Since $\mathrm{char}(K) \neq 2$ we can write $F = x_0^2 + x_1^2 + x_2^2 = x_1^2 - (x_0 + ix_2)(ix_2 - x_0)$ with $i \in K$ such that $i^2 = -1$. By means of the change of coordinates $y_0 = x_0 + ix_2$, $y_1 = x_1$ and $y_2 = ix_2 - x_0$ we can suppose that the equation of $\mathcal{C}$ is

$$x_1^2 - x_0 x_2 = 0. \tag{11.2}$$

It is easy to see that

$$x_0 = s^2,\ x_1 = st,\ x_2 = t^2, \qquad \text{with } [s,t] \in \mathbb{P}^1(K) \tag{11.3}$$

are parametric equations of $\mathcal{C}$ so that to find the points of $\mathcal{C} \cap \mathcal{D}$ is equivalent to solving the equation of degree $2n$ in s and t ($[s,t] \in \mathbb{P}^1(K)$)

$$H(s,t) := G(s^2, st, t^2) = 0. \tag{11.4}$$

Claim $H(s,t) \equiv 0$ if and only if F divides G in $K[x_0, x_1, x_2]$.

To prove the above claim consider $F = F(x_0, x_1, x_2) = x_1^2 - x_0 x_2$ and $G = G(x_0, x_1, x_2)$ as polynomials in x_1 with coefficients in $K[x_0, x_2]$. In the euclidean domain $K[x_0, x_2][x_1]$ we have (since F is monic with respect to x_1)

$$G(x_0, x_1, x_2) = Q(x_0, x_1, x_2) F(x_0, x_1, x_2) + R(x_0, x_1, x_2),$$

where $Q = Q(x_0, x_1, x_2) \in K[x_0, x_1, x_2]$ and $R(x_0, x_1, x_2) = a(x_0, x_2)x_1 + b(x_0, x_2)$ with $a(x_0, x_2), b(x_0, x_2) \in K[x_0, x_2]$. By (11.3) we get

$$\begin{aligned} 0 \equiv H(s,t) = G(s^2, st, t^2) &= Q(s^2, st, t^2) \cdot 0 + a(s^2, t^2)st + b(s^2, t^2) \\ &= a(s^2, t^2)st + b(s^2, t^2), \end{aligned}$$

i.e. $a(s^2, t^2)st + b(s^2, t^2) \equiv 0$. Since $a(s^2, t^2)st$ contains only powers of odd degree in s and t while $b(s^2, t^2)$ contains only powers of even degree in s and t, we have $a(s,t) \equiv 0$ and $b(s,t) \equiv 0$ which proves the claim.

By assumption F does not divide G, hence $H(s,t)$ is not identically zero. Since we know that $\mathcal{V}_+(H(s,t))$ is in a bijective correspondence with $\mathcal{C} \cap \mathcal{D}$ and $0 < \#(\mathcal{V}_+(H(s,t))) \leq 2n$ by Proposition 10.5, then $0 < \#(\mathcal{C} \cap \mathcal{D}) \leq 2n$. The assumption $\mathrm{char}(K) \neq 2$ is essential since the inverse of the projectivity $y_0 = x_0 + ix_2$, $y_1 = x_1$, $y_2 = ix_2 - x_0$ is the projectivity $y_0 = \frac{1}{2}x_0 + \frac{1}{2}x_2$, $y_1 = x_1$, $y_2 = \frac{1}{2i}x_0 - \frac{1}{2i}x_2$.

(iii) In this case the degenerate conic is the union of the lines $L_1 := x_0 + ix_1 = 0$ and $L_2 := x_0 - ix_1 = 0$. By i) $\#(\mathcal{V}_+(L_1) \cap \mathcal{D}) \leq n$ and $\#(\mathcal{V}_+(L_2) \cap \mathcal{D}) \leq n$ so that $\#(\mathcal{C} \cap \mathcal{D}) \leq 2n$.

(iv) In this case $\mathcal{C} = \mathcal{V}_+(x_0^2) = \mathcal{V}_+(x_0)$, i.e. $\mathcal{C} \cap \mathcal{D} = \mathcal{V}_+(x_0) \cap \mathcal{D}$, hence by i) $\#(\mathcal{V}_+(x_0) \cap \mathcal{D}) \leq n < 2n$.

11.1.1 Two Applications of Bézout's Theorem

Proposition 11.2 *Let $P_1, \ldots, P_5$ five points of $\mathbb{P}^2(K)$. Then:*

(i) *There exists at least one conic $\mathcal{C}$ passing through $P_1, \ldots, P_5$.*
(ii) *A conic $\mathcal{C}$ passing through $P_1, \ldots, P_5$ is non-degenerate if and only if no three of $P_1, \ldots, P_5$ are collinear. Moreover such a non-degenerate conic is unique.*
(iii) *A degenerate conic $\mathcal{C}$ passing through $P_1, \ldots, P_5$ is unique if and only if no four of $P_1, \ldots, P_5$ are collinear.*
(iv) *There exists at most one conic $\mathcal{C}$ passing through $P_1, \ldots, P_5$ if no four of $P_1, \ldots, P_5$ are collinear.*

Proof

(i) It follows from Proposition 10.45.
(ii) Since a degenerate conic of $\mathbb{P}^2(K)$ is a pair of distinct lines or a double line, at least three points among $P_1, \ldots, P_5$ are collinear, hence if no three of $P_1, \ldots, P_5$ are collinear every conic through $P_1, \ldots, P_5$ is non-degenerate. Suppose now that $\mathcal{D} = \widehat{G}$ is another conic passing through $P_1, \ldots, P_5$. We have just proved that $\mathcal{D}$ is non-degenerate. By Bézout's theorem, $F \mid G$ and $G \mid F$ so that $\widehat{G} = \widehat{F}$, i.e. $\mathcal{C} = \mathcal{D}$.
(iii) Let $\mathcal{C} = \widehat{F}$ and $\mathcal{D} = \widehat{G}$ be two conics passing through $P_1, \ldots, P_5$, then by ii), both the conics are degenerate, say

$$F = L_1 \cdot L_2, \quad G = L_1' \cdot L_2',$$
$$L_1, L_2, L_1', L_2' \text{ homogeneous polynomials of degree 1 in } x_0, x_1, x_2$$

Since $\#(\mathcal{C} \cap \mathcal{D}) \geq 5$, by Theorem 11.1), F and G have a common factor of degree 1, e.g. $F = L_1 \cdot L_2$ and $G = L_1 \cdot L_2'$. Therefore $\mathcal{C} \neq \mathcal{D}$ if and only if the lines $r : L_2 = 0$ and $s : L_2' = 0$ are distinct if and only if $r \cap s$ is a point if and only if the common line $L_1 = 0$ passes through at least four points among $P_1, \ldots, P_5$.
(iv) It follows from (ii) and (iii).

□

Corollary 11.3 *Through four non-collinear points of $\mathbb{P}^2(K)$ there passes a unique pencil of conics.*

Definition 11.4 If $m \geq 3$, an *m-gon* $\mathcal{P}_m$ is a set of m distinct points cyclically ordered, i.e. $P_1, P_2, \ldots, P_{m+1} = P_1$, called *vertices* of $\mathcal{P}_m$, and m lines $r_i = P_i P_{i+1}$, $i = 1, \ldots, m$ called *sides* of $\mathcal{P}_m$ in such a way that no three vertices

are collinear and no three sides are concurrent. Reversing the ordering of the vertices does not change the m-gon. Obviously an m-gon is determined by its cyclically ordered vertices, and also by its cyclically ordered sides. The 3-gons are just triangles, a 4-gon is a quadrilateral and for $m = 5, 6 \ldots$, m-gons are called pentagons, hexagons, etc. If m is even, by definition,

$$(P_1, P_{(m/2)+1}),\ (P_2, P_{(m/2)+2}), \ldots, (P_{m/2}, P_m)$$

are the pairs of *opposite vertices* and

$$(r_1, r_{(m/2)+1}),\ (r_2, r_{(m/2)+2}), \ldots, (r_{m/2}, r_m)$$

are the pairs of *opposite sides* of $\mathcal{P}_m$. An m-gon $\mathcal{P}_m$ is said to be *inscribed* in a non-degenerate conic $\mathcal{C}$ if and only if all its vertices belong to $\mathcal{C}$, in which case it is equivalently said that $\mathcal{C}$ is *circumscribed* to $\mathcal{P}_m$. If all sides of $\mathcal{P}_m$ are tangent to a non-degenerate conic $\mathcal{C}$, then it is said that $\mathcal{P}_m$ is circumscribed to $\mathcal{C}$ and also that $\mathcal{C}$ is inscribed in $\mathcal{P}_m$.

In particular, if $P_1, P_2, P_3, P_4, P_5, P_6$ are the vertices of an hexagon, then the three pairs of vertices (P_1, P_4), (P_2, P_5), (P_3, P_6) are the opposite vertices and

$$(P_1P_2, P_4P_5), \quad (P_2P_3, P_5P_6), \quad (P_3P_4, P_6P_1),$$

are the opposite sides. The three lines P_1P_4, P_2P_5 and P_3P_6 joining opposite vertices are called the *diagonals* of the hexagon.

We now use Bézout's theorem to prove a classical (projective) Pascal's theorem:

Theorem 11.5 (Pascal) *Let $\mathcal{C}$ be a non-degenerate conic of $\mathbb{P}^2(K)$, and $\mathcal{P}_6 = (P_1, P_2, \ldots, P_6)$ be a hexagon inscribed in $\mathcal{C}$. Then the points $A := P_1P_2 \cap P_4P_5$, $B := P_3P_4 \cap P_1P_6$ and $C := P_2P_3 \cap P_5P_6$ are collinear (see Fig. 11.1). In other words, the intersection points of the pairs of the opposite sides of a hexagon inscribed in a non-degenerate conics are collinear (Fig. 11.2).*

Proof (Plücker) Let $L_{ij} = 0$ be the equation of the line P_iP_j, $i, j \in \{1, 2, \ldots, 6\}$, $i \neq j$. Consider the polynomials

$$F_t := L_{12}L_{34}L_{56} + tL_{16}L_{23}L_{45}, \qquad t \in K.$$

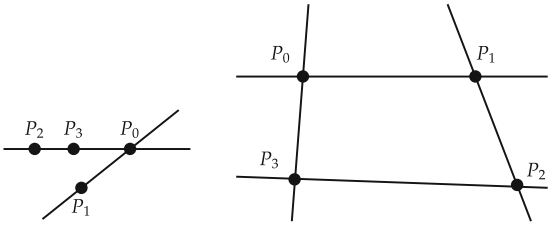


Fig. 11.1 Degenerate conics

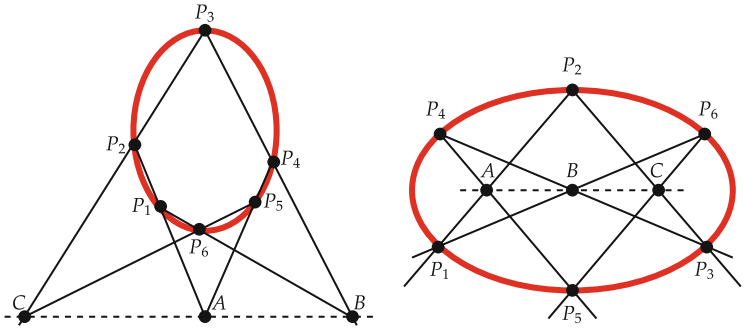


Fig. 11.2 Projective Pascal's theorem

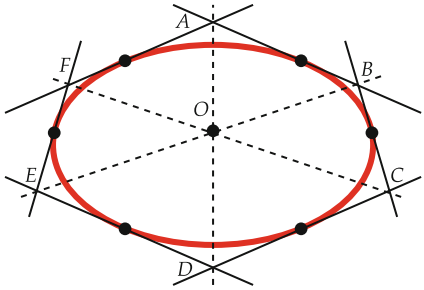


Fig. 11.3 Brianchon's theorem

For every $t \in K$, F_t is a polynomial of degree 3, indeed F_t is not identically 0 for all $t \in K$. If F_t is identically 0 for some $t \in K$, then we should have $L_{12}L_{34}L_{56} = -tL_{16}L_{23}L_{45}$, in contradiction with the $K[x_0, x_1, x_2]$'s factoriality.

First observe that $P_1, \ldots, P_6, A, B, C \in \mathcal{V}_+(F_t)$ and that A, B, C do not lie on $\mathcal{C}$ (if, e.g. $A \in \mathcal{C}$, then P_1P_2 meets $\mathcal{C}$ at three points, contradicting Theorem 11.1). Fix a point $P \in \mathcal{C}$, $P \neq P_j$, $j = 1, \ldots, 6$. We can find $t_P \in K$ such that $F_{t_P}(P) = 0$. Consider the cubic $\mathcal{D} = \widehat{F_{t_P}}$. Then

$$\mathcal{D} \cap \mathcal{C} \supset \{P_1, P_2, \ldots, P_6, P\}.$$

By Theorem 11.1, if $\mathcal{C} = \widehat{F}$, we have $F_{t_P} = F \cdot L$, where L is a homogeneous polynomial of degree 1. Since $A, B, C \in \mathcal{V}_+(F_{t_P}) \setminus \mathcal{V}_+(F)$, we must have $L(A) = L(B) = L(C) = 0$, i.e. A, B and C belong to the line of equation $L = 0$.

□

Theorem 11.6 (Brianchon) *In a projective plane $\mathbb{P}^2(K)$, consider a hexagon whose six sides are tangent to a non-degenerate conic $\mathcal{C}$. Then the three diagonals of the hexagon intersect at the same point* (*see Fig. 11.3*).

Proof Brianchon's theorem is the dual statement of Pascal's theorem. By Proposition 10.71 and Theorem 10.53, i) we may suppose that $\mathcal{C}$ is defined by the equation $x_0^2 + x_1^2 + x_2^2 = 0$. In this case the polarity $\pi^{\mathcal{C}}\colon \mathbb{P}^2(K) \to \mathbb{P}^2(K)^*$ is given by

$$\pi^{\mathcal{C}}([a_o, a_1, a_2]) = \mathcal{V}_+(a_0x_0 + a_1x_1 + a_2x_2), \qquad \forall [a_o, a_1, a_2] \in \mathbb{P}^2(K).$$

By Theorem 10.74 $\pi^{\mathcal{C}}$ is a projective isomorphism that transforms the circumscribed Brianchon hexagon of $\mathbb{P}^2(K)$ into the inscribed Pascal hexagon of $\mathbb{P}^2(K)^*$ and conversely. □

11.2 The General Bézout's Theorem and Further Applications

Since Bézout's theorem is a fundamental result in the theory of projective curves of $\mathbb{P}^2(K)$ (where K is an algebraically closed field of arbitrary characteristic), we wish to prove its general form, which requires more powerful tools. We shall also discuss some important applications.

11.3 The Resultant of Two Polynomials

Let A be a factorial domain, and let $f, g \in A[X]$ be two non-constant polynomials:

$$f = a_0X^n + a_1X^{n-1} + \cdots + a_n \quad \text{and} \quad g = b_0X^m + b_1X^{m-1} + \cdots + b_m, \tag{11.5}$$

with $n, m \geq 1$ and $a_0b_0 \neq 0$.

Proposition 11.7 *The polynomials f and g have a non-constant common divisor in $A[X]$ if and only if there exist two non-zero polynomials $f_1, g_1 \in A[X]$ such that*

$$fg_1 + gf_1 = 0, \quad \textit{with} \quad \deg(f_1) \leq n-1 \quad \textit{and} \quad \deg(g_1) \leq m-1. \tag{11.6}$$

Proof Assume that there exists a polynomial $h \in A[X]$ of degree ≥ 1 such that $f = hf_1$ and $g = -hg_1$, with $f_1, g_1 \in A[X]$. Then $\deg(f_1) \leq n-1$ and $\deg(g_1) \leq m-1$, and clearly $fg_1 + gf_1 = 0$.

Conversely, assume that (11.6) holds. Since by Gauss' theorem the ring $A[X]$ is factorial, we can write $f = aF_1^{s_1} \cdots F_p^{s_p}$, with $a \in A$, $F_1, \ldots, F_p$ pairwise coprime irreducible polynomials in $A[X]$ of degrees ≥ 1, and $s_i \geq 1$, $i = 1, \ldots, p$. Since $A[X]$ is factorial, it follows that F_i is prime in $A[X]$, $i = 1, \ldots, p$. Then we claim that there exists an $i \in \{1, \ldots, p\}$ such that F_i divides g. Indeed, otherwise from

the equality $fg_1 = -gf_1$ we deduce that $F_i^{s_i}$ divides f_1, $i = 1, \ldots, p$, hence $F = F_1^{s_1} \cdots F_p^{s_p}$ divides f_1, which is absurd because $\deg(f_1) < n = \deg(F)$. □

Now write

$$\begin{aligned} f_1 &= p_0 X^{n-1} + p_1 X^{n-2} + \cdots + p_{n-1} \\ \text{and } \ g_1 &= q_0 X^{m-1} + q_1 X^{m-2} + \cdots + q_{m-1}. \end{aligned} \tag{11.7}$$

Then (11.6), (11.5) and (11.7) imply

$$\begin{cases} a_0 q_0 + b_0 p_0 & = 0 \\ a_1 q_0 + a_0 q_1 + b_1 p_0 + b_0 p_1 & = 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots & \ldots \\ a_n q_{m-1} + b_m p_{n-1} & = 0. \end{cases} \tag{11.8}$$

Then (11.8) can be considered as a system of $n + m$ homogeneous equations in the $n + m$ indeterminates $q_0, \ldots, q_{m-1}, p_0, \ldots, p_{n-1}$ having the $(m + n) \times (m + n)$ matrix

$$\Omega = \underbrace{\begin{pmatrix} a_0 & 0 & 0 & \ldots & 0 \\ a_1 & a_0 & 0 & \ldots & 0 \\ a_2 & a_1 & a_0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & a_n \end{pmatrix}}_{m \text{ columns}} \underbrace{\begin{pmatrix} b_0 & 0 & 0 & \ldots & 0 \\ b_1 & b_0 & 0 & \ldots & 0 \\ b_2 & b_1 & b_0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \ldots & b_m \end{pmatrix}}_{n \text{ columns}}$$

Now we define

$$\mathfrak{R}(f, g) := \det(\Omega^t) = \begin{vmatrix} a_0 & a_1 & a_2 & \ldots & \ldots & \ldots & 0 \\ 0 & a_0 & a_1 & \ldots & \ldots & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & a_0 & a_1 & \ldots & a_n \\ b_0 & b_1 & b_2 & \ldots & b_m & 0 & \ldots \\ 0 & b_0 & b_1 & b_2 & \ldots & b_m & \ldots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ 0 & 0 & 0 & \ldots & \ldots & \ldots & b_m \end{vmatrix},$$

where Ω^t is the transpose of Ω and where the entries $a_0, \ldots, a_n$ occur only in the first m rows, while the entries $b_0, \ldots, b_m$ occur in the last n rows. The entries on the principal diagonal are a_0 (repeated m times) and b_m (repeated n times). For example, we have

$$f(x) = a_0x + a_1,\ \ g(x) = b_0x + b_1,$$
$$f(x) = a_0x^3 + a_1x^2 + a_2x + a_3,\ \ g(x) = b_0x^2 + b_1x + b_2,$$
$$\mathfrak{R}(f, g) = \begin{vmatrix} a_0 & a_1 \\ b_0 & b_1 \end{vmatrix} = a_0b_1 - a_1b_0,$$
$$\mathfrak{R}(f, g) = \begin{vmatrix} a_0 & a_1 & a_2 & a_3 & 0 \\ 0 & a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & 0 & 0 \\ 0 & b_0 & b_1 & b_2 & 0 \\ 0 & 0 & b_0 & b_1 & b_2 \end{vmatrix}.$$

Definition 11.8 Under the above notation the determinant $\mathfrak{R}(f, g) \in A$ is called the *Euler–Sylvester resultant*, or simply the *resultant*, of the polynomials $f, g \in A[X]$. The resultant has the following symmetry property:

$$\mathfrak{R}(g, f) = (-1)^{mn}\mathfrak{R}(f, g). \tag{11.9}$$

Furthermore, if $\lambda, \mu \in A$, $\lambda\mu \neq 0$, we have

$$\mathfrak{R}(\lambda f, \mu g) = \lambda^m\mu^n\mathfrak{R}(f, g). \tag{11.10}$$

Proposition 11.7 and the above discussion yield (see Theorem 1.30):

Corollary 11.9 *Under the above assumptions, the polynomials $f, g \in A[X]$ have a non-constant common factor if and only if $\mathfrak{R}(f, g) = 0$. In particular, if A is a field, f and g have a common root in some field $K \supset A$ if and only if $\mathfrak{R}(f, g) = 0$.*

Proposition 11.10 *Under the above assumptions there exist two polynomials $P, Q \in A[X]$ such that*

$$\mathfrak{R}(f, g) = Pf + Qg. \tag{11.11}$$

Moreover, one can choose P and Q such that $\deg(P) \leq m - 1$ *and* $\deg(Q) \leq n - 1$.

Proof We need the following simple fact: fix a positive integer t and let

$$P_i = \sum_{j=0}^{t} a_{ij}X^j \in A[X],\quad i = 1, \ldots, t + 1$$

be $t+1$ polynomials of degree $\leq t$. Then

$$\begin{vmatrix} a_{10} & a_{11} & \dots & a_{1t} \\ a_{20} & a_{21} & \dots & a_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ a_{t+1,0} & a_{t+1,1} & \dots & a_{t+1,t} \end{vmatrix} = \begin{vmatrix} P_1 & a_{11} & \dots & a_{1,t} \\ P_2 & a_{21} & \dots & a_{2,t} \\ \vdots & \vdots & \ddots & \vdots \\ P_{t+1} & a_{t+1,1} & \dots & a_{t+1,t} \end{vmatrix} = \sum_{i=1}^{t+1} h_i P_i,$$

with $h_i \in A$. Taking in particular, $t = n + m - 1$ and

$$P_i := X^{i-1} f, \quad i = 1, 2, \dots, m, \quad \text{and} \quad P_{m+j} := X^{j-1} g, \quad j = 1, \dots, n,$$

we get

$$\mathfrak{R}(f, g) = h_1 f + \cdots + h_m X^{m-1} f + h_{m+1} g + \cdots + h_{n+m} X^{n-1} g.$$

Putting $P := h_1 + \cdots + h_m X^{m-1}$ and $Q := h_{m+1} + \cdots + h_{n+m} X^{n-1}$ we get $\mathfrak{R}(f, g) = Pf + Qg$, with $P, Q \in A[X]$. □

Definition 11.11 The determinant $\mathfrak{R}(f, g)$ is a sum of terms of the following type:

$$\pm \; a_{i_1} \cdots a_{i_m} b_{j_1} \cdots b_{j_n}.$$

We shall say that this term has *weight* $i_1 + \cdots + i_m + j_1 + \cdots + j_n$. In particular the weight of each a_i is i, and the weight of each b_j is j.

Proposition 11.12 *Every term in the development of the determinant* $\mathfrak{R}(f, g)$ *has weight* mn.

Proof Let us denote by (i, j) the entry on i-th line and j-th column in the $(n+m) \times (n+m)$ matrix

$$\Omega^t = \begin{pmatrix} a_0 & a_1 & a_2 & \dots & \dots & \dots & \dots & 0 \\ 0 & a_0 & a_1 & \dots & \dots & \dots & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & a_0 & a_1 & \dots & \dots & a_n \\ b_0 & b_1 & b_2 & \dots & b_m & 0 & \dots & 0 \\ 0 & b_0 & b_1 & b_2 & \dots & b_m & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \dots & \dots & \dots & \dots & b_m \end{pmatrix}.$$

If we adopt the convention that $a_i = 0$ if $i < 0$ or if $i > n$ (respectively, $b_j = 0$ if $j < 0$ or if $j > m$), then clearly,

$$(i, j) := \begin{cases} a_{j-i} \text{ if } i \leq m \\ b_{j-i+m} \text{ if } i \geq m+1. \end{cases} \tag{11.12}$$

Now a general term in the development of the determinant of $\mathfrak{R}(f, g)$ is of the form

$$\pm\ (1, j_1)(2, j_2)\dots(n+m, j_{n+m}), \tag{11.13}$$

where

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n+m \\ j_1 & j_2 & \cdots & j_{n+m} \end{pmatrix}$$

is an arbitrary permutation of $\{1, 2, \dots, n+m\}$. Now, using (11.12), the weight of (11.13) is given by

$$\begin{aligned} &[(j_1 - 1) + \cdots + (j_m - m)] \\ &\quad + \underbrace{[(j_{m+1} - (m+1) + m) + \cdots + (j_{n+m} - (n+m) + m)]}_{n \text{ parentheses}} = \\ &= \sum_{i=1}^{n+m} j_i - \sum_{i=1}^{n+m} i + mn = mn, \end{aligned}$$

because $\sum_{i=1}^{n+m} i = \sum_{i=1}^{n+m} j_i$ (since σ is a permutation). □

Corollary 11.13 *Let A be a factorial domain. If*

$$\begin{aligned} F(x_0, \dots, x_n) &= a_0 x_0^m + a_1(x_1, \dots, x_n) x_0^{m-1} + \cdots + a_m(x_1, \dots, x_n), \\ G(x_0, \dots, x_n) &= b_0 x_0^p + b_1(x_1, \dots, x_n) x_0^{p-1} + \cdots + b_p(x_1, \dots, x_n) \end{aligned}$$

are two homogeneous polynomials of degrees $m \geq 1$, $p \geq 1$, respectively, where $a_i(x_1, \dots, x_n)$ and $b_j(x_1, \dots, x_n)$ are zero or homogeneous of degrees i, j respectively, such that $a_0 b_0 \neq 0$, their resultant $\mathfrak{R}(F, G)$ is either 0 or an homogeneous polynomial of $A[x_1, \dots, x_n]$ of degree mp.

Furthermore, $\mathfrak{R}(F, G)(x_1, \dots, x_n)$ is identically zero if and only if F and G have a nonconstant homogeneous common factor.

Proof The requirement $a_0 b_0 \neq 0$ implies that the degrees of F and G as polynomials in $B[x_0]$, where $B = A[x_1, \dots, x_n]$, are m and p respectively. We have only to observe that the weight of every term $\neq 0$ in the development of

the determinant $\mathfrak{R}(F, G)$ is its degree as a polynomial of $A[x_1, \ldots, x_n]$ and apply Proposition 11.12 and Corollary 11.9. □

Proposition 11.14 *Let A be a factorial domain. Given two polynomials*

$$f(X) = \prod_{i=1}^{m}(X - \lambda_i), \qquad g(X) = \prod_{j=1}^{p}(X - \mu_i)$$

where λ_i and μ_j belong to some field $K \supset A$, then

$$\mathfrak{R}(f, g) = \prod_{i=1}^{m}\prod_{j=1}^{p}(\lambda_i - \mu_j). \tag{11.14}$$

Proof Let us consider the polynomials f and g in the indeterminates $X, \lambda_1, \ldots, \lambda_m, \mu_1, \ldots .\mu_p$

$$f(X, \lambda_1, \ldots, \lambda_m) = \prod_{i=1}^{m}(X - \lambda_i),$$

$$g(X, \mu_1, \ldots .\mu_p) = \prod_{j=1}^{p}(X - \mu_i) \in A[X, \lambda_1, \ldots, \lambda_m, \mu_1, \ldots .\mu_p],$$

which are homogeneous of degrees m and p respectively. Then $\mathfrak{R}(f, g)$ is non-zero and homogeneous of degree mp in the indeterminates $\lambda_1, \ldots, \lambda_m, \mu_1, \ldots .\mu_p$. If we replace λ_i with μ_j in $f(X, \lambda_1, \ldots .\lambda_p)$, we get a polynomial f_1, which has the factor $X - \mu_j$ common with $g(X, \mu_1, \ldots, \mu_m)$. Therefore $\mathfrak{R}(f_1, g) = 0$ and, since $\mathfrak{R}(f_1, g)$ is obtained from $\mathfrak{R}(f, g)$ by changing λ_i with μ_j, i.e.

$$\begin{aligned}&\mathfrak{R}(f, g_1)(\lambda_1, \ldots, \lambda_{i-1}, \lambda_{i+1}, \ldots, \lambda_m, \mu_1, \ldots, \ldots, \mu_p)\\ &= \mathfrak{R}(f, g)(\lambda_1, \ldots, \underbrace{\mu_j}_{i\text{-th place}}, \ldots, \lambda_m, \mu_1, \ldots, \mu_p),\end{aligned}$$

then $\mathfrak{R}(f, g)(\lambda_1, \ldots, \underbrace{\mu_j}_{i\text{-th place}}, \ldots, \lambda_m, \mu_1, \ldots, \mu_p) = 0$. It means that the polynomial

$$\mathfrak{R}(f, g)(\lambda_i) \in A[\lambda_1, \ldots, \lambda_{i-1}, \lambda_{i+1}, \ldots, \lambda_m, \mu_1, \ldots, \ldots, \mu_p][\lambda_i]$$

vanishes for $\lambda_i = \mu_j$ and $\lambda_i - \mu_j$ divides $\mathfrak{R}(f, g)$ by Theorem 1.10. That occurs for every $1 \le i \le m$, $1 \le j \le p$, therefore, for degree reasons,

$$\mathfrak{R}(f, g)(\lambda_1, \ldots, \lambda_m, \mu_1, \ldots .\mu_p) = \lambda \prod_{i=1}^{m} \prod_{j=1}^{p} (\lambda_i - \mu_j), \qquad \lambda \in A, \lambda \neq 0. \tag{11.15}$$

To calculate λ we evaluate the polynomial $\mathfrak{R}(f, g)(\lambda_1, \ldots, \lambda_m, \mu_1, \ldots, \mu_p)$ at a point

$$(0, \ldots, 0, \mu_1, \ldots, \mu_p), \qquad \mu_j \neq 0, \ j = 1, \ldots, p.$$

Then

$$\mathfrak{R}(f, g)(0, \ldots, 0, \mu_1, \ldots, \mu_p) = \begin{vmatrix} 1 & 0 & 0 & \ldots & \ldots & \ldots & 0 \\ 0 & 1 & 0 & \ldots & \ldots & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & 0 & \ldots & 0 \\ 1 & b_1 & b_2 & \ldots & b_p & 0 & \ldots \\ 0 & 1 & b_1 & \ldots & \ldots & b_p & \ldots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ 0 & 0 & 0 & \ldots & \ldots & \ldots & b_p \end{vmatrix}.$$

Therefore the matrix is lower triangular and its determinant is the product

$$b_p^m = ((-1)^p \mu_1 \cdots \mu_p)^m = (-1)^{pm} (\mu_1 \cdots \mu_p)^m.$$

On the right-hand side of (11.15) the coefficient of $(\mu_1 \cdots \mu_p)^m$ is $(-1)^{pm}\lambda$, so that we conclude that $\lambda = 1$. □

Remark 11.15 Let $f(X) = a_0 \prod_{i=1}^{m} (X - \lambda_i)$, $g(X) = b_0 \prod_{j=1}^{p} (X - \mu_i)$. From (11.10) we get

$$\mathfrak{R}(f, g) = a_0^p b_0^m \prod_{i=1}^{m} \prod_{j=1}^{p} (\lambda_i - \mu_j) = a_0^p \prod_{i=1}^{m} g(\lambda_i) = (-1)^{mp} b_0^m \prod_{j=1}^{p} f(\mu_j). \tag{11.16}$$

Corollary 11.16 *Let $f(X)$, $g(X)$ and $h(X) \in A[X]$. Then we have*

$$\mathfrak{R}(f, gh) = \mathfrak{R}(f, g)\mathfrak{R}(f, h), \qquad \mathfrak{R}(fg, h) = \mathfrak{R}(f, h)\mathfrak{R}(g, h). \tag{11.17}$$

Proof If

$$f(X) = a_0 \prod_{i=1}^{m} (X - \lambda_i), \qquad g(X) = b_0 \prod_{j=1}^{p} (X - \mu_i), \qquad h(X) = c_0 \prod_{k=1}^{q} (X - \gamma_k)$$

then

$$\mathfrak{R}(f, g) = (-1)^{pm} b_0^m \prod_{j=1}^{p} f(\mu_j), \qquad \mathfrak{R}(f, h) = (-1)^{mq} c_0^m \prod_{k=1}^{q} f(\gamma_k).$$

Since

$$g(X)h(X) = b_0 c_0 \prod_{j=1}^{p} (X - \mu_i) \prod_{k=1}^{q} (X - \gamma_k)$$

we get

$$\begin{aligned} \mathfrak{R}(f, g)\mathfrak{R}(f, h) &= (-1)^{mp+mq} b_0^m c_0^m \prod_{j=1}^{p} f(\mu_j) \prod_{k=1}^{q} f(\gamma_k) \\ &= (-1)^{m(p+q)} (b_0 c_0)^m \prod_{j=1}^{p} f(\mu_j) \prod_{k=1}^{q} f(\gamma_k) = \mathfrak{R}(f, gh). \end{aligned}$$

Finally, we have by (11.9)

$$\begin{aligned} \mathfrak{R}(fg, h) &= (-1)^{(m+p)q} \mathfrak{R}(h, fg) \\ &= (-1)^{mq} \mathfrak{R}(h, f)(-1)^{pq} \mathfrak{R}(h, g) = \mathfrak{R}(f, h)\mathfrak{R}(g, h). \end{aligned}$$

□

Corollary 11.17 *We assume the notation of Remark 11.15. Let* $f(X) = g(X)q(X) + r(X)$. *Then we have*

$$\mathfrak{R}(f, g) = (-1)^{p(m-q)} b_0^{m-q} \mathfrak{R}(r, g), \tag{11.18}$$

where b_0 *is the leading coefficient of* $g(X)$ *and* $q = \deg r(X)$.

Proof Since $f(\mu_j) = r(\mu_j)$ for $j = 1, \ldots, p$ we have

$$\begin{aligned}\mathfrak{R}(f, g) &= (-1)^{pm} b_0^m \prod_{j=1}^{p} f(\mu_j) = (-1)^{p(m-q)} b_0^{m-q} (-1)^{pq} b_0^q \prod_{j=1}^{p} r(\mu_j) \\ &= (-1)^{p(m-q)} b_0^{m-q} \mathfrak{R}(r, g)\end{aligned}$$

by (11.16). □

Corollary 11.18 *We assume the notation of Remark 11.15. with $m \leq p$. Let $h(X) \in A[X]$ of degree $p - m$, then*

$$\mathfrak{R}(f, g) = \mathfrak{R}(f, fh + g).$$

Proof Since $(fh + g)(\lambda_i) = g(\lambda_i)$ for $i = 1, \ldots, m$ we have

$$\mathfrak{R}(f, g) = a_0^p \prod_{i=1}^{m} g(\lambda_i) = a_0^p \prod_{i=1}^{m} (fh + g)(\lambda_i) = \mathfrak{R}(f, fh + g). \qquad \square$$

Now we are ready to prove Theorem 11.1.

Proof Let $F(x_0, x_1, x_2)$ and $G(x_0, x_1, x_2)$ be the equations of $\mathcal{C}$ and $\mathcal{D}$ respectively, with F and G non-zero homogeneous polynomials of degrees n and m respectively. We claim that we can find a projective coordinate system $[x_0, x_1, x_2]$ of $\mathbb{P}^2$ such that the point $[1, 0, 0]$ does not belong to $\mathcal{C} \cup \mathcal{D}$. In fact, by Lemma 10.7, formula (10.2) and Exercise 8.3, we can choose a point $Q \in \mathbb{P}^2 \setminus (\mathcal{C} \cup \mathcal{D})$ and a projective linear automorphism σ of $\mathbb{P}^2$ such that $\sigma(Q) = [1, 0, 0]$. The new projective coordinate system will be given by the projective linear automorphism σ^{-1} and with respect to it the coordinates of the point Q will be $[1, 0, 0]$. Notice that, as the equations σ are linear and homogeneous, the homogeneous polynomials $\sigma \cdot F$ and $\sigma \cdot G$ (which are the equations of $\mathcal{C}$ and $\mathcal{D}$ in the new system of coordinates) are still of degrees n and m respectively. We notice that Bézout's theorem holds for $\mathcal{C}$ and $\mathcal{D}$ if and only if it holds for their images $\sigma(\mathcal{C})$ and $\sigma(\mathcal{D})$.

With this choice of the coordinate system the equations F and G are of the form

$$\begin{aligned}F(x_0, x_1, x_2) &= a_0 x_0^n + a_1(x_1, x_2) x_0^{n-1} + \cdots + a_n(x_1, x_2), \\ G(x_0, x_1, x_2) &= b_0 x_0^m + b_1(x_1, x_2) x_0^{m-1} + \cdots + b_m(x_1, x_2),\end{aligned}$$

where $a_i(x_1, x_2) \in K[x_1, x_2]$ is a homogeneous polynomial of degree i, $i = 0, \ldots, n$, and $b_j(x_1, x_2) \in K[x_1, x_2]$ is a homogeneous polynomial of degree j, $j = 0, \ldots, m$. The fact that $P_0 \notin \mathcal{C} \cup \mathcal{D}$ implies that $a_0 b_0 \neq 0$. Now, setting $A := K[x_1, x_2]$ we know by Theorem 1.9 that A is factorial. Consider F and G as polynomials of $A[x_0]$. Then by Corollary 11.13 the resultant $\mathfrak{R}(F, G) = \mathfrak{R}(x_1, x_2)$ is a non-zero homogeneous polynomial of degree mn of $K[x_1, x_2]$ because F and

G have no non-constant common factor. Let $[c_1, c_2] \in \mathbb{P}^1$ be a zero of $\mathfrak{R}(x_1, x_2)$. Then the polynomials

$$h(x_0) := F(x_0, c_1, c_2) = a_0 x_0^n + a_1(c_1, c_2)x_0^{n-1} + \cdots + a_n(c_1, c_2) \in K[x_0],$$

$$k(x_0) := G(x_0, c_1, c_2) = b_0 x_0^m + b_1(c_1, c_2)x_0^{n-1} + \cdots + b_m(c_1, c_2) \in K[x_0]$$

have by Corollary 11.9 a common root $c_0 \in K$ (K is algebraically closed). Therefore the point $[c_0, c_1, c_2] \in \mathbb{P}^2$ belongs to $\mathcal{C} \cap \mathcal{D}$. Conversely, if a point $[c_0, c_1, c_2] \in \mathbb{P}^2$ belongs to $\mathcal{C} \cap \mathcal{D}$ then the polynomials $h(x_0)$ and $k(x_0)$ have the common root c_0, hence by Corollary 11.9, $\mathfrak{R}(c_1, c_2) = 0$. Since for a fixed $[c_1, c_2] \in \mathbb{P}^1$ such that $\mathfrak{R}(c_1, c_2) = 0$ there are only finitely many possibilities for c_0 such that $[c_0, c_1, c_2] \in \mathcal{C} \cap \mathcal{D}$ and since by Proposition 10.5 the equation $\mathfrak{R}(x_1, x_2) = 0$ has at most mn zeros (and at least one) on $\mathbb{P}^1$, we infer that $\mathcal{C} \cap \mathcal{D}$ is finite and nonempty.

Now, consider the (finite) set $\mathcal{A}$ of lines of $\mathbb{P}^2$ joining any two different points of the intersection $\mathcal{C} \cap \mathcal{D}$ ($\mathcal{A}$ empty if $\mathcal{C} \cap \mathcal{D}$ consists only of one point). We may arrange the above projective linear automorphism σ in such a way that no line of $\mathcal{A}$ passes through the point P_0. Set $P = [c_0, c_1, c_2] \in \mathcal{C} \cap \mathcal{D}$. Then the line P_0P has the equation

$$c_2 x_1 - c_1 x_2 = 0. \tag{11.19}$$

On the other hand, if for a given $[c_1, c_2] \in \mathcal{V}_+(\mathfrak{R}(x_1, x_2))$ there are two distinct elements $c_0, c_0' \in K$ such that

$$[c_0, c_1, c_2],\ [c_0', c_1, c_2] \in \mathcal{C} \cap \mathcal{D},$$

then these two points are on the line of equation (11.19), i.e. P_0 and these two points are collinear. In other words, if $P \in \mathcal{C} \cap \mathcal{D}$ then the line P_0P does not contain any point of $\mathcal{C} \cap \mathcal{D}$ other than P. Consequently, there is a bijective correspondence between $\mathcal{C} \cap \mathcal{D}$ and the set of all lines P_0P, with $P \in \mathcal{C} \cap \mathcal{D}$. It follows that the set $\mathcal{C} \cap \mathcal{D}$ is in a bijective correspondence with $\mathcal{V}_+(\mathfrak{R}(x_1, x_2))$, and in particular, $\#(\mathcal{C} \cap \mathcal{D}) = \#\mathcal{V}_+(\mathfrak{R}(x_1, x_2))$. Finally, by Proposition 10.5, $\#(\mathcal{V}_+(\mathfrak{R}(x_1, x_2))) \leq nm$. □

11.4 Intersection Multiplicity of Two Curves

We are about to introduce the intersection multiplicity of two curves which permits us to reformulate Bézout's theorem in its strongest form. We closely follow [19].

Theorem 11.19 *Given two projective curves $\widehat{F}$ and $\widehat{G}$ there exists a uniquely determined $\mathfrak{m}_P(\widehat{F}, \widehat{G}) \in \mathbb{N} \cup \{0, \infty\}$ satisfying the following properties:*

(i) $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \mathfrak{m}_P(\widehat{G}, \widehat{F})$.
(ii) $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \infty$ *if and only if P lies on a common component of $\widehat{F}$ and $\widehat{G}$.*
(iii) $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = 0$ *if and only if $P \notin \widehat{F} \cap \widehat{G}$.*
(iv) $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = 1$ *if $\widehat{F}$ and $\widehat{G}$ are lines and P is their unique point of intersection.*
(v) *If $F = F_1 F_2$ then* $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \mathfrak{m}_P(\widehat{F_1}, \widehat{G}) + \mathfrak{m}_P(\widehat{F_2}, \widehat{G})$.
(vi) *If $\widehat{F}$ and $\widehat{G}$ have degrees n and m respectively with $n \leq m$ and H is a homogeneous polynomial of degree $m - n$, then*

$$\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \mathfrak{m}_P(\widehat{F}, \widehat{FH+G}).$$

Moreover, if $\widehat{F}$ and $\widehat{G}$ do not have common components and we select a projective coordinate system such that the following conditions are satisfied:

(a) $[1, 0, 0] \notin \widehat{F} \cup \widehat{G}$,
(b) $[1, 0, 0]$ *does not lie on any line connecting two distinct points of $\widehat{F} \cap \widehat{G}$,*
(c) $[1, 0, 0]$ *does not lie on the tangent line to $\widehat{F}$ or $\widehat{G}$ at any point of $\widehat{F} \cap \widehat{G}$,*

then, setting $P = [a, b, c] \in \widehat{F} \cap \widehat{G}$, we have

$$\mathfrak{m}_P(\widehat{F}, \widehat{G}) = m_{[b,c]}\mathfrak{R}(F, G)(x_1, x_2),$$

i.e. the largest integers k such that $(bx_2 - cx_1)^k$ divides the resultant $\mathfrak{R}(F, G)(x_1, x_2)$.

Proof First we prove uniqueness, i.e. the conditions (i)–(vi) determine $\mathfrak{m}_P(\widehat{F}, \widehat{G})$ completely. We can assume that $P = [0, 0, 1]$ since these conditions are independent of the choice of coordinates. Furthermore we may assume that $\widehat{F}$ and $\widehat{G}$ are irreducible by (i) and (v), that $\mathfrak{m}_P(\widehat{F}, \widehat{G})$ is finite by (ii) and that $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = k > 0$ by (iii). By induction on k we assume that every intersection multiplicity $< k$ can be calculated using only the conditions (i)–(vi) (intersection multiplicity 0 is given by (iii)).

We consider the polynomials $f(x_0) := F(x_0, 0, 1) \in K[x_0]$ and $g(x_0) := G(x_0, 0, 1) \in K[x_0]$ of degrees r and s respectively. By (i) we assume that $r \leq s$.

- Case 1. If $r = 0$, the polynomial $f(x_0)$ is constant and hence $= 0$ because $F(0, 0, 1) = 0$. Since F is homogeneous the polynomial $F(x_0, 0, x_2)$ is identically zero. Thus there exists a homogeneous polynomial $Q(x_0, x_1, x_2)$ such that

$$F(x_0, x_1, x_2) = x_1 Q(x_0, x_1, x_2).$$

We can also find two homogeneous polynomials $S(x_0, x_1, x_2)$, $T(x_0, x_2)$ with $T(0, 1) \neq 0$ and some integer q such that

$$G(x_0, x_1, x_2) = G(x_0, 0, x_2) + x_1 S(x_0, x_1, x_2) = x_0^q T(x_0, x_2) + x_1 S(x_0, x_1, x_2).$$

Since $G(0, 0, 1) = 0$ the integer q is > 0. Hence the point $P = [0, 0, 1]$ does not lie on the curve $\widehat{T}$ of equation $T(x_0, x_2) = 0$ and by (iii) and (iv) we have

$$\mathfrak{m}_P(\widehat{x_1}, \widehat{T}) = 0, \qquad \mathfrak{m}_P(\widehat{x_0}, \widehat{x_1}) = 1, \tag{11.20}$$

where $\widehat{x_0}$, $\widehat{x_1}$ are the lines $x_0 = 0$ and $x_1 = 0$ respectively. From (v) we obtain

$$\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \mathfrak{m}_P(\widehat{Q}, \widehat{G}) + \mathfrak{m}_P(\widehat{x_1}, \widehat{G}) \tag{11.21}$$

and from (vi) we have

$$\mathfrak{m}_P(\widehat{x_1}, \widehat{G}) = \mathfrak{m}_P(\widehat{x_1}, \widehat{x_0^q T + x_1 S}) = \mathfrak{m}_P(\widehat{x_1}, \widehat{x_1 S + x_0^q T}) = \mathfrak{m}_P(\widehat{x_1}, \widehat{x_0^q T}). \tag{11.22}$$

Iterating (v) q times we get by (11.20)

$$\mathfrak{m}_P(\widehat{x_1}, \widehat{x_0^q T}) = \mathfrak{m}_P(\widehat{x_1}, \widehat{x_0^q}) + \mathfrak{m}_P(\widehat{x_1}, \widehat{T}) = q\mathfrak{m}_P(\widehat{x_0}, \widehat{x_1}) + \mathfrak{m}_P(\widehat{x_1}, \widehat{T}) = q. \tag{11.23}$$

Hence from (11.21), (11.22) and (11.23) we have

$$\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \mathfrak{m}_P(\widehat{Q}, \widehat{G}) + q. \tag{11.24}$$

Since $q > 0$ from (11.24) we obtain that $\mathfrak{m}_P(\widehat{Q}, \widehat{G}) < \mathfrak{m}_P(\widehat{F}, \widehat{G}) = k$ and our inductive hypothesis tells us that $\mathfrak{m}_P(\widehat{Q}, \widehat{G})$ can be calculated by using only conditions (i)–(vi) so that the same happens to $\mathfrak{m}_P(\widehat{F}, \widehat{G})$.

- Case 2. If $r > 0$, multiplying by suitable constants we make the polynomials $f(x_0)$ and $g(x_0)$ monic. The polynomial

$$L(x_0, x_1, x_2) = x_2^{n+s-r} G(x_0, x_1, x_2) - x_0^{s-r} x_2^m F(x_0, x_1, x_2)$$

is homogeneous and not identically zero because $F(x_0, x_1, x_2)$ and $G(x_0, x_1, x_2)$ are irreducible and distinct. Moreover

$$l(x_0) := L(x_0, 0, 1) = g(x_0) - x_0^{s-r} f(x_0) = x_0^s + \cdots - x_0^{s-r}(x_0^r + \cdots)$$

has degree $t < s$. By (vi) and (v) we get

$$\mathfrak{m}_P(\widehat{F}, \widehat{L}) = \mathfrak{m}_P(\widehat{F}, \widehat{x_2^{n+s-r}\overline{G - x_0^{s-r}x_2^m F}}) = \mathfrak{m}_P(\widehat{F}, \widehat{x_2^{n+s-r}G})$$
$$= \mathfrak{m}_P(\widehat{F}, \widehat{x_2^{n+s-r}}) + \mathfrak{m}_P(\widehat{F}, \widehat{G}) = \mathfrak{m}_P(\widehat{F}, \widehat{G}),$$

since $P \notin \widehat{x_2}$ and hence $\mathfrak{m}_P(\widehat{F}, \widehat{x_2^{n+s-r}}) = 0$. Now, we can replace F and G by F and L (or by L and F if $t < r$) and continue until we reach the situation $r = 0$. □

Remark 11.20 We need the assumption (b) in the statement of Theorem 11.19 in order to have a bijective correspondence between the points of $\widehat{F} \cap \widehat{G}$ and the zeros of $\mathfrak{R}(F, G)(x_1, x_2)$, otherwise a single root of the resultant might give more points of intersection (see the proof of Theorem 11.1 above).

The condition (c) allows us to prove the following result.

Proposition 11.21 *Given two projective curves $\widehat{F}$, $\widehat{G}$ and a point $P \in \widehat{F} \cap \widehat{G}$, we have $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = 1$ if and only if P is not a singular point of $\widehat{F}$ and $\widehat{G}$ and the tangent lines to $\widehat{F}$ and $\widehat{G}$ are distinct.*

One can find a proof in [19], pp. 63–66 if $K = \mathbb{C}$.

We now prove the existence. We give the following:

Definition 11.22 Given two projective curves $\widehat{F}$ and $\widehat{G}$, their *intersection multiplicity* $\mathfrak{m}_P(\widehat{F}, \widehat{G})$ at a point $P \in \mathbb{P}^2(K)$ is defined as follows:

(α) $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = \infty$ if P belongs to a common component of $\widehat{F}$ and $\widehat{G}$.
(β) $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = 0$ if $P \notin \widehat{F} \cap \widehat{G}$.
(γ) If $P \in \widehat{F} \cap \widehat{G}$ but it does not lie in a common component of $\widehat{F}$ and $\widehat{G}$, then removing any of their common components (obtaining coprime polynomials F_0 and G_0) and choosing a projective coordinate system such that the conditions (a)–(c) are satisfied and $P = [a, b, c]$ gives

$$\mathfrak{m}_P(\widehat{F}, \widehat{G}) = m_{[b,c]}(\mathfrak{R}(F_0, G_0))), \tag{11.25}$$

i.e. the greatest integer k such that $(bx_2 - cx_1)^k$ divides $\mathfrak{R}(F_0, G_0)$.

We have to show that the conditions (i)–(vi) of Theorem 11.19 are satisfied:

(i) It follows at once from (11.9) for (γ) and it is obvious for (α) and (β).
(ii) It follows from definition and Corollary 11.13 ($\mathfrak{R}(F, G)$ is identically zero).
(iii) We have only to prove the implication $\mathfrak{m}_P(\widehat{F}, \widehat{G}) = 0 \implies P \notin \widehat{F} \cap \widehat{G}$. By contradiction, suppose that $P = [a, b, c] \in \widehat{F} \cap \widehat{G}$ i.e. $F(a, b, c) = G(a, b, c) = 0$. Then the resultant $\mathfrak{R}(F, G)(x_1, x_2)$ vanishes when $(x_1, x_2) = (b, c)$ by Corollary 10.6. Hence $bx_2 - cx_1$ divides $\mathfrak{R}(F, G)(x_1, x_2)$ and $m_{[b,c]}(\mathfrak{R}(F, G)) > 0$ because by (a) $(b, c) \neq (0, 0)$ (or $m_{[b,c]}(\mathfrak{R}(F, G)) = \infty$ if $\mathfrak{R}(F, G)(x_1, x_2)$ is identically zero).

(iv) If $F = a_0x_0 + a_1x_1 + a_2x_2$ and $G = b_0x_0 + b_1x_1 + b_2x_2$ represent two distinct lines with $a_0b_0 \neq 0$, their intersection point is

$$[a_1b_2 - a_2b_1, a_2b_0 - a_0b_2, a_0b_1 - a_1b_0] \neq [1, 0, 0]$$

by condition (a). Then the polynomial

$$\mathfrak{R}(F, G)(x_1, x_2) = \begin{vmatrix} a_0 & a_1x_1 + a_2x_2 \\ b_0 & b_1x_1 + b_2x_2 \end{vmatrix} = (a_0b_1 - a_1b_0)x_1 + (a_0b_2 - a_2b_0)x_2$$

is not identically zero and has $[a_0b_2 - a_2b_0, -(a_0b_1 - a_1b_0)]$ as a root of multiplicity 1.

(v) It follows immediately from Corollary 11.16.

(vi) It follows immediately from Corollary 11.18.

The proof of Theorem 11.19 is now complete. □

Remark 11.23 The uniqueness (in Proposition 11.19) shows that the intersection multiplicity $\mathfrak{m}_P(\widehat{F}, \widehat{G})$ (Definition 11.22) is independent of the coordinates chosen and is intrinsic to P, $\widehat{F}$ and $\widehat{G}$. Moreover (iii) and (v) imply that $\mathfrak{m}_P(\widehat{F}, \widehat{G})$ depends only on those components of $\widehat{F}$ and $\widehat{G}$ which contain P.

Theorem 11.24 (Bézout) *Under the assumptions of Theorem 11.1, $\mathcal{C} \cap \mathcal{D}$ is finite, non-empty and*

$$\sum_{P \in \mathcal{C} \cap \mathcal{D}} \mathfrak{m}_P(\mathcal{C}, \mathcal{D}) = mn.$$

Proof From the proof of Theorem 11.1, Propositions 11.19 and 10.6 we have

$$\sum_{P=[a,b,c] \in \mathcal{C} \cap \mathcal{D}} \mathfrak{m}_P(\mathcal{C}, \mathcal{D}) = \sum_{[b,c] \in \mathcal{V}_+(\mathfrak{R}(F,G))} m_{[b,c]}(\mathfrak{R}(F, G)) = mn. \qquad \square$$

Remark 11.25 The proof of Theorem 11.1 works also if $\mathrm{char}(K) = 2$. In particular, Theorem 11.5 and Proposition 11.2 hold true if $\mathrm{char}(K) = 2$.

Remark 11.26 We want to compare the intersection multiplicity between a curve $\widehat{F}$ and a line r given in Definition 10.18 with the one between two curves given above. We assume that r is not a component of $\widehat{F}$ and suppose that

$$F(x_0, x_1, x_2) = a_0x_0^m + a_1(x_1, x_2)x_0^{m-1} + \cdots + a_m(x_1, x_2), \ r : x_0 = 0,$$
$$P = [0, 0, 1].$$

Then we have

$$\mathfrak{R}(F,r)(x_1,x_2)=\begin{vmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ \vdots & \vdots & \vdots & 1 & 0\\ a_0 & a_1(x_1,x_2) & \ldots\ldots & a_m(x_1,x_2)\end{vmatrix}=a_m(x_1,x_2)$$

with $a_m(x_1,x_2)$ not identically zero. Since $F(0,0,1)=0$ we have $a_m(0,1)=0$ so that

$$a_m(x_1,x_2)=\lambda_1 x_1^m+\cdots\lambda_k x_1^k x_2^{m-k}=x_1^k(\lambda_1 x_1^{m-k}+\cdots\lambda_k x_2^{m-k}).$$

Hence $k=m_{[0,1]}(\mathfrak{R}(F,r))\geq 1$ and $\mathfrak{m}_P(\widehat{F},r)=k$, according to Definition 11.22.

Choose $Q=[0,1,0]$ and write $r: uP+vQ=[0,u,v]$, so that $F(0,u,v)=a_m(u,v)=u^k(\lambda_1 u^{m-k}+\cdots\lambda_k v^{m-k})$ and the two intersection multiplicities coincide.

11.5 Applications of Bézout's Theorem

11.5.1 Points of Inflection

Definition 11.27 Let $\mathcal{C}=\widehat{F}\subset\mathbb{P}^2(K)$ be a curve of degree $d\geq 2$ such that the characteristic of K does not divide $d!$. By Euler's identities (10.4) and (10.5) we easily get

$$\begin{aligned}d(d-1)F=&F''_{x_0x_0}x_0^2+F''_{x_1x_1}x_1^2+F''_{x_2x_2}x_2^2\\&+2(F''_{x_0x_1}x_0x_1+F''_{x_0x_2}x_0x_2+F''_{x_1x_2}x_1x_2),\end{aligned}\tag{11.26}$$

where $F''_{x_ix_j}$ denotes the second partial derivative of F with respect the variables x_i and x_j, $0\leq i,j\leq 2$. The identity (11.26) can be put in the matrix form

$$d(d-1)F=(x_0,x_1,x_2)\,\mathcal{H}_F\,(x_0,x_1,x_2)^t,$$

where

$$\mathcal{H}_F=\begin{pmatrix}F''_{x_0x_0} & F''_{x_0x_1} & F''_{x_0x_2}\\ F''_{x_1x_0} & F''_{x_1x_1} & F''_{x_1x_2}\\ F''_{x_2x_0} & F''_{x_2x_1} & F''_{x_2x_2}\end{pmatrix}$$

is the *Hessian matrix* of F. The determinant $H_F := \det(\mathcal{H}_F)$ is called the *Hessian form* (or simply the *Hessian*) of F and it is a homogeneous form of degree $3(d-2)$. The associated curve $\widehat{H_F}$ is called the *Hessian* of $\mathcal{C}$.

If $(y_0, y_1, y_2)^t = A(x_0, x_1, x_2)^t$ is a new system of projective coordinates (with A a non-singular 3×3 matrix) then one can easily check that

$$\mathcal{H}_F = A\,\overline{\mathcal{H}_F}\,A^t,$$

where $\overline{\mathcal{H}_F}$ is the Hessian matrix with respect to (y_0, y_1, y_2), then the two Hessian forms are equal up to a constant factor.

If P is an arbitrary point of $\mathbb{P}^2(K)$ we can associate to the symmetric matrix

$$\mathcal{H}_F(P) = \begin{pmatrix} F''_{x_0x_0}(P) & F''_{x_0x_1}(P) & F''_{x_0x_2}(P) \\ F''_{x_1x_0}(P) & F''_{x_1x_1}(P) & F''_{x_1x_2}(P) \\ F''_{x_2x_0}(P) & F''_{x_2x_1}(P) & F''_{x_2x_2}(P) \end{pmatrix}$$

the conic $\gamma_P := \widehat{G_P}$ of equation

$$\begin{aligned} G_P := {} & F''_{x_0x_0}(P)x_0^2 + F''_{x_1x_1}(P)x_1^2 + F''_{x_2x_2}(P)x_2^2 \\ & + 2F''_{x_0x_1}(P)x_0x_1 + 2F''_{x_1x_2}(P)x_1x_2 + 2F''_{x_2x_0}(P)x_2x_0 = 0. \end{aligned} \tag{11.27}$$

Proposition 11.28 *Under the hypothesis and notation of Definition 11.27, if $P \in \mathbb{P}^2(K)$ is an arbitrary point we have:*

(i) $G_P(P) = d(d-1)F(P)$, *hence $P \in \mathcal{C}$ if and only if $P \in \gamma_P$.*
(ii) $H_F(P) = 0$ *if and only if the conic γ_P is degenerate, thus $P \in \mathcal{C} \cap \widehat{H_F}$ if and only if P belongs to the degenerate conic γ_P.*
(iii) *P is a singular point of $\mathcal{C}$ if and only if P is a singular point of γ_P.*
(iv) $\mathrm{Sing}(\mathcal{C}) \subset \widehat{H_F}$.
(v) *For each non singular point $P \in \mathcal{C} \cap \widehat{H_F}$ the tangent line $T_P(\mathcal{C})$ to $\mathcal{C}$ at P is an irriducible component of γ_P.*

Proof Statements (i) and (ii) trivially follow from (11.26) and from the above definitions. Euler's identities for derivatives F_{x_0}, F_{x_1} and F_{x_2} give the identities

$$2(d-1)\frac{\partial F}{\partial x_i}(P) = \frac{\partial G_P}{\partial x_i}(P), \qquad i = 0, 1, 2 \tag{11.28}$$

from which the assertion (iii) follows. If P is singular for $\mathcal{C}$ it is singular for γ_P by (11.28), hence γ_P is degenerate and (iv) follows from (ii). By (11.28) the equations of the tangent lines to $\mathcal{C}$ and γ_P at P are the same and this common tangent must be an irriducible component of the degenerate conic γ_P. □

Definition 11.29 Let $\mathcal{C} = \widehat{F} \subset \mathbb{P}^2(K)$ be a curve of degree $d \geq 3$ and suppose that the characteristic of K does not divide $d!$. A non-singular point $P \in \mathcal{C}$ is called

a *flex* (or *point of inflection*) if

$$\mathfrak{m}_P(\mathcal{C}, T_P(\mathcal{C})) \geq 3.$$

If $\mathcal{C}$ has a line L as a component, a non-singular point P of $\mathcal{C}$ such that $P \in L$ (then $L = T_P(\mathcal{C})$) is trivially a flex. To avoid this degenerate case we suppose that the curves considered in this subsection have no line as a component.

Lemma 11.30 *Let $F(x_0, x_1, x_2)$ be a homogeneous polynomial of degree $d > 1$ such that the characteristic of K does not divide $d!$. Then*

$$x_0^2 H_F(x_0, x_1, x_2) = (d-1)\begin{vmatrix} dF & (d-1)F'_{x_1} & (d-1)F'_{x_2} \\ F'_{x_1} & F''_{x_1x_1} & F''_{x_1x_2} \\ F'_{x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix}; \tag{11.29}$$

Proof Euler's identities

$$\begin{aligned} (d-1)F'_{x_0} &= x_0F''_{x_0x_0} + x_1F''_{x_0x_1} + x_2F''_{x_0x_2}, \\ (d-1)F'_{x_1} &= x_0F''_{x_1x_0} + x_1F''_{x_1x_1} + x_2F''_{x_1x_2}, \\ (d-1)F'_{x_2} &= x_0F''_{x_2x_0} + x_1F''_{x_2x_1} + x_2F''_{x_2x_2}, \\ dF &= x_0F'_{x_0} + x_1F'_{x_1} + x_2F'_{x_2} \end{aligned} \tag{11.30}$$

and the basic property of determinants give

$$\begin{aligned} x_0H_F(x_0, x_1, x_2) &= \begin{vmatrix} x_0F''_{x_0x_0} & F''_{x_0x_1} & F''_{x_0x_2} \\ x_0F''_{x_0x_1} & F''_{x_1x_1} & F''_{x_1x_2} \\ x_0F''_{x_0x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix} \\ &= \begin{vmatrix} (d-1)F'_{x_0} - x_1F''_{x_0x_1} - x_2F''_{x_0x_2} & F''_{x_0x_1} & F''_{x_0x_2} \\ (d-1)F'_{x_1} - x_1F''_{x_1x_1} - x_2F''_{x_1x_2} & F''_{x_1x_1} & F''_{x_1x_2} \\ (d-1)F'_{x_2} - x_1F''_{x_1x_2} - x_2F''_{x_2x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix} \\ &= \begin{vmatrix} (d-1)F'_{x_0} & F''_{x_0x_1} & F''_{x_0x_2} \\ (d-1)F'_{x_1} & F''_{x_1x_1} & F''_{x_1x_2} \\ (d-1)F'_{x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix} \\ &= (d-1)\begin{vmatrix} F'_{x_0} & F''_{x_0x_1} & F''_{x_0x_2} \\ F'_{x_1} & F''_{x_1x_1} & F''_{x_1x_2} \\ F'_{x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix}. \end{aligned} \tag{11.31}$$

Therefore

$$x_0^2 H_F(x_0, x_1, x_2) = (d-1)\begin{vmatrix} x_0 F'_{x_0} & x_0 F''_{x_0x_1} & x_0 F''_{x_0x_2} \\ F'_{x_1} & F''_{x_1x_1} & F''_{x_1x_2} \\ F'_{x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix}. \tag{11.32}$$

Applying the same procedure to the right side of (11.32) we obtain

$$x_0^2 H_F(x_0, x_1, x_2) = (d-1)\begin{vmatrix} dF & (d-1)F'_{x_1} & (d-1)F'_{x_2} \\ F'_{x_1} & F''_{x_1x_1} & F''_{x_1x_2} \\ F'_{x_2} & F''_{x_1x_2} & F''_{x_2x_2} \end{vmatrix}. \qquad \square$$

Proposition 11.31 *Under the above assumptions and notation, let P be a non-singular point of $\mathcal{C}$ such that the tangent line $T_P(\mathcal{C})$ is not contained in $\mathcal{C}$, then Hessian polynomial H_F is not identically zero and*

$$\mathfrak{m}_P(\widehat{H_F}, T_P(\mathcal{C})) = \mathfrak{m}_P(\mathcal{C}, T_P(\mathcal{C})) - 2.$$

Proof We can choose a system of projective coordinates such that $P = [1, 0, 0]$ and $T_P(\mathcal{C})$ has the equation $x_2 = 0$. The affine curve $\mathcal{C}_0 := \mathcal{C} \cap \mathbb{U}_0$ (see Definition 9.9) has the equation $f(x, y) = 0$, where $f(x, y) := F(1, x, y)$ and $r_0 := T_{P_0}(\mathcal{C}_0)$ is the line $y = 0$ where $P_0 = (0, 0)$. By Proposition 10.40 we have

$$2 \leq m := \mathfrak{m}_P(\mathcal{C}, T_P(\mathcal{C})) = \mathfrak{m}_{P_0}(\mathcal{C}_0, r_0)$$

so that we can write

$$f(x, y) = x^m \varphi(x) + y\psi(x, y), \tag{11.33}$$

with $\varphi(x) \in K[x]$, $\varphi(0) \neq 0$, and $\psi(x, y) \in K[x, y]$ $\psi(0, 0) \neq 0$. By differentiating $f(x, y)$ we obtain

$$\begin{aligned} f'_x &= x^{m-1}h(x) + y\psi'_x(x, y), && \text{with } h(x) = m\varphi(x) + x\varphi'(x) \\ f'_y &= \psi(x, y) + y\psi'_y(x, y) \\ f''_{xx} &= x^{m-2}k(x) + y\psi''_{xx}(x, y), && \text{with } k(x) = (m-1)h(x) + xh'(x) \\ f''_{xy} &= \psi'_x(x, y) + y\psi''_{xy}(x, y) \\ f''_{yy} &= 2\psi'_y(x, y) + y\psi''_{yy}(x, y). \end{aligned} \tag{11.34}$$

In particular we have $h(0) \neq 0$ and $k(0) \neq 0$.

From Lemma 11.30 we get

$$H_F(1,x,y) = (d-1)\begin{vmatrix} df & (d-1)f'_x & (d-1)f'_y \\ f'_x & f''_{xx} & f''_{xy} \\ f'_y & f''_{xy} & f''_{yy} \end{vmatrix}$$
$$= (d-1)\begin{vmatrix} d(x^m\varphi + y\psi) & (d-1)(x^{m-1}h + y\psi'_x) & (d-1)(\psi + y\psi'_y) \\ x^{m-1}h + y\psi'_x & x^{m-2}k + y\psi''_{xx} & \psi'_x + y\psi''_{xy} \\ \psi + y\psi'_y & \psi'_x + y\psi''_{xy} & 2\psi'_y + y\psi''_{yy} \end{vmatrix}. \tag{11.35}$$

In order to determine the intersection multiplicity $\mathfrak{m}_{P_0}(\widehat{H_F}, r_0)$ we have to calculate $H_F(1,x,0)$:

$$H_F(1,x,0) = (d-1)\begin{vmatrix} dx^m\varphi & (d-1)x^{m-1}h & (d-1)\psi(x,0) \\ x^{m-1}h & x^{m-2}k & \psi'_x(x,0) \\ \psi(x,0) & \psi'_x(x,0) & 2\psi'_y(x,0) \end{vmatrix} = x^{m-2}g(x).$$

We observe that $g(0) = -(d-1)k(0)\psi(0,0)^2 \neq 0$ so the Hessian polynomial H_F is not identically zero and $\mathfrak{m}_{P_0}(\widehat{H_F}, r_0) = m-2$. □

Corollary 11.32 *Under the above assumptions and notation, a non-singular curve $\mathcal{C}$ cannot be a component of $\widehat{H_F}$.*

Proof If $\mathcal{C}$ were an irreducible component of $\widehat{H_F}$ then, for $P \in \mathcal{C}$,

$$\mathfrak{m}_P(\mathcal{C}, T_P(\mathcal{C})) \leq \mathfrak{m}_P(\widehat{H_F}, T_P(\mathcal{C})),$$

but that is false by Proposition 11.31. □

Now we are able to prove the following result:

Theorem 11.33 *Under the above assumptions and notation, a non-singular point $P \in \mathcal{C}$ is a flex if and only if $P \in \widehat{H_F}$. In particular, a non-singular curve $\mathcal{C}$ of degree d can have at most $3d(d-2)$ flexes and at least one flex.*

Proof Let $P = [p_0, p_1, p_2]$ be a non-singular point of $\mathcal{C}$ and $T_P(\mathcal{C})$ the tangent line to $\mathcal{C}$ at P of equation $F'_{x_0}(P)x_0 + F'_{x_1}(P)x_1 + F'_{x_2}(P)x_2 = 0$. Let $Q = [q_0, q_1, q_2] \in T_P(\mathcal{C})$ be a variable point different to P. We have

$$F(sP+tQ) = \sum_{l=0}^{d} s^{d-l}t^l \left(\sum_{i+j+k=l} \frac{1}{i!j!k!} \cdot \frac{\partial^l F}{\partial x_0{}^i \partial x_1{}^j \partial x_2{}^k}(P)\, q_0^i q_1^j q_2^k \right)$$
$$= s^d F(P) + s^{d-1}t(F'_{x_0}(P)q_0 + F'_{x_1}(P)q_1 + F'_{x_2}(P)q_2)$$

$$+\frac{1}{2}s^{d-2}t^2G_P(Q)+t^3(\cdots)$$
$$=+\frac{1}{2}s^{d-2}t^2G_P(Q)+t^3(\cdots), \tag{11.36}$$

since $F(P)=0$ and $F'_{x_0}(P)q_0+F'_{x_1}(P)q_1+F'_{x_2}(P)q_2=0$. Then P is a point of inflection of $\mathcal{C}$ if and only if $G_P(Q)=0$ for every $Q\in T_P(\mathcal{C})$. It means that $T_P(\mathcal{C})$ is contained in the conic γ_P so that γ_P is degenerate and $H_F(P)=0$, i.e. $P\in\widehat{H_F}$ by Proposition 11.28.

Conversely, if $P\in\mathcal{C}\cap\widehat{H_F}$ is non-singular, the tangent line $T_P(\mathcal{C})$ is a component of degenerate conic γ_P by Proposition 11.28, so $G_P(Q)=0$ for every $Q\in T_P(\mathcal{C})$ and P is a flex of $\mathcal{C}$.

Since the curve $\mathcal{C}$ is non-singular it is also irreducible by Corollary 10.17 so it has no common component with $\widehat{H_F}$ and it cannot be an irreducible component of $\widehat{H_F}$ by Corollary 11.32. By Bézout's Theorem 11.1, we have $1\le\#(\mathcal{C}\cap\widehat{H_F})\le\deg(F)\deg(H_F)\le 3d(d-2)$. □

Corollary 11.34 *Let $\mathcal{C}\subset\mathbb{P}^2(K)$ be a nonsingular curve of degree 3 over an algebraically closed field of characteristic $\neq 2, 3$. Then $\mathcal{C}$ has at least 1 and at most 9 flexes (Fig. 11.4).*

11.5.2 Legendre Form of Cubics

Theorem 11.35 (Legendre) *Let K be an algebraically closed field of characteristic $\neq 2, 3$. Every non-singular cubic $\mathcal{C}$ in $\mathbb{P}^2(K)$ is equivalent under a projective*

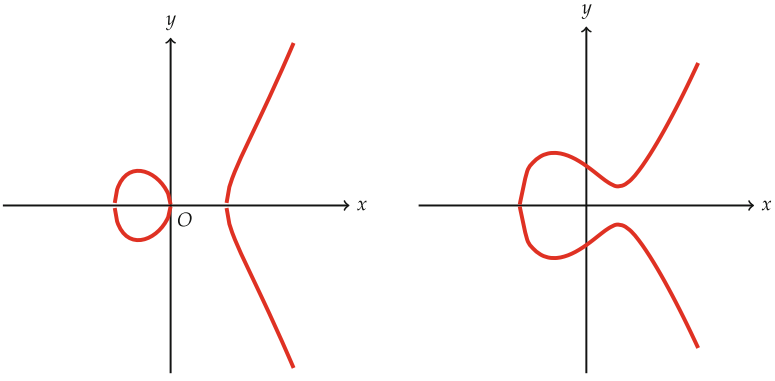


Fig. 11.4 Non-singular cubics $y^2=x^3-x$; $y^2=x^3-x+\frac{1}{2}$

transformation to the curve defined by

$$y^2z = x(x-z)(x-\lambda y) \tag{11.37}$$

for some $\lambda \in K \setminus \{0,1\}$.

Proof By Corollary 11.33 $\mathcal{C}$ has a point of inflection P. By a suitable change of coordinates we may assume that $P = [0,1,0]$ and the tangent line to $\mathcal{C}$ at P is $x_2 = 0$. Let $\mathcal{C} := \widehat{F}$, where

$$\begin{aligned} F(x_0,x_1,x_2) &= ax_0^3 + bx_1^3 + cx_2^3 + dx_0^2x_1 + ex_0x_1^2 + fx_0^2x_2 + gx_0x_2^2 \\ &\quad + hx_1^2x_2 + kx_1x_2^2 + lx_0x_1x_2, \end{aligned} \tag{11.38}$$

with $a, b, \ldots, l \in K$. Since

$$\begin{aligned} \frac{\partial F}{\partial x_0} &= 3ax_0^2 + 2dx_0x_1 + ex_1^2 + 2fx_0x_2 + gx_2^2 + lx_1x_2, \\ \frac{\partial F}{\partial x_1} &= 3bx_1^2 + dx_0^2 + 2ex_0x_1 + 2hx_1x_2 + kx_2^2 + lx_0x_2, \\ \frac{\partial F}{\partial x_2} &= 3cx_2^2 + fx_0^2 + 2gx_0x_2 + hx_1^2 + 2kx_1x_2 + lx_0x_1, \end{aligned}$$

from our assumptions

$$F(0,1,0) = 0, \quad \frac{\partial F}{\partial x_0}(0,1,0) = \frac{\partial F}{\partial x_1}(0,1,0) = 0, \quad \frac{\partial F}{\partial x_2}(0,1,0) \neq 0,$$

we deduce $b = 0, e = 0, h \neq 0$. As in Lemma 11.30 we can obtain

$$x_2^2 H_F(x_0,x_1,x_2)) = 4 \begin{vmatrix} F''_{x_0x_0} & F'_{x_0} & F''_{x_0x_2} \\ F'_{x_0} & \frac{3}{2}F & F'_{x_2} \\ F''_{x_0x_2} & F'_{x_2} & F''_{x_2x_2} \end{vmatrix},$$

so, because $H_F(0,1,0) = 0$, we have

$$\begin{aligned} 0 = H_F(0,1,0)) &= 4 \begin{vmatrix} F''_{x_0x_0}(0,1,0)) & 0 & F''_{x_0x_2}(0,1,0)) \\ 0 & 0 & F'_{x_2}(0,1,0) \\ F''_{x_0x_2}(0,1,0)) & F'_{x_2}(0,1,0)) & F''_{x_2x_2}(0,1,0)) \end{vmatrix} \\ &= -4(F'_{x_2}(0,1,0))^2 F''_{x_0x_0}(0,1,0)). \end{aligned}$$

Thus $F''_{x_0x_0}(0,1,0)) = 0$. Since

$$\frac{\partial^2 F}{\partial x_0^2} = 6ax_0 + 2dx_1 + 2fx_2,$$

deduce that $d = 0$. Therefore F has the form

$$\begin{aligned} F(x_0,x_1,x_2) &= ax_0^3 + cx_2^3 + fx_0^2x_2 + gx_0x_2^2 + hx_1^2x_2 + kx_1x_2^2 + lx_0x_1x_2 \\ &= x_1x_2(lx_0 + hx_1 + kx_2) + \phi(x_0,x_2), \end{aligned}$$

where $\phi(x_0,x_2)$ is homogeneous of degree 3 in x_0 and x_2. After the change of coordinates

$$[x_0,x_1,x_2] \mapsto [x,y,z], \qquad x_0 = x,\ x_1 = y - \frac{lx+kz}{2h},\ x_2 = z,$$

we obtain

$$\begin{aligned} F(x,y,z) &= z\left(y - \frac{lx+kz}{2h}\right)\left(lx + h\left(y - \frac{lx+kz}{2h}\right) + kz\right) + \phi(x,z) \\ &= z\left(y - \frac{lx+kz}{2h}\right)\left(hy + \frac{lx+kz}{2}\right) + \phi(x,z) \\ &= hz\left(y - \frac{lx+kz}{2h}\right)\left(y + \frac{lx+kz}{2h}\right) + \phi(x,z) \\ &= hz\left(y^2 - \left(\frac{lx+kz}{2h}\right)^2\right) + \phi(x,z) = hy^2z + \psi(x,z), \end{aligned}$$

where $\psi(x,z)$ is homogeneous of degree 3 in x and z. Since K is algebraically closed,

$$\psi(x,z) = (a_1x - b_1z)(a_2x - b_2z)(a_3x - b_3z) \tag{11.39}$$

and z does not divide $\psi(x,z)$, otherwise F would be reducible and $\mathcal{C}$ singular, thus $a_1a_2a_3 \neq 0$. We write

$$\psi(x,z) = a_1a_2a_3\left(x - \frac{b_1}{a_1}z\right)\left(x - \frac{b_2}{a_2}z\right)\left(x - \frac{b_3}{a_3}z\right),$$

so

$$F(x,y,z) = hy^2z + a_1a_2a_3\left(x - \frac{b_1}{a_1}z\right)\left(x - \frac{b_2}{a_2}z\right)\left(x - \frac{b_3}{a_3}z\right).$$

After the change of coordinates $x = X$, $y = \delta Y$, $z = Z$, where $\delta^2 = -\dfrac{a_1 a_2 a_3}{h}$, we put $\mathcal{C}$ into the form

$$Y^2 Z = (X - \alpha Z)(X - \beta Z)(X - \gamma Z).$$

The elements $\alpha = \dfrac{b_1}{a_1}$, $\beta = \dfrac{b_2}{a_2}$, $\gamma = \dfrac{b_3}{a_3}$ are distinct, otherwise $\mathcal{C}$ would be singular. If we apply the change of coordinates (using again the letters x, y and z)

$$X = (\beta - \alpha)x + \alpha z, \quad Y = \eta y, \quad Z = z,$$

where $\eta^2 = (\beta - \alpha)^3$, we get

$$y^2 z = x(x - z)\left(x - \frac{\gamma - \alpha}{\beta - \alpha} z\right),$$

i.e. (11.37) with $\lambda = \dfrac{\gamma - \alpha}{\beta - \alpha} \in K \setminus \{0, 1\}$. □

Definition 11.36 Equation (11.37) is called *Legendre form* of the cubic $\mathcal{C}$.

Remark 11.37 The proof of Theorem 11.35 also shows that an irreducible singular curve admits an equation of the form (11.37). In this case the roots of the polynomial $\psi(x, z)$ are not distinct.

An Application of Legendre Form

Theorem 11.38 (Plücker) *A non-singular cubic $\mathcal{C} = \widehat{F}$ of $\mathbb{P}^2(K)$ has exactly nine distinct points of inflection and the line joining two of them passes through a third inflection point.*

Proof By Theorem 11.35 we can suppose that $F(x, y, z) = y^2 z - x(x - z)(x - \lambda z)$ with $\lambda \neq 0, 1$. Because

$$\frac{1}{8} H_F(x, y, z) = \begin{vmatrix} -3x + (\lambda + 1)z & 0 & (\lambda + 1)x - \lambda z \\ 0 & z & y \\ (\lambda + 1)x - \lambda z & y & -\lambda x \end{vmatrix}$$

$$= (3\lambda - (\lambda + 1)^2)x^2 z + \lambda(\lambda + 1)xz^2 - \lambda^2 z^3 + 3xy^2 - (\lambda + 1)y^2 z$$

the flexes of $\mathcal{C}$ are the solutions of the system of equations

$$\begin{cases} (3\lambda - (\lambda + 1)^2)x^2 z + \lambda(\lambda + 1)xz^2 - \lambda^2 z^3 + 3xy^2 - (\lambda + 1)y^2 z = 0 \\ y^2 z = x(x - z)(x - \lambda z). \end{cases} \tag{11.40}$$

Since $P = [0, 1, 0]$ is a flex and the tangent line to $\mathcal{C}$ at P is $z = 0$, all other flexes belong to $\mathbb{U}_z = \{[x, y, z] \in \mathbb{P}^2(K) : z \neq 0\}$ so that we can suppose that they have the third coordinate $z = 1$. Therefore the system (11.40) becames

$$\begin{cases} (3\lambda - (\lambda+1)^2)x^2 + \lambda(\lambda+1)x - \lambda^2 + 3xy^2 - (\lambda+1)y^2 = 0 \\ y^2 = x(x-1)(x-\lambda), \end{cases} \tag{11.41}$$

from which we deduce the equation of fourth degree

$$3x^4 - 4(\lambda+1)x^3 + 6\lambda x^2 - \lambda^2 = 0. \tag{11.42}$$

If $\lambda = -1$ Eq. (11.42) is $3x^4 - 6x^2 - 1 = 0$ whose 4 roots are all distinct. We assume now $\lambda \neq -1$. We have to prove that Eq. (11.42) has 4 simple roots, i.e. the system

$$\begin{cases} 3x^4 - 4(\lambda+1)x^3 + 6\lambda x^2 - \lambda^2 = 0 \\ x^3 - (\lambda+1)x^2 + \lambda x = 0 \end{cases} \tag{11.43}$$

has no solution. We observe that $x = 0$ is not a solution of (11.43) so we may consider

$$\begin{cases} 3x^4 - 4(\lambda+1)x^3 + 6\lambda x^2 - \lambda^2 = 0 \\ x^2 = (\lambda+1)x - \lambda. \end{cases} \tag{11.44}$$

Substituting the second equation in the first one twice we get the equivalent system

$$\begin{cases} (\lambda-1)^2(\lambda+1)x = \lambda(\lambda-1)^2 \\ x^2 = (\lambda+1)x - \lambda. \end{cases}$$

The only possible solution is $x = \dfrac{\lambda}{\lambda+1}$ but then from the second equation of (11.44) we obtain $\left(\dfrac{\lambda}{\lambda+1}\right)^2 = 0$ which implies $\lambda = 0$, a contradiction.

Let a_1, a_2, a_3, a_4 be the roots of (11.42). To each root a_i we associate $\pm b_i$ such that $(\pm b_i)^2 = a_i$, $i = 1, 2, 3, 4$. In this way we get the points $P = [0, 1, 0]$, $[a_1, \pm b_1, 1]$, $[a_2, \pm b_2, 1]$, $[a_3, \pm b_3, 1]$, $[a_4, \pm b_4, 1]$ which are the nine inflection points of $\mathcal{C}$.

The last statement follows very easily. Fix two flexes P_1, P_2 of $\mathcal{C}$ and suppose that $P_1 = [0, 1, 0]$. From the above discussion we have $P_2 = [a, b, 1]$, with a, $b \in K$. Hence $P_3 := [a, -b, 1]$ is a flex of $\mathcal{C}$ collinear with P_1, P_2. □

Remark 11.39 Theorem 11.38 agrees with Bézout's theorem. Since both the curves $\mathcal{C}$ and H_F are of degree 3, the number of intersection points of two

curves reaches the maximum possible of nine as stated by Bézout's theorem. In particular, all these intersection points (i.e. points of inflection of $\mathcal{C}$) have intersection multiplicity 1.

11.5.3 Max Noether's Theorem

Let K be an algebraically closed field of arbitrary characteristic.

Theorem 11.40 (Max Noether) *Let $\mathcal{C}_1 = \widehat{F_1}$ and $\mathcal{C}_2 = \widehat{F_2}$ be two curves of degree n in $\mathbb{P}^2(K)$ that intersect in exactly n^2 points. Fix a natural number $m < n$ and suppose that an irreducible curve $\mathcal{D} = \widehat{G}$ of degree m contains exactly mn of these points, then there is a curve of degree $n - m$ which passes through the remaining $n(n - m)$ points.*

Proof Choose a point $P = [a, b, c] \in \mathcal{D}$, $P \notin \mathcal{C}_1 \cap \mathcal{C}_2$. Since $\deg(F_1) = \deg(F_2) = n$ the curve $\mathcal{C}$ of degree n defined by

$$F_2(a, b, c)\, F_1(x_0, x_1, x_2) - F_1(a, b, c)\, F_2(x_0, x_1, x_2)$$

meets $\mathcal{D}$ in almost $mn + 1$ points, namely P and mn points of $\mathcal{C}_1 \cap \mathcal{C}_2$. By Bézout's theorem, $\mathcal{C}$ and $\mathcal{D}$ have a common irreducible component which must be $\mathcal{D}$ itself because $\mathcal{D}$ is irreducible.Therefore

$$F_2(a, b, c)\, F_1(x_0, x_1, x_2) - F_1(a, b, c)\, F_2(x_0, x_1, x_2) = G(x_0, x_1, x_2) H(x_0, x_1, x_2)$$

for some homogeneous polynomial $H \in K[x_0, x_1, x_2]$ of degree $n - m$. If $[u, v, w] \in \mathcal{C}_1 \cap \mathcal{C}_2$ then either $G(u, v, w) = 0$ or $H(u, v, w) = 0$. Therefore the $n(n - m)$ points of $\mathcal{C}_1 \cap \mathcal{C}_2$ which do not lie on $\mathcal{D}$ must all lie on $\widehat{H}$. □

Corollary 11.41 *If two cubics intersect in exactly nine points and an irreducible conic contains six of these points then the remaining three points are collinear. Conversely, if three points of these nine points are collinear then the remaining six points lie on a conic.*

Proof Apply Theorem 11.40 twice. □

11.5.4 Conics Passing Through a Finite Number of Points

We closely follow [30]. Throughout this subsection we shall denote the K-vector space ${}_{\mathrm{h}}K[x_0, x_1, x_2]_d$ by S_d. By Exercise 10.1 we know that

$$\dim_K S_d = \binom{d+2}{d} = \frac{(d+1)(d+2)}{2}. \tag{11.45}$$

For $P_1, \ldots, P_n \in \mathbb{P}^2(K)$, we define

$$S_d(P_1, \ldots, P_n) := \{F \in S_d \mid F(P_1) = \cdots = F(P_n) = 0\}.$$

It is immediately seen that $S_d(P_1, \ldots, P_n)$ is a subspace of S_d.

Proposition 11.42 *Under the above notation we have*

$$\dim_K S_d(P_1, \ldots, P_n) \geq \frac{(d+1)(d+2)}{2} - n. \tag{11.46}$$

In particular, $\dim_K S_2(P_1, \ldots, P_n) \geq 6 - n$.

Proof Each of the conditions $F(P_i) = 0$ is a single linear condition on F, so that (11.46) follows from (11.45). □

Corollary 11.43 *Let $P_1, \ldots, P_n$ be $n \leq 5$ points no four of which are collinear, then*

$$\dim_K S_2(P_1, \ldots, P_n) = 6 - n.$$

Proof If $n = 5$ Proposition 11.2-iv) implies that $\dim_K S_2(P_1, \ldots, P_5) \leq 1$, so that $\dim_K S_2(P_1, \ldots, P_5) = 1$ by Proposition 11.42. If $n \leq 4$ we can add $P_{n+1}, \ldots, P_5$ such that no four of $P_1, \ldots, P_5$ are collinear. Since $\dim_K S_2(P_1, \ldots, P_5) = 1$ and each point imposes at most one linear condition on F we get

$$1 = \dim_K S_2(P_1, \ldots, P_5) \geq \dim_K S_2(P_1, \ldots, P_n) - (5 - n),$$

i.e. $\dim_K S_2(P_1, \ldots, P_n) \leq 6 - n$ so that $\dim_K S_2(P_1, \ldots, P_n) = 6 - n$ by Proposition 11.42. □

Lemma 11.44 *Let $F \in S_d$, $F \neq 0$.*

(i) *Let $\mathcal{L} = \widehat{L}$ be a line of $\mathbb{P}^2(K)$. If $\mathcal{V}_+(\widehat{F}) \supset \mathcal{L}$, then $F = L \cdot G$ with $G \in S_{d-1}$.*
(ii) *Let $\mathcal{C} = \widehat{C}$ be a non-degenerate conic of $\mathbb{P}^2(K)$. If $\mathcal{V}_+(\widehat{F}) \supset \mathcal{C}$, then $F = C \cdot H$ with $H \in S_{d-2}$.*

Proof Immediate by Theorem 11.1. □

Corollary 11.45 *Let $\mathcal{L} = \widehat{L}$ be a line, and let $\mathcal{C} = \widehat{C}$ be a non-degenerate conic of $\mathbb{P}^2(K)$. Suppose that points $P_1, \ldots, P_n \in \mathbb{P}^2(K)$ are given and fix $d \geq 1$.*

(i) *If $P_1, \ldots, P_a \in \mathcal{L}$, $P_{a+1}, \ldots, P_n \notin \mathcal{L}$ and $a > d$, then*

$$S_d(P_1, \ldots, P_n) = L \cdot S_{d-1}(P_{a+1}, \ldots, P_n) := \{L \cdot D \mid D \in S_{d-1}(P_{a+1}, \ldots, P_n)\}.$$

(ii) *If $P_1, \ldots, P_a \in \mathcal{C}$, $P_{a+1}, \ldots, P_n \notin \mathcal{C}$ and $a > 2d$, then*

$$S_d(P_1, \ldots, P_n) = C \cdot S_{d-2}(P_{a+1}, \ldots, P_n) := \{C \cdot E \mid E \in S_{d-2}(P_{a+1}, \ldots, P_n)\}.$$

Proof

(i) If $F \in S_d(P_1, \ldots, P_n)$ $(F \neq 0)$, by Theorem 11.1 we have $\mathcal{L} \subset \mathcal{C}$ (since $a > d$). Then Lemma 11.44, i) implies $F = L \cdot D$, with $D \in S_{d-1}(P_{a+1}, \ldots, P_n)$, hence $S_d(P_1, \ldots, P_n) \subset L \cdot S_{d-1}(P_{a+1}, \ldots, P_n)$. The inclusion $L \cdot S_{d-1}(P_{a+1}, \ldots, P_n) \subset S_d(P_1, \ldots, P_n)$ is obvious so that we get i).

(ii) is exactly the same.

□

Lemma 11.46 *Let $P_1, \ldots, P_8 \in \mathbb{P}^2(K)$ be distinct points such that no four of them are collinear and no seven of them lie on a non-degenerate conic. Then*

$$\dim_K S_3(P_1, \ldots, P_8) = 2.$$

Proof By Proposition 11.42 we have $\dim_K S_3(P_1, \ldots, P_8) \geq 2$.

Case 1 No three points of $P_1, \ldots, P_8$ are collinear and no six lie on a non-degenerate conic.

Suppose by contradiction that $\dim_K S_3(P_1, \ldots, P_8) \geq 3$. Let P_9, P_{10} be two distinct points on the line $L = P_1P_2$. Then

$$\dim_K S_3(P_1 \ldots, P_{10}) \geq \dim_K S_3(P_1, \ldots, P_8) - 2 \geq 1,$$

so that there exists $0 \neq F \in S_3(P_1, \ldots, P_{10})$. By Corollary 11.45 ($a = 4$ since $P_1, P_2, P_9, P_{10} \in L$) we have $F = L \cdot C$, with $C \in S_2(P_3, \ldots, P_8)$. Since the conic $\mathcal{C} := \widehat{C}$ passes through $P_3, \ldots, P_8$ then by assumption $\mathcal{C}$ is degenerate, i.e. $\mathcal{C}$ is a pair of distinct lines or a double line. In both cases at least three points of $P_1, \ldots, P_8$ are collinear, contradicting our hypothesis.

Case 2 Suppose that P_1, P_2, P_3 lie on the line $\mathcal{L} = \widehat{L}$. Let P_9 be a fourth point on the line $\mathcal{L}$ distinct from all points $P_1, \ldots, P_8$. By Corollary 11.45 we have

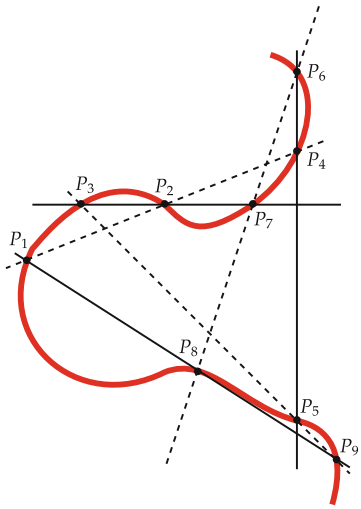
$$S_3(P_1, \ldots, P_9) = L \cdot S_2(P_4, \ldots, P_8).$$

Also, since no four of $P_4, \ldots, P_8$ are collinear, by Corollary 11.43, $\dim_K S_2(P_4, \ldots, P_8) = 6 - 5 = 1$, hence $\dim_K S_3(P_1, \ldots, P_9) = 1$ which implies $\dim_K S_3(P_1, \ldots, P_8) \leq 2$ so that $\dim_K S_3(P_1, \ldots, P_8) = 2$.

Case 3 Suppose that $P_1, \ldots, P_6$ belong to a non-degenerate conic $\mathcal{C} = \widehat{C}$. Choose a point $P_9 \in \mathcal{C}$ distinct from $P_1, \ldots, P_6$. By Corollary 11.45-(ii) we have

$$S_3(P_1, \ldots, P_9) = C \cdot S_1(P_7, P_8).$$

Fig. 11.5 Nine points



Moreover, $S_1(P_7, P_8)$ is generated by L (where $\mathcal{L} = \widehat{L}$ is the line P_7P_8) so that $S_3(P_1, \ldots, P_9)$ is the 1-dimensional space spanned by $C \cdot L$. Therefore $\dim_K S_3(P_1, \ldots, P_8) \leq 2$ and $\dim_K S_3(P_1, \ldots, P_8) = 2$.

□

Proposition 11.47 *Let $\mathcal{C}_1 = \widehat{C}_1, \mathcal{C}_2 = \widehat{C}_2$ be two cubics of $\mathbb{P}^2(K)$ whose intersection consists of nine distinct points $P_1, \ldots, P_9$. Then a cubic $\mathcal{D} = \widehat{D}$ through $P_1, \ldots, P_8$ also passes through P_9 (see Fig. 11.5).*

Proof If four points of $P_1, \ldots, P_8$ belong to a line L, then $L \subset \mathcal{C}_1 \cap \mathcal{C}_2$ which is absurd by Bézout's theorem. For exactly the same reason, no seven points of $P_1, \ldots, P_8$ can lie on a non-degenerate conic. By Lemma 11.46 we get $\dim_K S_3(P_1, \ldots, P_8) = 2$. Therefore the polynomials C_1 and C_2 form a basis of $S_3(P_1, \ldots, P_8)$, and hence $D = \alpha_1 C_1 + \alpha_2 C_2$, with $\alpha_1, \alpha_2 \in K$. Now $C_1(P_9) = C_2(P_9) = 0$ so that $D(P_9) = 0$, i.e. $P_9 \in \mathcal{D}$. □

From Proposition 11.47 we the deduce the converse of Pascal's Theorem 11.5.

Corollary 11.48 (Converse of Pascal's Theorem) *Consider Fig. 11.2. If the opposite sides of a hexagon $P_1P_2P_3P_4P_5P_6$ can be extended until they meet in three collinear points A, B, C, and the nine points and the six lines are all distinct; then $P_1, \ldots, P_6$ lie on a non-degenerate conic.*

Proof Assume the notation of Theorem 11.5 and consider Fig. 11.2. The intersection of the cubics $\mathcal{C}_1 := P_1P_2 \cup P_3P_4 \cup P_5P_6$ and $\mathcal{C}_2 := P_2P_3 \cup P_4P_5 \cup P_6P_1$ consists of nine points

$$\mathcal{C}_1 \cap \mathcal{C}_2 = \{P_1, P_2, P_3, P_4, P_5, P_6, A, B, C\}.$$

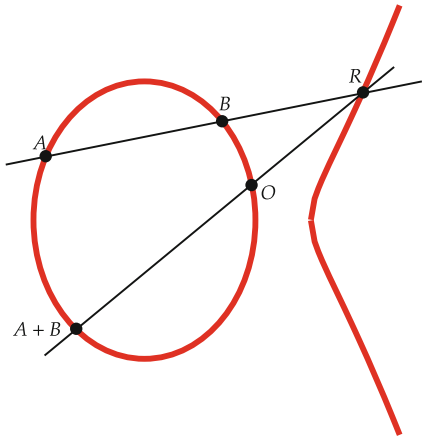


Fig. 11.6 Abelian group law on an elliptic curve (with identity O)

By Proposition 5.27-(i) there exists a conic $\mathcal{D}$ passing through points $P_1, \ldots, P_5$. Let $\mathcal{L}$ be the line containing A, B, C, then the cubic $\mathcal{D} \cup \mathcal{L}$ passes through the points $P_1, P_2, P_3, P_4, P_5, A, B, C$. By Proposition 11.47 $P_6 \in \mathcal{D} \cup \mathcal{L}$. Since P_6 does not lie on $\mathcal{L}$, by assumption, then $P_6 \in \mathcal{D}$. □

Elliptic Curves

Definition 11.49 A non-singular cubic of $\mathbb{P}^2(K)$ is called an *elliptic curve*. Elliptic curves are very important objects in Algebraic Geometry and in Number Theory with many remarkable applications.

Proposition 11.47 is a basic result to prove the following fundamental theorem.

Theorem 11.50 *Let $\mathcal{C}$ an elliptic curve of $\mathbb{P}^2(K)$, where K is an algebraically closed field of characteristic 0. Fix an arbitrary flex $O \in \mathcal{C}$ (O exists by Theorem 11.33). Then there exists an abelian group law on $\mathcal{C}$ whose identity is O.*

The reader can find a proof in [30], pp. 33–41, in [26], pp. 235–240 or in [21], pp. 67–74 (where there is a complete proof of associativity). The group law is illustrated in Fig. 11.6. Two remarks:

1. If $A = B$ the line AB is the tangent to $\mathcal{C}$ at A.
2. We have to note that the choice of a point of inflection as identity is obliged since the tangent to $\mathcal{C}$ at O meets $\mathcal{C}$ only at O so that $O + O = O$.

11.6 Exercises

Exercise 11.1 Let $\mathcal{C}$ be a non-singular curve of $\mathbb{P}^2(K)$ (where K is an algebraically closed field of characteristic 0), let $P \in \mathcal{C}$ be a flex of $\mathcal{C}$ and $\sigma \in \mathrm{PGL}_2(K)$ a projective automorphism. Show that $\sigma(P)$ is a flex of $\sigma(\mathcal{C})$.

Exercise 11.2 Let $\mathcal{C} = \widehat{f}$ be an affine curve of equation $y^2 = x^3 + px + q$ over an algebraically closed field of characteristic $\neq 2, 3$. Prove that the projective closure $\widehat{x_0(f)}$ is non-singular if and only if $x^3 + px + q$ has only simple roots.

Exercise 11.3 Prove Proposition 11.42. (Hint: By induction on n taking into account that each point imposes at most one linear condition on F.)

Exercise 11.4 Let $\mathcal{C}$ be the projective curve of equation $x_1^2 x_2 - x_0^3 + x_0 x_2^2 = 0$ over an algebraically closed field of characteristic $\neq 2, 3$. Show that $\mathcal{C}$ is non-singular and $[0, 1, 0]$ is a flex of $\mathcal{C}$.

Exercise 11.5 (Group Law on a Projective Conic) Let $\widehat{F}$ be a non-degenerate conic of $\mathbb{P}^2(K)$. Let ℓ be a line and let N be a point on $\mathcal{V}_+(\widehat{F}) \smallsetminus \ell$.

1. For $A, B \in \mathcal{V}_+(\widehat{F})$, $A \neq B$, let $P_{AB} = AB \cap \ell$. We define

$$A \circ B := \begin{cases} R & \text{if } R = P_{AB}N \cap \mathcal{V}_+(\widehat{F}) \\ N & \text{if } P_{AB}N \text{ is tangent to } \widehat{F} \text{ at } N. \end{cases}$$

2. If $A = B \in \mathcal{V}_+(\widehat{F})$ and t_A is the tangent line at A to $\widehat{F}$, let $P_A = t_A \cap \ell$ we define

$$A \circ B := \begin{cases} R & \text{if } R = P_A N \cap \mathcal{V}_+(\widehat{F}) \\ N & \text{if } P_A N \text{ is tangent to } \widehat{F} \text{ at } N. \end{cases}$$

Prove that $\mathcal{V}_+(\widehat{F})$ equipped with the operation $\circ$ is an abelian group. (Hint: Use Pascal's Theorem 11.5 for associativity.) Further show that if ℓ is tangent to $\widehat{F}$, the group $(\mathcal{V}_+(\widehat{F}), \circ)$ is isomorphic to the additive group of K while if ℓ intersects the conic in two distinct points, then $(\mathcal{V}_+(\widehat{F}), \circ)$ is isomorphic to the multiplicative group K^* of K.

Chapter 12
Absolute Plane Geometry



12.1 Elements of Absolute Plane Geometry

This chapter is an introduction to the absolute plane geometry closely following [23]. In particular, we shall give an example of a non-euclidean absolute plane geometry.

Definition 12.1 An incidence structure $\mathcal{P} := (\mathcal{X}, \mathcal{D})$ (see Definition 6.1) is called an *absolute plane* if the following group of axioms $a_1), \ldots, a_{10})$ are satisfied:
Incidence axioms.

(a_1) Each line of $\mathcal{P}$ contains at least two points.
(a_2) There exist three non-collinear points of $\mathcal{X}$.

Order axioms.

(a_3) There are two total order relations on each line, say $\leq$ and the opposite $\leq_0$ defined by $A \leq_0 B$ iff $B \leq A$.

Let $A, B, C \in \mathcal{X}$ be three distinct collinear points. We say that C is *between* A and B if $A \leq C \leq B$ or $B \leq C \leq A$. The betweenness relation does not depend on the order chosen on the line d containing A, B and C; indeed

$$A \leq_0 C \leq_0 B \iff B \leq C \leq A, \qquad B \leq_0 C \leq_0 A \iff A \leq C \leq B.$$

Let A and B be two distinct points of $\mathcal{X}$. We define the *segment with endpoints A and B* to be the set

$$[AB] := \{C \in \mathcal{X} : \ C \in AB, \ C \text{ is between } A \text{ and } B\}.$$

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We obviously have $[AB] = [BA]$. Let $A, B, C \in \mathcal{X}$ three distinct non-collinear points, we call the segments $[BC]$, $[CA]$ and $[AB]$ the *sides of triangle* $\triangle ABC$. Now we can state the second order axiom:

(a_4) (Pasch's axiom). If a line $d \in \mathcal{D}$ meets one side of $\triangle ABC$, then d also meets another side of $\triangle ABC$.

Distance axioms. There exists a function $d\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}_+ := [0, \infty)$, called the *distance function*, which satisfies the following axioms:

(a_5) For every $A, B \in \mathcal{X}$, $d(A, B) = d(B, A)$.
(a_6) If $A, B \in \mathcal{X}$, then $d(A, B) = 0$ if and only if $A = B$.
(a_7) A point $C \in AB$ lies on the segment $[AB]$ if and only if $d(A, B) = d(A, C) + d(C, B)$.

Let $\leq$ be a total order on a line d (by axiom a_3)). Fix a point $O \in d$. The sets

$$d_O^+ := \{A \in d \mid A \geq O\}, \qquad d_O^- := \{A \in d \mid A \leq O\}$$

are called *opposite rays of origin* O of d or simply O*-rays*. The opposite rays of origin O with respect to the order $\leq$ coincide with the opposite rays of origin O with respect to the order $\leq_0$. From now on we shall equally denote d_O^+ or d_O^- simply by d_O. A ray with origin O will be also denoted by $(\overrightarrow{OX})$ and the line which contains $(\overrightarrow{OX})$ by OX.

(a_8) For every line $d \in \mathcal{D}$, for every O-ray d_O and for every real number $x \geq 0$, there exists $A \in d_O$ such that $d(O, A) = x$.

Let $f\colon d_O \to \mathbb{R}_+ = [0, \infty)$ be the function defined by $f(A) := d(O, A)$ for all $A \in \mathcal{X}$. Then f is bijective by axiom a_8) (surjectivity), and by axioms a_6) and a_7) (injectivity).

Superposition axioms. We need a preliminary definition. Let $\mathcal{P} = (\mathcal{X}, \mathcal{D})$ be an incidence structure satisfying axioms $(a_1), \ldots, (a_8)$.

Definition 12.2 A map $\sigma\colon \mathcal{X} \to \mathcal{X}$ is called an *automorphism* of $\mathcal{P}$ if σ is bijective, $\sigma(d) \in \mathcal{D}$ for every $d \in \mathcal{D}$ and

$$d(\sigma(A), \sigma(B)) = d(A, B), \qquad \forall\, A, B \in \mathcal{X}.$$

It is immediately seen that the set $\mathrm{Aut}(\mathcal{P})$ of all automorphisms of $\mathcal{P}$ is a group with respect to composition of automorphisms.

Now we can state two superposition axioms.

(a_9) For each $d \in \mathcal{D}$ there is a unique automorphism $s_d \in \mathrm{Aut}(\mathcal{P})$, $s_d \neq \mathrm{id}_{\mathcal{X}}$, which fixes every point of d, i.e. $s_d(P) = P$ for all $P \in d$. We call s_d a *symmetry of axis* d.

(a_{10}) Let d and $d' \in \mathcal{D}$ be two lines such that $d \cap d' = \{O\}$. Let d_O and d'_O be two of the O-rays of d and d'. Then there exists a line $e \in \mathcal{D}$ such that $s_e(d_O) = d'_O$.

The absolute planes are the subject of the *absolute plane geometry*. By abuse of notation we shall also use $\mathcal{P}$ instead of $\mathcal{X}$.

Proposition 12.3 *The standard euclidean plane $\mathcal{P} = \mathbb{E}^2(\mathbb{R})$ is an absolute plane.*

Proof All axioms are easily verified. Furthermore $\mathrm{Aut}(\mathcal{P})$ coincides with the group $\mathrm{O}(\mathbb{E}^n(\mathbb{R}))$ of the isometries of $\mathbb{E}^2(\mathbb{R})$ by Proposition 4.24. By Exercise 5.6 we know that an isometry of $\mathbb{E}^2(\mathbb{R})$ which fixes every point of a line d is necessarily an orthogonal symmetry through d. Finally, the line e of axiom a_{10}) is one of the bisector lines of d and d' (Definition 4.20). □

Definition 12.4 Two lines d and d' of $\mathcal{P}$ are called *parallel* if either $d = d'$, or $d \cap d' = \varnothing$.

If $\mathcal{P} = \mathbb{E}^2(\mathbb{R})$, two lines are parallel according to Definition 12.4 if and only if they are parallel according to Definition 3.68.
Euclid's axiom or parallel postulate:

($\star$) Let d be a line of $\mathcal{P}$, and let $A \in \mathcal{P}$ be a point not lying on d. Then there exists *at most* a line d' through A which is parallel to d.

We shall see in Proposition 12.40 that there always exists *at least* a line d' through A which is parallel to d.

Propositions 12.3 and 3.74 imply:

Corollary 12.5 *The standard euclidean plane $\mathcal{P} = \mathbb{E}^2(\mathbb{R})$ satisfies Euclid's axiom* ($\star$).

An absolute plane satisfying axiom ($\star$) will be called a *euclidean* (*absolute*) *plane* and its geometry *euclidean* (*absolute*) *geometry*; in the opposite case, it will be called a *non-euclidean* (*absolute*) *plane* and its geometry *non-euclidean* (*absolute*) *geometry*.

At the end of this chapter we shall treat the Poincaré hyperbolic half-plane, which is an absolute plane not satisfying axiom ($\star$). In particular, we shall show that Euclid's axiom ($\star$) is independent of the axioms $a_1), \ldots, a_{10})$. First of all we wish to investigate some fundamental properties which follow only from the axioms $a_1), \ldots, a_{10})$, i.e. common to euclidean and non-euclidean geometry.

From now on $\mathcal{P} = (\mathcal{X}, \mathcal{D})$ shall denote an arbitrary absolute plane. Therefore $\mathcal{P}$ satisfies all axioms $a_1), \ldots, a_{10})$ but does not necessarily satisfy Euclid's axiom ($\star$).

Proposition 12.6 *If a line d of $\mathcal{P}$ meets all sides of a triangle $\triangle ABC$, then it passes through one vertex of $\triangle ABC$.*

Proof Suppose by contradiction

$$d \cap [BC] = M, \quad d \cap [CA] = N, \quad d \cap [AB] = P \qquad \text{and} \quad A, B, C \notin d.$$

Then M, N and P are pairwise distinct. By axiom a_3) suppose $M \in [NP]$. Then the line BC meets $[NP]$ but does not meet either $[AN]$ or $[AP]$, contradicting Pasch's axiom (a_4) applied to the triangle $\triangle ANP$. □

Proposition 12.7 *Let d be a line of an absolute plane $\mathcal{P}$. The relation $\mathcal{R}$ on $\mathcal{P} \setminus d$ defined by*

$$A \mathcal{R} B \quad \Longleftrightarrow \quad [AB] \cap d = \varnothing$$

is an equivalence relation. Furthermore $\#(\mathcal{P} \setminus d)/\mathcal{R}) = 2$.

Proof Reflexivity and symmetry of $\mathcal{R}$ are evident. If A, B and C are three points such that $A \mathcal{R} B$ and $B \mathcal{R} C$, i.e. $[AB]$ and $[BC]$ do not meet d, then $[AC] \cap d = \varnothing$ by Pasch's axiom, hence $A \mathcal{R} C$.

There exist two points $A, B \in \mathcal{P}$ such that $[AB] \cap d \neq \varnothing$. Indeed, take one point $O \in d$ and a line e through O, $e \neq d$. By axiom a_8) there exists a point $A \in e_O^+$ and $B \in e_O^-$ (for instance we can take $d(O, A) = d(O, B) = 1$). Since $[AB] \cap d = \{O\}$, A is not equivalent to B. Let $C \in \mathcal{P} \setminus d$. If $C \in e$, then $C \in e_O^+$ or e_O^- but $C \neq O$. If $C \in e_O^+$ then $C \mathcal{R} A$, while if $C \in e_O^-$ then $C \mathcal{R} B$. If $C \notin e$, then the line d does not pass through any vertex of $\triangle ABC$ and $[AB] \cap d = \{O\}$ so that by Pasch's axiom, either $[AC] \cap d \neq \varnothing$ or $[BC] \cap d \neq \varnothing$. By Proposition 12.6 only one of $[AC] \cap d \neq \varnothing$ and $[BC] \cap d \neq \varnothing$ is possibile, hence $C \mathcal{R} A$ or $C \mathcal{R} B$. □

Definition 12.8 If $\widehat{A}$ and $\widehat{B}$ are the (2) elements of $(\mathcal{P} \setminus d)/\mathcal{R}$ we define $\mathbb{H}_d^{\pm}$ as being the sets $\{P \in \mathcal{P} \setminus d : P \mathcal{R} A\}$ and $\{P \in \mathcal{P} \setminus d : P \mathcal{R} B\}$. We call the sets $\mathbb{H}_d^{\pm}$ *open half-planes separated by d*. The sets $\overline{\mathbb{H}}_d^{\pm} := \mathbb{H}_d^{\pm} \cup d$ are called the *closed half-planes* separated by d. The line d is the *boundary* of $\mathbb{H}_d^{\pm}$ and of $\overline{\mathbb{H}}_d^{\pm}$.

Definition 12.9 A line d of $\mathcal{P}$ is *oriented* if one of the total order relations (given by axiom (a_3)) is chosen.

Proposition 12.10 *For every line d of $\mathcal{P}$ and for every point $O \in d$, there is a unique bijective map $f : d \to \mathbb{R}$ such that $f(O) = 0$, $|f(A) - f(B)| = d(A, B)$, $\forall\, A, B \in d$ which is an increasing function, i.e. $f(A) \leq f(B)$ if $A \leq B$.*

Proof Since d is oriented we have $d_O^+ = \{P \in d : O \leq P\}$ and $d_O^- = \{P \in d : P \leq O\}$. Thus we can define

$$f(P) = \begin{cases} d(O, P) & \text{if } P \in d_O^+, \\ -d(O, P) & \text{if } P \in d_O^-. \end{cases}$$

In particular, $f(O) = 0$. The requested properties of f are easily proved. □

Definition 12.11 Let d be an oriented line of $\mathcal{P}$ and let $O \in d$ be a fixed point. The pair (d, O) is called *pointed oriented line*. If (d, O) is a pointed oriented line, we call the real number $f(P)$, where f is given by Proposition 12.10, *abscissa of* $P \in d$.

Corollary 12.12 *For every pair of distinct points* $\{A, B\}$ *of* $\mathcal{P}$*, there exists a unique point* $C \in [AB]$ *such that* $d(A, C) = d(B, C)$*. We call* C *the midpoint of the segment* $[AB]$*. If* $A = B$ *the midpoint of* $[AA]$ *is* A*.*

Proof Suppose $A \leq B$. By axiom a_8) there exists a unique point C of the ray emanating from A and containing B such that $d(A, C) = \frac{1}{2}d(A, B)$. Since by axiom (a_7) one has $d(A, B) = d(A, C) + d(B, C)$, then $d(A, C) = d(B, C)$. □

Proposition 12.13 *Let* $\sigma : \mathcal{P} \to \mathcal{P}$ *be an automorphism of* $\mathcal{P}$*.*

(i) $\sigma([AB]) = [\sigma(A)\sigma(B)]$.
(ii) *Every ray with origin* O *is transformed by* σ *into a ray with origin* $\sigma(O)$.
(iii) *Every half-plane with boundary* d *is transformed by* σ *into a half-plane with boundary* $\sigma(d)$.

Proof (i) If $A, B, C \in \mathcal{P}$ are three distinct collinear points then $\sigma(A)$, $\sigma(B)$ and $\sigma(C)$ are collinear. Therefore, by axiom a_7), if $C \in [AB]$, then $d(A, C) + d(C, B) = d(A, B)$ which implies $d(\sigma(A), \sigma(C)) + d(\sigma(C), \sigma(B)) = d(\sigma(A), \sigma(B))$, i.e. $\sigma(C) \in [\sigma(A)\sigma(B)]$. Therefore if d and $\sigma(d)$ are oriented, then the restriction $\sigma_{|d} : d \to \sigma(d)$ is bijective and monotone. Now (ii) and (iii) easily follow. □

Proposition 12.14 *Let* $\sigma : \mathcal{P} \to \mathcal{P}$ *be an automorphism of* $\mathcal{P}$*.*

(i) *If* σ *has two distinct fixed points* A *and* B*, then* σ *fixes every point of the line* AB*.*
(ii) *If* σ *permutes two distinct points* A *and* B*, then* σ *fixes the midpoint of the segment* $[AB]$*.*
(iii) *If* σ *fixes three distinct non-collinear points* A*,* B *and* C*, then* $\sigma = \mathrm{id}_{\mathcal{P}}$*.*

Proof

(i) Any point P of AB is uniquely determined by its distances from A and B taking into account that $d(A, \sigma(P)) = d(A, P)$ and $d(B, \sigma(P)) = d(B, P)$.
(ii) Let C be the midpoint of $[AB]$. Since $\sigma(A) = B$, $\sigma(B) = A$ and $d(\sigma(C), \sigma(A)) = d(A, C) = d(C, B) = d(\sigma(C), \sigma(B))$ it follows that $\sigma(C)$ is the midpoint of $[\sigma(A)\sigma(B)] = [BA]$ so that $\sigma(C) = C$.
(iii) By (i), σ fixes all points of the lines BC, CA and AB. Let $M \in \mathcal{P} \setminus (AB \cup BC \cup CA)$ and let $D \in [BC]$ with $D \neq B, C$. By Pasch's axiom a_4) the line DM meets a second side of $\triangle ABC$ at a point E. Since D and E are distinct fixed points and $M \in DE$, then M is fixed by σ.

□

Definition 12.15 Let $\mathcal{A}$ and $\mathcal{B}$ be two subsets of $\mathcal{P}$ (more precisely, if $\mathcal{A}, \mathcal{B} \subset \mathcal{X}$). We say that $\mathcal{A}$ and $\mathcal{B}$ are *congruent*, written $\mathcal{A} \equiv \mathcal{B}$, if there exists an automorphism $\sigma \in \mathrm{Aut}(\mathcal{P})$ such that $\sigma(\mathcal{A}) = \mathcal{B}$. Taking into account that $\mathrm{Aut}(\mathcal{P})$ is a group, it immediately follows that the congruence relation $\equiv$ is an equivalence relation on the set of all subsets of $\mathcal{P}$.

Remark 12.16 The proofs of Propositions 12.13 and 12.14 do not require the superposition axioms. We now show some consequences of axioms (a_9) and (a_{10}).

Proposition 12.17 *If s_d is a symmetry of $\mathcal{P}$ of axis d, then:*

(i) $s_d^2 = \mathrm{id}_{\mathcal{P}}$.
(ii) *The locus of fixed points of s_d is the line d.*
(iii) $\sigma_d(\mathbb{H}_d^{\pm}) = \mathbb{H}_d^{\mp}$ *and* $\sigma_d(\overline{\mathbb{H}}_d^{\pm}) = \overline{\mathbb{H}}_d^{\mp}$.

Proof

(i) The automorphism s_d^2 is different from s_d and it fixes all points of d, then by axiom a_9) we have $s_d^2 = \mathrm{id}_{\mathcal{P}}$.
(ii) It follows from Proposition 12.14.
(iii) $A \in \mathcal{P} \setminus d$. By Proposition 12.14-(ii) and by (i) the midpoint O of the segment $[As_d(A)]$ is fixed by s_d. Hence $O \in d$, so that the points A and $s_d(A)$ do not belong to the same half-plane separated by d. □

Proposition 12.18 *Let A and B two distinct points of $\mathcal{P}$. Then there exists a unique line d such that $s_d(A) = B$.*

Proof Let O be the midpoint of the segment $[AB]$ and let $(\overrightarrow{OA})$ (resp. $(\overrightarrow{OB})$) be the ray with origin O containing A (resp. B). By axiom a_{10}) there exists a line d such that $s_d((\overrightarrow{OA})) = (\overrightarrow{OB})$. Since $d(O, A) = d(O, B)$, we get $s_d(A) = B$.

By contradiction, let d' be a line different from d such that $s_{d'}(A) = B$ (hence $s_d \cap s'_d = O$ by Proposition 12.14-(ii)). Then s_d and $s_{d'}$ are automorphisms which permute A and B. In particular, s_d and $s_{d'}$ fix the line $r := AB$ ($r \neq d$ and $r \neq d'$) and they are distinct from s_r. Hence by Proposition 12.13-(iii), s_d and $s_{d'}$ transform a half-plane with boundary r into a half-plane with the same boundary r. Let $\mathbb{H}_r$ be any of the half-planes separated by r. If $P \in \mathbb{H}_r \cap d$ then $s_d(P) = P \in \mathbb{H}_r$ so that s_d preserves the half-planes separated by r. The same happens for $s_{d'}$ and so for their composition $s_d \circ s_{d'}$. The composition $s_d \circ s_{d'}$ fixes A and B so that it fixes every point of r by Proposition 12.14-(i) but does not permute the half-planes with boundary r, hence $s_d \circ s_{d'} \neq s_r$. Therefore by axiom a_9), $s_d \circ s_{d'} = \mathrm{id}_{\mathcal{X}}$. Since $s_{d'}^{-1} = s_{d'}$, we have $s_d = s_{d'}$ and then $d = d'$ (Proposition 12.17). □

Definition 12.19 The unique line d given by Proposition 12.18 is called *the symmetry axis of segment* $[AB]$.

Corollary 12.20 *Let $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$ be two rays of $\mathcal{P}$ with the same origin O. Then there exists a unique line d such that $s_d((\overrightarrow{OX})) = (\overrightarrow{OY})$.*

Proof By Proposition 12.13-(ii), if $s_d((\overrightarrow{OX})) = (\overrightarrow{OY})$, then $s_d(O) = O$. In particular, such a line d necessarily passes through O. First suppose that $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$ do not lie on the same line. Let $A \in (\overrightarrow{OX})$ and $B \in (\overrightarrow{OY})$ such that $d(O, A) = d(O, B) > 0$. Then the equality $s_d((\overrightarrow{OX})) = (\overrightarrow{OY})$ implies $s_d(A) = B$, so that d is the line joining O and the midpoint of $[AB]$. If $(\overrightarrow{OX}) = (\overrightarrow{OY})$, then $d = OX$. If $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$ lie on the same line but $(\overrightarrow{OX}) \neq (\overrightarrow{OY})$, then d is the symmetry axis of the segment $[AB]$. □

Definition 12.21 The unique line d given by Corollary 12.20 is called the *bisector line* of the rays $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$.

Proposition 12.22 *Let A and B two distinct points of $\mathcal{P}$. Then $\{M \in \mathcal{P} : d(M, A) = d(M, B\}$ coincides with the symmetry axis of $[AB]$.*

Proof Let d be the symmetry axis of $[AB]$. If $M \in d$, then $d(M, A) = d(s_d(M), s_d(A)) = d(M, B)$ (since $M \in d$ is fixed by s_d). Conversely, let $M \in \mathcal{P}$ such that $d(M, A) = d(M, B)$. Let e be the bisector line of the rays (MA) and (MB). Then $s_e(A) = B$ and $e = d$, by Proposition 12.18. □

Definition 12.23 Let d, e be two lines of $\mathcal{P}$. We say that d is *perpendicular* to e, written $d \perp e$, if $d \neq e$ and $s_d(e) = e$. Two rays $((\overrightarrow{OX})$ and $(\overrightarrow{OY}))$ are perpendicular if OX and OY are perpendicular.

Proposition 12.24 *We have $d \perp e \iff e \perp d$. Moreover, if $d \perp e$ then $d \cap e \neq \varnothing$.*

Proof Let $A \in e \setminus d$. Since $s_d(e) = e$, $B := s_d(A) \in e$ and $B \neq A$. The midpoint O of $[AB]$ is fixed by s_d so that $O \in d \cap e$. Moreover, d is the symmetry axis of $[AB]$ (Proposition 12.22). Since s_e preserves distances, for every point $M \in d$ we have

$$\begin{aligned} d(s_e(M), A) &= d(s_e(M), s_e(A)) = d(M, A) = d(M, B) \\ &= d(s_e(M), s_e(B)) = d(s_e(M), B). \end{aligned}$$

Hence $s_e(M)$ belongs to the symmetry axis of $[AB]$ so that $e \perp d$. □

Remark 12.25 From the proof of Proposition 12.24 we see that the symmetry axis of a segment $[AB]$ $(A \neq B)$ is the line perpendicular to AB passing through the midpoint O of $[AB]$.

Proposition 12.26 *Let A be a point and let d be a line of $\mathcal{P}$. Then there exists one and only one line e perpendicular to d and passing through A.*

Proof Suppose $A \notin d$. the perpendicular line e necessarily passes through $B := s_d(A)$ so that $e = AB$ which implies the uniqueness of e. Moreover, $s_d(AB) = s_d(A)s_d(B) = BA$, hence $d \perp e$. If $A \in d$, then e is the bisector line of the rays d_A^+ and d_A^- (Corollary 12.20 and Definition 12.21). □

Proposition 12.27 *Let d, d' be two distinct lines perpendicular to a line e. Then $d \cap d' = \varnothing$.*

Proof Suppose by contradiction that there exists $A \in d \cap d'$. Then through A there would be two lines perpendicular to e, contradicting Proposition 12.26. □

Corollary 12.28 *Let d, e be two lines of $\mathcal{P}$ and let σ be an automorphism of $\mathcal{P}$. Then $d \perp e$ if and only if $\sigma(d) \perp \sigma(e)$.*

Proof Let $d \perp e$, $O = d \cap e$, and let $A, B \in e$, be such that the midpoint of $[AB]$ is O. Then $d = \{M \in \mathcal{P} : d(M, A) = d(M, B\}$. Hence $\sigma(d) = \{N \in \mathcal{P} : d(N, \sigma(A)) = d(N, \sigma(B))\}$ so that $\sigma(d) \perp \sigma(e)$. □

Proposition 12.29 *Let $\triangle ABC$ and $\triangle A'B'C'$ be two triangles of $\mathcal{P}$. Assume that*

$$d(A, B) = d(A', B'), \quad d(B, C) = d(B', C'), \quad d(A, C) = d(A', C').$$

Then there exists a unique automorphism $\sigma \in \mathrm{Aut}(\mathcal{P})$ such that

$$\sigma(A) = A', \qquad \sigma(B) = B', \qquad \sigma(C) = C'. \tag{12.1}$$

Moreover, σ is the composition of two or three axial symmetries.

Proof Let $\sigma, \tau \in \mathrm{Aut}(\mathcal{P})$ be two automorphisms which satisfy the assumptions. Then $\alpha := \sigma^{-1} \circ \tau$ fixes three non-collinear points A, B and C, hence by Proposition 12.14-(iii), $\alpha = \mathrm{id}_{\mathcal{P}}$, i.e. $\sigma = \tau$ and uniqueness is proved.

Let s_1 be an axial symmetry such that $s_1(A) = A'$ (s_1 is uniquely determined if $A \neq A'$) and let $B_1 := s_1(B)$, $C_1 := s_1(C)$. Then $d(A', B_1) = d(A, B) = d(A', B')$ so that there is an axial symmetry s_2 such that $s_2(A') = A'$ and $s_2(B_1) = B'$. Let $C_2 := s_2(C_1)$, then

$$d(A', C_2) = d(A', C_1) = d(A, C) = d(A', C'),$$
$$d(B', C_2) = d(B_1, C_1) = d(B, C) = d(B', C').$$

There are two cases. If $C_2 = C'$ then $\sigma := s_2 \circ s_1$ satisfies (12.1). If $C_2 \neq C'$, then the line $A'B'$ is the symmetry axis of $[C'C_2]$ (indeed $d(A', C_2) = d(A', C')$ and $d(B', C_2) = d(B', C')$). If s_3 is the symmetry of axis $A'B'$, then $\sigma := s_3 \circ s_2 \circ s_1$ is the required automorphism (indeed $s_3(C_2) = C'$). □

Corollary 12.30 *Every automorphism of $\mathcal{P}$ is the composition of two or three axial symmetries.*

Definition 12.31 An automorphism $\sigma \in \mathrm{Aut}(\mathcal{P})$ is called a *rotation* if either $\sigma = \mathrm{id}_{\mathcal{P}}$ or if σ has a unique fixed point O called the *centre* of σ. By convention, every point of $\mathcal{P}$ is the centre of $\mathrm{id}_{\mathcal{P}}$.

Proposition 12.32 *Let $\mathcal{P}$ be an absolute plane.*

(i) *The composition of two symmetries whose axes meet at a point O is a rotation with centre O.*
(ii) *Conversely, for every rotation σ with centre O and for every line d passing through O, there exist two lines d_1 and d_2 passing through O such that $\sigma = s_{d_1} \circ s_d = s_d \circ s_{d_2}$.*

Proof

(i) Let d, d_1 be two lines passing through O and let $\sigma := s_d \circ s_{d_1}$. Then O is fixed by σ. If $d = d_1$ then $\sigma = \mathrm{id}_{\mathcal{P}}$, which is a rotation. Let $d \neq d_1$. We have to prove that O is the only fixed point of σ. Suppose that a point $A \neq O$ is fixed by σ. Take $B := s_{d_1}(A)$, then $s_d(B) = \sigma(A) = A$ and $s_{d_1}(B) = A$. By Proposition 12.18 we conclude that $A = B$ or $d = d_1$. Then $A = B$, hence $A \in d_1 \cap d = \{O\}$, i.e. $A = O$.
(ii) If $\sigma = \mathrm{id}_{\mathcal{P}}$ we can take $d_1 = d_2 = d$. Let $\sigma \neq \mathrm{id}_{\mathcal{P}}$. Take $A \in d \setminus \{O\}$ and $B := \sigma(A)$. Then $d(O, A) = d(O, B)$ so that there exists a symmetry whose axis d_1 passes through O (and the midpoint of $[AB]$) such that $s_{d_1}(A) = B$. Hence $s_{d_1} \circ \sigma$ is an automorphism $\neq \mathrm{id}_{\mathcal{P}}$ which fixes O and A. By Proposition 12.14-(ii) every point of $d = OA$ is fixed by $s_{d_1} \circ \sigma$. By axiom a_9), we have $s_{d_1} \circ \sigma = s_d$, i.e. $\sigma = s_{d_1} \circ s_d$. In a completely analogous way one proves that there exists a line d_2 through O such that $\sigma = s_d \circ s_{d_2}$. □

Corollary 12.33 *If σ is a rotation with centre O and d is a line passing through O, then the automorphisms $s_d \circ \sigma$ and $\sigma \circ s_d$ are symmetries whose axes pass through O.*

Proof By Proposition 12.32 we have $\sigma = s_{d_1} \circ s_d$ i.e. $\sigma \circ s_d = s_{d_1}$ and $\sigma = s_d \circ s_{d_2}$ i.e. $s_d \circ \sigma = s_{d_2}$. □

Proposition 12.34 *The set $\mathcal{R}_O$ of all rotations with centre O is an abelian subgroup of* $\mathrm{Aut}(\mathcal{P})$.

Proof First we prove that $\mathcal{R}_O$ is a subgroup of $\mathrm{Aut}(\mathcal{P})$. Let $r, r' \in \mathcal{R}_O$ be two rotations. We have to show that $r' \circ r^{-1} \in \mathcal{R}_O$. Fix a line d passing through O. By Corollary 12.33, there exist two symmetries d_1, d_2 through O such that $r = s_{d_1} \circ s_d$ and $r' = s_{d_2} \circ s_d$. Then

$$r' \circ r^{-1} = s_{d_2} \circ s_d \circ (s_{d_1} \circ s_d)^{-1} = s_{d_2} \circ s_{d_1},$$

so that by Proposition 12.32-(i), $r' \circ r^{-1} \in \mathcal{R}_O$.

Now we have to show that $r \circ r' = r' \circ r$ for every $r, r' \in \mathcal{R}_O$. Let $A \in \mathcal{P} \setminus \{O\}$ and $s := s_{OA}$. By Corollary 12.33, $t := r \circ s$ is the symmetry whose axis passes through O such that $t(A) = r(A) := B$. Hence $r \circ s(B) = A$. Analogously, $t' := r' \circ s$ is the symmetry whose axis passes through O such that $t'(A) = r'(A) := B'$, then $r' \circ s(B') = A$. Since $\mathcal{R}_O$ is a subgroup of $\mathrm{Aut}(\mathcal{P})$, $r \circ r', r' \circ r \in \mathcal{R}_O$. By Corollary 12.33, $u := r' \circ r \circ s$ is the symmetry whose axis passes through O such that $u(B) = B'$; analogously, $v := r \circ r' \circ s$ is the symmetry whose axis passes through O such that $v(B') = B$. Since u and v fix the midpoint of $M = [BB']$ (Proposition 12.14) their axes coincide with the line OM, hence (by axiom a_9)) $u = v$, i.e. $r' \circ r \circ s = r \circ r' \circ s$ which implies $r' \circ r = r \circ r'$. □

Corollary 12.35 *For every pair of rays $(\vec{OX})$ and $(\vec{OY})$ with the same origin O, there exists a unique rotation σ with centre O such that $\sigma((\vec{OX})) = (\vec{OY})$.*

Proof Let s be a symmetry whose axis contains $(\vec{OX})$ and let s' be the axial symmetry interchanging the rays $(\vec{OX})$ and $(\vec{OY})$ (Corollary 12.20). Then $\sigma := s' \circ s$ is a rotation with centre O (Proposition 12.32-i)) such that $\sigma((\vec{OX})) = (\vec{OY})$. Let σ' be another rotation with centre O such that $\sigma'((\vec{OX})) = (\vec{OY})$. Let $A \in (\vec{OX})$, $A \neq O$. Then $\sigma(A) = \sigma'(A)$, so that A is a fixed point of the rotation $\sigma^{-1} \circ \sigma'$ with centre O, which implies $\sigma = \sigma'$. □

Proposition 12.36 *Let O and O' be two distinct points of $\mathcal{P}$, and let s be an axial symmetry which interchanges O and O'. Then the map*

$$f : \mathcal{R}_O \to \mathcal{R}_{O'}, \qquad f(r) = s \circ r \circ s, \quad \forall r \in \mathcal{R}_O,$$

is a group isomorphism.

Proof Exercise for the reader. □

Definition 12.37 Let O be a point of $\mathcal{P}$. The *symmetry through O* (or the *central symmetry through O*) is the map $\sigma_O : \mathcal{P} \to \mathcal{P}$ defined by the following condition: O is the midpoint of the segment $[M\sigma_O(M)]$, $\forall M \in \mathcal{P}$.

Proposition 12.38 *Let σ_O be the central symmetry through O, then there exist two axial symmetries s_d and s_e with centre O, whose axes d and e are perpendicular, such that*

$$\sigma_O = s_d \circ s_e = s_e \circ s_d.$$

In particular, the central symmetry through O is the unique involutive rotation with centre O, different from $\mathrm{id}_{\mathcal{P}}$.

Proof Let d, e be two perpendicular lines passing through O, then the axial symmetries s_d, s_e commute. Indeed, take $A \in e \setminus \{O\}$, then $B = s_d(A)$ belongs to e by Definition 12.23 so that $s_e \circ s_d(A) = s_e(B) = B$ and $s_d \circ s_e(A) = s_d(A) = B$.

By Corollary 12.35 we have the equality $s_d \circ s_e = s_e \circ s_d := r$, the unique rotation with centre O such that $r((\overrightarrow{OA})) = (\overrightarrow{OB})$. Therefore

$$(s_e \circ s_d)^2 = (s_e \circ s_d) \circ (s_d \circ s_e) = s_e \circ s_d^2 \circ s_e = s_e^2 = \mathrm{id}_{\mathcal{P}},$$

i.e. r is an involutive rotation. Furthermore $r \neq \mathrm{id}_{\mathcal{P}}$ because $d \neq e$.

We shall now prove that each rotation $r \neq \mathrm{id}_{\mathcal{P}}$ with centre O such that $r^2 = \mathrm{id}_{\mathcal{P}}$ coincides with the central symmetry σ_O. Let $A \in \mathcal{P}$. If Q is the midpoint of the segment $[Ar(A)]$, then $r(Q)$ is the midpoint of the segment $[r(A)r(r(A))] = [r(A)A]$, hence Q is fixed by r. Since r is a rotation, then $Q = O$ and $r = \sigma_O$. □

Proposition 12.39 *Let d be a line and let O be a point of an absolute plane $\mathcal{P}$ such that $O \notin d$. If σ_O is the symmetry through O then $d \cap \sigma_O(d) = \varnothing$.*

Proof Assume by contradiction that there exists $A \in d \cap \sigma_O(d)$, then $B := \sigma_O(A) \in d \cap \sigma_O(d)$. Since $B \neq A$ ($A \neq O$) the lines d and $\sigma_O(d)$ should coincide with the line AB. In particular, O (which is the midpoint of $[AB]$) should belong to d. □

Proposition 12.40 *Let d be a line and let A be a point of an absolute plane $\mathcal{P}$ such that $A \notin d$. The there exists at least a line d' passing through A such that $d \cap d' = \varnothing$, i.e. $d' \parallel d$.*

Proof Let B be an arbitrary point of d and let O be the midpoint of of the segment $[AB]$. By Proposition 12.39, the line $d' := \sigma_O(d)$ satisfies the thesis. □

12.1.1 Angles in an Absolute Plane

Let $\mathcal{P}$ be an absolute plane, let O be a point of $\mathcal{P}$, and let $\mathcal{M}_O$ be the set of pairs $((\overrightarrow{OX}), (\overrightarrow{OY}))$ of rays emanating from O. Let $d = OX$ be the line containing $(\overrightarrow{OX})$ and $e = OY$ be the line containing $(\overrightarrow{OY})$. Denote by:

(a) $\mathbb{H}_d((\overrightarrow{OY}))$ (resp. $\overline{\mathbb{H}}_d((\overrightarrow{OY}))$) the open (resp. closed) half-plane delimited by d which contains $(\overrightarrow{OY})$,

(b) $\mathbb{H}_e((\overrightarrow{OX}))$ (resp. $\overline{\mathbb{H}}_e((\overrightarrow{OX}))$) the open (resp. closed) half-plane delimited by e which contains $(\overrightarrow{OX})$.

With each pair $((\overrightarrow{OX}), (\overrightarrow{OY})) \in \mathcal{M}_O$ is associated an *open angular sector* $\overset{\circ}{S}((\overrightarrow{OA}), (\overrightarrow{OY}))$ of sides $(\overrightarrow{OA})$ and $(\overrightarrow{OY})$ and a *closed angular sector*

$S((\overrightarrow{OA}), (\overrightarrow{OY}))$ defined by

$$\overset{\circ}{S}((\overrightarrow{OA}), (\overrightarrow{OY})) := \mathbb{H}_d((\overrightarrow{OY})) \cap \mathbb{H}_e((\overrightarrow{OX})),$$

$$S((\overrightarrow{OA}), (\overrightarrow{OY})) := \overline{\mathbb{H}}_d((\overrightarrow{OY})) \cap \overline{\mathbb{H}}_e((\overrightarrow{OX})),$$

respectively.

Definition 12.41 Two pairs $((\overrightarrow{OX}), (\overrightarrow{OY}))$ and $((\overrightarrow{O'X'}), (\overrightarrow{O'Y'}))$ of $\mathcal{M}_O$ are called *congruent* if there exists an automorphism $\sigma \in \mathrm{Aut}(\mathcal{P})$ such that $\sigma((\overrightarrow{OX})) = (\overrightarrow{O'X'})$ and $\sigma((\overrightarrow{OY})) = (\overrightarrow{O'Y'})$. Since $\mathrm{Aut}(\mathcal{P})$ is a group the congruence relation is an equivalence relation on $\mathcal{M}_O$ and its congruence classes are called *non-oriented angles*. The congruence class of $((\overrightarrow{OX}), (\overrightarrow{OY})) \in \mathcal{M}_O$ will be denoted by $\widehat{XOY}$.

Since there exists an axial symmetry s which interchanges $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$ (Corollary 12.20) we have $\widehat{XOY} = \widehat{YOX}$. If $(\overrightarrow{OX'})$ and $(\overrightarrow{OY'})$ are the opposite rays of $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$ respectively, it is also true that $\widehat{X'OY'} = \widehat{XOY}$. Indeed, the central symmetry σ_O interchanges $(\overrightarrow{OX})$ and $(\overrightarrow{OX'})$ as well as $(\overrightarrow{OY})$ and $(\overrightarrow{OY'})$. The congruence class $\widehat{XOX}$ of two equal rays $(\overrightarrow{OX})$ is called the *zero angle*, denoted by 0; the congruence class $\widehat{XOX'}$ of two opposite rays $(\overrightarrow{OX})$ and $(\overrightarrow{OX'})$ is said a *straight angle*, denoted by ω.

The study of angles will be simplified by representing them as a pair $((\overrightarrow{OX}), (\overrightarrow{OY}))$, where $(\overrightarrow{OX}))$ is fixed and $(\overrightarrow{OY})) \subset \overline{\mathbb{H}}_d((\overrightarrow{OY}))$ with $d = OX$.

Proposition 12.42 *Let $(\overrightarrow{OX})$ be a ray of $\mathcal{P}$, and let Π be a closed half-plane delimited by the line OX. Then every non-oriented angle of $\mathcal{P}$ has a unique representation of the form $((\overrightarrow{OX}), (\overrightarrow{OY}))$, where $(\overrightarrow{OY})$ is a ray contained in Π. In other words, If $\mathcal{S}_{O,\Pi}$ is the set of rays emanating from O and contained in Π and $\mathcal{A}$ the set of non-oriented angles of $\mathcal{P}$, the map $\mathcal{S}_{O,\Pi} \to \mathcal{A}$, defined by $(\overrightarrow{OY}) \to \widehat{XOY}$, is bijective.*

Proof We have to show that for any non-oriented angle α there exists a unique ray $(\overrightarrow{OY})$ contained in Π such that $\alpha = \widehat{XOY}$. If $\alpha = \widehat{X'O'X'} = 0$ there exists an axial symmetry s_1 such that $s_1(O') = O$; therefore $s_1((\overrightarrow{O'X'})) = (\overrightarrow{OZ})$ for some $Z \in \mathcal{P}$ (Proposition 12.18). Moreover there exists an axial symmetry s_2 such that $s_2((\overrightarrow{OZ})) = (\overrightarrow{OX})$ (Corollary 12.20). Hence $\sigma := s_2 \circ s_1$ is an automorphism such that $\sigma((\overrightarrow{O'X'})) = (\overrightarrow{OX})$. Thus the thesis is proved if $\alpha = 0$. The same proof works for the straight angle ω.

Hence we can assume $\alpha \neq 0, \omega$. Uniqueness is immediate. Indeed,if $\widehat{XOY} = \widehat{XOZ}$ (with $(\overrightarrow{OX}), (\overrightarrow{OZ}) \subset \Pi$), then there exists $\sigma \in \mathrm{Aut}(\mathcal{P})$ such that $\sigma((\overrightarrow{OX})) = (\overrightarrow{OX})$ (hence σ fixes every point of the line $d = OX$) and $\sigma((\overrightarrow{OY})) = (\overrightarrow{OZ})$. By axiom a_9) we have $\sigma = \mathrm{id}_{\mathcal{P}}$ or $\sigma = s_d$. We claim that $\sigma = \mathrm{id}_{\mathcal{P}}$. Indeed, the rays $(\overrightarrow{OY})$ and $(\overrightarrow{OZ})$ are both contained in the closed half-plane Π (but not contained in d since $\alpha \neq 0, \omega$), while s_d interchanges the open half-planes delimited by d (Proposition 12.17).

Let $\alpha = \widehat{UAV} \neq 0, \omega$. It remains to prove that there exists a ray $(\overrightarrow{OY}) \subset \Pi$ such that $\alpha = \widehat{XOY}$. As above there exists an axial symmetry s_1 such that $s_1(A) = O$. Put $s_1((\overrightarrow{AU})) = (\overrightarrow{OU_1})$ and $s_1((\overrightarrow{AV})) = (\overrightarrow{OV_1})$. There exists a symmetry s_2 whose axis passes through O such that $s_2((\overrightarrow{OU_1})) = (\overrightarrow{OX})$. If $s_2((\overrightarrow{OV_1})) \subset \Pi$, the ray $(\overrightarrow{OY}) := s_2((\overrightarrow{OV_1}))$ verifies the thesis. Otherwise, we can take $(\overrightarrow{OY}) = s_d(s_2((\overrightarrow{OV_1})))$, where d is the line OX. □

Corollary 12.43 *Let $((\overrightarrow{OX}), (\overrightarrow{OY}))$ and $((\overrightarrow{O'X'}), (\overrightarrow{O'Y'}))$ be two pairs of perpendicular rays. Then $((\overrightarrow{OX}), (\overrightarrow{OY}))$ and $((\overrightarrow{O'X'}), (\overrightarrow{O'Y'}))$ are congruent. We call the non-oriented angle $\widehat{XOY}$ a right angle and denote it by δ.*

Proof Let $\Pi = \overline{\mathbb{H}}_d((\overrightarrow{OY}))$ be the closed half-plane delimited by $d = OX$ containing $(\overrightarrow{OY})$. The couple $((\overrightarrow{O'X'}), (\overrightarrow{O'Y'}))$ is congruent to $((\overrightarrow{OX}), (\overrightarrow{OZ}))$, with $(\overrightarrow{OZ}) \subset \Pi$ (Proposition 12.42). Since $(\overrightarrow{O'X'})$ and $(\overrightarrow{O'Y'})$ are perpendicular, the same holds for $(\overrightarrow{OX})$ and $(\overrightarrow{OZ})$ by Corollary 12.28. By assumption $(\overrightarrow{OX})$ and $(\overrightarrow{OY})$ are perpendicular so that $(\overrightarrow{OY}) = (\overrightarrow{OZ})$ by Proposition 12.26. □

Order Relation on the Set of All Angles

Definition 12.44 We fix a ray $(\overrightarrow{OX})$ and a closed half-plane Π delimited by OX. By Proposition 12.42 every angle α is of the form $\widehat{XOY}$ (where the ray $(\overrightarrow{OY}) \subset \Pi$ is uniquely determined), therefore we can associate α with the closed angular sector S_α delimited by $(\overrightarrow{OX})$ and $(\overrightarrow{OY}X)$; for instance, $S_0 = (\overrightarrow{OX})$ and $S_\omega = \Pi$. Let $\mathcal{A}$ be the set of all angles of $\mathcal{P}$. We put on $\mathcal{A}$ the following order relation:

$$\alpha \leq \beta \quad \Longleftrightarrow \quad S_\alpha \subseteq S_\beta, \quad \forall\, \alpha, \beta \in \mathcal{A}.$$

Therefore 0 is the smallest element of $\mathcal{A}$, and ω is the biggest element of $\mathcal{A}$. In order to prove that $\leq$ is a total order relation on $\mathcal{A}$ we need the following result.

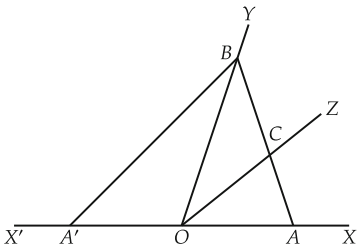


Fig. 12.1 Order relation on the set of angles: I

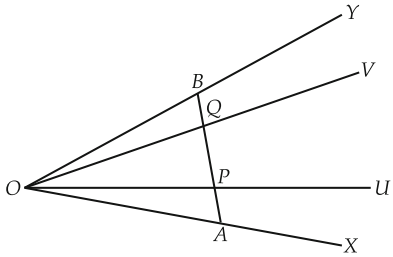


Fig. 12.2 Order relation on the set of angles: II

Proposition 12.45 *Assume the above notation and assumptions. Let* $(\overrightarrow{OX'})$ *be the opposite ray of* $(\overrightarrow{OX})$*. Let* $(\overrightarrow{OY})$ *be a ray contained in* Π*, distinct from* $(\overrightarrow{OX})$ *and* $(\overrightarrow{OX'})$*. Let* $A \in (\overrightarrow{OX})$*,* $B \in (\overrightarrow{OY})$ *and* $A' \in (\overrightarrow{OX'})$ *(Fig. 12.1).*

(i) *Every ray* $(\overrightarrow{OZ})$ *contained in* Π*, such that* $(\overrightarrow{OZ}) \neq (\overrightarrow{OX}), (\overrightarrow{OY}), (\overrightarrow{OX'})$*, meets only one of the segments*

(a) $[AB]$, *if* $S((\overrightarrow{OX}), (\overrightarrow{OZ})) \subset S((\overrightarrow{OX}), (\overrightarrow{OY}))$,

(b) $[A'B]$, *if* $S((\overrightarrow{OX}), (\overrightarrow{OZ})) \supset S((\overrightarrow{OX}), (\overrightarrow{OY}))$.

(ii) *Let* $(\overrightarrow{OU})$ *and* $(\overrightarrow{OV})$ *be two rays contained in* $S((\overrightarrow{OX}), (\overrightarrow{OY}))$ *which meet the segment* $[AB]$ *at the points* P *and* Q *respectively (Fig. 12.2). Then*

$$S((\overrightarrow{OX}), (\overrightarrow{OU})) \subset S((\overrightarrow{OX}), (\overrightarrow{OV})) \iff P \in [AQ].$$

Proof (i) By Proposition 12.6 and Pasch's axiom applied to the triangle $\triangle ABA'$, the line OZ meets only one between the sides $[AB]$ and $[A'B]$ at a point which necessarily belongs to the ray $(\overrightarrow{OZ})$. Since $A, B \in \overline{\mathbb{H}}_d((\overrightarrow{OX}))$ (with $d = OY$) as well as $A, B \in \overline{\mathbb{H}}_e((\overrightarrow{OY}))$ (with $e = OX$), we have

$$[AB] \subset \overline{\mathbb{H}}_d((\overrightarrow{OX})) \cap \overline{\mathbb{H}}_e((\overrightarrow{OY})) = S((\overrightarrow{OX}), (\overrightarrow{OY})).$$

In the same way one sees that $[A'B] \subset S((\overrightarrow{OY}), (\overrightarrow{OX'}))$. Therefore $(\overrightarrow{OZ}) \subset S((\overrightarrow{OX}), (\overrightarrow{OY}))$ or $(\overrightarrow{OZ}) \subset S((\overrightarrow{OY}), (\overrightarrow{OX'}))$. In other words $S((\overrightarrow{OX}), (\overrightarrow{OY}))$ (resp. $S((\overrightarrow{OY}), (\overrightarrow{OX'}))$) is the union of all rays with origin O which meet $[AB]$ (resp. $[A'B]$).

If $C := OZ \cap [AB]$, by the same argument as above we see that $S((\overrightarrow{OX}), (\overrightarrow{OZ}))$ is the union of all rays with origin O which meet $[AC]$. In particular, $S((\overrightarrow{OX}), (\overrightarrow{OZ})) \subset S((\overrightarrow{OX}), (\overrightarrow{OY}))$.

If OZ meets $[A'B]$ in the same way we see that $S((\overrightarrow{OX'}), (\overrightarrow{OZ})) \subset S((\overrightarrow{OX'}), (\overrightarrow{OY}))$ (and $\overset{\circ}{S}((\overrightarrow{OX'}), (\overrightarrow{OZ})) \subset \overset{\circ}{S}((\overrightarrow{OX'}), (\overrightarrow{OY}))$) so that

$$\begin{aligned} S((\overrightarrow{OX}), (\overrightarrow{OZ})) &= \Pi \setminus \overset{\circ}{S}((\overrightarrow{OX'}), (\overrightarrow{OZ})) \supset \Pi \setminus \overset{\circ}{S}((\overrightarrow{OX'}), (\overrightarrow{OY})) \\ &= S((\overrightarrow{OX}), (\overrightarrow{OY})). \end{aligned}$$

(ii) By (i) we have $S((\overrightarrow{OX}), (\overrightarrow{OU})) \subset S((\overrightarrow{OX}), (\overrightarrow{OV}))$ if and only if $\overrightarrow{OU} \subset S((\overrightarrow{OX}), (\overrightarrow{OV}))$, i.e. $P \in AQ$. □

Corollary 12.46 *The order relation $\leq$ on $\mathcal{A}$ of Definition 12.44 is a total order relation such that every non-empty subset of $\mathcal{A}$ has a supremum and an infimum.*

Proof The first assertion follows from Proposition 12.45-i). Let $\alpha := \widehat{XOY}$ be as in Proposition 12.45. The function $f : \mathcal{A} \to [AB] \cup [BA']$ defined by

$$f(\widehat{AOZ}) = \begin{cases} C = OZ \cap [AB] & \text{if } 0 \leq \widehat{AOZ} \leq \widehat{XOY} \\ C = OZ \cap [BA'] & \text{if } \widehat{XOY} \leq \widehat{AOZ} \leq \omega \end{cases}$$

is a strictly increasing and bijective. Therefore there exists a bijective strictly increasing map $\mathcal{A} \to [a, b]$, where $[a, b]$ is a bounded closed interval of $\mathbb{R}$ of which every non-empty subset has a supremum and an infimum. □

Sum of Angles

We say that a non-oriented angle γ is the *sum* of two non-oriented angles α and β, written $\gamma = \alpha + \beta$, if there exist three rays with origin O, $(\overrightarrow{OX})$, $(\overrightarrow{OY})$ and $(\overrightarrow{OZ})$ which lie on the same closed half-plane delimited by OX such that $\alpha = \widehat{XOY}$, $\beta = \widehat{YOZ}$ and $\gamma = \widehat{XOZ}$, and satisfying the following conditions:

(a) $S((\overrightarrow{OX}), (\overrightarrow{OZ})) = S((\overrightarrow{OX}), (\overrightarrow{OY})) \cup S((\overrightarrow{OY}), (\overrightarrow{OZ}))$, or

(b) the ray $(\overrightarrow{OZ})$ is the opposite of $(\overrightarrow{OX})$ (in this case we say that α and β are *supplementary* and $\alpha + \beta = \omega$).

If $\gamma = \alpha + \beta$ we can write $\alpha = \gamma - \beta$ and $\beta = \gamma - \alpha$.

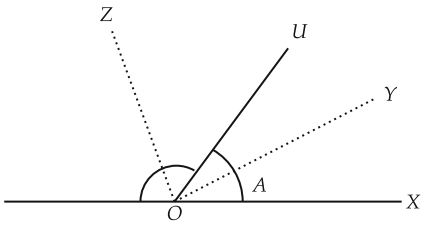


Fig. 12.3 Bisector

The sum of two angles may not be defined, e.g. $\omega + \omega$ is not defined. For every angle α, $\alpha + 0$ is always defined and we have $\alpha + 0 = \alpha$. If n is a positive integer then $n \cdot \widehat{XOY} := \underbrace{\widehat{XOY} + \cdots + \widehat{XOY}}_{n \text{ times}}$ whenever the sum is defined. Moreover $0 \cdot \alpha := 0$ for every angle α. The sum of two angles (whenever it is defined) is clearly commutative and associative. The following implication $\alpha + \beta = \alpha + \gamma \implies \beta = \gamma$ is immediate.

Bisector of an Angle

In Definition 12.21 we introduced the notion of bisector of two rays $(\vec{OY})$ and $(\vec{OZ})$ as being the unique line d passing through O such that $s_d((\vec{OY})) = (\vec{OZ})$. If $(\vec{OY})$ and $(\vec{OZ})$ are not the opposite rays, we call the *interior bisector* of $(\vec{OY})$ and $(\vec{OZ})$ the ray $(\vec{OU})$ of d contained in $S((\vec{OY}), (\vec{OZ}))$ (Fig. 12.3).

Clearly we have $\widehat{YOU} = \widehat{UOZ}$. Since $\widehat{YOZ} = \widehat{YOU} + \widehat{UOZ}$, then $\widehat{YOZ} = 2 \cdot \widehat{YOU}$ so we can say that the angle $\widehat{YOU} = \widehat{UOZ}$ *bisects* $\widehat{YOZ}$ and write $\widehat{YOU} = \frac{1}{2} \cdot \widehat{YOZ}$. The bisector of a straight angle $\widehat{XOX'}$ is the ray emanating from O, lying on Π and perpendicular at O to OX. Therefore the right angle δ is half of ω, i.e. $\delta = \frac{1}{2} \cdot \omega$. An angle is called *acute* if $< \frac{\omega}{2}$, *obtuse* if $> \frac{1}{2}\omega$.

By induction, for every non-oriented angle α and for every integer $n \geq 1$ we can define the angle α_n by means of the formula $\alpha = 2^n \cdot \alpha_n$; thus α_n will be denoted by $2^{-n} \cdot \alpha$.

Measure of Angles

Let $\alpha = \widehat{XOY}$, where the rays $(\vec{OX})$ and $(\vec{OY})$ are contained in a closed half-plane Π delimited by OX.

Fo every integer $n \geq 1$ consider the angle $2^{-n} \cdot \omega$ and

$$\widehat{XOY}_n^q := q2^{-n} \cdot \omega, \qquad q = 0, 1, \ldots, 2^n.$$

For n fixed we clearly have

$$\bigcup_{q=0}^{2^n-1} S((\overrightarrow{OY_n^q}), (\overrightarrow{OY_n^{q+1}})) = \Pi.$$

As $n \to +\infty$ then the covering $\{S((\overrightarrow{OY_n^q}), (\overrightarrow{OY_n^{q+1}}))\}$ becomes finer and finer so that we have:

Proposition 12.47 *For any α and for any integer $n \geq 1$ there exists one unique integer $0 \leq q_n \leq 2^n$ such that*

$$q_n 2^{-n} \cdot \omega \leq \alpha < (q_n + 1)2^{-n} \cdot \omega.$$

Archimedes' axiom holds true for angles.

Proposition 12.48 *For every angle $\alpha \neq 0$ there exists a natural number $n \geq 1$ such that $2^{-n} \cdot \omega \leq \alpha$.*

Proof Let $A := \{2^{-n} \cdot \omega \mid n \geq 1\} \subset \mathcal{A}$. By Corollary 12.46 A has an infimum ε. Let $f : \mathcal{A} \to [0, \delta]$ defined by $f(\beta) = \frac{1}{2} \cdot \beta$. It easy to see that f is bijective, strictly increasing on the interval $[0, \delta]$ of $\mathcal{A}$ and $f(A) \subset A$. Hence $\inf A \leq \inf f(A)$, i.e. $\varepsilon \leq \frac{1}{2}\varepsilon$, which implies $\varepsilon = 0$. □

Corollary 12.49 *For every angle α let $\mathcal{E}_\alpha := \{q2^{-n} \cdot \omega \mid n \geq 1,\ 0 \leq q \leq 2^n,\ q2^{-n} \cdot \omega \leq \alpha\}$. Then*

$$\alpha = \sup \mathcal{E}_\alpha.$$

Proof We obviously have $\alpha \geq \gamma$ for each $\gamma \in \mathcal{E}_\alpha$. For any angle β with $\beta < \alpha$, there exists an integer $n \geq 1$ such that $2^{-n} \cdot \omega < \alpha - \beta$ (Proposition 12.48). For the above n, by Proposition 12.47, there is an integer q_n such that $q_n 2^{-n} \cdot \omega \leq \alpha < (q_n + 1)2^{-n} \cdot \omega$. Hence

$$\beta < \alpha - 2^{-n} \cdot \omega < (q_n + 1)2^{-n} \cdot \omega - 2^{-n} \cdot \omega = q_n 2^{-n} \cdot \omega \leq \alpha.$$

Since $q_n 2^{-n} \cdot \omega \in \mathcal{E}_\alpha$, α is the supremum of $\mathcal{E}_\alpha$. □

Definition 12.50 A *measure on the set of non-oriented angles* of $\mathcal{P}$ is any strictly increasing function $\varphi : \mathcal{A} \to \mathbb{R}_+$ such that

$$\varphi(\alpha + \beta) = \varphi(\alpha) + \varphi(\beta), \tag{12.2}$$

for all $\alpha, \beta \in \mathcal{P}$ whose sum $\alpha + \beta$ is defined.

Theorem 12.51 *For every real number $k > 0$ there exists a unique measure φ such that $\varphi(\omega) = k$ and φ is a bijective map $\mathcal{A} \to [0, k]$.*

Proof Suppose that φ exists. By (12.2), for every pair of natural numbers (n, q) with $q \leq 2^n$, we have

$$\varphi(q2^{-n} \cdot \omega) = q2^{-n}k. \tag{12.3}$$

By Corollary 12.49 we also have

$$\varphi(\alpha) = \sup_{\beta \in \mathcal{E}_\alpha} \varphi(\beta). \tag{12.4}$$

Then (12.3) and (12.4) imply the uniqueness of φ. We observe that the relations (12.3) and (12.4) define a measure as the reader can easily check. Finally, let

$$D_\xi = \{q2^{-n}k \,:\, q \leq 2^n,\ q2^{-n}k \leq \xi\}$$

for a fixed number $\xi \in [0, k]$, then $\alpha := \sup\left\{\frac{x}{k} \cdot \omega \mid x \in D_\xi\right\}$ is the only angle which satisfies the equality $\varphi(\alpha) = \xi$ (if $\xi = k$ then $\alpha = \omega$). □

Definition 12.52 If in Theorem 12.51 we take $k = \pi$ the measure $\varphi(\alpha)$ is called the *radian measure* of α.

12.1.2 Triangles

Definition 12.53 Let $\triangle ABC$ be a triangle in an absolute plane $\mathcal{P}$. We denote the angle $\widehat{BAC}$ determined by the rays $(\overrightarrow{AB})$ and $(\overrightarrow{AC})$ simply by $\widehat{A}$. The supplementary angle $\omega - \widehat{BAC}$ of $\widehat{BAC}$ is the angle $\widehat{DAB}$ where the ray $(\overrightarrow{AD})$ is the opposite of $(\overrightarrow{AC})$; it will be called the *exterior angle* of $\triangle ABC$ adjacent to A.

Proposition 12.54 *Let $\triangle ABC$ be a triangle in an absolute plane $\mathcal{P}$. Then*

$$d(A, B) = d(A, C) \iff \widehat{B} = \widehat{C}.$$

Proof If $d(A, B) = d(A, C)$, there exists a symmetry σ whose axis passes through A and such that $\sigma(B) = C$, hence $\widehat{ABC} = \widehat{ACB}$.

Conversely, let $\widehat{ABC} = \widehat{ACB}$ and A' be the symmetric of A with respect to the axis d of $[BC]$, then the equalities $\widehat{A'BC} = \widehat{ACB} = \widehat{ABC}$ imply that the rays $(\overrightarrow{BA})$ and $(\overrightarrow{BA'})$ coincide as well as $(\overrightarrow{CA})$ and $(\overrightarrow{CA'})$. Hence $A' = A$, $A \in d$ and, in particular, $d(A, B) = d(A, C)$. □

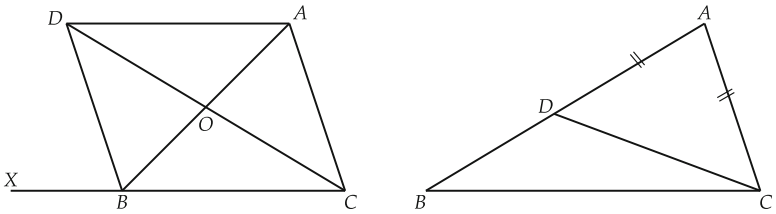


Fig. 12.4 (**a**) Exterior angle. (**b**) Sides and angles

Proposition 12.55 *Let $\triangle ABC$ be a triangle in an absolute plane $\mathcal{P}$. Then the exterior angle of $\triangle ABC$ adjacent to B is strictly greater than $\widehat{A}$ and $\widehat{C}$, i.e.*

$$\omega - \widehat{B} > \widehat{A}, \widehat{C}.$$

Proof We prove that $\omega - \widehat{B} > \widehat{A}$ for instance. Let $(\overrightarrow{BX})$ be the ray opposite to $(\overrightarrow{BC})$. We have to show that $\widehat{ABX} > \widehat{A}$. Let σ_O be the symmetry with centre the midpoint O of the side $[AB]$ and let $D = \sigma_O(C)$ (Fig. 12.4a). Since σ_O is an automorphism we get $\widehat{BAC} = \widehat{ABD}$. Since $D \in S((\overrightarrow{BX}), (\overrightarrow{BA}))$ we conclude that $\widehat{ABX} > \widehat{ABD} = \widehat{BAC}$. □

Corollary 12.56 *The sum of two angles of a triangle is strictly smaller than the straight angle ω.*

Proof Assume the notation of Proposition 12.55. Then we have $\widehat{BAC} + \widehat{CBA} < \widehat{ABX} + \widehat{CBA} = \omega$. □

Proposition 12.57 *Let $\triangle ABC$ be a triangle in an absolute plane $\mathcal{P}$. Then*

$$d(A, B) > d(A, C) \iff \widehat{ACB} > \widehat{ABC}.$$

Proof Suppose $d(A, B) > d(A, C)$. Let $D \in [AB]$ such that $d(A, D) = d(A, C)$ by axiom a_8) (Fig. 12.4b). By Proposition 12.54, $\widehat{ADC} = \widehat{ACD}$, and by Proposition 12.55 we get $\widehat{ABC} < \widehat{ADC}$ which implies $\widehat{ABC} < \widehat{ACB}$.

Conversely, let $\widehat{ACB} > \widehat{ABC}$. Assume by contradiction that $d(A, B) \leq d(A, C)$ then $d(A, B) \neq d(A, C)$ by Proposition 12.54, and the inequality $d(A, B) < d(A, C)$ implies (according to that just proved above) $\widehat{ABC} > \widehat{ACB}$. □

Proposition 12.58 *Let $\triangle ABC$ be a triangle in an absolute plane $\mathcal{P}$. Then we have the inequality*

$$d(B, C) < d(A, B) + d(A, C).$$

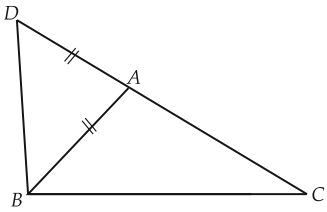


Fig. 12.5 Triangular inequality

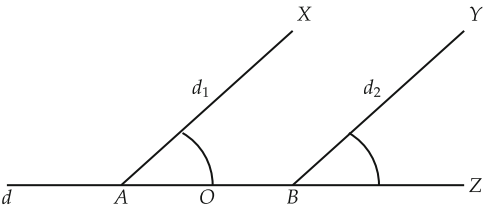


Fig. 12.6 Parallels

Proof Let $D \in (\overrightarrow{AC})$ such that $d(A, D) = d(A, B)$ (Fig. 12.5). By Proposition 12.54 we have $\widehat{BDC} = \widehat{DBA} < \widehat{DBC}$, hence by Proposition 12.55 we get $d(B, C) < d(D, C)$, i.e. $d(B, C) < d(A, B) + d(A, C)$. □

Corollary 12.59 *For any three points A, B, C of $\mathcal{P}$ we have the inequality*

$$d(B, C) \leq d(A, B) + d(A, C).$$

Moreover, $d(B, C) = d(A, B) + d(A, C)$ if and only if $A \in [BC]$.

Corollary 12.59 states that any absolute plane is a metric space according to Definition 1.153.

Proposition 12.60 *Let $d = AB$, $d_1 = AX$, $d_2 = BY$ be three distinct lines in an absolute plane $\mathcal{P}$ such that $\widehat{XAB} = \widehat{YBZ}$ as in* Fig. 12.6*. Then d_1 and d_2 are parallel.*

Proof If O is the midpoint of $[AB]$, then $d_2 = \sigma_O(d_1)$, so that by Proposition 12.39 we end the proof. □

12.2 The Poincaré Hyperbolic Plane

In this section we give an example (due to Henri Poincarè) of an absolute plane not satisfying Euclid's axiom ($\star$).

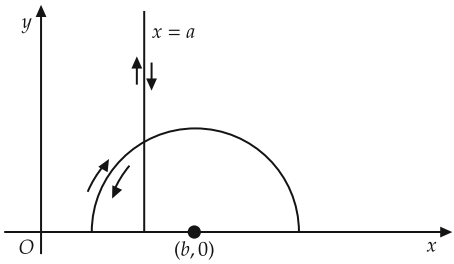


Fig. 12.7 Hyperbolic geodesics

Definition 12.61 Let

$$\mathbb{H} := \{(x, y) \in \mathbb{R}^2 \mid y > 0\},$$
$$\mathcal{D} := \{r_a : a \in \mathbb{R}\} \cup \{C_{b,r} : b, r \in \mathbb{R}, r > 0\},$$

where $r_a := \{(a, y) \in \mathbb{R}^2 : y > 0\}$ and $C_{b,r} := \{(x, y) \in \mathbb{R}^2 : (x - b)^2 + y^2 = r^2,\ y > 0\}$. We call the couple $\mathcal{P} = (\mathbb{H}, \mathcal{D})$ the *Poincaré hyperbolic half-plane* and the elements of $\mathcal{D}$ *hyperbolic lines* (see Fig. 12.7). We shall denote a point $(x, y) \in \mathbb{H}$ also by the corresponding complex number $z = x + iy$.

The main result of this section is the following:

Theorem 12.62 (Poincaré) *The couple $\mathcal{P} = (\mathbb{H}, \mathcal{D})$ of Definition 12.61 is an absolute plane not satisfying Euclid's axiom* ($\star$).

Proof It is a very easy exercise to prove that $\mathcal{P}$ is an incidence structure that satisfies axioms a_1) and a_2). If $z_1 = a + iy_1 \in \mathbb{H}$ and $z_2 = a + iy_2 \in \mathbb{H}$ only the line $x = a$ contains z_1 and z_2. If $z_1 = a + iy_1 \in \mathbb{H}$ and $z_2 = b + iy_2 \in \mathbb{H}$ with $a \neq b$, there is one and only one circle passing through z_1 and z_2 having its centre on the real axis $y = 0$. On the lines r_a we have the natural order $z_1 \leq z_2$ iff $y_1 \leq y_2$ and the opposite order. On the lines $C_{b,r}$ we have the counter-clockwise and the clockwise orders (see Fig. 12.7), so that $\mathcal{P}$ satisfies axioms a_3).

If $d = r_a$, the half-planes are

$$\mathbb{H}_d^+ = \{z \in \mathbb{H} : \mathrm{Re}(z) > a\},\ \mathbb{H}_d^- = \{z \in \mathbb{H} : \mathrm{Re}(z) < a\},$$
$$\overline{\mathbb{H}}_d^+ = \{z \in \mathbb{H} : \mathrm{Re}(z) \geq a\},\ \overline{\mathbb{H}}_d^- = \{z \in \mathbb{H} : \mathrm{Re}(z) \leq a\}.$$

If $d = C_{b,r}$ we have

$$\mathbb{H}_d^+ = \{z \in \mathbb{H} : |z - b| > r\},\ \mathbb{H}_d^- = \{z \in \mathbb{H} : |z - b| < r\},$$
$$\overline{\mathbb{H}}_d^+ = \{z \in \mathbb{H} : |z - b| \geq r\},\ \overline{\mathbb{H}}_d^- = \{z \in \mathbb{H} : |z - b| \leq r\}.$$

Two points z_1 and z_2 belong to the same above half-plane if and only if the segment $[z_1 z_2]$ does not meet d. Thus Pasch's axiom a_4) is easily verified.

Euclid's axiom ($\star$) is clearly not satisfied by $\mathcal{P}$; through a point P which does not belong to a line d there are infinitely many hyperbolic lines parallel to d (for instance take $d = r_0 : x = 0$ and $P = (2, 1)$, then the lines $C_{b,r}$ with $r^2 = (2-b)^2 + 1$ and $b > r$ (i.e. $b > \frac{5}{4}$) are parallel to d). □

Lemma 12.63 *Let z_1, $z_2 \in \mathbb{H}\,(z_1 \neq z_2)$, and let the line d joining z_1 and z_2 have end-points z_1^*, $z_2^* \in \mathbb{R} \cup \{\infty\}$, chosen in such a way that z_1 lies between z_1^* and z_2. Then there exists a unique element $\tau \in \mathrm{PSL}_2(\mathbb{R})$ such that*

$$\tau(z_1^*) = 0, \quad \tau(z_2^*) = \infty, \qquad \tau(z_1) = i. \tag{12.5}$$

Also $\tau(z_2) = ri$ $(r > 1)$ so that the cross-ratio $\rho(z_2, z_1, z_1^*, z_2^*) = r$.

Proof Assume that neither z_1^* nor z_2^* is ∞. We may suppose $z_1^* > z_2^*$, otherwise we just relabel. Let

$$\sigma(z) = \frac{z - z_1^*}{z - z_2^*},$$

then $\sigma \in \mathrm{PSL}_2(\mathbb{R})$, $\sigma(z_1^*) = 0$, $\sigma(z_2^*) = \infty$, so that σ maps d to the imaginary axis. If $\sigma(z_1) = ki$ then

$$\tau = \upsilon_{1/k} \circ \sigma, \qquad \text{where} \quad \upsilon_{1/k}(z) = \frac{1}{k} z,$$

is the required transformation. Uniqueness follows from Exercise 9.7, and as z_1 lies between z_1^* and z_2, $\tau(z_1) = i$ lies between $\tau(z_2^*) = 0$ and $\tau(z_2)$, so that $\tau(z_2) = ri$, $r > 1$, hence $\rho(z_2, z_1, z_1^*, z_2^*) = r$ (see Definition 9.35).

If $\mathrm{Re}(z_1) = \mathrm{Re}(z_2) = \alpha \in \mathbb{R}$, so that $z_1^* = \alpha$ and $z_2^* = \infty$, $z_1 = \alpha + ki$. Then

$$\tau(z) = \frac{1}{k}(z - \alpha)$$

is the required transformation. □

Definition 12.64 (Hyperbolic Distance) Let z_1, z_2 be two points of $\mathbb{H}$ and let $\gamma(t) = (x(t), y(t))$, $t_0 \leq t \leq t_1$ be any regular parametrization of hyperbolic segment $[z_1 z_2]$. The *hyperbolic distance* $d(z_1, z_2)$ is given by

$$d(z_1, z_2) = \left| \int_{t_0}^{t_1} \frac{\sqrt{\dot{x}(t)^2 + \dot{y}(t)^2}}{y(t)}\, dt \right|. \tag{12.6}$$

It is well known from elementary calculus that the right–hand side of (12.6) is independent of the regular parametrization chosen so that $d(z_1, z_2)$ depends only on z_1 and z_2.

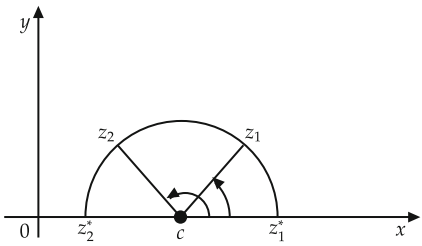


Fig. 12.8 Hyperbolic distance

(i) If $z_1 = \alpha + iy_1$ and $z_2 = \alpha + iy_2$ ($\alpha \in \mathbb{R}$), choose the regular parametrization $x = \alpha$, $y = t$ so that

$$d(z_1, z_2) = \left| \int_{y_1}^{y_2} \frac{dt}{t} \right| = \left| \log\left(\frac{y_2}{y_1}\right) \right|. \tag{12.7}$$

(ii) If z_1, z_2, z_1^*, z_2^* are defined as in Lemma 12.63 with $\mathrm{Re}(z_1) \neq \mathrm{Re}(z_2)$ we choose the regular parametrization

$$x = c + r\cos\theta, \quad y = r\sin\theta, \qquad \theta \in (0, \pi),\ r = \frac{z_1^* - z_2^*}{2},\ c = \frac{z_1^* + z_2^*}{2}.$$

According to Fig. 12.8 let

$$z_1 = c + r\cos\theta_1 + ir\sin\theta_1,\ z_2 = c + r\cos\theta_2 + ir\sin\theta_2. \tag{12.8}$$

Since $\dot{x}(\theta) = -r\sin\theta$ and $\dot{y}(\theta) = r\cos\theta$ we get

$$d(z_1, z_2) = \left| \int_{\theta_1}^{\theta_2} \frac{d\theta}{\sin\theta} \right| = \left| \left[\log\left(\mathrm{tg}\,\frac{\theta}{2}\right) \right]_{\theta_1}^{\theta_2} \right|. \tag{12.9}$$

Taking into account the relations

$$\mathrm{tg}\,\frac{\theta}{2} = \frac{\sin\theta}{1 + \cos\theta} = \sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}}, \qquad \theta \in (0, \pi), \tag{12.10}$$

if $z = c + r\cos\theta + i(r\sin\theta) \in C_{c,r}$ we easily get

$$\mathrm{tg}\,\frac{\theta}{2} = \left| \frac{z_1^* - z}{z_2^* - z} \right|.$$

Hence

$$\left|\left[\log\left(\operatorname{tg}\frac{\theta}{2}\right)\right]_{\theta_1}^{\theta_2}\right| = \left[\log\left|\frac{z_1^*-z}{z_2^*-z}\right|\right]_{z_1}^{z_2} = \log\left|\frac{z_1^*-z_2}{z_2^*-z_2}\right| - \log\left|\frac{z_1^*-z_1}{z_2^*-z_1}\right|$$
$$= \log\left(\left|\frac{z_1^*-z_2}{z_2^*-z_2}\right| : \left|\frac{z_1^*-z_1}{z_2^*-z_1}\right|\right). \tag{12.11}$$

Since z_1, z_2, z_1^* and z_2^* lie on the same circle, the cross-ratio (Definition 9.35)

$$\rho(z_2, z_1, z_1^*, z_2^*) = \frac{z_2 - z_1^*}{z_2 - z_2^*} : \frac{z_1 - z_1^*}{z_1 - z_2^*}$$

is a real number (Exercise 9.9). Moreover By Lemma 12.63 there is a Moebius transform τ such that

$$\tau(z_2^*) = \infty,\ \tau(z_1^*) = 0, \quad \tau(z_1) = i,\ \tau(z_2) = ri, \qquad r = \rho(z_2, z_1, z_1^*, z_2^*) > 1. \tag{12.12}$$

Hence from and (12.9) and (12.11) we have $d(z_1, z_2) = \log(\rho(z_2, z_1, z_1^*, z_2^*))$. Since

$$\rho(z_2, z_1, z_2^*, z_1^*) = \frac{1}{\rho(z_2, z_1, z_1^*, z_2^*)}, \quad \rho(z_1, z_2, z_1^*, z_2^*) = \frac{1}{\rho(z_2, z_1, z_1^*, z_2^*)},$$
$$\rho(z_1, z_2, z_2^*, z_1^*) = \rho(z_2, z_1, z_1^*, z_2^*)$$

the formula

$$d(z_1, z_2) = |\log(\rho(z_2, z_1, z_1^*, z_2^*))| \tag{12.13}$$

holds true, interchanging z_1 and z_2 as well as z_1^* and z_2^*. Also formula (12.7) can be put in the form (12.13). If $z_1 = \alpha + iy_1$, $z_2 = \alpha + iy_2$, $z_1^* = \alpha$ and $z_2^* = \infty$, then

$$|\log(\rho(z_2, z_1, z_1^*, z_2^*))| = \left|\log\left(\frac{y_2}{y_1}\right)\right|.$$

In both cases i) and ii) (of Definition 12.64) we have the inequality

$$d(z_1, z_2) \geq \left|\log\left(\frac{y_2}{y_1}\right)\right|. \tag{12.14}$$

In case i) the inequality (12.14) coincides with the equality (12.7). In case ii) since

$$\frac{1}{\sin\theta} \geq |\operatorname{ctg}\theta|, \qquad \forall\,\theta \in (0, \pi),$$

we have ($z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$ as in (12.8))

$$d(z_1, z_2) = \left| \int_{\theta_1}^{\theta_2} \frac{d\theta}{\sin\theta} \right| \geq \left| \int_{\theta_1}^{\theta_2} |\operatorname{ctg}\theta|\, d\theta \right| \geq \left| \int_{\theta_1}^{\theta_2} \operatorname{ctg}\theta\, d\theta \right|$$
$$= \left| \log \frac{\sin\theta_2}{\sin\theta_1} \right| = \left| \log \frac{r\sin\theta_2}{r\sin\theta_1} \right| = \left| \log\left(\frac{y_2}{y_1}\right) \right|.$$

Thus axioms a_5), a_6) are clearly satisfied. Now we check axiom a_7). If $z_1 = \alpha + iy_1$, $z_2 = \alpha + iy_2$, $z_3 = \alpha + iy_3$ are three distinct points on a hyperbolic line $x = \alpha \in \mathbb{R}$ such that $z_1 < z_2 < z_3$, i.e. $y_1 < y_2 < y_3$. Then

$$d(z_1, z_2) = \log\left(\frac{y_2}{y_1}\right), \quad d(z_2, z_3) = \log\left(\frac{y_3}{y_2}\right), \quad d(z_1, z_3) = \log\left(\frac{y_3}{y_1}\right),$$

therefore

$$d(z_1, z_2) + d(z_2, z_3) = \log\left(\frac{y_2}{y_1}\right) + \log\left(\frac{y_3}{y_2}\right) = \log\left(\frac{y_2}{y_1}\frac{y_3}{y_2}\right) = \log\left(\frac{y_3}{y_1}\right)$$
$$= d(z_1, z_3).$$

Conversely, if $z_1 < z_3$ and $z_2 \in z_1 z_3$ with $z_2 \neq z_1, z_3$ are such that $d(z_1, z_2) + d(z_2, z_3) = d(z_1, z_3)$, then $z_2 \in [z_1 z_3]$, i.e. $z_1 < z_2 < z_3$. Otherwise, if $z_3 < z_1 < z_2$, we have just showed that $d(z_3, z_1) + d(z_1, z_2) = d(z_3, z_2)$ which implies $d(z_1, z_2) + d(z_2, z_3) + d(z_1, z_2) = d(z_2, z_3)$, i.e. $d(z_1, z_2) = 0$ contradicting the hypothesis. Similarly, one sees that $z_1 < z_3 < z_2$ contradicts the hypothesis.

Assume the notation of (12.8). Let $z_3 = c + r\cos\theta_3 + ir\sin\theta_3 \in C_{c,r}$ be between z_1 and z_2, i.e. $\theta_1 < \theta_3 < \theta_2$. Let τ be defined as in (12.12), then $\tau(z_3) = ki$ with $1 < k < r$. Therefore

$$\begin{aligned} d(z_1, z_3) + d(z_3, z_2) &= |\log \rho(z_3, z_1, z_1^*, z_3^*)| + |\log \rho(z_2, z_3, z_3^*, z_2^*)| \\ &= |\log \rho(z_3, z_1, z_1^*, z_2^*)| + |\log \rho(z_2, z_3, z_1^*, z_2^*)| \\ &= |\log \rho(\tau(z_3), \tau(z_1), \tau(z_1^*), \tau(z_2^*))| \\ &\quad + |\log \rho(\tau(z_2), \tau(z_3), \tau(z_1^*), \tau(z_2^*))| \\ &= \log k + \log \frac{r}{k} \\ &= \log r = |\log \rho(z_2, z_1, z_1^*, z_2^*)| = d(z_1, z_2). \end{aligned}$$

The converse can be proved as above.

By (12.14) we have

$$\lim_{y_2 \to 0^+} d(z_1, z_2) = \lim_{y_2 \to +\infty} d(z_1, z_2) = +\infty.$$

The map $r \to \mathbb{R}_+, z_2 \to d(z_1, z_2)$ from any line r emanating from z_1, is continuous with respect to z_2 (on r and $\mathbb{R}_+$ there is the euclidean topology) and bijective, i.e. $d(z_1, z_2)$ verifies axiom (a_8).

Finally, we have to prove superposition axioms (a_9) and (a_{10}). By Exercise 9.10, Moebius transforms of $\mathrm{PSL}_2(\mathbb{R})$ preserve the half-plane $\mathbb{H}$ as well as the real line $\mathrm{Im}(z) = 0$. Also the maps

$$\tau(z) = \frac{a\overline{z} + b}{c\overline{z} + d}, \qquad ad - bc < 0, \tag{12.15}$$

which are the composition of Moebius transforms of $\mathrm{GL}_2^-(\mathbb{R}) := \{A \in \mathrm{GL}_2(\mathbb{R}) : \det(A) < 0\}$ with the automorphism $z \to \overline{z}$ of $\mathbb{C}$, preserve the half-plane $\mathbb{H}$ as well as the real line $\mathrm{Im}(z) = 0$.

Lemma 12.65 *The Moebius transforms of* $\mathrm{PSL}_2(\mathbb{R})$ *and the transforms* (12.15) *are automorphisms of* $\mathcal{P}$*, i.e. they map hyperbolic lines to hyperbolic lines and preserve the distances.*

Proof The assertions concerning the Moebius transforms of $\mathrm{PSL}_2(\mathbb{R})$ follow from Exercise 9.10, Corollary 9.36 and formula (12.13). The reader can proceed as in Exercise 9.8 taking into account that the transforms (12.15) are compositions of $z \to a\overline{z} + b$ $(a, b \in \mathbb{R}, a < 0)$ and of $z \to \dfrac{1}{\overline{z}}$. Moreover one can easily check that the transforms $\tau(z)$ of the form (12.15) change the cross-ratio into its complex conjugate

$$\rho(\tau(z_1), \tau(z_2), \tau(z_3), \tau(z_4)) = \overline{\rho(z_1, z_2, z_3, z_4)}.$$

Hence

$$\begin{aligned} d(\tau(z_1), \tau(z_2)) &= |\log \rho(\tau(z_2), \tau(z_1), \tau(z_1)^*, \tau(z_2)^*)| = |\log \overline{\rho(z_2, z_1, z_1^*, z_2^*)}| \\ &= |\log \rho(z_2, z_1, z_1^*, z_2^*)| = d(z_1, z_2). \end{aligned}$$

□

Corollary 12.66 *Let* $a, k \in \mathbb{R}$, $k > 0$. *Let*

$$M_1(z) = -\overline{z} + 2a, \qquad M_2(z) = \frac{k}{\overline{z} - a} + a. \tag{12.16}$$

Then $M_1(z)$ *is the euclidean symmetry of axis the line* $\mathrm{Re}(z) = a$*, and* $M_2(z)$ *is the symmetry of axis the circle of centre* a *and radius* $r = \sqrt{k}$*. These transforms, restricted to* $\mathbb{H}$*, are called hyperbolic symmetries.*

Lemma 12.67 *For all* $z, w \in \mathbb{H}$, $\tau(z,w) := \left|\dfrac{z-w}{z-\overline{w}}\right| \in [0,1)$ *is invariant with respect to* $\mathrm{PSL}_2(\mathbb{R})$, *i.e.*

$$\tau(\sigma(z),\sigma(w)) = \tau(z,w), \qquad \forall\, \sigma \in \mathrm{PSL}_2(\mathbb{R}).$$

Proof This follows from the formula

$$|\sigma(z)-\sigma(w)| = |z-w|\sqrt{|\sigma'(z)\sigma'(w)|}$$

(here σ' denotes the derivative of σ) which is easily proved by direct computation. □

Lemma 12.68 *Let* $z, w \in \mathcal{P}$. *Then*

$$d(z,w) = \log\left(\frac{1+\left|\dfrac{z-w}{z-\overline{w}}\right|}{1-\left|\dfrac{z-w}{z-\overline{w}}\right|}\right).$$

Proof We may suppose that z, w and r are as z_1, z_2 and r (respectively) in Lemma 12.63 so that $d(z,w) = \log r$. Also

$$\tau(z,w) = \tau(i,ri) = \left|\frac{1-r}{1+r}\right| = \frac{r-1}{r+1} = \frac{e^{d(z,w)}-1}{e^{d(z,w)}+1}$$

and therefore

$$d(z,w) = \log\left(\frac{1+\tau(z,w)}{1-\tau(z,w)}\right) = \log\left(\frac{1+\left|\dfrac{z-w}{z-\overline{w}}\right|}{1-\left|\dfrac{z-w}{z-\overline{w}}\right|}\right).$$ □

Lemma 12.69 *For any hypertbolic line* d *there exists a hyperbolic symmetry* s_d.

Proof The existence of s_d is given by Corollary 12.66. Let $\sigma \neq \mathrm{id}_{\mathcal{P}}$ be an automorphism of $\mathcal{P}$ which fixes any point of d. By Proposition 12.14-(ii) (which does not require superposition axioms), σ does not have any other fixed point. By Lemma 12.63 there exists an automorphism τ of $\mathcal{P}$ such that $\tau(d)$ is the imaginary half-axis $\mathrm{Re}(z) = 0$, $\mathrm{Im}(z) > 0$. Then $\gamma := \tau \circ \sigma \circ \tau^{-1}$ fixes any point of $\tau(d)$.

Since $d(\gamma(z), \gamma(w)) = d(z, w)$, we have

$$\log\left(\frac{1+\left|\dfrac{\gamma(z)-\gamma(w)}{\gamma(z)-\gamma(\overline{w})}\right|}{1-\left|\dfrac{\gamma(z)-\gamma(w)}{\gamma(z)-\gamma(\overline{w})}\right|}\right) = \log\left(\frac{1+\left|\dfrac{z-w}{z-\overline{w}}\right|}{1-\left|\dfrac{z-w}{z-\overline{w}}\right|}\right).$$

For every $w \in \tau(d)$ we have

$$\log\left(\frac{1+\left|\dfrac{\gamma(z)-w}{\gamma(z)-\overline{w}}\right|}{1-\left|\dfrac{\gamma(z)-w}{\gamma(z)-\overline{w}}\right|}\right) = \log\left(\frac{1+\left|\dfrac{z-w}{z-\overline{w}}\right|}{1-\left|\dfrac{z-w}{z-\overline{w}}\right|}\right).$$

By Lemma 12.68, taking into account that $t \to \log\left(\dfrac{1+t}{1-t}\right), 0 \leq t < 1$, is injective, we get

$$\left|\frac{\gamma(z)-w}{\gamma(z)-\overline{w}}\right| = \left|\frac{z-w}{z-\overline{w}}\right|. \tag{12.17}$$

Since the symmetry $s_{\tau(d)}(z) = -\overline{z}$ satisfies (12.17) and the triangles $\triangle z w \overline{w}$ and $\triangle \gamma(z) w \overline{w}$ are congruent (with a common side) we have that $\gamma(z) = s_{\tau(d)}(z)$ and $\sigma(z) = s_d(z)$. □

Thus axiom a_9) is satisfied by Lemma 12.69.

Lemma 12.70 *Axiom* (a_{10}) *is satisfied.*

Proof As in the proof of Lemma 12.69, we may assume that one of the hyperbolic rays lies on a euclidean line r parallel to the imaginary axis and the other is the arc $(\overrightarrow{AP})$ on the semicircle Γ as in Fig. 12.9.

Since the line r meets Γ, the point H lies between P and Q. Consider the hyperbolic rays $(\overrightarrow{AH})$ and $(\overrightarrow{AZ})$ lying on r and emanating from A. Let p and q the abscissas of P and Q respectively so that the centre of Γ is $(\dfrac{p+q}{2}, 0)$ and its

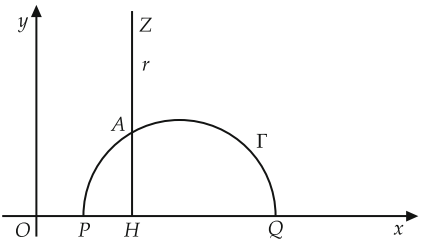


Fig. 12.9 Hyperbolic bisector

radius is $\dfrac{q-p}{2}$. Furthermore $A = h + i\sqrt{(q-h)(h-p)}$ and

$$[AH] : z = h + it, \quad 0 \le t \le \sqrt{(q-h)(h-p)},$$
$$(\overrightarrow{AZ}) : z = h + it, \quad \sqrt{(q-h)(h-p)} \le t < +\infty.$$

Let

$$s_e(z) = \frac{(q-p)(q-h)}{\overline{z}-q} + q,$$
$$s_f(z) = \frac{(q-p)(h-p)}{\overline{z}-p} + p.$$

Then $s_e(H) = P$ and

$$\begin{aligned} s_e(A) &= \frac{(q-p)(q-h)}{h-q-i\sqrt{(q-h)(h-p)}} + q \\ &= \frac{(q-p)(q-h)[h-q+i\sqrt{(q-h)(h-p)}] + q(h-q)(p-q)}{(h-q)(p-q)} \\ &= h + i\sqrt{(q-h)(h-p)} = A. \end{aligned}$$

The hyperbolic line e is the semicircle of centre $(q, 0)$ and radius $\sqrt{(q-h)(q-p)}$. In a similar way we have $s_f(A) = A$. Also $\lim_{t\to+\infty} s_f(h+it) = p$, hence $s_e((\overrightarrow{AH})) = (\overrightarrow{AP})$ and $s_f((\overrightarrow{AZ})) = (\overrightarrow{AP})$. The hyperbolic line f is the semicircle of centre $(p, 0)$ and radius $\sqrt{(h-p)(q-p)}$. □

Thus the proof of Theorem 12.62 is complete. □

12.3 Exercises

Exercise 12.1 Let $\mathcal{P}$ be an absolute plane. Use Corollary 12.59 to prove that a map $\sigma : \mathcal{P} \to \mathcal{P}$ is an automorphism of $\mathcal{P}$ if and only if σ is bijective and $d(\sigma(A), \sigma(B)) = d(A, B)$ for all $A, B \in \mathcal{P}$.

Exercise 12.2 Let d be a line of an absolute plane $\mathcal{P}$, and let $A \in \mathcal{P} \setminus d$. Let $M = d \cap e$, where e is the perpendicular to d passing through A. Prove the following assertions:

1. $d(A, N) \ge d(A, M)$ for all $N \in d$.

2. For any ray $d_M^{\pm}$ the map $d_M^{\pm} \to \mathbb{R}$, $N \to d(A, N)$, is continuous (with respect to euclidean topologies), stricltly increasing and such that $\lim_{d(M,N)\to+\infty} d(A, N) = +\infty$.

The *distance* $d(A, d)$ from A to d is defined by $d(A, d) := d(A, M)$.

Exercise 12.3 Let d be a line of an absolute plane $\mathcal{P}$. The circle $\mathcal{C}(O, r)$ of centre O and radius r is defined in the usual way,

$$\mathcal{C}(O, r) := \{P \in \mathcal{P} : d(P, O) = r\}.$$

Prove that (use Exercise 12.2)

$$\mathcal{C}(O, r) \cap d \neq \varnothing \quad \Longleftrightarrow \quad d(O, d) \leq r.$$

Furthermore $d \cap \mathcal{C}(O, r)$ is only one point iff $d(O, d) = r$, and two distinct points iff $d(O, d) < r$.

Exercise 12.4 Let $\mathcal{C}(O, r)$ and $\mathcal{C}(O', r')$ be two circles in an absolute plane $\mathcal{P}$. Then

$$\mathcal{C}(O, R) \cap \mathcal{C}(O', R') \neq \varnothing \quad \Longleftrightarrow \quad |r - r'| \leq d(O, O') \leq r + r'.$$

Furthermore $\mathcal{C}(O, R) \cap \mathcal{C}(O', R')$ is only one point iff $d(O, O') = |r - r'|$ or $d(O, O') = r + r'$, and two distinct points iff $|r - r'| < d(O, O') < r + r'$.

Chapter 13
Cayley–Klein Geometries



13.1 Euclidean Metric from a Projective Point of View

In Chap. 12 we introduced the Poincaré half-plane as a model of the plane hyperbolic geometry. It is a remarkable fact that hyperbolic distance may be expressed by the cross-ratio (12.13), i.e. by means of a *projective quantity*. In the first half of 1800 J. V. Poncelet (1788–1867) discovered that some euclidean notions might be formulated in terms of projective geometry. In order to show these connections we have to consider the euclidean plane immersed into the complex projective plane.

We fix the immersion $j_0 \colon \mathbb{A}^2(\mathbb{R}) \to \mathbb{P}^2(\mathbb{C})$, $j_0(x, y) = [1, x, y]$ for all $(x, y) \in \mathbb{A}^2(\mathbb{R})$. We recall from Definition 10.102 that $I = [0, 1, i]$, $J = [0, 1, -i]$ are the cyclic points (with respect to j_0), and that the lines of the pencils with base locus I and J are the isotropic lines. We extend the standard euclidean metric to a "complex-valued distance"

$$d(A, B) = \left[\left(\frac{a_1}{a_0} - \frac{b_1}{b_0} \right)^2 + \left(\frac{a_2}{a_0} - \frac{b_2}{b_0} \right)^2 \right]^{1/2},$$
$$A = [a_0, a_1, a_2], \ B = [b_0, b_1, b_2], \ a_0 b_0 \neq 0, \tag{13.1}$$

where $\sqrt{z} > 0$ if $z > 0$. The locus of the points which have distance 0 from a fixed point $A = [a_0, a_1, a_2]$ is determined by equation

$$(x_1 a_0 - x_0 a_1)^2 + (x_2 a_0 - x_0 a_2)^2 = 0.$$

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Such a locus consists of the lines

$$x_1a_0 - x_0a_1 + i(x_2a_0 - x_0a_2) = 0,$$
$$x_1a_0 - x_0a_1 - i(x_2a_0 - x_0a_2) = 0,$$

which are the isotropic lines of the pencil with base locus A. Every pencil (whose base locus is a proper point) contains two isotropic lines which are the locus of the points whose distances from the base locus is 0. Two points P, Q lying on an isotropic line but distinct from cyclic points have distance 0, and the distance of two points is not defined if one of them is cyclic. Cyclic points and isotropic lines play an essential role in the projective setting of metrical notions. E. N. Laguerre (1834–1886) proved that the angle $\measuredangle(r, s)$ of two euclidean lines r and s might be represented as the complex logarithm of the cross-ratio $\rho(r, s, \xi, \sigma)$, where ξ and σ are the isotropic lines of the pencil determined by r and s. We may assume $r : m_1x_1 - x_2 = 0$ and $s : m_2x_1 - x_2 = 0$, $m_1 \neq m_2$, so that $r \cap s = O = [1, 0, 0]$. Therefore $\xi : x_1 + ix_2 = 0$ and $\sigma : x_1 - ix_2 = 0$. Then we have

$$\rho(r, s, \xi, \sigma) = \frac{1 + m_1m_2 + i(m_1 - m_2)}{1 + m_1m_2 - i(m_1 - m_2)}.$$

If $z = a + ib \in \mathbb{C}$, $z \neq 0$, we have the obvious relations

$$\frac{z}{\overline{z}} = \frac{a^2 - b^2}{a^2 + b^2} + \frac{2ab}{a^2 + b^2}\, i = \sin(2\theta) + i\sin(2\theta) = e^{2i\theta}, \qquad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2},$$

and

$$\operatorname{tg}(\theta) = \left(\frac{\sin(2\theta)}{1 + \cos(2\theta)}\right) = \frac{b}{a}, \qquad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Let $z = 1 + m_1m_2 + i(m_1 - m_2)$, then

$$\rho(r, s, \xi, \sigma) = e^{2i\theta} \quad \text{with} \quad \theta = \operatorname{arctg}\left(\frac{m_1 - m_2}{1 + m_1m_2}\right) = \measuredangle(r, s),$$

so that

$$2i\theta = \log \rho(r, s, \xi, \sigma), \tag{13.2}$$

where if $z \neq 0$, we define the multi-valued logarithm

$$\log z = \log|z| + i \arg z. \tag{13.3}$$

We define the *principal value* of the logarithm as the complex number

$$\log|z| + i \arg z, \qquad -\pi < \arg z \leq \pi. \tag{13.4}$$

Since $\rho(r, s, \xi, \sigma)$ is a complex number of module 1, $\log \rho(r, s, \xi, \sigma)$ is a pure imaginary number uniquely determined up to integer multiples of $2\pi i$. From formula (13.2) we get *Laguerre's formula* immediately:

$$\measuredangle(r, s) \equiv -\frac{i}{2} \log \rho(r, s, \xi, \sigma) \pmod{\pi}. \tag{13.5}$$

In particular from (13.5) we see that two lines r and s are perpendicular if and only if the pairs (r, s) and (ξ, σ) harmonically separate, where ξ and σ are the isotropic lines of the pencil determined by r and s.

Also the congruence of segments of a line may be projectively described. Let $[AB]$ and $[PQ]$ be two segments of an affine line $r \to \mathbb{P}^1(\mathbb{R}) \subset \mathbb{P}^1(\mathbb{C})$, $x \to [1, x]$, and $\overline{r}$ be its projective closure. Denote the euclidean distance of A and B by $|A - B|$. We claim that $|A - B| = |P - Q|$ if and only if there exists a parabolic projectivity $\varphi\colon \overline{r} \to \overline{r}$ whose fixed point is the point at infinity of r, and such that $\varphi(A) = P, \varphi(B) = Q$. Choose $P_0 = [0, 1]$ (the point at infinity of r) and $Q_0 = [1, 0]$ as base points of $\overline{r}$. Let $A = [x_A, y_A]$, $B = [x_B, y_B]$, $P = [x_P, y_P]$, $Q = [x_Q, y_Q]$, all distinct from P_0. Then

$$|A - B| = \left| \frac{y_A}{x_A} - \frac{y_B}{x_B} \right|.$$

The parabolic projectivities of r which fix P_0 are associated with the matrices $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$, $k \in \mathbb{R} \setminus \{0\}$, with respect to the above coordinate system. Therefore

$$|P - Q| = \left| \frac{y_A + kx_A}{x_A} - \frac{y_B + kx_B}{x_B} \right| = \left| \frac{y_A}{x_A} - \frac{y_B}{x_B} \right| = |A - B|.$$

Conversely, if $|A - B| = |P - Q|$ with $A, B, P, Q \neq P_0$, we have

$$\left| \frac{y_P}{x_P} - \frac{y_Q}{x_Q} \right| = \left| \frac{y_A}{x_A} - \frac{y_B}{x_B} \right| \quad \text{i.e,} \quad \left| \frac{y_P}{x_1} - \frac{y_Q}{x_2} \right| = \left| \frac{y_A}{x_1} - \frac{y_B}{x_2} \right|$$

having assumed $x_A = x_P = x_1$ and $x_B = x_Q = x_2$. Hence

$$y_P = y_A + \alpha x_1,$$
$$y_Q = y_B + \alpha x_2,$$

i.e. $P = \varphi(A)$ and $Q = \varphi(B)$, where φ is the parabolic projectivity whose matrix is $\begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix}$.

As A. Cayley [10] and F. Klein [20] discovered, Euclidean and non-Euclidean geometries can be introduced as geometries living in a projective space which is endowed with a projective metric. It is the aim of this chapter to present these geometries and for this purpose we need the following definition.

Definition 13.1 Let x, y and z be three distinct points of a metric space (X, d). We say that x, y and z are *collinear in the metric sense* if $d(x, y) + d(y, z) = d(x, z)$.

Let R be an open set of $\mathbb{P}^n(\mathbb{R})$ (with respect to euclidean topology on $\mathbb{P}^n(\mathbb{R})$). A metric on R is called a *projective metric* if it induces the euclidean topology on R (as a subspace of $\mathbb{P}^n(\mathbb{R})$) and it satisfies the *strong triangular inequality*, i.e. any three distinct points are collinear if and only if they are collinear in the metric sense.

13.2 Projective Metrics on $\mathbb{P}^1(\mathbb{R})$

Remark 13.2 We need to extend the definition of cross-ratio allowing two (but not three) of four points to coincide. We put

$$\rho(A, B, C, D) := \begin{cases} 1 & \text{if } C = D,\ A \neq B \text{ or } A = B,\ C \neq D \\ 0 & \text{if } B = D,\ A \neq C \text{ or } A = C,\ B \neq D \\ \infty & \text{if } A = D,\ B \neq C \text{ or } B = C,\ A \neq D. \end{cases} \tag{13.6}$$

Definition 13.3 Let $\widehat{F}$ be a non-degenerate conic of $\mathbb{P}^1(\mathbb{R})$, i.e. a pair of $\{\xi, \eta\}$ of distinct points ξ, η of $\mathbb{P}^1(\mathbb{R})$ or complex conjugate $\xi, \eta \in \mathbb{P}^1(\mathbb{C})$. We fix $\{\xi, \eta\}$ and we call it *the absolute* of $\mathbb{P}^1(\mathbb{R})$. The projectivities of $\mathbb{P}^1(\mathbb{R})$ which preserve the absolute $\{\xi, \eta\}$ will be called *motions* of $\mathbb{P}^1(\mathbb{R})$. Let $\ell\colon \mathbb{P}^1(\mathbb{R}) \times \mathbb{P}^1(\mathbb{R}) \to \mathbb{R}$ be a map which verifies the following conditions:

(i) $\ell(x, y) = \ell(x', y')$ if there exists a motion φ such that $\varphi(x) = x'$ and $\varphi(y) = y'$,
(ii) $\ell(x, y) + \ell(y, z) = \ell(x, z)$ for every $x, y, z \in \mathbb{P}^1(\mathbb{R})$.

Assumption (i) implies that ℓ is a function of cross-ratio $\rho(x, y, \xi, \eta)$ while assumption (ii) shows that ℓ behaves as a logarithm taking into account the relation $\rho(x, y, \xi, \eta)\rho(y, z, \xi, \eta) = \rho(x, z, \xi, \eta)$ (see Proposition 9.25 and Exercise 9.1). Therefore we obtain

$$\ell(x, y) = k \log \rho(x, y, \xi, \eta), \tag{13.7}$$

where k is a suitable complex constant $\neq 0$ (having extended $\rho(A, B, C, D)$ according to Remark 13.2). We have to determine an open subset $A \subset \mathbb{P}^1(\mathbb{R})$ (with

respect to the euclidean topology on $\mathbb{P}^1(\mathbb{R}) \cong \mathbb{S}^1$), such that $\ell(x, y) \in \mathbb{R} \setminus \{0\}$. We can see $\ell(x, y)$ as the length of the *oriented segment* with endpoints x and $y \in A$ and define the metric $d(x, y) = |\ell(x, y)|$ on A. The isometries of the metric space (A, d) will be the restriction to A of projectivities of $\mathbb{P}^1(\mathbb{R})$ which preserve A and the absolute $\{\xi, \eta\}$.

(a) If ξ, η of $\mathbb{P}^1(\mathbb{R})$, then $\log \rho(x, y, \xi, \eta)$ must be a real number for all $x, y \in A$, so that $\rho(x, y, \xi, \eta) > 0$ must be satisfied. This occurs if and only if the pairs (x, y) and (ξ, η) do not separate, i.e. A is one of the two open segments $]\eta\xi[$ of $\mathbb{P}^1(\mathbb{R})$.

(b) If $\xi, \eta \in \mathbb{P}^1(\mathbb{C})$, ξ, η complex conjugate, then $|\rho(x, y, \xi, \eta)| = 1$ which implies $\log \rho(x, y, \xi, \eta) = ir$ with $r \in \mathbb{R}$, $r \neq 0$. Therefore $A = \mathbb{P}^1(\mathbb{R})$.

The following lemma gives an explicit formula for $\ell(x, y)$ relative to a line of $\mathbb{P}^n(\mathbb{R})$.

Lemma 13.4 *Let $F(z, w)$ be a symmetric non-degenerate bilinear form of $\mathbb{R}^{n+1}$, and let $\widehat{F}$ be the associated hyperquadric of $\mathbb{P}^n(\mathbb{R})$. Let $x, y \in \mathbb{P}^n(\mathbb{R}) \setminus \mathcal{V}_+(\widehat{F})$, $x \neq y$ such that the line r joining x and y meets $\mathcal{V}_+(\widehat{F})$ at two distinct points ξ and η are possibly complex conjugate.*

(a) *If ξ and η are complex conjugate, then*

$$\ell(x, y) = k \log \rho(x, y, \xi, \eta) = 2k \arccos\left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}}\right). \tag{13.8}$$

(b) *If ξ, η are real and they do not separate the pair x, y, then*

$$\ell(x, y) = k \log \rho(x, y, \xi, \eta) = 2k \operatorname{arccosh}\left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}}\right). \tag{13.9}$$

Proof Since both x and y are distinct from ξ and η we have

$$\xi = x + \lambda_0 y, \quad \eta = x + \lambda_1 y, \qquad \lambda_0\lambda_1 \neq 0,$$

where λ_0, λ_1 are the solutions of $F(x + ty, x + ty) = 0$, i.e. $F(x, x) + 2tF(x, y) + t^2F(y, y) = 0$, hence

$$\lambda_0 = \frac{-F(x, y) - \sqrt{F(x, y)^2 - F(x, x)F(y, y)}}{F(y, y)},$$

$$\lambda_1 = \frac{-F(x, y) + \sqrt{F(x, y)^2 - F(x, x)F(y, y)}}{F(y, y)},$$

assuming that $\sqrt{z} > 0$ if $z > 0$. Choosing x and y as base points to determine $\rho(x, y, \xi, \eta)$ (also in $\mathbb{P}^n(\mathbb{C})$) we obtain

$$\rho(x, y, x + \lambda_0 y, x + \lambda_1 y) = \frac{\lambda_0}{\lambda_1} = \frac{F(x, y) + \sqrt{F(x, y)^2 - F(x, x)F(y, y)}}{F(x, y) - \sqrt{F(x, y)^2 - F(x, x)F(y, y)}}. \tag{13.10}$$

(a) The roots λ_0, λ_1 are complex conjugate if and only if $F(x, y)^2 < F(x, x)F(y, y)$. After changing the sign of the coordinates of y such that $F(x, y) \geq 0$ (if necessary) we get

$$0 \leq \frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} < 1,$$

and

$$\begin{aligned}
&\log \rho(x, y, x + \lambda_0 y, x + \lambda_1 y) \\
&= \log \left(\frac{F(x, y) + \sqrt{F(x, y)^2 - F(x, x)F(y, y)}}{F(x, y) - \sqrt{F(x, y)^2 - F(x, x)F(y, y)}} \right) \\
&= \log \left(\frac{\dfrac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} + \sqrt{\dfrac{F^2(x, y)}{F(x, x)F(y, y)} - 1}}{\dfrac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} - \sqrt{\dfrac{F^2(x, y)}{F(x, x)F(y, y)} - 1}} \right) \\
&= \log \left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} + \sqrt{\frac{F^2(x, y)}{F(x, x)F(y, y)} - 1} \right)^2 \\
&= 2 \log \left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} + \sqrt{\frac{F^2(x, y)}{F(x, x)F(y, y)} - 1} \right).
\end{aligned} \tag{13.11}$$

Denote $\dfrac{F(x, y)}{\sqrt{F(x, x)F(y, y)}}$ by α so that

$$\ell(x, y) = 2k \log(\alpha + \sqrt{\alpha^2 - 1}), \qquad 0 \leq \alpha < 1.$$

By the well-known identity (see [31], p. 164)

$$i \arccos(z) = \log(z + \sqrt{z^2 - 1}), \qquad z \in \mathbb{C} \setminus \{x \in \mathbb{R} : |x| \geq 1\}, \tag{13.12}$$

we end the proof of (13.8).

(b) The roots λ_0, λ_1 are real if and only if

$$F(x, y)^2 - F(x, x)F_\alpha(y, y) > 0. \tag{13.13}$$

If ξ, η do not separate the pair x, y, i.e. $\rho(x, y, \xi, \eta) > 0$, by (13.10) we have

$$\frac{F(x, y) + \sqrt{F(x, y)^2 - F(x, x)F(y, y)}}{F(x, y) - \sqrt{F(x, y)^2 - F(x, x)F(y, y)}} > 0. \tag{13.14}$$

The inequality (13.14) is equivalent to

$$\begin{aligned} 0 &< \left(F(x, y) + \sqrt{F(x, y)^2 - F(x, x)F(y, y)}\right) \\ &\quad \times \left(F(x, y) - \sqrt{F(x, y)^2 - F(x, x)F(y, y)}\right) \\ &= F(x, x)F(y, y). \end{aligned} \tag{13.15}$$

Then by (13.13) and (13.15) we get

$$\frac{F^2(x, y)}{F(x, x)F(y, y)} > 1.$$

As in case (a), changing the sign of the coordinates of y such that $F(x, y) > 0$, we obtain

$$\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} > 1.$$

In the same way as (13.11) one has

$$\begin{aligned} &\log \rho(x, y, x + \lambda_0 y, x + \lambda_1 y) \\ &\quad = 2 \log\left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}} + \sqrt{\frac{F^2(x, y)}{F(x, x)F(y, y)} - 1}\right). \end{aligned}$$

Taking into account the relation

$$\operatorname{arccosh}(z) = \log(z + \sqrt{z^2 - 1}), \qquad z \in \mathbb{R},\ z > 1, \tag{13.16}$$

we end the proof of case (b). □

Corollary 13.5 *Under the notation and hypothesis of* Lemma 13.4 *in case* (a)

$$d_{\mathrm{ell}}(x, y) := 2k_e \arccos\left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}}\right), \qquad k_e > 0, \tag{13.17}$$

defines a metric on $\mathbb{P}^1(\mathbb{R})$ *called an* elliptic metric. *In particular we have* $0 \leq d_{\mathrm{ell}}(x, y) \leq \pi k_e$.

In case (b)

$$d_{\mathrm{hyp}}(x, y) = 2k_h \operatorname{arccosh}\left(\frac{F(x, y)}{\sqrt{F(x, x)F(y, y)}}\right), \qquad k_h > 0, \tag{13.18}$$

defines a metric on each of the open segments with endpoints ξ *and* η *called a* hyperbolic metric. *In this case* $0 \leq d_{\mathrm{hyp}}(x, y) < +\infty$. *Both formulas are valid if* $x = y$, *being the distances* $= 0$. *Furthermore we can assume that* $d_{\mathrm{hyp}}(x, \xi) = d_{\mathrm{hyp}}(x, \eta) = \infty$ *for* $x \neq \xi, \eta$.

Proof In case (a), since $\log(\alpha + \sqrt{\alpha^2 - 1})$ is a pure imaginary complex number by (13.12), we may take $k = -ik_e$ with $k_e > 0$, so that $d_{\mathrm{ell}}(x, y)$ is a non-negative real number. In case (b), since the right-hand side of (13.9) is non-negative we put $k_h = k > 0$. The last assertion follows from the relations $\rho(x, \xi, \xi, \eta) = \infty$ and $\rho(x, \eta, \xi, \eta) = 0$ by Remark 13.2. □

Example 13.6 Consider the symmetric bilinear form $F\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$F(x, y) = \begin{cases} x_0y_0 + x_1y_1 & \text{elliptic case,} \\ x_0y_0 - x_1y_1 & \text{hyperbolic case,} \end{cases} \qquad x = (x_0, x_1),\ y = (y_0, y_1) \in \mathbb{R}^2.$$

Then the elliptic metric on $\mathbb{P}^1(\mathbb{R})$ is given by

$$d_{\mathrm{ell}}(x, y) = 2k_e \arccos\left(\frac{x_0y_0 + x_1y_1}{\sqrt{(x_0^2 + x_1^2)(y_0^2 + y_1^2)}}\right). \tag{13.19}$$

The elliptic distance (13.19) coincides with the euclidean length of arc β as in Fig. 13.1. In particular the length of elliptic line is $2\pi k_l$.

Let $F(x, x) = x_0^2 - x_1^2$. Denote by I each of the two segments of endpoints $[1, 1]$ and $[1, -1]$. If x, y belong to the same I and $x \neq [1, 1], [1, -1]$, $y \neq [1, 1]$, $[1, -1]$, then the hyperbolic metric on $I \smallsetminus \{[1, 1], [1, -1]\}$ is given by

$$d_{\mathrm{hyp}}(x, y) = 2k_h \operatorname{arccosh}\left(\frac{x_0y_0 - x_1y_1}{\sqrt{(x_0^2 - x_1^2)(y_0^2 - y_1^2)}}\right). \tag{13.20}$$

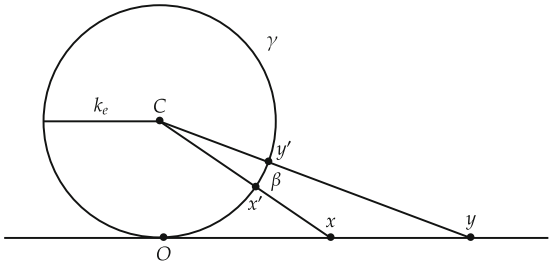


Fig. 13.1 Elliptic metric

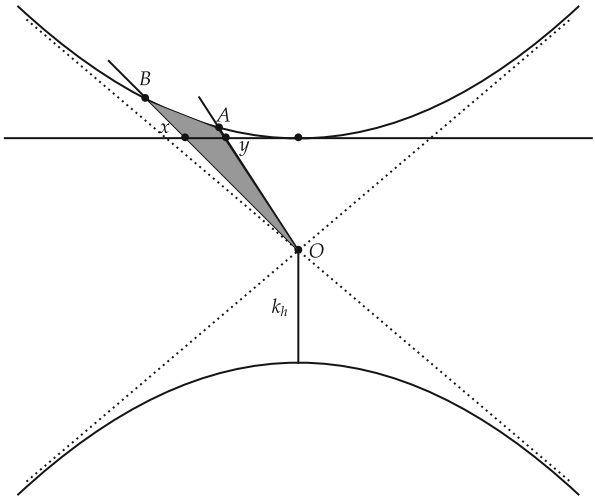


Fig. 13.2 Hyperbolic metric

The reader can prove that the hyperbolic distance $d_{\text{hyp}}(x, y)$ is the euclidean area of the hyperbolic sector as in Fig. 13.2. In particular the length of a hyperbolic line is infinite.

Definition 13.7 (Euclidean Metric) The euclidean metric on $\mathbb{A}^1(\mathbb{R})$ should be associated with a degenerate conic $x_0^2 = 0$, i.e. with $\xi = \eta = [0, 1]$, but, in this case, we have $\rho(x, y, \xi, \eta) = 1$, i.e. $\ell(x, y) = 0$ for all x, y with $x \neq y$. Nevertheless we may obtain the euclidean (or parabolic) metric in the following way. Consider the non-degenerate quadratic form

$$F_\varepsilon(z_0, z_1) = z_0^2 + \varepsilon z_1^2, \quad \varepsilon \in \mathbb{R} \setminus \{0\},$$

and let

$$d_\varepsilon^+(x,y) = 2k_\varepsilon \arccos\left(\frac{x_0y_0+\varepsilon x_1y_1}{\sqrt{(x_0^2+\varepsilon x_1^2)(y_0^2+\varepsilon y_1^2)}}\right),\ \varepsilon>0,$$

$$d_\varepsilon^-(x,y) = 2k_\varepsilon \operatorname{arccosh}\left(\frac{x_0y_0+\varepsilon x_1y_1}{\sqrt{(x_0^2+\varepsilon x_1^2)(y_0^2+\varepsilon y_1^2)}}\right),\ \varepsilon<0,$$

be the corresponding elliptic ($\varepsilon > 0$) and hyperbolic ($\varepsilon < 0$) metrics. Choose $k_\varepsilon = \dfrac{1}{2\sqrt{|\varepsilon|}}$. From the relation

$$\arccos\alpha = \arcsin\sqrt{1-\alpha^2},\quad 0\le\alpha\le 1, \tag{13.21}$$

we have for $\varepsilon > 0$

$$d_\varepsilon^+(x,y) = \frac{1}{\sqrt{\varepsilon}}\arcsin\left(\sqrt{\varepsilon}\sqrt{\frac{(x_0y_1-x_1y_0)^2}{(x_0^2+\varepsilon x_1^2)(y_0^2+\varepsilon y_1^2)}}\right).$$

Since $\lim\limits_{\alpha\to 0}\dfrac{\arcsin(\alpha)}{\alpha} = 1$, we get

$$\lim_{\varepsilon\to 0^+} d_\varepsilon^+(x,y) = \sqrt{\left(\frac{x_1}{x_0}-\frac{y_1}{y_0}\right)^2} = \left|\frac{x_1}{x_0}-\frac{y_1}{y_0}\right|. \tag{13.22}$$

Similarly, from the formula

$$\operatorname{arccosh}\alpha = \operatorname{arcsinh}\sqrt{\alpha^2-1},\quad 1\le\alpha, \tag{13.23}$$

we have for $\varepsilon < 0$

$$d_\varepsilon^-(x,y) = \frac{1}{\sqrt{|\varepsilon|}}\operatorname{arcsinh}\left(\sqrt{|\varepsilon|}\sqrt{\frac{(x_0y_1-x_1y_0)^2}{(x_0^2+\varepsilon x_1^2)(y_0^2+\varepsilon y_1^2)}}\right).$$

Since $\lim\limits_{\alpha\to 0}\dfrac{\operatorname{arcsinh}(\alpha)}{\alpha} = 1$, we get

$$\lim_{\varepsilon\to 0^-} d_\varepsilon^-(x,y) = \sqrt{\left(\frac{x_1}{x_0}-\frac{y_1}{y_0}\right)^2} = \left|\frac{x_1}{x_0}-\frac{y_1}{y_0}\right|. \tag{13.24}$$

Therefore both relations (13.22) and (13.24) give the euclidean metric on $\mathbb{A}^1(\mathbb{R})$ According to Klein [20] the euclidean metric is the "tangent metric" at $\varepsilon = 0$ to the metrics associated with the absolutes $z_0^2 + \varepsilon z_1^2 = 0$, $\varepsilon \in \mathbb{R} \setminus \{0\}$.

13.3 Projective Metrics of $\mathbb{P}^2(\mathbb{R})$ and $\mathbb{P}^3(\mathbb{R})$

Elliptic, hyperbolic and parabolic (i.e. euclidean) metrics on $\mathbb{P}^1(\mathbb{R})$ are the only projective metrics of dimension 1. Each of them is determined by the choice of an absolute and by the corresponding group of projectivities which fix the absolute with respect to them the metric is invariant. This procedure may be generalized to $\mathbb{P}^n(\mathbb{R})$ with $n \geq 2$.

Definition 13.8 We rename a conic of $\mathbb{P}^2(\mathbb{R})$ or a quadric of $\mathbb{P}^3(\mathbb{R})$ by 2-*quadric* or 3-*quadric* respectively A non-singular n-quadric $\widehat{F}$ (with $n = 2, 3$) will be called an *absolute n-quadric* of $\mathbb{P}^n(\mathbb{R})$. We can define a *distance* on a suitable subset $\mathcal{X}$ of $\mathbb{P}^n(\mathbb{R}) \setminus \mathcal{V}_+(\widehat{F})$ as follows:

$$d(x, y) := k \log \rho(x, y, \xi, \eta), \qquad k \in \mathbb{C}^*, \qquad x, y \in \mathcal{X}, \tag{13.25}$$

where ξ and η are the (possibly complex-conjugate) intersections of the line xy with $\mathcal{V}_+(\widehat{F})$ and k is fixed. The relative group of isometries consists of projectivities of $\mathbb{P}^n(\mathbb{R})$ which fix $\widehat{F}$ (see (10.64)).

If $n = 2$, the lines through a point $P \in \mathcal{X}$ tangent to $\widehat{F}$ are called the *isotropic lines* of the pencil with base locus P. They may be real or complex-conjugate lines. If $x, y \in \mathcal{X}$ lie on an isotropic line then $\xi = \eta$, so that $d(x, y) = 0$. If $n = 3$ we define an *isotropic cone* with vertex $P \in \mathcal{X}$ as the tangent cone $C_P^{\mathrm{p}}(\widehat{F})$. It may be a real cone or an imaginary cone (having the only vertex real). Furthermore, if $x \in \mathcal{X}$ and $y \in \mathcal{V}_+(\widehat{F})$, then $d(x, y) = \infty$. We have to choose a suitable set of *lines*

$$\mathrm{L}_{\mathcal{X}} \subset \{r \cap \mathcal{X} : r \text{ projective line}\},$$

and if $n = 3$, also a suitable set of *planes*

$$\mathrm{P}_{\mathcal{X}} \subset \{\sigma \cap \mathcal{X} : \sigma \text{ projective plane}\}.$$

Let $r, s \in \mathrm{L}_{\mathcal{X}}$ be two lines which meet at a point P of $\mathcal{X}$ and let $\sigma, \tau \in \mathrm{P}_{\mathcal{X}}$ be two planes passing through a line t of $\mathcal{X}$. The *angle* $\measuredangle(r, s)$ (o more properly the *measure of the angle*) between r and s

$$\measuredangle(r, s) = k' \log \rho(r, s, u, v), \qquad k' \in \mathbb{C}^*, \tag{13.26}$$

where u and v are the isotropic lines emanating from P lying on the plane σ joining r and s, i.e. $\{u, v\} = \sigma \cap C_P^{\mathrm{p}}(\widehat{F})$. Analogously the *angle* $\measuredangle(\sigma, \tau)$ between σ and τ

is defined by

$$\measuredangle(\sigma,\tau) = k'' \log \rho(\sigma,\tau,\eta,\varrho), \qquad k'' \in \mathbb{C}^*, \tag{13.27}$$

where η and ϱ are the plane tangent to $\widehat{F}$ passing through t. In particular, if $r \cap s \in \mathcal{V}_+(\widehat{F})$ (or $\sigma \cap \tau = t$ where t a line tangent to the absolute) then $\measuredangle(r,s) = 0$ (or $\measuredangle(\sigma,\tau) = 0$); we say that r and s are *parallel lines* (or σ and τ are *parallel planes*). In other words (according to Definition 10.79) we have

$$\text{if } n = 2,\ \measuredangle(r,s) = d((\pi^{\widehat{F}})^*(r), (\pi^{\widehat{F}})^*(s)),$$
$$\text{if } n = 3,\ \measuredangle(\sigma,\tau) = d((\pi^{\widehat{F}})^*(\sigma), (\pi^{\widehat{F}})^*(\tau)).$$

We say that two incident lines r and s (two planes σ and τ) are *perpendicular* if they are conjugate with respect to the absolute $\widehat{F}$, i.e. $\rho(r,s,u,v) = -1$ ($\rho(\sigma,\tau,\eta,\varrho) = -1$ resp.) (Proposition 10.90). Hence $\log \rho(r,s,u,v)$ has πi among its values so that, taking $2k' = -i$, r and s are perpendicular if and only if $\measuredangle(r,s) = \dfrac{\pi}{2}$. A completely analogue fact holds true for planes. A plane σ and a line r are perpendicular if r contains the pole of σ with respect to $\widehat{F}$.

F. Klein, guided by some earlier research by A. Cayley [10], determined 7 regions of $\mathbb{P}^2(\mathbb{R})$, each of them equipped with a projective metric (Maßbestimmung) in [20]. He considered also *degenerate absolutes* in order to include the euclidean metric. D. M. Y. Sommerville in [33] completed Klein's work, classifying the notion of absolute to the projective space $\mathbb{P}^n(\mathbb{R})$ of arbitrary dimension n. Since in $\mathbb{P}^n(\mathbb{R})$ there exist n first-kind forms (range of points, pencil of lines, pencil of planes and so on) and on each form we can put euclidean, elliptic or hyperbolic metrics we have 3^n regions of $\mathbb{P}^n(\mathbb{R})$ called *actual spaces* by Sommerville. Points, lines, planes, ... of the actual spaces will be called *actual points, lines, planes, ...*. The sets of actual points, lines, planes will be denoted by $\mathcal{X}$, $\mathrm{L}_\mathcal{X}$, $\mathrm{P}_\mathcal{X}$ respectively. In Sect. 13.4 we shall briefly expose the theory of general absolutes given in [37]. We have to state two points:

(a) On every first-kind form of the same type we put the same metric.
(b) Not all metrics are projective metrics according to Definition 13.1. We can have real isotropic lines and some metrics can take values in $\mathbb{C} \cup \{\infty\}$. Nevertheless all metrics are considered as *projective metrics* and called *Cayley–Klein metrics.*

Since incidence and cross-ratio are invariant with respect to projectivities we may examine various cases of absolutes given in canonical form. In all the following figures the real absolute is red, the real isotropic lines are blue, the imaginary absolute is green and the complex-conjugate isotropic lines are orange. We choose the same constants k_e, k'_e, k_h, k'_h..., for each type of first-kind form (range of points, pencil of lines, pencil of planes).

13.3.1 Non-degenerate Absolute

Riemann Elliptic Plane $\mathcal{S}_2$ Let $\mathbf{i}(\widehat{F}) = 2$ i.e. $F = x_0^2 + x_1^2 + x_2^2$. Then $\mathcal{X} = \mathrm{Int}(\widehat{F}) = \mathbb{P}^2(\mathbb{R})$. By Corollary 13.5 we have the elliptic metric on the whole $\mathbb{P}^2(\mathbb{R})$: for any $x = [x_0, x_1, x_2]$, $y = [y_0, y_1, y_2] \in \mathbb{P}^2(\mathbb{R})$

$$d_{\mathrm{ell}}(x, y) = 2k_e \arccos\left(\frac{x_0y_0 + x_1y_1 + x_2y_2}{\sqrt{(x_0^2 + x_1^2 + x_2^2)(y_0^2 + y_1^2 + y_2^2)}}\right). \tag{13.28}$$

The lines of Riemann's elliptic plane are projective lines. The dual conic $\widehat{F^*}$ of $\widehat{F}$ is $u_0^2 + u_1^2 + u_2^2 = 0$. The angle $\measuredangle(r, s)$ between the lines $r : u_0x_0 + u_1x_1 + u_2x_2 = 0$ and $s : v_0x_0 + v_1x_1 + v_2x_2 = 0$ is the elliptic angle

$$\measuredangle(r, s) = 2k'_e \arccos\left(\frac{u_0v_0 + u_1v_1 + u_2v_2}{\sqrt{(u_0^2 + u_1^2 + u_2^2)(v_0^2 + v_1^2 + v_2^2)}}\right). \tag{13.29}$$

Absolute geometry is inconsistent with elliptic geometry: in that theory, there are no parallel lines at all, but in absolute geometry parallel lines do exist (Proposition 12.40). Nevertheless the elliptic plane $\mathcal{S}_2$ is an incidence structure and the elliptic metric is a projective metric according to Definition 13.1 (left to the reader as an exercise).

Riemann Elliptic Space $\mathcal{S}_3$ Let $\mathbf{i}(\widehat{F}) = 3$ i.e. $F = x_0^2 + x_1^2 + x_2^2 + x_3^2$. Then $\mathcal{X} = \mathrm{Int}(\widehat{F}) = \mathbb{P}^3(\mathbb{R})$. The lines and the planes of the Riemann's elliptic space are the projective lines and planes. If $x = [x_0, x_1, x_2, x_3]$, $y = [y_0, y_1, y_2, y_3] \in \mathbb{P}^3(\mathbb{R})$, we have the elliptic metric

$$d_{\mathrm{ell}}(x, y) = 2k_e \arccos\left(\frac{x_0y_0 + x_1y_1 + x_2y_2 + x_3y_3}{\sqrt{x_0^2 + x_1^2 + x_2^2 + x_3^2}\sqrt{y_0^2 + y_1^2 + y_2^2 + y_3^2}}\right). \tag{13.30}$$

Mutatis mutandis, we have the formula for the angles between two planes.

Lobachevskii Hyperbolic Plane ${}^1\mathcal{S}_2$ Let $\mathbf{i}(\widehat{F}) = 1$ i.e. $F = x_0^2 + x_1^2 - x_2^2$. We have the *hyperbolic metric* on $\mathcal{X} = \mathrm{Int}(\widehat{F}) = \{[x_0, x_1, x_2] \in \mathbb{P}^2(\mathbb{R}) : x_0^2 + x_1^2 - x_2^2 < 0\}$: for $x = [x_0, x_1, x_2]$, $y = [y_0, y_1, y_2] \in \mathrm{Int}(\widehat{F})$

$$d_{\mathrm{hyp}}(x, y) = 2k_h \operatorname{arccosh}\left(\frac{x_0y_0 + x_1y_1 - x_2y_2}{\sqrt{(x_0^2 + x_1^2 - x_2^2)(y_0^2 + y_1^2 - y_2^2)}}\right), \tag{13.31}$$

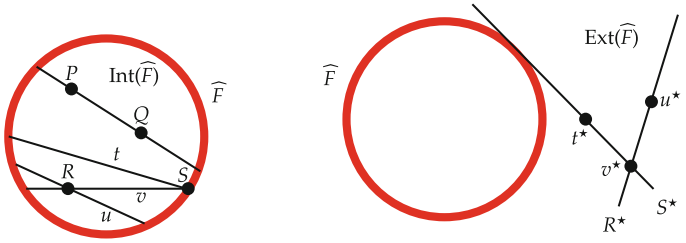


Fig. 13.3 Lobachevskii hyperbolic plane

The lines of $\mathrm{Int}(\widehat{F})$ are the intersections of projective lines with $\mathrm{Int}(\widehat{F})$. The dual conic is $F^* : u_0^2 + u_1^2 - u_2^2$. The angle $\measuredangle(r, s)$ between the lines $r : u_0x_0 + u_1x_1 + u_2x_2 = 0$ and $s : v_0x_0 + v_1x_1 + v_2x_2 = 0$ of $\mathrm{Int}(\widehat{F})$ which meet at $P \in \mathrm{Int}(\widehat{F})$ is elliptic and given by (Corollary 13.5)

$$\measuredangle(r, s) = 2k'_e \arccos\left(\frac{u_0v_0 + u_1v_1 - u_2v_2}{\sqrt{(u_0^2 + u_1^2 - u_2^2)(v_0^2 + v_1^2 - v_2^2)}}\right). \tag{13.32}$$

If $r \cap s \in \mathcal{V}_+(\widehat{F})$ we put $\measuredangle(r, s) = 0$ and we call them parallel. If $r \cap s \in \mathrm{Ext}(\widehat{F})$ they are called *hyperparallel* and $\measuredangle(r, s)$ is not defined. On the left-hand side of Fig. 13.3 some points and lines of the Lobachevskii hyperbolic plane are represented. On the right-hand side of Fig. 13.3 corresponding polars and poles ($P^* := \pi^{\widehat{F}}(P)$ and $r^* := (\pi^{\widehat{F}})^*(r)$) are shown. The Lobachevskii hyperbolic plane is a model of hyperbolic geometry introduced in Chap. 12 by means of a Poincarè upper half plane and thus it is an absolute plane. The reader should compare those two models.

Now we have to give an analytic proof of the strong triangular inequality (Definition 13.1) for d_{hyp}. Let P, Q, R three distinct points of $\mathrm{Int}(\widehat{F})$. We may assume (up to a change of coordinates)

$$P = [0, 0, -1], \quad Q = [0, y, z],$$
$$R = [x_0, x_1, x_2], \qquad y^2 - z^2 < 0,\ x_0^2 + x_1^2 - x_2^2 < 0.$$

Hence assuming $k_h = 1$ we get from (13.31)

$$d_{\mathrm{hyp}}(P, Q) = \operatorname{arccosh}\left(\frac{z}{\sqrt{-(y^2 - z^2)}}\right), \quad z > 0,$$

$$d_{\mathrm{hyp}}(P, R) = \operatorname{arccosh}\left(\frac{x_2}{\sqrt{-(x_0^2 + x_1^2 - x_2^2)}}\right), \quad x_2 > 0,$$

$$d_{\text{hyp}}(Q, R) = \operatorname{arccosh}\left(\frac{x_1 y + x_2 z}{\sqrt{(y^2 - z^2)(x_0^2 + x_1^2 - x_2^2)}}\right), \quad x_1 y + x_2 z > 0.$$

Since (see [31], § 9, Ch. 1)

$$\sinh(\operatorname{arccosh}(w)) = \sinh(\log(w + \sqrt{w^2 - 1})) = \sqrt{w^2 - 1}, \qquad (w > 1),$$

by the addition formula for hyperbolic cosine we have

$$\begin{aligned}
\cosh(d_{\text{hyp}}(P, Q) + d_{\text{hyp}}(P, R)) &= \cosh(d_{\text{hyp}}(P, Q)) \cosh(d_{\text{hyp}}(P, R)) \\
&\quad + \sinh(d_{\text{hyp}}(P, Q)) \sinh(d_{\text{hyp}}(P, R)) \\
&= \frac{z}{\sqrt{-(y^2 - z^2)}} \frac{x_2}{\sqrt{-(x_0^2 + x_1^2 - x_2^2)}} \\
&\quad + \sqrt{\frac{z^2}{-(y^2 - z^2)} - 1} \sqrt{\frac{x_2^2}{-(x_0^2 + x_1^2 - x_2^2)} - 1} \\
&= \frac{x_2 z + |y| \sqrt{x_0^2 + x_1^2}}{\sqrt{(y^2 - z^2)((x_0^2 + x_1^2 - x_2^2)}} \\
&\geq \frac{x_2 z + y x_1}{\sqrt{(y^2 - z^2)((x_0^2 + x_1^2 - x_2^2)}} \\
&= \cosh d_{\text{hyp}}(Q, R).
\end{aligned}$$

Since the function $u = \cosh(t)$ is strictly increasing on the interval $[0, +\infty)$, we have the inequality $d_{\text{hyp}}(P, Q) + d_{\text{hyp}}(P, R) \geq d_{\text{hyp}}(Q, R)$. Furthermore $d_{\text{hyp}}(P, Q) + d_{\text{hyp}}(P, R) = d_{\text{hyp}}(Q, R)$ if and only if $x_0 = 0$ i.e. P, Q and R are collinear.

We can define a metric also on $\text{Ext}(\widehat{F})$ by means of (13.25) and (13.26) but some care is requested. Since we want to have only one metric (elliptic or hyperbolic) on first-kind forms (range of points, pencil of lines) we cannot define a distance between all pairs of points of $\text{Ext}(\widehat{F})$ and not all lines of $\text{Ext}(\widehat{F})$ are lines of the geometries we are going to describe.

Doubly Hyperbolic Plane ${}^1\mathcal{S}_2'$ $\mathbf{i}(\widehat{F}) = 1$. We put $\mathcal{X} = \text{Ext}(\widehat{F})$ and $\text{L}_\mathcal{X}$ is defined as being the set of the traces on $\mathcal{X}$ of projective lines which meet $\mathcal{V}_+(\widehat{F})$ at real points. The lines ξ_P and σ_P, tangent to $\widehat{F}$ emanating from $P \in \text{Ext}(\widehat{F})$ are the isotropic lines of the pencil with base locus P. If $x, y \in \mathcal{X}$ belong to an isotropic line, then $d(x, y) = 0$; if $x \in \mathcal{V}_+(\widehat{F})$ and $y \in \mathcal{X}$ then $d(x, y) = \infty$. On the non-isotropic lines of $\text{L}_\mathcal{X}$ we have the hyperbolic metric (13.31). Hence the distance of two points of

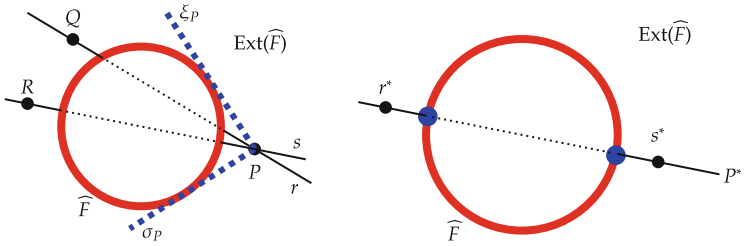


Fig. 13.4 Doubly hyperbolic plane

Ext($\widehat{F}$) not lying on a line of $\mathrm{L}_\mathcal{X}$ is not defined. The angle $\measuredangle(r, s)$ between the lines $r : u_0x_0 + u_1x_1 + u_2x_2 = 0$ and $s : v_0x_0 + v_1x_1 + v_2x_2 = 0$ of $\mathrm{L}_\mathcal{X}$ which meet at $P \in \mathrm{Ext}(\widehat{F})$ is hyperbolic and given by (Corollary 13.5)

$$\measuredangle(r, s) = 2k'_h \operatorname{arccosh}\left(\frac{u_0v_0 + u_1v_1 - u_2v_2}{\sqrt{(u_0^2 + u_1^2 - u_2^2)(v_0^2 + v_1^2 - v_2^2)}}\right). \tag{13.33}$$

If $r \cap s \in \mathcal{V}_+(\widehat{F})$ we put $\measuredangle(r, s) = 0$. The doubly hyperbolic plane is not an incidence structure since two points which lie on a projective line not intersecting $\mathcal{V}_+(\widehat{F})$ at real points are not joined by a line of this geometry (Fig. 13.4).

Dual Hyperbolic Plane ${}^1\mathbb{S}_2^*$ $\mathbf{i}(\widehat{F}) = 1$ We put $\mathcal{X} = \mathrm{Ext}(\widehat{F})$. The set $\mathrm{L}_\mathcal{X}$ of lines consists of the traces on Ext($\widehat{F}$) of the projective lines which do not meet $\mathcal{V}_+(\widehat{F})$ at real points. The lines ξ_P and σ_P, tangent to $\widehat{F}$ emanating from $P \in \mathrm{Ext}(\widehat{F})$ are the isotropic lines of the pencil with base locus P. On the non-isotropic lines of $\mathrm{L}_\mathcal{X}$ we have the elliptic metric: for $x = [x_0, x_1, x_2]$, $y = [y_0, y_1, y_2] \in \mathrm{Ext}(\widehat{F})$ such that the line xy meets $\widehat{F}$ at complex-conjugate points

$$d(x, y) = 2k_e \arccos\left(\frac{x_0y_0 + x_1y_1 - x_2y_2}{\sqrt{(x_0^2 + x_1^2 - x_2^2)(y_0^2 + y_1^2 - y_2^2)}}\right). \tag{13.34}$$

Hence the distance of two points of Ext($\widehat{F}$) not lying on a line of $\mathrm{L}_\mathcal{X}$ is not defined. The angle $\measuredangle(r, s)$ between the lines $r : u_0x_0+u_1x_1+u_2x_2 = 0$ and $s : v_0x_0+v_1x_1+v_2x_2 = 0$ of $\mathrm{L}_\mathcal{X}$ which meet at $P \in \mathrm{Ext}(\widehat{F})$ is hyperbolic as given by (13.33). As in preceding case the dual hyperbolic plane is not an incidence structure (Fig. 13.5).

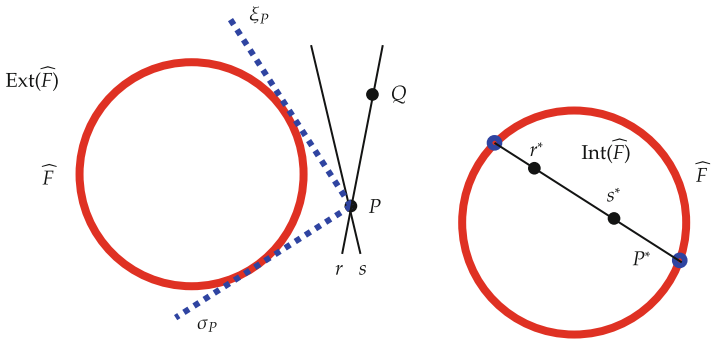


Fig. 13.5 Dual hyperbolic plane

13.3.2 Non-ruled Quadric (i.e. $i(\widehat{F}) = 2$) $F = x_0^2 + x_1^2 + x_2^2 - x_3^2$. We Get Four Actual Spaces

Hyperbolic Space ${}^1\mathcal{S}_3$ We have the hyperbolic metric on $\mathcal{X} = \text{Int}(\widehat{F}) = \{[x_0, x_1, x_2, x_3] \in \mathbb{P}^3(\mathbb{R}) : x_0^2 + x_1^2 + x_2^2 - x_3^2 < 0\}$: for $x = [x_0, x_1, x_2, x_3]$, $y = [y_0, y_1, y_2, y_3] \in \text{Int}(\widehat{F})$

$$d_{\text{hyp}}(x, y) = 2k_h \ \text{arccosh}\left(\frac{x_0y_0 + x_1y_1 + x_2y_2 - x_3y_3}{\sqrt{(x_0^2 + x_1^2 + x_2^2 - x_3^2)(y_0^2 + y_1^2 + y_2^2 - y_3^2)}}\right). \tag{13.35}$$

The lines and the planes of ${}^1\mathcal{S}_3$ are the traces of projective lines and planes on $\mathcal{X}$. On pencils of lines and pencils of planes we have an elliptic metric.

In the last three actual spaces associated with the absolute $F = x_0^2 + x_1^2 + x_2^2 - x_3^2$ the set of points is $\mathcal{X} = \text{Ext}(\widehat{F}) = \{[x_0, x_1, x_2, x_3] \in \mathbb{P}^3(\mathbb{R}) : x_0^2 + x_1^2 + x_2^2 - x_3^2 > 0\}$.

Totally Hyperbolic Space ${}^1\mathcal{S}'_3$ Lines and planes are the traces on $\mathcal{X}$ of the projective lines and projective planes which intersect $\mathcal{V}_+(\widehat{F})$ at real points. The set $\mathcal{X}$, the pencils of lines and the pencils of planes are equipped with a hyperbolic metric.

Dual Hyperbolic Space ${}^1\mathcal{S}^*_3$ The planes of $\text{P}_\mathcal{X}$ are the traces on $\mathcal{X}$ of the projective planes which intersect $\mathcal{V}_+(\widehat{F})$ in real points. The lines are the traces on $\mathcal{X}$ of the projective lines (lying on the planes of $\text{P}_\mathcal{X}$) which do not intersect $\mathcal{V}_+(\widehat{F})$ at real points so that $\mathcal{X}$ is endowed with an elliptic metric. Hence the pencils of lines and the pencils of planes are equipped with a hyperbolic metric.

Space ${}^1\mathcal{S}''_3$ The planes of $\text{P}_\mathcal{X}$ are the traces on $\mathcal{X}$ of the projective planes which do not intersect $\mathcal{V}_+(\widehat{F})$ in real points. The lines are the traces on $\mathcal{X}$ of the projective

lines (lying on the planes of $\mathrm{P}_{\mathcal{X}}$) which do not intersect $\mathcal{V}_+(\widehat{F})$ at real points On $\mathcal{X}$ and on the pencils of lines there is the elliptic metric, while on the pencils of planes we have a hyperbolic metric.

13.3.3 Ruled Quadric (i.e. $i(\widehat{F}) = 1$) $F = x_0^2 + x_1^2 - x_2^2 - x_3^2$. We Get Three Actual Spaces On $\mathcal{X} = \mathbb{P}^3(\mathbb{R}) \setminus \mathcal{V}_+(\widehat{F})$

For each point $P \in \mathcal{X}$ there are infinitely many planes tangent to $\widehat{F}$ joining P to the lines lying on $\widehat{F}$ and infinitely many lines tangent to $\widehat{F}$ which are the isotropic lines.

Space ${}^2\mathcal{S}_3$ The actual lines are those projective lines r such that $\#(r \cap \mathcal{V}_+(\widehat{F})) = 2$ and the actual planes σ are those projective planes which pass through the actual lines. The conics $C = \sigma \cap \mathcal{V}_+(\widehat{F})$ are non-degenerate with real points. The interior points of the conics C are the actual points. We have a hyperbolic metric on $\mathcal{X}$, an elliptic metric on $\mathrm{L}_{\mathcal{X}}$ and a hyperbolic metric on $\mathrm{P}_{\mathcal{X}}$ (see Proposition 10.101).

Space ${}^2\mathcal{S}_3'$ The actual lines are those projective lines r such that $\#(r \cap \mathcal{V}_+(\widehat{F})) = 2$ and the actual planes σ are those projective planes which pass through the actual lines. The exterior points of the conics $\sigma \cap \mathcal{V}_+(\widehat{F})$ are the actual points. We have a hyperbolic metric on $\mathcal{X}$, on $\mathrm{L}_{\mathcal{X}}$ and on $\mathrm{P}_{\mathcal{X}}$.

Space ${}^2\mathcal{S}_3''$ The actual lines are those projective lines r such that $r \cap \mathcal{V}_+(\widehat{F}) = \varnothing$ and the actual planes σ are those projective planes which pass through the actual lines. The actual points are the points of the actual lines. We have an elliptic metric on $\mathcal{X}$, a hyperbolic metric on the pencils of lines and an elliptic metric on the pencils of planes (see Proposition 10.101)).

13.3.4 Degenerate Absolute

Euclidean Plane $\mathcal{R}_2$ The procedure of Definition 13.7 can be extended to the case $n = 2$. Consider the absolute

$$F_\varepsilon(z_0, z_1, z_2) = z_0^2 + \varepsilon z_1^2 + \varepsilon z_2^2, \qquad \varepsilon \in \mathbb{R}^*.$$

If $x = [x_0, x_1, x_2]$, $y = [y_0, y_1, y_2] \in \mathbb{A}^2(\mathbb{R})$ i.e $x_0 y_0 \neq 0$, we put

$$d_\varepsilon^+(x, y) := 2k_\varepsilon \arccos\left(\frac{x_0 y_0 + \varepsilon x_1 y_1 + \varepsilon x_2 y_2}{\sqrt{(x_0^2 + \varepsilon x_1^2 + \varepsilon x_2^2)(y_0^2 + \varepsilon y_1^2 + \varepsilon y_2^2)}}\right), \qquad \varepsilon > 0$$

and

$$d_\varepsilon^-(x,y) := 2k_\varepsilon \arccos\left(\frac{x_0y_0+\varepsilon x_1y_1+\varepsilon x_2y_2}{\sqrt{(x_0^2+\varepsilon x_1^2+\varepsilon x_2^2)(y_0^2+\varepsilon y_1^2+\varepsilon y_2^2)}}\right), \qquad \varepsilon<0.$$

Take $k_\varepsilon = \dfrac{1}{2\sqrt{|\varepsilon|}}$, then from (13.21) we get

$$d_\varepsilon^+(x,y) = \frac{1}{\sqrt{\varepsilon}}\arcsin \times\left(\sqrt{\varepsilon}\sqrt{\frac{(x_0y_1-x_1y_0)^2+(x_0y_2-x_2y_0)^2+\varepsilon(x_1y_2-x_2y_1)^2}{(x_0^2+\varepsilon x_1^2+\varepsilon x_2^2)(y_0^2+\varepsilon y_1^2+\varepsilon y_2^2)}}\right), \quad \varepsilon>0. \tag{13.36}$$

From (13.23) we get

$$d_\varepsilon^-(x,y) = \frac{1}{\sqrt{|\varepsilon|}}\operatorname{arcsinh} \times\left(\sqrt{|\varepsilon|}\sqrt{\frac{(x_0y_1-x_1y_0)^2+(x_0y_2-x_2y_0)^2+\varepsilon(x_1y_2-x_2y_1)^2}{(x_0^2+\varepsilon x_1^2+\varepsilon x_2^2)(y_0^2+\varepsilon y_1^2+\varepsilon y_2^2)}}\right), \quad \varepsilon<0. \tag{13.37}$$

Hence

$$\lim_{\varepsilon\to0^+} d_\varepsilon^+(x,y) = \lim_{\varepsilon\to0^-} d_\varepsilon^+(x,y) = \sqrt{\frac{(x_0y_1-x_1y_0)^2+(x_0y_2-x_2y_0)^2}{x_0^2y_0^2}} = \sqrt{\left(\frac{y_1}{y_0}-\frac{x_1}{x_0}\right)^2+\left(\frac{y_2}{y_0}-\frac{x_2}{x_0}\right)^2},$$

which is the *euclidean distance* $d_{\mathrm{eucl}}(x,y)$ on $\mathbb{A}^2(\mathbb{R})$. A projectivity of $\mathbb{P}^2(\mathbb{R})$ which fixes the point-conic $F_\varepsilon = 0$ associated with the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}, \qquad \varepsilon \neq 0,$$

also preserves the line-conic $\varepsilon u_0^2 + u_1^2 + u_2^2 = 0$ (see Definition 10.79) whose associated matrix is

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \varepsilon^{-1} & 0 \\ 0 & 0 & \varepsilon^{-1} \end{pmatrix}.$$

The projectivities of $\mathbb{P}^2(\mathbb{R})$ which preserve $d_{\text{eucl}}(x, y)$ are those which fix the line at infinity, $x_0 = 0$. Furthermore, if you let $\varepsilon \to 0$ they fix the degenerate conic $u_1^2 + u_2^2 = 0$ so that they preserve the intersections of $x_0 = 0$ with the lines $x_2 = \pm i x_1$ i.e. the circular points $[0, 1, i]$ and $[0, 1, -i]$ (possibly interchanged). According to Definition 13.12 we say that the euclidean metric *corresponds to the degenerate absolute* $\Omega : \mathcal{Q}_0 \supset T_0 \supset \mathcal{Q}_1$, where

$$\mathcal{Q}_0 = \widehat{F_0}, \quad F_0(x_0, x_1, x_2) = x_0^2; \qquad T_0 : x_0 = 0,$$

and $\mathcal{Q}_1 = \widehat{F_1}$, $F_1(x_1, x_2) = x_1^2 + x_2^2$, is the non-degenerate imaginary conic of T_0, i.e. $\mathcal{Q}_1$ is the pair of circular points $[0, 1, i]$ and $[0, 1, -i]$. Hence $\mathcal{X} = \mathbb{A}^2(\mathbb{R}) = \mathbb{P}^2(\mathbb{R}) \setminus T_0$ is the affine plane ($\mathrm{L}_{\mathcal{X}}$ is the set of affine lines) equipped with the euclidean metric (Definition 13.13)

$$\begin{aligned} d(x, y) &= \sqrt{F_1(x - y, x - y)} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}, \qquad x = [1, x_1, x_2],\ y = [1, y_1, y_2]. \end{aligned}$$

The angle between the lines $u\colon [u_0, u_1, u_2]$ and $v\colon [v_0, v_1, v_2]$ (where $u, v \neq T_0$) is elliptic and given by (Example 13.6)

$$\measuredangle(u, v) = 2k_e \arccos\left(\frac{u_1 v_1 + u_2 v_2}{\sqrt{u_1^2 + u_2^2}\sqrt{v_1^2 + v_2^2}} \right).$$

The dashed orange lines are the complex-conjugate isotropic lines (Figs. 13.6 and 13.7).

Euclidean Space R_3 Consider the absolute

$$\mathcal{Q}_0 = \widehat{F_0}, \quad F_0(x_0, x_1, x_2, x_3) = x_0^2; \qquad T_0 : x_0 = 0,$$



Fig. 13.6 The absolute of a euclidean plane

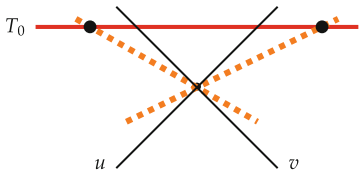


Fig. 13.7 Angles in a euclidean plane

and $\mathcal{Q}_1 = \widehat{F_1}$, $F_1(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$ is the non-degenerate imaginary conic of T_0 (see Definition 10.102). The actual space $\mathcal{X}$ is the affine space $\mathbb{A}^3(\mathbb{R}) = \mathbb{P}^3(\mathbb{R}) \setminus T_0$. The actual lines and planes are the affine lines and planes. We have the usual euclidean metric on $\mathcal{X}$

$$\begin{aligned} d(x, y) &= \sqrt{F_1(x - y, x - y)} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}, \\ x &= [1, x_1, x_2, x_3], \quad y = [1, y_1, y_2, y_3]. \end{aligned}$$

Let r and $s \in \mathrm{L}_{\mathcal{X}}$, let σ be the plane generated by r and s. In the pencil with base locus $r \cap s$ on the projective closure $\overline{\sigma}$ of σ take the complex conjugate lines u and v passing through the points $\mathcal{Q}_1 \cap \overline{\sigma}$. Then we have the elliptic metric $\measuredangle(r, s) = k_e \log \rho(r, s, u, v)$. Let $\alpha : ax + by + cz + d = 0$ and $\beta : a'x + b'y + c'z + d' = 0$ be two planes of $\mathrm{P}_{\mathcal{X}}$, let $r = \overline{\alpha} \cap T_0$ and $s = \overline{\beta} \cap T_0$. In the pencil generated by r and s take the complex conjugate lines u and v tangent to $\mathcal{Q}_1$. Then we have the elliptic metric (see (13.29))

$$\begin{aligned} \measuredangle(\alpha, \beta) &= k_e \log \rho(r, s, u, v) \\ &= 2k'_e \arccos\left(\frac{aa' + bb' + cc'}{\sqrt{(a^2 + b^2 + c^2)((a')^2 + (b')^2 + (c')^2)}} \right). \end{aligned}$$

From now onwards we shall examine the remaining degenerate absolutes of projective plane.

Dual Euclidean Plane $\mathcal{S}_2^1$ Interchanging the distances (elliptic between points and euclidean between lines) we obtain the *dual euclidean plane* (Fig. 13.8). Consider the absolute $\Omega : \mathcal{Q}_0 \supset T_0$, where

$$\begin{aligned} &\mathcal{Q}_0 = \widehat{F_0}, \quad F_0(x_0, x_1, x_2) = x_0^2 + x_1^2; \\ &T_0 \ : \ \begin{cases} x_0 = 0 \\ x_1 = 0. \end{cases} \end{aligned}$$

Hence $\mathcal{X} = \mathbb{P}^2(\mathbb{R}) \setminus T_0$ and $\mathrm{L}_{\mathcal{X}}$ consists of the projective lines not passing through T_0. If the line joining $x = [x_0, x_1, x_2]$ and $y = [y_0, y_1, y_2]$ does not pass through T_0

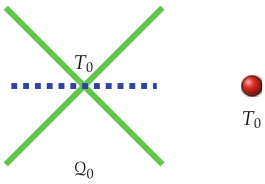


Fig. 13.8 The absolute of a dual euclidean plane



Fig. 13.9 The absolute for angles in a dual euclidean plane

we have the elliptic distance

$$d(x,y) = 2k_e \arccos\left(\frac{F_0(x,y)}{\sqrt{F_0(x,x)}\sqrt{F_0(y,y)}}\right)$$
$$= 2k_e \arccos\left(\frac{x_0y_0 + x_1y_1}{\sqrt{x_0^2+x_1^2}\sqrt{y_0^2+y_1^2}}\right).$$

All lines through T_0 are isotropic. The angle between the lines $u = [u_0, u_1, u_2]$ and $v = [v_0, v_1, v_2]$ not passing through T_0, i.e. with $u_2, v_2 \neq 0$, is euclidean,

$$\measuredangle(u,v) = \sqrt{G_1(u-v,u-v)} = \sqrt{(u_0-v_0)^2+(u_1-v_1)^2}.$$

This metric corresponds to the absolute $\Omega^* : \mathcal{D}_0 \supset T_0^* \supset \mathcal{D}_1$ of $\mathrm{L}\chi$, where (Fig. 13.9)

$$\mathcal{D}_0 = \widehat{G_0}, \quad G_0(u_0,u_1,u_2) = u_2^2;$$
$$T_0^* : u_2 = 0;$$
$$\mathcal{D}_1 = \widehat{G_1}, \quad G_1(u_0,u_1) = u_0^2+u_1^2,$$
$$\mathcal{D}_1 \text{ the non-degenerate imaginary conic of } T_0^*.$$

Minkowski Plane ${}^1\mathcal{R}_2$ Consider the absolute $\Omega : \mathcal{Q}_0 \supset T_0 \supset \mathcal{Q}_1$, where

$$\mathcal{Q}_0 = \widehat{F_0}, \quad F_0(x_0,x_1,x_2) = x_0^2;$$
$$T_0 : x_0 = 0;$$
$$\mathcal{Q}_1 = \widehat{F_1}, \quad F_1(x_1,x_2) = x_1^2 - x_2^2, \quad \mathcal{Q}_1 \text{ the non-degenerate real conic of } T_0.$$

Hence $\mathcal{X} = \mathbb{A}^2(\mathbb{R}) = \mathbb{P}^2(\mathbb{R}) \setminus T_0$ The distance between $x = [x_0, x_1, x_2]$ and $y = [x_0, y_1, y_2] \in \mathbb{A}^2(\mathbb{R})$ is euclidean,

$$d(x, y) = \sqrt{|F_1(x - y, x - y)|} = \sqrt{|(x_1 - y_1)^2 - (x_2 - y_2)^2|}. \tag{13.38}$$

By lines of $\mathrm{L}_{\mathcal{X}}$ we understand all projective lines which intersect T_0 in points of some fixed segment bounded by $[0, 1, 1]$ and $[0, -1, 1]$. On $\mathrm{L}_{\mathcal{X}}$ the absolute is $\Omega^* : \mathcal{D}_0 \supset T_0^*$, where

$$\mathcal{D}_0 = \widehat{G_0}, \quad G_0(u_0, u_1, u_2) = u_1^2 - u_2^2;$$

$$T_0^* : \begin{cases} u_1 = 0 \\ u_2 = 0. \end{cases}$$

The angle between two lines $r : [u_0, u_1, u_2]$ and $s : [v_0, v_1, v_2]$ of $\mathrm{L}_{\mathcal{X}}$ is hyperbolic and given by

$$\begin{aligned} \measuredangle(r, s) &= 2k_h \operatorname{arccosh}\left(\frac{G_0(u, v)}{\sqrt{G_0(u, u)G_0(v, v)}}\right) \\ &= 2k_h \operatorname{arccosh}\left(\frac{u_1v_1 - u_2v_2}{\sqrt{(u_1^2 - u_2^2)(v_1^2 - v_2^2)}}\right). \end{aligned}$$

Furthermore $\measuredangle(r, s) = 0$ if r and s meet at a point of $T_0 \setminus \{[0, 1, 1], [0, 1, -1]\}$. The lines through $[0, 1, 1]$ or $[0, -1, 1]$ are isotropic.

We could consider all projective lines r and s not intersecting $\mathcal{Q}_1$ and define $\measuredangle(r, s) = k_h \log|\rho(r, s, \xi, \eta)|$, where ξ and η are the isotropic lines emanating from $r \cap s$ or also the complex angle $k_h \log \rho(r, s, \xi, \eta)$.

H. Minkowski (1864–1909) used this geometry for the description of phenomena which occur in the special theory of relativity (Figs. 13.10 and 13.11).

Dual Minkowski Plane ${}^1S_2^1$ Let $\Omega \supset T_0 \supset \mathcal{Q}_1$ be the absolute (Figs. 13.12 and 13.13)

$$\mathcal{Q}_0 = \widehat{F_0}, \quad F_0(x_0, x_1, x_2) = x_0^2 - x_1^2;$$

$$T_0 : \begin{cases} x_0 = 0 \\ x_1 = 0. \end{cases}$$



Fig. 13.10 The absolute of a Minkowski plane

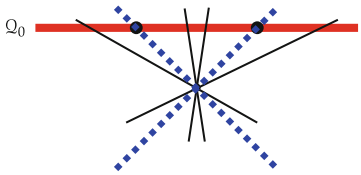


Fig. 13.11 Angles in a Minkowski plane

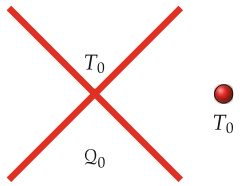


Fig. 13.12 Absolute of the dual Minkowski plane



Fig. 13.13 The absolute for the lines in a dual Minkowski plane

Choose the line $r : x_0 + x_1 = 0$ as the improper line so that $\mathcal{E} := \mathbb{P}^2(\mathbb{R}) \setminus \Omega_0$ is the affine plane with the line $s : x_0 - x_1 = 0$ missing. Let

$$\mathcal{X} := \{[x_0, x_1, x_2] \in \mathcal{E} : x_0^2 - x_1^2 > 0\}$$

(or $\mathcal{X} := \{[x_0, x_1, x_2] \in \mathcal{E} : x_0^2 - x_1^2 < 0\}$). Hence $\mathcal{X}$ is a certain pair of " vertically opposite angles" formed by the absolute lines. The lines of $\mathrm{L}_{\mathcal{X}}$ are the traces on $\mathcal{X}$ of the projective lines not passing through T_0, equipped with the hyperbolic metric

$$d(x, y) = 2k_h \operatorname{arccosh}\left(\frac{x_0 y_0 - x_1 y_1}{\sqrt{(x_0^2 - x_1^2)(y_0^2 - y_1^2)}}\right).$$

$$x = [x_0, x_1, x_2],\ y = [y_0, y_1, y_2] \in \mathcal{X}.$$

The lines passing through T_0 are the isotropic lines. On $\mathrm{L}_{\mathcal{X}}$ the absolute is $\Omega^* : \mathcal{D}_0 \supset T_0^* \supset \mathcal{D}_1$, where

$$\mathcal{D}_0 = \widehat{G_0}, \quad G_0(u_0, u_1, u_2) = u_2^2;$$

$$T_0^* : u_2 = 0;$$

$$\mathcal{D}_1 = \widehat{G_1}, \quad G_1(u_1, u_2) = u_0^2 - u_1^2, \quad \mathcal{D}_1 \text{ the non-degenerate real conic of } T_0^*.$$

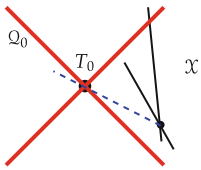


Fig. 13.14 Dual Minkowski plane

The angle between two lines $r : [u_0, u_1, u_2]$ and $s : [v_0, v_1, v_2]$ which meet at a point of $\mathcal{X}$ is euclidean (Fig. 13.14),

$$\measuredangle(r, s) = \sqrt{|(u_0 - v_0)^2 - (u_1 - v_1)^2|}.$$

Galilean Plane F_2 Consider the flag space $\Omega\colon \Omega_0 \supset T_0 \supset \Omega_1$ where

$$\Omega_0 : x_0^2 = 0; \quad T_0 : x_0 = 0; \quad \Omega_1 : \begin{cases} x_0 = 0, \\ x_1^2 = 0. \end{cases}$$

Hence $\mathcal{X}$ is the affine plane $\mathbb{P}^2(\mathbb{R}) \setminus T_0$ equipped with the metric $(F_1(x_1, x_2) = x_1^2)$

$$d(x, y) = \sqrt{F_1(x - y, x - y)} = |x_1 - y_1|, \quad x = [1, x_1, x_2], \ y = [1, y_1, y_2] \in \mathcal{X}.$$

The lines of $\mathrm{L}_{\mathcal{X}}$ are the affine lines which do not pass through $\mathcal{D}_1$. The lines through $\Omega_1 = [0, 0, 1]$ are isotropic. The absolute of $\mathrm{L}_{\mathcal{X}}$ is given by $\Omega^*\colon \mathcal{D}_0 \supset T_0^* \supset \mathcal{D}_1$, where

$$\mathcal{D}_0 : u_2^2 = 0; \quad T_0^* : u_2 = 0; \quad \mathcal{D}_1 : \begin{cases} u_2 = 0, \\ u_1^2 = 0. \end{cases}$$

The angle between two lines $u\colon [u_0, u_1, 1]$ and $v\colon [v_0, v_1, 1]$ not passing through $\Omega_1 = [0, 0, 1]$ is euclidean and given by

$$\measuredangle(u, v) = \sqrt{G_1(u - v, u - v)} = |u_1 - v_1|,$$

where $G_1(u_0, u_1) = u_1^2$.

This name is explained by the close connection between the motions of this geometry and the Galilean transformations in classical kinematics (Figs. 13.15 and 13.16).

The following table summarizes the classification of plane projective geometries given above. Columns show the distance between points, while the rows the angles between lines.



Fig. 13.15 The absolute of a Galilean plane

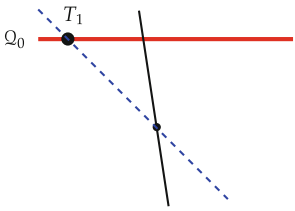


Fig. 13.16 Galilean plane

	Elliptic	Parabolic	Hyperbolic
Elliptic	S_2 : Elliptic plane	R_2 : Euclidean plane	${}^1S_2$: Hyperbolic plane
Euclidean	S_2^1 : Dual euclidean plane	F_2 : Galilean plane	${}^1S_2^1$: Dual Minkowski plane
Hyperbolic	${}^1S_2$: Dual hyperbolic plane	${}^1R_2$: Minkowski plane	${}^1S_2$: Double hyperbolic plane

(13.39)

The reader is suggested to examine the remaining 19 actual spaces of $\mathbb{P}^3(\mathbb{R})$ (see Exercise 13.2).

Remark 13.9 The distance $d(x, y) = k \log \rho(x, y, \xi, \eta)$ associated with the absolute $\widehat{F}$, where ξ and η are the intersections of the line xy with $\widehat{F}$, may be extended to $\mathbb{P}^2(\mathbb{R})$ in the following way (see (13.3)):

$$d(x, y) = \begin{cases} \text{not defined} & \text{both } x, y \in \mathcal{V}_+(\widehat{F}) \\ \infty & \text{only one between } x \text{ and } y \text{ belongs to } \mathcal{V}_+(\widehat{F}) \\ \in \mathbb{C} & \text{if } \rho(x, y, \xi, \eta) < 0. \end{cases} \tag{13.40}$$

Assume (13.3), then for $z_1, z_2 \in \mathbb{C}^*$,

$$\log(\frac{z_1}{z_2}) = \log z_1 - \log z_2 + 2\pi i \eta(z_1, z_2),$$

$$\eta(z_1, z_2) = \begin{cases} -1 & \pi < \arg z_1 - \arg z_2 \leq 2\pi \\ 0 & -\pi < \arg z_1 - \arg z_2 \leq \pi \\ 1 & -2\pi < \arg z_1 - \arg z_2 \leq -\pi. \end{cases}$$

If we disregard the order of the couples (x, y) and (ξ, η) the distance $d(x, y)$ is determined up to its sign and it is unique $\pmod{2k\pi i}$.

Remark 13.10 The *circles* of the nine geometries of $\mathbb{P}^2(\mathbb{R})$ are the loci of the points whose distances from a fixed point $\overline{y}$ are constant. If the absolute $\widehat{F}$ is not degenerate by (13.17) and (13.18) the circle of centre $\overline{y}$ and radius r is the set of points x such that

$$F(x, \overline{y})^2 = R^2 F(x, x) F(\overline{y}, \overline{y}),$$

where $R = \cos\left(\frac{r}{2k_e}\right)$ or $R = \cosh\left(\frac{r}{2k_h}\right)$. Hence the circles are conics Γ_R. The intersection $\mathcal{V}_+(\widehat{F}) \cap \Gamma_R$ is given by the system $F(x, x) = F(x, \overline{y})^2 = 0$ which has two double solutions. Hence the circles Γ_R centred at $\overline{y}$ are tangent to the absolute $\widehat{F}$ at the intersections of $\widehat{F}$ with the polar line $F(x, \overline{y}) = 0$ of the centre $\overline{y}$.

Remark 13.11 In the central column of Table (13.39) we find the Euclidean geometry, the Galilean geometry and the Minkowskian geometry. The field of complex numbers $\mathbb{C}$ gives an analytic model of the euclidean geometry. If we associate to the point $P = (x, y)$ the complex number $z = x + iy$, we can express various geometric objects and quantities in terms of complex numbers.

(1) The euclidean distance between the points z_1 and z_2 by $d^2(z_1, z_2) = |z_1 - z_2|^2$.
(2) The angle between the lines $r = z_0 z_1$ and $s = z_0 z_2$ by $\measuredangle(r, s) = \arg \frac{z_2 - z_0}{z_1 - z_0}$.
(3) An equation of the line $z_1 z_2$ by $(\overline{z}_1 - \overline{z}_2)z - (z_1 - z_2)\overline{z} + (z_1\overline{z}_2 + \overline{z}_1 z_2) = 0$.
(4) The circle centred at z_0 with radius r by $A|z|^2 + Bz - \overline{B}\overline{z} + C = 0$, $(A \neq 0$, $\mathfrak{Re}(A) = \mathfrak{Re}(C) = 0)$, where

$$\overline{z}_0 = -\frac{B}{A}, \quad |z_0|^2 - r^2 = \frac{C}{A}.$$

(5) Any euclidean isometry by

$$\begin{cases} z' &= pz + q \\ z' &= p\overline{z} + q, \end{cases} \qquad |p| = 1.$$

We want to introduce some number systems in order to extend (1)–(5) to Galilean geometry and Minkowskian geometry. A solution is given by the *dual numbers* and *double numbers* which appeared in works of Study and Clifford respectively for the first time. The reader can find an accurate description of these number systems in [35].

13.4 General Absolutes

Definition 13.12 We closely follow [37] changing only notation. Let

$$\mathcal{Q}_0 = \widehat{F}_0, \qquad F_0(x_0, \dots, x_n) = x_0^2 + \cdots + x_{p_0}^2 - x_{p_0+1}^2 - \cdots - x_{r_0}^2, \tag{13.41}$$

where $r_0 = \operatorname{rank}(\mathcal{Q}_0) - 1$ and $p_0 = \mathbf{i}(\mathcal{Q}_0) \geq \dfrac{r_0 - 1}{2}$. The hyperquadric $\mathcal{Q}_0$ contains the subspace

$$T_0 = \{[x_0, \dots, x_n] \mid x_0 = x_1 = \cdots = x_{r_0} = 0\}$$

of dimension $n - r_0 - 1$ (with coordinates $x_{r_0+1}, \dots, x_n$).

In T_0 consider the hyperquadric $\mathcal{Q}_1$

$$\mathcal{Q}_1 = \widehat{F}_1, \qquad F_1(x_{r_0+1}, \dots, x_n) = x_{r_0+1}^2 + \cdots + x_{p_1}^2 - x_{p_1+1}^2 - \cdots - x_{r_1}^2, \tag{13.42}$$

where $r_1 - r_0 = \operatorname{rank}(\mathcal{Q}_1)$ and $p_1 - (r_0 + 1) = \mathbf{i}(\mathcal{Q}_1)$. Furthermore

$$\mathbf{i}(\mathcal{Q}_1) \geq \frac{\operatorname{rank}(\mathcal{Q}_1)}{2} \quad \Longrightarrow \quad p_1 \geq \frac{r_1 + r_0 + 2}{2}.$$

The hyperquadric $\mathcal{Q}_1$ contains the subspace

$$T_1 = \{[x_0, \dots, x_n] \mid x_0 = x_1 = \cdots = x_{r_1} = 0\}$$

of dimension $n - r_1 - 1$ (with coordinates $x_{r_1+1}, \dots, x_n$).

In general, for $i \geq 0$, (defining $T_{-1} = \mathbb{P}^n(\mathbb{R})$, $r_{-1} = -1$) consider the hyperquadric $\mathcal{Q}_i$ of the subspace $T_{i-1} : x_0 = x_1 = \cdots = x_{r_{i-1}} = 0$, $\dim T_{i-1} = n - r_{i-1} - 1$

$$\mathcal{Q}_i = \widehat{F}_i, \qquad F_i(x_{r_{i-1}+1}, \dots, x_n) = x_{r_{i-1}+1}^2 + \cdots + x_{p_i}^2 - x_{p_i+1}^2 - \cdots - x_{r_i}^2, \tag{13.43}$$

where $r_i - r_{i-1} = \operatorname{rank}(\mathcal{Q}_i)$ and $p_i - (r_{i-1}+1) = \mathbf{i}(\mathcal{Q}_i)$, with

$$p_i \geq \frac{r_i + r_{i-1} + 2}{2}.$$

The hyperquadric $\mathcal{Q}_i$ contains the subspace $T_i = \{(x_0, \ldots, x_n) \mid x_0 = x_1 = \cdots = x_{r_i} = 0\}$ of dimension $n - r_i - 1$.

Thus we have constructed a finite descending chain Ω of hyperquadrics $\mathcal{Q}_i$ and of subspaces T_i

$$\Omega : \mathcal{Q}_0 \supset T_0 \supset \mathcal{Q}_1 \supset T_1 \supset \cdots \supset \mathcal{Q}_{m-1} \supset T_{m-1} \supset \mathcal{Q}_m, \tag{13.44}$$

where $\mathcal{Q}_m$ is a non-degenerate hyperquadric of T_{m-1} ($r_m = n$), i.e.

$$\mathcal{Q}_m = \widehat{F_m}, \qquad F_m(x_{r_{m-1}+1}, \ldots, x_n) = x^2_{r_{m-1}+1} + \cdots + x^2_{p_m} - x^2_{p_m+1} - \cdots - x^2_n = 0.$$

If T_{m-1} is a point we agree that $\mathcal{Q}_m$ is the "twice-counted point" (omitted if not necessary). We say that the couple $(\mathbb{P}^n(\mathbb{R}), \Omega)$ is a *space of general absolute* Ω, $\mathcal{Q}_i$ are the *absolute hyperquadrics* and T_i are the *absolute vertices*.

- If $r_0 > 0$ we denote $(\mathbb{P}^n(\mathbb{R}), \Omega)$ by ${}^{p_0 p_1 \cdots p_m}\mathcal{S}_n^{r_0 r_1 \cdots r_{m-1}}$ (we omit $r_m = n$) and we call ${}^{p_0 p_1 \cdots p_m}\mathcal{S}_n^{r_0 r_1 \cdots r_{m-1}}$ the *semi-non-euclidean space*. If $m = 1$ we have the *quasi-non-euclidean space* ${}^{p_0 p_1}\mathcal{S}_n^{r_0}$. Consider the following particular cases:
  - (a) If $p_i = r_i$, $i = 0, \ldots, m$ then $\mathcal{Q}_i$ is the imaginary cone $x^2_{r_{i-1}+1} + \cdots + x^2_{r_i} = 0$ with real vertex T_i or $x_0 = x_1 = \cdots = x^2_{r_i} = 0$ if $r_i = r_{i-1} + 1$, i.e. a couple of T_i. In denoting this absolute we shall omit the indices $p_0, \ldots, p_m$; we simply write $\mathcal{S}_n^{r_0 r_1 \cdots r_{m-1}}$ and we call it a *semi-elliptic space*. If $m = 1$ we have the *quasi-elliptic space* $\mathcal{S}_n^{r_0}$. If $r_0 = n$, i.e. $\mathcal{Q}_0$ is non-degenerate and $p_0 = n$, the semi-elliptic space is called an *elliptic space* and simply denoted by $\mathcal{S}_n$.
  - (b) If $p_i < r_i$ for some index i, ${}^{p_0 p_1 \cdots p_m}\mathcal{S}_n^{r_0 r_1 \cdots r_{m-1}}$ is called a *semi-hyperbolic space*. If $m = 1$ we have the *quasi-hyperbolic space* ${}^{p_0 p_1}\mathcal{S}_n^{r_0}$. If $r_0 = n$ and $p_0 < n$, we have the *hyperbolic space of index* p_0 denoted by ${}^{p_0}\mathcal{S}_n$.
- For $r_0 = 0$ (i.e. if $\mathcal{Q}_0$ is the double hyperplane $x_0^2 = 0$) we have the case of a *semi-euclidean space* ${}^{p_1 \cdots p_m}\mathcal{R}_n^{r_1 \cdots r_{m-1}}$ (we omit $p_0 = r_0 = 0$ and $r_m = n$).
  - (c) If $p_i = r_i$, $i = 0, \ldots, m$, we simply denote ${}^{p_1 \cdots p_m}\mathcal{R}_n^{r_1 \cdots r_{m-1}}$ by $\mathcal{R}_n^{r_1 \cdots r_{m-1}}$.
  - (d) For $m = 1$ we have the *quasi-euclidean space* ${}^{p_1}\mathcal{R}_n$; it is called a *euclidean space* if $p_1 = n$ and denoted by $\mathcal{R}_n : \mathcal{Q}_0 \supset T_0 \supset \mathcal{Q}_1$, where

$$\mathcal{Q}_0 : x_0^2 = 0, \ T_0 : x_0 = 0, \quad \mathcal{Q}_1 : \begin{cases} x_0 = 0 \\ x_1^2 + \cdots + x_n^2 = 0. \end{cases}$$

The space ${}^{1}\mathcal{R}_4$ is the *Minkowski pseudo-euclidean space*, which plays a fundamental role in the theory of relativity.

(e) Let $r_i = i$ for $i = 0, 1, \ldots, m-1$ with $m < n$, then $p_i = r_i$, $i = 0, \ldots, m-1$. Semi-euclidean spaces ${}^{p_1 \cdots p_m}\mathcal{R}_n^{r_1 \cdots r_{m-1}}$ will be called *Galilean spaces* and denoted by ${}^{p_m}\mathcal{R}_n^{123\cdots m-1}$. Hence $\mathcal{Q}_0 \colon x_0^2 = 0$ and

$$\mathcal{Q}_{i+1} = \{(x_0, \ldots, x_n) \mid x_0 = x_1 = \cdots = x_i = 0, x_{i+1}^2 = 0\},\ i < m-1$$
$$\mathcal{Q}_m = \{(x_0, \ldots, x_n) \mid x_m^2 + \cdots + x_{p_m}^2 - x_{p_m+1}^2 + \cdots + x_n^2 = 0\}$$

The semi-euclidean spaces ${}^{p_m}\mathcal{R}_n^{123\cdots m-1}$ are called Galilean spaces since $\mathcal{R}_4^1$ (i.e. $\mathcal{Q}_0 \colon x_0^2 = 0$, $T_0 \colon x_0 = 0$, $T_1 \colon x_0 = x_1 = 0$, $\mathcal{Q}_1 \colon x_0 = x_1 = 0, x_2^2 + x_3^2 + x_4^2 = 0$) serves a geometrical interpretation of space-time in the classical mechanics of Galileo and Newton.
If $m = n$ the Galilean space $\mathcal{R}_n^{12\cdots n-1}$ consists of a descending chain

$$\begin{aligned}
&\mathcal{Q}_0 : x_0^2 = 0,\\
&T_0 : x_0 = 0,\\
&\mathcal{Q}_1 : x_0 = 0, x_1^2 = 0,\\
&T_1 : x_0 = 0, x_1 = 0,\\
&\quad\vdots \qquad \vdots \qquad \vdots\\
&T_{n-1} : x_0 = 0, x_1 = 0, \ldots, x_{n-1} = 0,\\
&\mathcal{Q}_n : x_0 = 0, x_1 = 0, \ldots, x_{n-1} = 0, x_{n-1}^2 = 0.
\end{aligned}$$

We denote $\mathcal{R}_n^{12\cdots n-1}$ by $\mathcal{F}_n$ and call it a *flag space*. It is sufficient to give the sequence of subspaces T_i to determine a flag space (being the hyperquadrics $\mathcal{Q}_i$ uniquely associated).

The space ${}^{p_0}\mathcal{S}_n^{n-1}$ ($r_0 = n-1$ and $m = 1$), i.e.

$$\mathcal{Q}_0 = \widehat{F_0}, \qquad F_0(x_0, \ldots, x_n) = x_0^2 + \cdots + x_{p_0}^2 - x_{p_0+1}^2 + \cdots + x_{n-1}^2 = 0,$$

with vertex the point $T_0 = [0, \ldots, 0, 1]$ and $\mathcal{Q}_1$ the twice-counted point T_0, is called the *dual space* of ${}^{p_0}\mathcal{R}_n$, which is defined by $\mathcal{Q}_0 : x_0^2 = 0$, $T_0 : x_0 = 0$,

$$\mathcal{Q}_1 : x_1^2 + \cdots + x_{p_1}^2 - x_{p_1+1}^2 + \cdots + x_n^2 = 0,$$

with $p_1 = 1 + \mathbf{i}(\mathcal{Q}_1)$.

Definition 13.13 (Metrics) Let $(\mathbb{P}^n(\mathbb{R}), \Omega)$ be a space of general absolute

$$\Omega : \Omega_0 \supset T_0 \supset \cdots \supset \Omega_i \supset T_i \supset \cdots \supset \Omega_{m-1} \supset T_{m-1} \supset \Omega_m,$$

where Ω_i are as in (13.43). Let x, y be two distinct points of $\mathbb{P}^n(\mathbb{R})$, and $r = xy$ be the line of $\mathbb{P}^n(\mathbb{R})$ joining them.

(i) If r is contained in Ω_i for some index i, the distance $d(x, y)$ is not defined.
(ii) If r is tangent to Ω_i at a point $\xi \notin T_i$ but $r \not\subset \Omega_i$ we get

$$d(x, y) = k \log \rho(x, y, \xi, \xi) = 0 \qquad \text{if} \quad x, y \notin \Omega_i, \quad k \in \mathbb{C}.$$

The lines tangent to Ω_i at a point $\xi \notin T_i$ not lying on Ω_i are called *isotropic lines*.

(iii) If $x, y \in T_{i-1} \setminus \Omega_i$ and $r \cap \Omega_i = \{p, q\}$ with $p \neq q$ and $p, q \notin T_i$ (hence $r \subset T_{i-1}$), we say that r is a *non-euclidean line of level* i, $0 \leq i \leq m$. We define the *non-euclidean metric of order* i as follows:

$$d(x, y) = k \log \rho(x, y, p, q), \qquad k \in \mathbb{C}, k \neq 0, \tag{13.45}$$

which can assume complex values and it is unique $\pmod{2\pi i}$ and up to the sign (depending on the order of the couples (x, y) and (p, q)).

By Corollary 13.5, if p, q are real and do not separate the couple x, y, (13.45) is the hyperbolic metric on the open segment with endpoints p and q

$$d(x, y) = k_h \operatorname{arccosh}\left(\frac{F_i(x, y)}{\sqrt{F_i(x, x)F_i(y, y)}}\right), \qquad k_h > 0.$$

If p, q are complex conjugate we have the elliptic metric on r

$$d(x, y) = k_l \arccos\left(\frac{F_i(x, y)}{\sqrt{F_i(x, x)F_i(y, y)}}\right), \qquad k_l > 0.$$

(iv) If $x, y \notin \Omega_i$ and $\varnothing \neq r \cap \Omega_i \subset T_i$ but $r \cap T_{i+1} = \varnothing$ we say that r is a *euclidean line of level* i, $0 \leq i \leq m$. For every $x, y \notin \Omega_i$ we may change the representatives of x and y such that $F_i(x, x) = F_i(y, y) = 1$. Since

$$F_i(x + ty, x + ty) = F_i(x, x) + 2t F_i(x, y) + t^2 F_i(y, y)$$

and since r intersects Ω_i at only one (double) point $z \in T_i$ (otherwise $r \subset \Omega_i$) we have $F_i(x, y) = \pm 1$ so that $t = -F_i(x, y)$ and

$$z = x - F_i(x, y)y.$$

Changing the signs of all coordinates of y we may have $z = x - y \in T_i$. The *euclidean metric of order i* on $\mathbb{A}^n(\mathbb{R}) \cong \mathbb{P}^n(\mathbb{R}) \setminus T_0$ is defined as follows:

$$d(x, y) = \sqrt{|F_{i+1}(x - y, x - y)|}. \tag{13.46}$$

As seen in the cases of doubly hyperbolic and dual hyperbolic planes of Definition 13.8 with one absolute we may associate several different geometries which may also be *degenerate*, i.e. they are not even incidence structures.

Remark 13.14 By (13.46) and (13.45) a metric on all first-kind forms is defined. We define the *euclidean h-angle* (*non-euclidean h-angle*) between two subspaces $\mathcal{X}$ and $\mathcal{Y}$ of dimension h passing through a subspace $\mathcal{Z}$ of dimension $h - 1$ as the euclidean (non-euclidean) distance between $\mathcal{X}$ and $\mathcal{Y}$ as elements of the pencil with base locus $\mathcal{Z}$. In this way we have trivially extended Definition 13.8.

Definition 13.15 Let Ω be an absolute of $\mathbb{P}^n(\mathbb{R})$ as in (13.44). A projectivity $\varphi\colon \mathbb{P}^n(\mathbb{R}) \to \mathbb{P}^n(\mathbb{R})$ which preserves Ω (i.e. $\varphi(\Omega_i)\ \Omega_i$ and $\varphi(T_i) = T_i$, $i = 0, \ldots, m$) is called the *similarity* of Ω.

Proposition 13.16 *Every similarity φ of ${}^{p_0 p_1 \cdots p_m}\mathcal{S}_n^{r_0 r_1 \cdots r_{m-1}}$ preserves non-euclidean distances.*

Proof A similarity φ preserves cross-ratio and incidence so that distance (13.45) is invariant with respect to φ. □

Remark 13.17 A similarity may not preserve a euclidean metric (13.46).

Definition 13.18 A similarity φ of ${}^{l_0 l_1 \cdots l_r}\mathcal{S}_n^{m_0 m_1 \cdots m_{r-1}}$ which preserves the euclidean distance is called the *isometry* or *motion* of ${}^{l_0 l_1 \cdots l_r}\mathcal{S}_n^{m_0 m_1 \cdots m_{r-1}}$.

Corollary 13.19 *The isometries of elliptic space $\mathcal{S}_n$ are exactly the projectivities of $\mathbb{P}^n(\mathbb{R})$ which fix the imaginary ellipsoid*

$$x_0^2 + x_0^2 + \cdots + x_n^2 = 0.$$

13.5 Exercises

Exercise 13.1 Let $x = [x_0, \ldots, x_n]$ and $x = [y_0, \ldots, y_n]$ be two points of $\mathbb{P}^n(\mathbb{R})$ and

$$\|x\| = \left(\sum_{i=0}^{n} x_i^2\right)^{1/2}.$$

Prove that

$$d(x, y) = \min \left\{ \sum_{i=0}^{n} \left| \frac{x_i}{\|x\|} - \frac{y_i}{\|y\|} \right|, \sum_{i=0}^{n} \left| \frac{x_i}{\|x\|} + \frac{y_i}{\|y\|} \right| \right\},$$

is a metric on $\mathbb{P}^n(\mathbb{R})$. Is $d(x, y)$ a projective metric?

Exercise 13.2 Complete the list of degenerate absolutes of $\mathbb{P}^3(\mathbb{R})$ and describe the remaining 19 actual spaces associated with them.

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